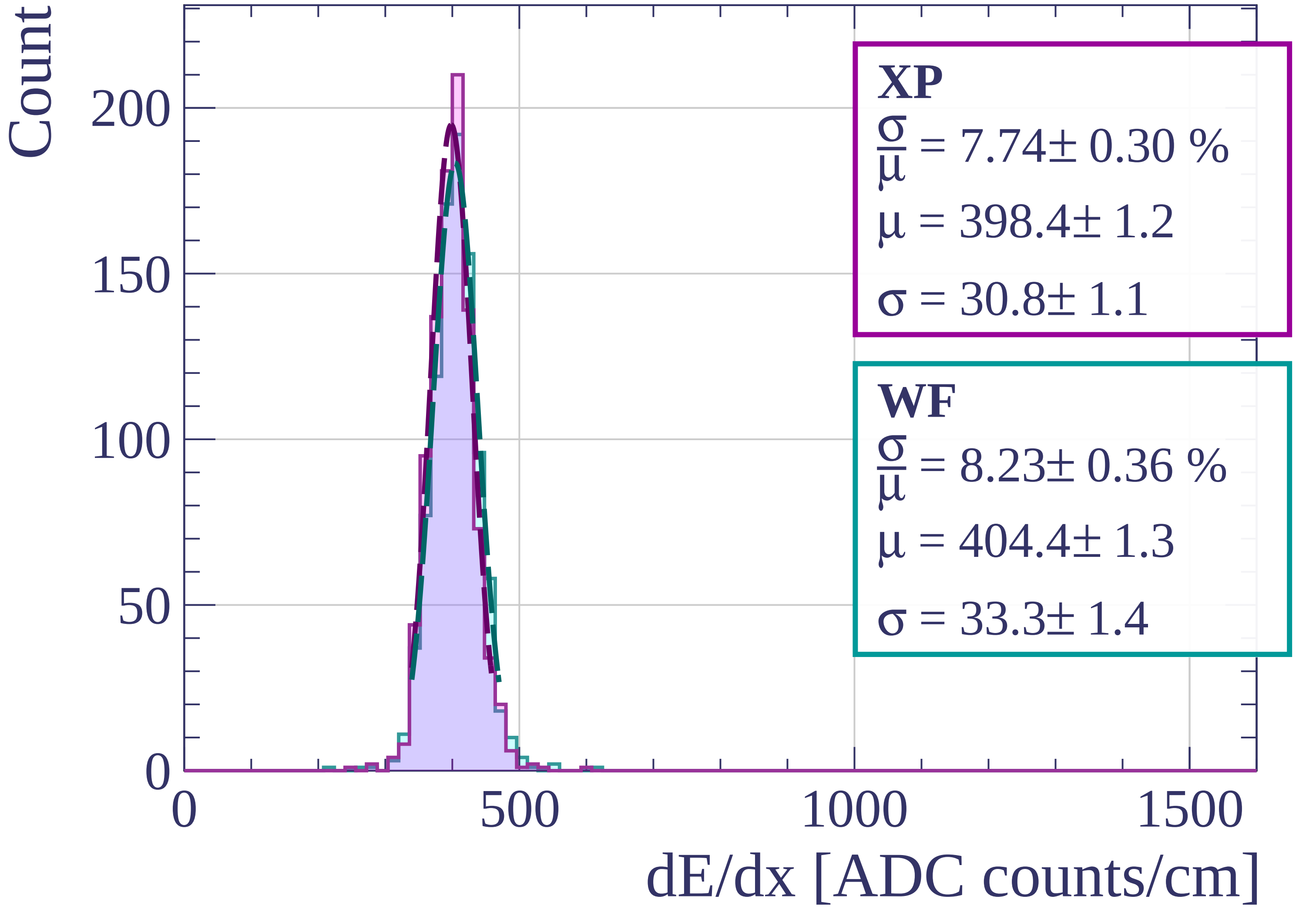
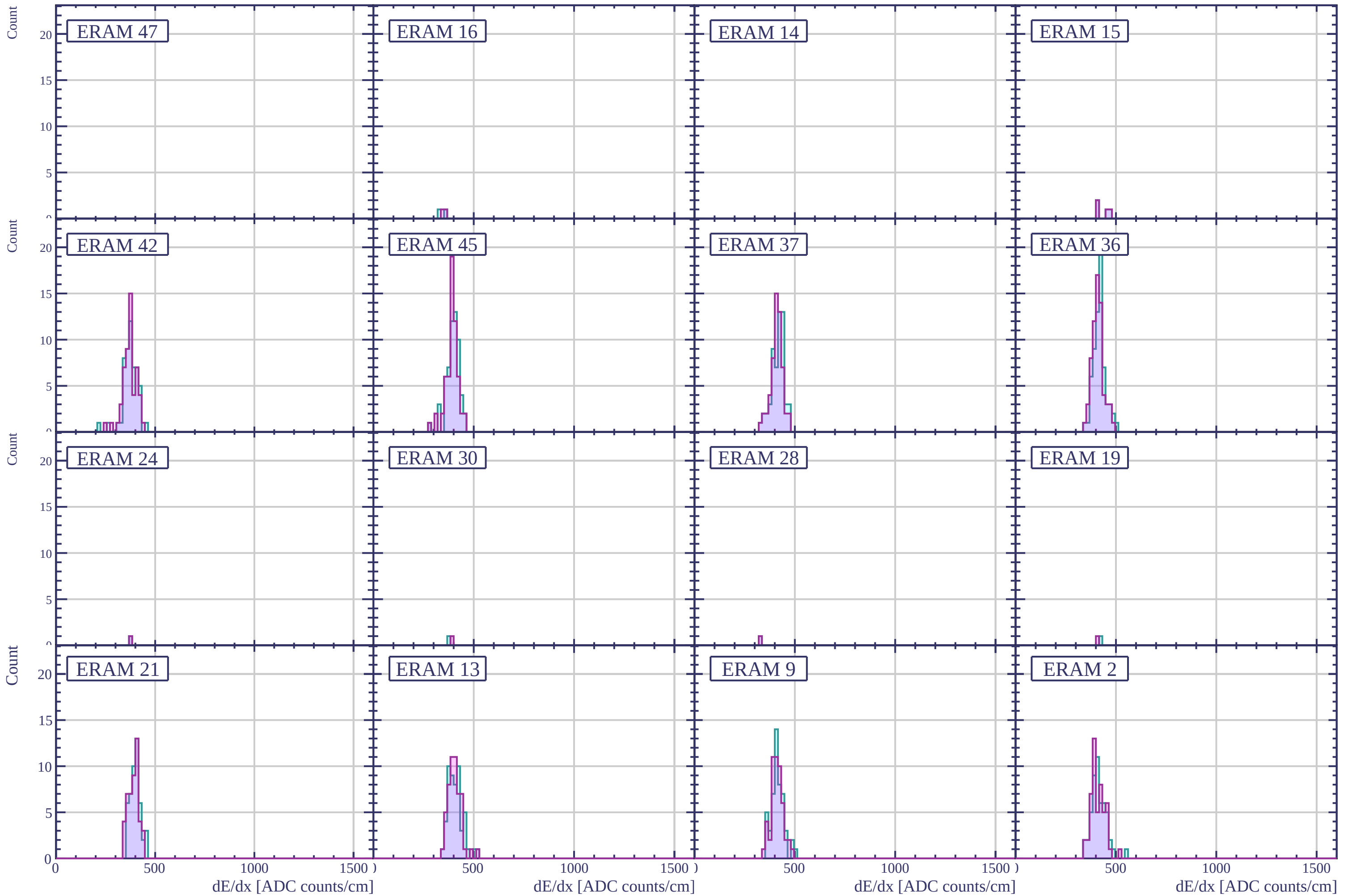
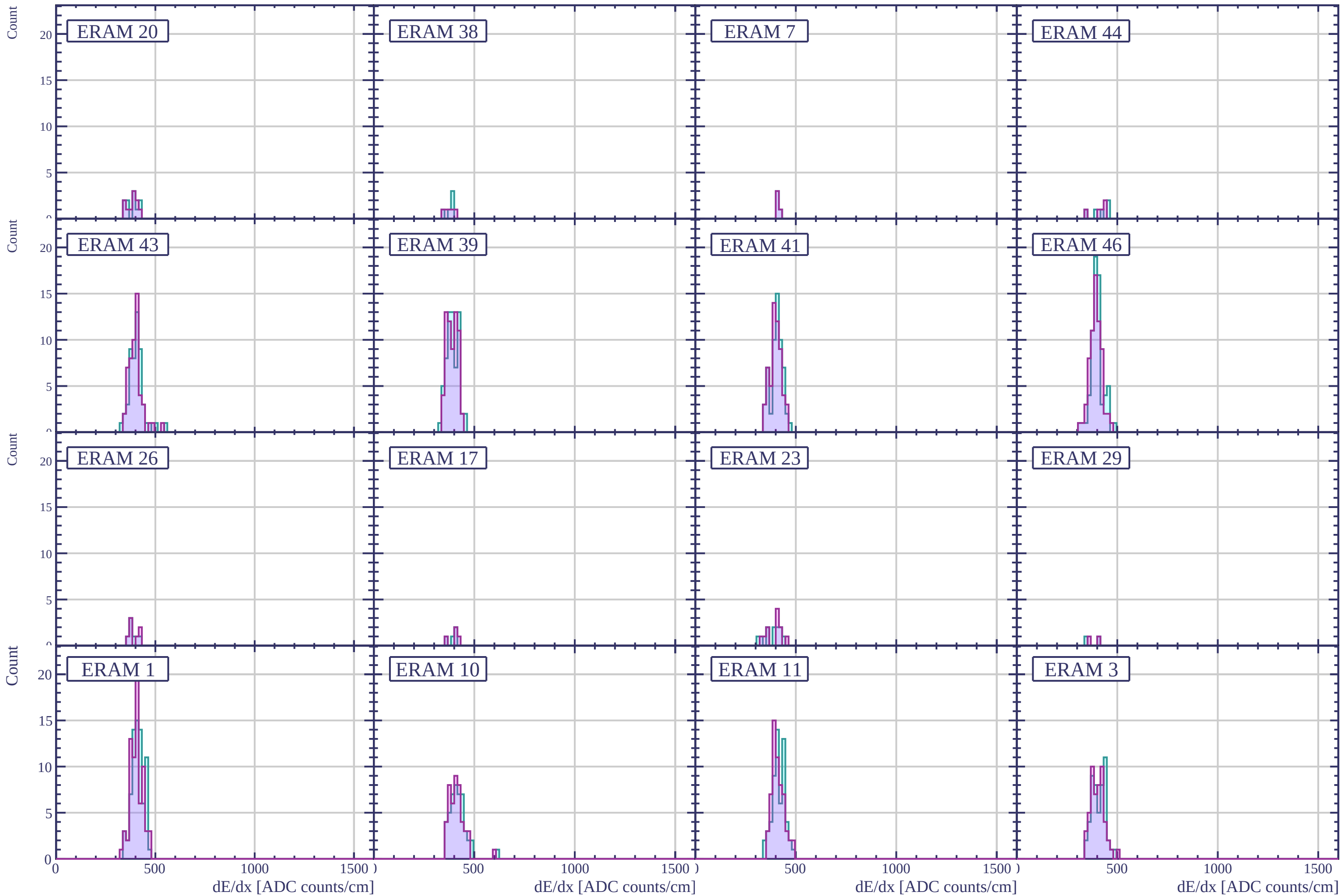
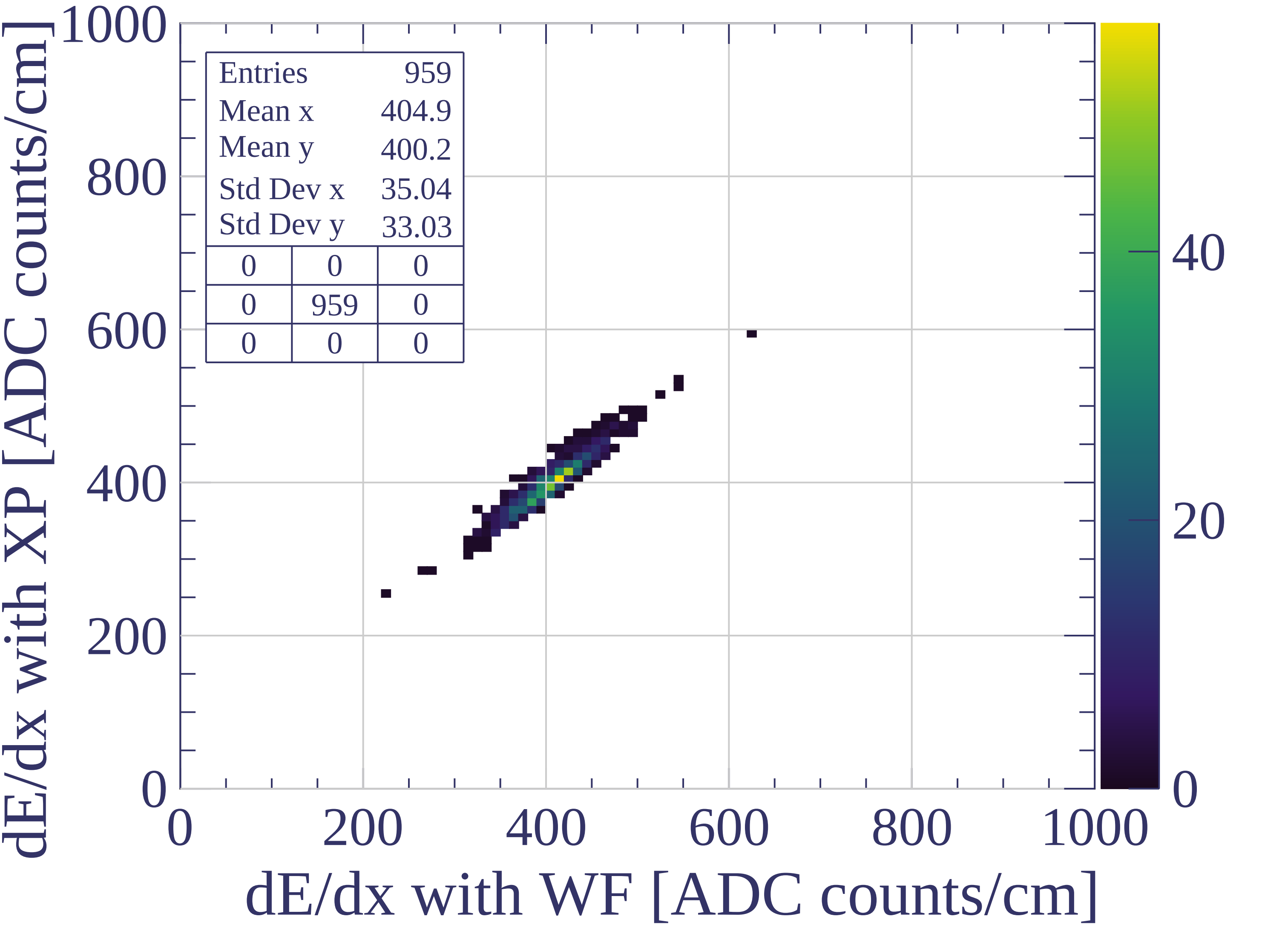
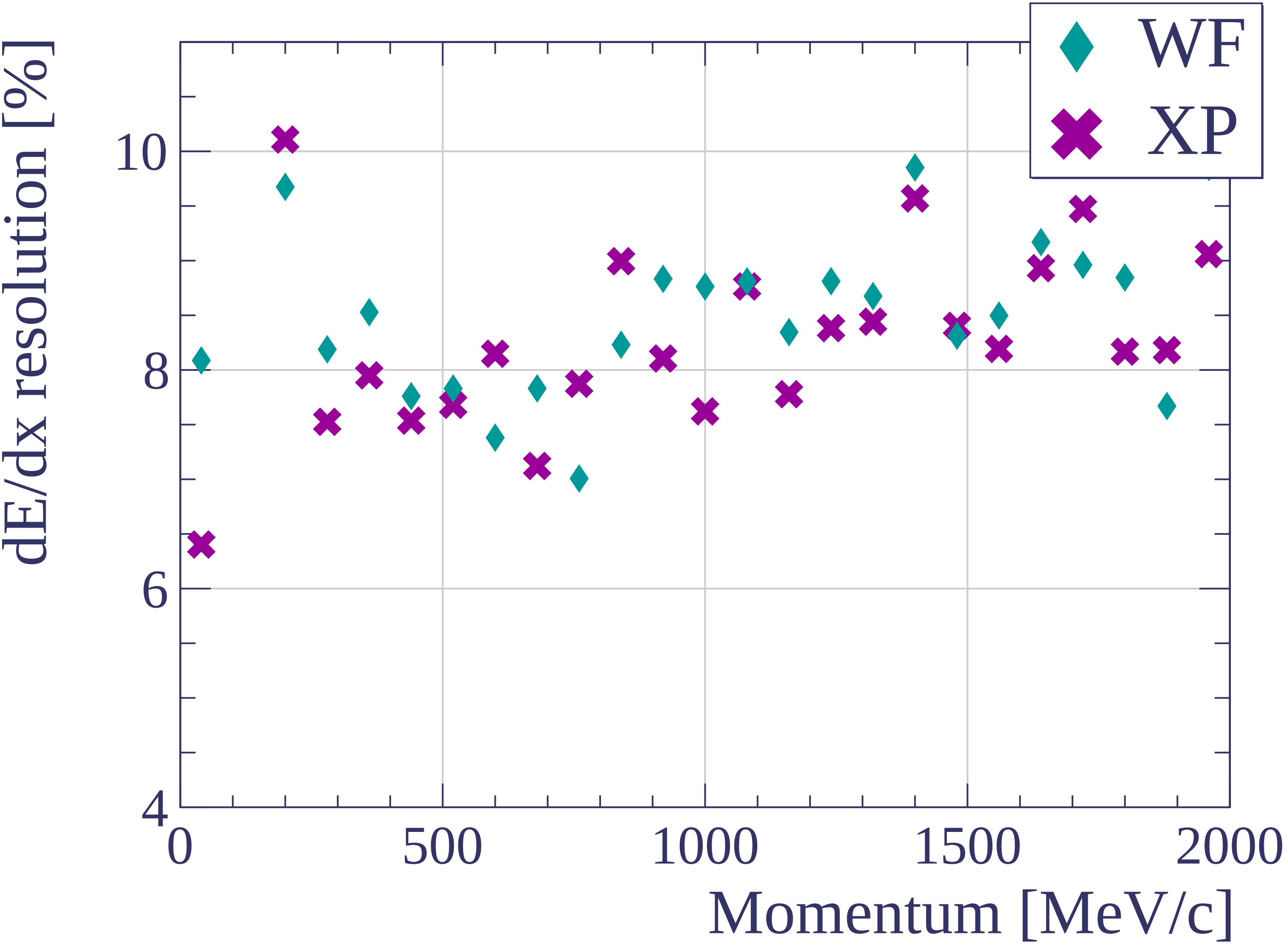


Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.74± 0.30 %
μ = 398.4± 1.2
σ = 30.8± 1.1
WF
σ/μ = 8.23± 0.36 %
μ = 404.4± 1.3
σ = 33.3± 1.4

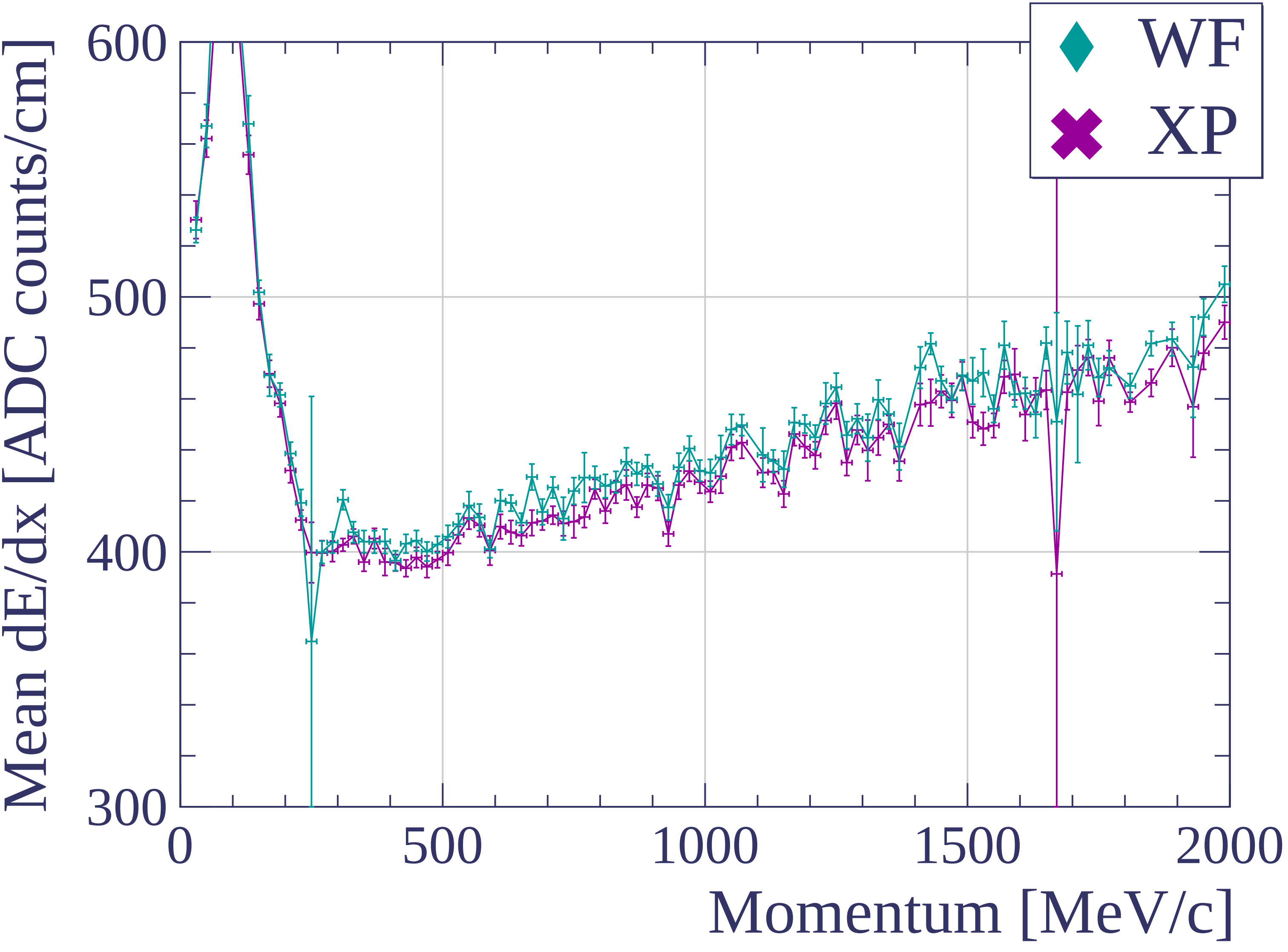

ERAM 47
ERAM 16
ERAM 14
ERAM 15
ERAM 42
ERAM 45
ERAM 37
ERAM 36
ERAM 24
ERAM 30
ERAM 28
ERAM 19
ERAM 21
ERAM 13
ERAM 9
ERAM 2
Count
dE/dx [ADC counts/cm]

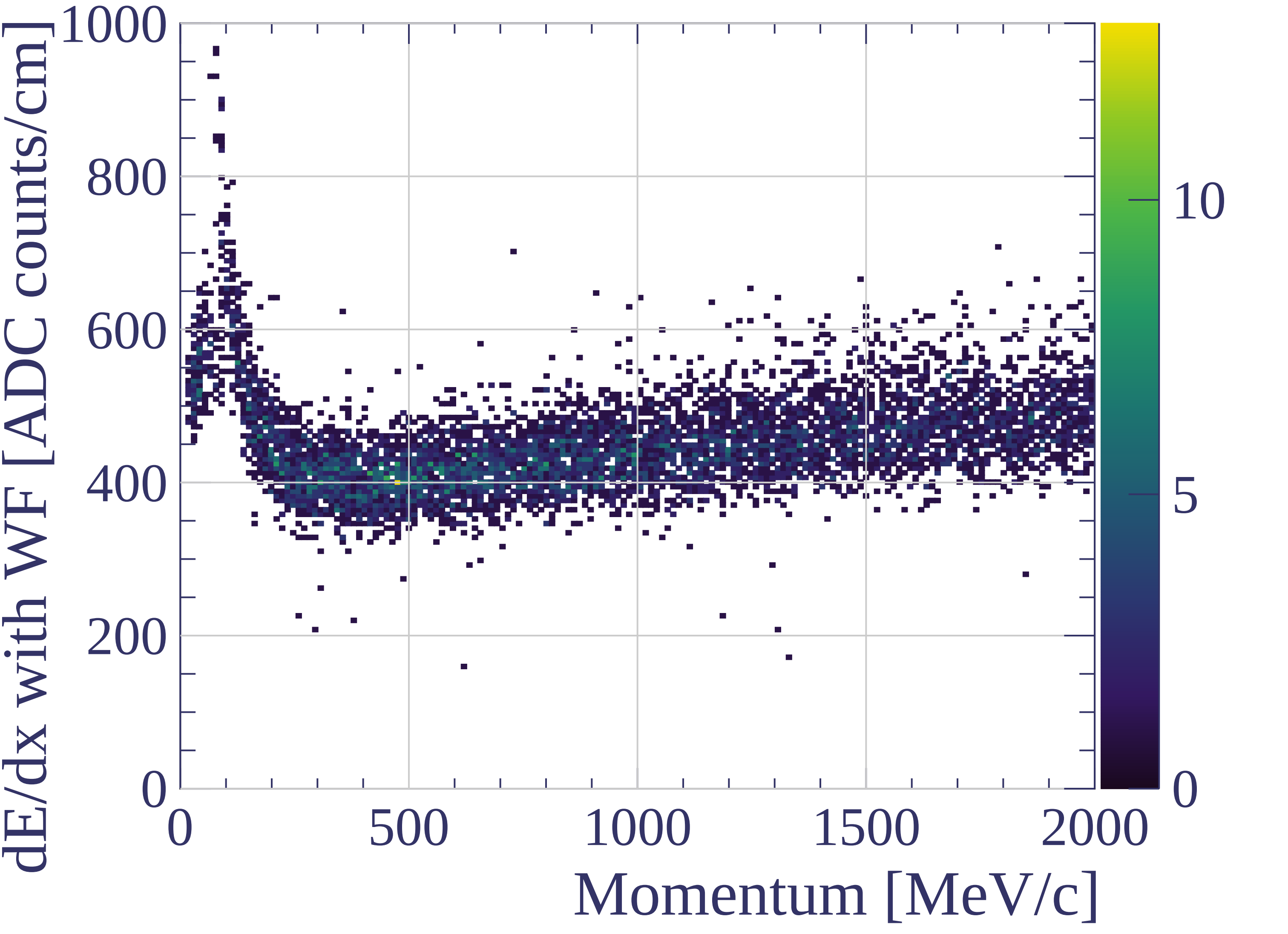

ERAM 20
ERAM 38
ERAM 7
ERAM 44
ERAM 43
ERAM 39
ERAM 41
ERAM 46
ERAM 26
ERAM 17
ERAM 23
ERAM 29
ERAM 1
ERAM 10
ERAM 11
ERAM 3
Count
dE/dx [ADC counts/cm]
0
5
10
15
20
500
1000
1500


Entries 959
Mean x 404.9
Mean y 400.2
Std Dev x 35.04
Std Dev y 33.03
0 0 0
0 959 0
0 0 0
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
20
40

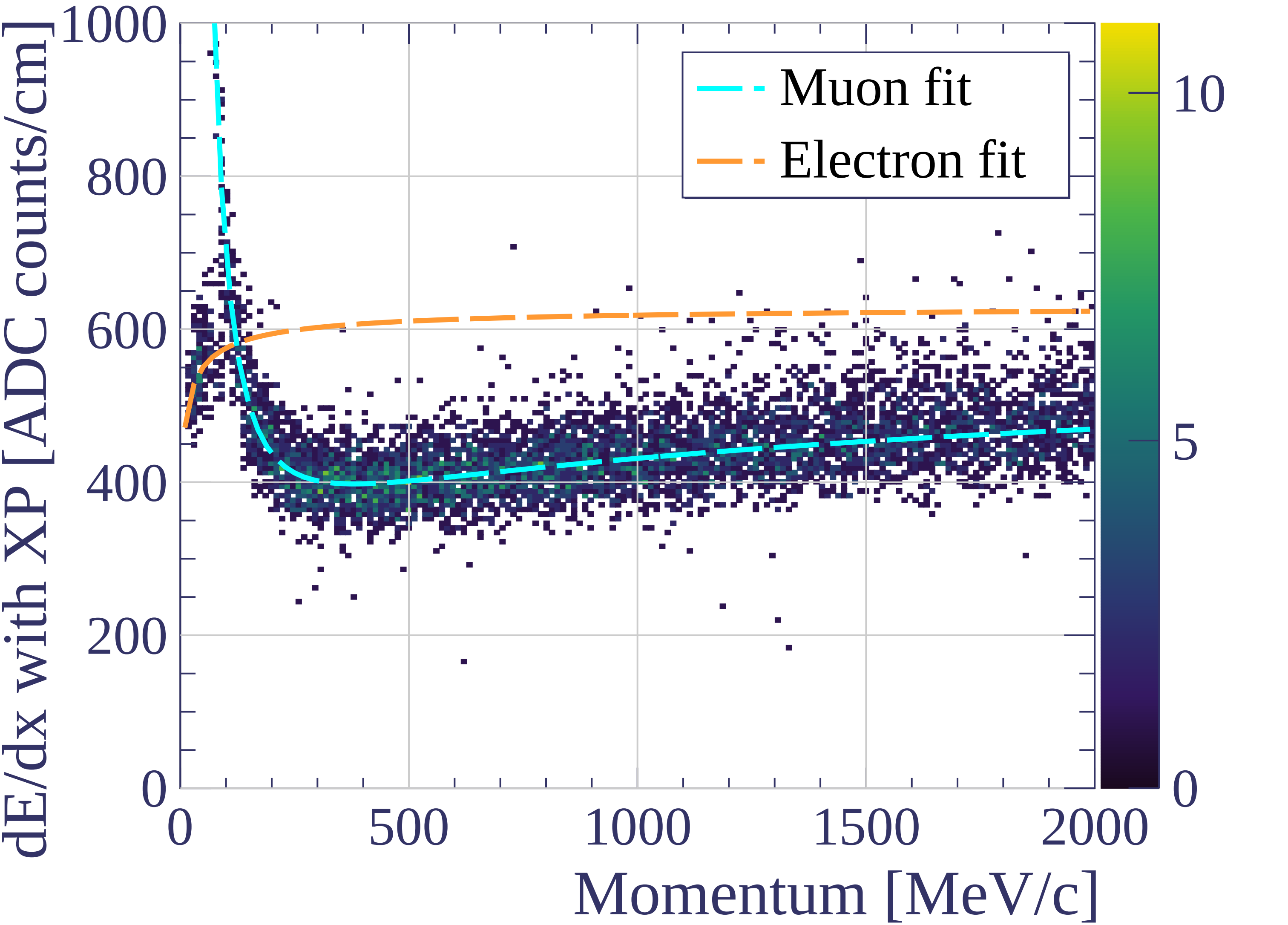


dE/dx resolution [%]
Momentum [MeV/c]
WF
XP
4
6
8
10
0
500
1000
1500
2000

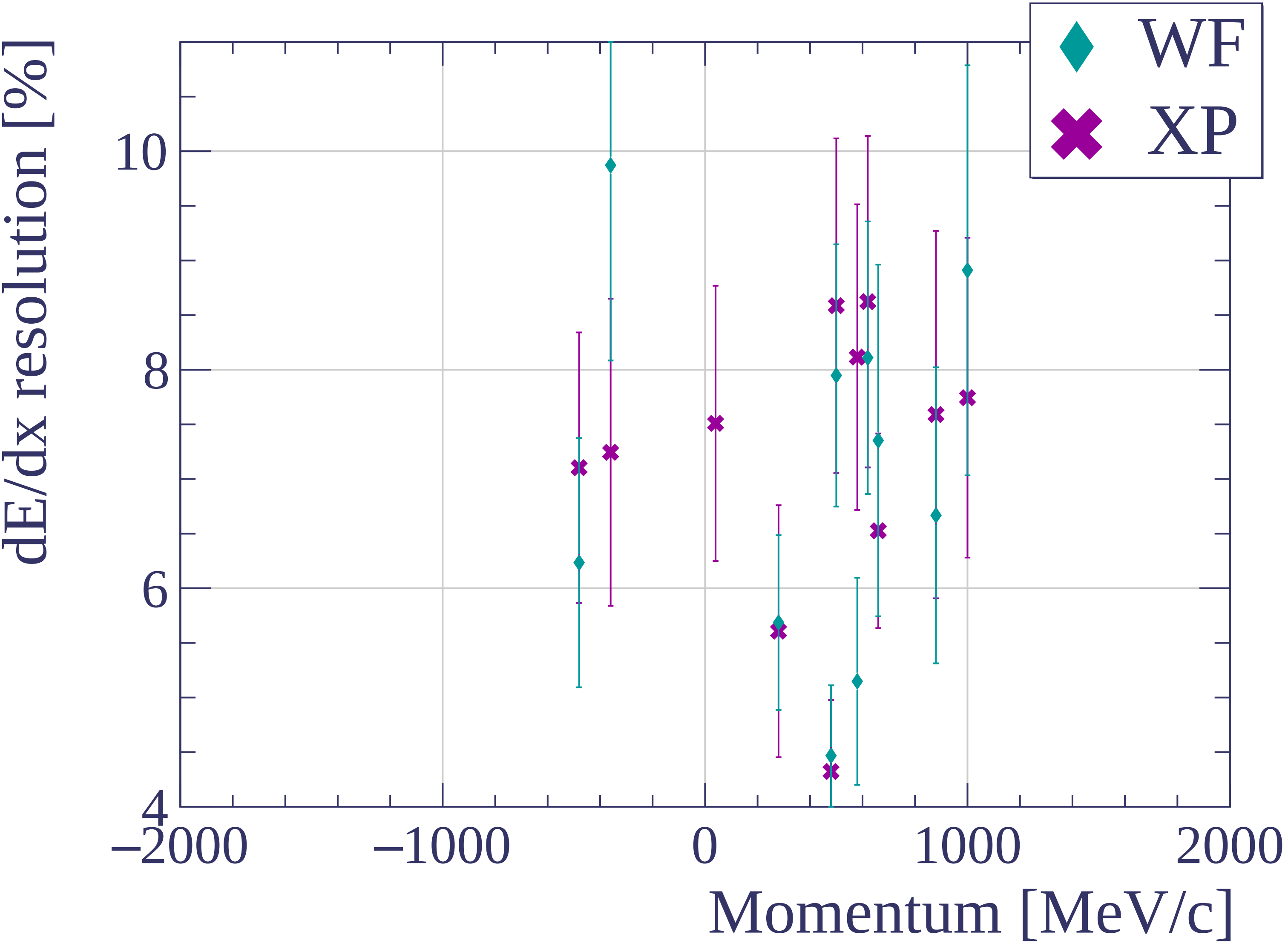

WF
XP
600
500
400
300
0
500
1000
1500
2000
Mean dE/dx [ADC counts/cm]
Momentum [MeV/c]

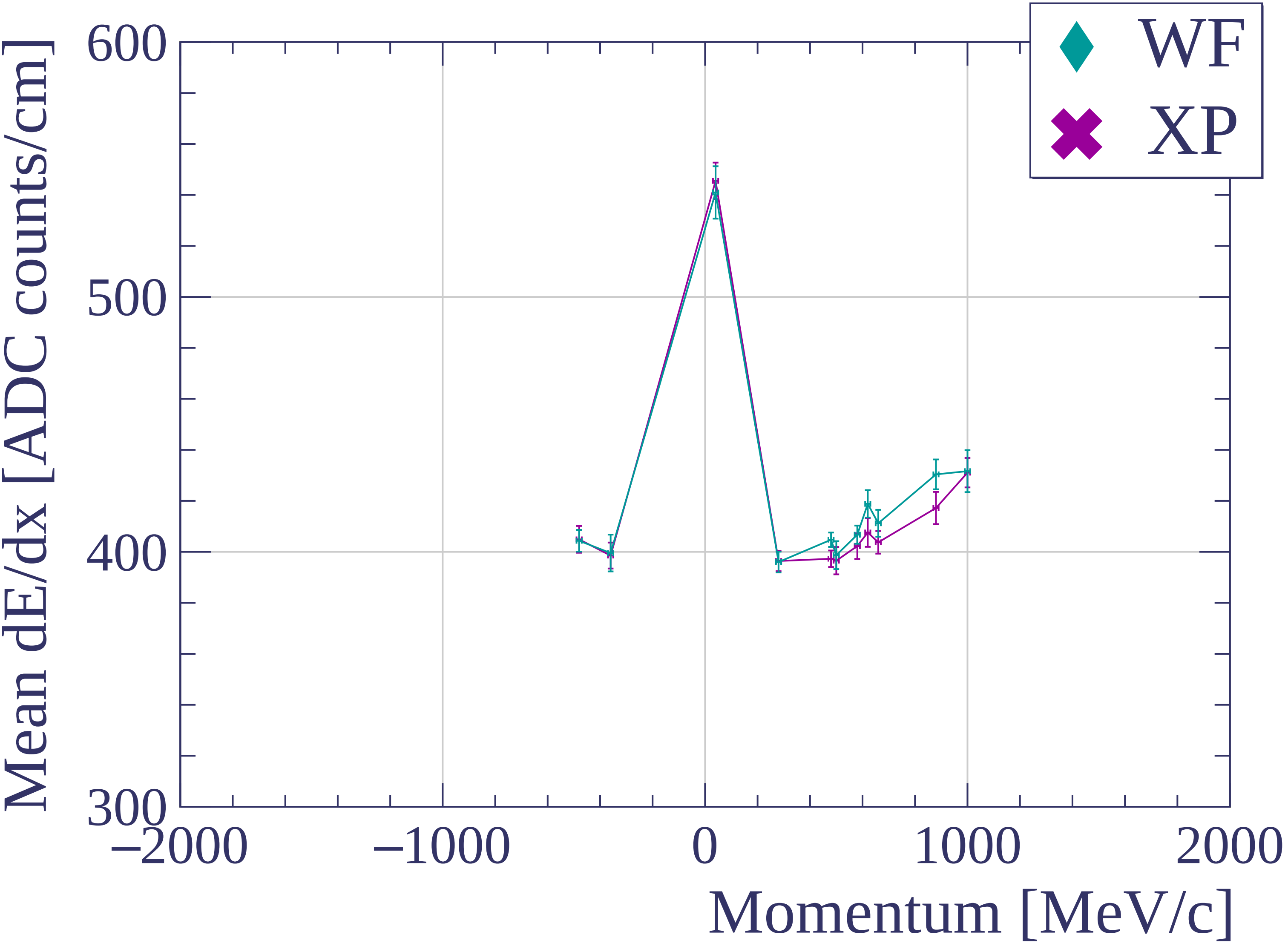


dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
0
500
1000
1500
2000
Momentum [MeV/c]
10
5
0

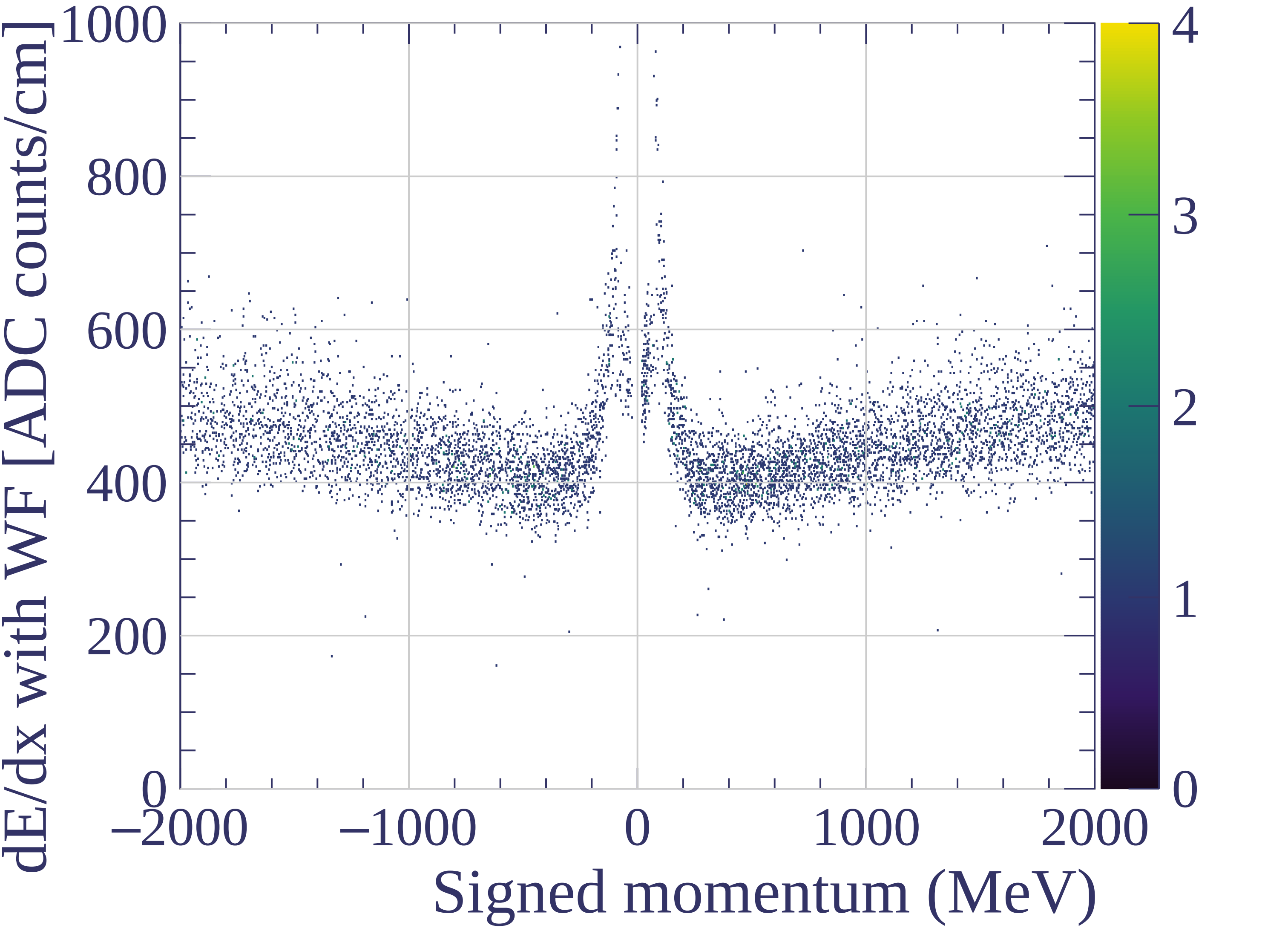


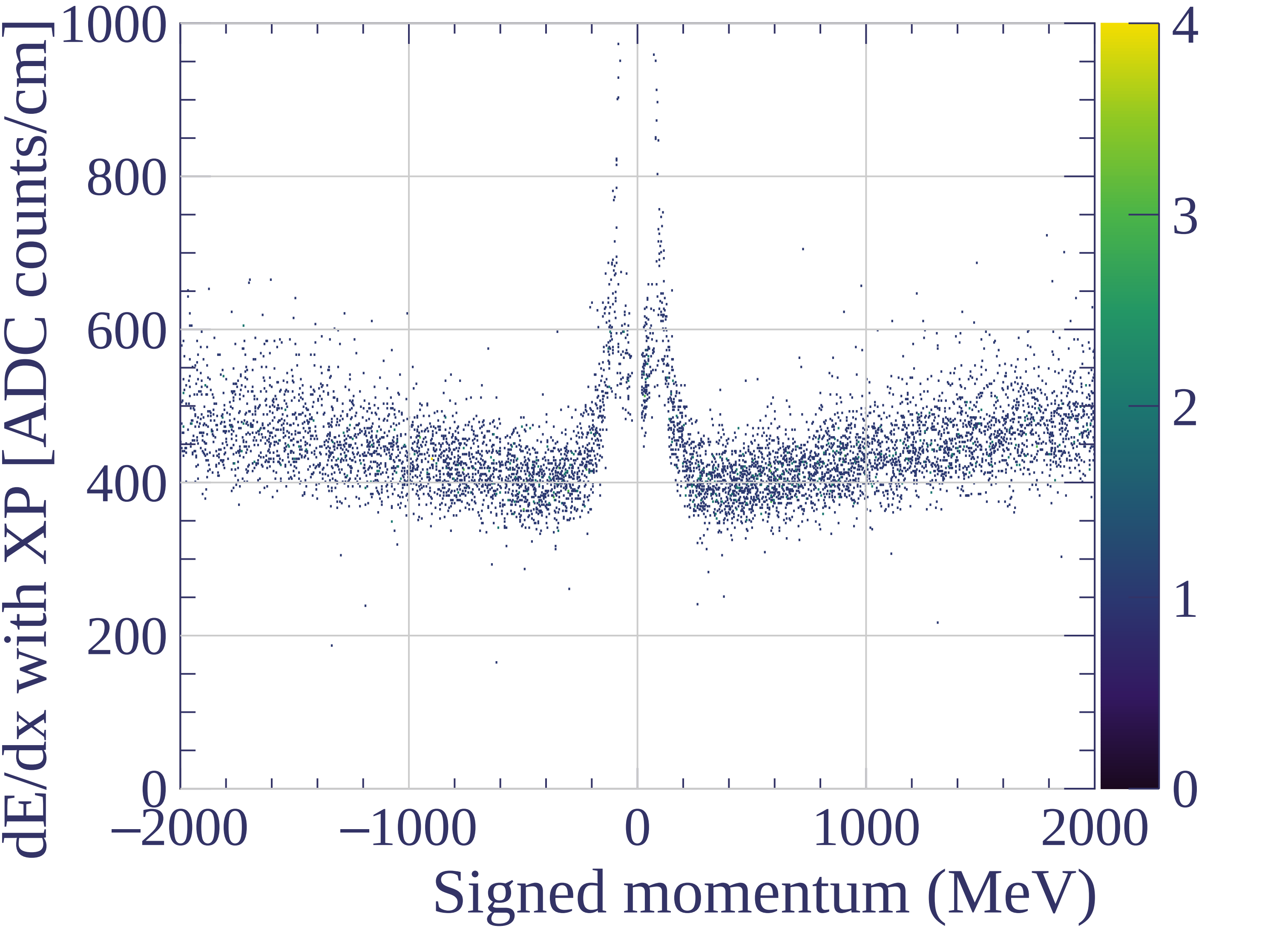
Muon fit
Electron fit
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
0
500
1000
1500
2000
0
5
10


WF
XP
dE/dx resolution [%]
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]

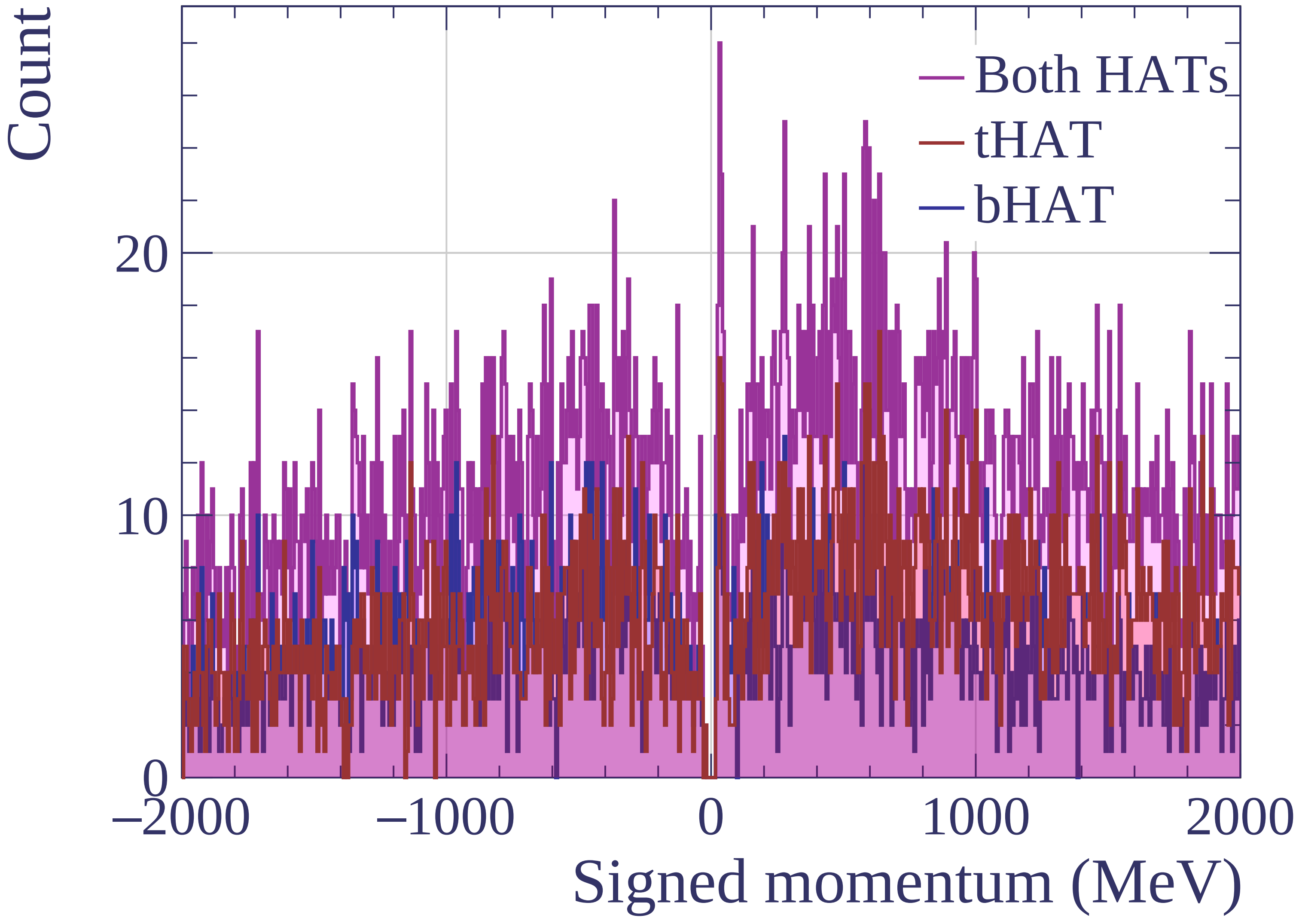


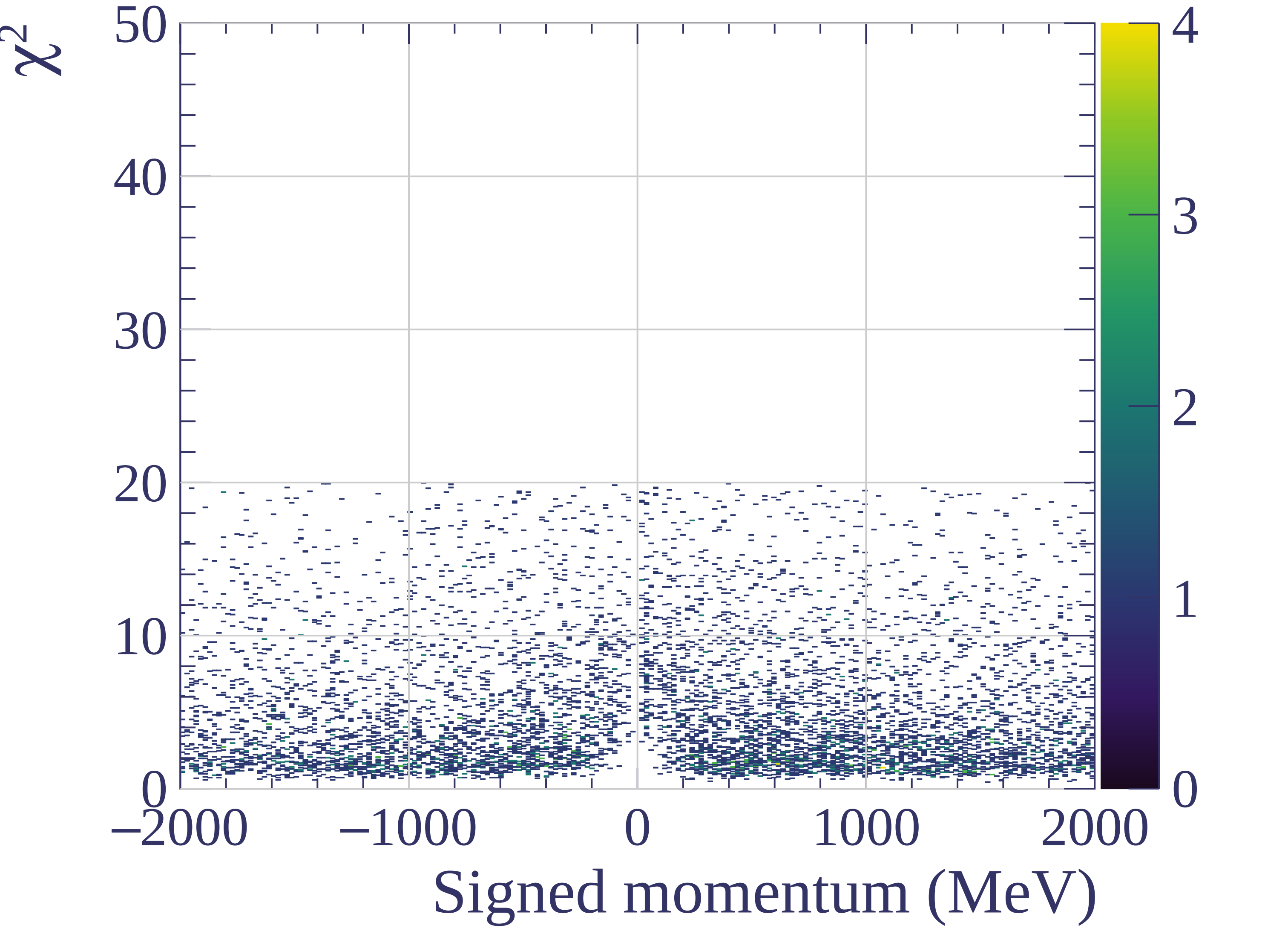
Mean dE/dx [ADC counts/cm]
600
500
400
300
−2000
−1000
0
1000
2000
Momentum [MeV/c]
WF
XP

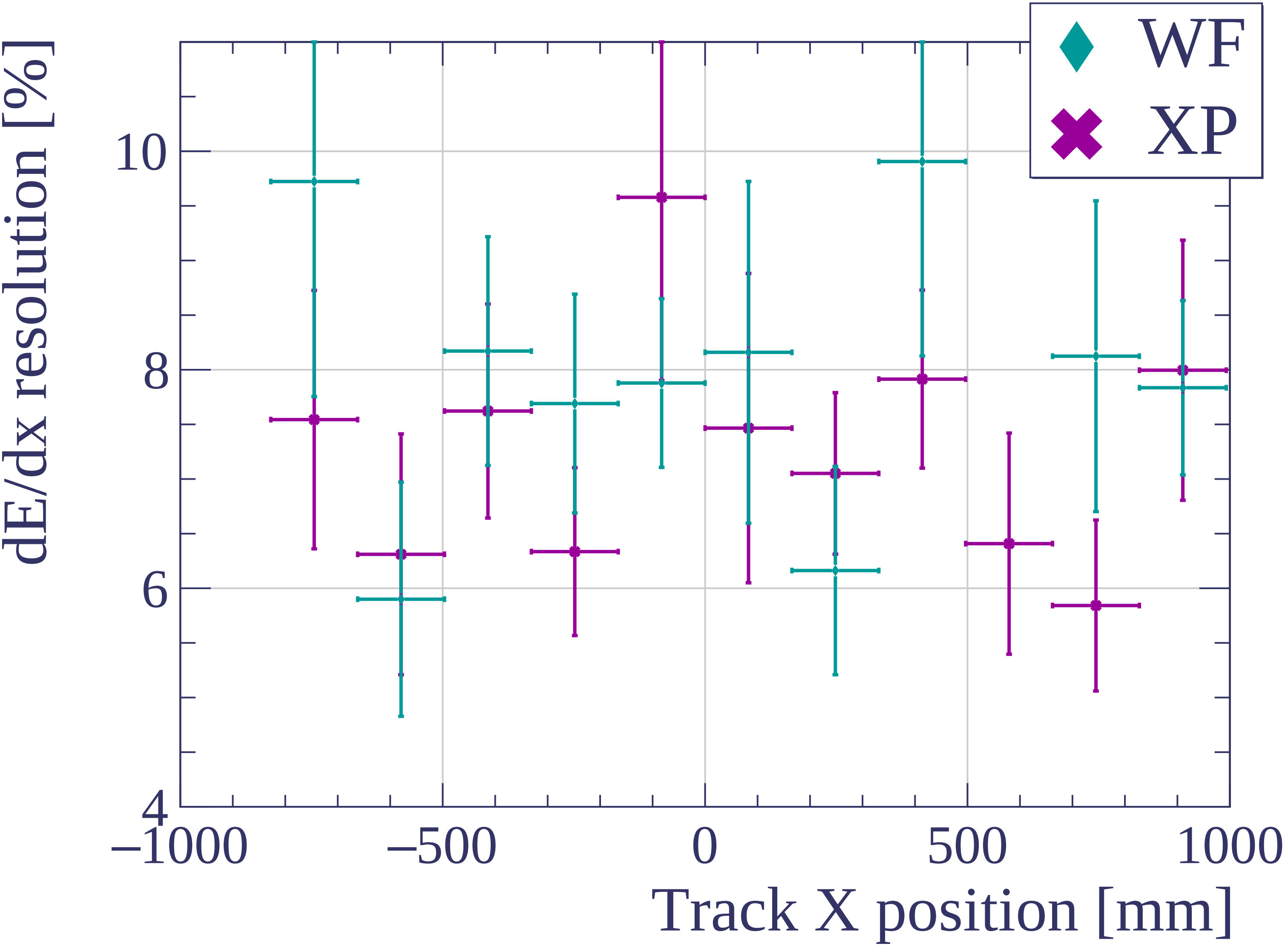

dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
4
3
2
1
0

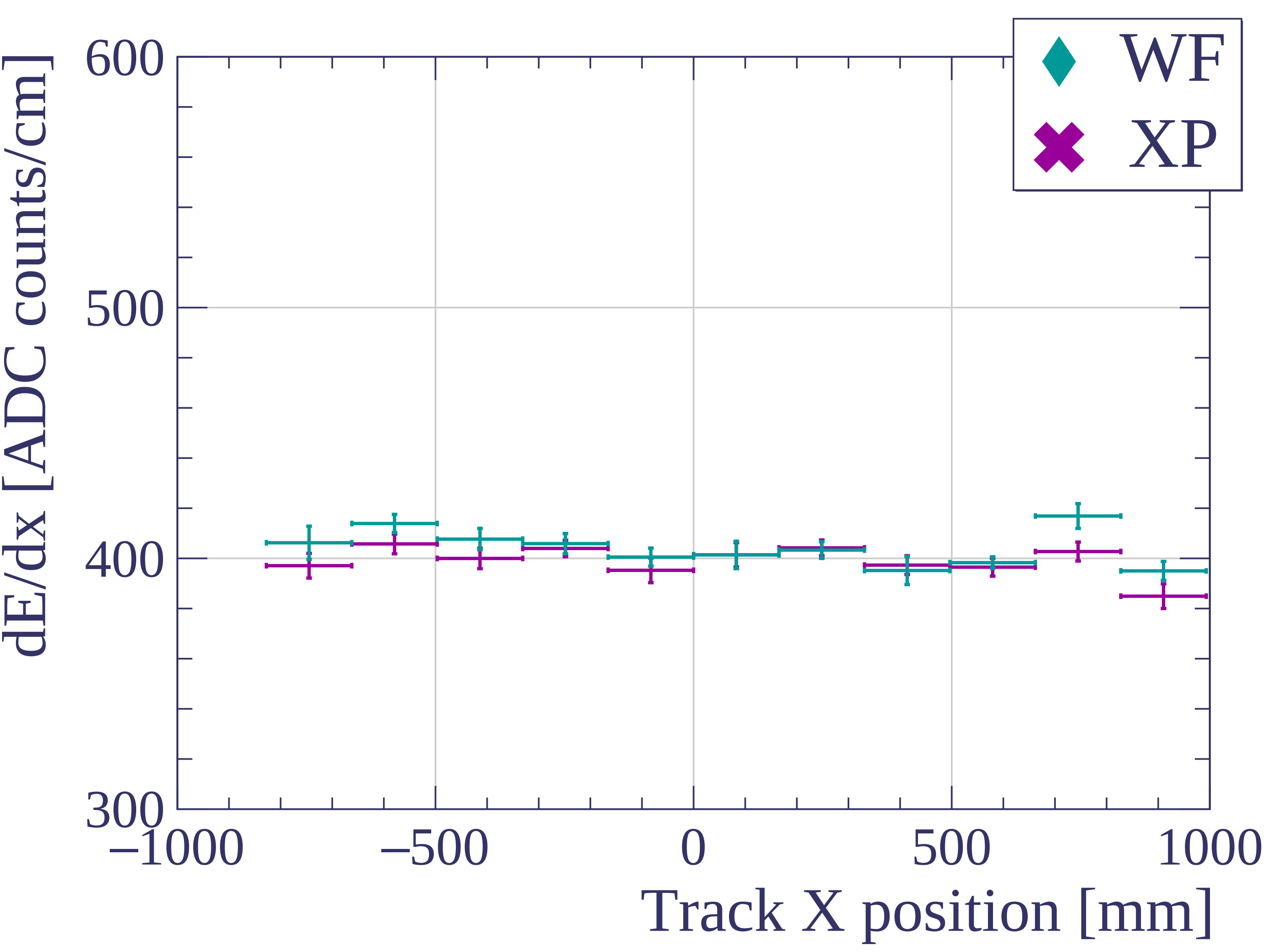


dE/dx with XP [ADC counts/cm]
Signed momentum (MeV)
0
200
400
600
800
1000
−2000
−1000
0
1000
2000
0
1
2
3
4

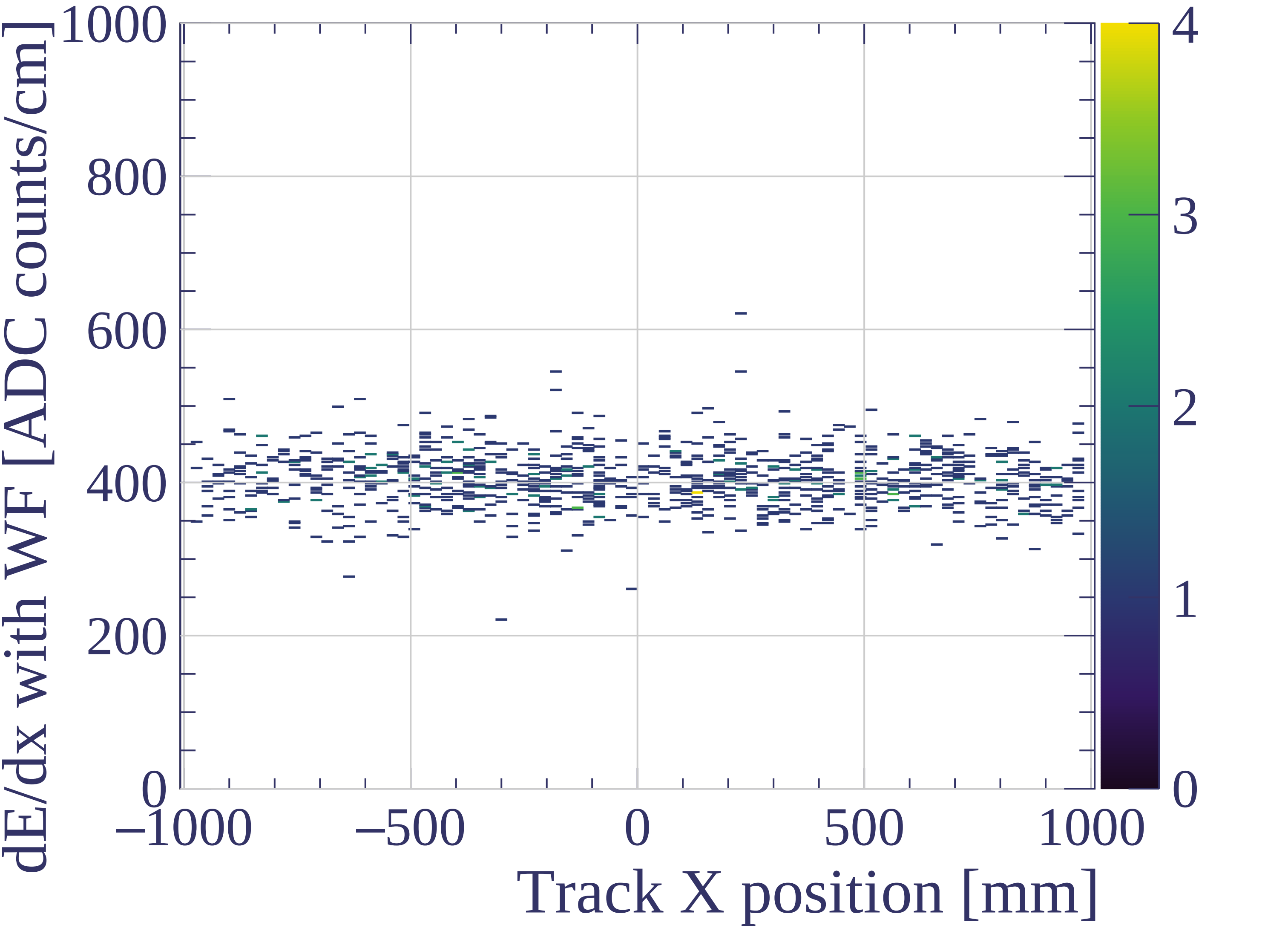

Count
Both HATs
tHAT
bHAT
0
10
20
−2000
−1000
0
1000
2000
Signed momentum (MeV)

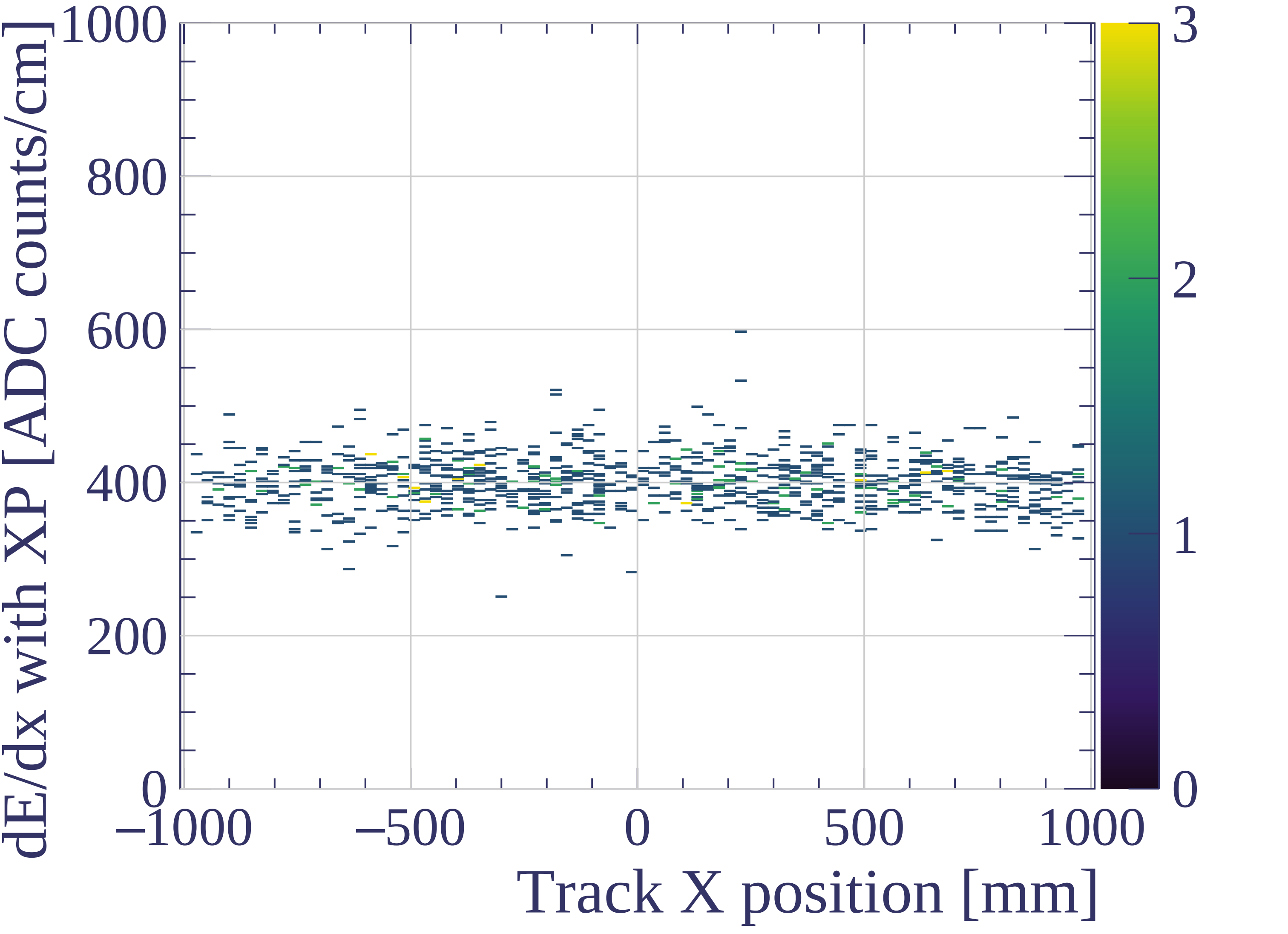

χ^2
50
40
30
20
10
0
4
3
2
1
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)


WF
XP
dE/dx resolution [%]
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]

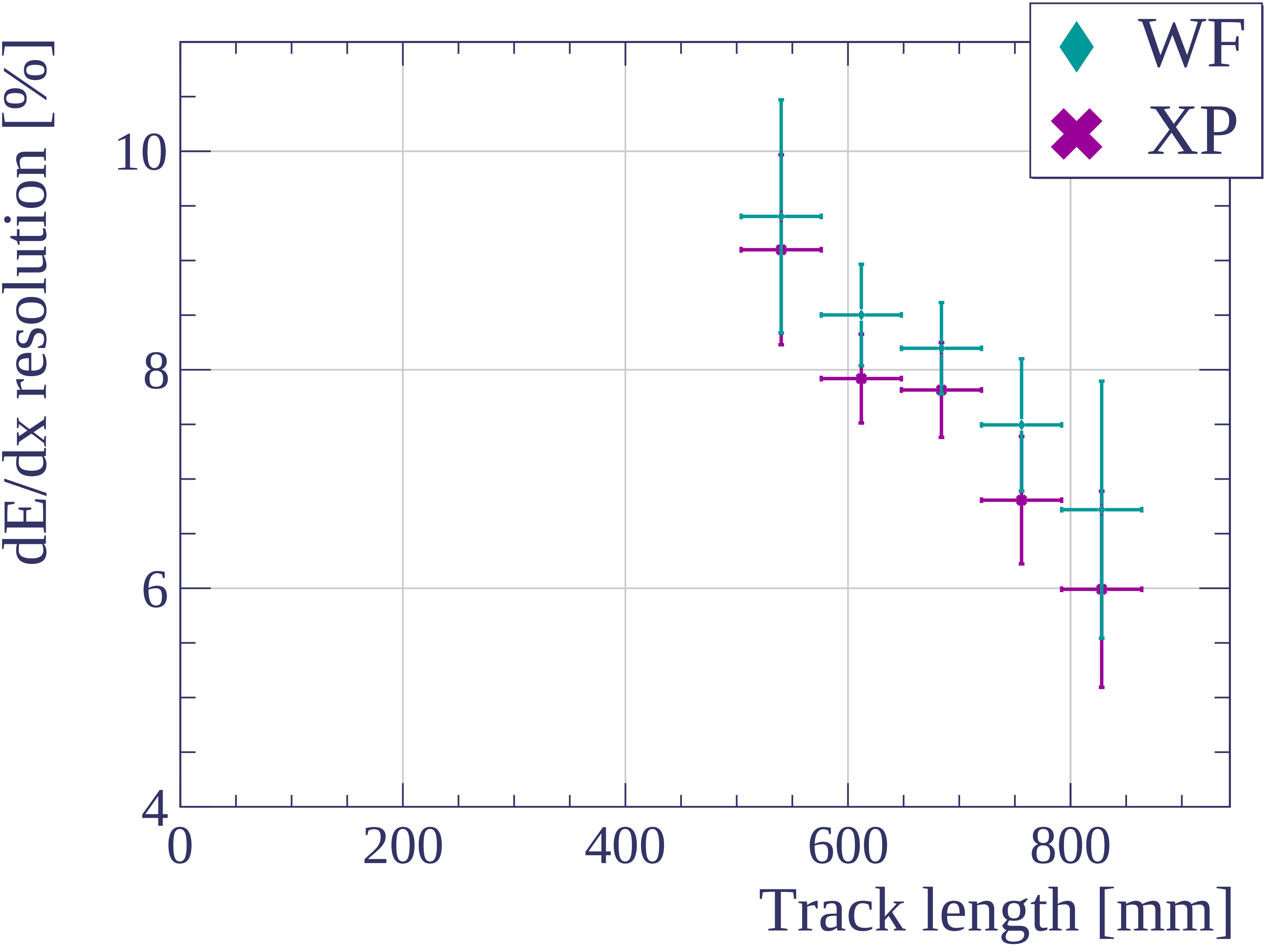


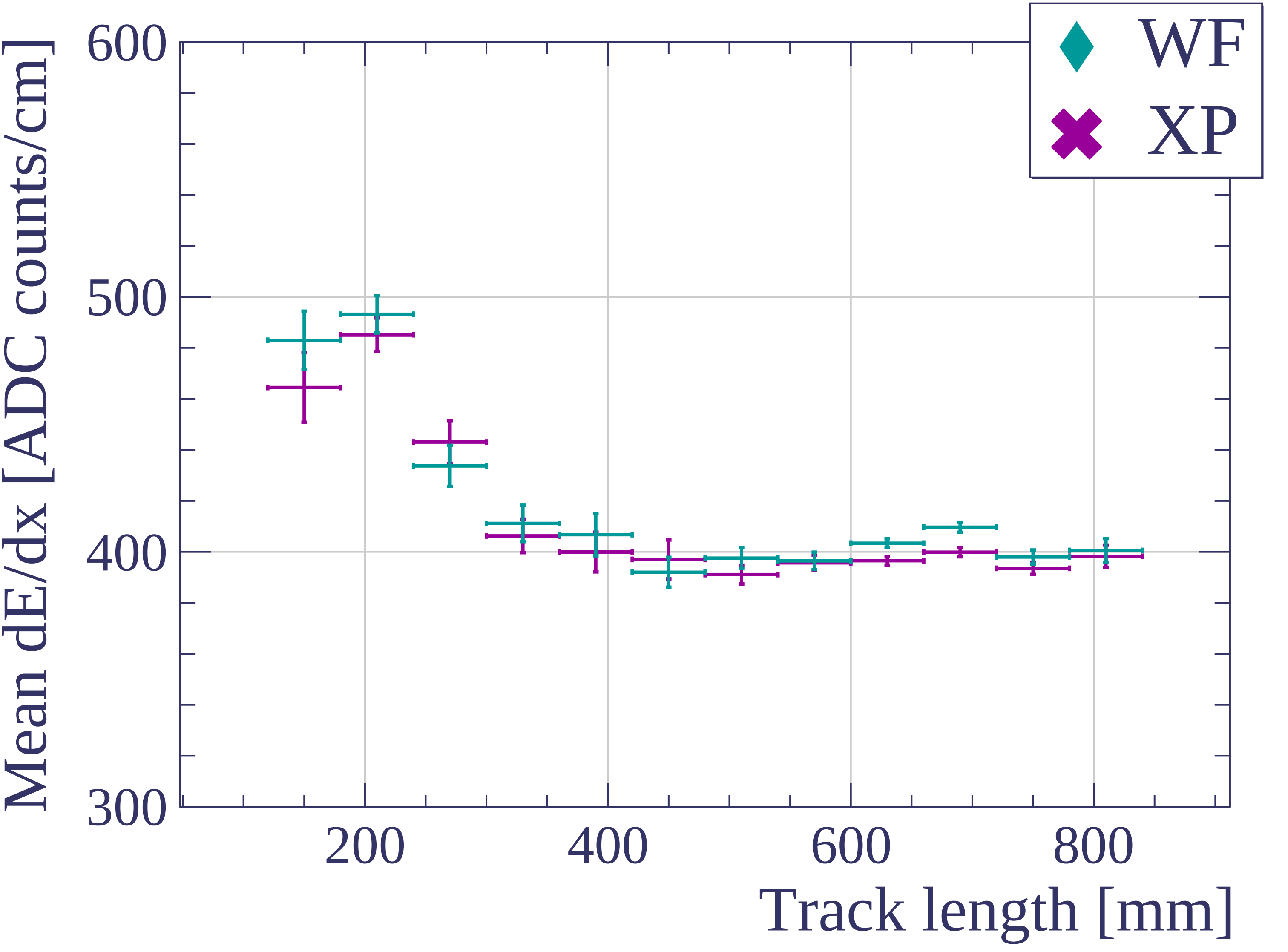
dE/dx [ADC counts/cm]
600
500
400
300
−1000
−500
0
500
1000
Track X position [mm]
WF
XP


dE/dx with WF [ADC counts/cm]
Track X position [mm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
4
3
2
1
0

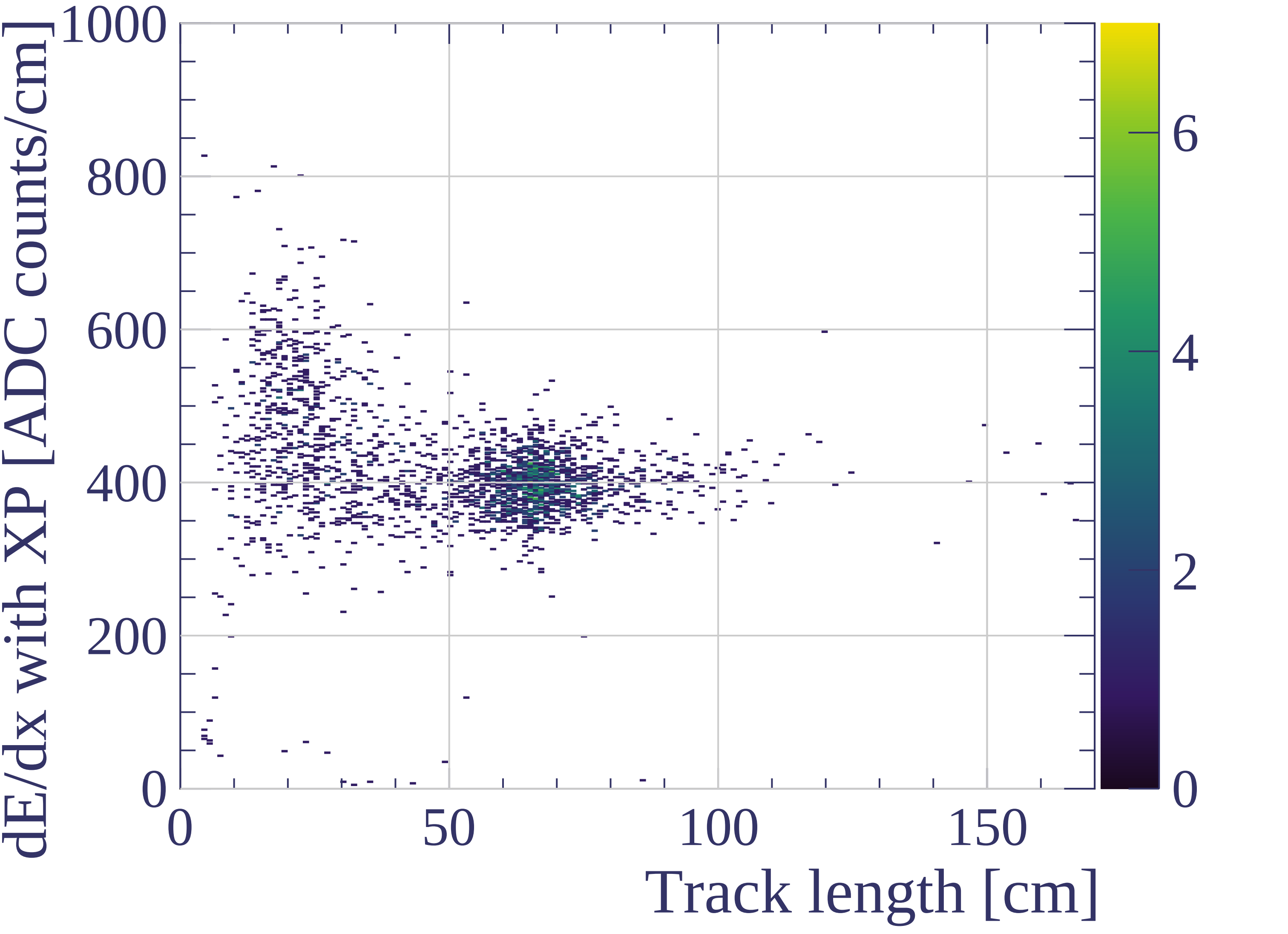


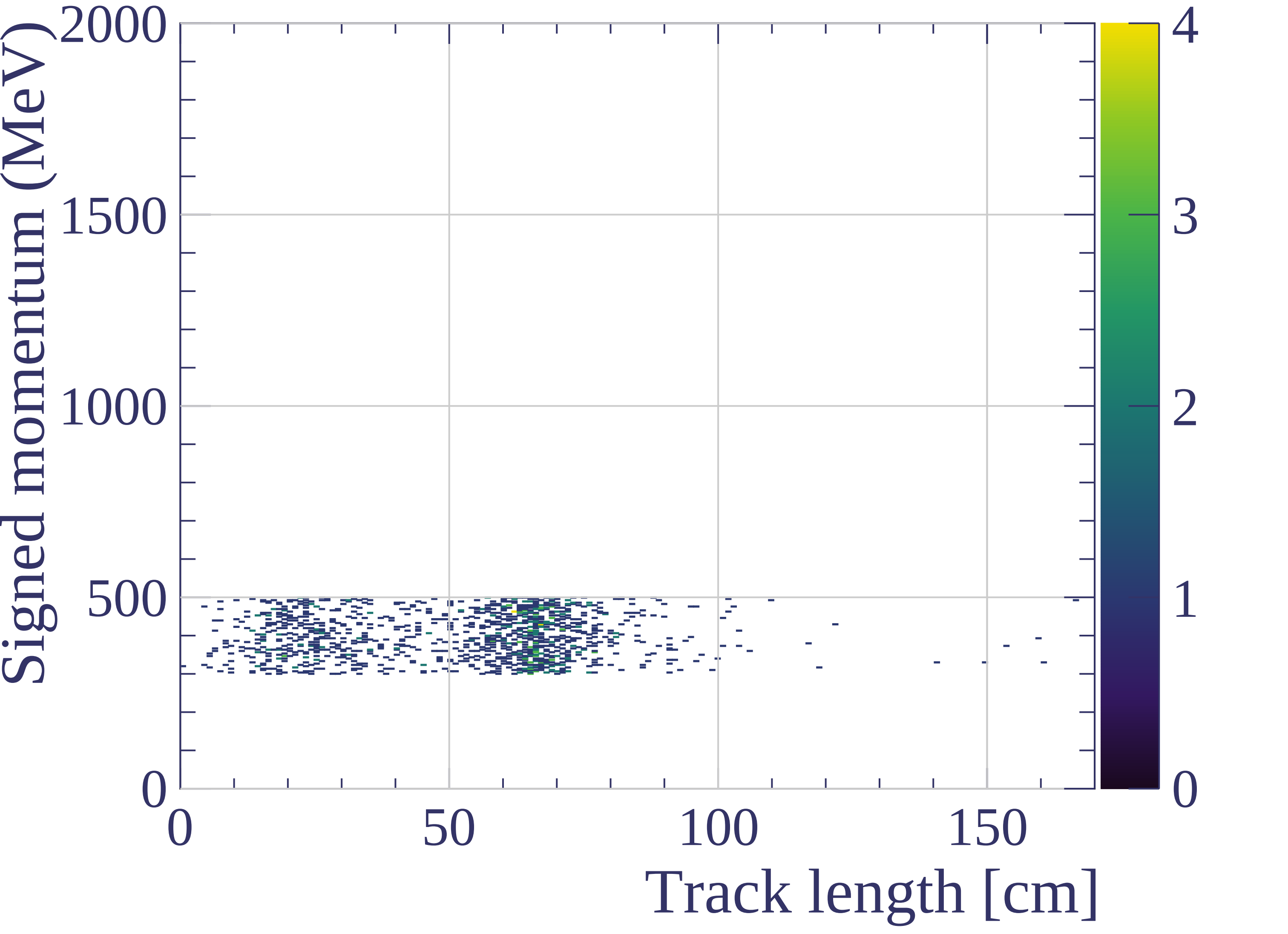
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
3
2
1
0

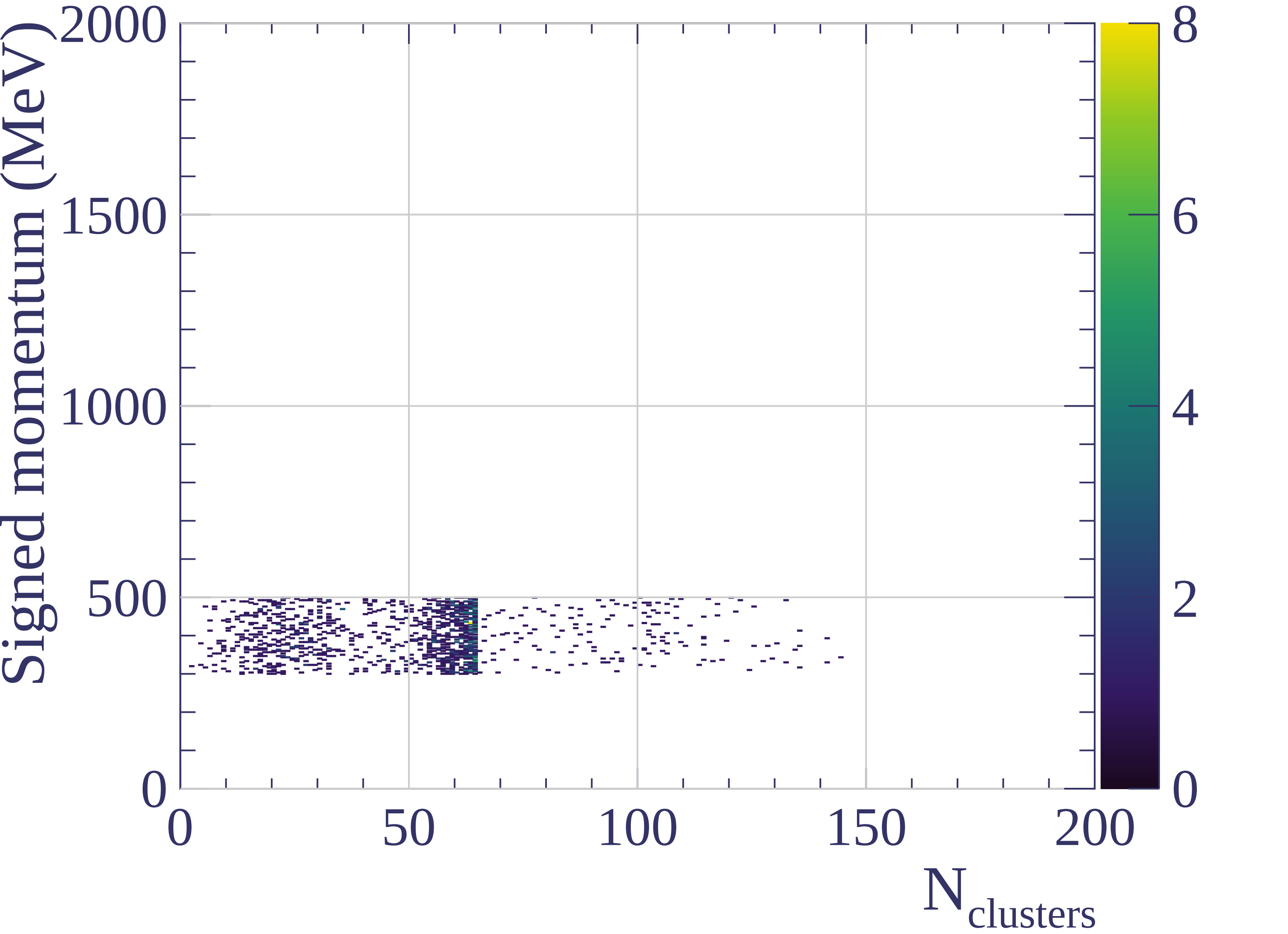

WF
XP
dE/dx resolution [%]
Track length [mm]
4
6
8
10
0
200
400
600
800



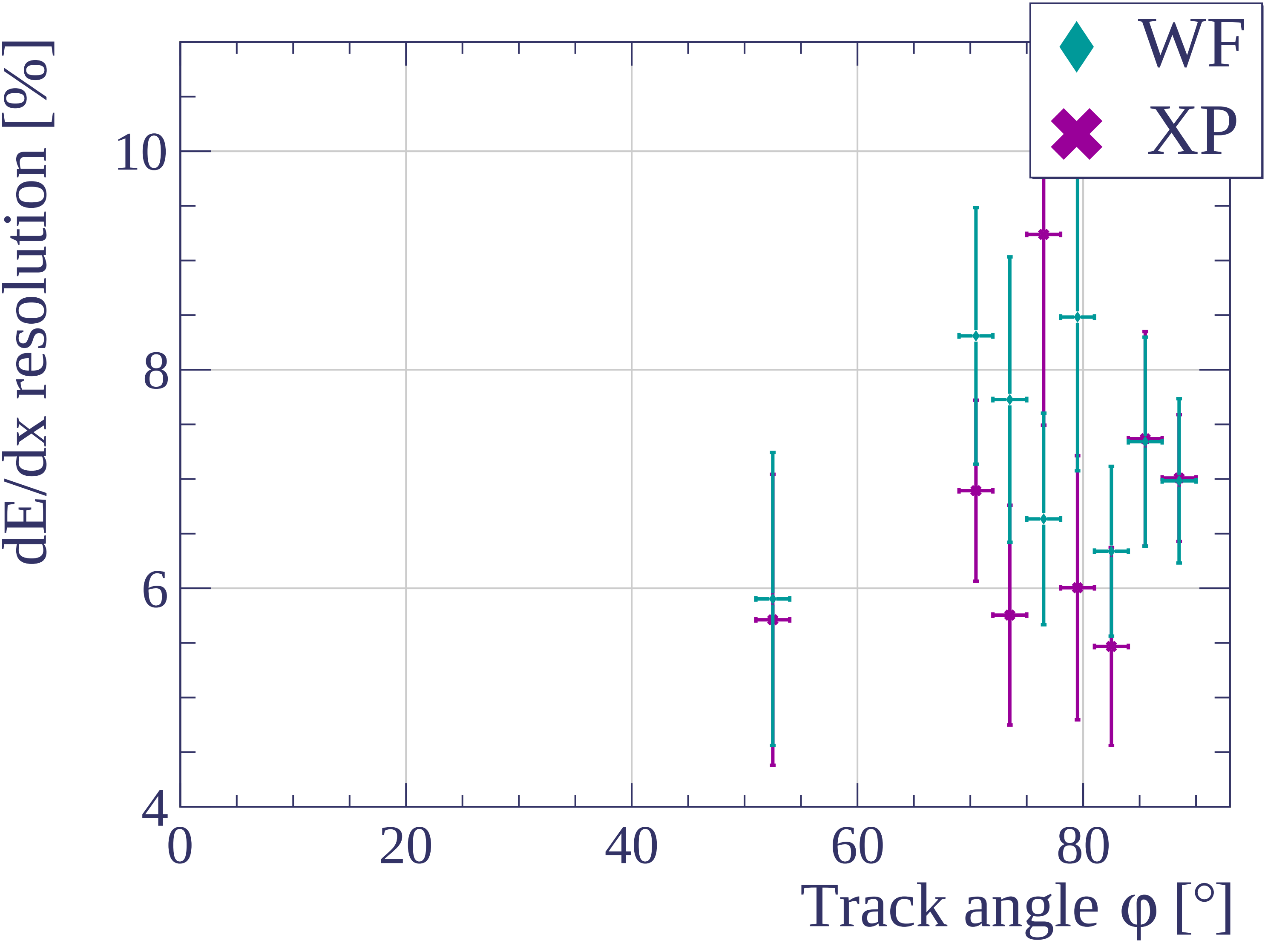
WF
XP
600
500
400
300
200
400
600
800
Mean dE/dx [ADC counts/cm]
Track length [mm]


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
0
50
100
150
Track length [cm]
6
4
2
0

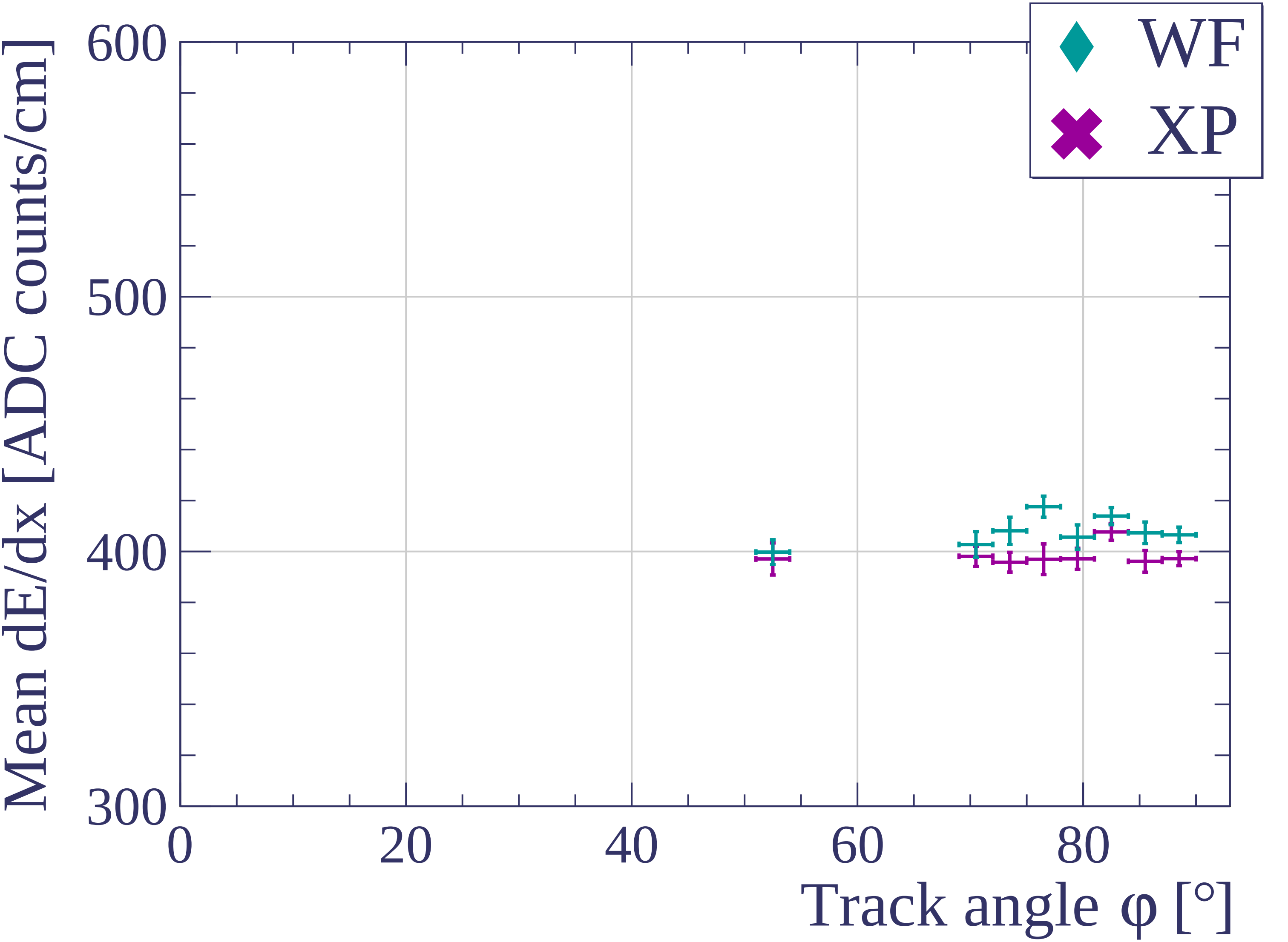

2000
1500
1000
500
0
Signed momentum (MeV)
0
50
100
150
Track length [cm]
4
3
2
1
0



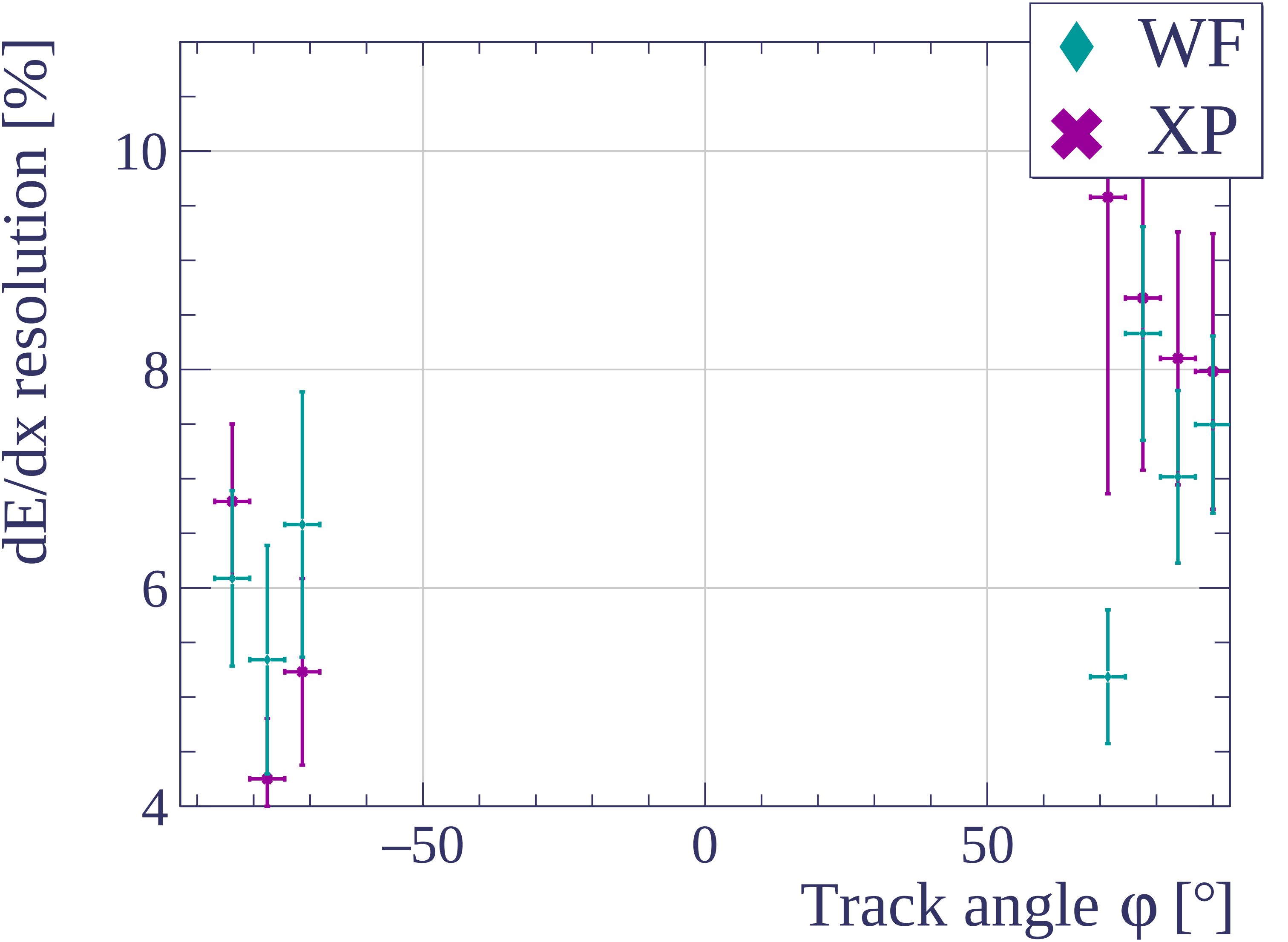
Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
$N_{clusters}$
8
6
4
2
0



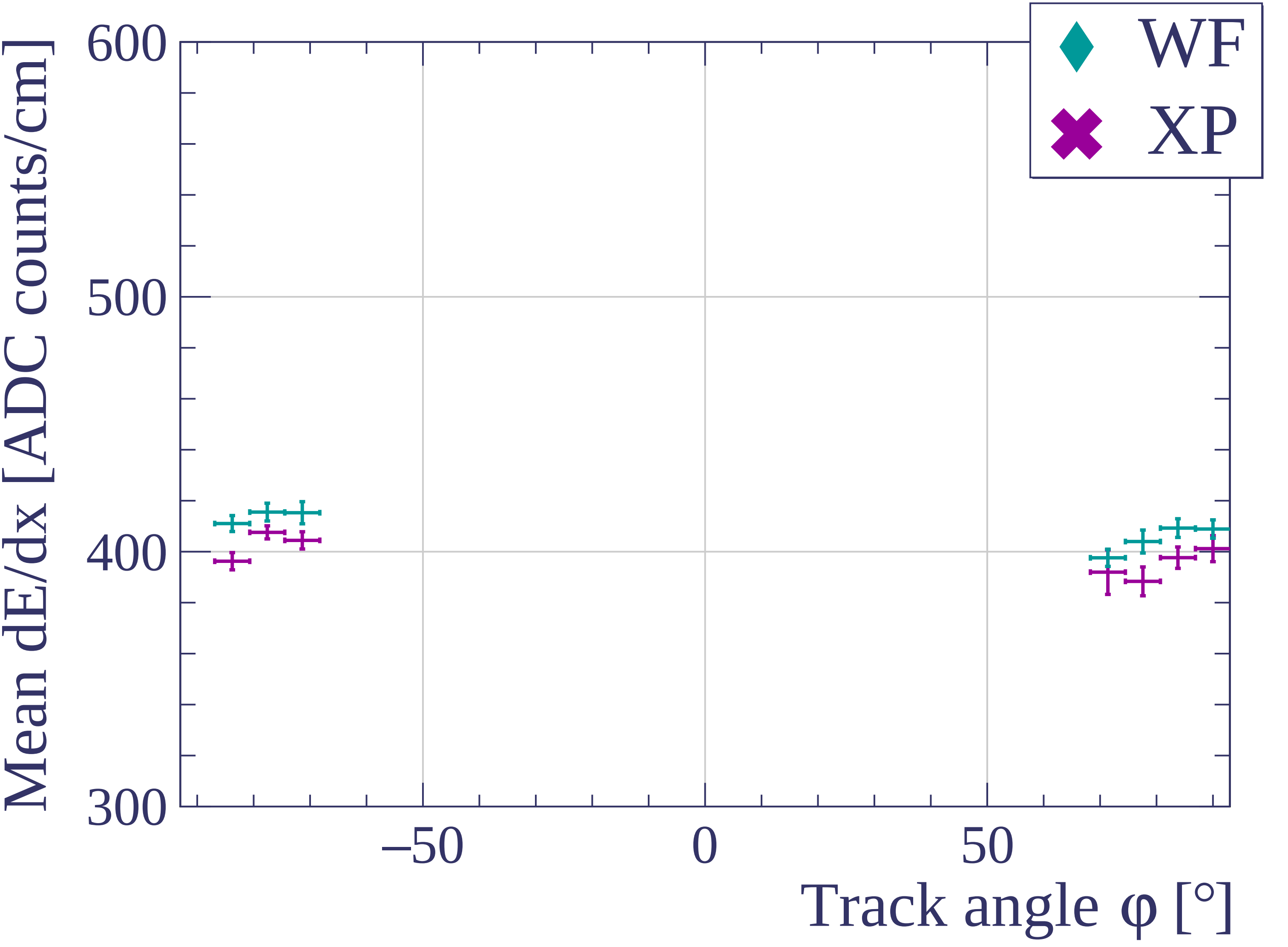
WF
XP
dE/dx resolution [%]
Track angle φ [°]
4
6
8
10
0
20
40
60
80



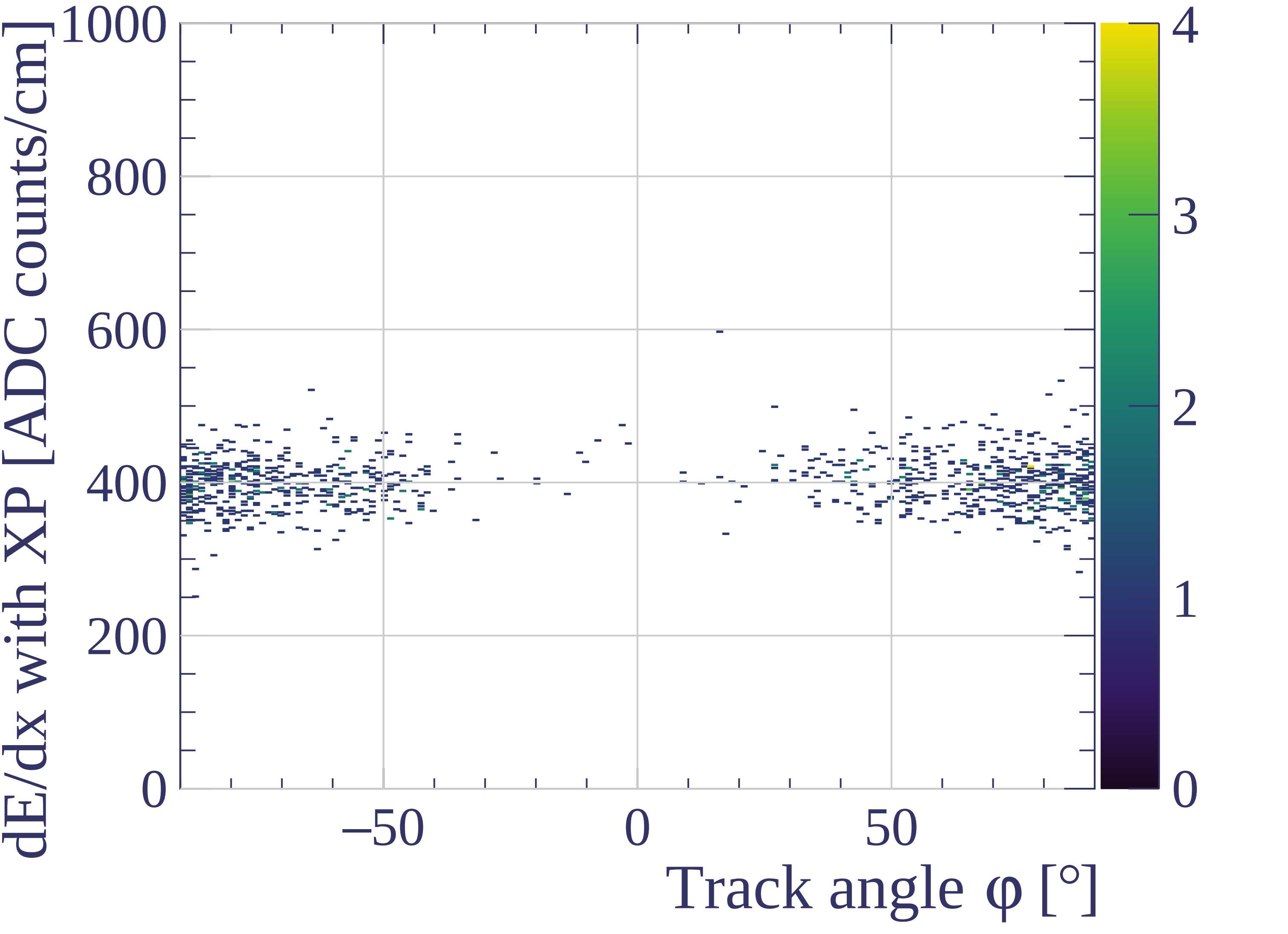
WF
XP
600
500
400
300
0
20
40
60
80
Mean dE/dx [ADC counts/cm]
Track angle φ [°]



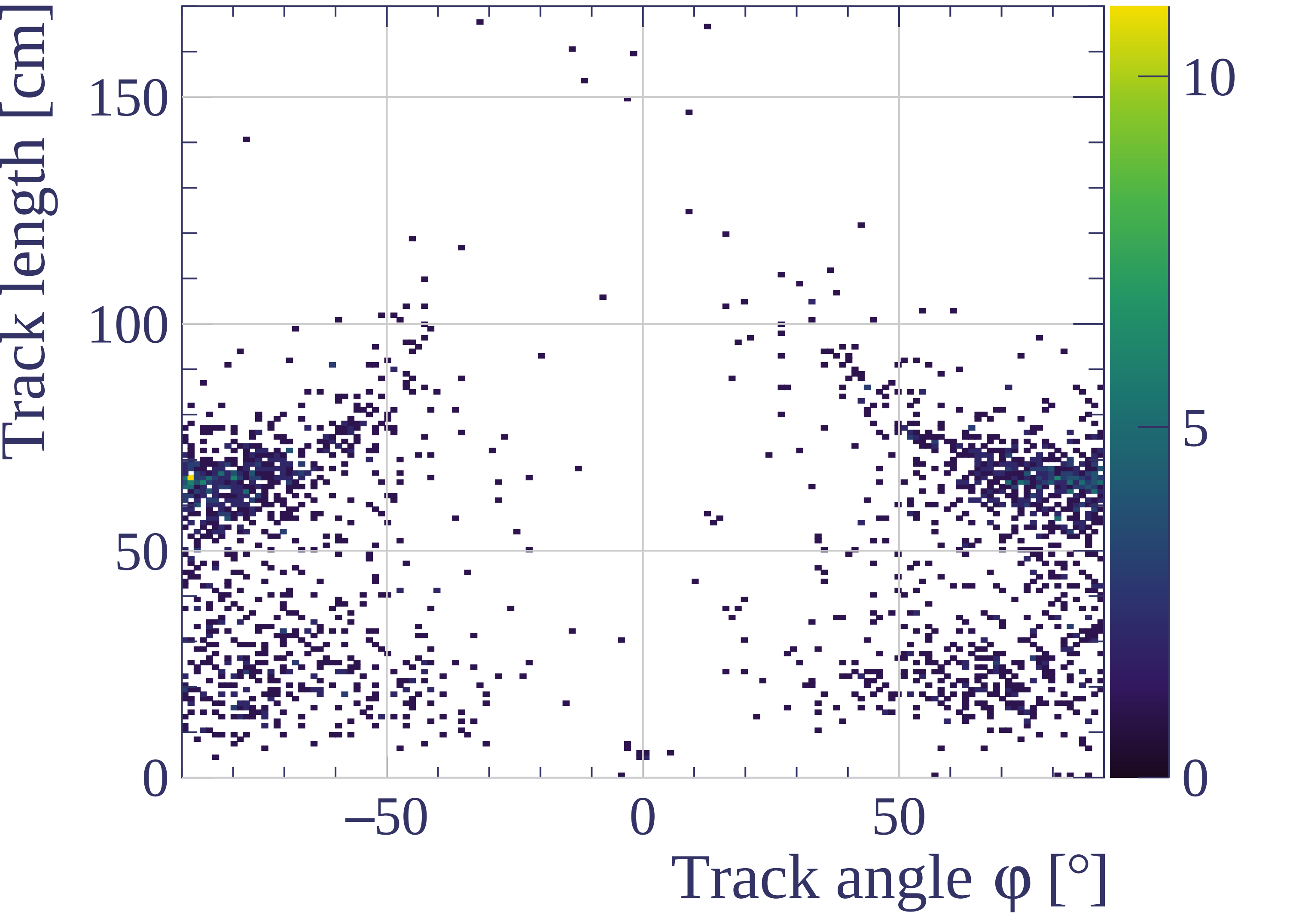
WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle φ [°]



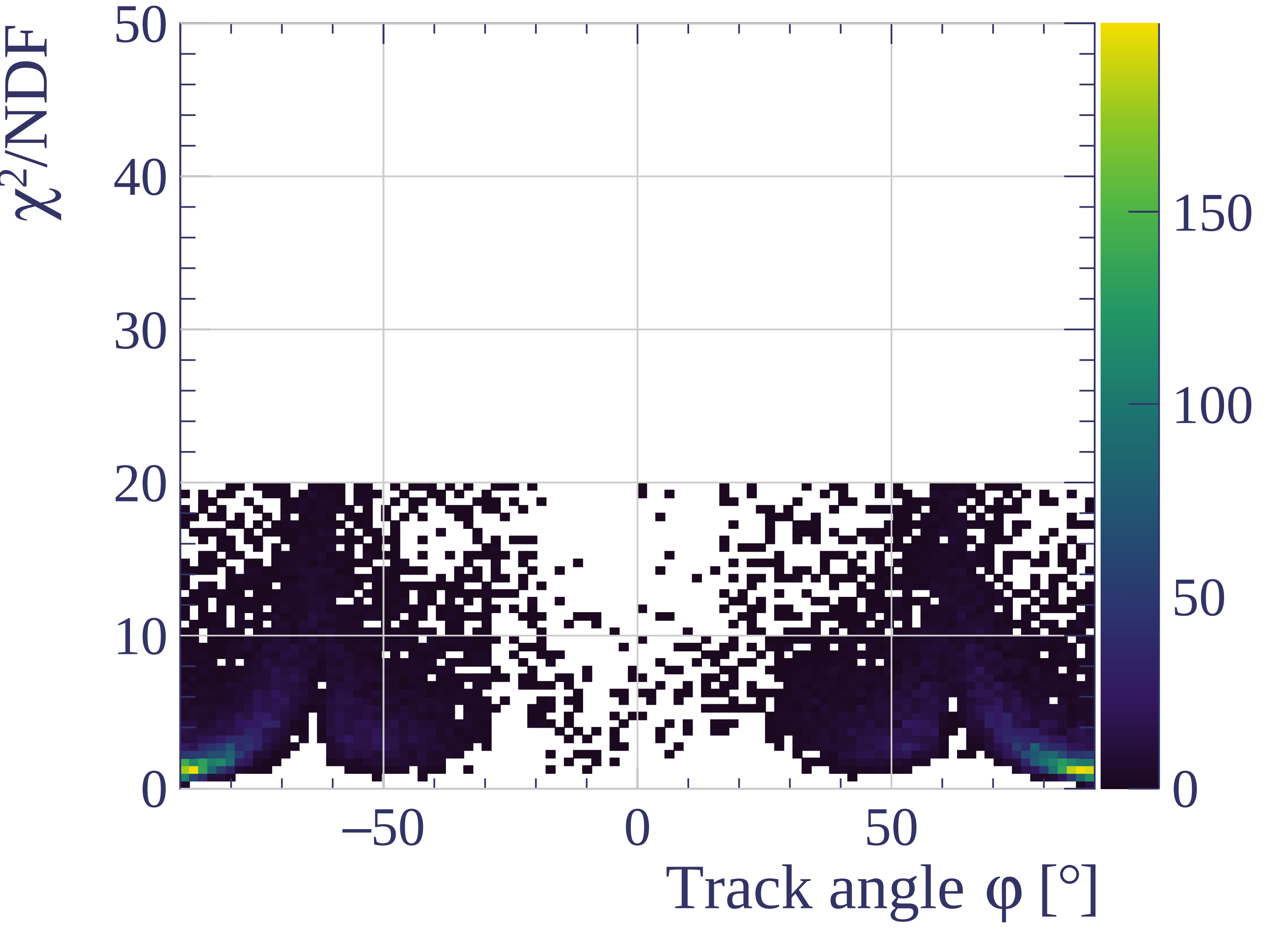
WF
XP
600
500
400
300
−50
0
50
Mean dE/dx [ADC counts/cm]
Track angle φ [°]



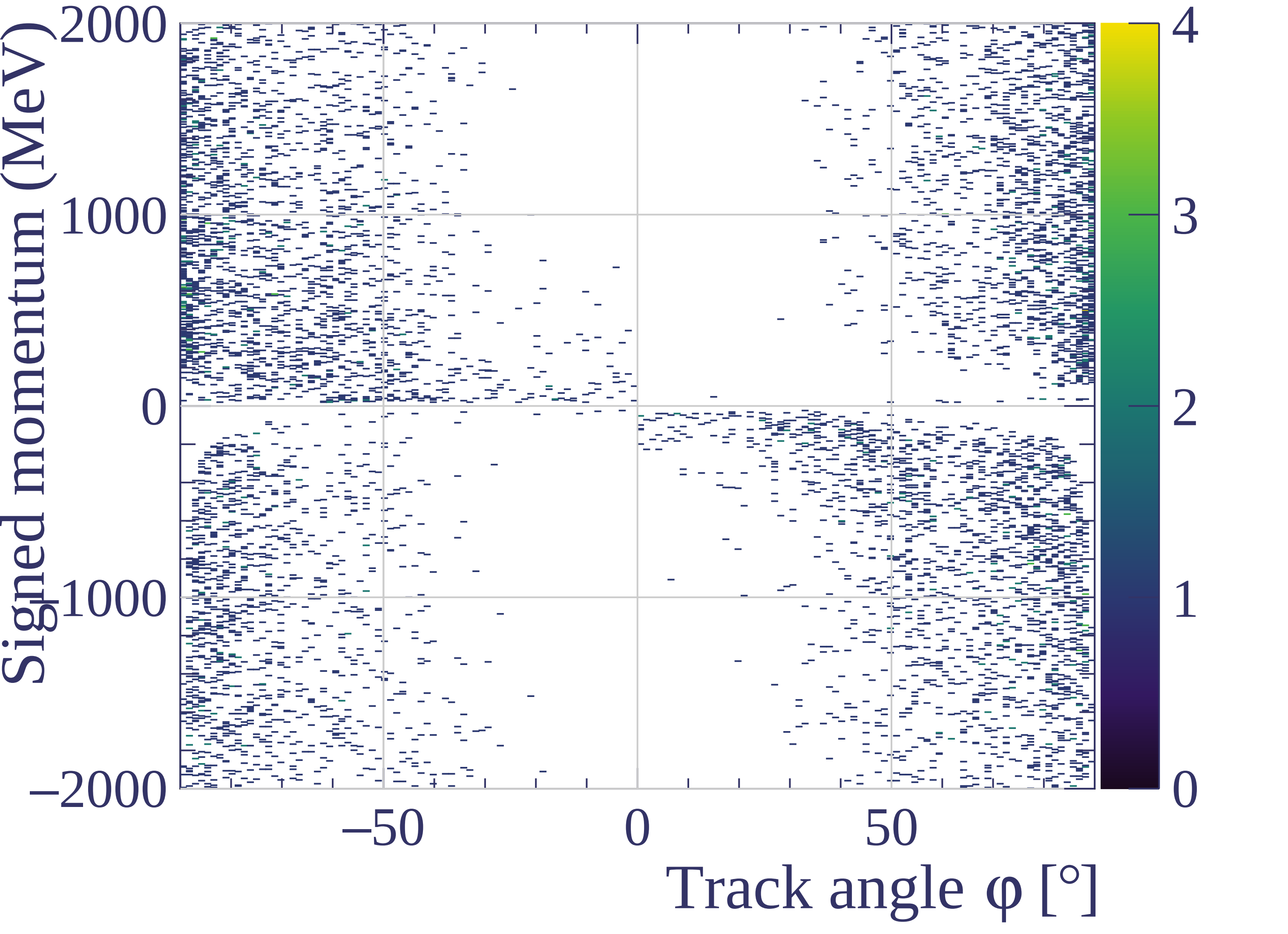
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle φ [°]
4
3
2
1
0



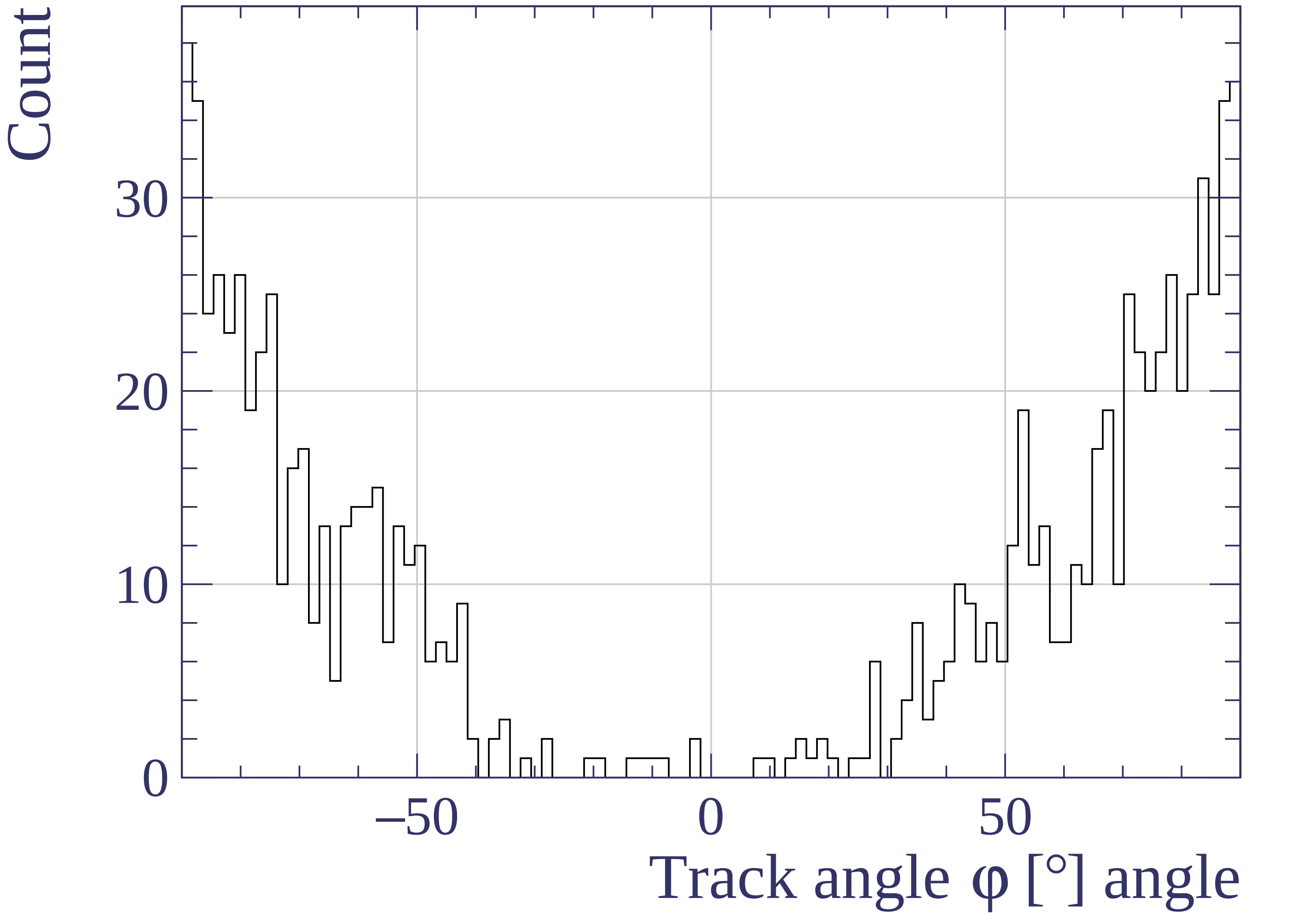
Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
10
5
0

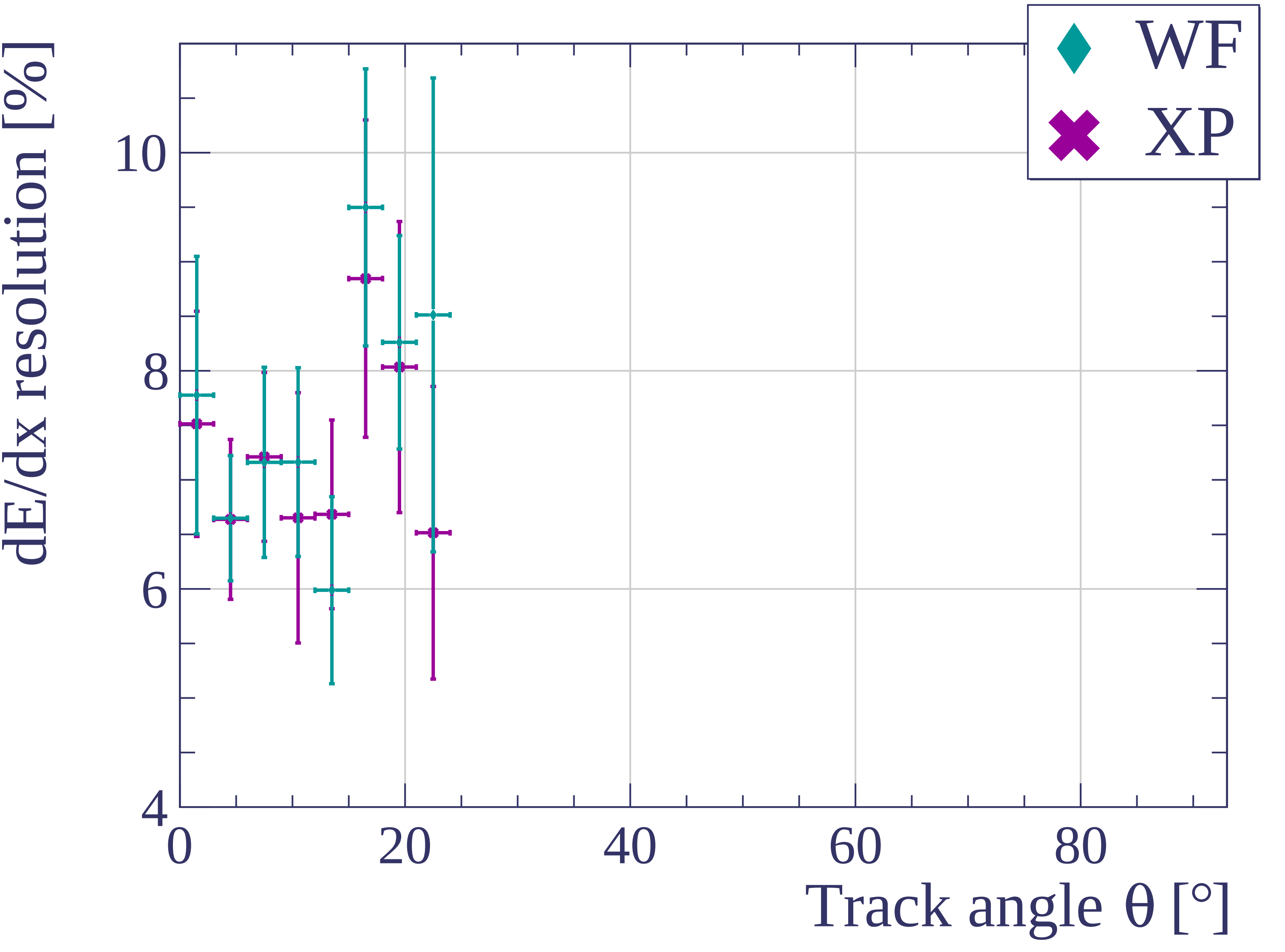


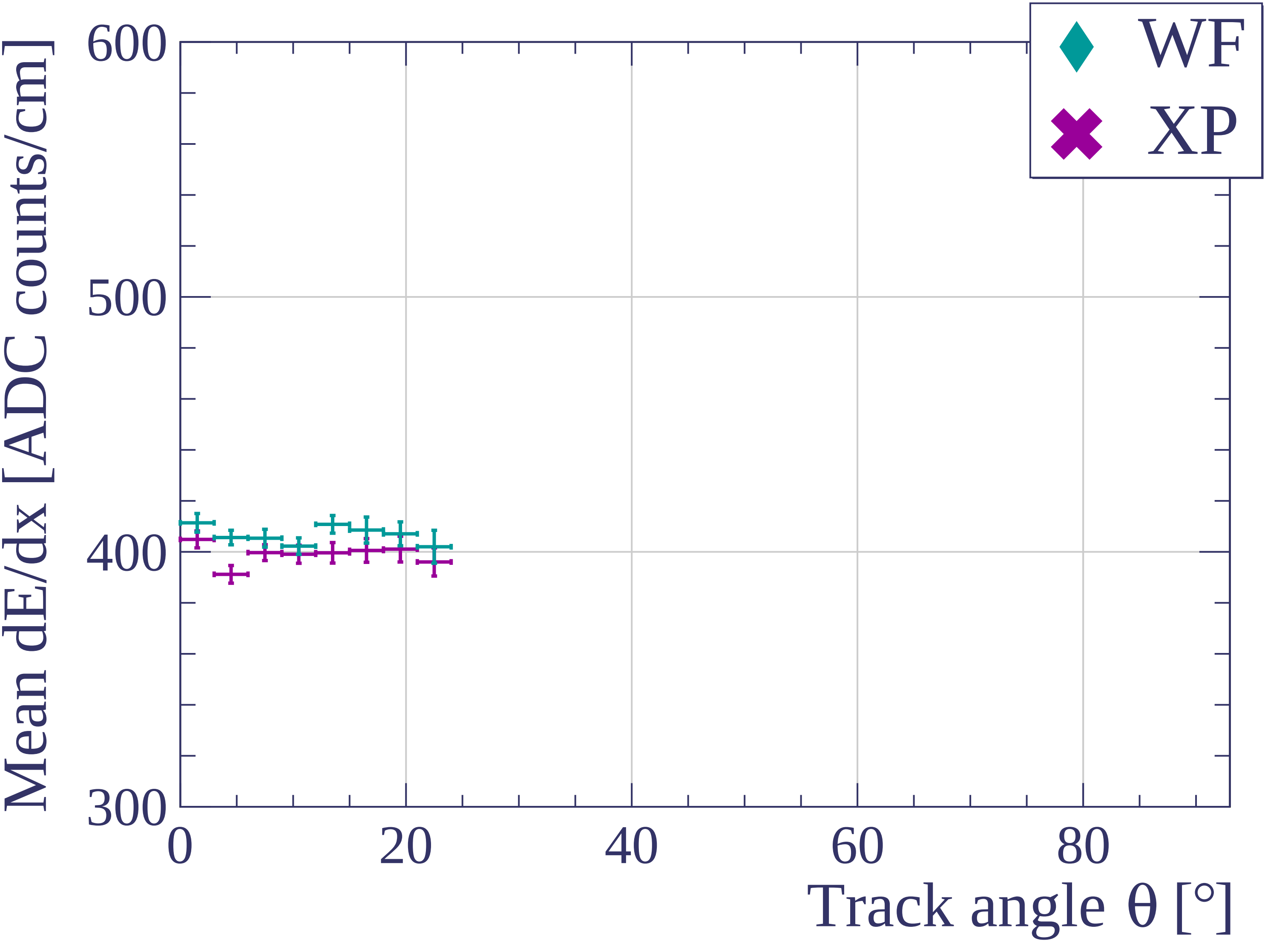
χ^2/NDF
50
40
30
20
10
0
−50
0
50
Track angle φ [°]
150
100
50
0



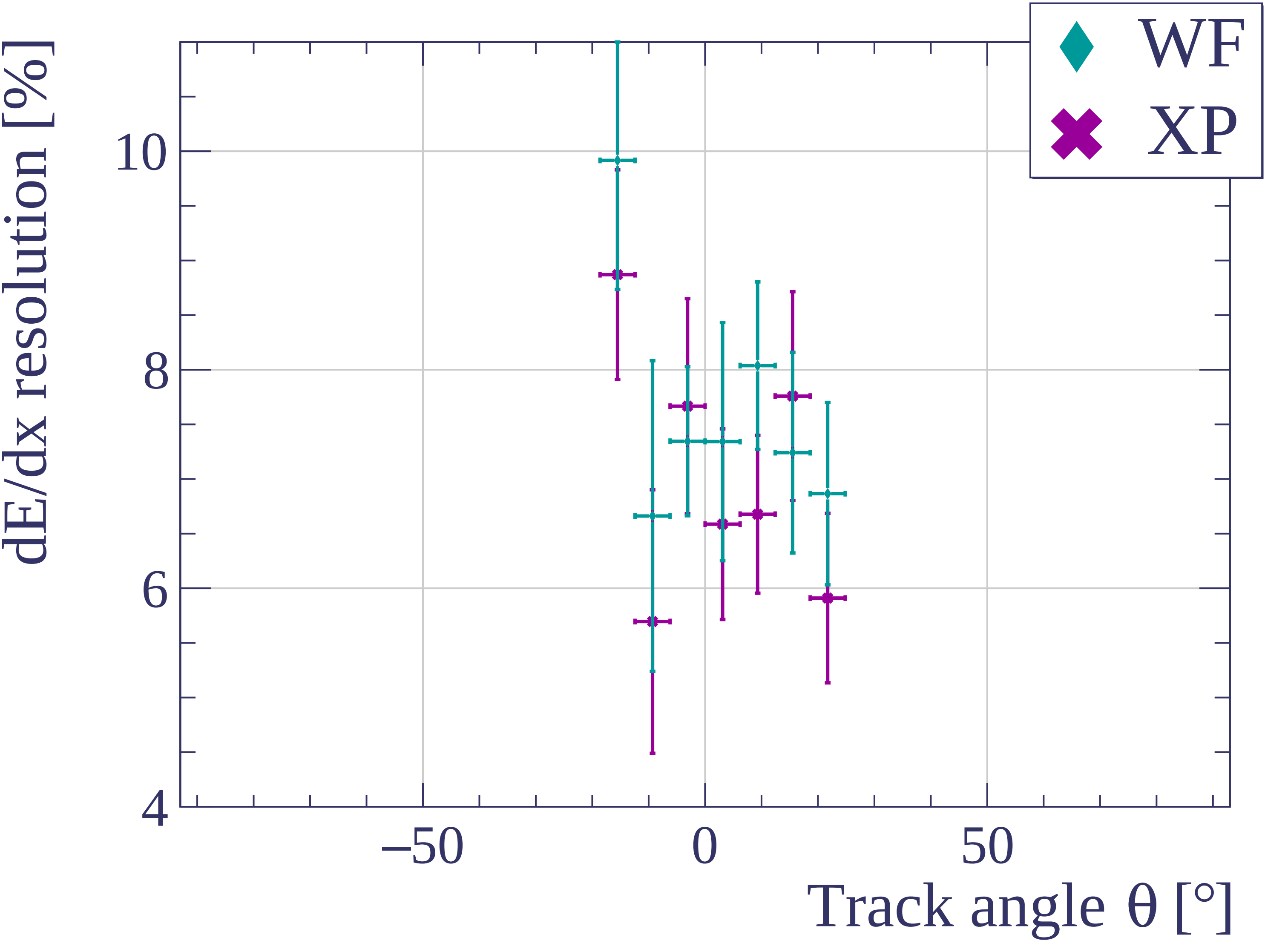
2000
1000
0
−1000
−2000
Signed momentum (MeV)
−50
0
50
Track angle φ [°]
0
1
2
3
4


Count
30
20
10
0
−50
0
50
Track angle φ [°] angle

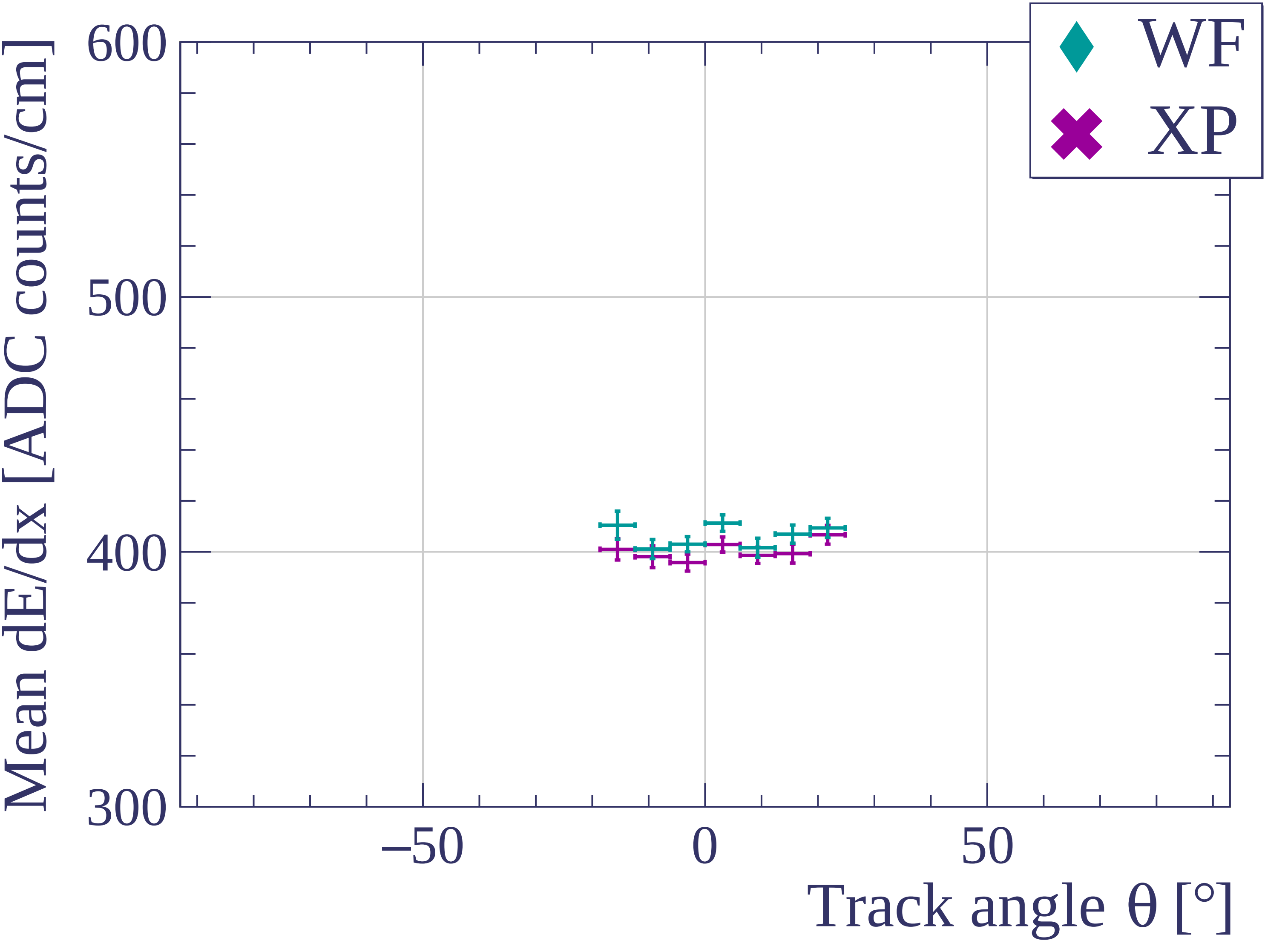

WF
XP
dE/dx resolution [%]
Track angle θ [°]
4
6
8
10
0
20
40
60
80



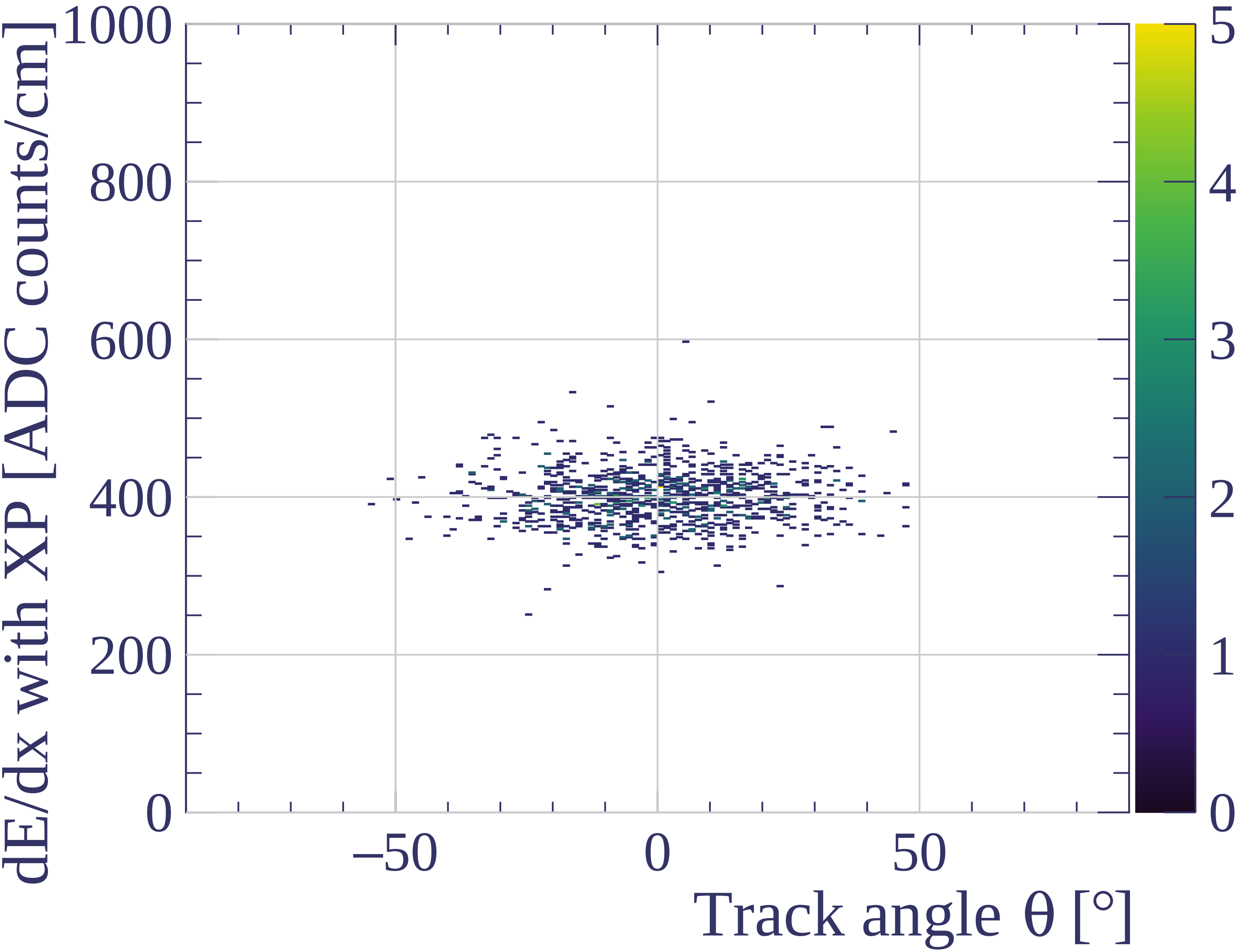
WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
0
20
40
60
80
Track angle θ [°]

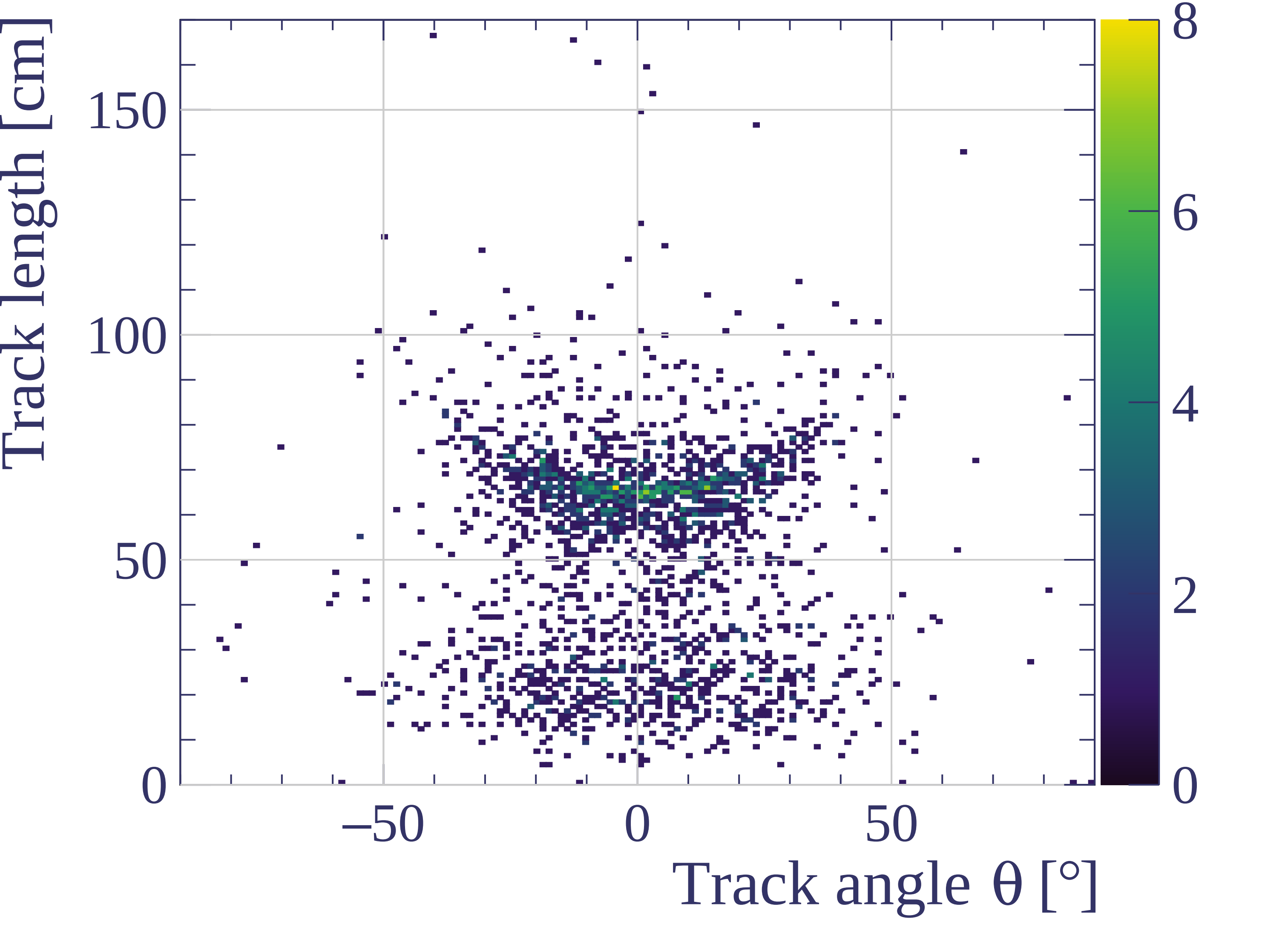


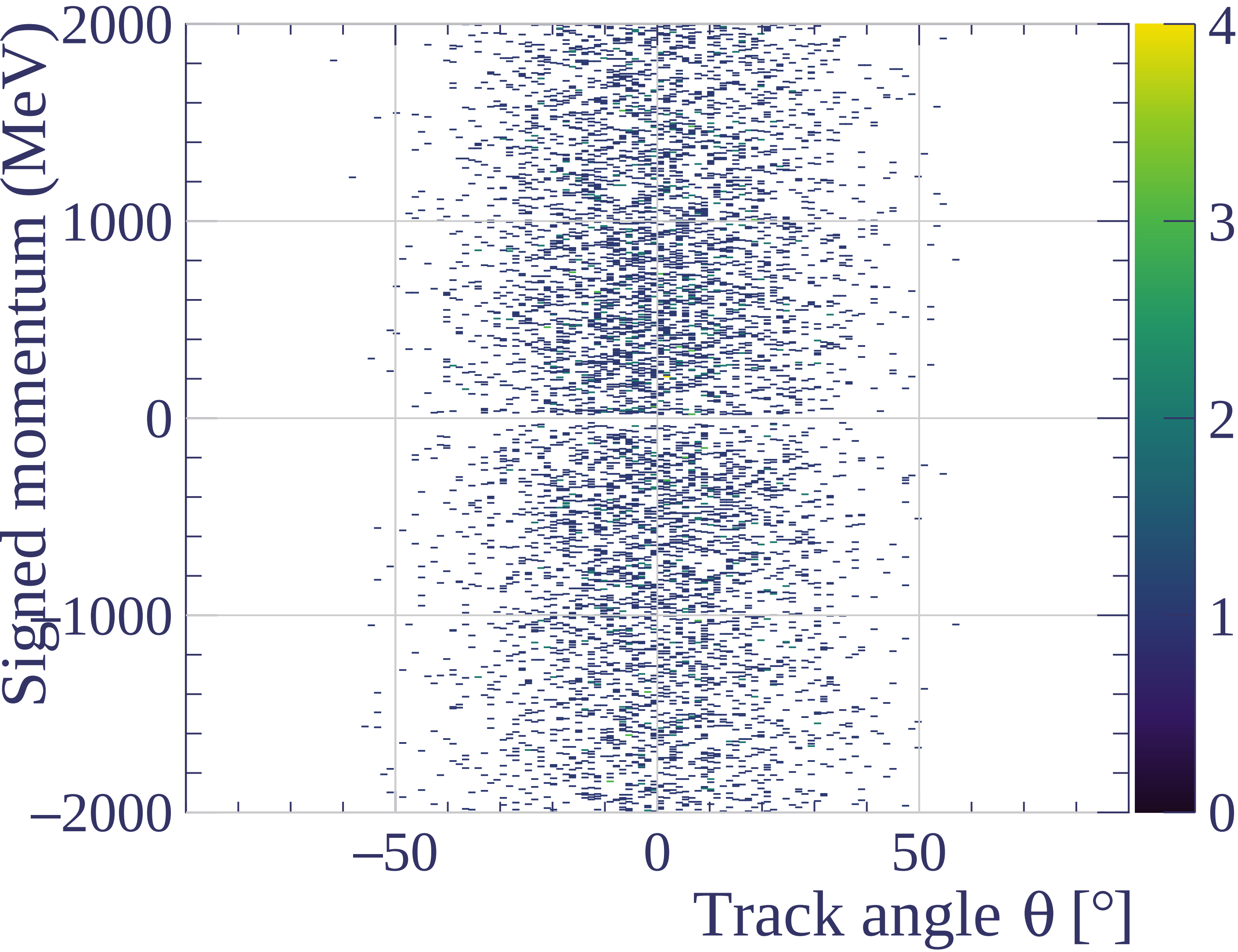
WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle θ [°]

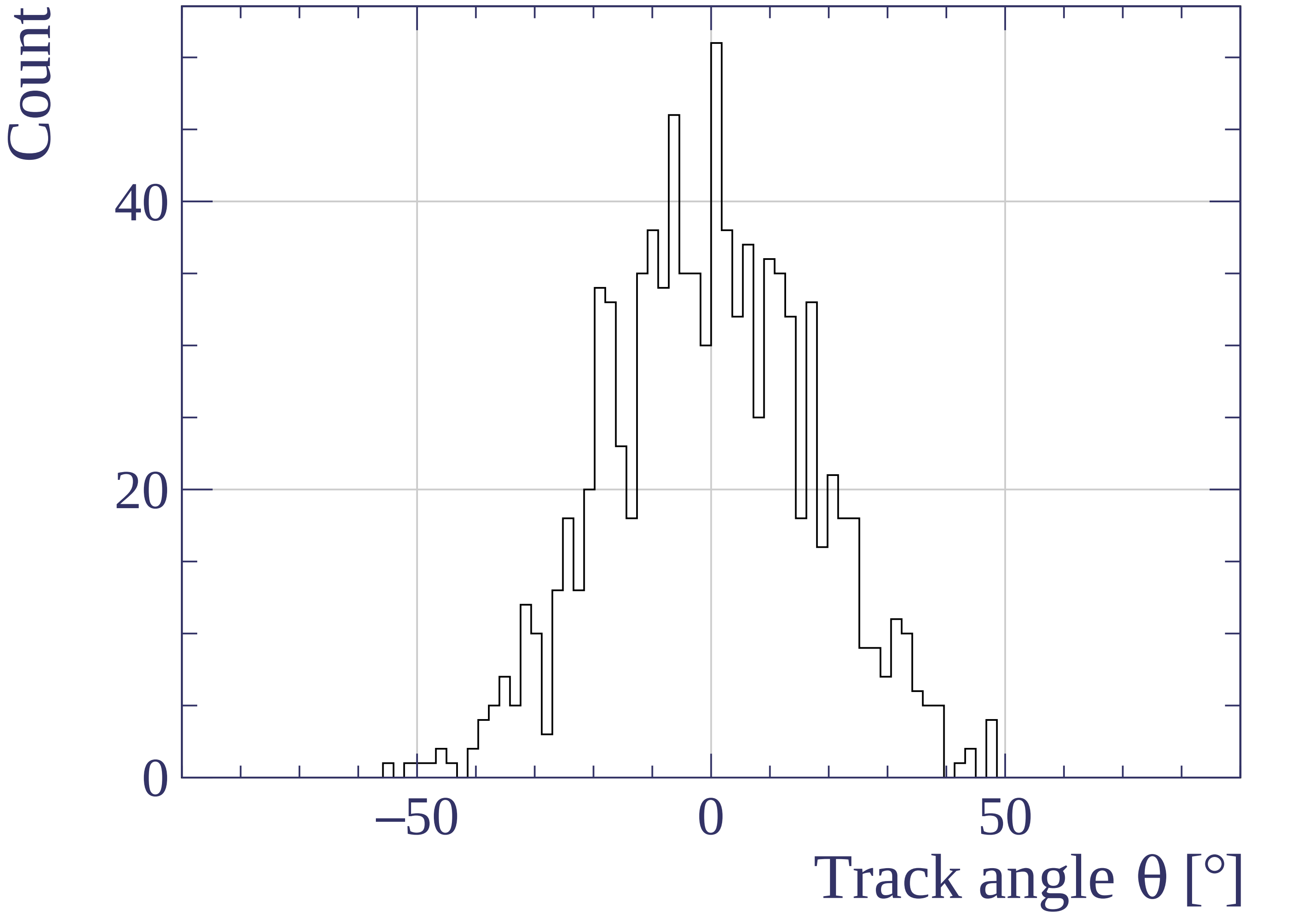


WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
−50
0
50
Track angle θ [°]

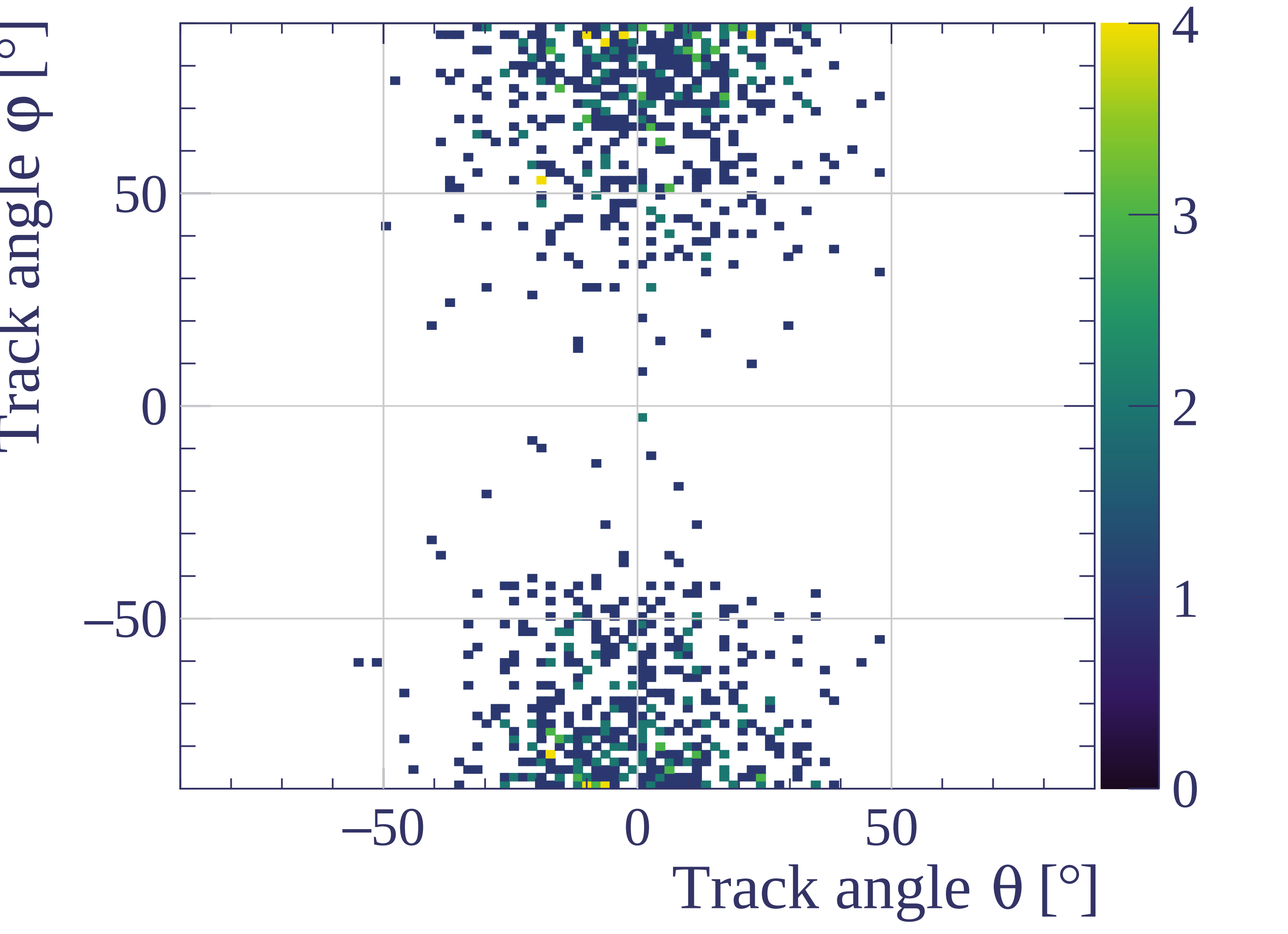

dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
5
4
3
2
1
0


Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
8
6
4
2
0

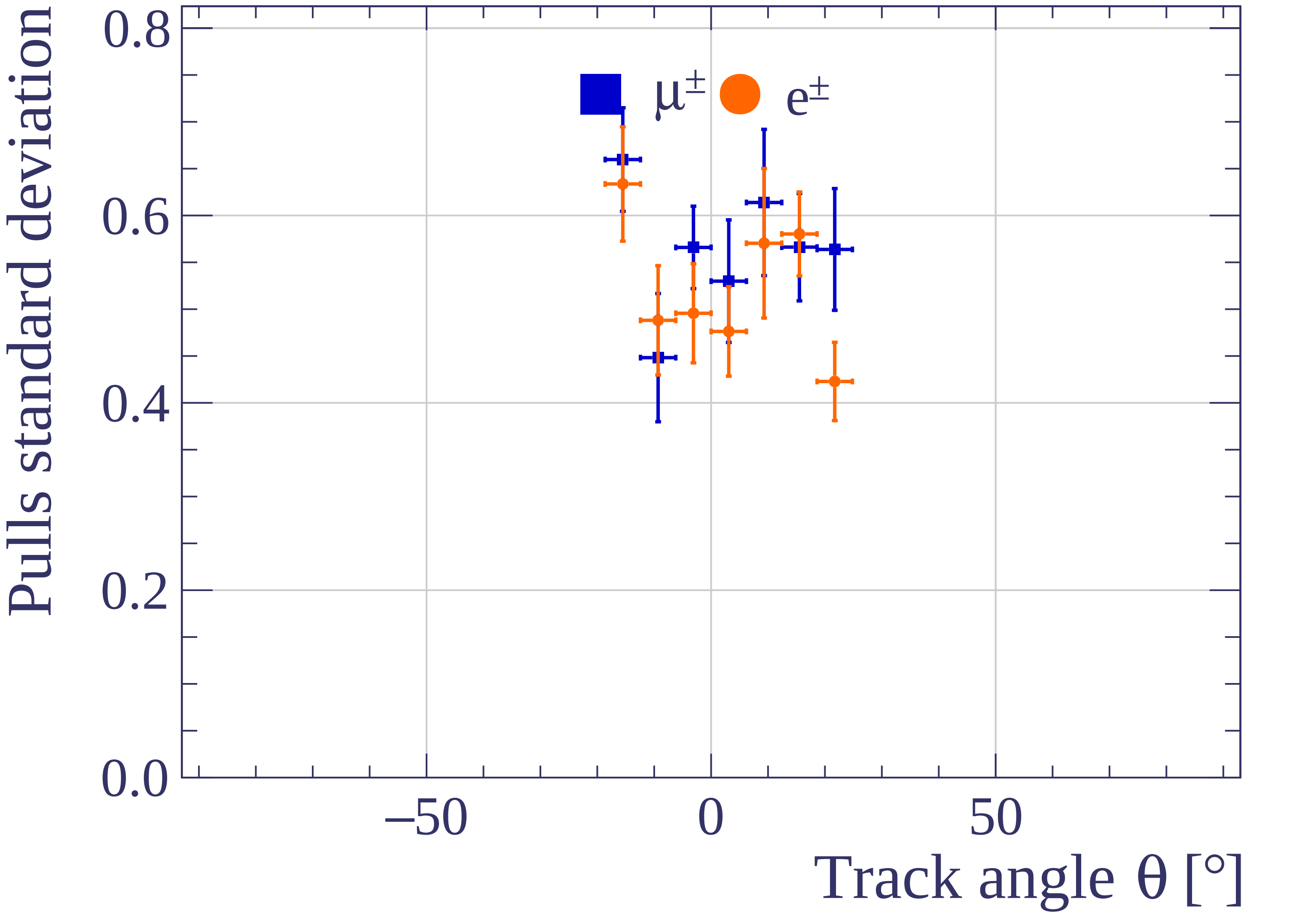

Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle θ [°]
4
3
2
1
0

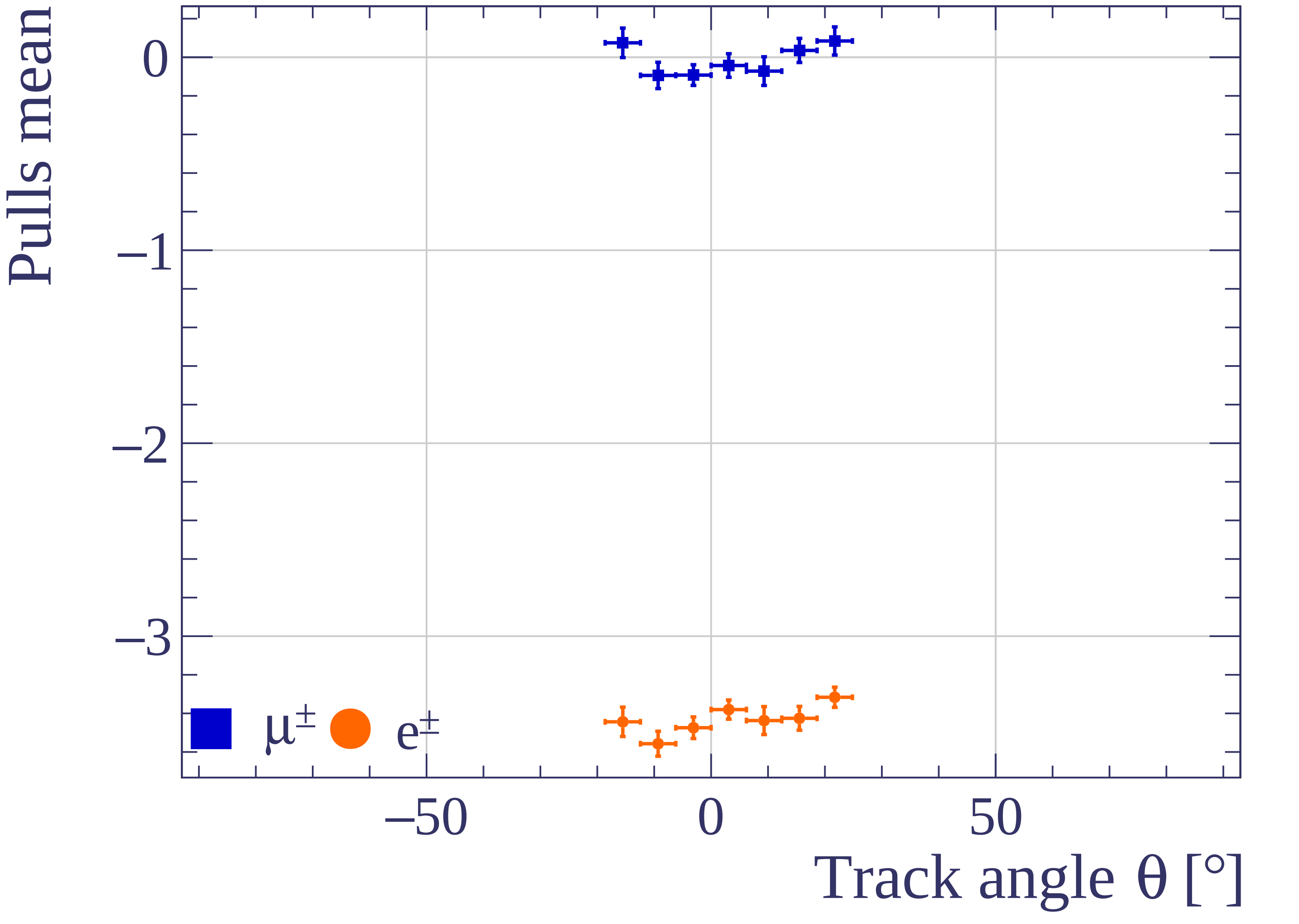


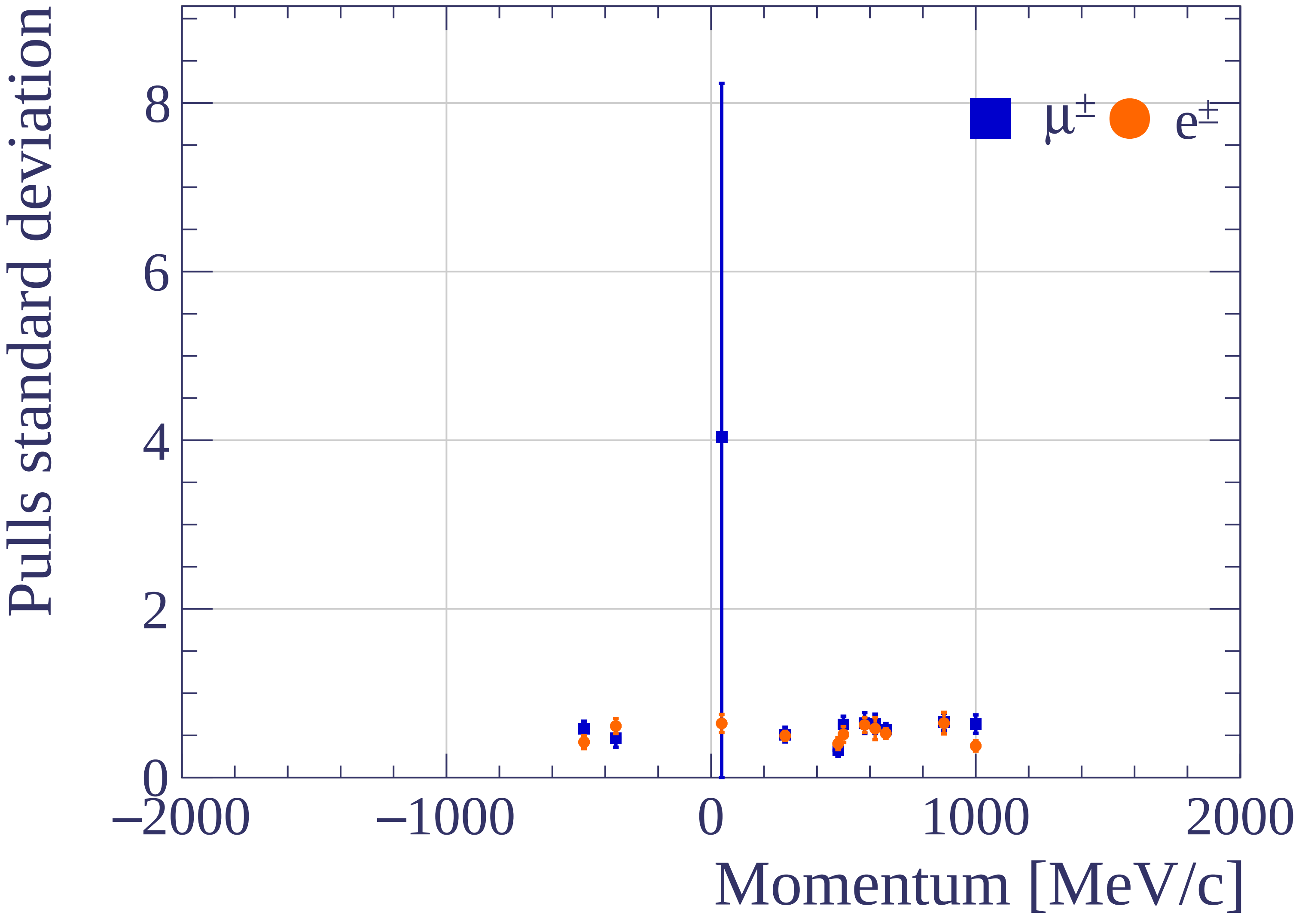
Count
40
20
0
−50
0
50
Track angle θ [°]



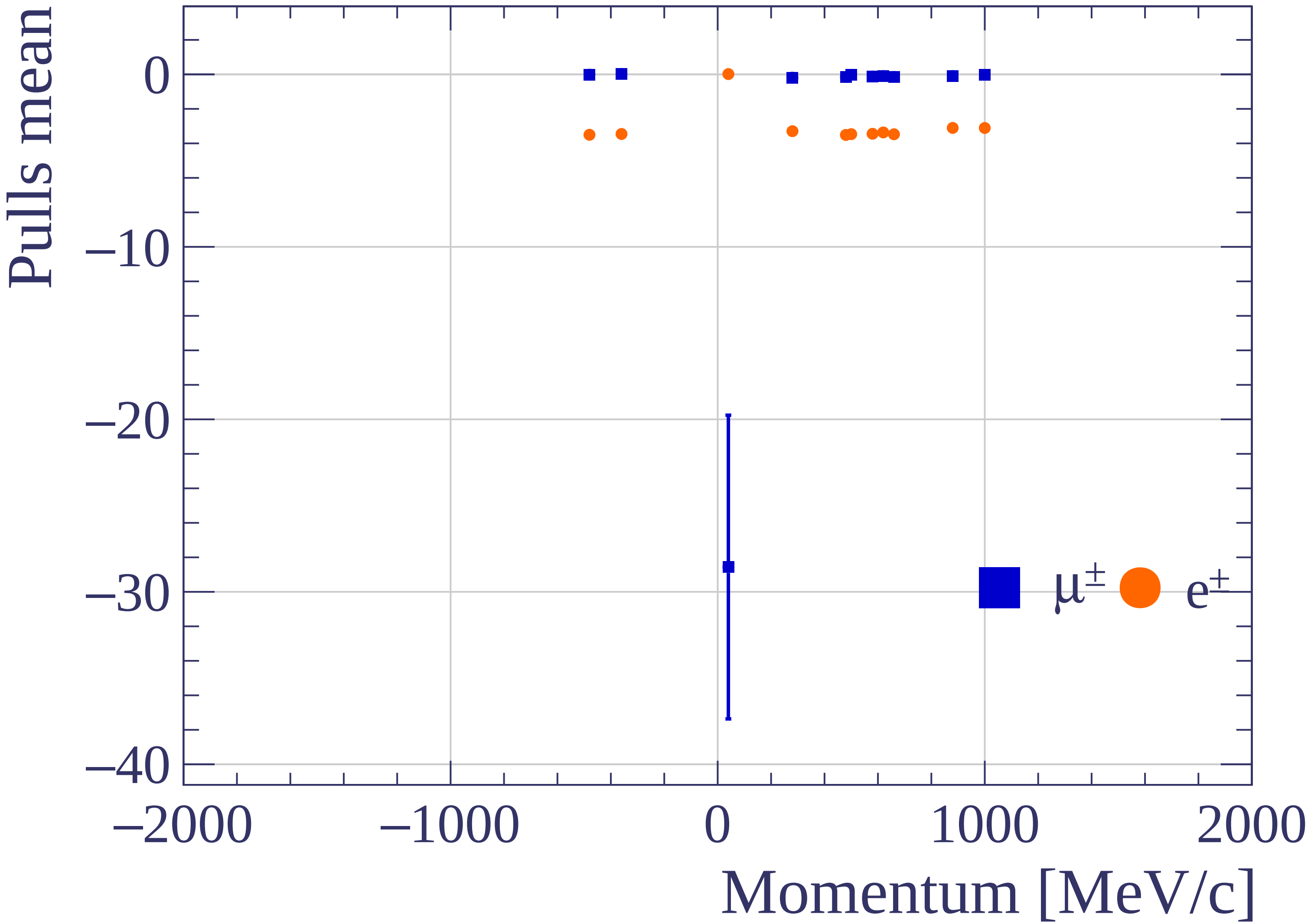
Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
4
3
2
1
0


μ±
e±
Pulls standard deviation
0.0
0.2
0.4
0.6
0.8
−50
0
50
Track angle θ [°]

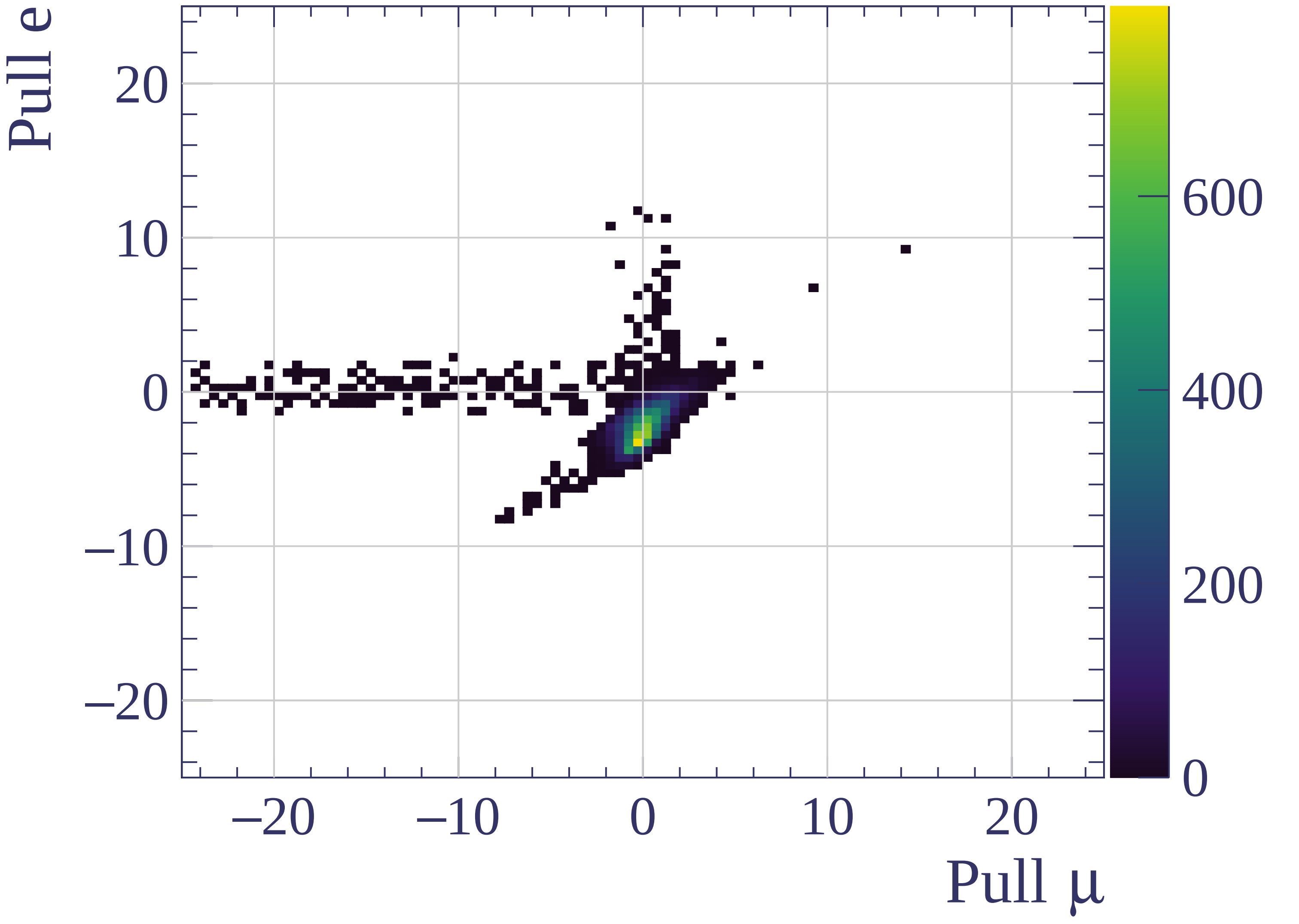

Pulls mean
0
−1
−2
−3
μ±
e±
−50
0
50
Track angle θ [°]



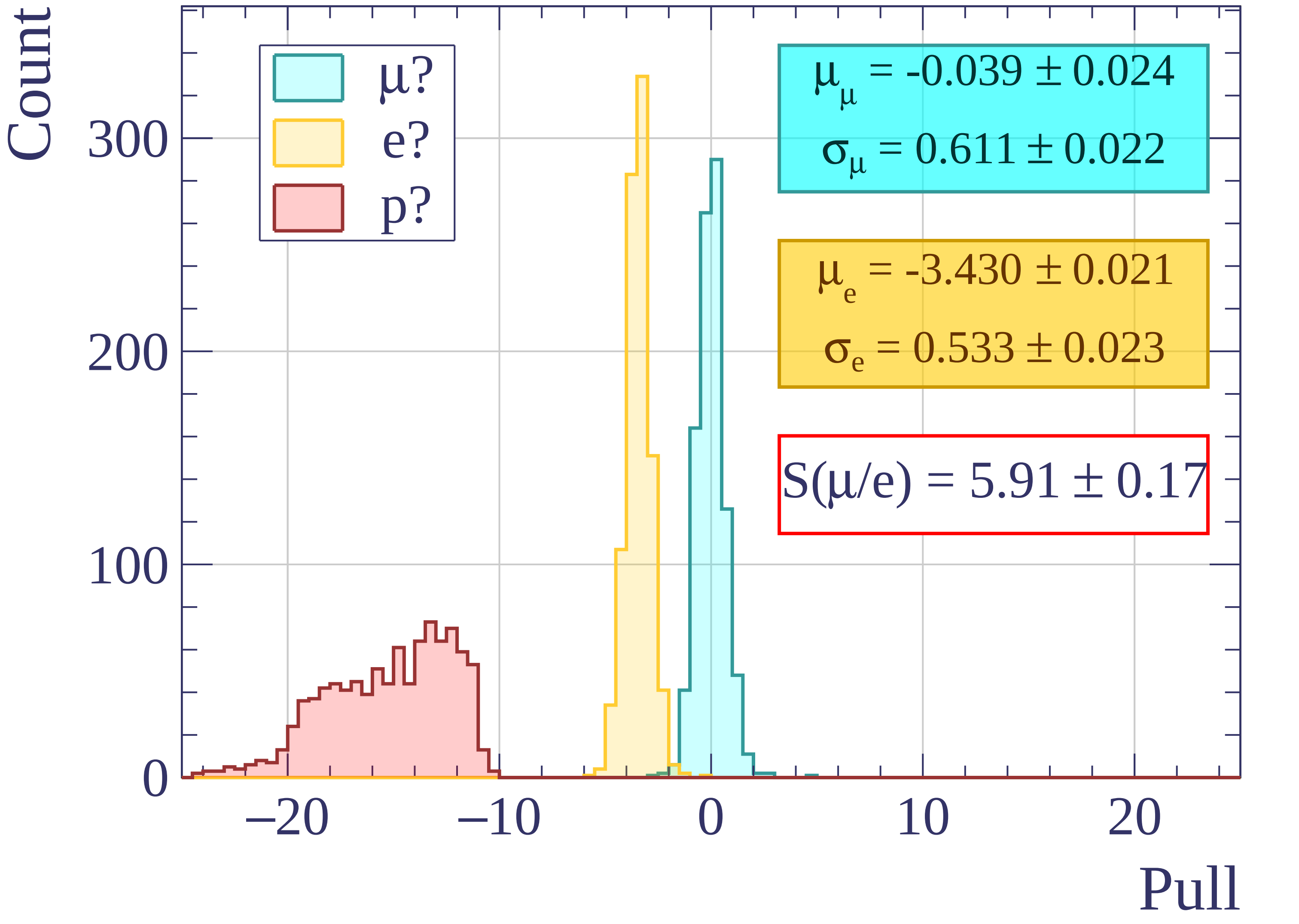
Pulls standard deviation
0
2
4
6
8
−2000
−1000
0
1000
2000
Momentum [MeV/c]
$\mu^{\pm}$
$e^{\pm}$



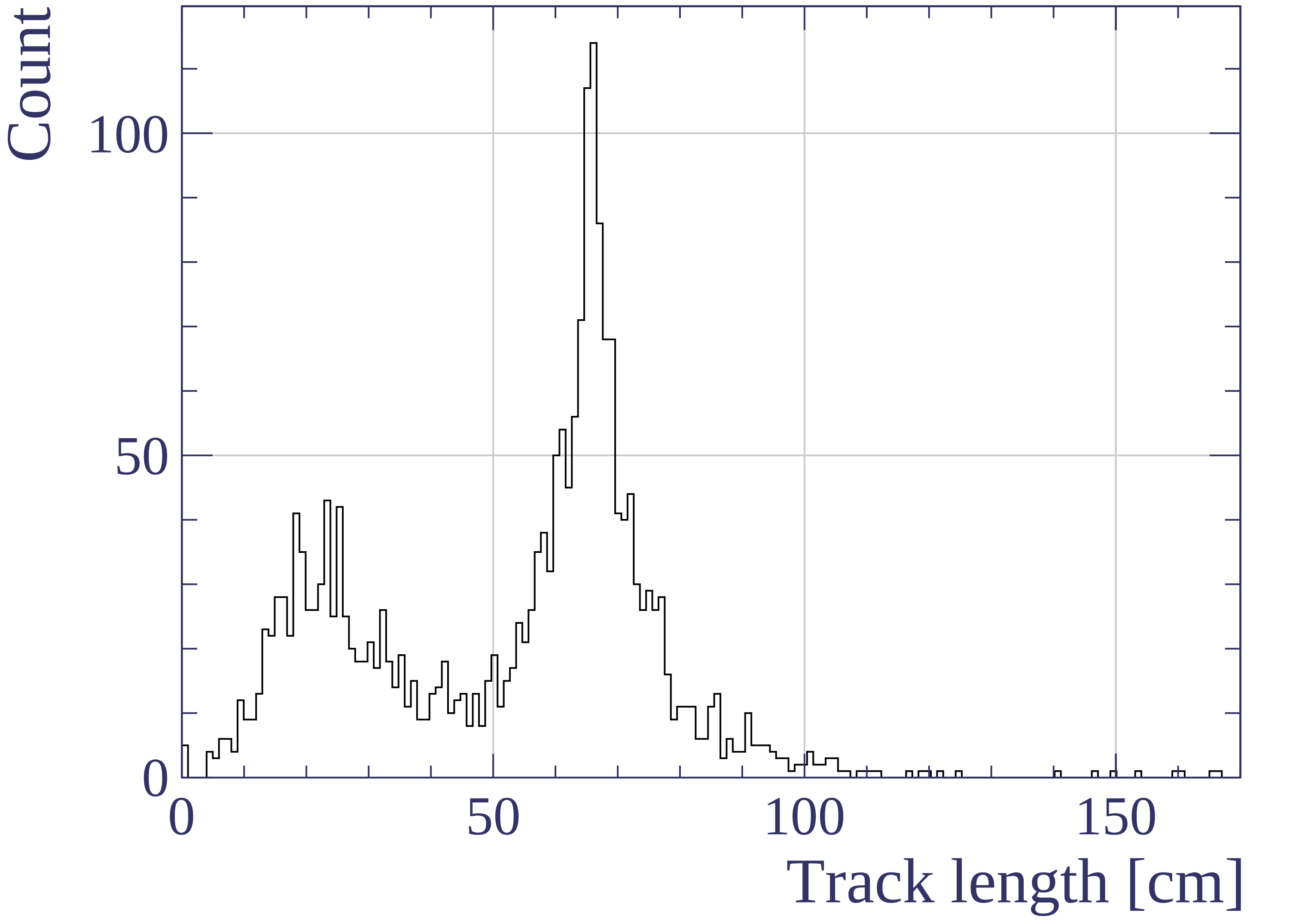
Pulls mean
0
−10
−20
−30
−40
−2000
−1000
0
1000
2000
Momentum [MeV/c]
μ±
e±

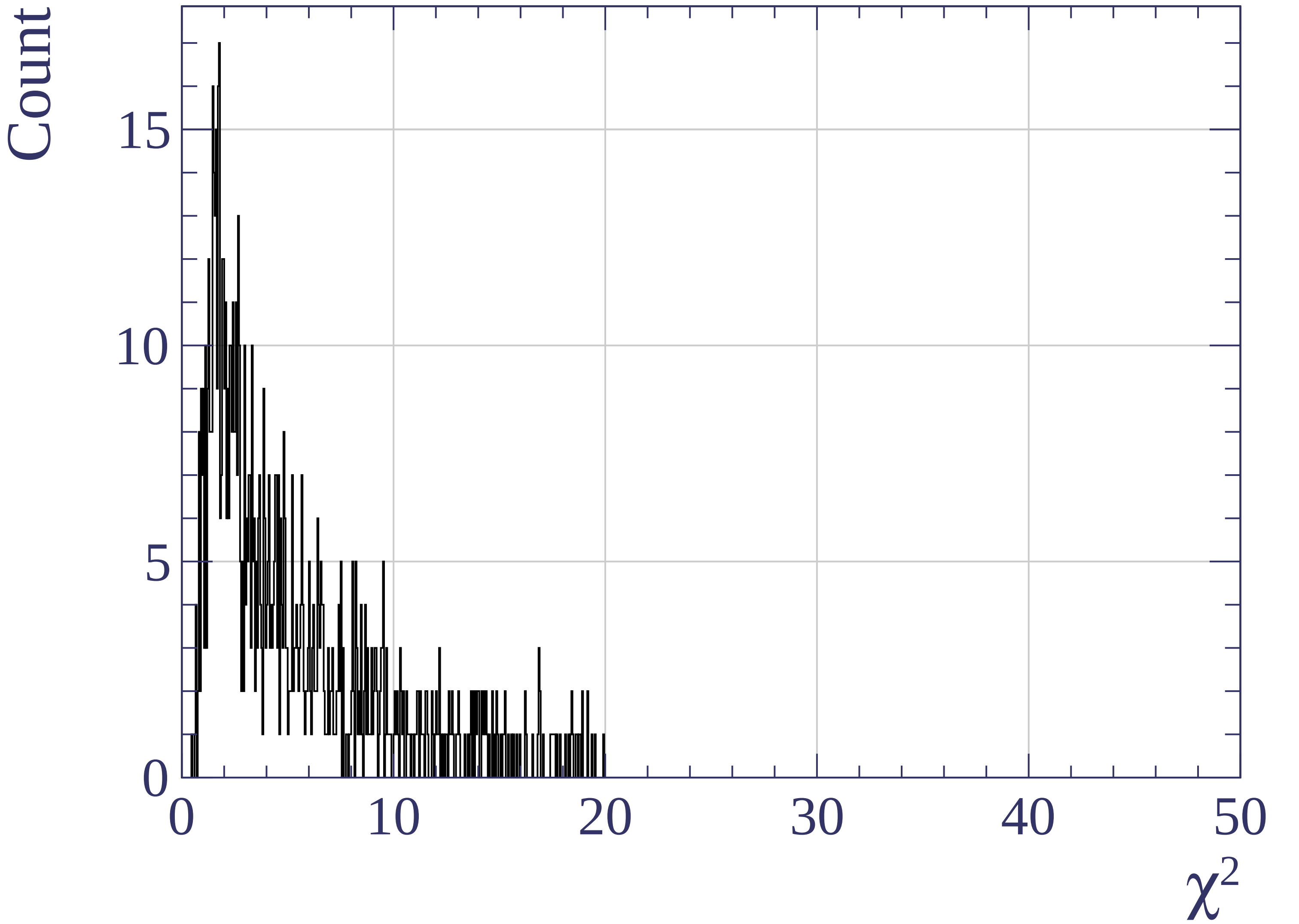


Pull e
20
10
0
−10
−20
−20
−10
0
10
20
Pull μ
600
400
200
0

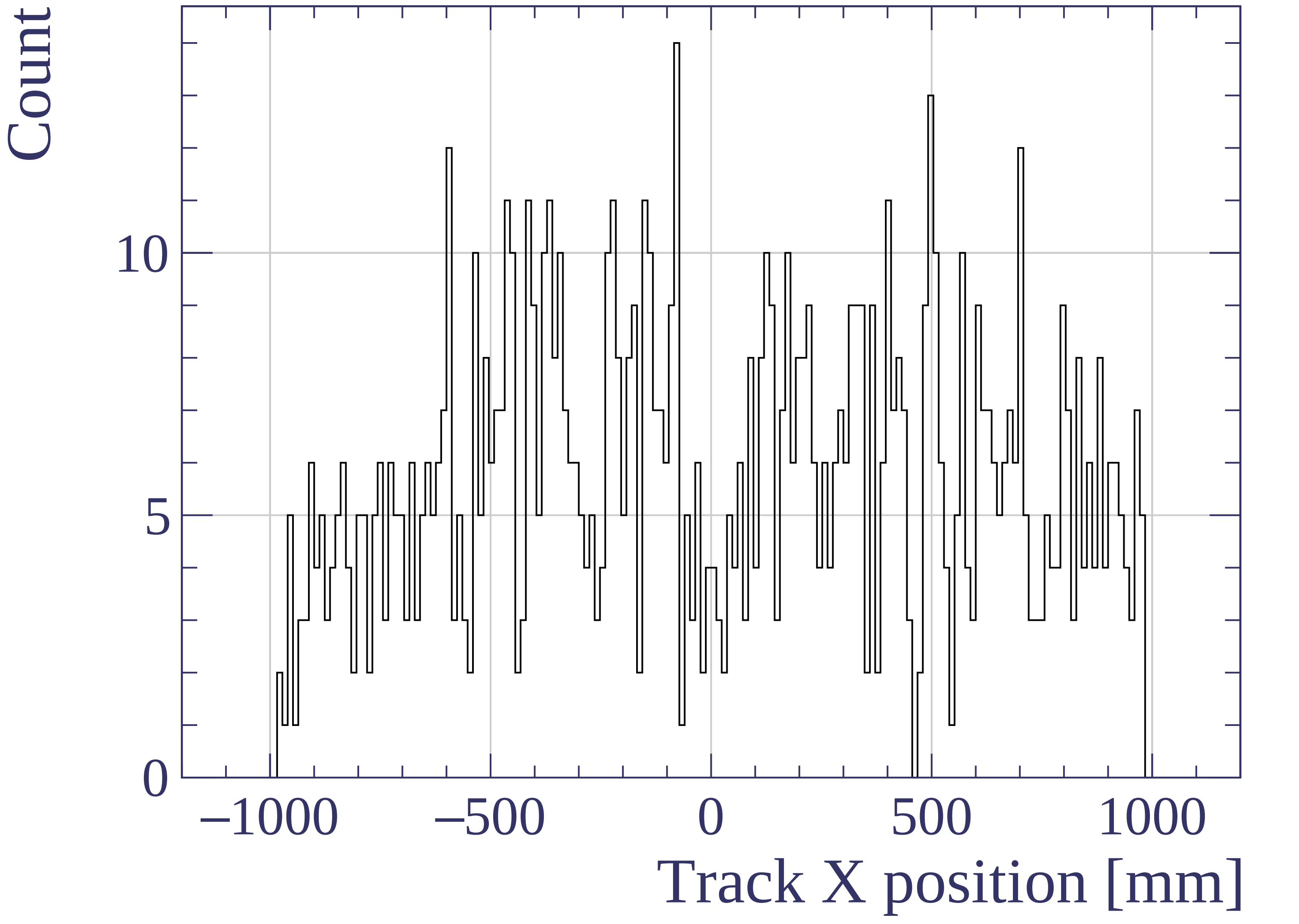


Count
300
200
100
0
−20
−10
0
10
20
Pull
μ?
e?
p?
μ_μ = -0.039 ± 0.024
σ_μ = 0.611 ± 0.022
μ_e = -3.430 ± 0.021
σ_e = 0.533 ± 0.023
S(μ/e) = 5.91 ± 0.17

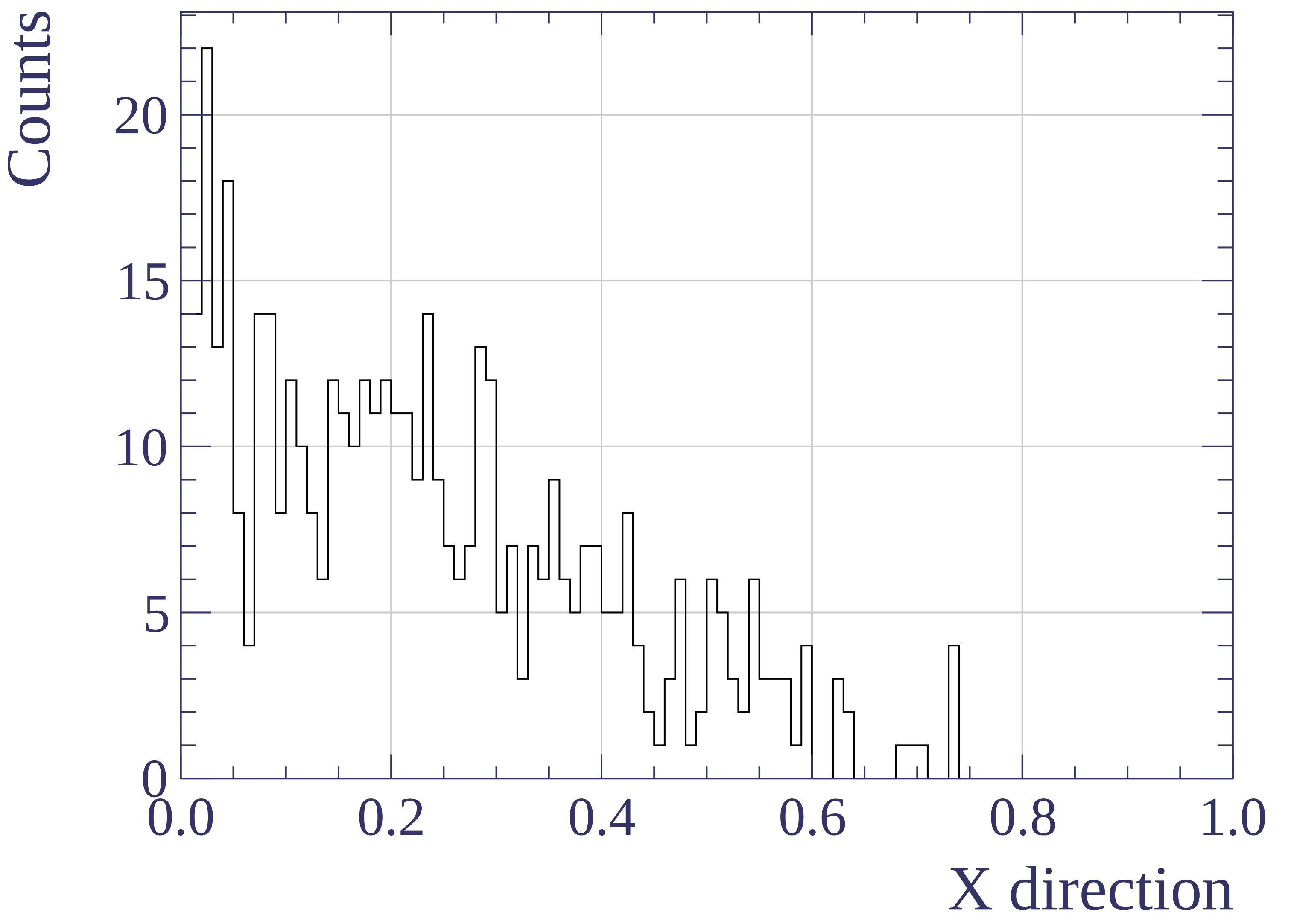

Count
100
50
0
0
50
100
150
Track length [cm]

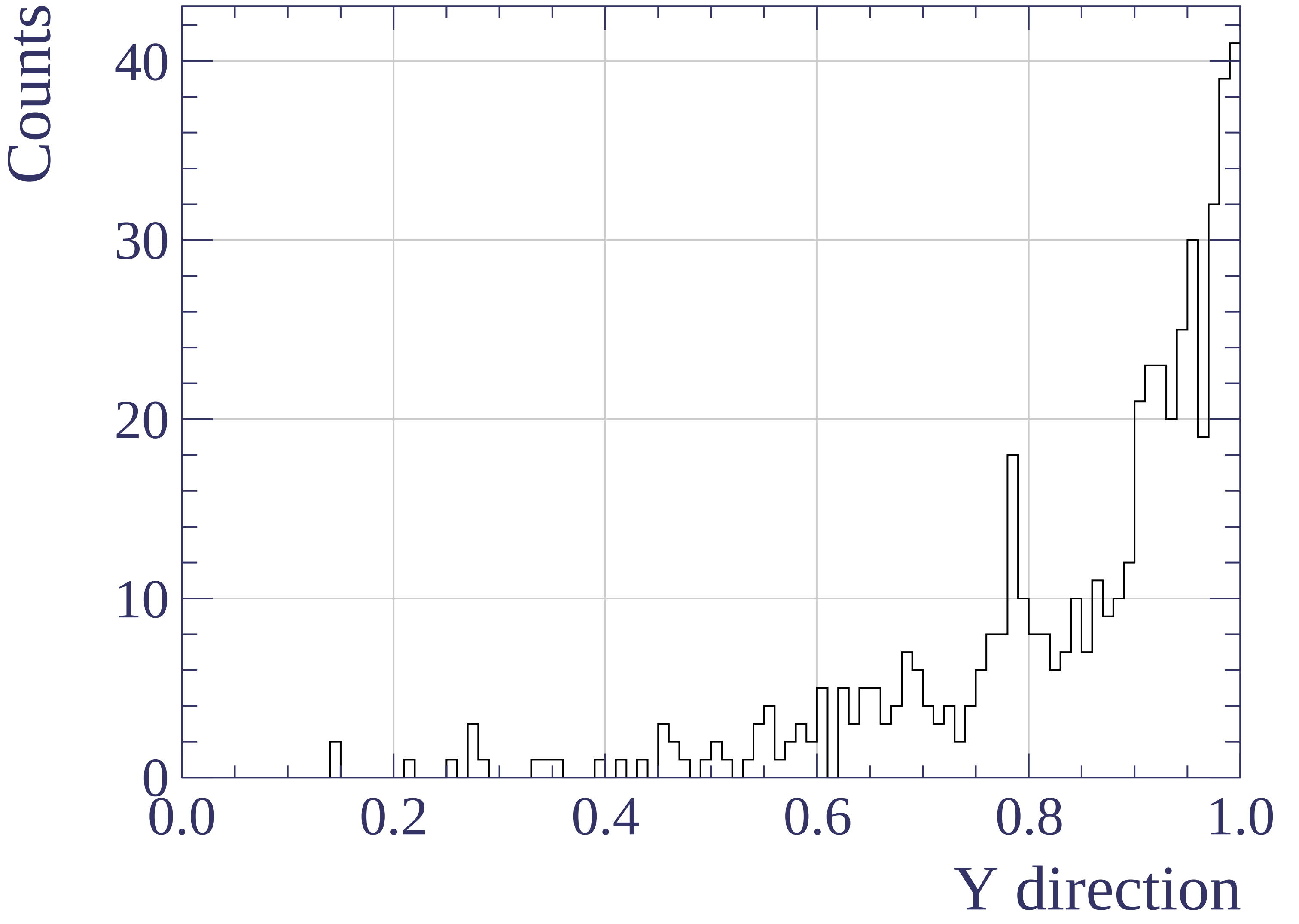


Count
0
5
10
15
0
10
20
30
40
50
χ^2

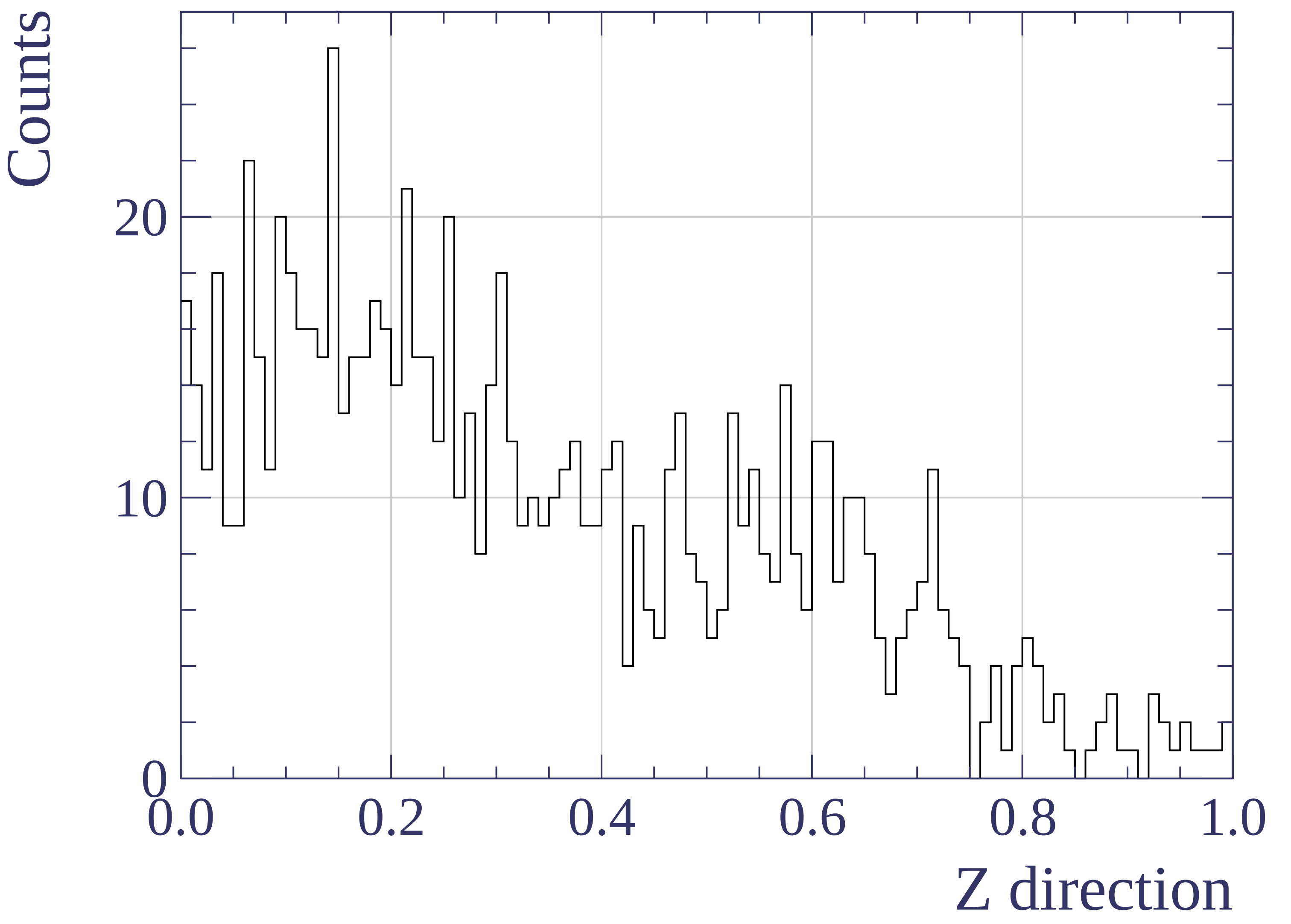


Count
0
5
10
−1000
−500
0
500
1000
Track X position [mm]

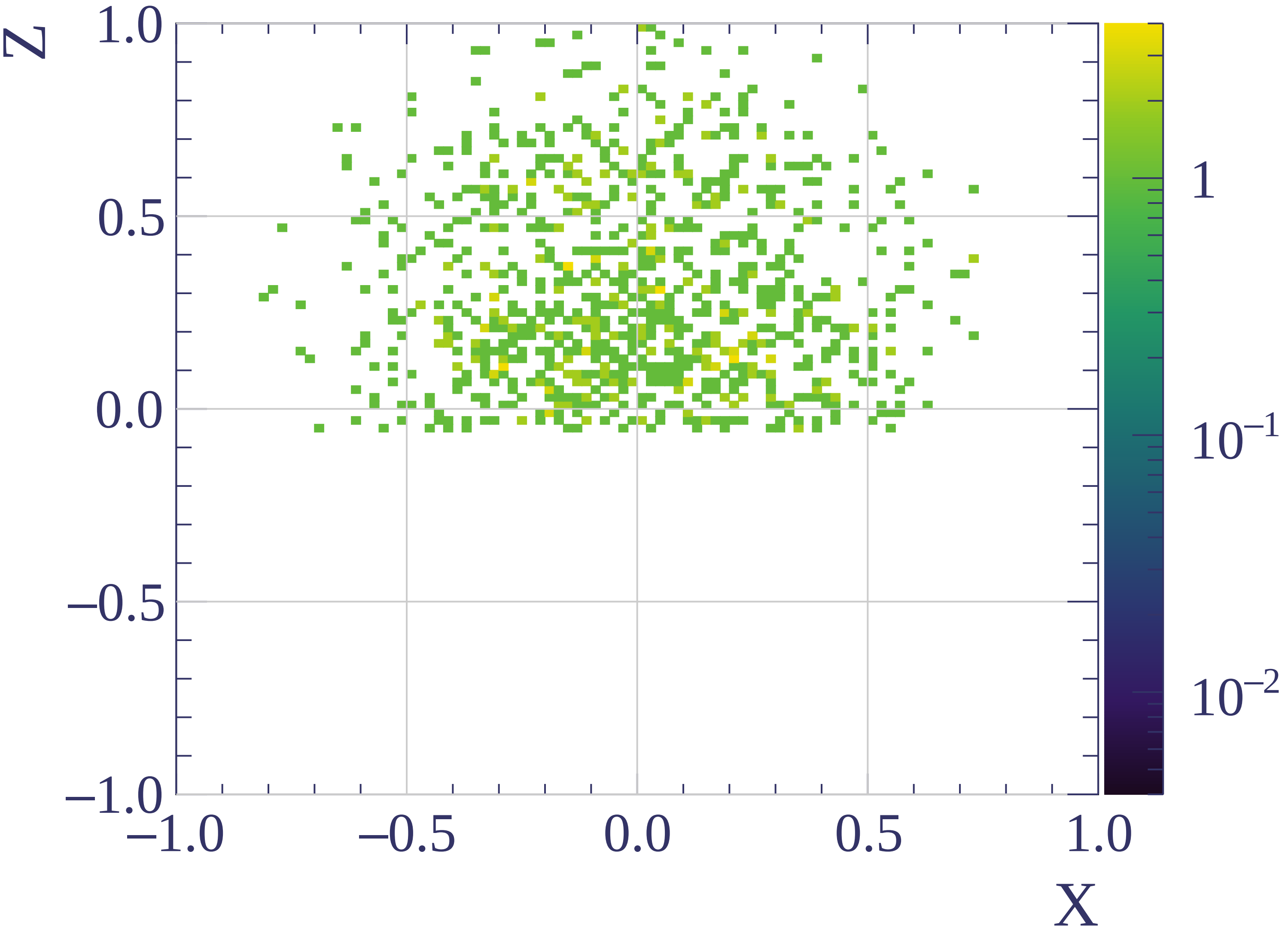

Counts
20
15
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction



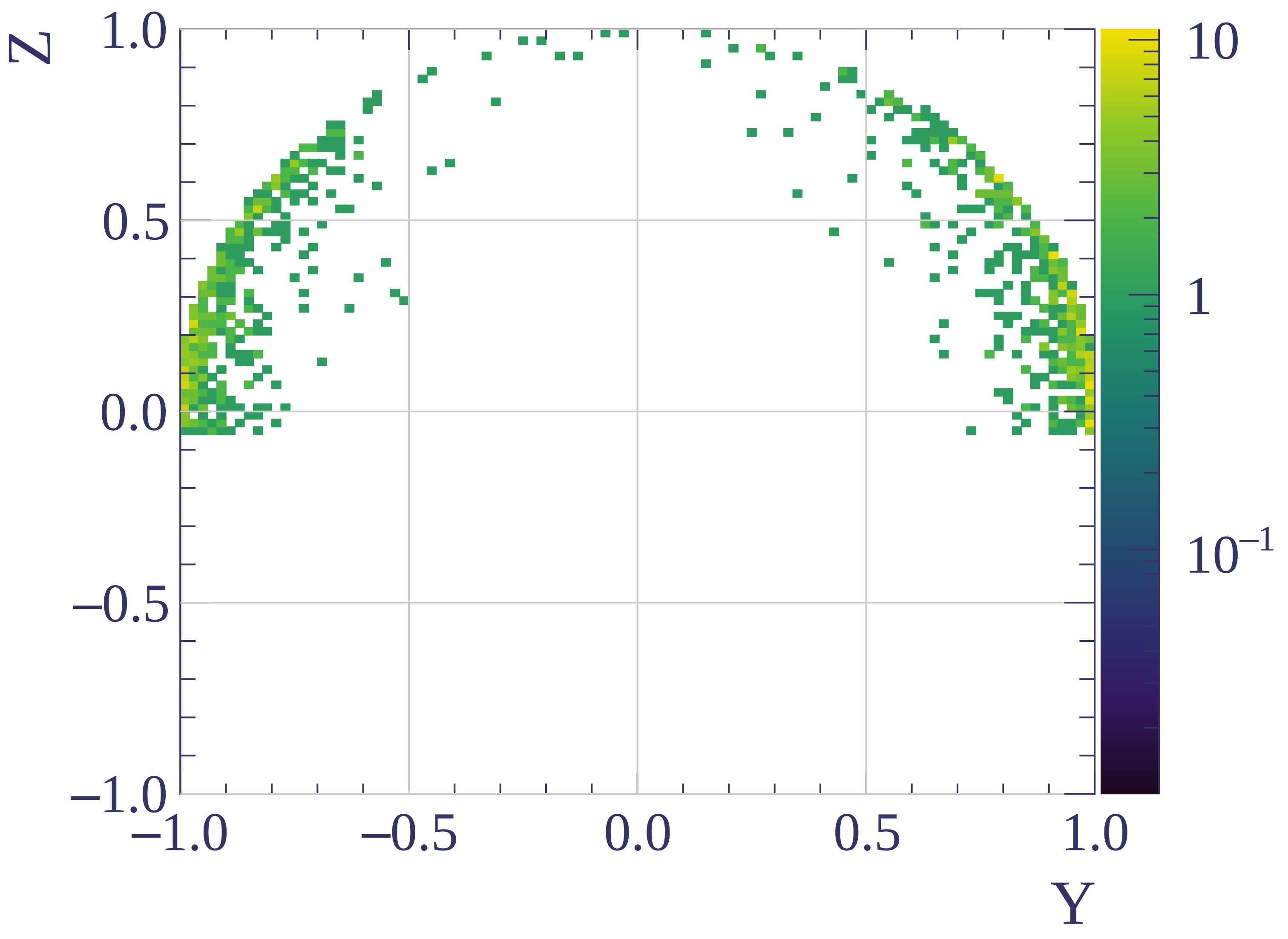
Counts
40
30
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



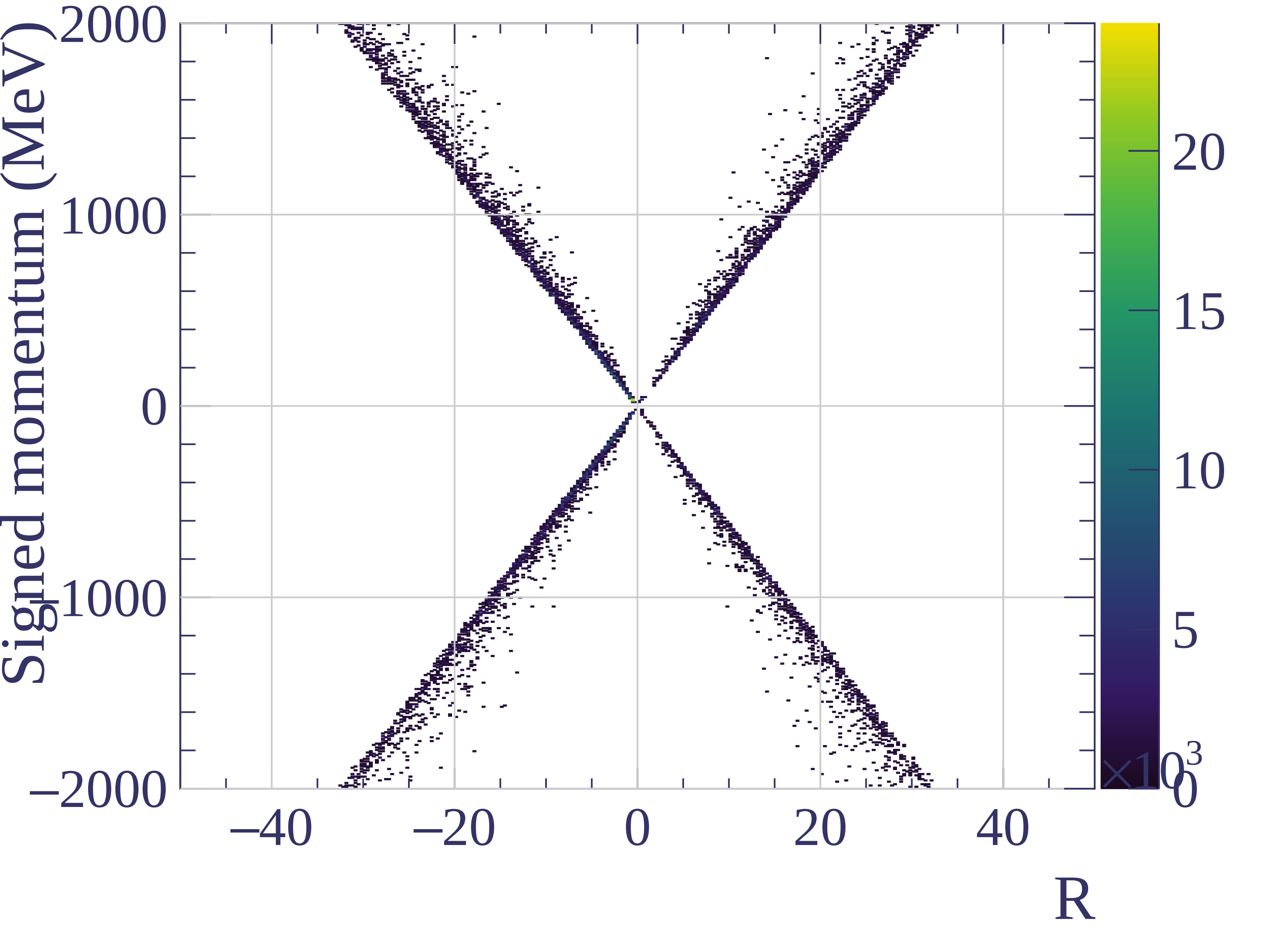
Counts
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction

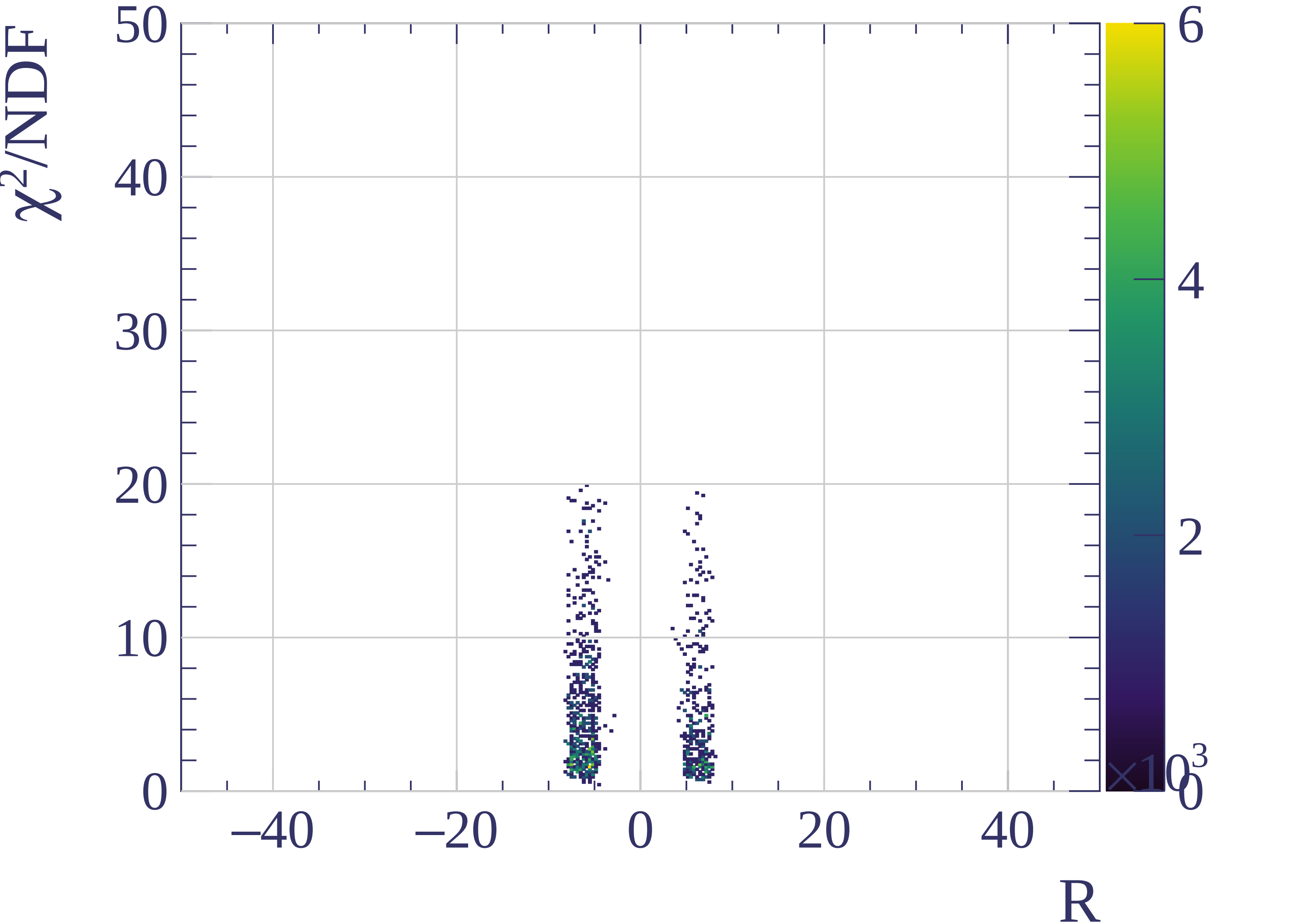


Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
1
10^{-1}
10^{-2}

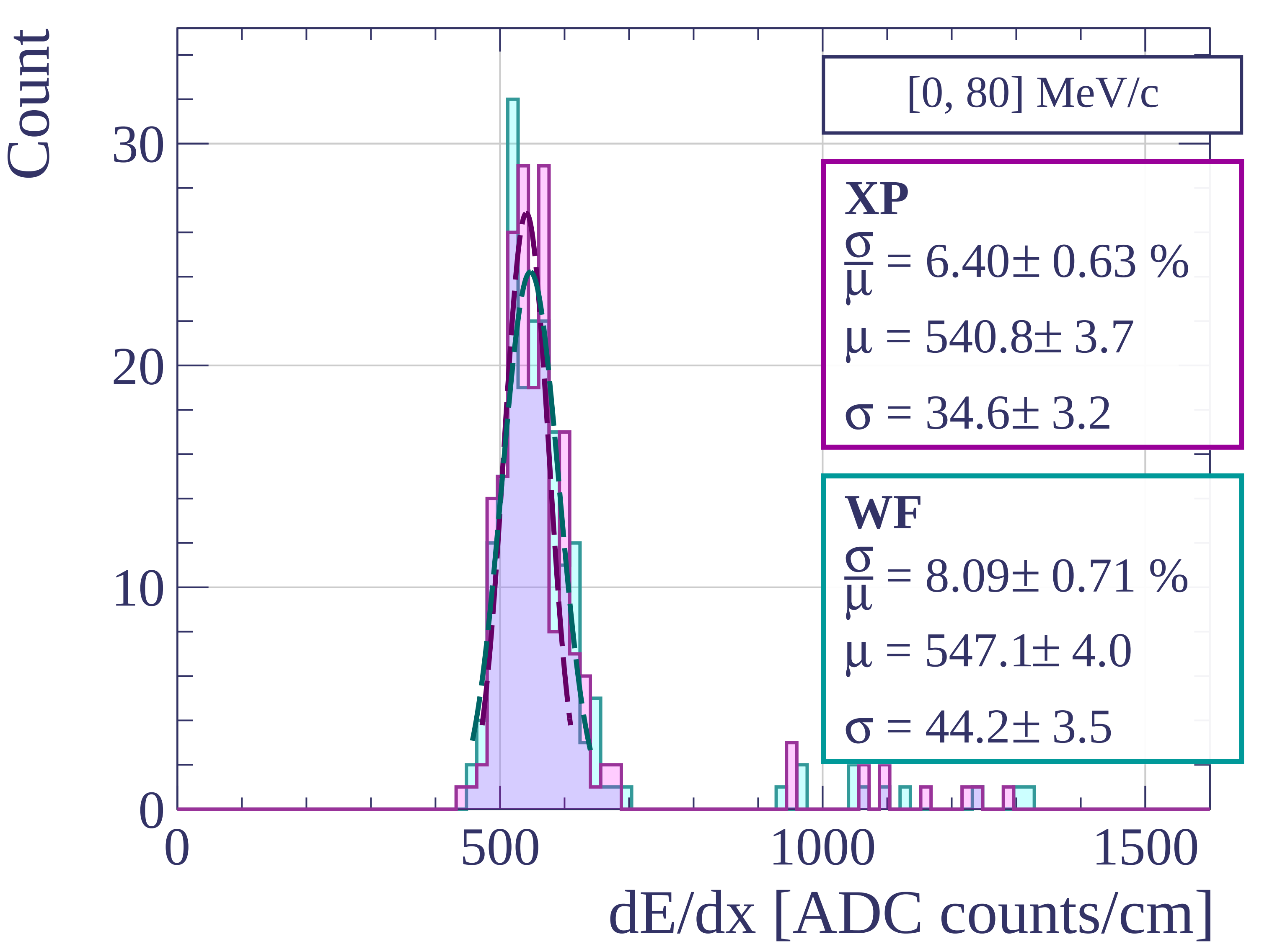


Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
10
1
10^{-1}

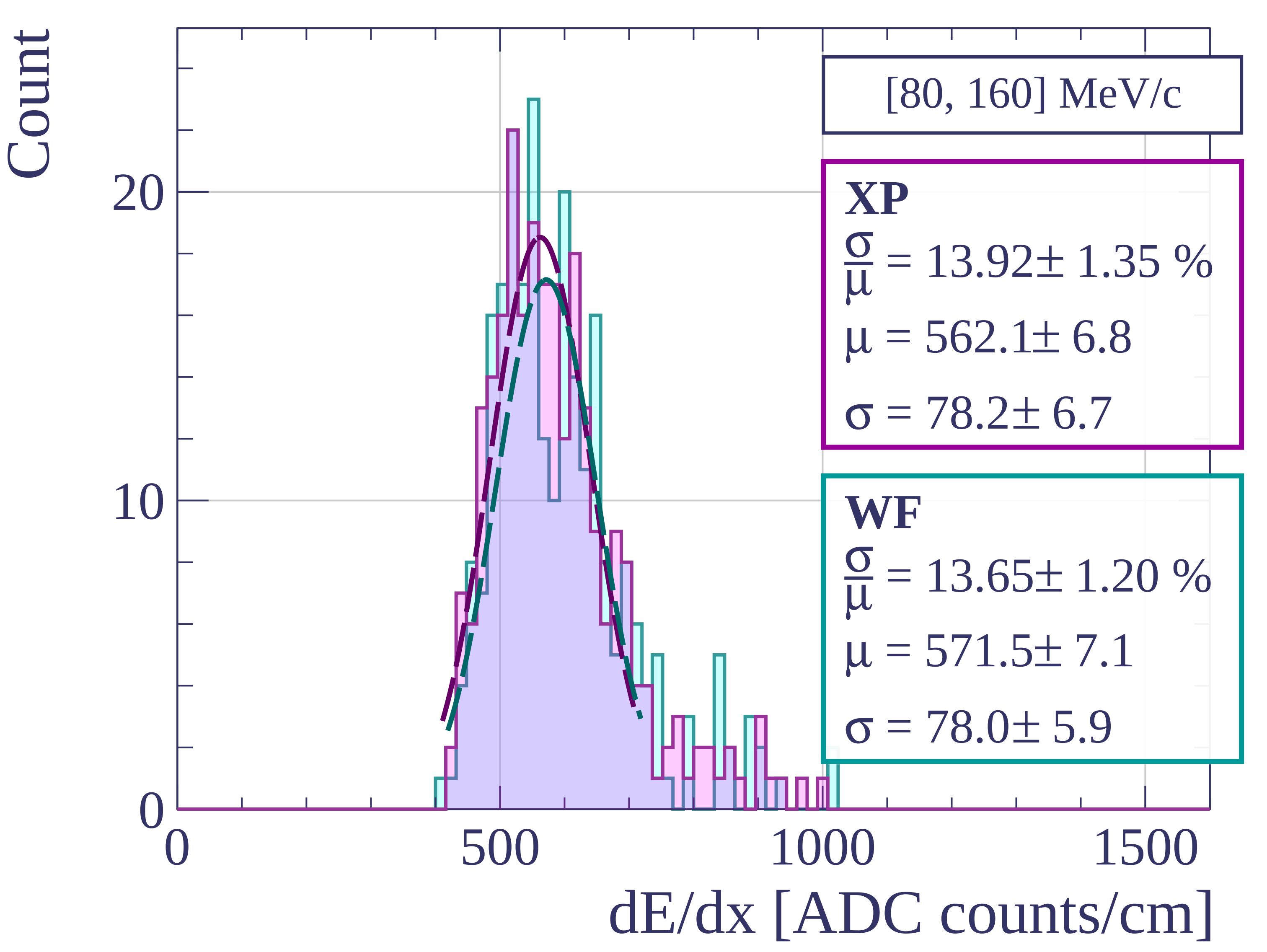

Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
20
15
10
5
0
$\times 10^3$



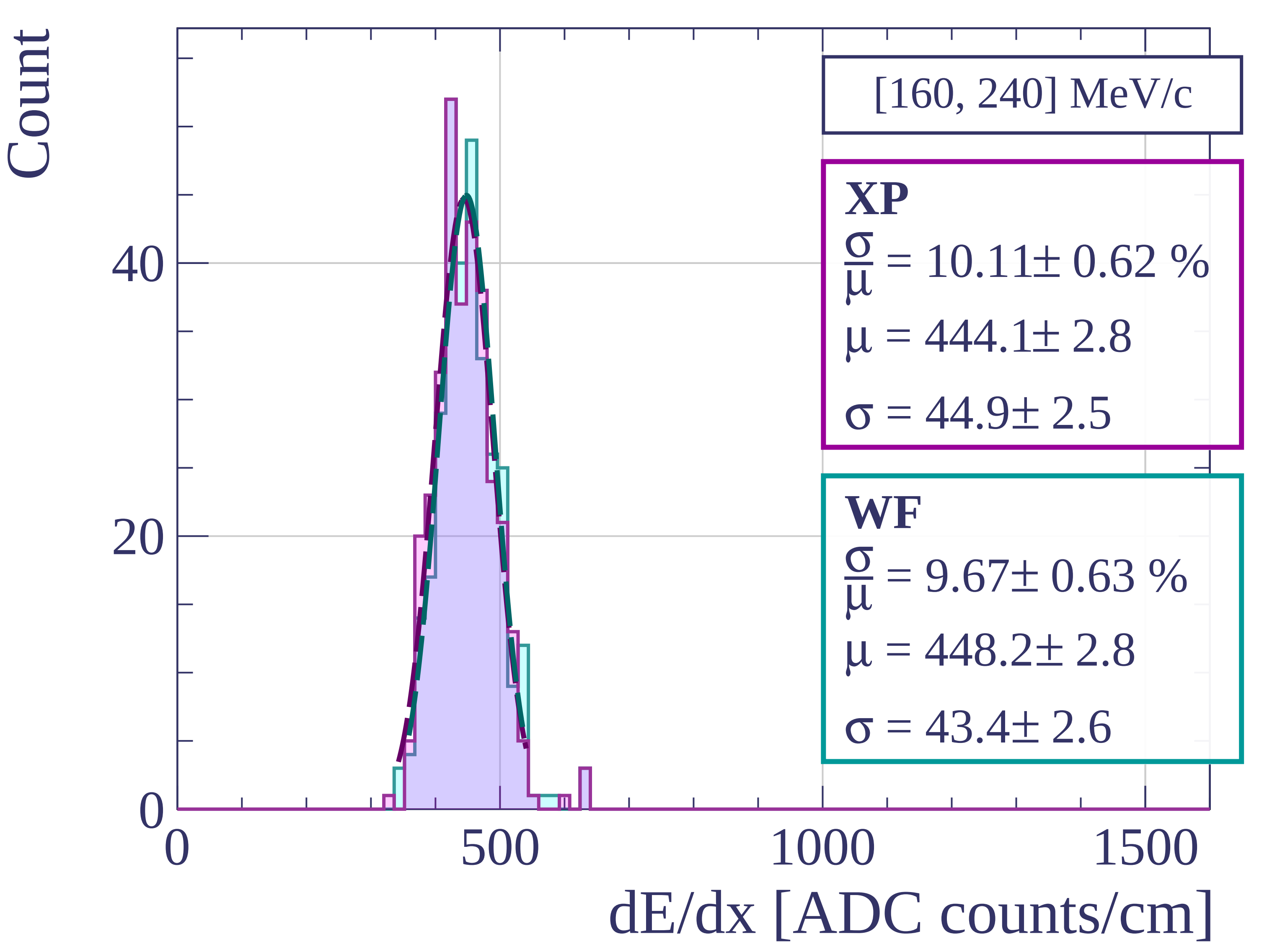
χ^2/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
6
4
2
0
$\times10^3$

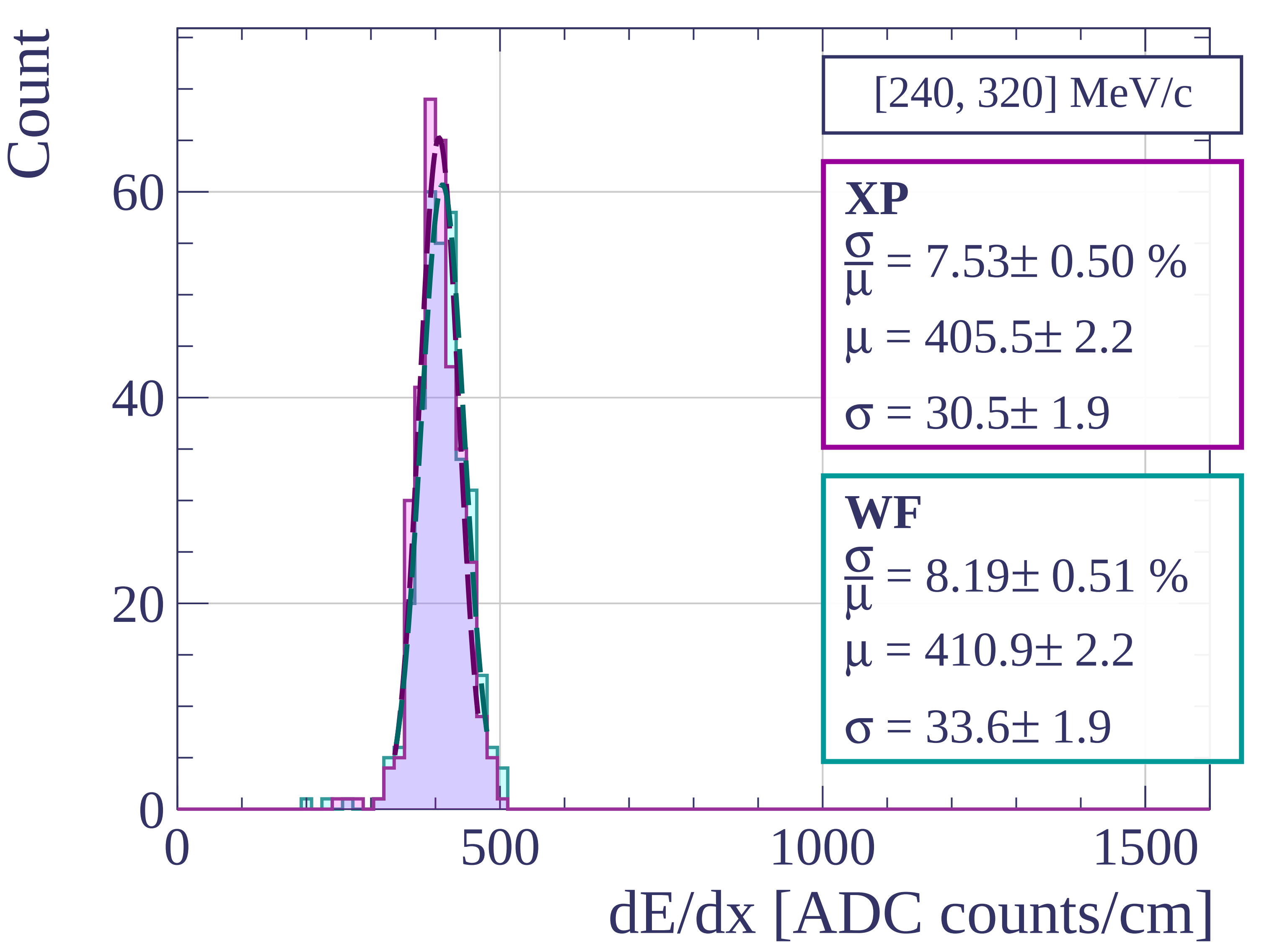


Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.40± 0.63 %
μ = 540.8± 3.7
σ = 34.6± 3.2
WF
$\frac{\sigma}{\mu}$ = 8.09± 0.71 %
μ = 547.1± 4.0
σ = 44.2± 3.5

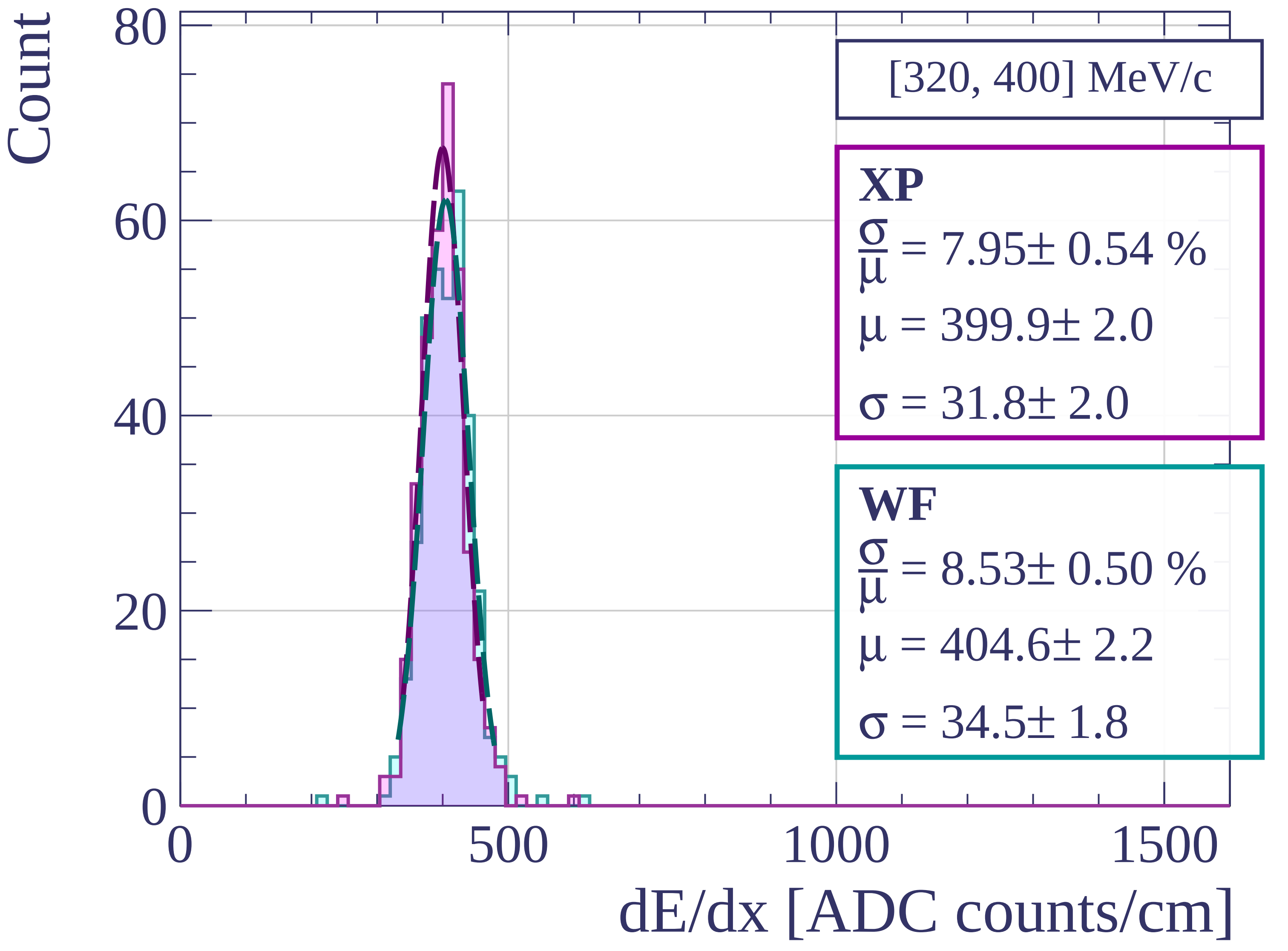


Count
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 160] MeV/c
XP
σ/μ = 13.92± 1.35 %
μ = 562.1± 6.8
σ = 78.2± 6.7
WF
σ/μ = 13.65± 1.20 %
μ = 571.5± 7.1
σ = 78.0± 5.9

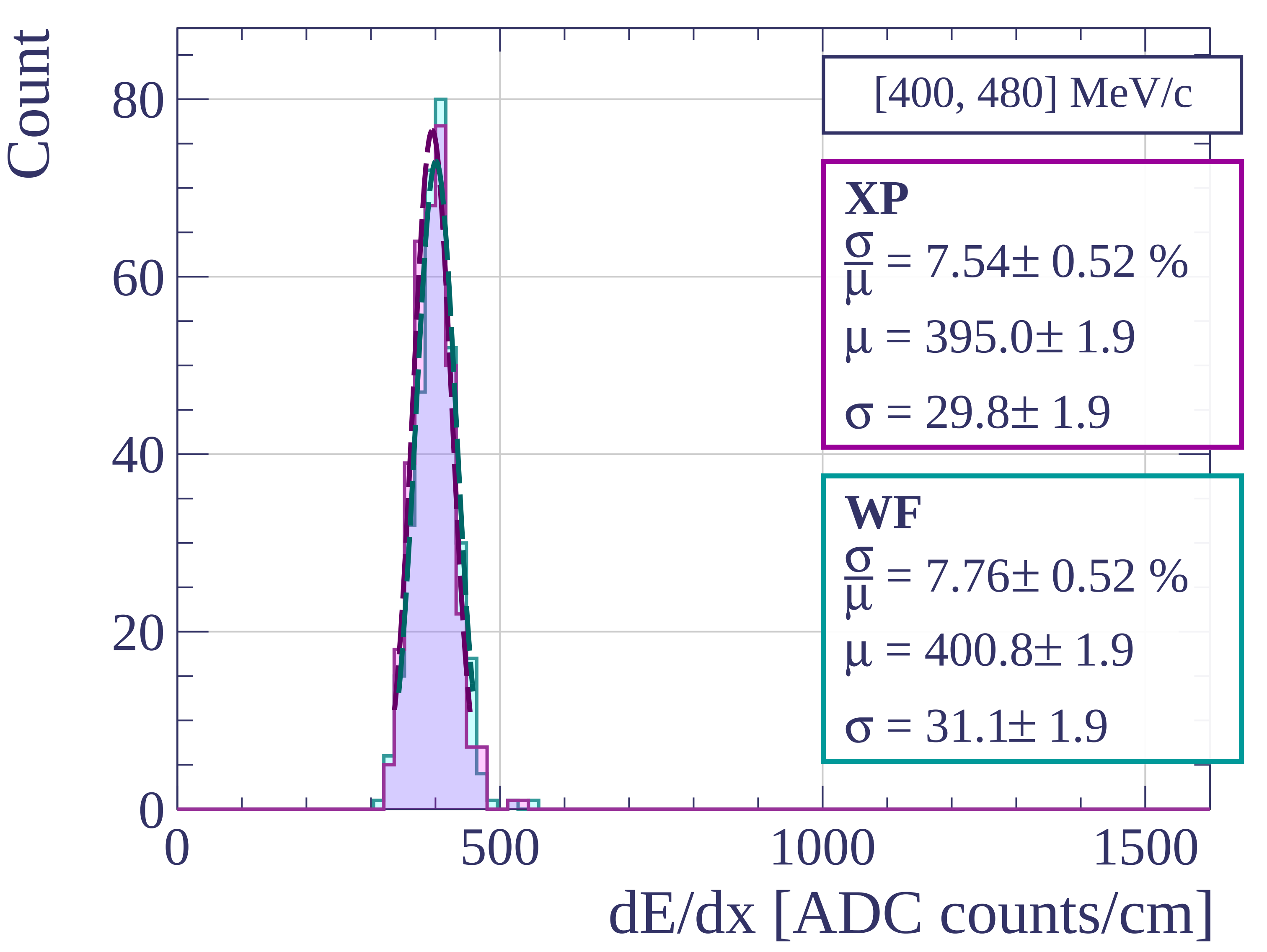

Count
0
20
40
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 240] MeV/c
XP
σ/μ = 10.11± 0.62 %
μ = 444.1± 2.8
σ = 44.9± 2.5
WF
σ/μ = 9.67± 0.63 %
μ = 448.2± 2.8
σ = 43.4± 2.6

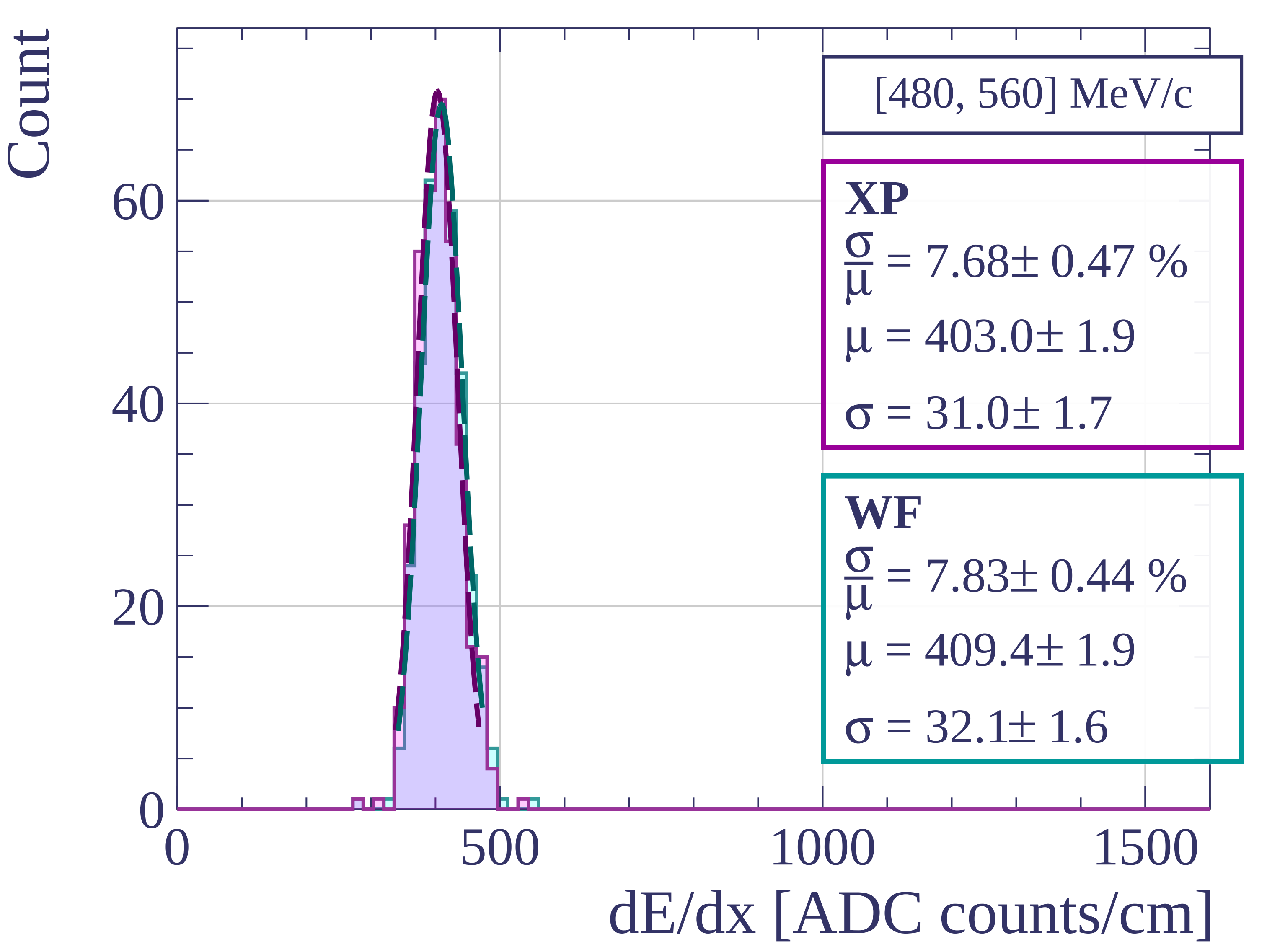


Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[240, 320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.53± 0.50 %
μ = 405.5± 2.2
σ = 30.5± 1.9
WF
$\frac{\sigma}{\mu}$ = 8.19± 0.51 %
μ = 410.9± 2.2
σ = 33.6± 1.9

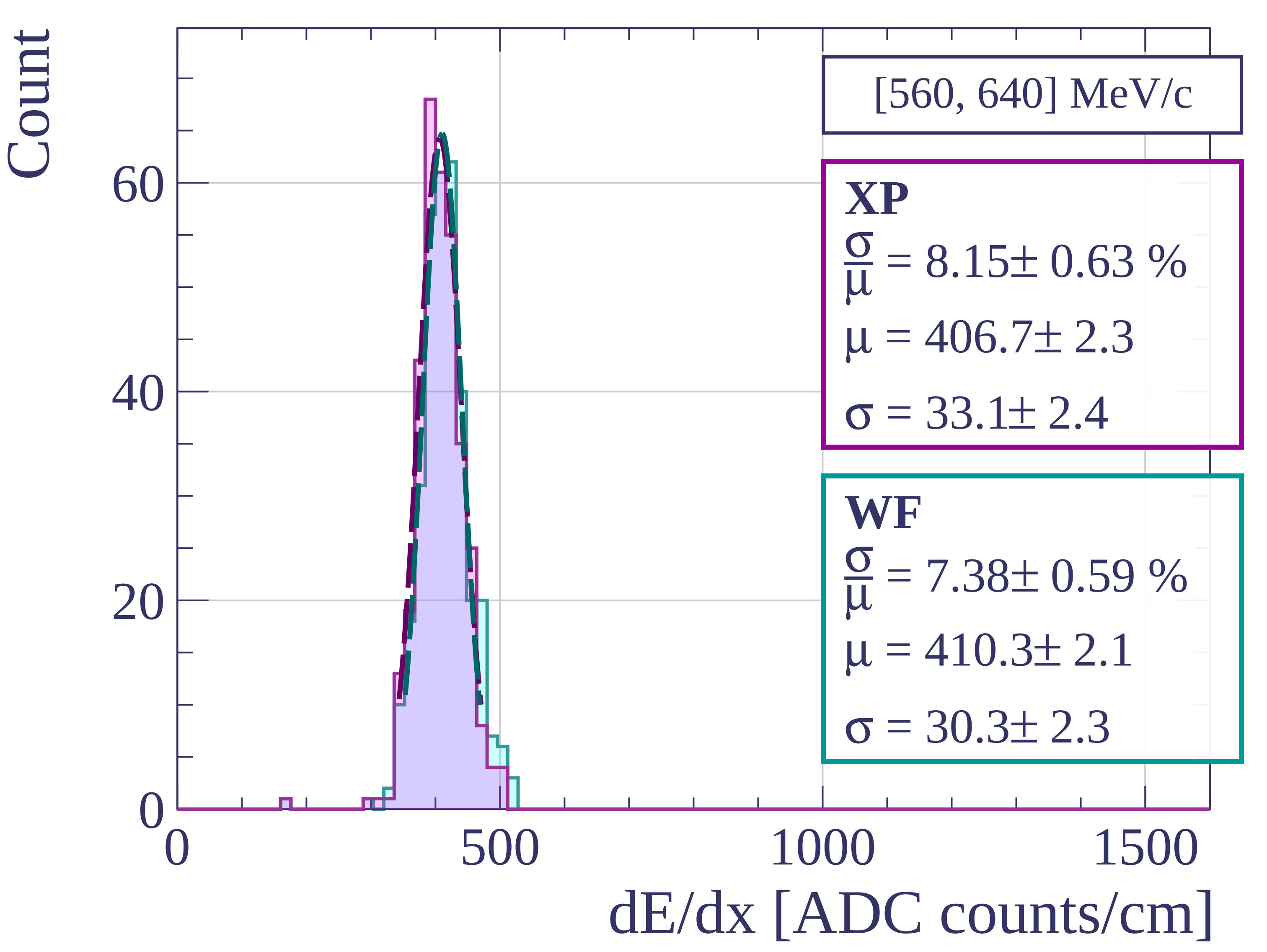


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[320, 400] MeV/c
XP
σ/μ = 7.95± 0.54 %
μ = 399.9± 2.0
σ = 31.8± 2.0
WF
σ/μ = 8.53± 0.50 %
μ = 404.6± 2.2
σ = 34.5± 1.8

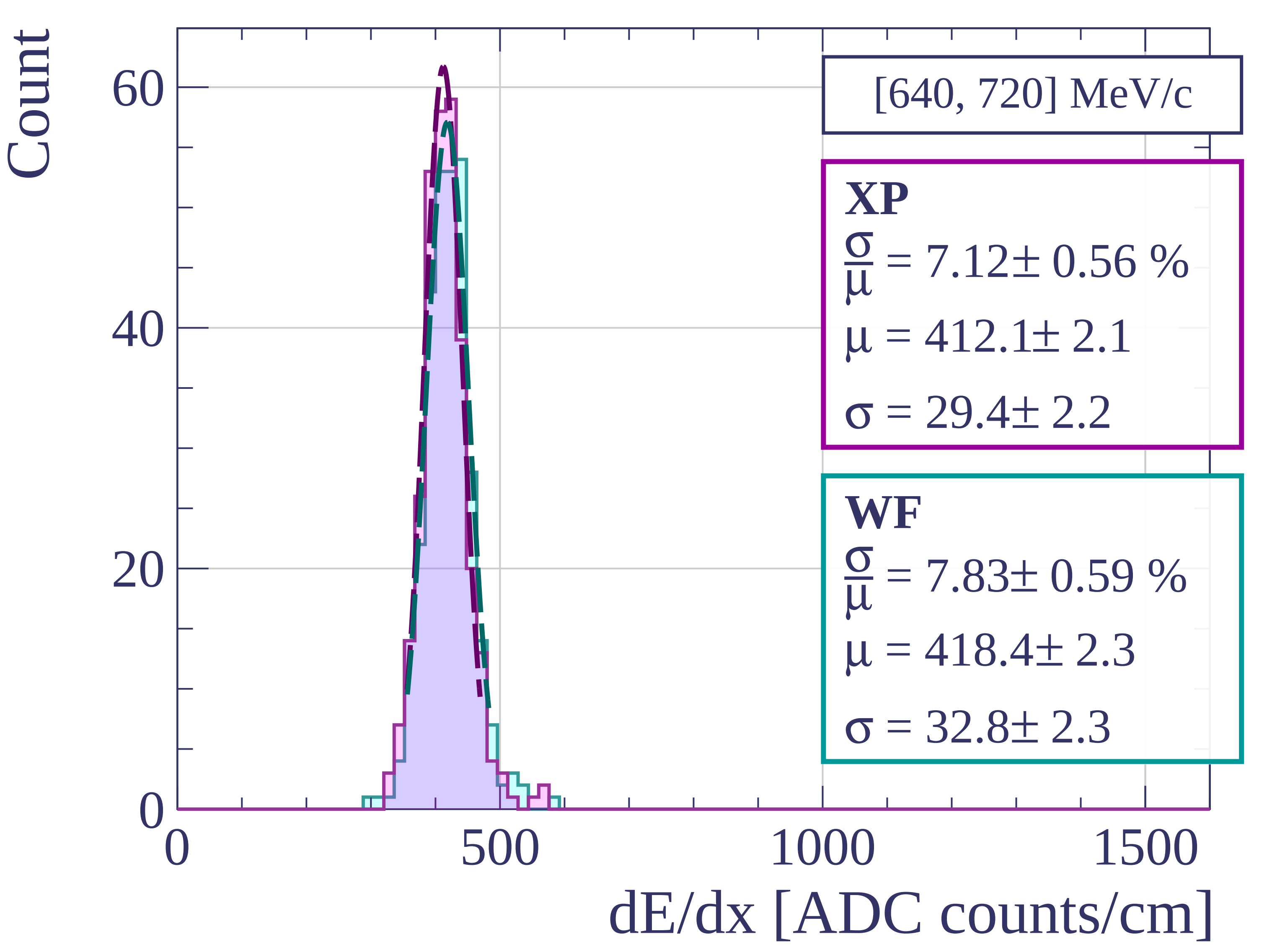

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.54± 0.52 %
μ = 395.0± 1.9
σ = 29.8± 1.9
WF
$\frac{\sigma}{\mu}$ = 7.76± 0.52 %
μ = 400.8± 1.9
σ = 31.1± 1.9



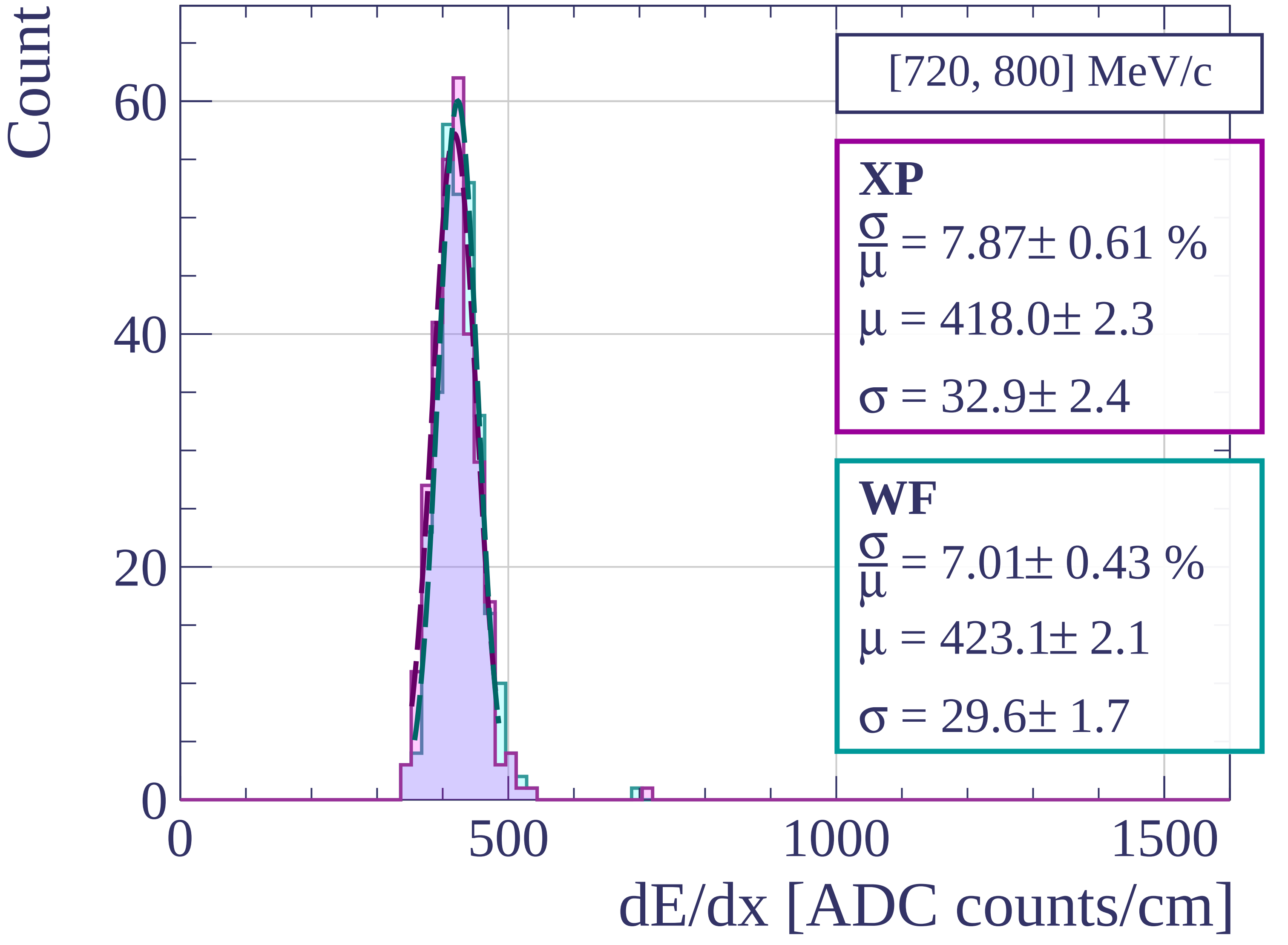
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.68± 0.47 %
μ = 403.0± 1.9
σ = 31.0± 1.7
WF
$\frac{\sigma}{\mu}$ = 7.83± 0.44 %
μ = 409.4± 1.9
σ = 32.1± 1.6



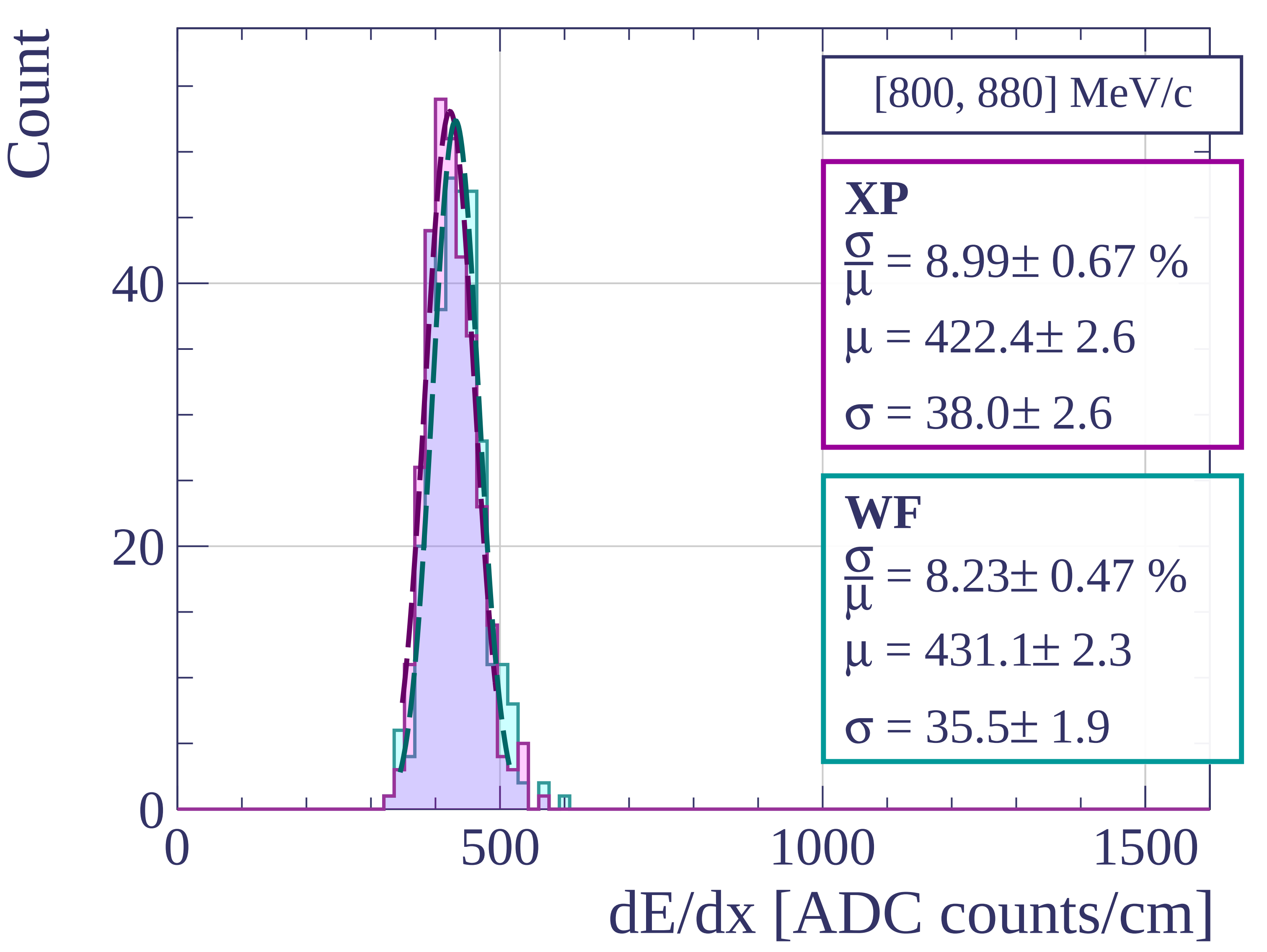
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 640] MeV/c
XP
σ/μ = 8.15± 0.63 %
μ = 406.7± 2.3
σ = 33.1± 2.4
WF
σ/μ = 7.38± 0.59 %
μ = 410.3± 2.1
σ = 30.3± 2.3

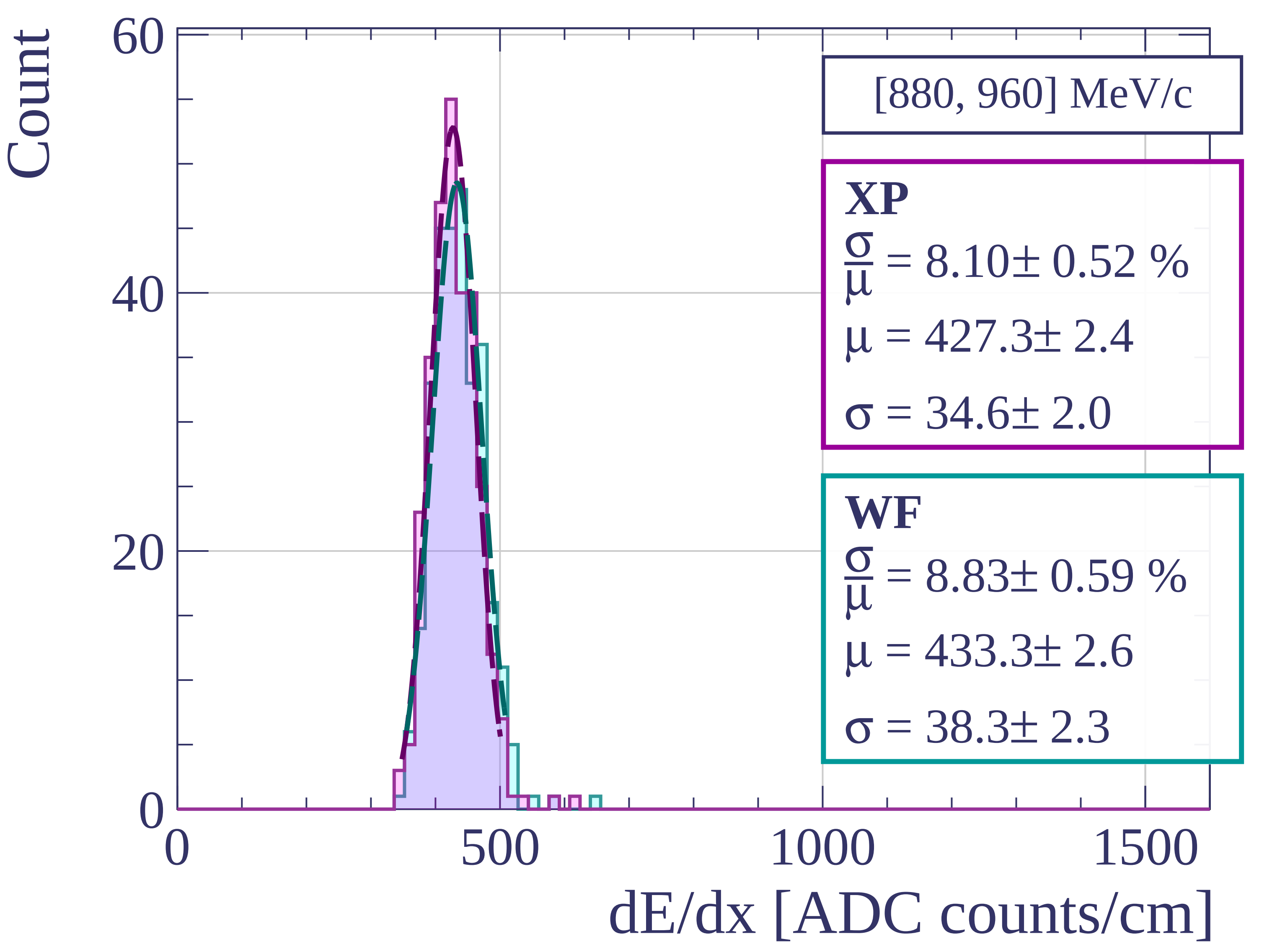


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[640, 720] MeV/c
XP
σ/μ = 7.12± 0.56 %
μ = 412.1± 2.1
σ = 29.4± 2.2
WF
σ/μ = 7.83± 0.59 %
μ = 418.4± 2.3
σ = 32.8± 2.3

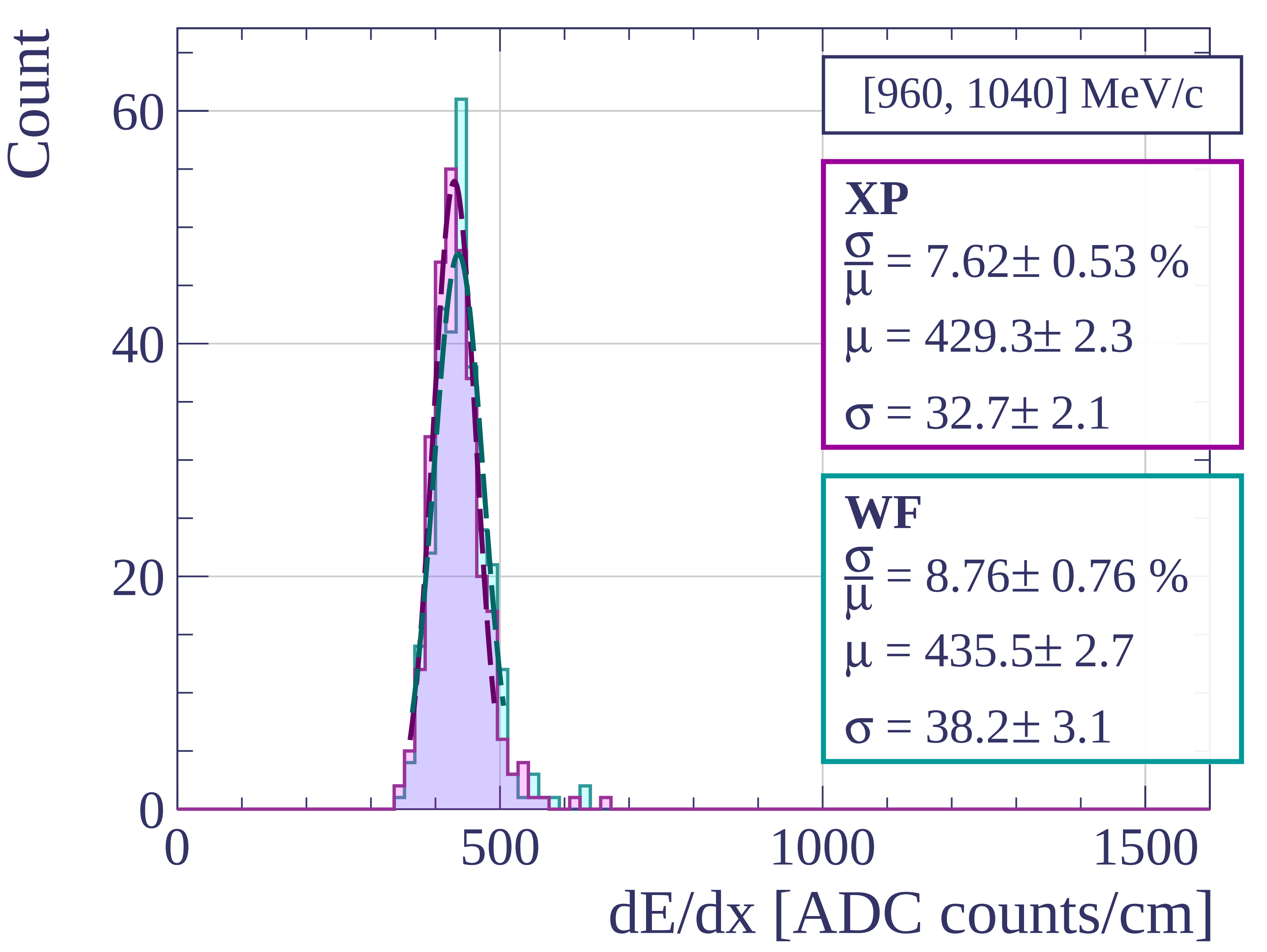


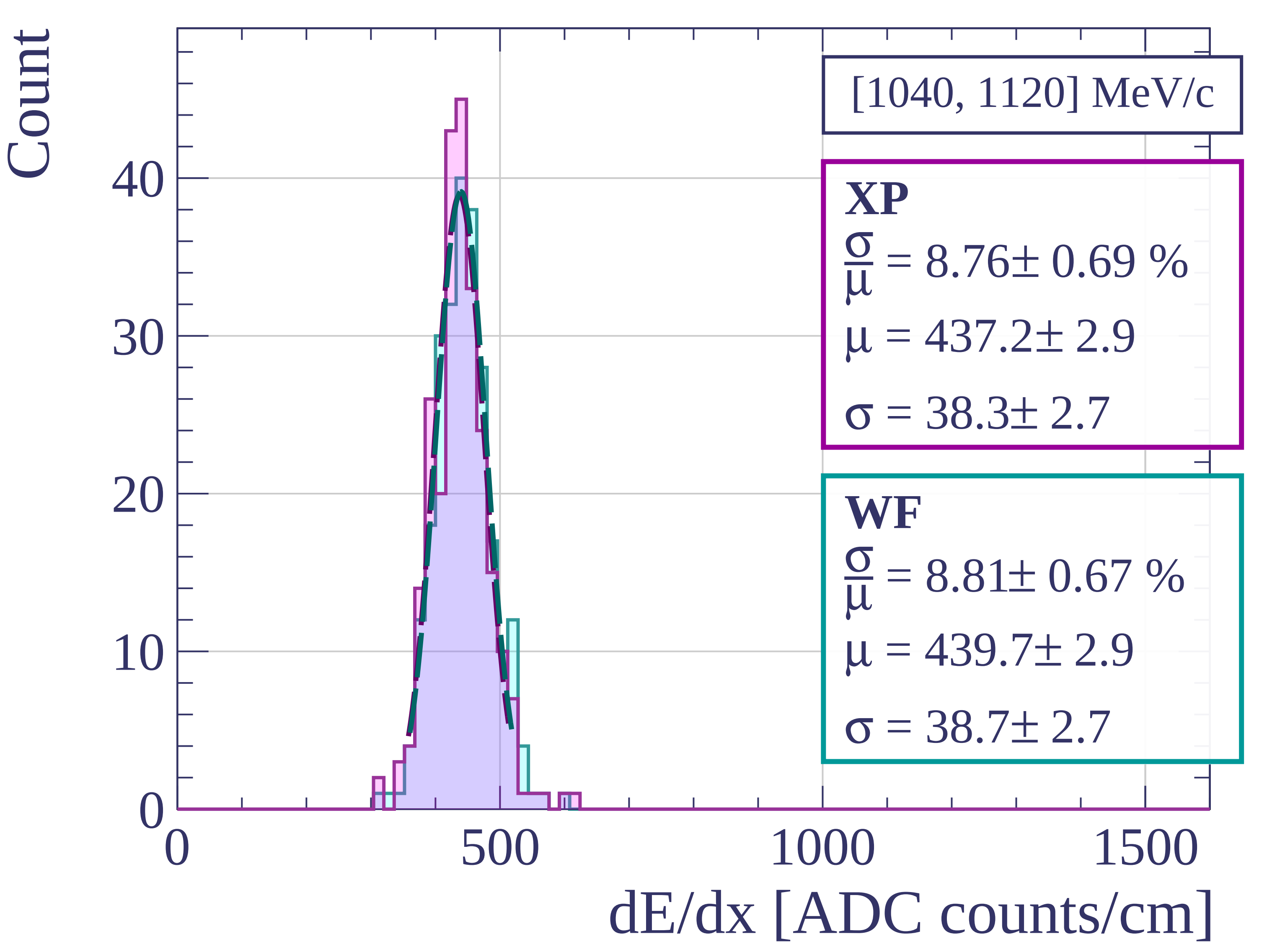
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.87± 0.61 %
μ = 418.0± 2.3
σ = 32.9± 2.4
WF
$\frac{\sigma}{\mu}$ = 7.01± 0.43 %
μ = 423.1± 2.1
σ = 29.6± 1.7

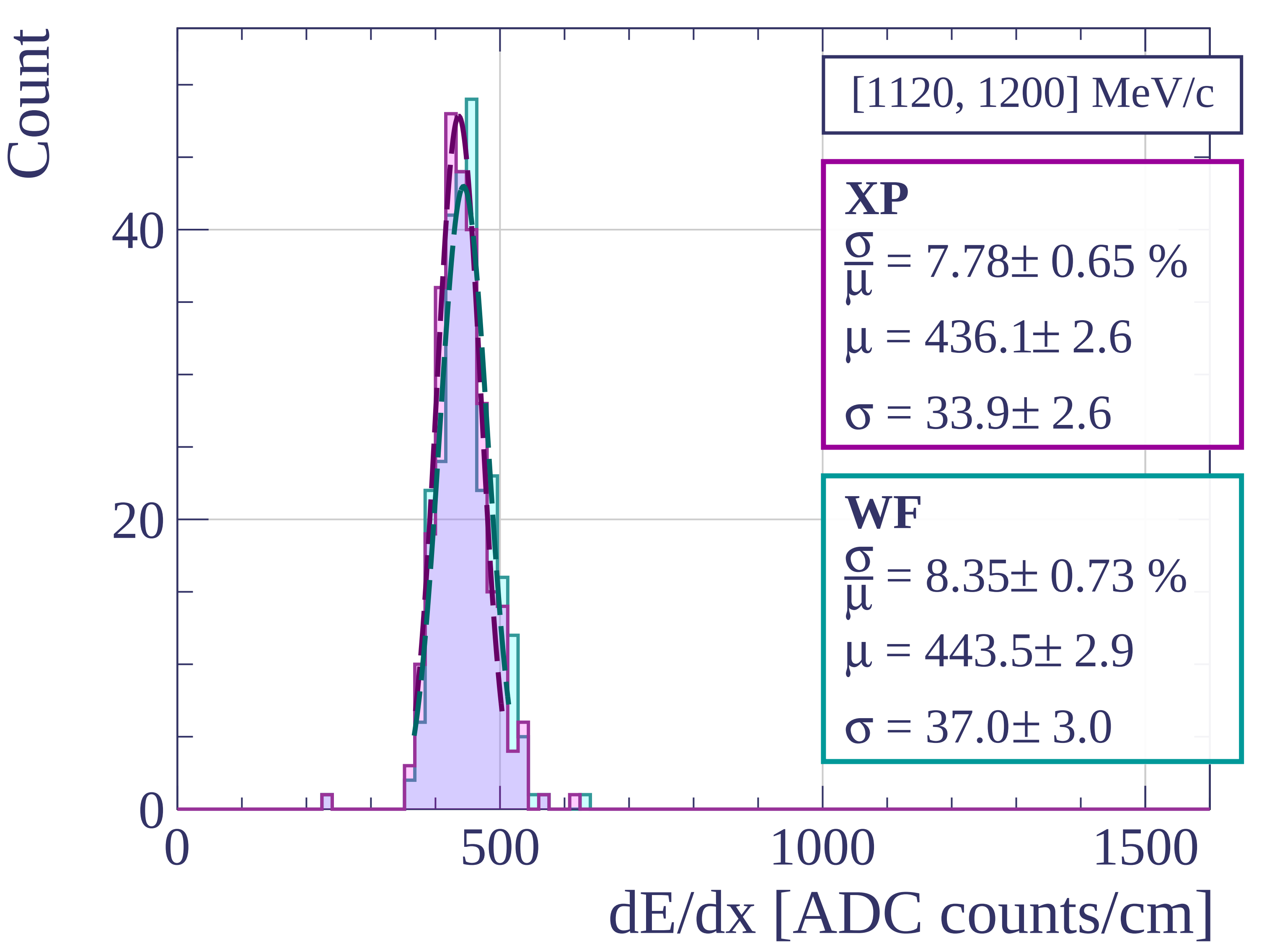

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.99± 0.67 %
μ = 422.4± 2.6
σ = 38.0± 2.6
WF
$\frac{\sigma}{\mu}$ = 8.23± 0.47 %
μ = 431.1± 2.3
σ = 35.5± 1.9



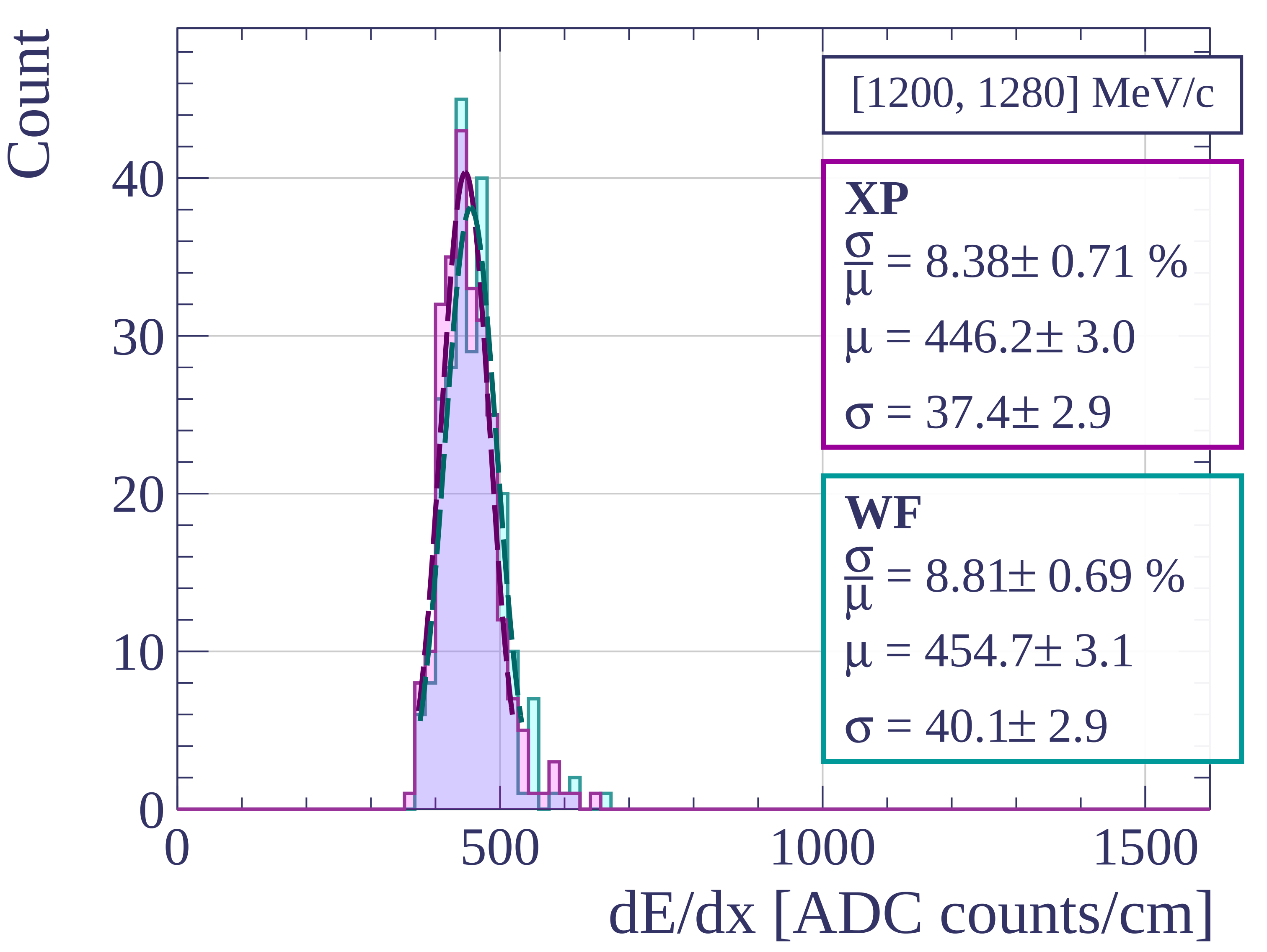
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[880, 960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.10± 0.52 %
μ = 427.3± 2.4
σ = 34.6± 2.0
WF
$\frac{\sigma}{\mu}$ = 8.83± 0.59 %
μ = 433.3± 2.6
σ = 38.3± 2.3


Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[960, 1040] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.62± 0.53 %
μ = 429.3± 2.3
σ = 32.7± 2.1
WF
$\frac{\sigma}{\mu}$ = 8.76± 0.76 %
μ = 435.5± 2.7
σ = 38.2± 3.1

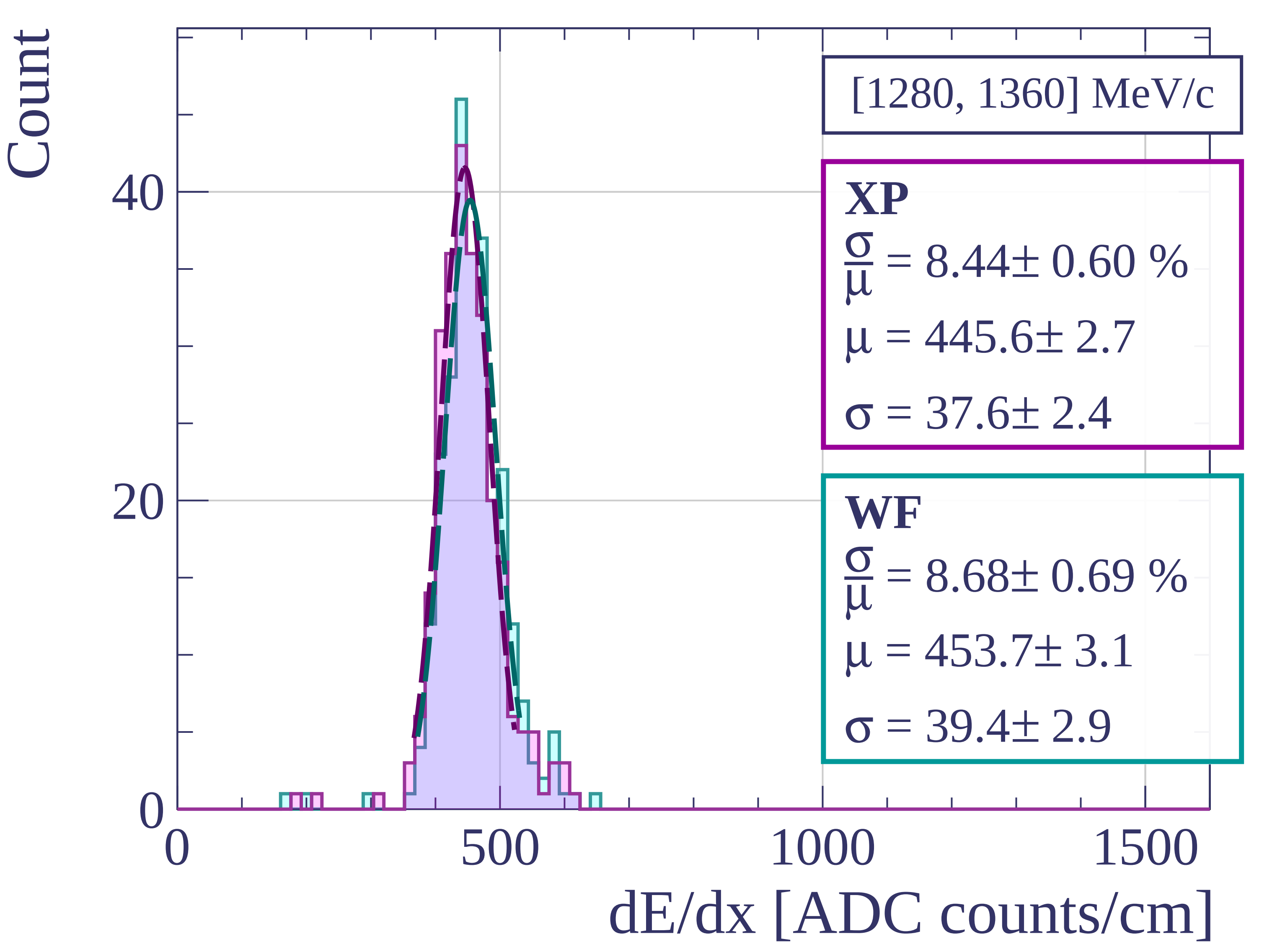

Count
dE/dx [ADC counts/cm]
[1040, 1120] MeV/c
XP
σ/μ = 8.76± 0.69 %
μ = 437.2± 2.9
σ = 38.3± 2.7
WF
σ/μ = 8.81± 0.67 %
μ = 439.7± 2.9
σ = 38.7± 2.7
0
10
20
30
40
500
1000
1500



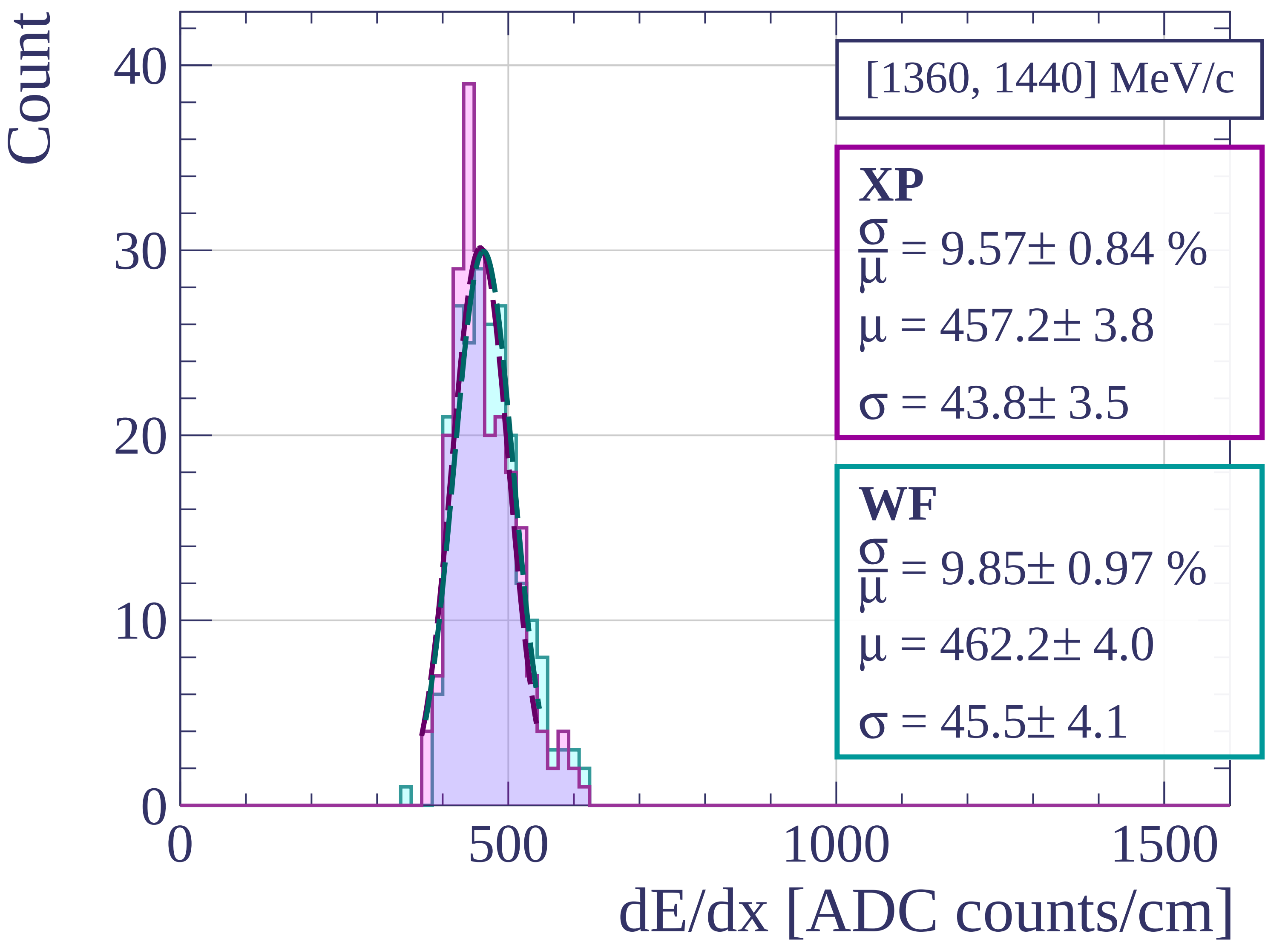
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1200] MeV/c
XP
σ/μ = 7.78± 0.65 %
μ = 436.1± 2.6
σ = 33.9± 2.6
WF
σ/μ = 8.35± 0.73 %
μ = 443.5± 2.9
σ = 37.0± 3.0

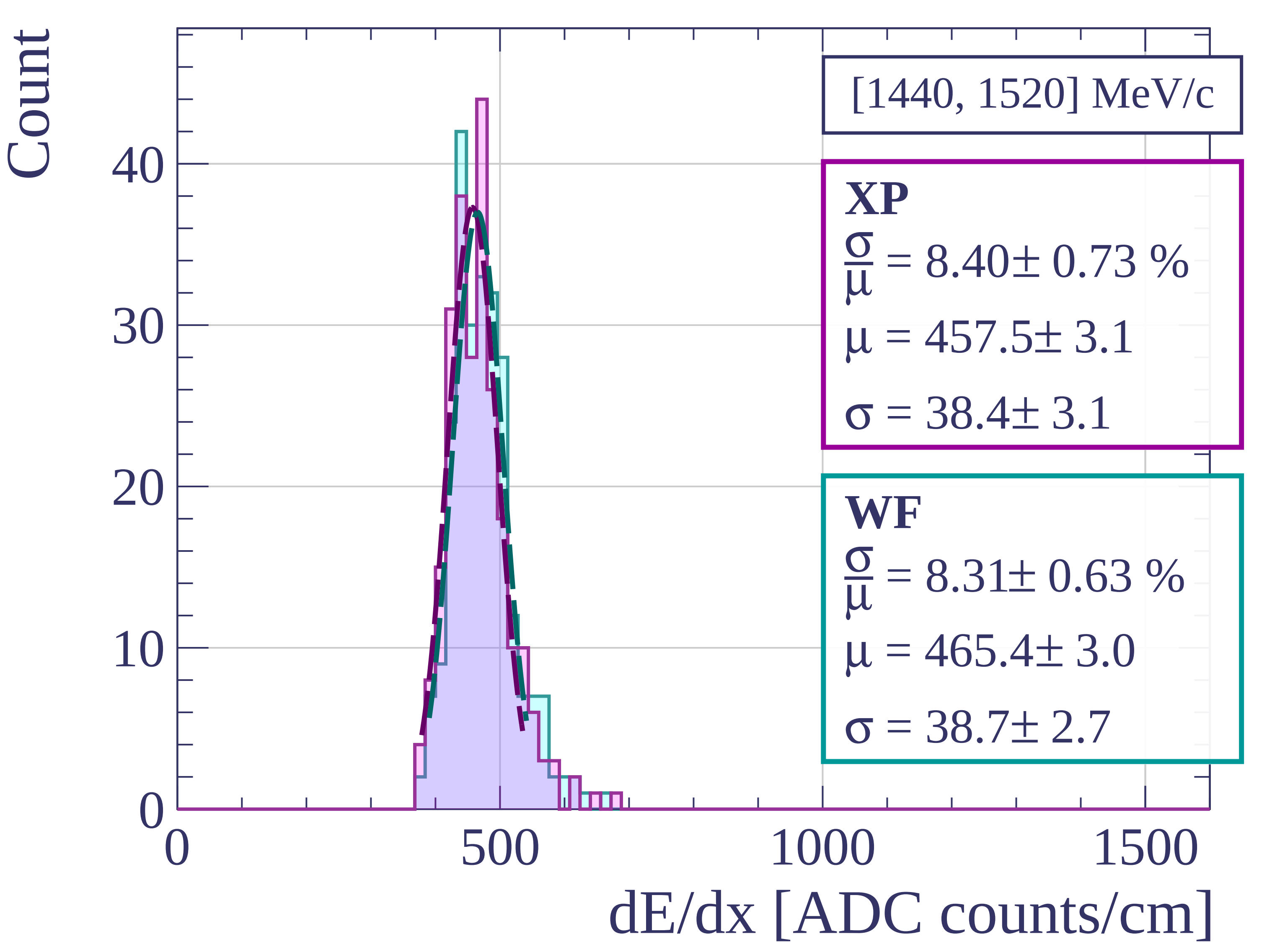


Count
0
10
20
30
40
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.38± 0.71 %
μ = 446.2± 3.0
σ = 37.4± 2.9
WF
$\frac{\sigma}{\mu}$ = 8.81± 0.69 %
μ = 454.7± 3.1
σ = 40.1± 2.9

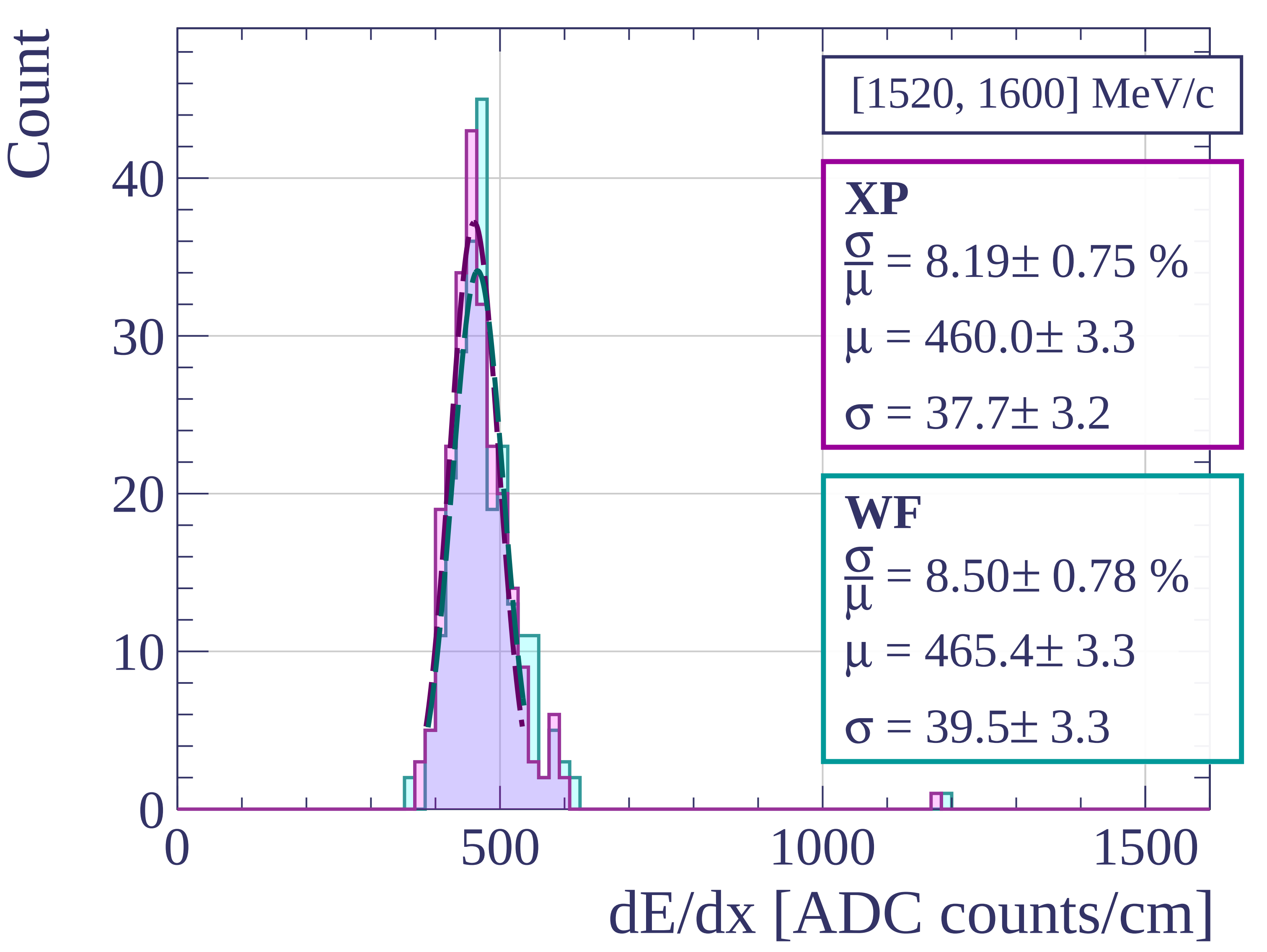


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.44± 0.60 %
μ = 445.6± 2.7
σ = 37.6± 2.4
WF
$\frac{\sigma}{\mu}$ = 8.68± 0.69 %
μ = 453.7± 3.1
σ = 39.4± 2.9

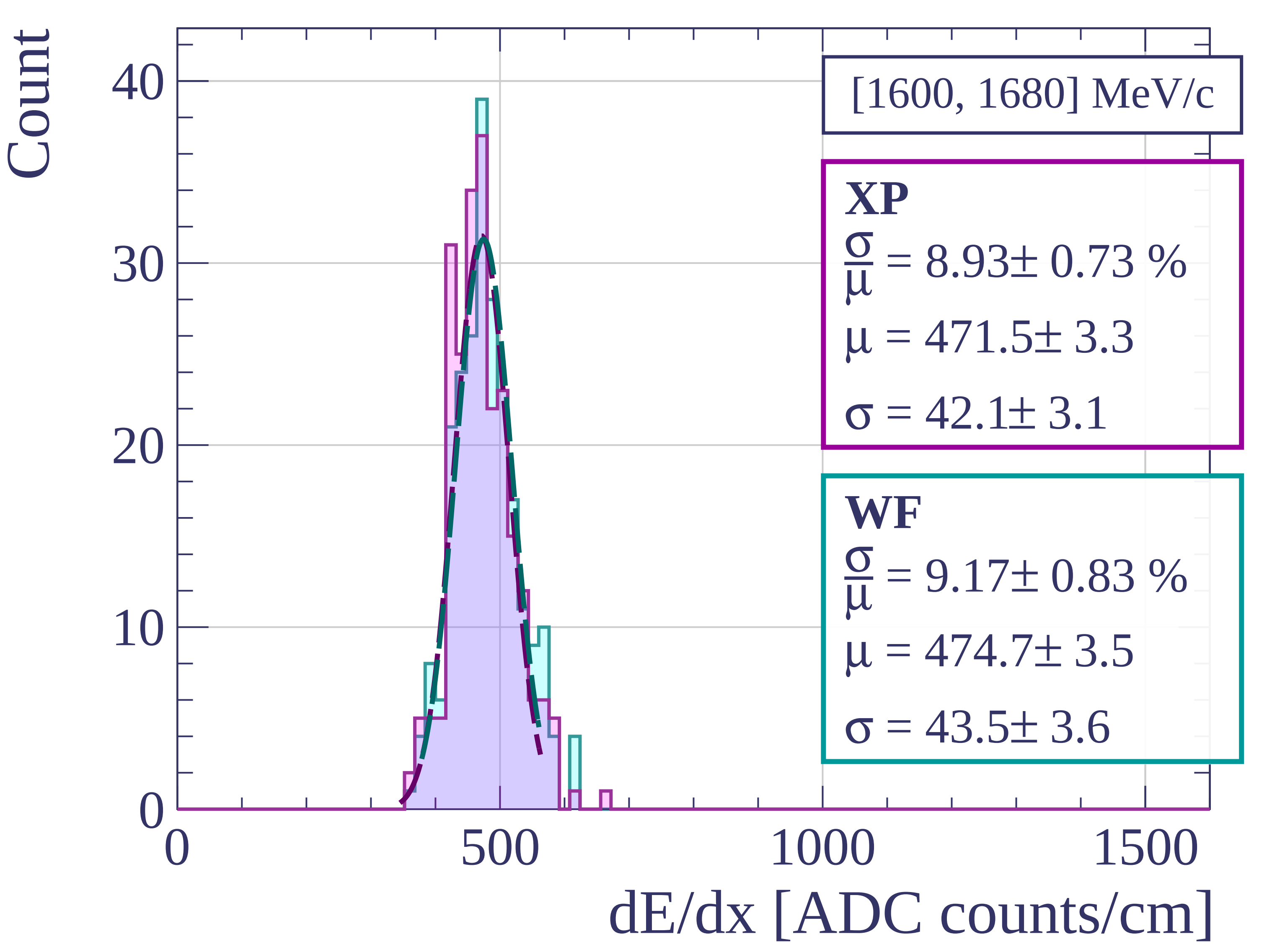

Count
dE/dx [ADC counts/cm]
0
10
20
30
40
0
500
1000
1500
[1360, 1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.57± 0.84 %
μ = 457.2± 3.8
σ = 43.8± 3.5
WF
$\frac{\sigma}{\mu}$ = 9.85± 0.97 %
μ = 462.2± 4.0
σ = 45.5± 4.1



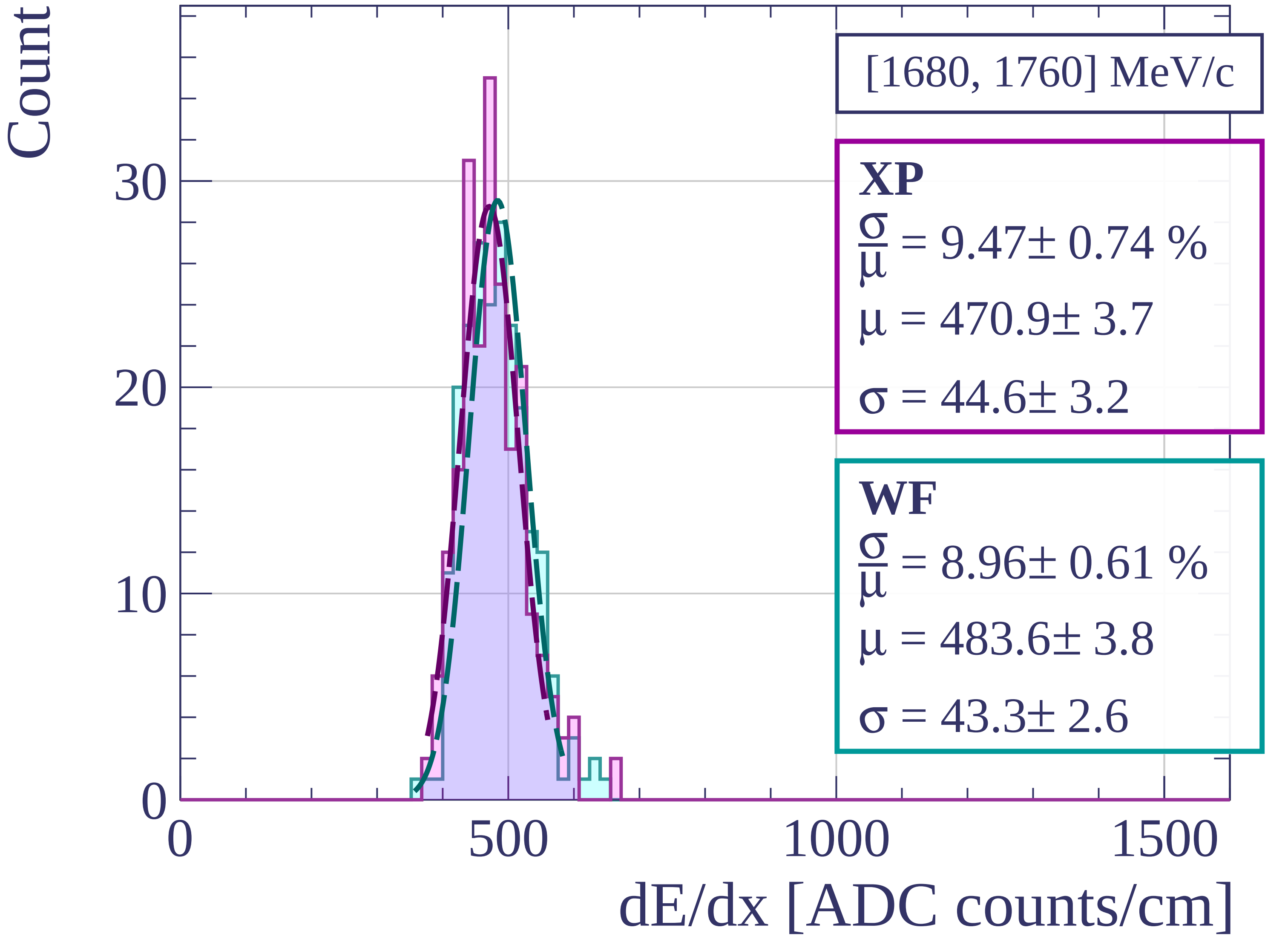
Count
dE/dx [ADC counts/cm]
[1440, 1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.40± 0.73 %
μ = 457.5± 3.1
σ = 38.4± 3.1
WF
$\frac{\sigma}{\mu}$ = 8.31± 0.63 %
μ = 465.4± 3.0
σ = 38.7± 2.7
0
10
20
30
40
0
500
1000
1500

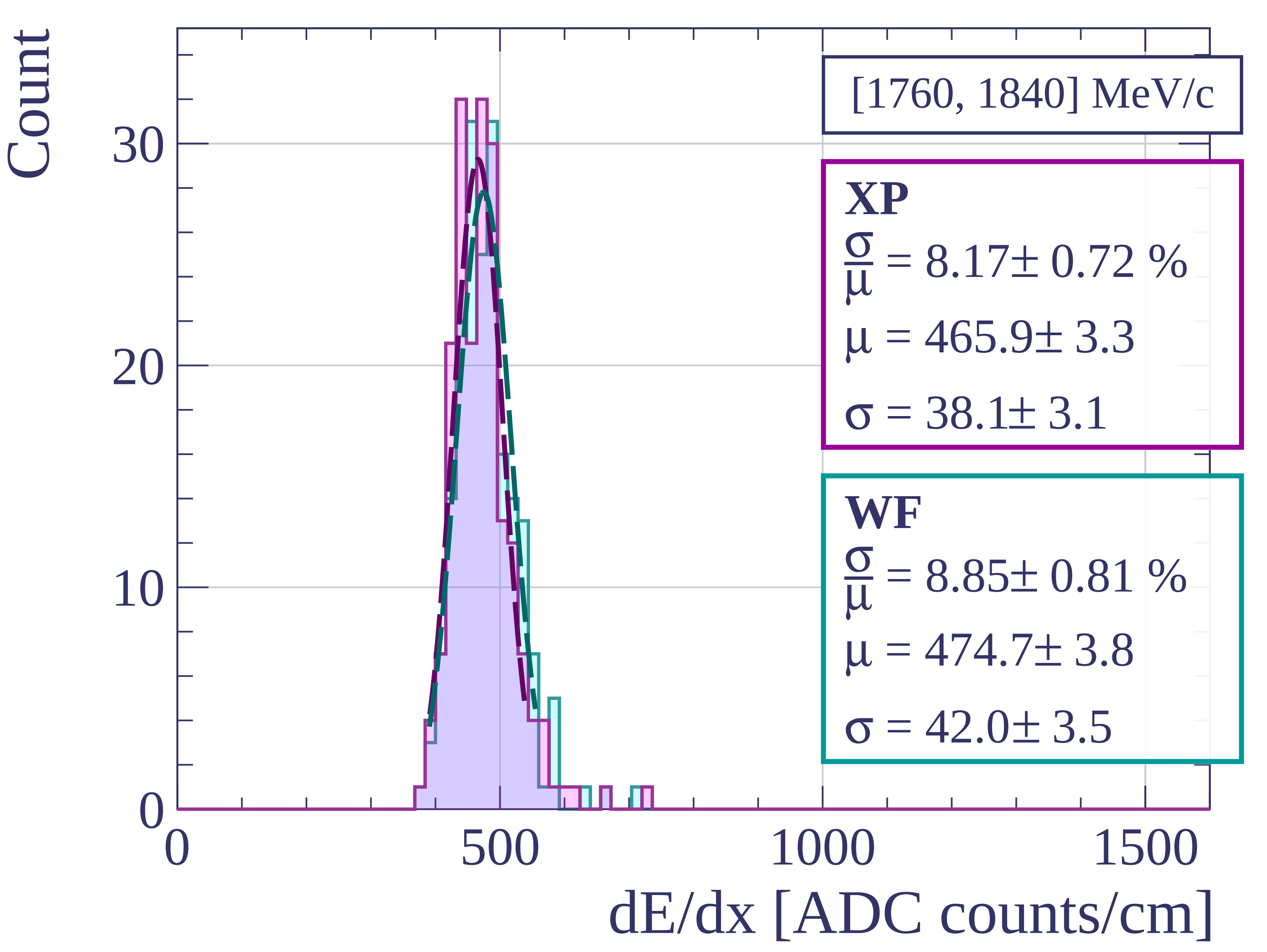


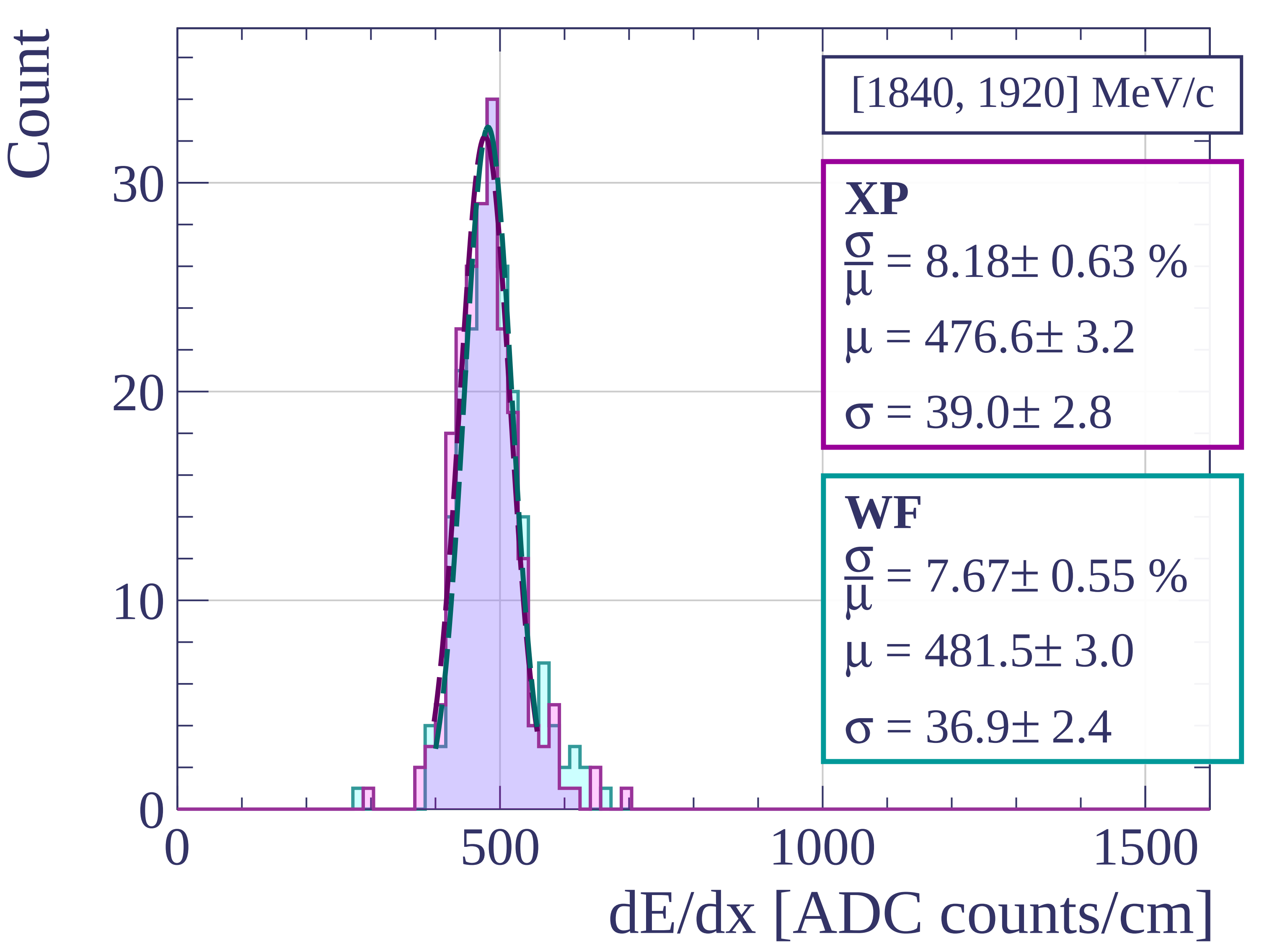
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
10
20
30
40
[1520, 1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.19± 0.75 %
μ = 460.0± 3.3
σ = 37.7± 3.2
WF
$\frac{\sigma}{\mu}$ = 8.50± 0.78 %
μ = 465.4± 3.3
σ = 39.5± 3.3

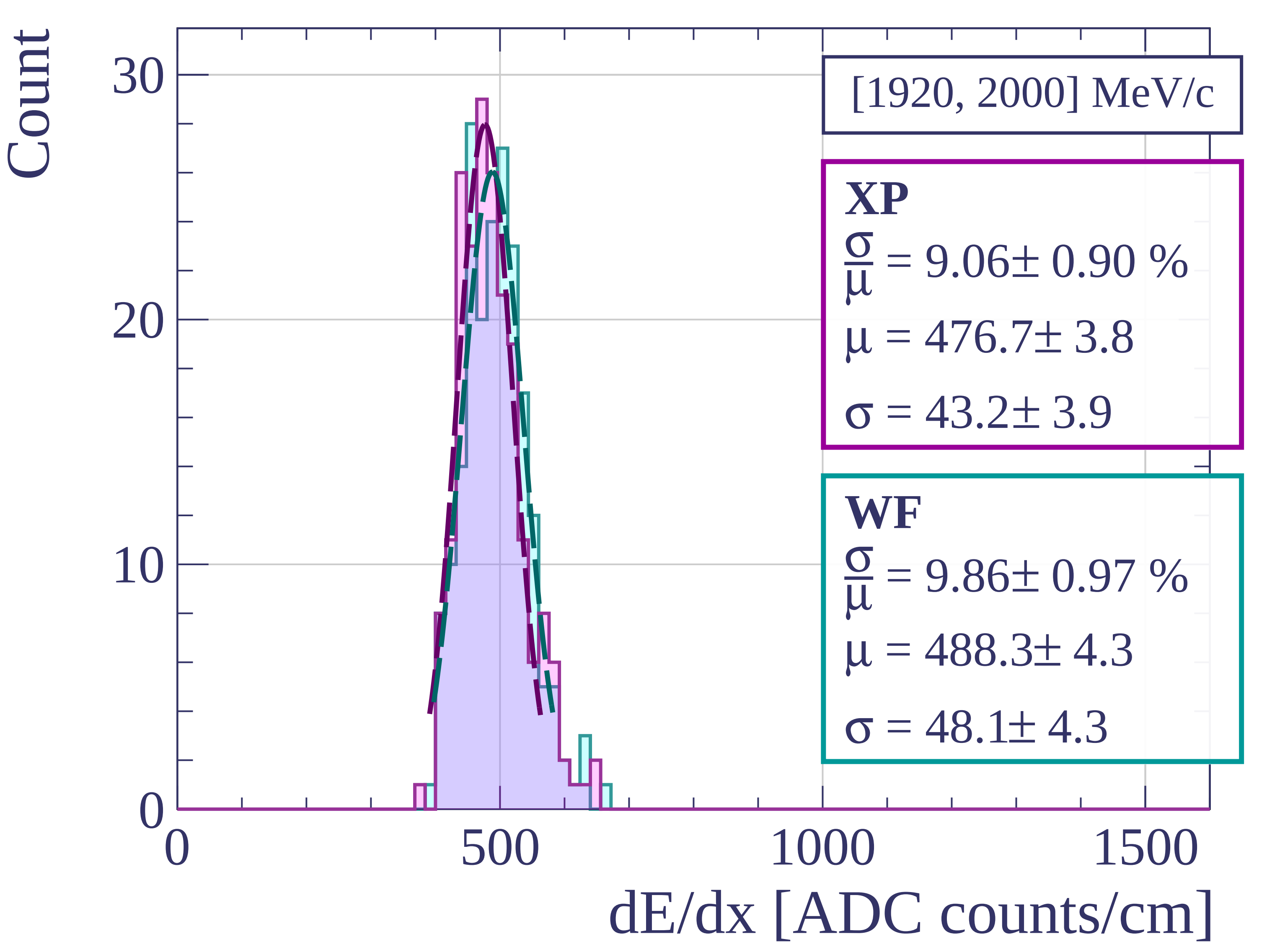


Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1680] MeV/c
XP
σ/μ = 8.93± 0.73 %
μ = 471.5± 3.3
σ = 42.1± 3.1
WF
σ/μ = 9.17± 0.83 %
μ = 474.7± 3.5
σ = 43.5± 3.6

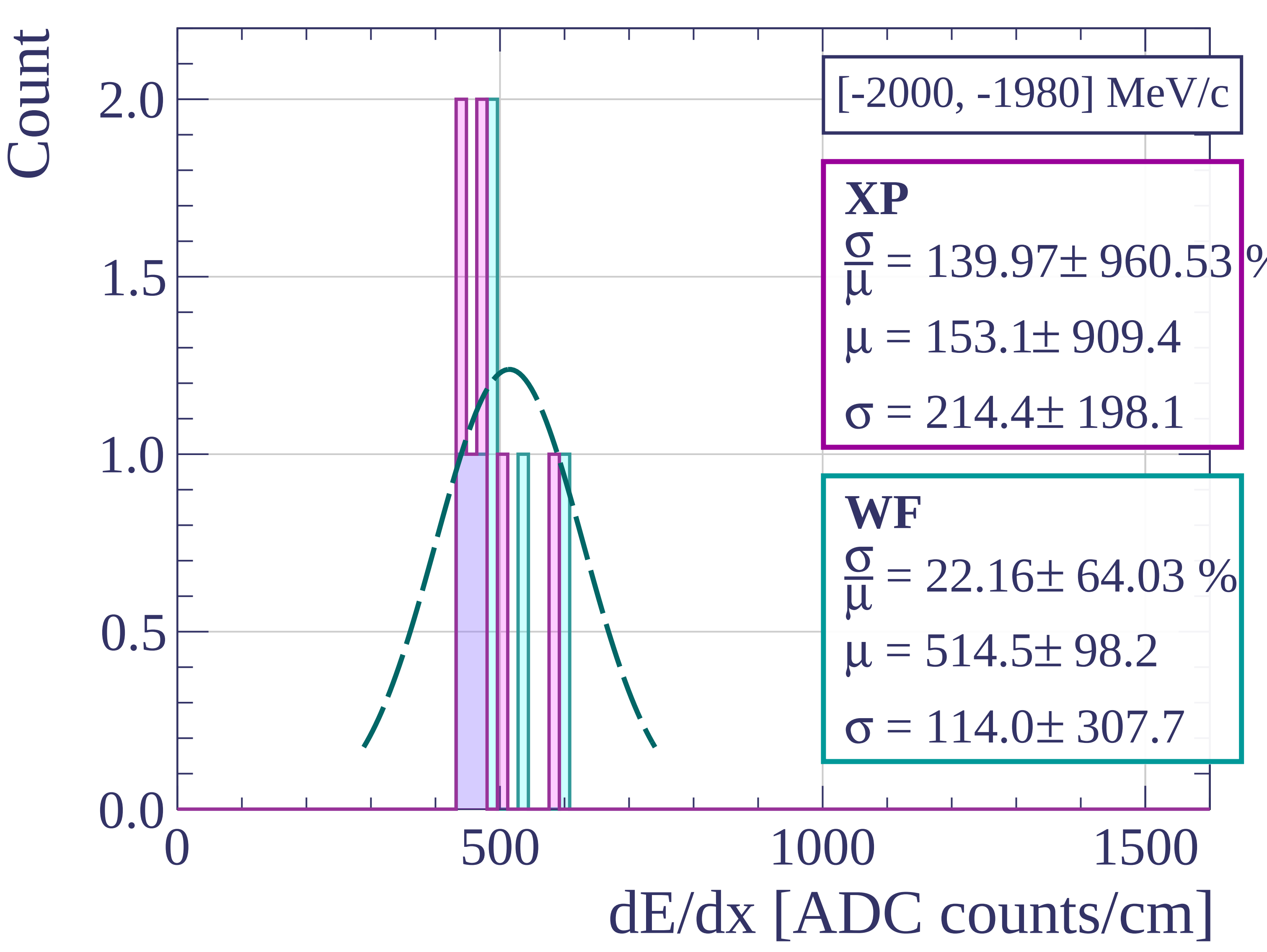

Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1680, 1760] MeV/c
XP
σ/μ = 9.47± 0.74 %
μ = 470.9± 3.7
σ = 44.6± 3.2
WF
σ/μ = 8.96± 0.61 %
μ = 483.6± 3.8
σ = 43.3± 2.6


Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
[1760, 1840] MeV/c
XP
σ/μ = 8.17± 0.72 %
μ = 465.9± 3.3
σ = 38.1± 3.1
WF
σ/μ = 8.85± 0.81 %
μ = 474.7± 3.8
σ = 42.0± 3.5

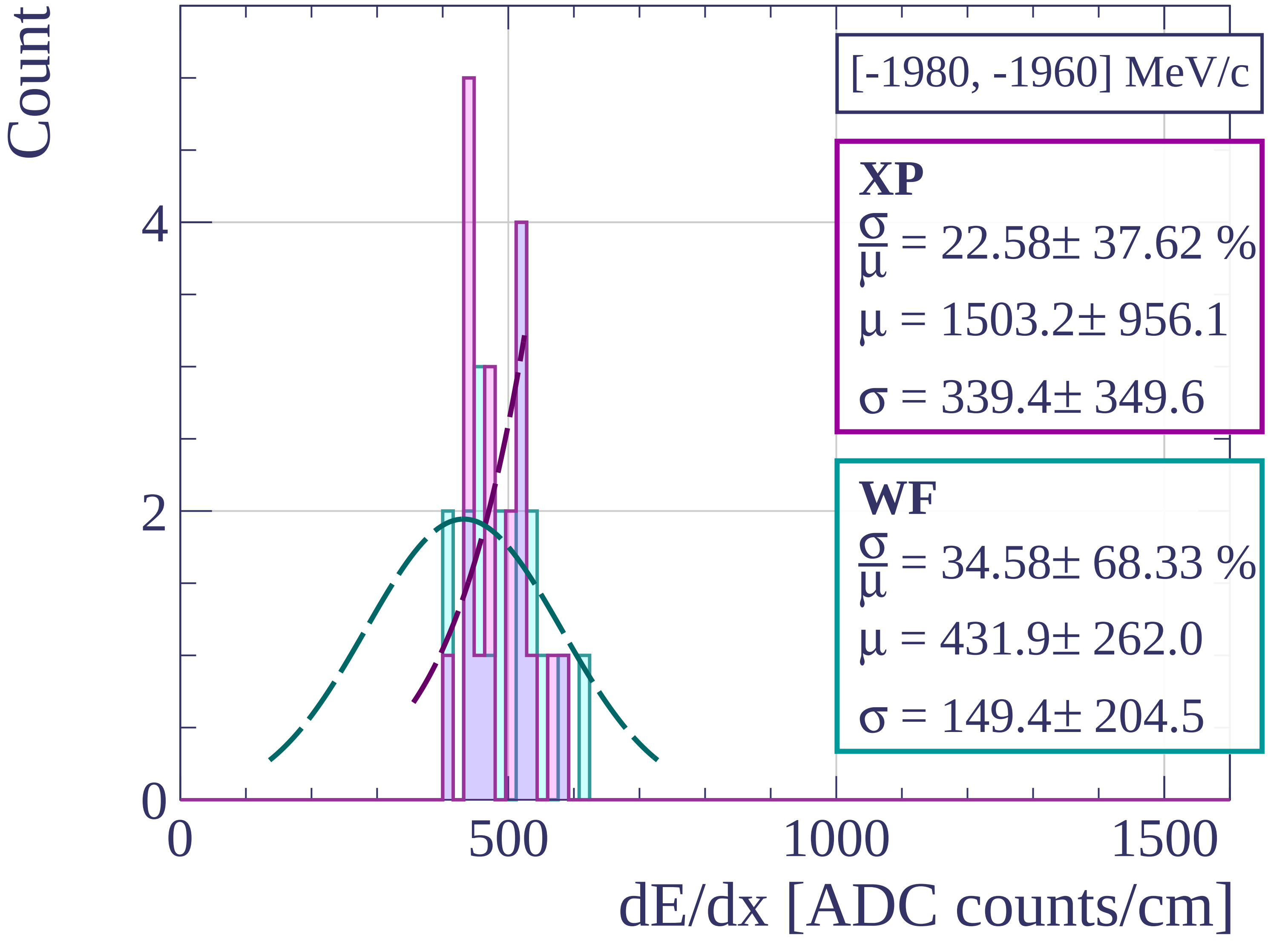

Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1840, 1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.18± 0.63 %
μ = 476.6± 3.2
σ = 39.0± 2.8
WF
$\frac{\sigma}{\mu}$ = 7.67± 0.55 %
μ = 481.5± 3.0
σ = 36.9± 2.4

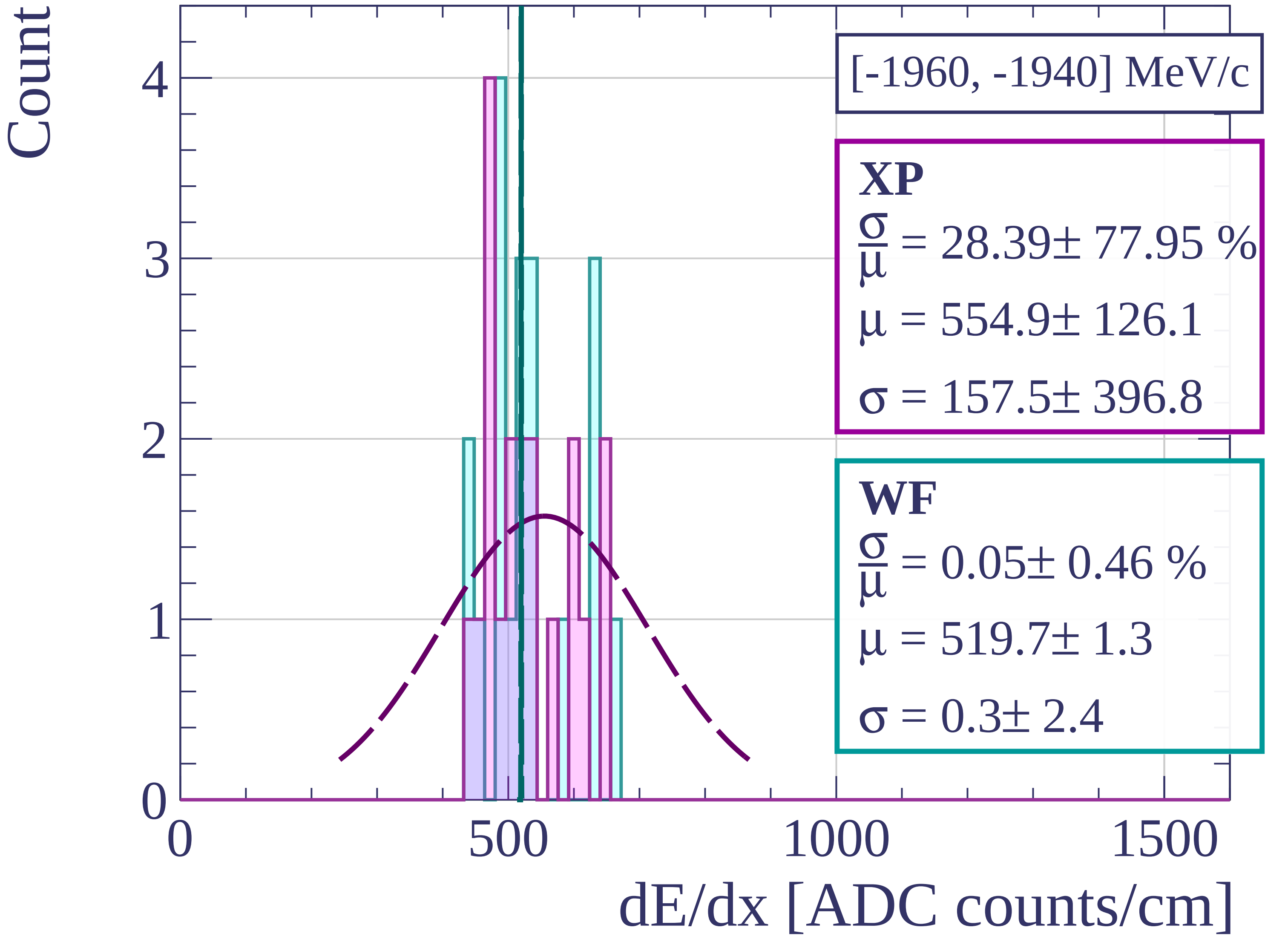


Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 2000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.06± 0.90 %
μ = 476.7± 3.8
σ = 43.2± 3.9
WF
$\frac{\sigma}{\mu}$ = 9.86± 0.97 %
μ = 488.3± 4.3
σ = 48.1± 4.3

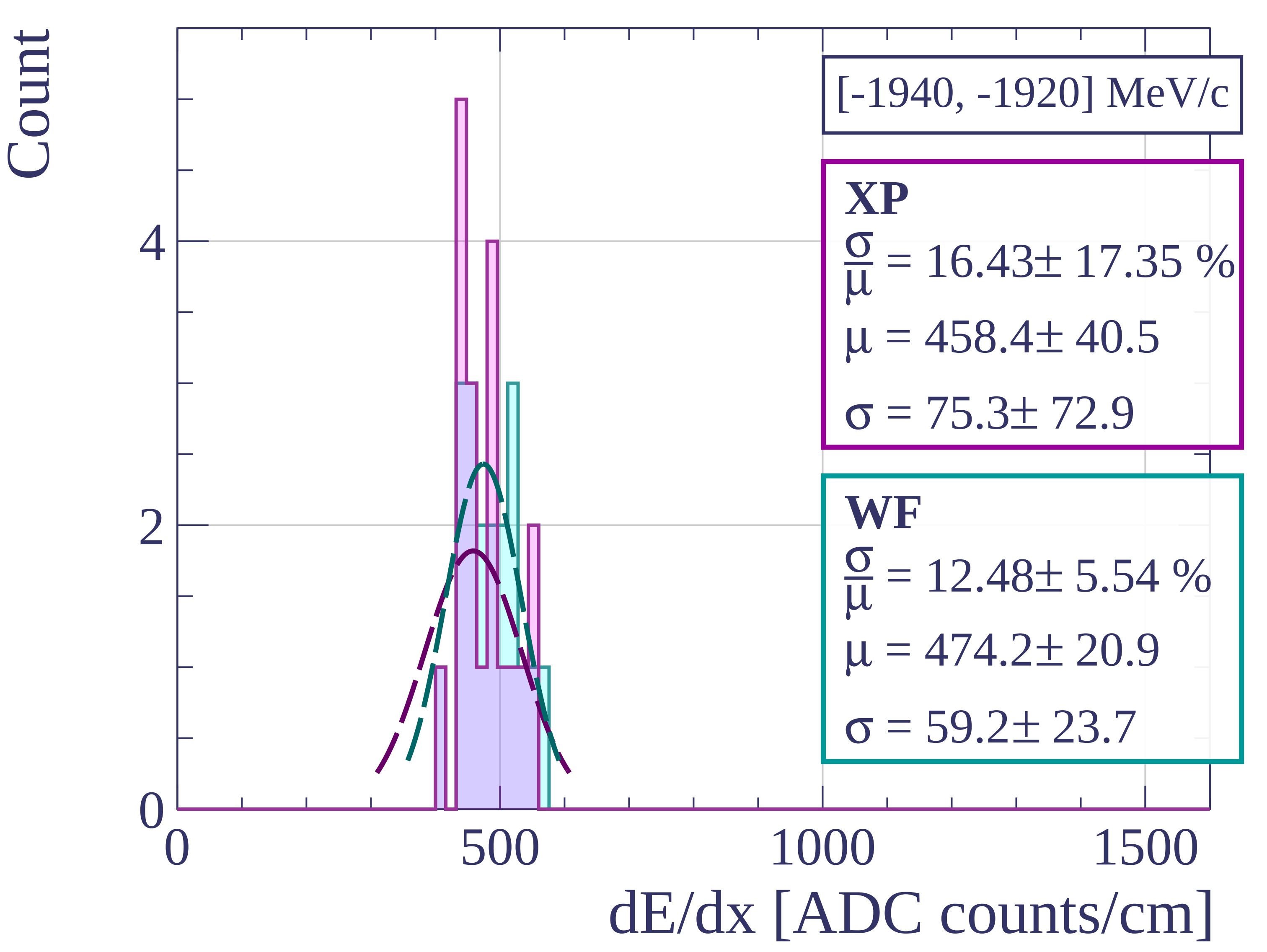


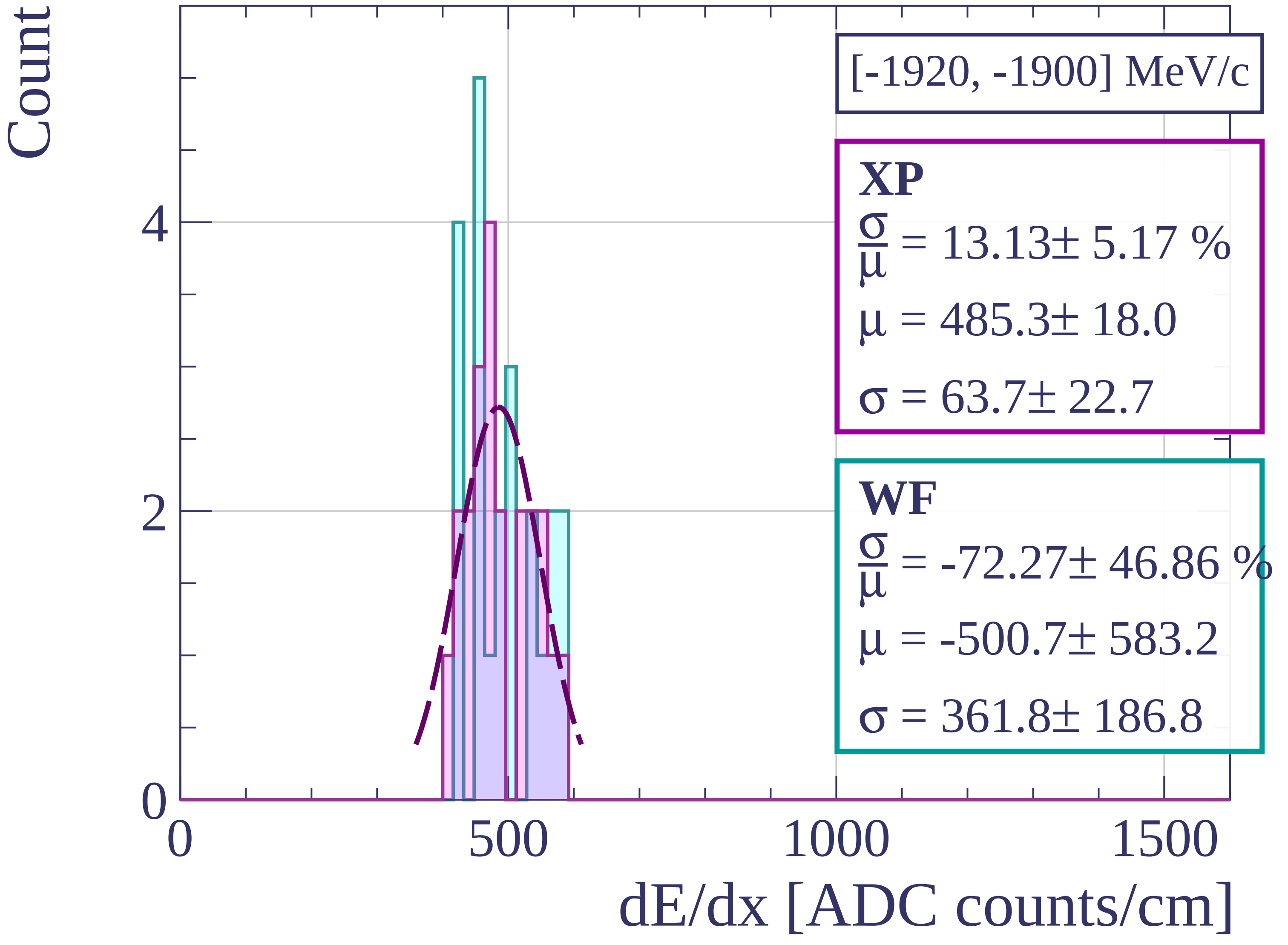
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-2000, -1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 139.97± 960.53 %
μ = 153.1± 909.4
σ = 214.4± 198.1
WF
$\frac{\sigma}{\mu}$ = 22.16± 64.03 %
μ = 514.5± 98.2
σ = 114.0± 307.7


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1980, -1960] MeV/c
XP
$\frac{\sigma}{\mu} = 22.58 \pm 37.62$ %
$\mu = 1503.2 \pm 956.1$
$\sigma = 339.4 \pm 349.6$
WF
$\frac{\sigma}{\mu} = 34.58 \pm 68.33$ %
$\mu = 431.9 \pm 262.0$
$\sigma = 149.4 \pm 204.5$

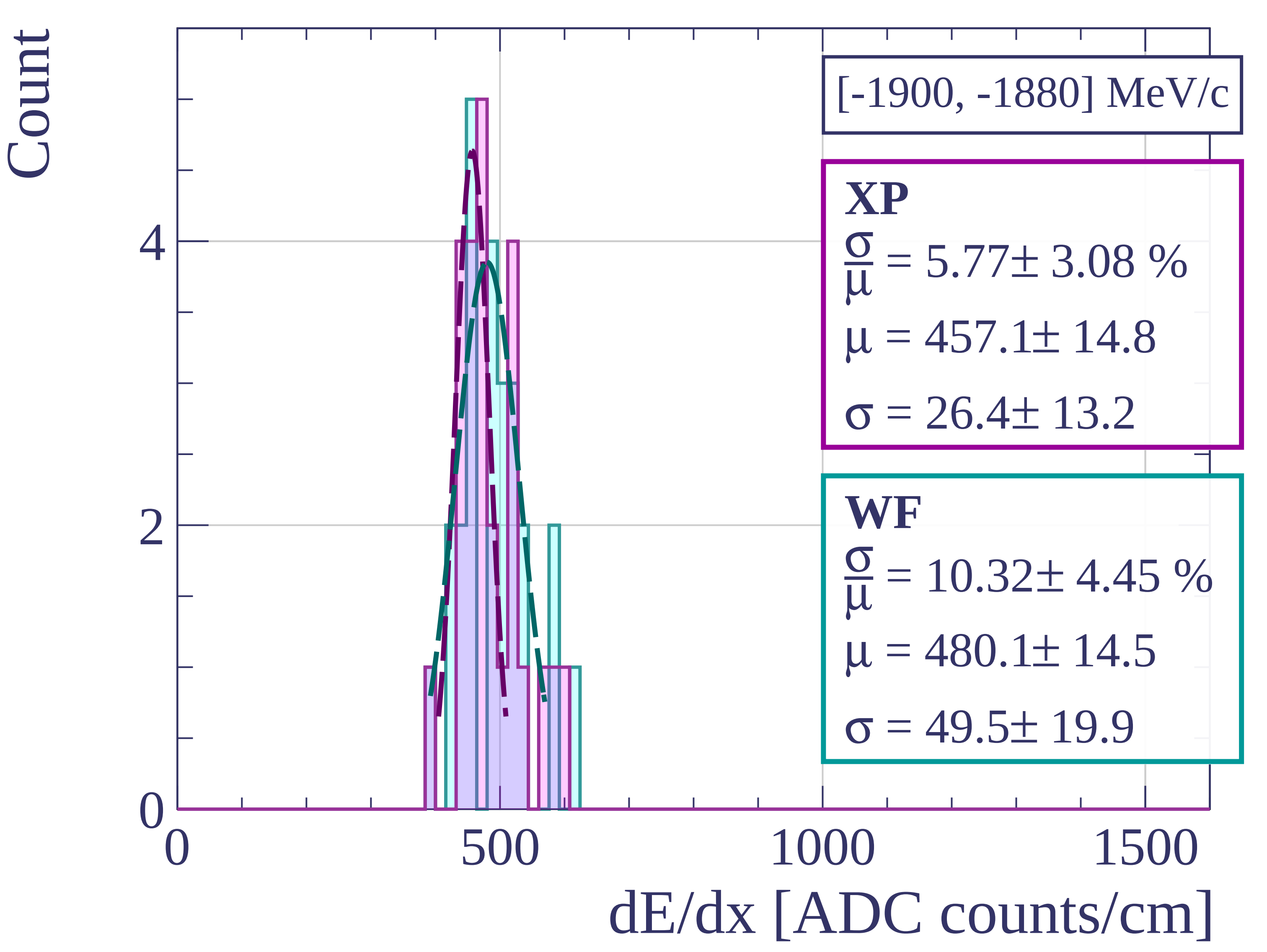


Count
0
1
2
3
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1960, -1940] MeV/c
XP
σ/μ = 28.39± 77.95 %
μ = 554.9± 126.1
σ = 157.5± 396.8
WF
σ/μ = 0.05± 0.46 %
μ = 519.7± 1.3
σ = 0.3± 2.4

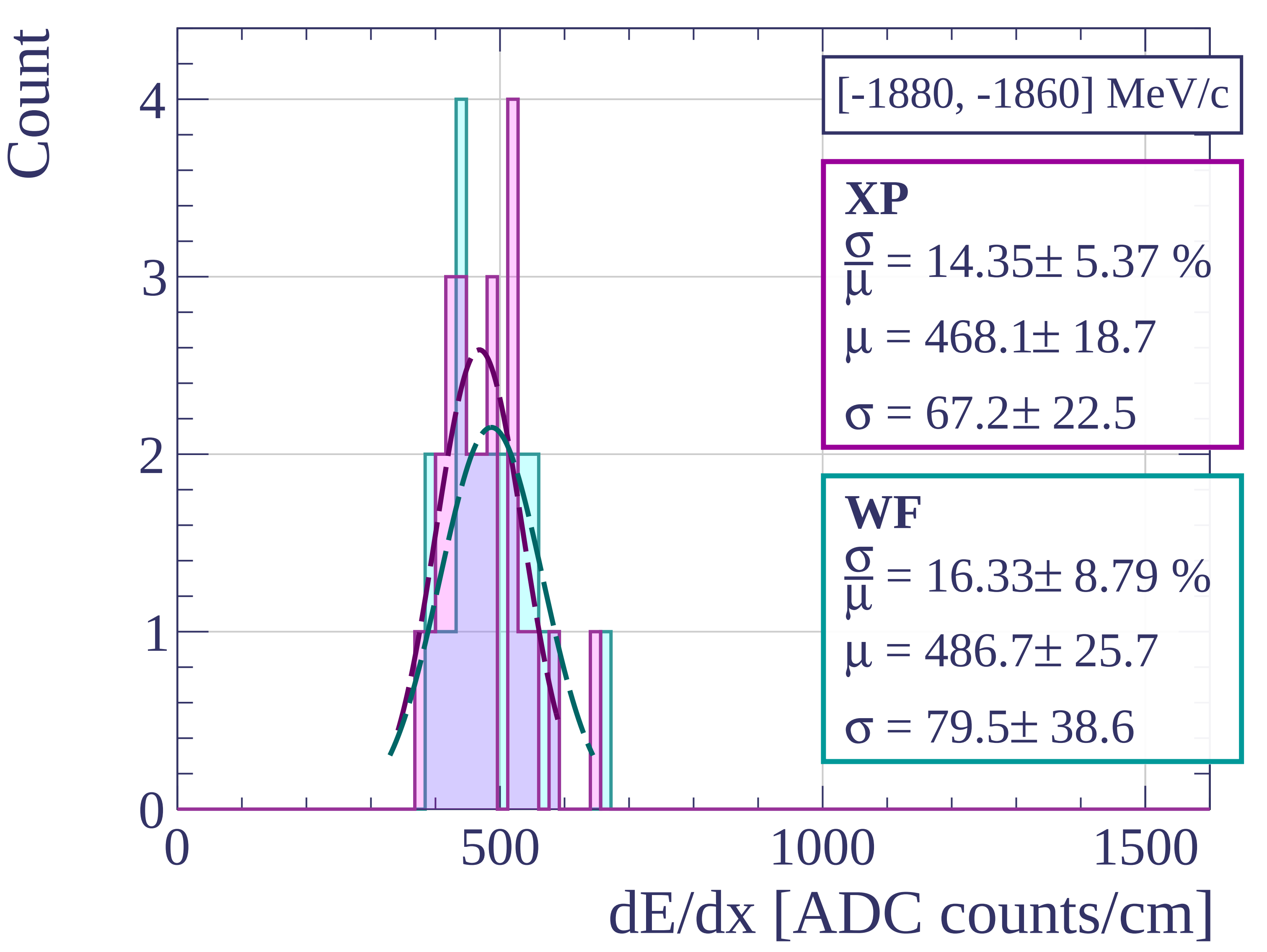

Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1940, -1920] MeV/c
XP
σ/μ = 16.43± 17.35 %
μ = 458.4± 40.5
σ = 75.3± 72.9
WF
σ/μ = 12.48± 5.54 %
μ = 474.2± 20.9
σ = 59.2± 23.7



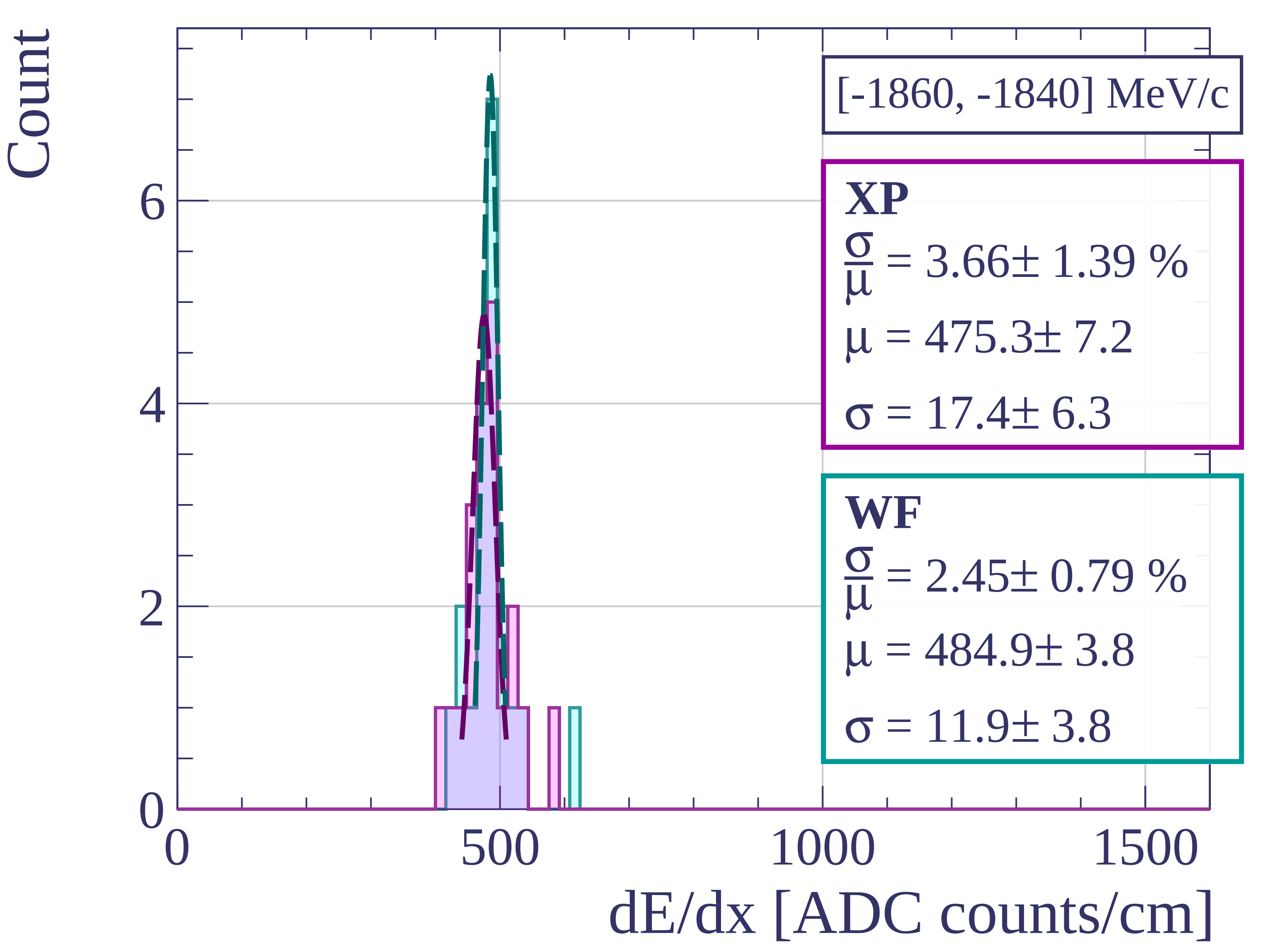
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[-1920, -1900] MeV/c
XP
σ/μ = 13.13± 5.17 %
μ = 485.3± 18.0
σ = 63.7± 22.7
WF
σ/μ = -72.27± 46.86 %
μ = -500.7± 583.2
σ = 361.8± 186.8

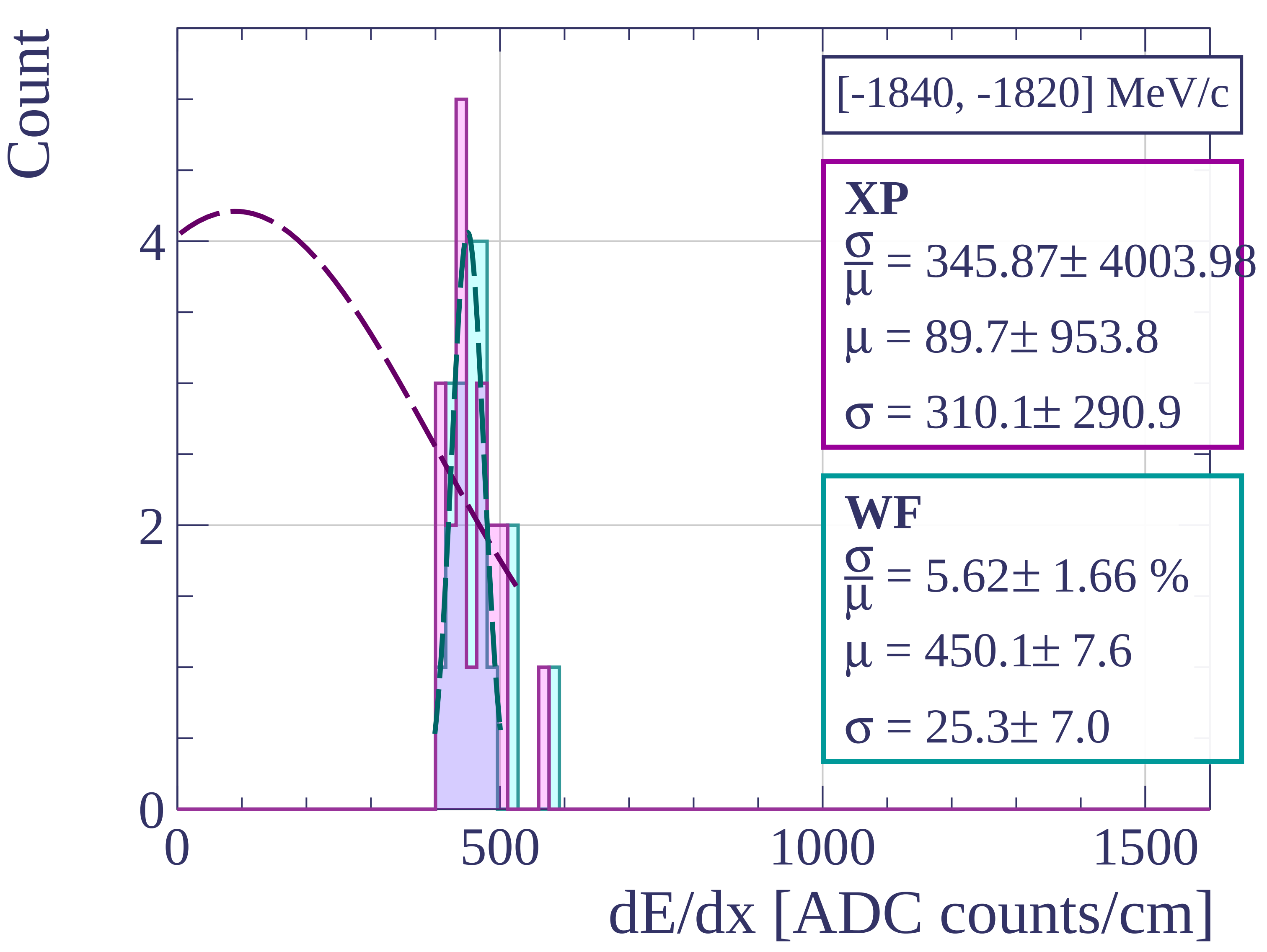


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1900, -1880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.77± 3.08 %
μ = 457.1± 14.8
σ = 26.4± 13.2
WF
$\frac{\sigma}{\mu}$ = 10.32± 4.45 %
μ = 480.1± 14.5
σ = 49.5± 19.9

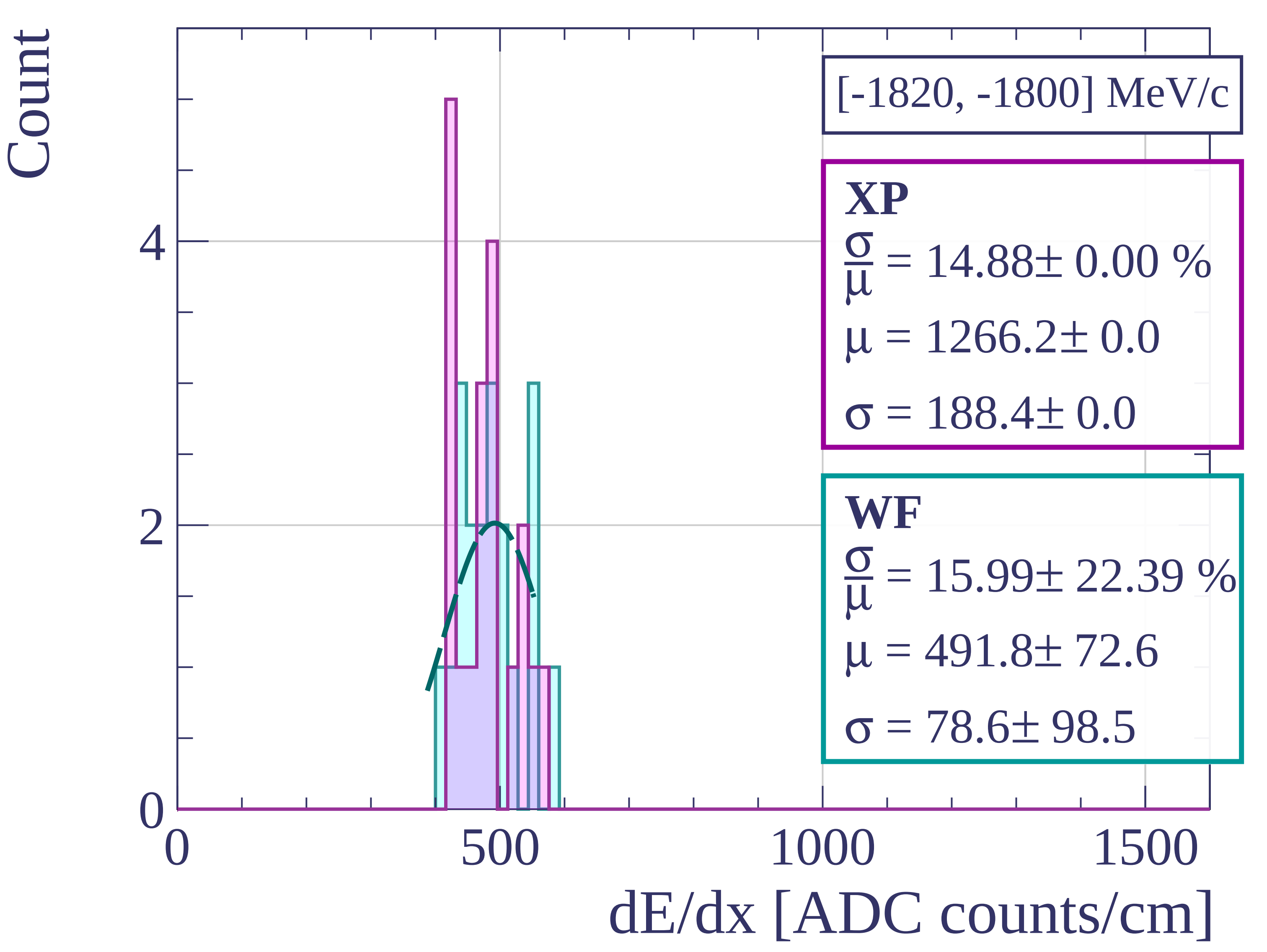


Count
dE/dx [ADC counts/cm]
[-1880, -1860] MeV/c
XP
σ/μ = 14.35± 5.37 %
μ = 468.1± 18.7
σ = 67.2± 22.5
WF
σ/μ = 16.33± 8.79 %
μ = 486.7± 25.7
σ = 79.5± 38.6
0
1
2
3
4
0
500
1000
1500

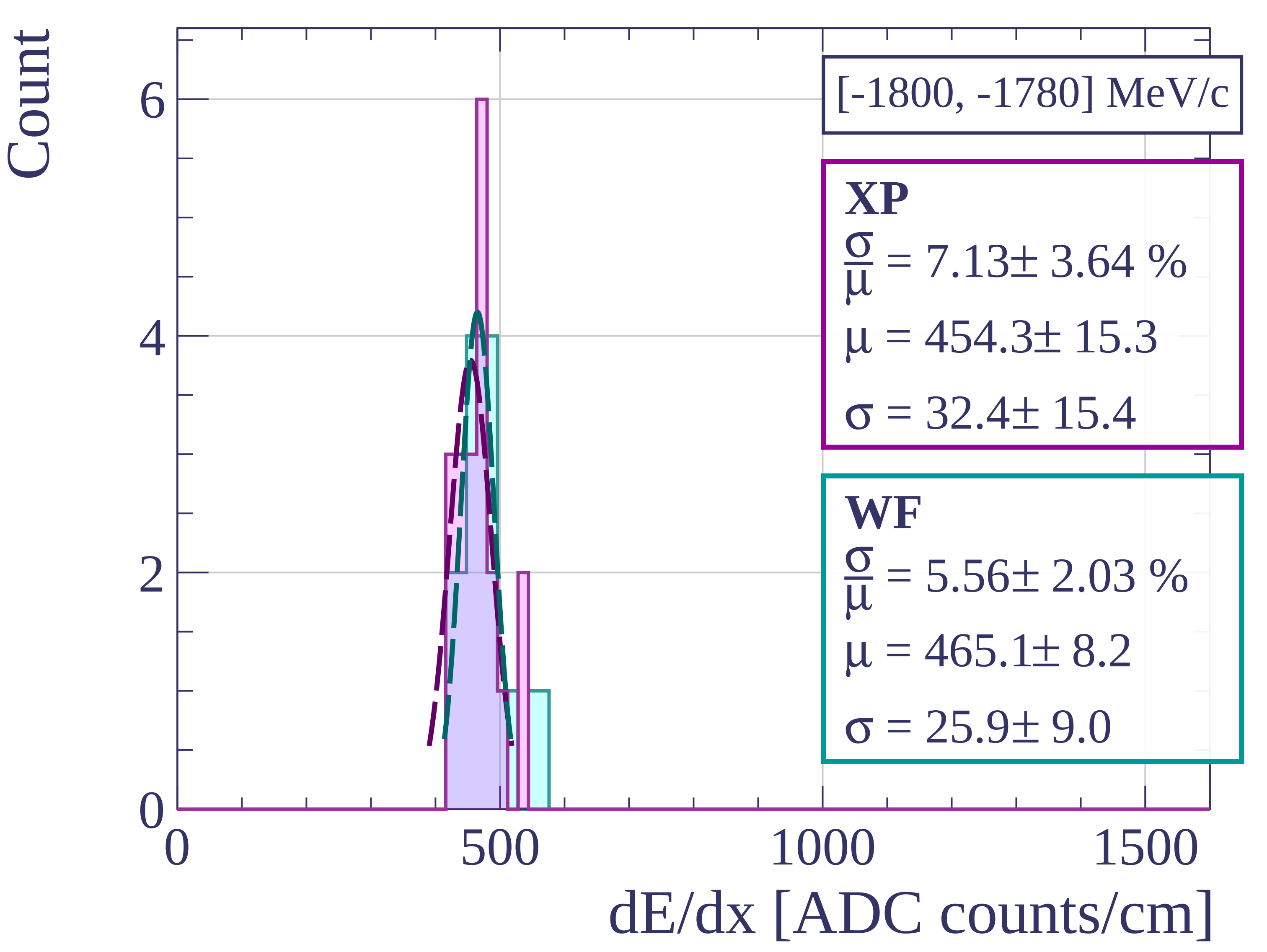

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1860, -1840] MeV/c
XP
σ/μ = 3.66± 1.39 %
μ = 475.3± 7.2
σ = 17.4± 6.3
WF
σ/μ = 2.45± 0.79 %
μ = 484.9± 3.8
σ = 11.9± 3.8

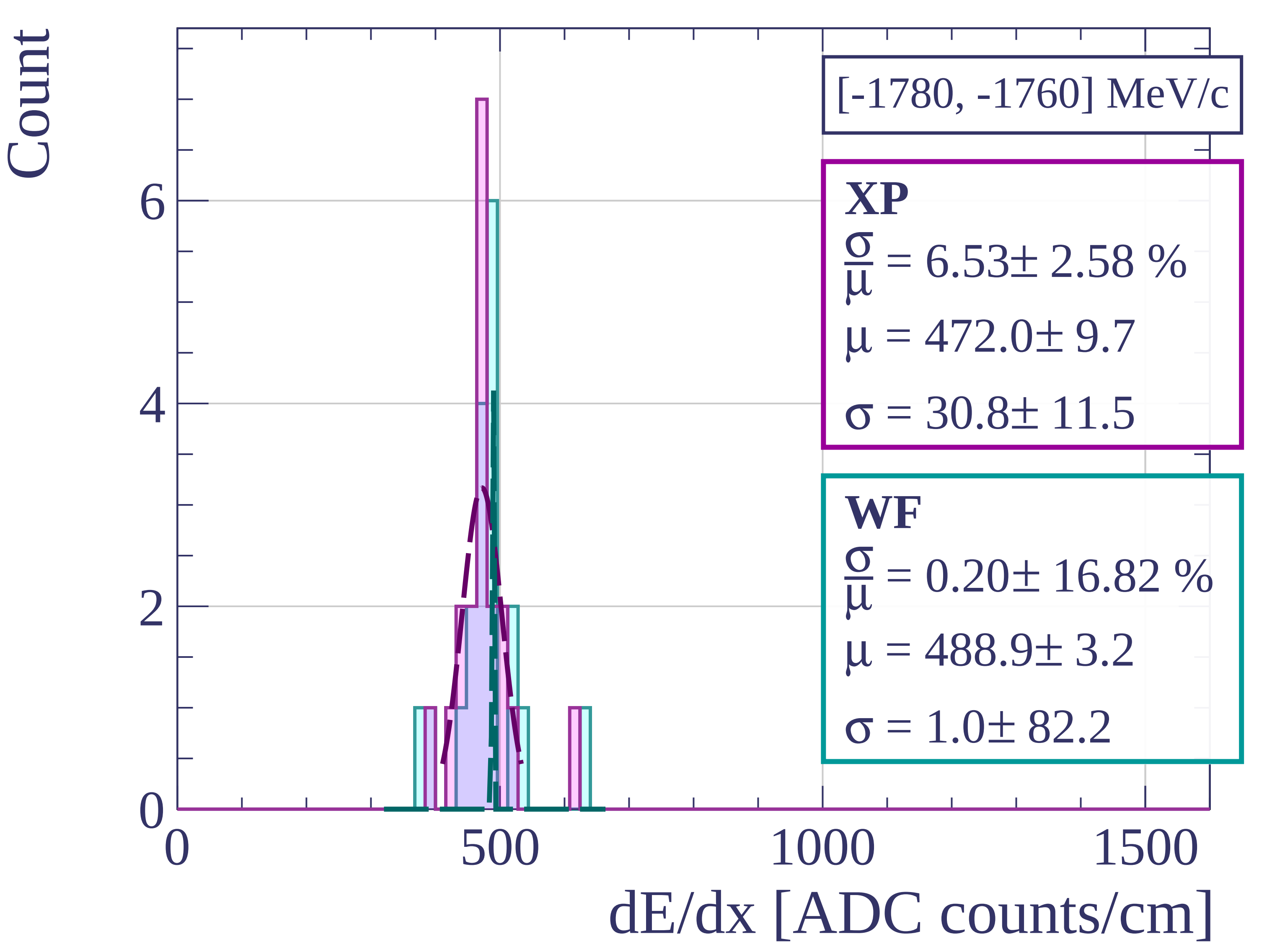


Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[-1840, -1820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 345.87± 4003.98
μ = 89.7± 953.8
σ = 310.1± 290.9
WF
$\frac{\sigma}{\mu}$ = 5.62± 1.66 %
μ = 450.1± 7.6
σ = 25.3± 7.0

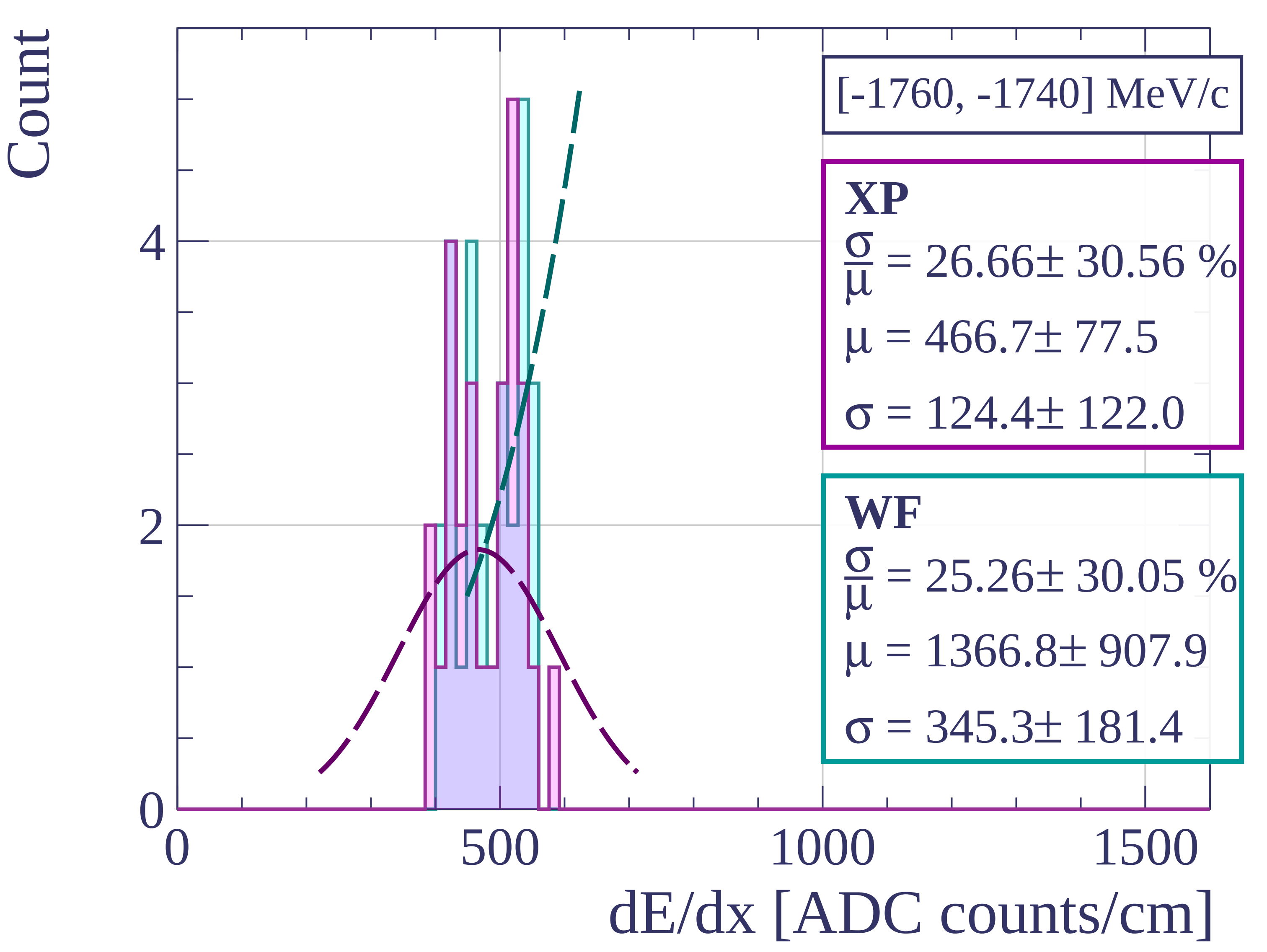


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1820, -1800] MeV/c
XP
σ/μ = 14.88± 0.00 %
μ = 1266.2± 0.0
σ = 188.4± 0.0
WF
σ/μ = 15.99± 22.39 %
μ = 491.8± 72.6
σ = 78.6± 98.5

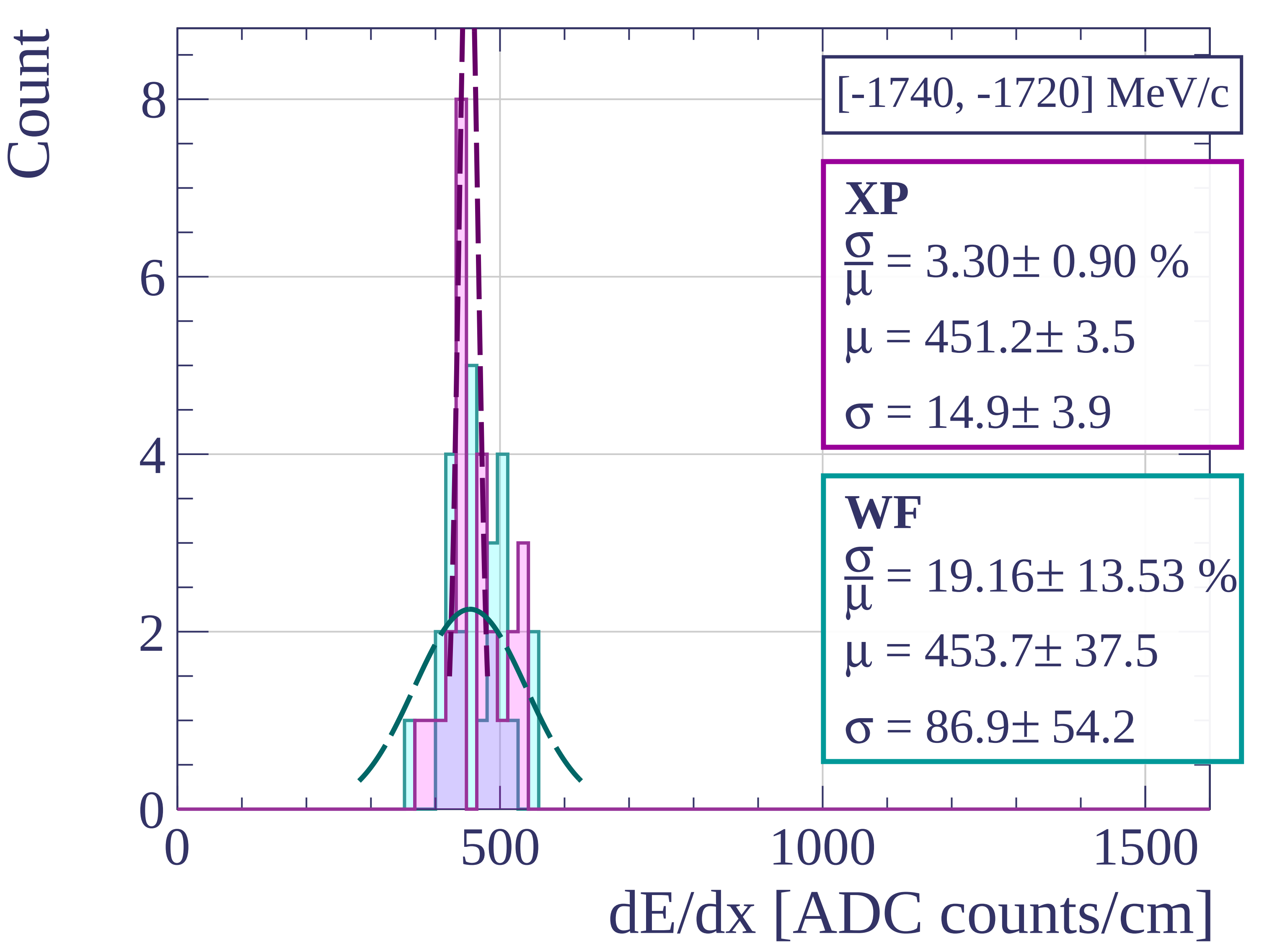

Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
[-1800, -1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.13± 3.64 %
μ = 454.3± 15.3
σ = 32.4± 15.4
WF
$\frac{\sigma}{\mu}$ = 5.56± 2.03 %
μ = 465.1± 8.2
σ = 25.9± 9.0



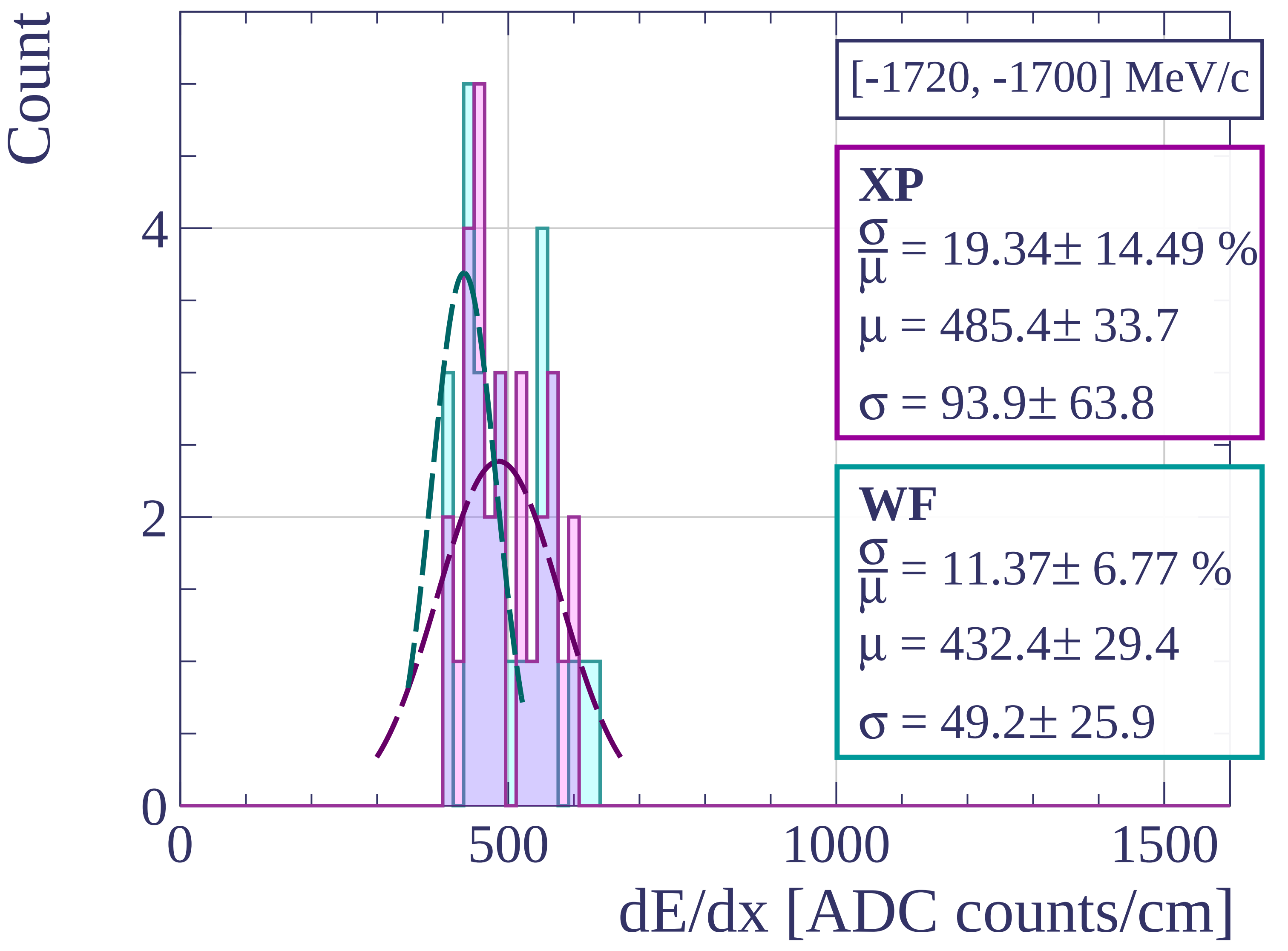
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1780, -1760] MeV/c
XP
σ/μ = 6.53± 2.58 %
μ = 472.0± 9.7
σ = 30.8± 11.5
WF
σ/μ = 0.20± 16.82 %
μ = 488.9± 3.2
σ = 1.0± 82.2



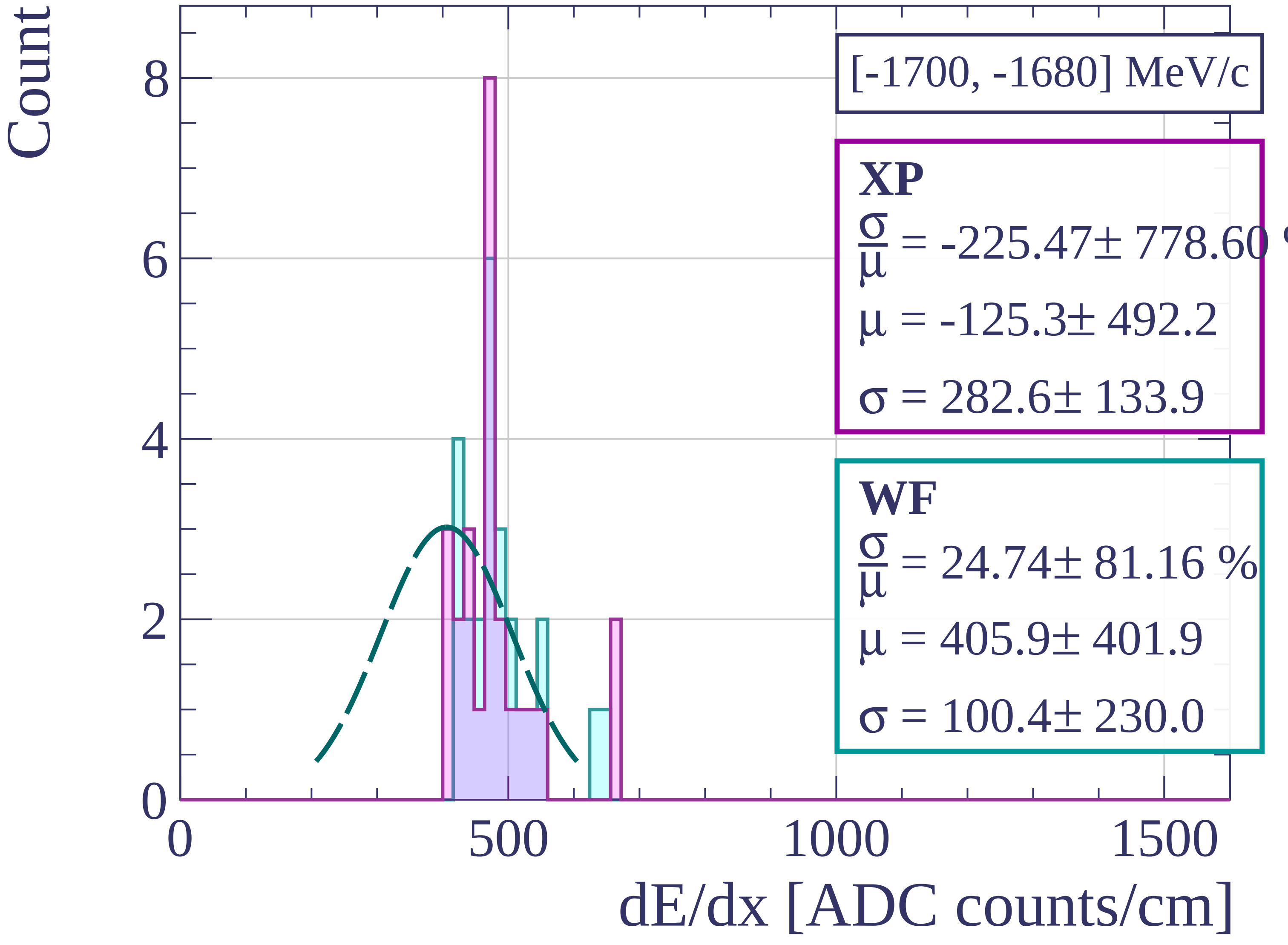
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[-1760, -1740] MeV/c
XP
σ/μ = 26.66± 30.56 %
μ = 466.7± 77.5
σ = 124.4± 122.0
WF
σ/μ = 25.26± 30.05 %
μ = 1366.8± 907.9
σ = 345.3± 181.4



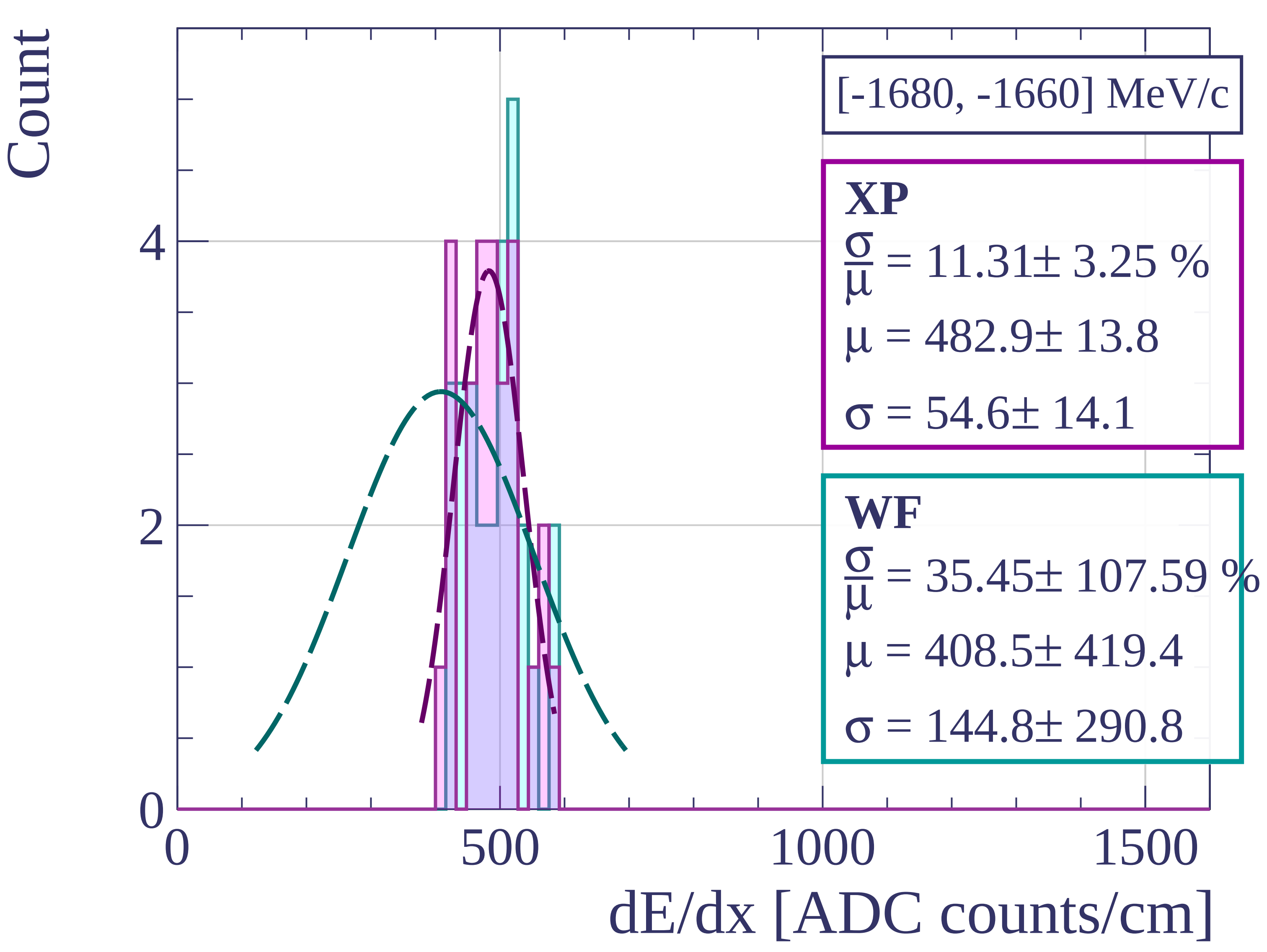
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1740, -1720] MeV/c
XP
σ/μ = 3.30± 0.90 %
μ = 451.2± 3.5
σ = 14.9± 3.9
WF
σ/μ = 19.16± 13.53 %
μ = 453.7± 37.5
σ = 86.9± 54.2



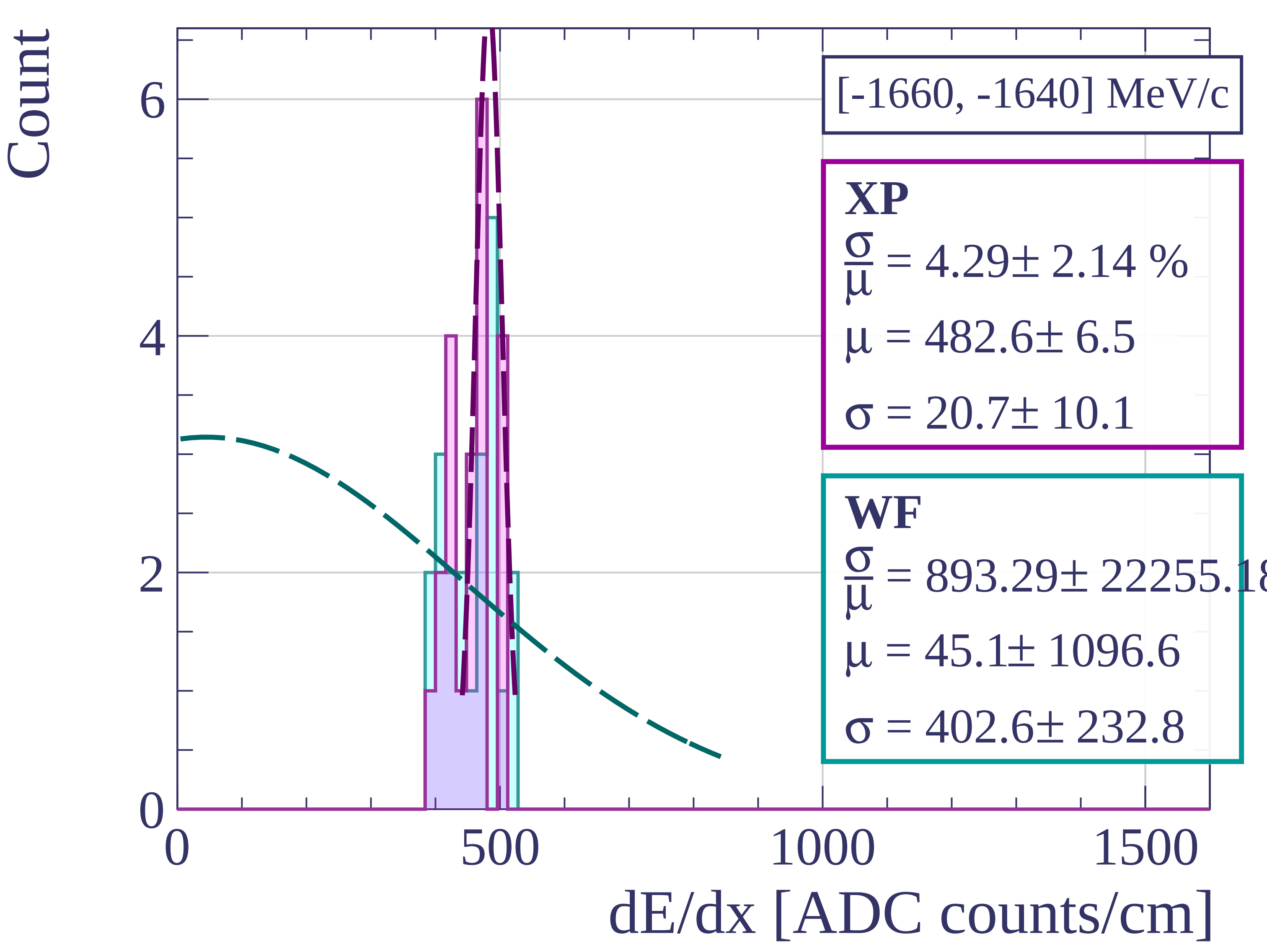
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[-1720, -1700] MeV/c
XP
σ/μ = 19.34± 14.49 %
μ = 485.4± 33.7
σ = 93.9± 63.8
WF
σ/μ = 11.37± 6.77 %
μ = 432.4± 29.4
σ = 49.2± 25.9



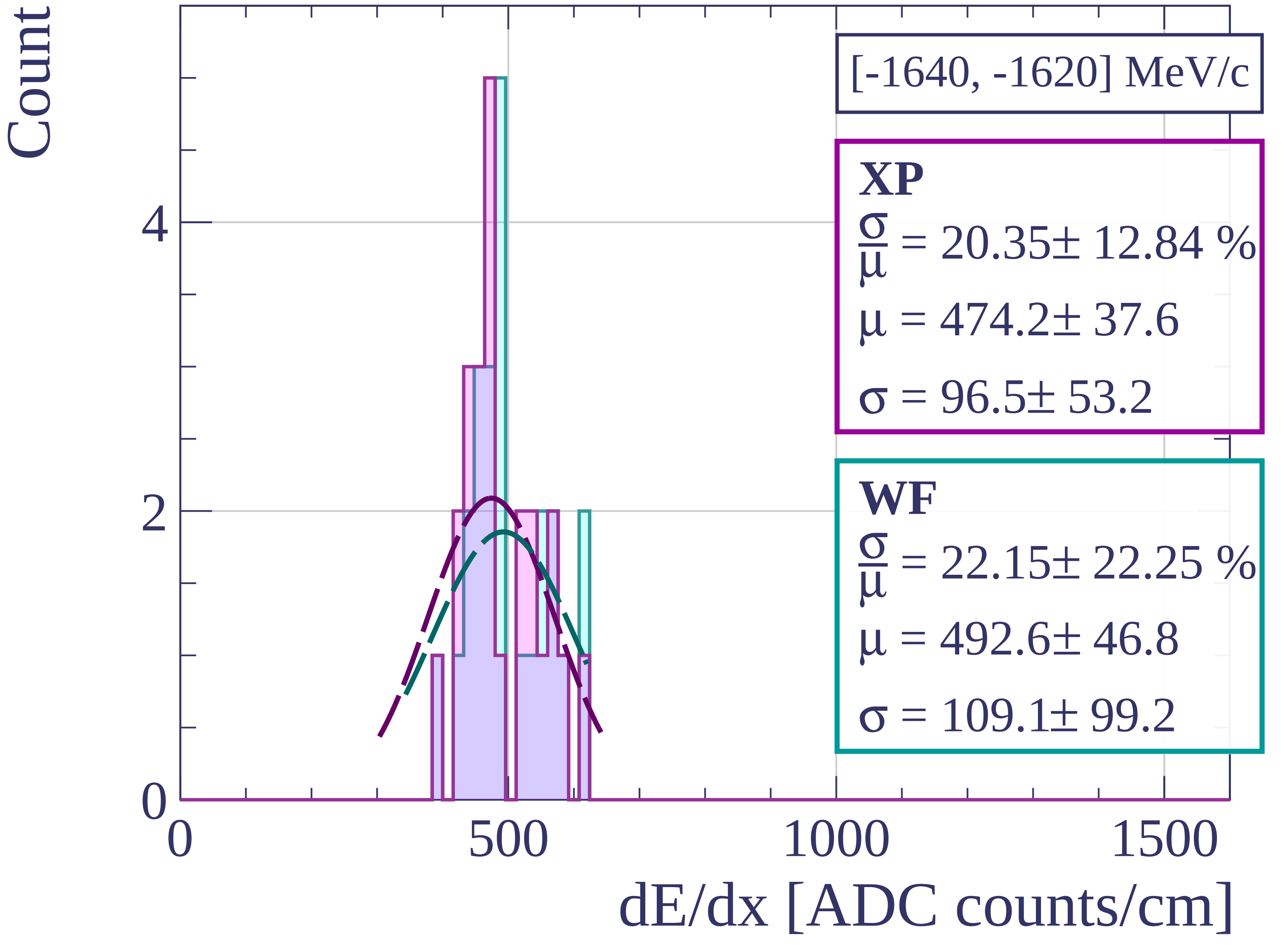
Count
[-1700, -1680] MeV/c
XP
$\frac{\sigma}{\mu}$ = -225.47± 778.60 %
μ = -125.3± 492.2
σ = 282.6± 133.9
WF
$\frac{\sigma}{\mu}$ = 24.74± 81.16 %
μ = 405.9± 401.9
σ = 100.4± 230.0
dE/dx [ADC counts/cm]



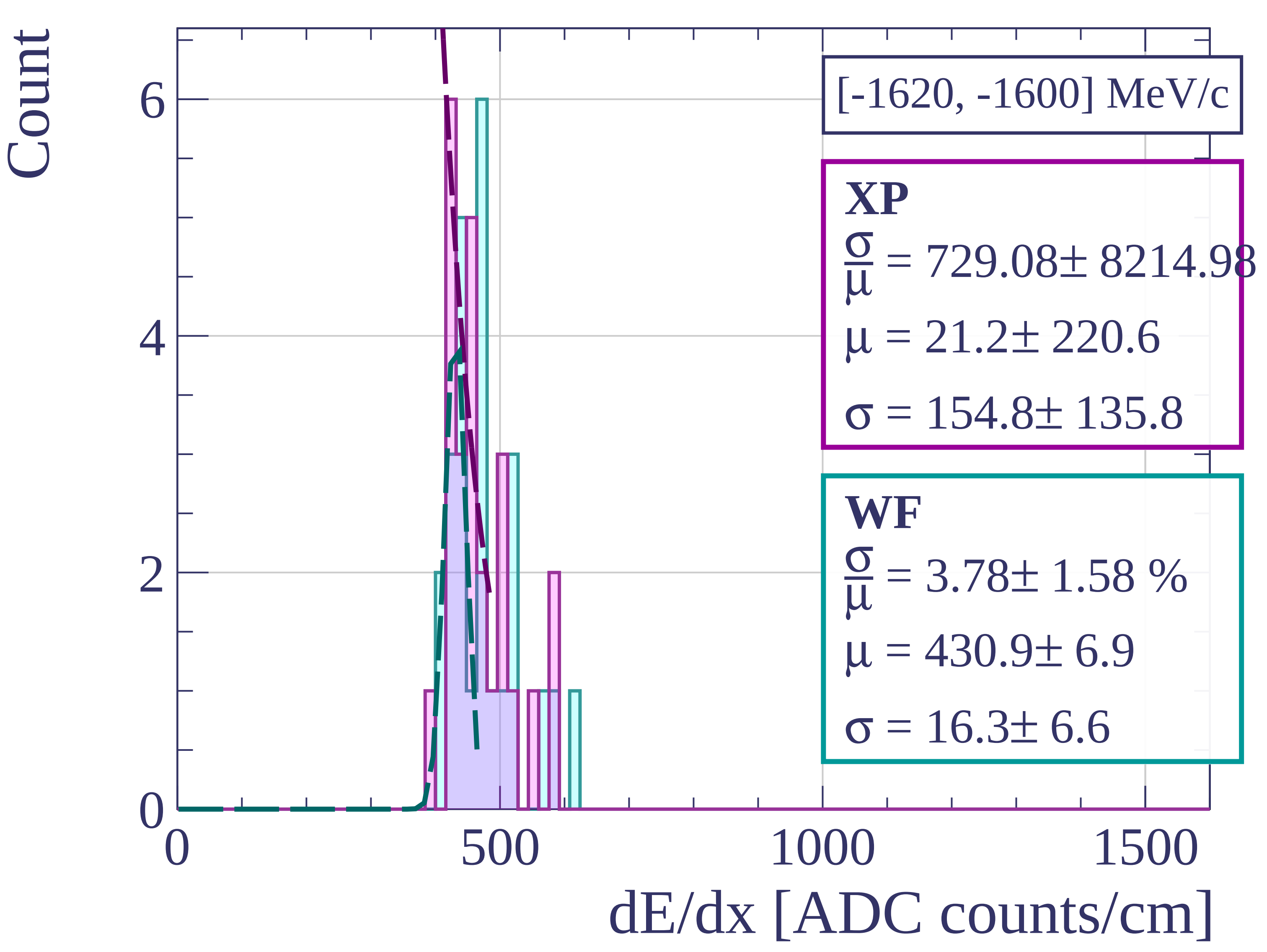
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1680, -1660] MeV/c
XP
σ/μ = 11.31± 3.25 %
μ = 482.9± 13.8
σ = 54.6± 14.1
WF
σ/μ = 35.45± 107.59 %
μ = 408.5± 419.4
σ = 144.8± 290.8

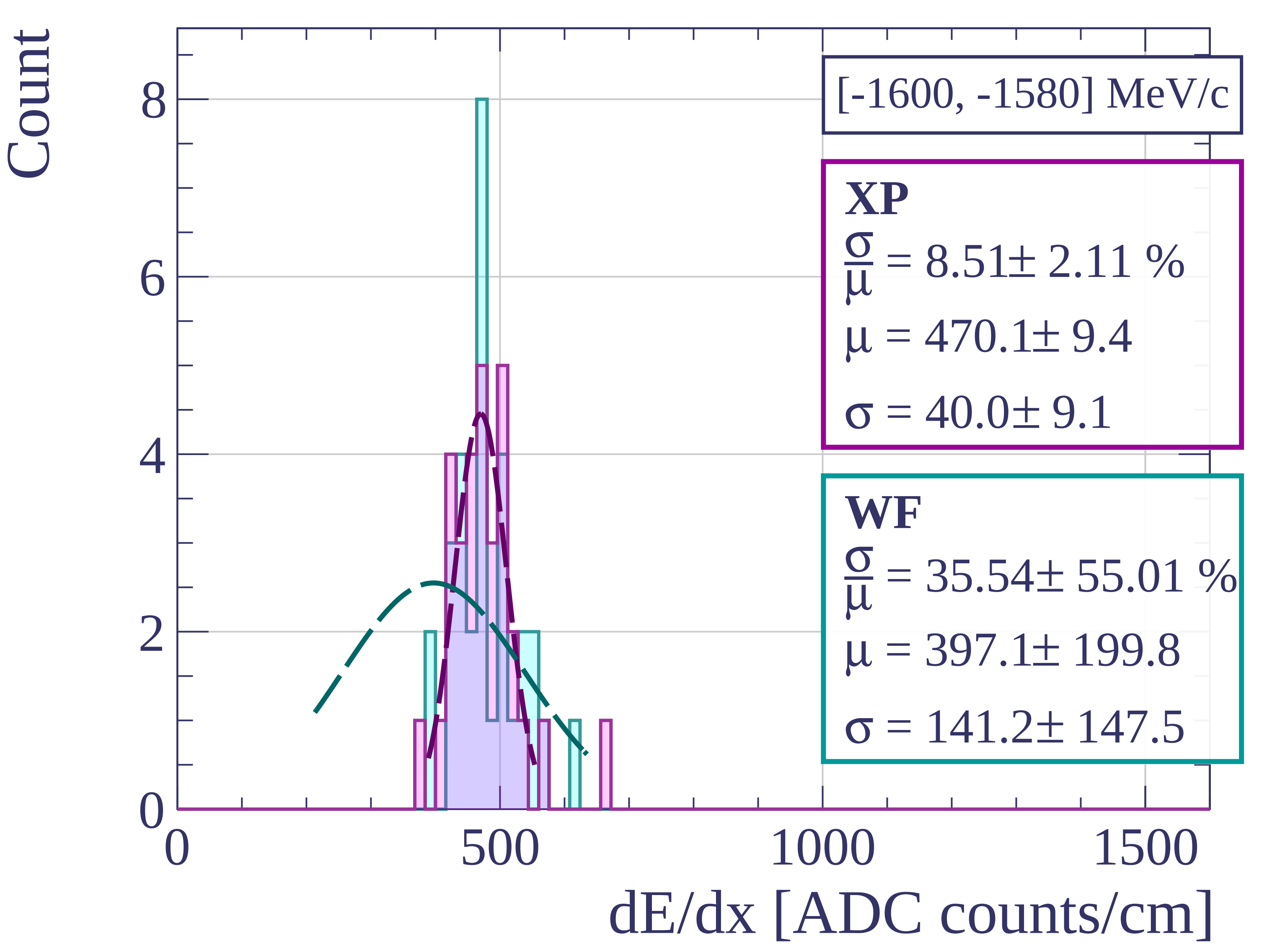


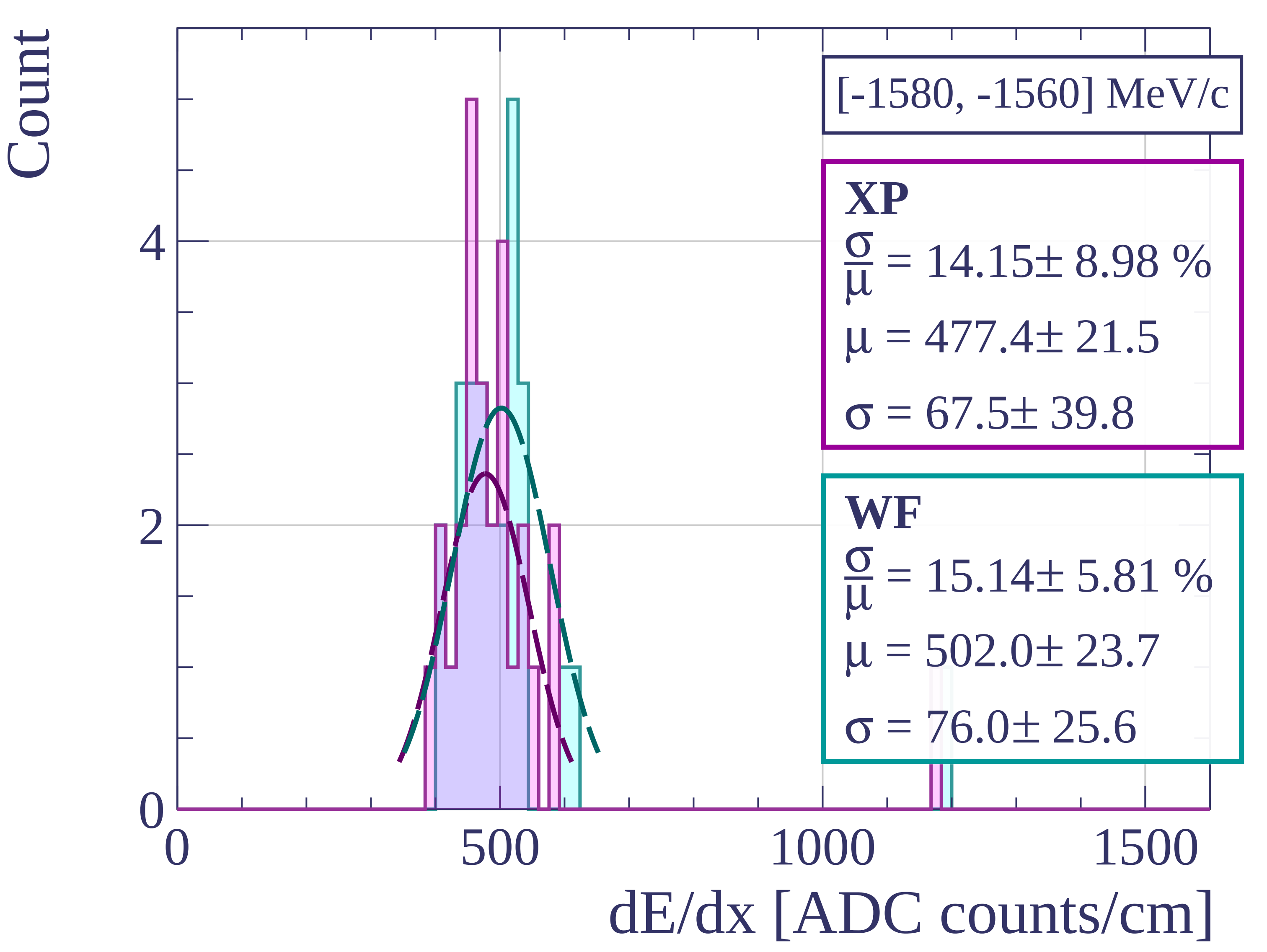
Count
dE/dx [ADC counts/cm]
[-1660, -1640] MeV/c
XP
σ/μ = 4.29± 2.14 %
μ = 482.6± 6.5
σ = 20.7± 10.1
WF
σ/μ = 893.29± 22255.18
μ = 45.1± 1096.6
σ = 402.6± 232.8
0
2
4
6
0
500
1000
1500

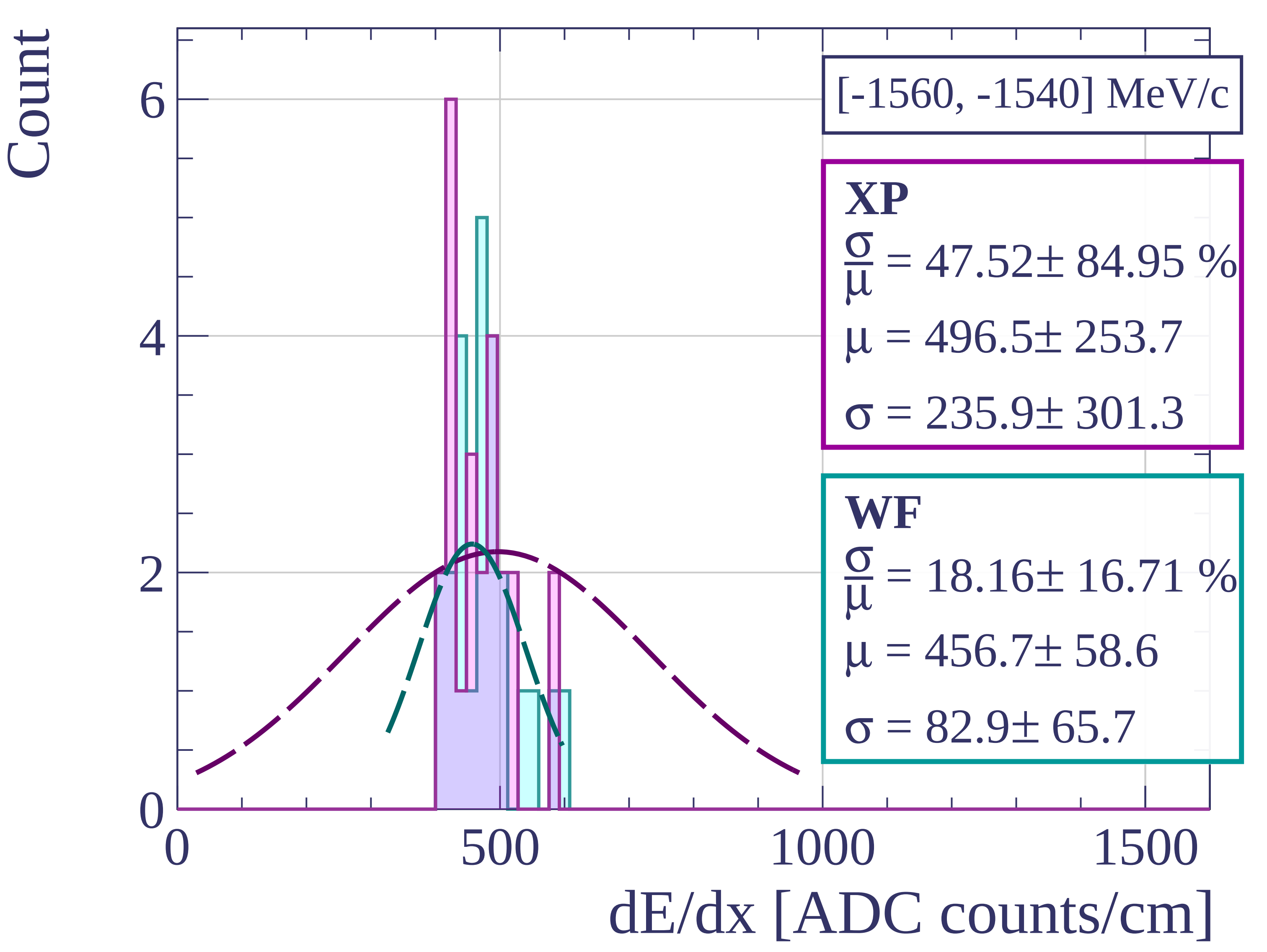


Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[-1640, -1620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 20.35± 12.84 %
μ = 474.2± 37.6
σ = 96.5± 53.2
WF
$\frac{\sigma}{\mu}$ = 22.15± 22.25 %
μ = 492.6± 46.8
σ = 109.1± 99.2

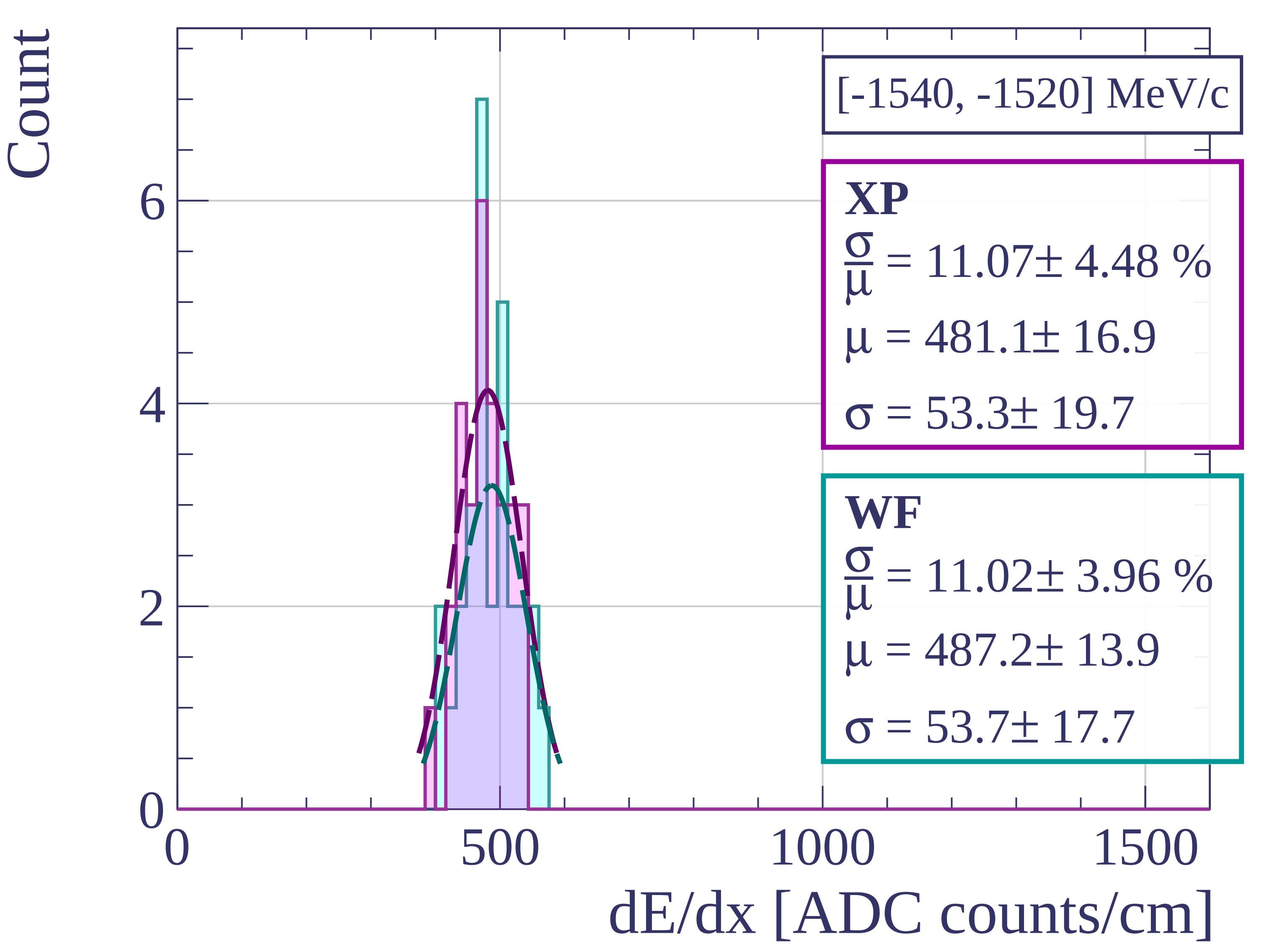

[-1620, -1600] MeV/c
XP
σ/μ = 729.08± 8214.98
μ = 21.2± 220.6
σ = 154.8± 135.8
WF
σ/μ = 3.78± 1.58 %
μ = 430.9± 6.9
σ = 16.3± 6.6
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500


Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-1600, -1580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.51± 2.11 %
μ = 470.1± 9.4
σ = 40.0± 9.1
WF
$\frac{\sigma}{\mu}$ = 35.54± 55.01 %
μ = 397.1± 199.8
σ = 141.2± 147.5

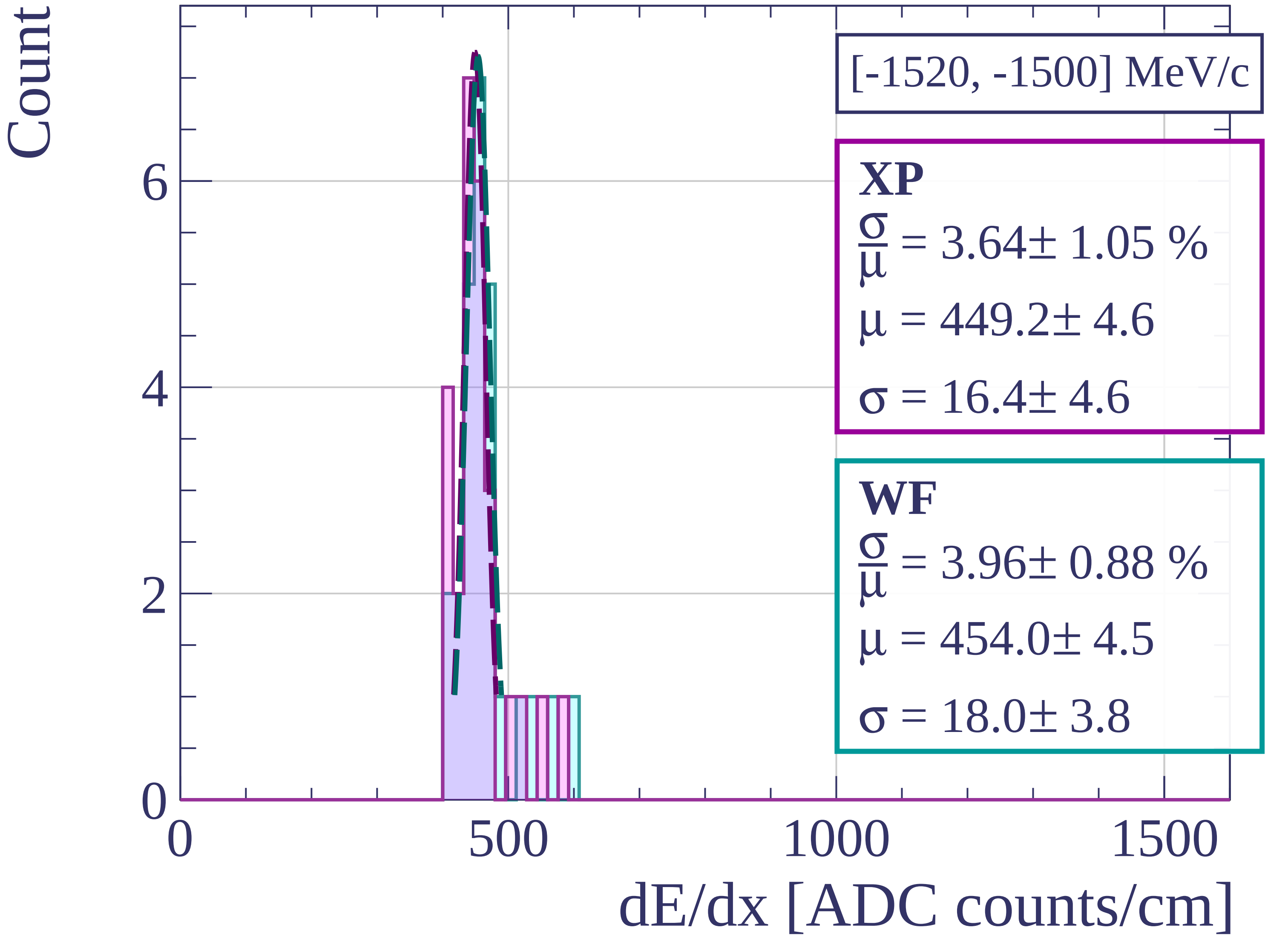

Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1580, -1560] MeV/c
XP
σ/μ = 14.15± 8.98 %
μ = 477.4± 21.5
σ = 67.5± 39.8
WF
σ/μ = 15.14± 5.81 %
μ = 502.0± 23.7
σ = 76.0± 25.6



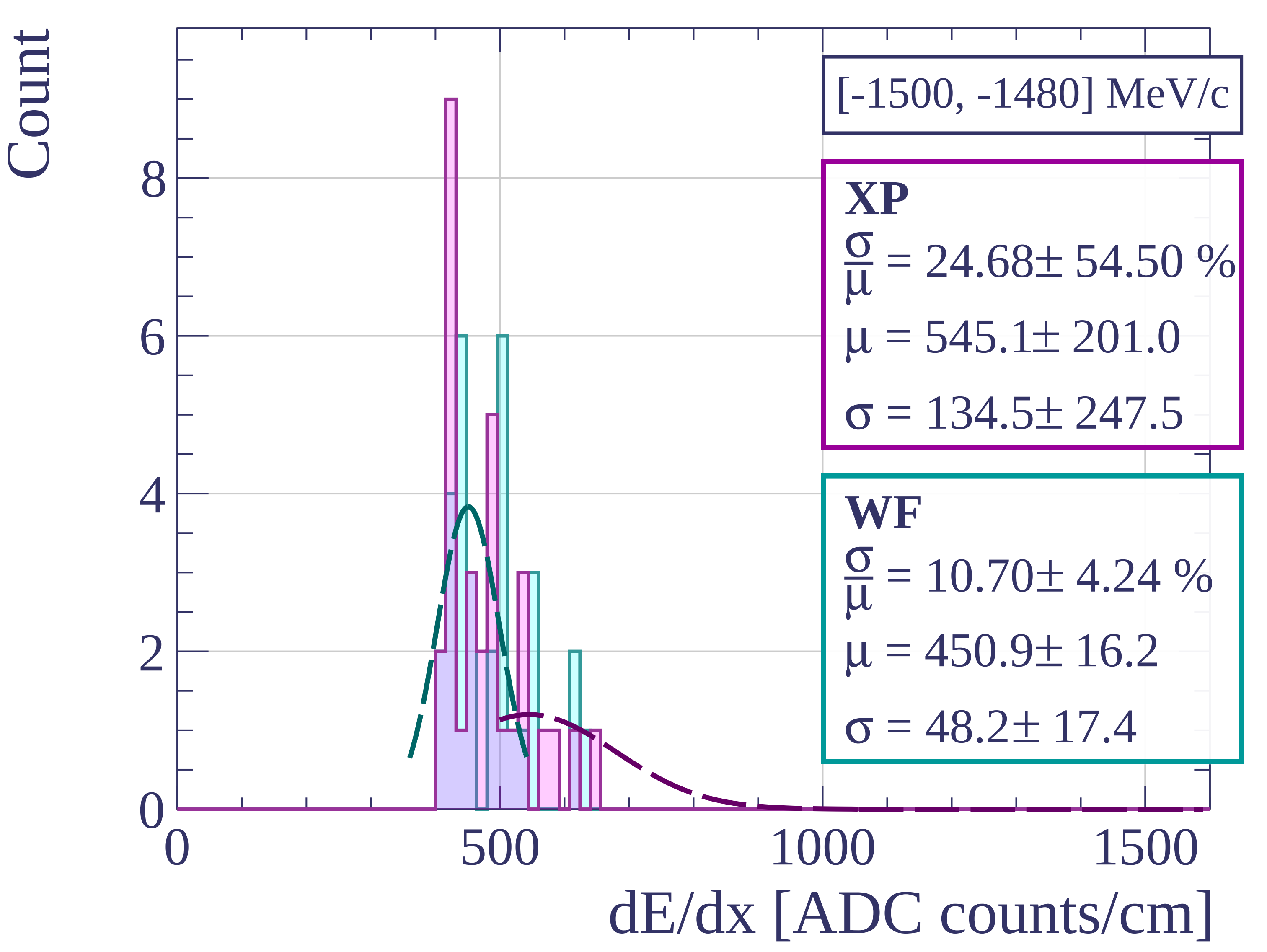
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1560, -1540] MeV/c
XP
σ/μ = 47.52± 84.95 %
μ = 496.5± 253.7
σ = 235.9± 301.3
WF
σ/μ = 18.16± 16.71 %
μ = 456.7± 58.6
σ = 82.9± 65.7

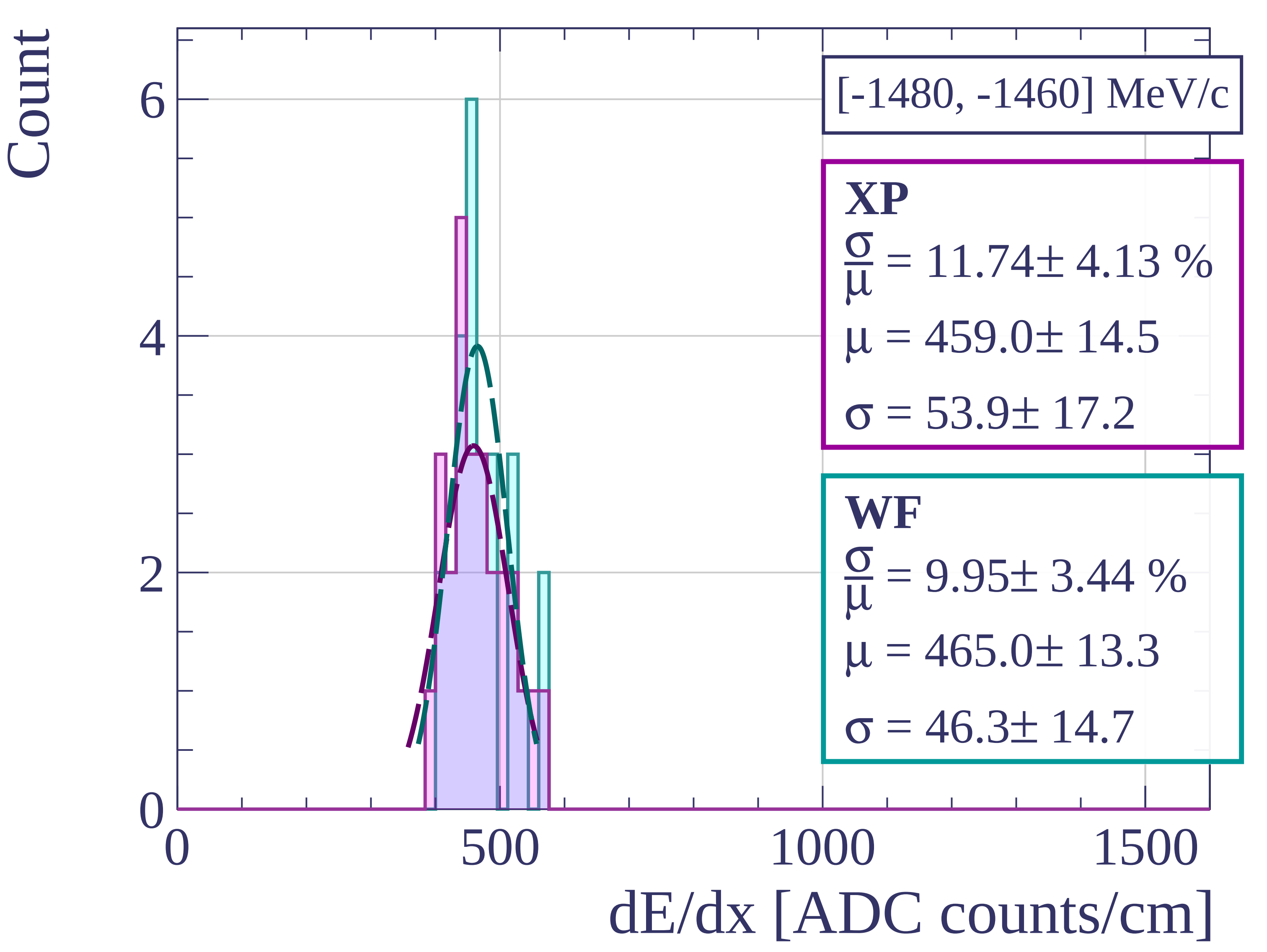


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1540, -1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.07± 4.48 %
μ = 481.1± 16.9
σ = 53.3± 19.7
WF
$\frac{\sigma}{\mu}$ = 11.02± 3.96 %
μ = 487.2± 13.9
σ = 53.7± 17.7

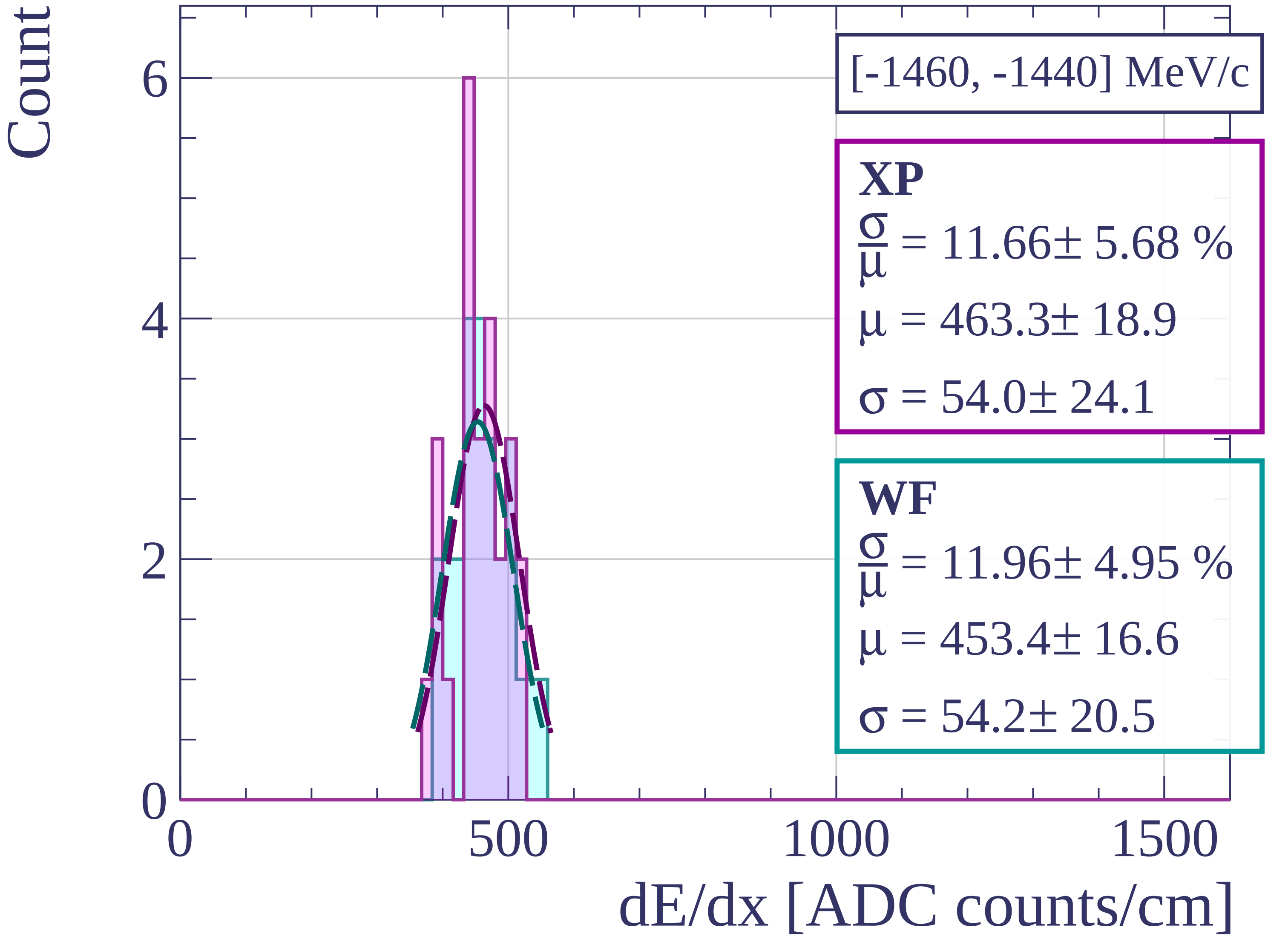


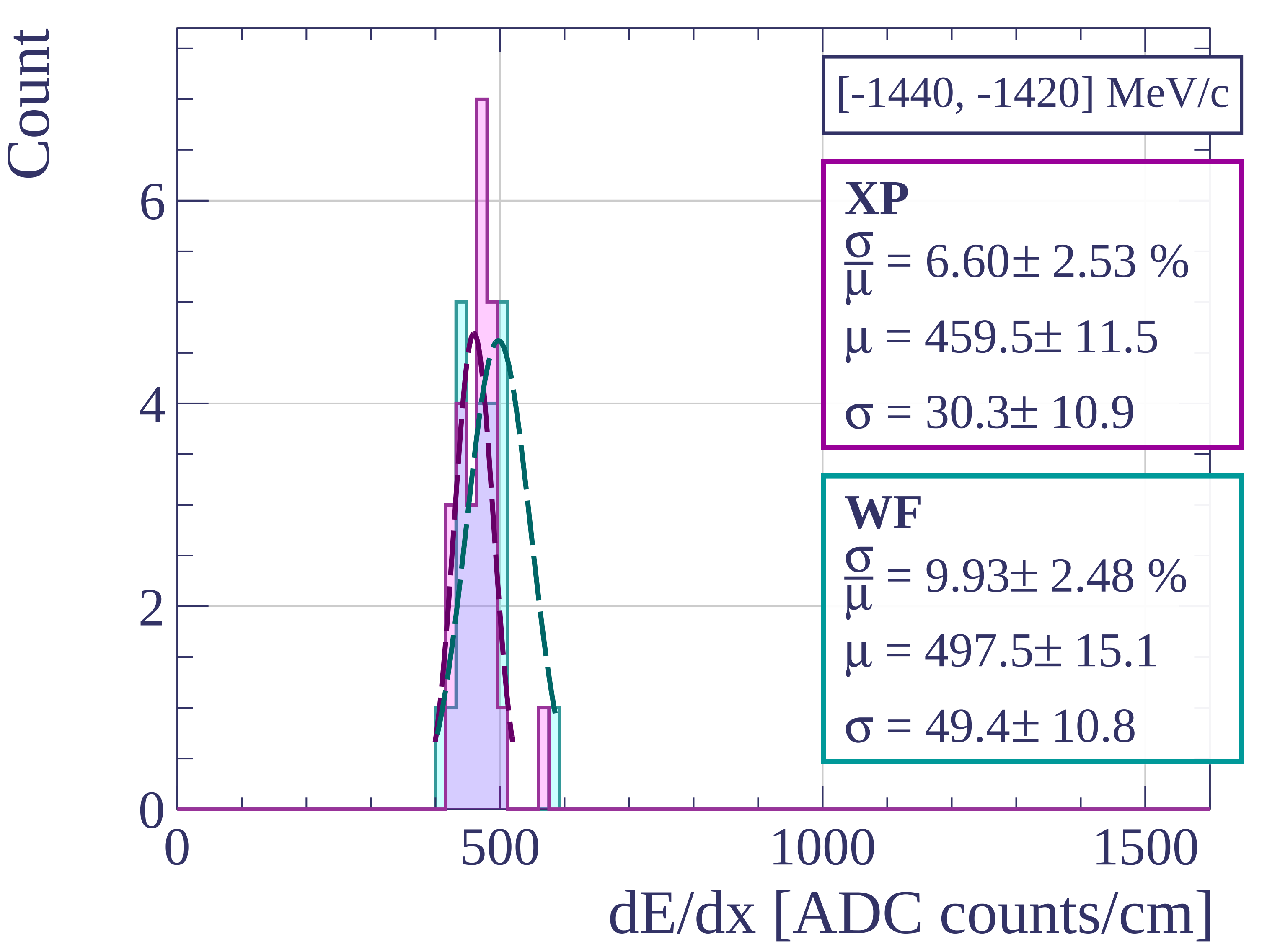
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1520, -1500] MeV/c
XP
σ/μ = 3.64± 1.05 %
μ = 449.2± 4.6
σ = 16.4± 4.6
WF
σ/μ = 3.96± 0.88 %
μ = 454.0± 4.5
σ = 18.0± 3.8

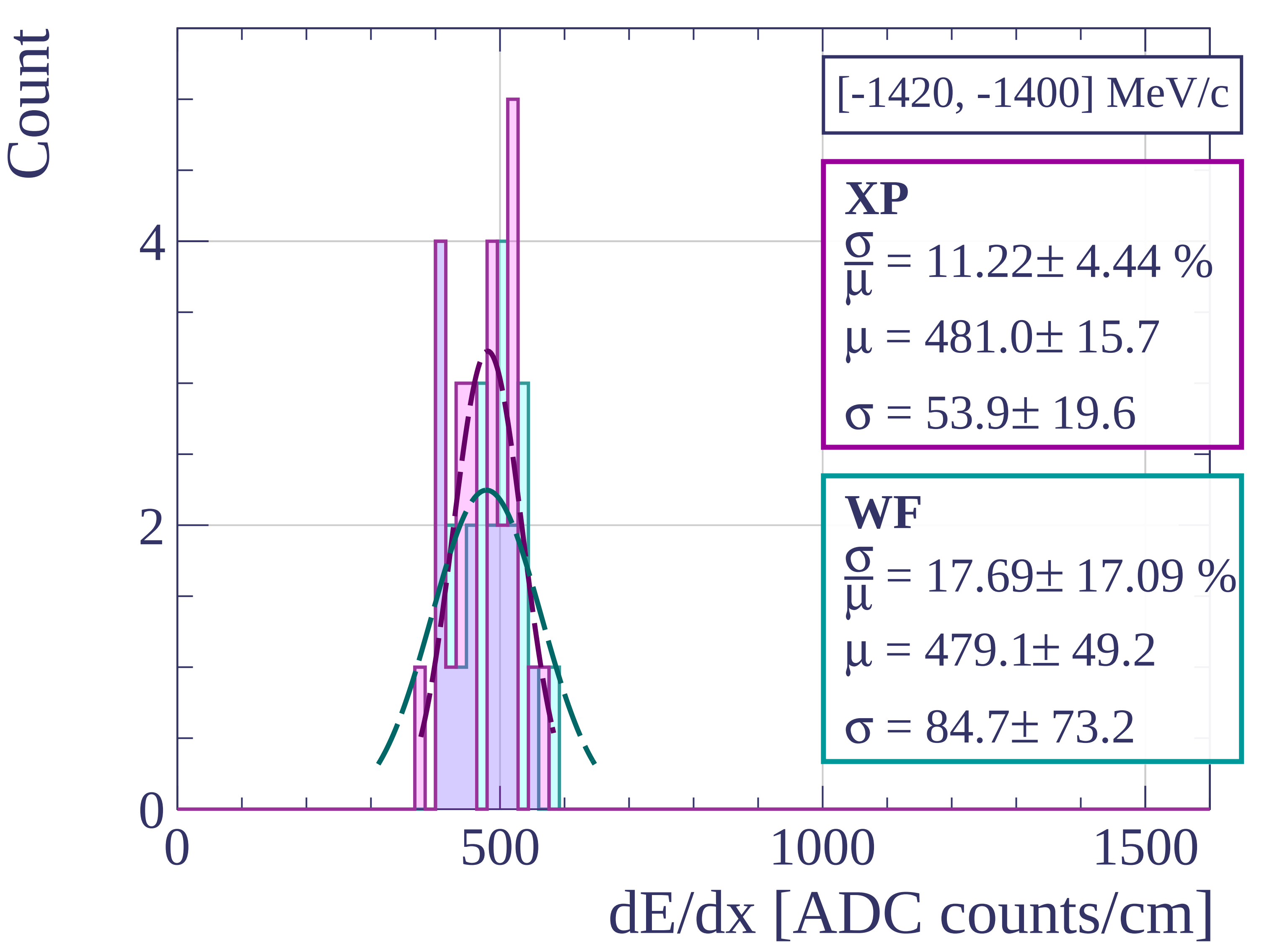

Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1500, -1480] MeV/c
XP
σ/μ = 24.68± 54.50 %
μ = 545.1± 201.0
σ = 134.5± 247.5
WF
σ/μ = 10.70± 4.24 %
μ = 450.9± 16.2
σ = 48.2± 17.4



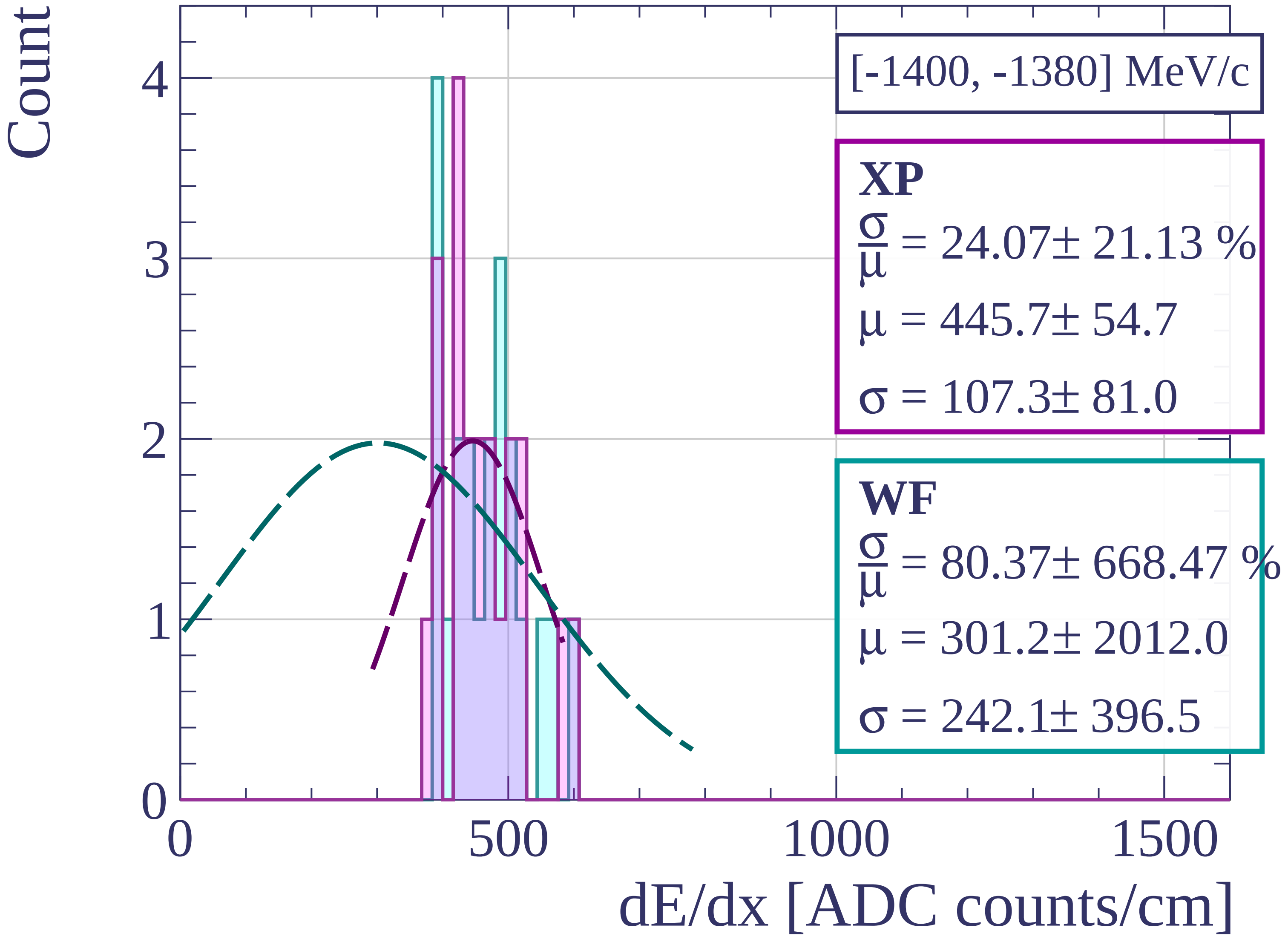
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1480, -1460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.74± 4.13 %
μ = 459.0± 14.5
σ = 53.9± 17.2
WF
$\frac{\sigma}{\mu}$ = 9.95± 3.44 %
μ = 465.0± 13.3
σ = 46.3± 14.7


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1460, -1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.66± 5.68 %
μ = 463.3± 18.9
σ = 54.0± 24.1
WF
$\frac{\sigma}{\mu}$ = 11.96± 4.95 %
μ = 453.4± 16.6
σ = 54.2± 20.5

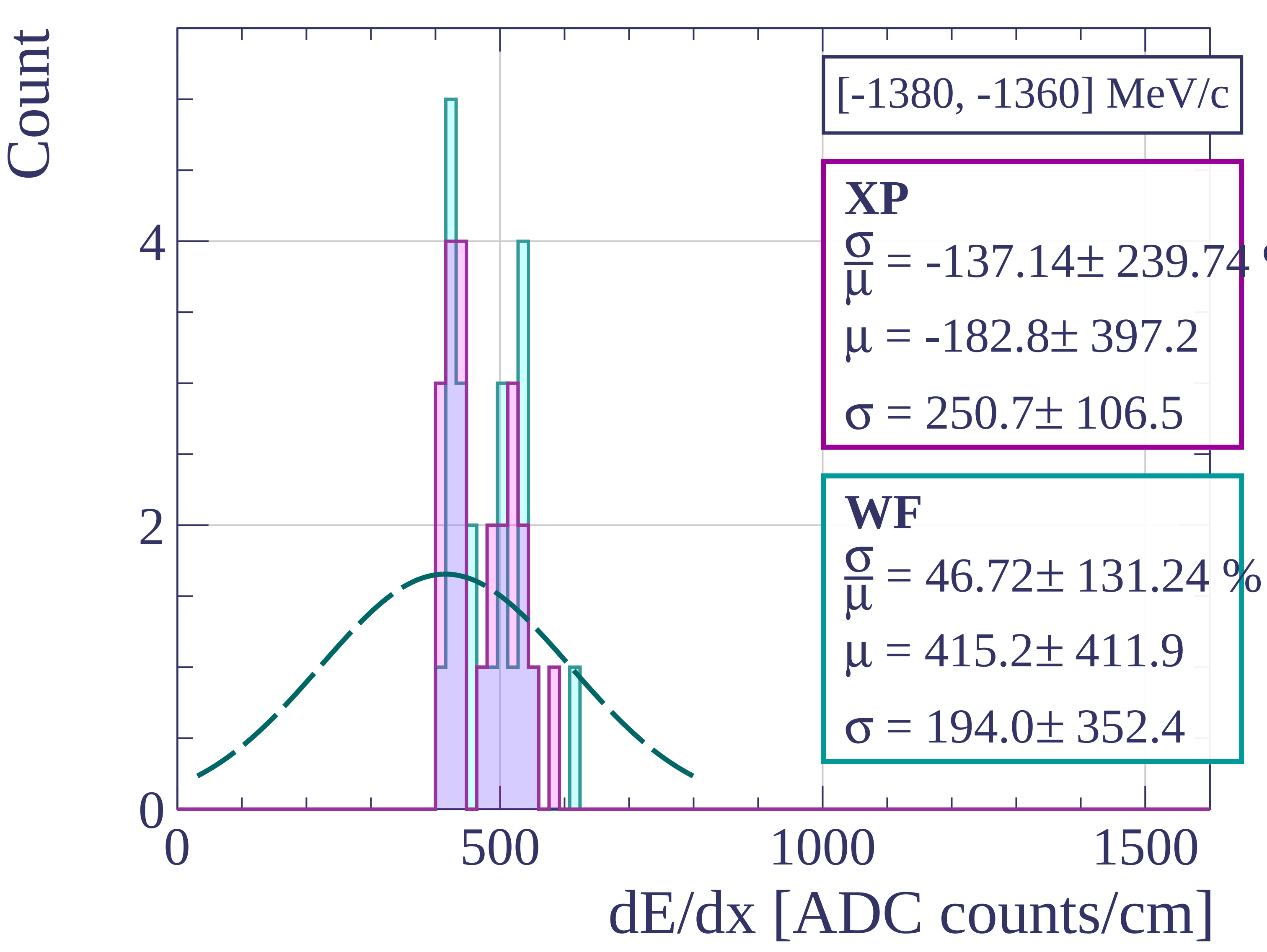

[-1440, -1420] MeV/c
XP
σ/μ = 6.60± 2.53 %
μ = 459.5± 11.5
σ = 30.3± 10.9
WF
σ/μ = 9.93± 2.48 %
μ = 497.5± 15.1
σ = 49.4± 10.8
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500

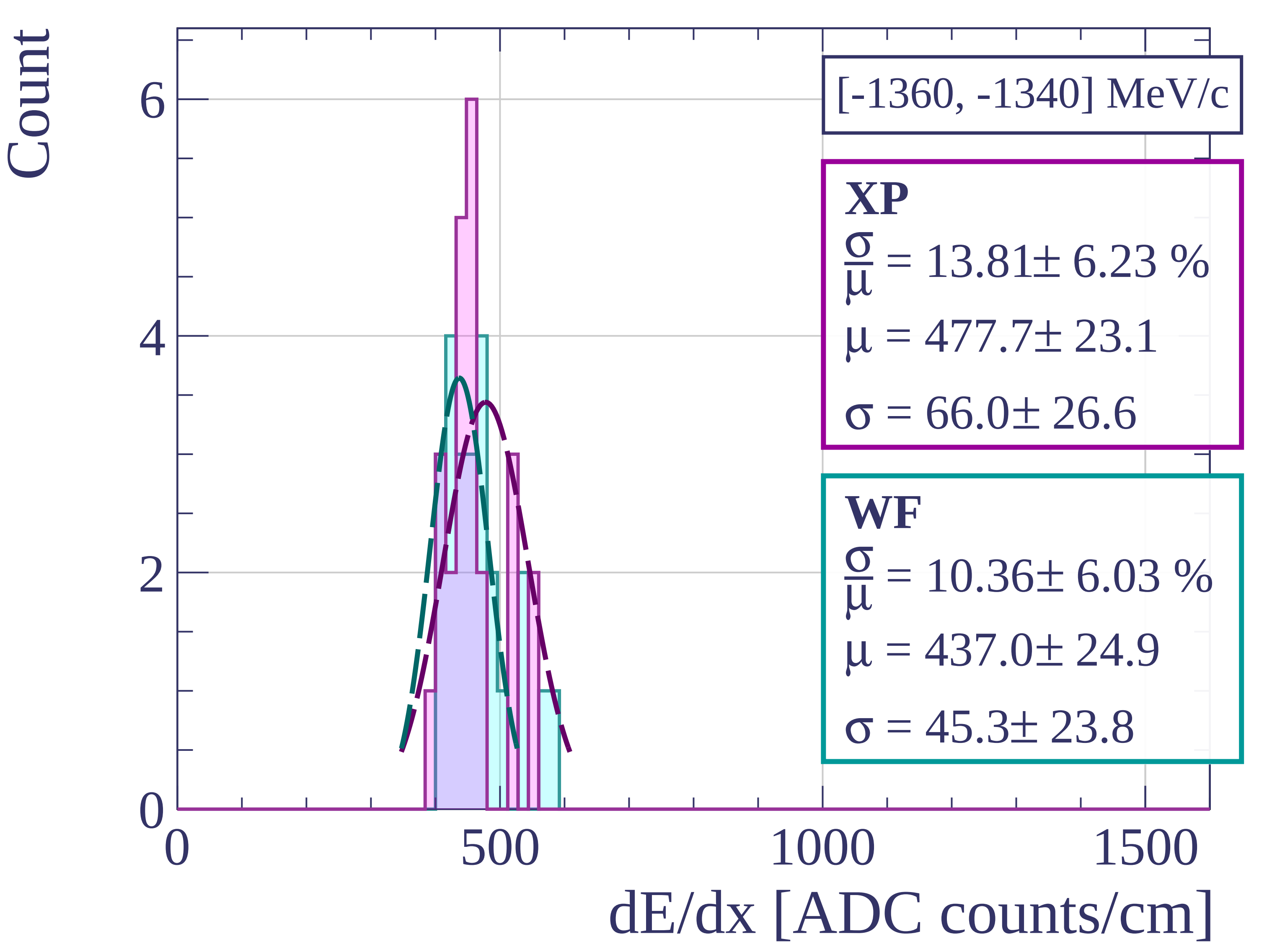


Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1420, -1400] MeV/c
XP
σ/μ = 11.22± 4.44 %
μ = 481.0± 15.7
σ = 53.9± 19.6
WF
σ/μ = 17.69± 17.09 %
μ = 479.1± 49.2
σ = 84.7± 73.2

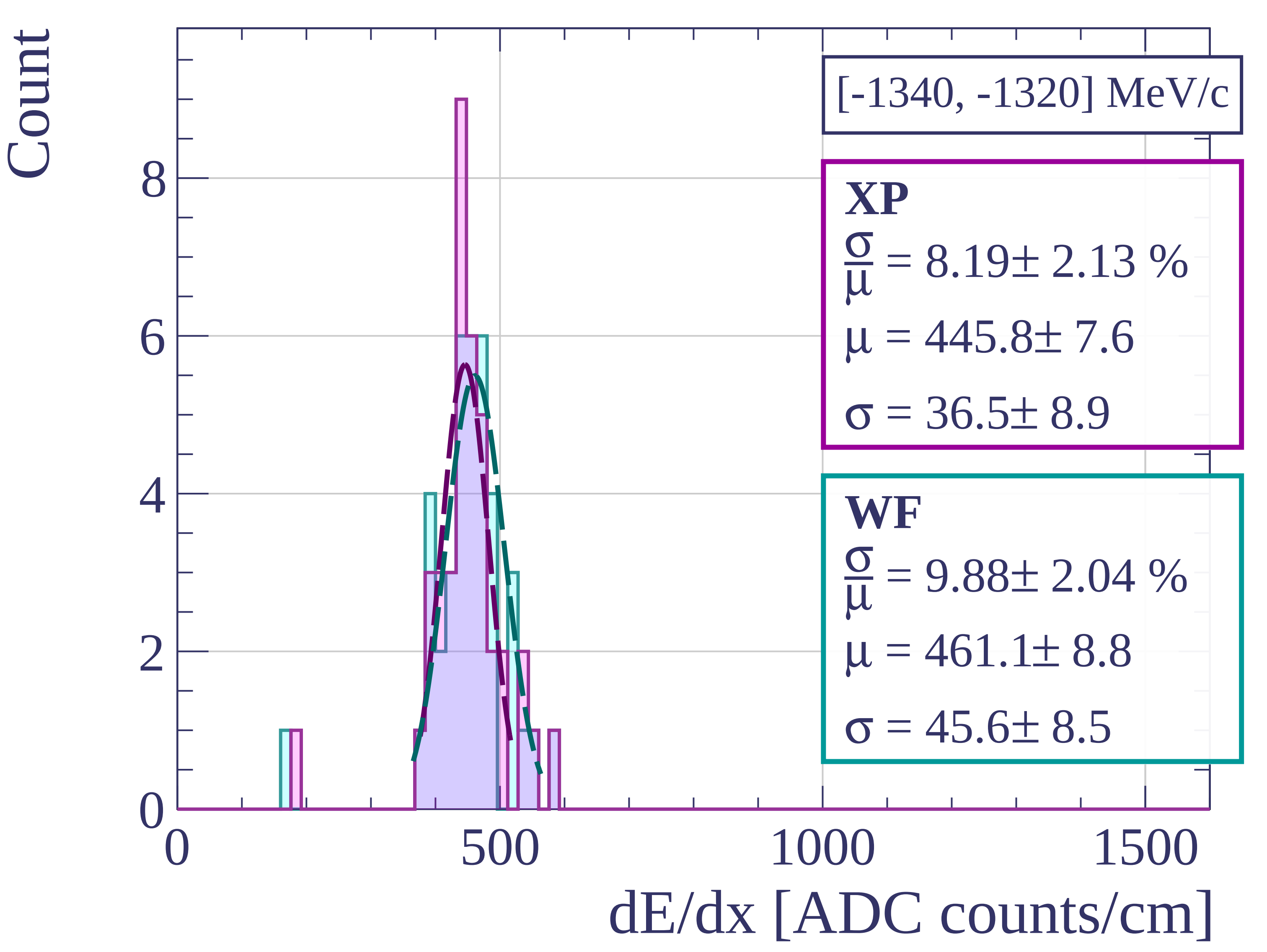


Count
4
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1400, -1380] MeV/c
XP
σ/μ = 24.07± 21.13 %
μ = 445.7± 54.7
σ = 107.3± 81.0
WF
σ/μ = 80.37± 668.47 %
μ = 301.2± 2012.0
σ = 242.1± 396.5

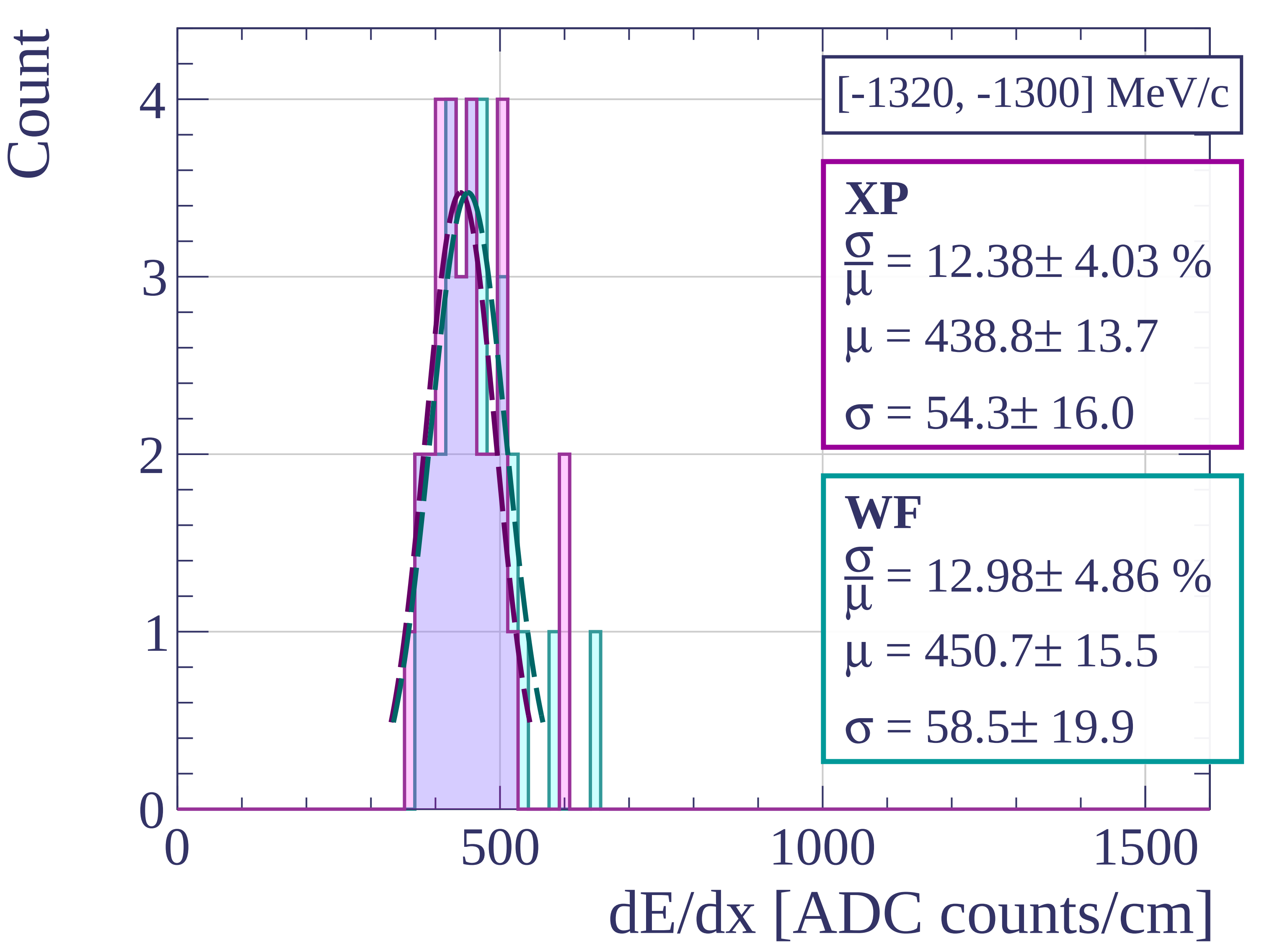

Count
dE/dx [ADC counts/cm]
[-1380, -1360] MeV/c
XP
σ/μ = -137.14± 239.74 %
μ = -182.8± 397.2
σ = 250.7± 106.5
WF
σ/μ = 46.72± 131.24 %
μ = 415.2± 411.9
σ = 194.0± 352.4
0
2
4
0
500
1000
1500

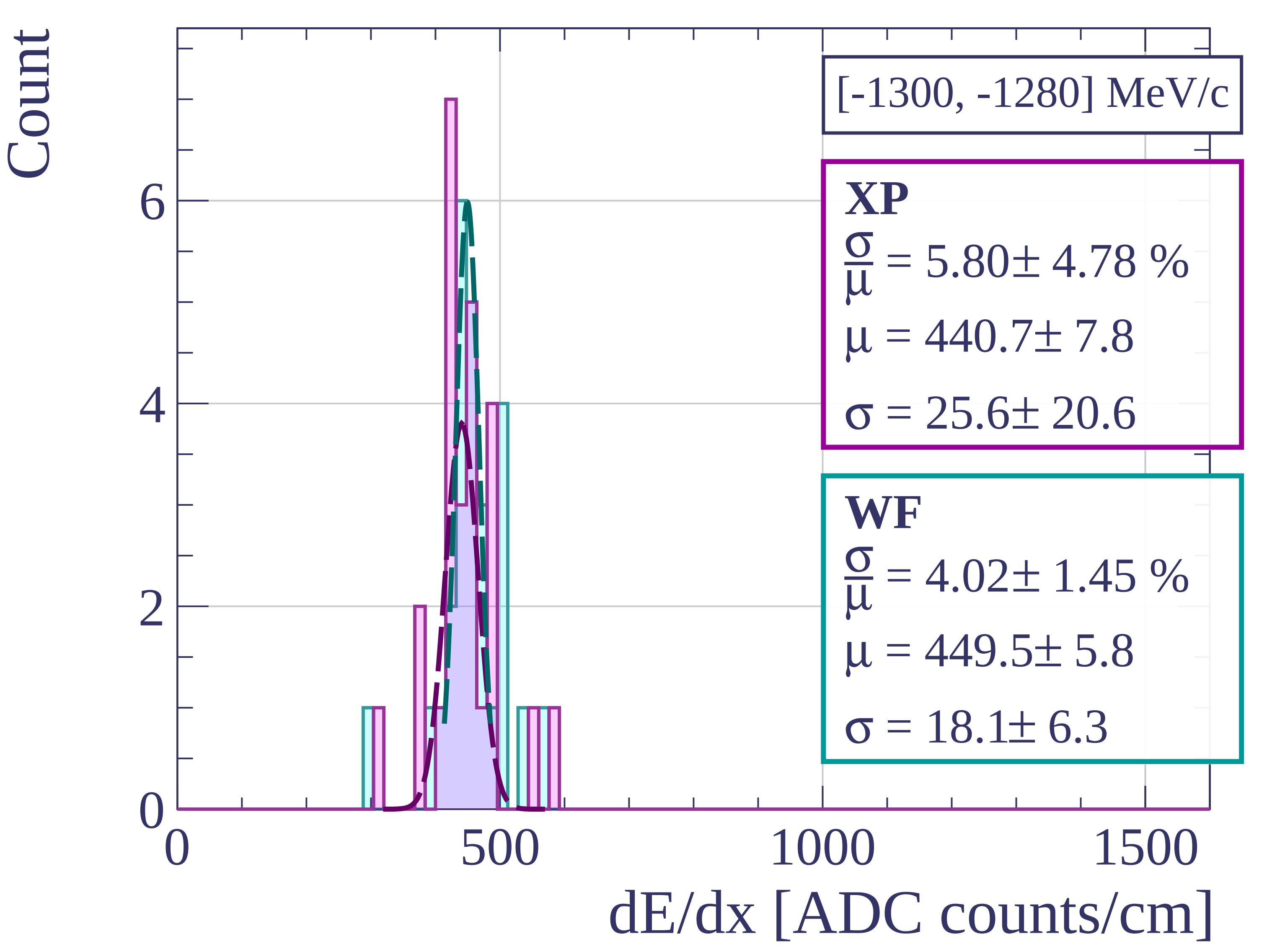


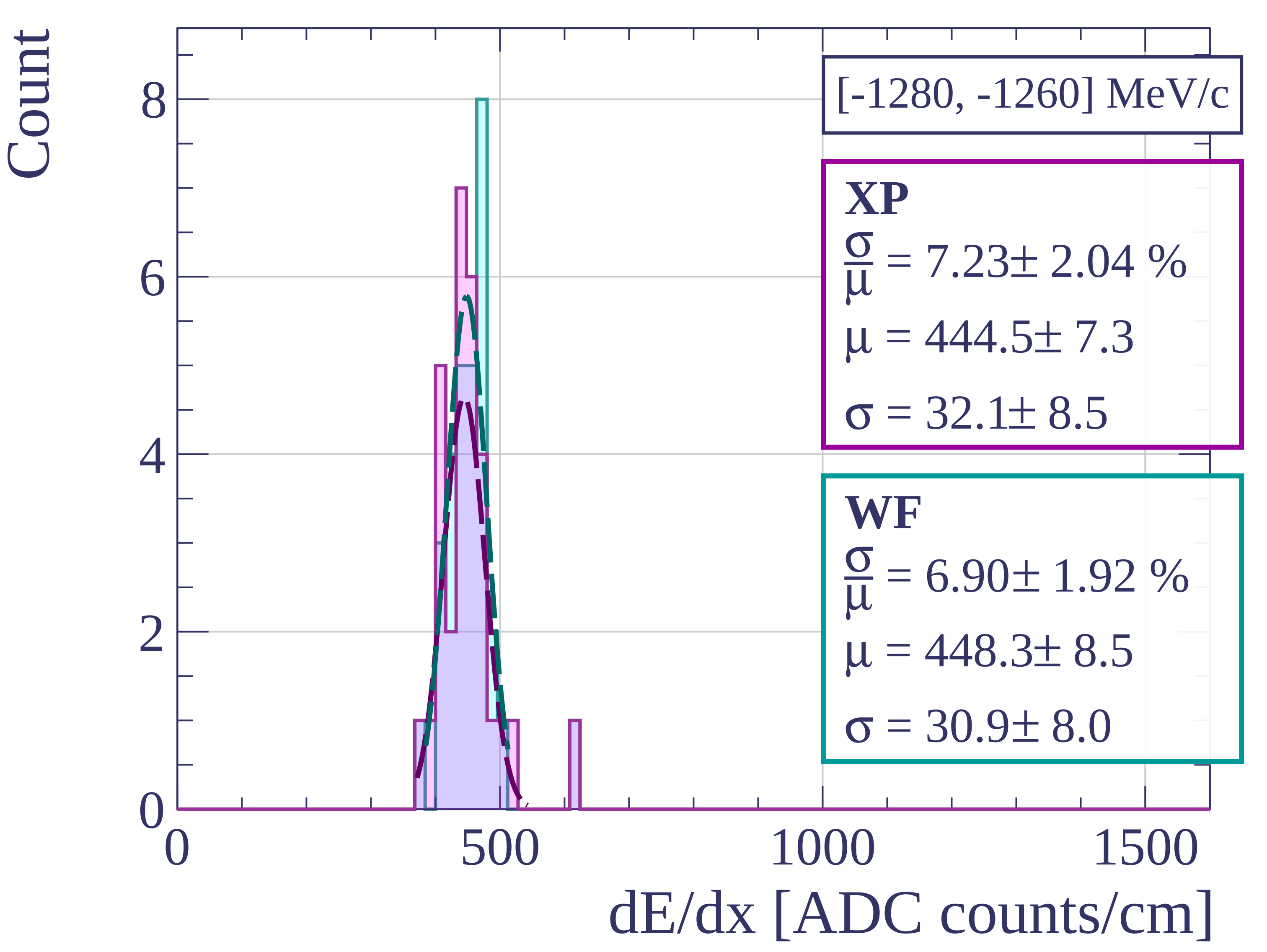
Count
[-1360, -1340] MeV/c
XP
σ/μ = 13.81± 6.23 %
μ = 477.7± 23.1
σ = 66.0± 26.6
WF
σ/μ = 10.36± 6.03 %
μ = 437.0± 24.9
σ = 45.3± 23.8
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]



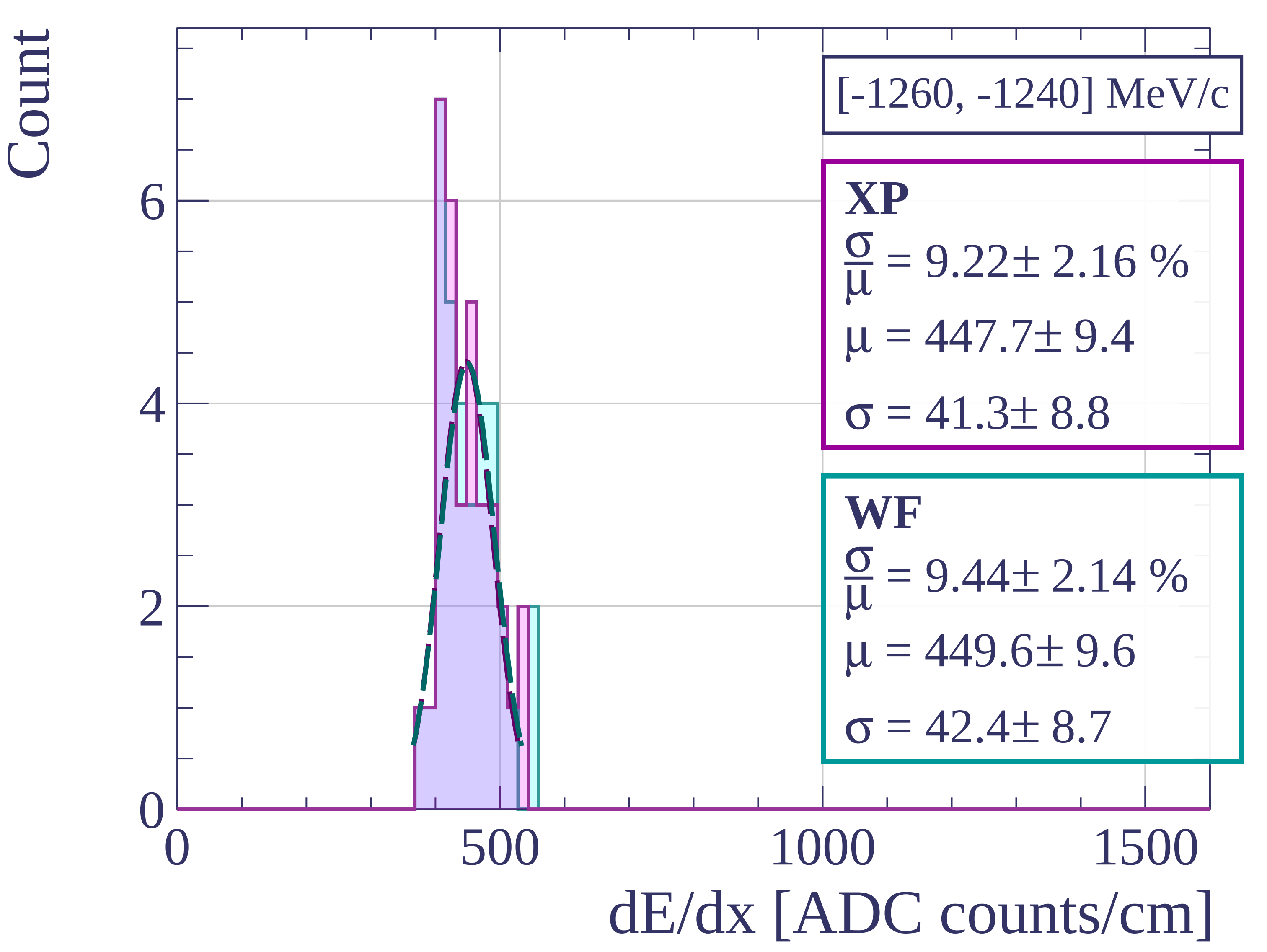
Count
dE/dx [ADC counts/cm]
[-1340, -1320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.19± 2.13 %
μ = 445.8± 7.6
σ = 36.5± 8.9
WF
$\frac{\sigma}{\mu}$ = 9.88± 2.04 %
μ = 461.1± 8.8
σ = 45.6± 8.5
0
2
4
6
8
0
500
1000
1500


Count
0
1
2
3
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1320, -1300] MeV/c
XP
σ/μ = 12.38± 4.03 %
μ = 438.8± 13.7
σ = 54.3± 16.0
WF
σ/μ = 12.98± 4.86 %
μ = 450.7± 15.5
σ = 58.5± 19.9

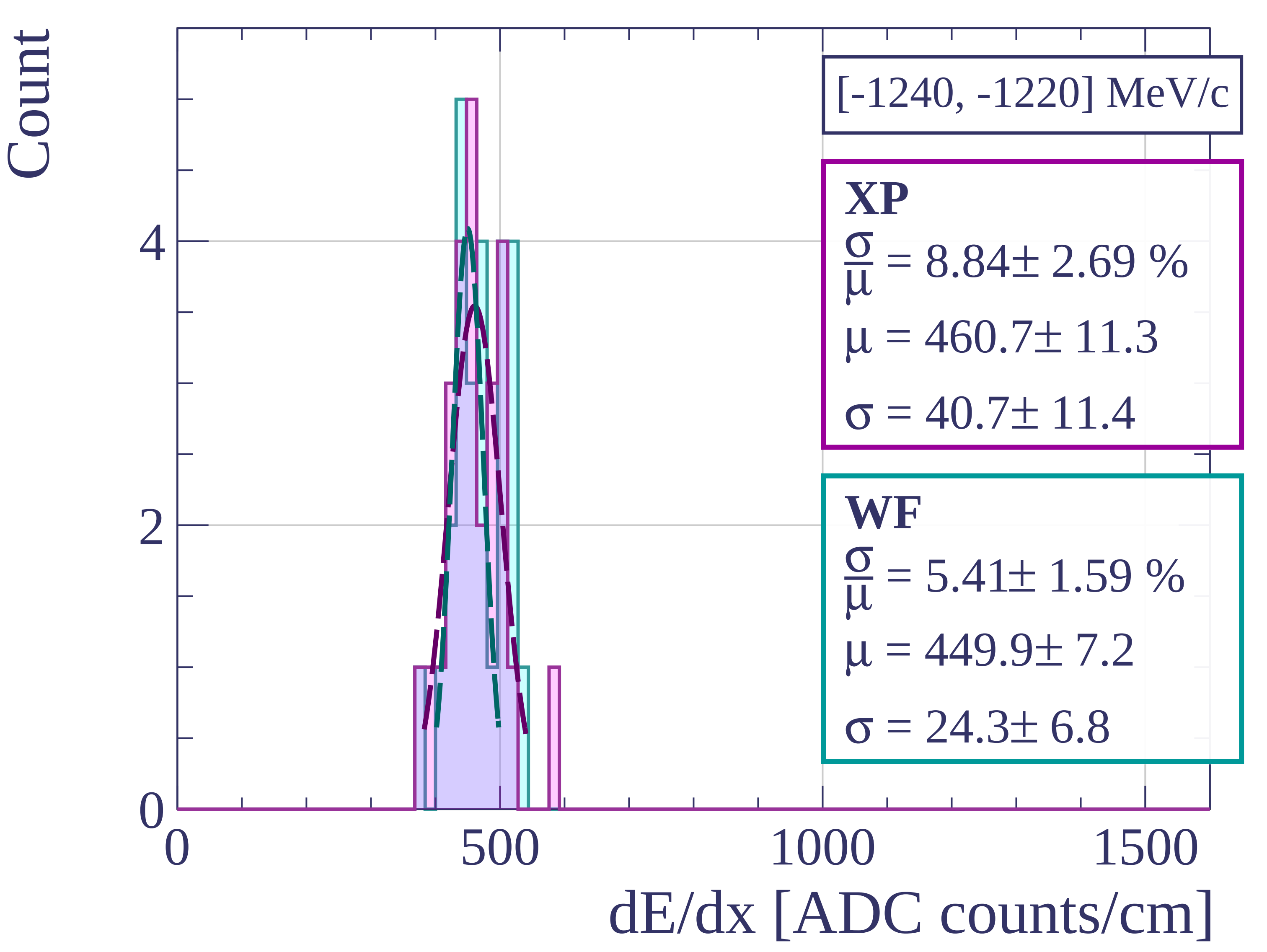

Count
dE/dx [ADC counts/cm]
[-1300, -1280] MeV/c
XP
σ/μ = 5.80± 4.78 %
μ = 440.7± 7.8
σ = 25.6± 20.6
WF
σ/μ = 4.02± 1.45 %
μ = 449.5± 5.8
σ = 18.1± 6.3
0
2
4
6
0
500
1000
1500



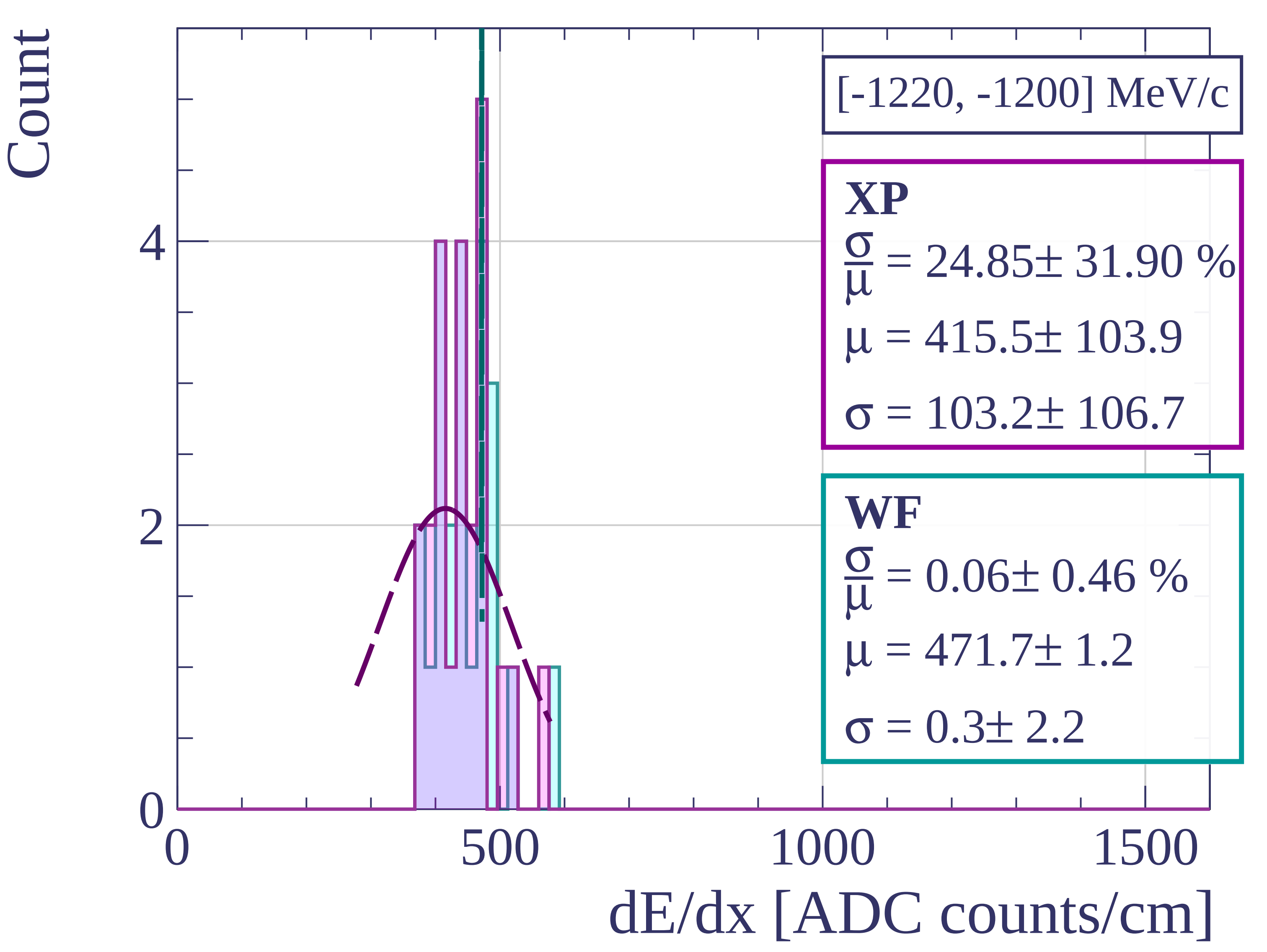
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1280, -1260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.23± 2.04 %
μ = 444.5± 7.3
σ = 32.1± 8.5
WF
$\frac{\sigma}{\mu}$ = 6.90± 1.92 %
μ = 448.3± 8.5
σ = 30.9± 8.0

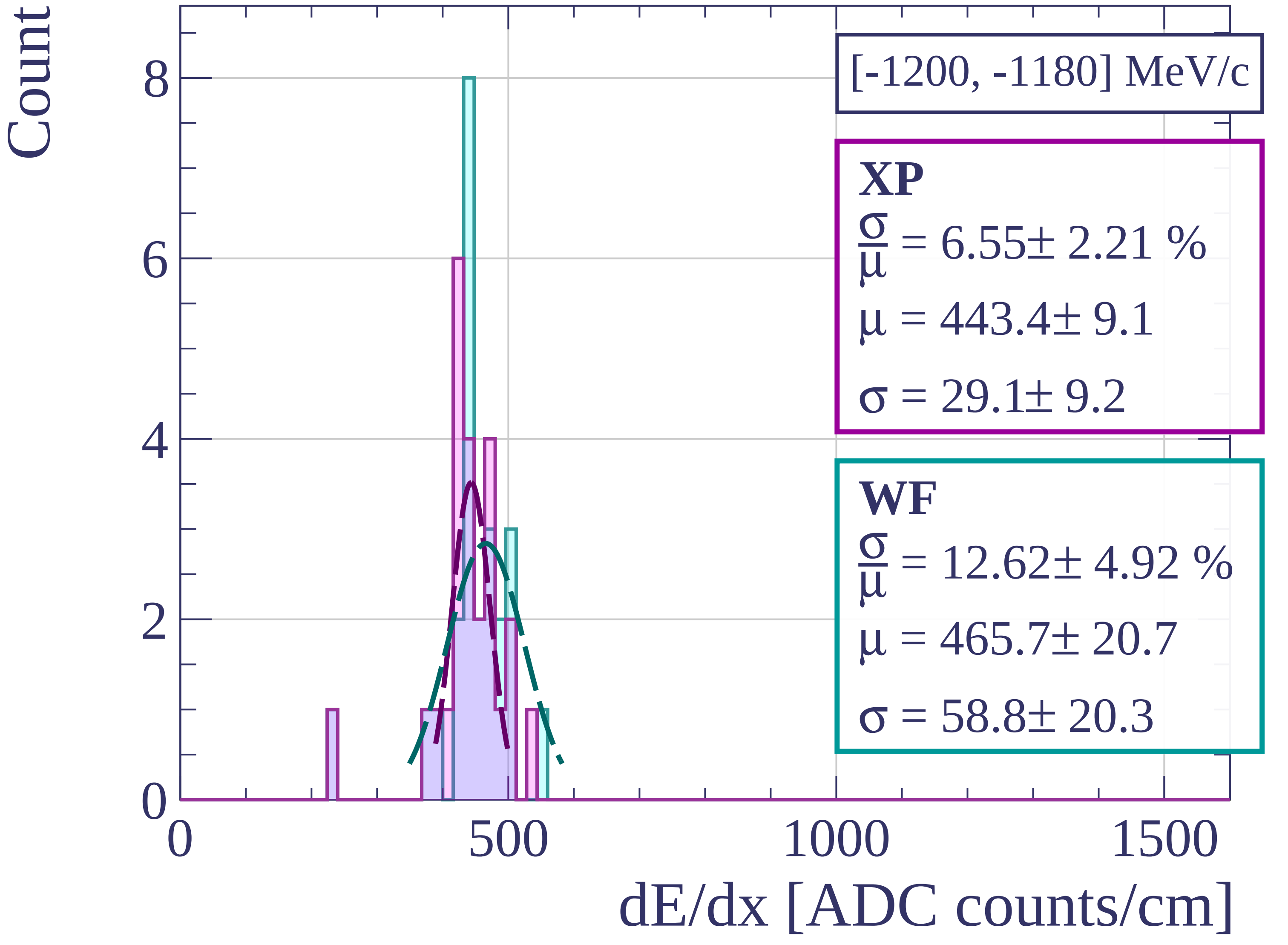


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1260, -1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.22± 2.16 %
μ = 447.7± 9.4
σ = 41.3± 8.8
WF
$\frac{\sigma}{\mu}$ = 9.44± 2.14 %
μ = 449.6± 9.6
σ = 42.4± 8.7

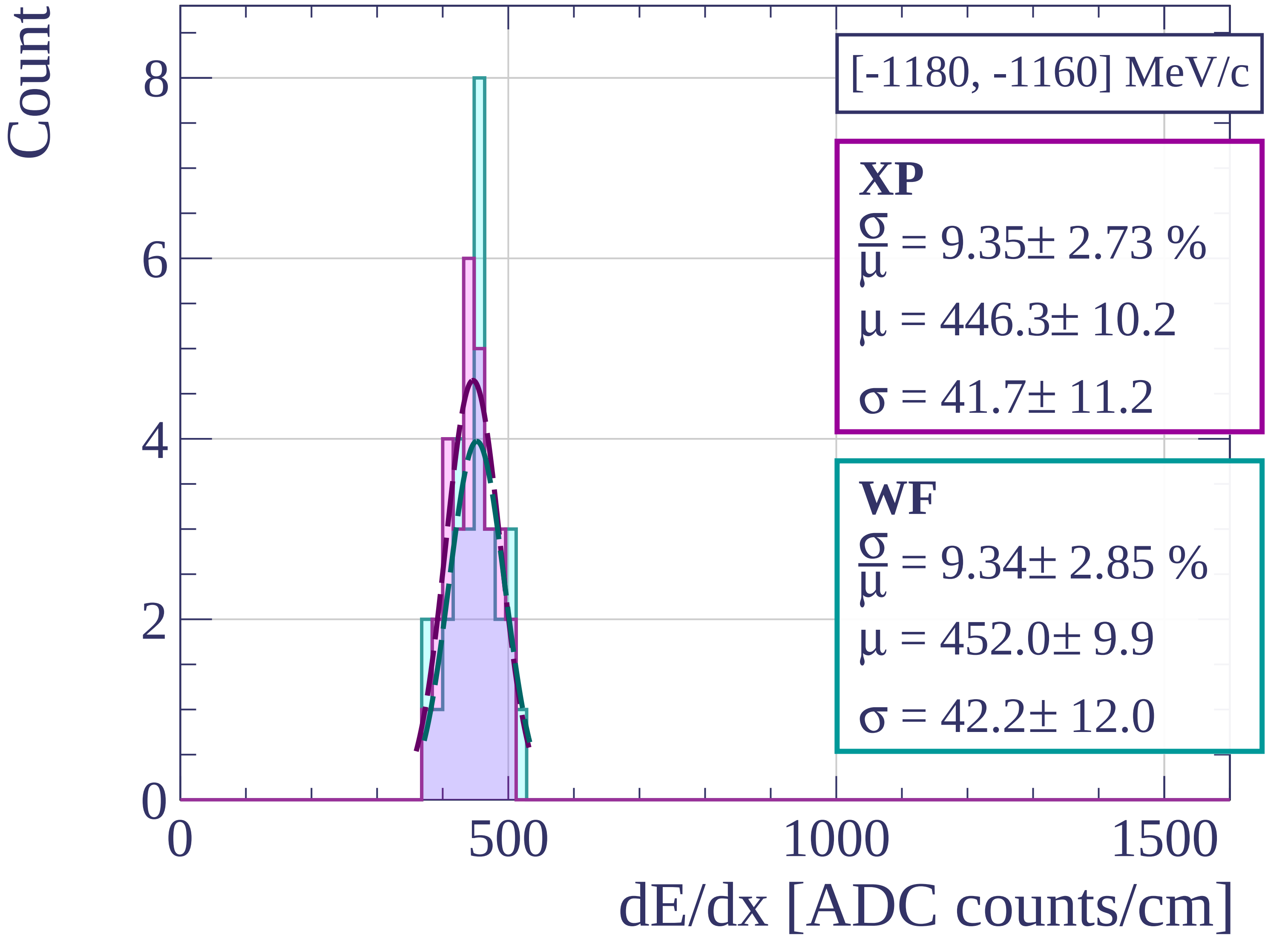


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1240, -1220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.84± 2.69 %
μ = 460.7± 11.3
σ = 40.7± 11.4
WF
$\frac{\sigma}{\mu}$ = 5.41± 1.59 %
μ = 449.9± 7.2
σ = 24.3± 6.8

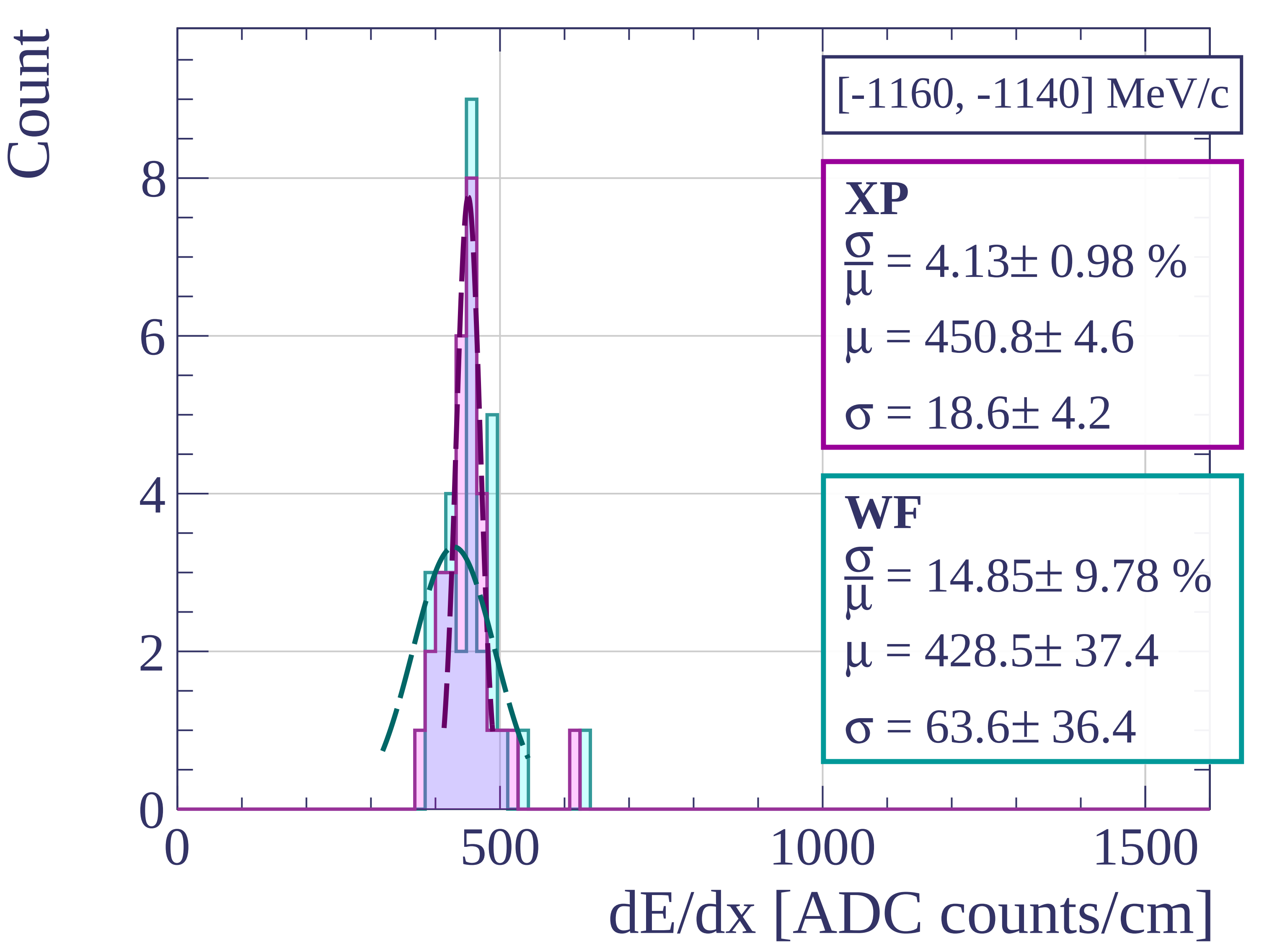

Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[-1220, -1200] MeV/c
XP
σ/μ = 24.85± 31.90 %
μ = 415.5± 103.9
σ = 103.2± 106.7
WF
σ/μ = 0.06± 0.46 %
μ = 471.7± 1.2
σ = 0.3± 2.2



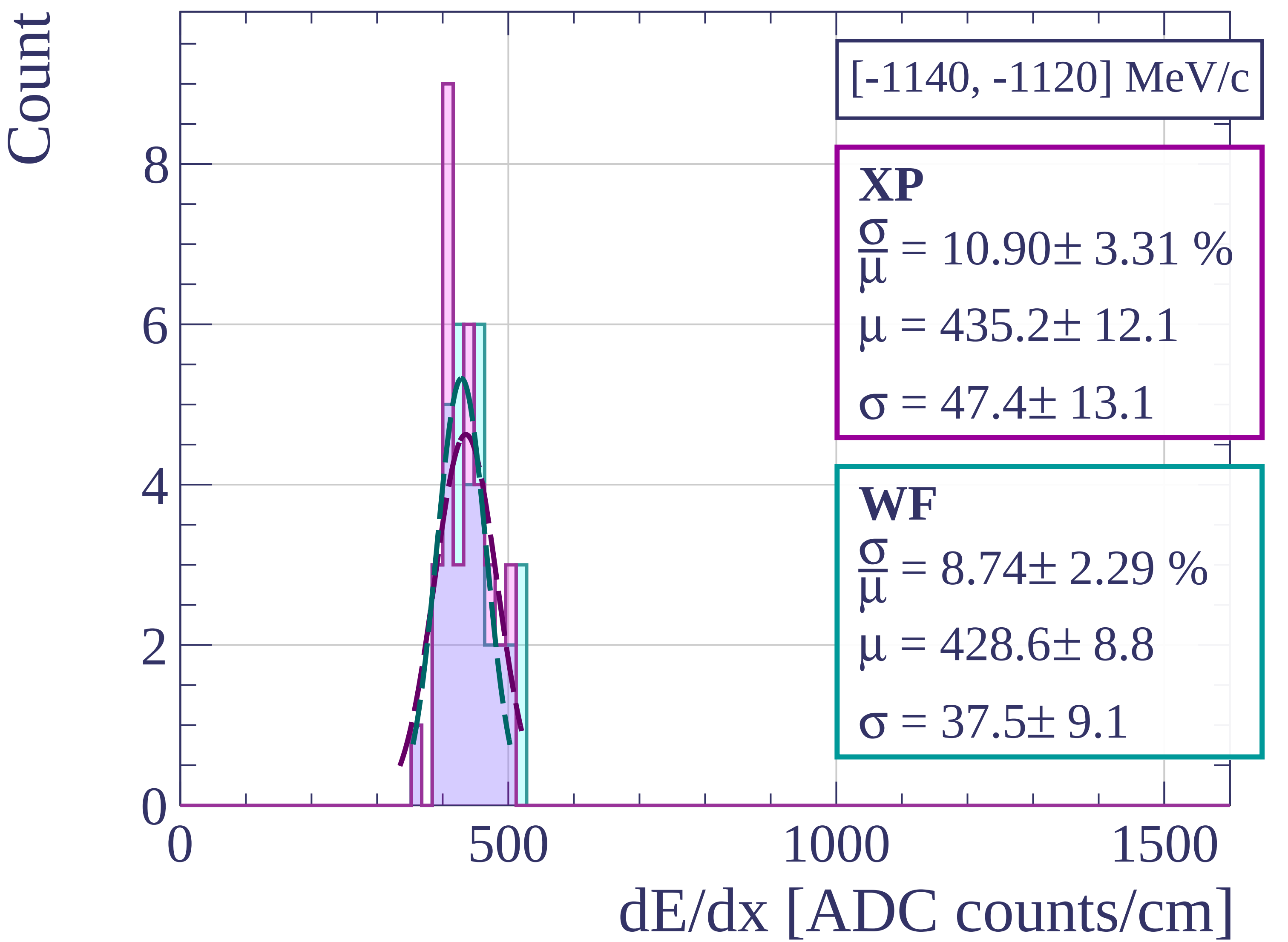
Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1200, -1180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.55± 2.21 %
μ = 443.4± 9.1
σ = 29.1± 9.2
WF
$\frac{\sigma}{\mu}$ = 12.62± 4.92 %
μ = 465.7± 20.7
σ = 58.8± 20.3



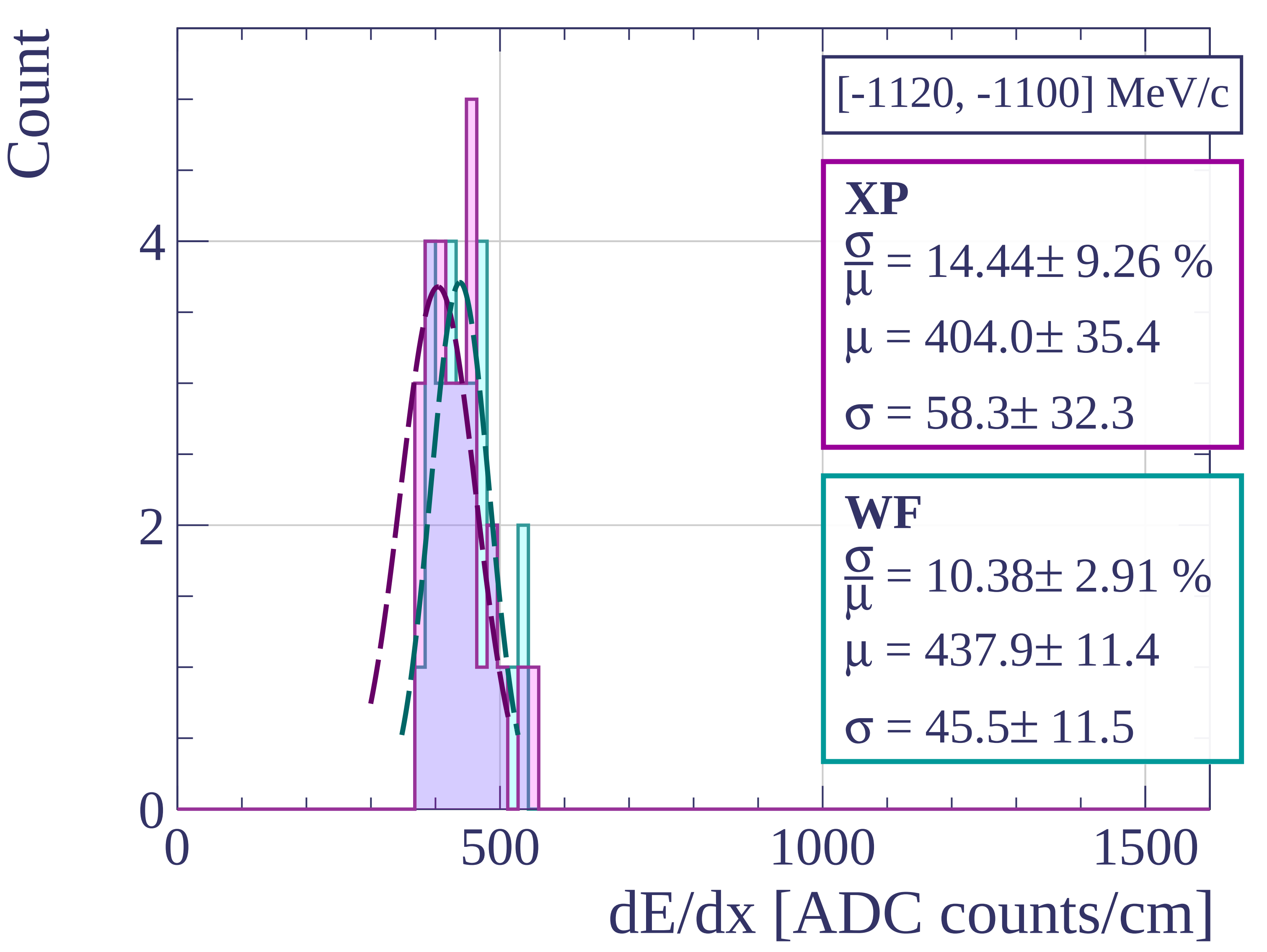
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-1180, -1160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.35± 2.73 %
μ = 446.3± 10.2
σ = 41.7± 11.2
WF
$\frac{\sigma}{\mu}$ = 9.34± 2.85 %
μ = 452.0± 9.9
σ = 42.2± 12.0

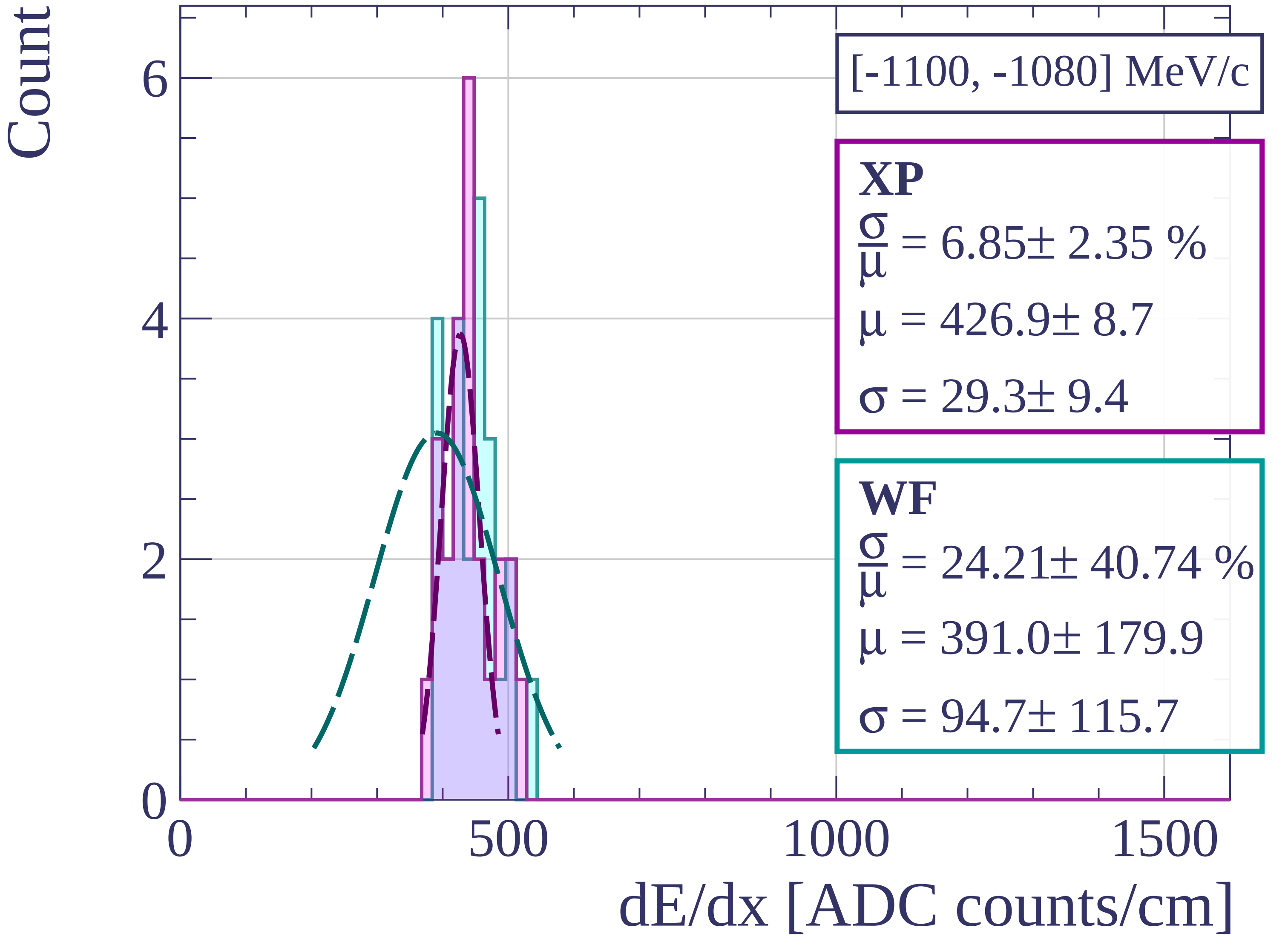


Count
dE/dx [ADC counts/cm]
[-1160, -1140] MeV/c
XP
σ/μ = 4.13± 0.98 %
μ = 450.8± 4.6
σ = 18.6± 4.2
WF
σ/μ = 14.85± 9.78 %
μ = 428.5± 37.4
σ = 63.6± 36.4
0
2
4
6
8
0
500
1000
1500

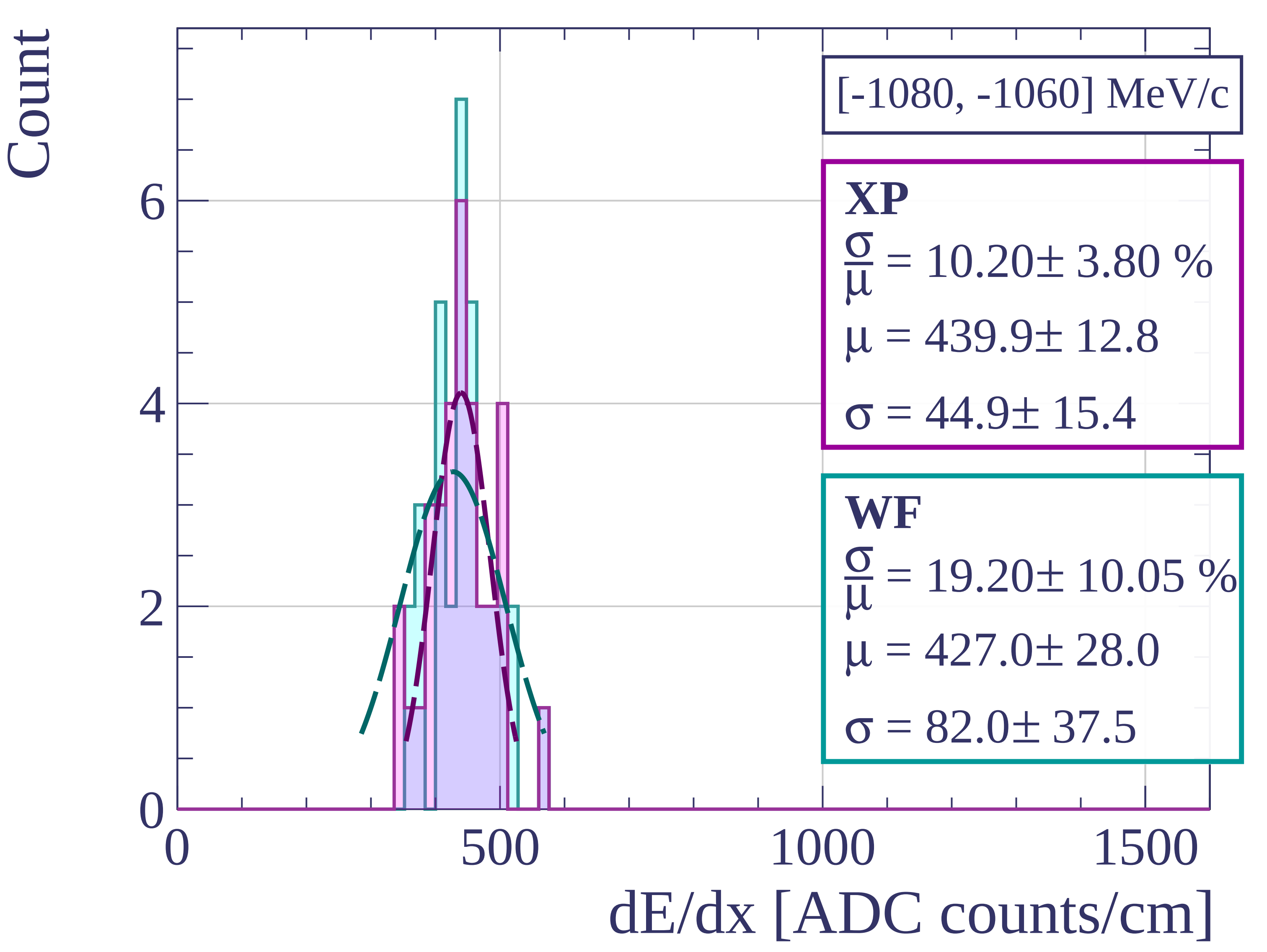


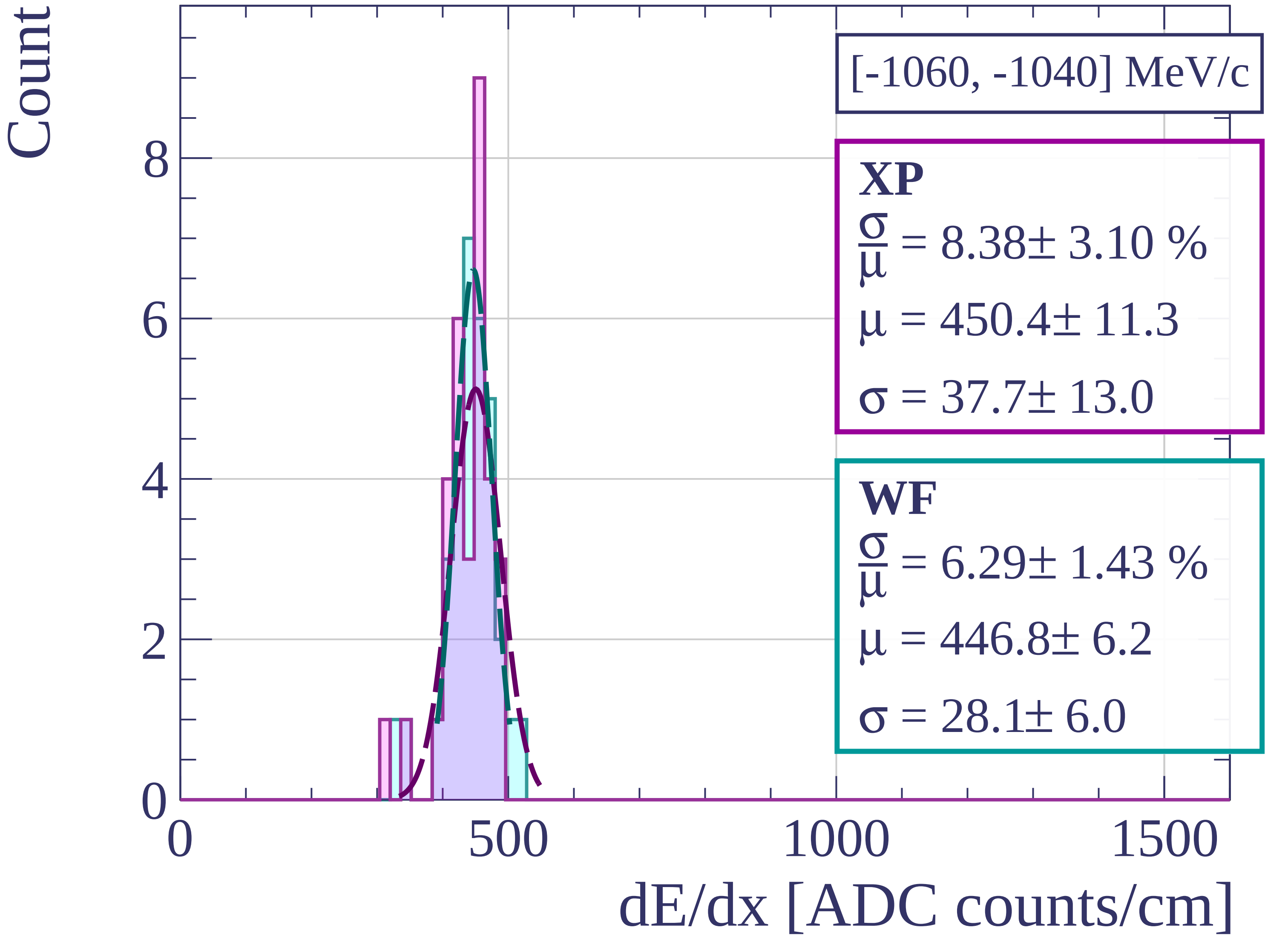
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1140, -1120] MeV/c
XP
σ/μ = 10.90± 3.31 %
μ = 435.2± 12.1
σ = 47.4± 13.1
WF
σ/μ = 8.74± 2.29 %
μ = 428.6± 8.8
σ = 37.5± 9.1


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1120, -1100] MeV/c
XP
σ/μ = 14.44± 9.26 %
μ = 404.0± 35.4
σ = 58.3± 32.3
WF
σ/μ = 10.38± 2.91 %
μ = 437.9± 11.4
σ = 45.5± 11.5

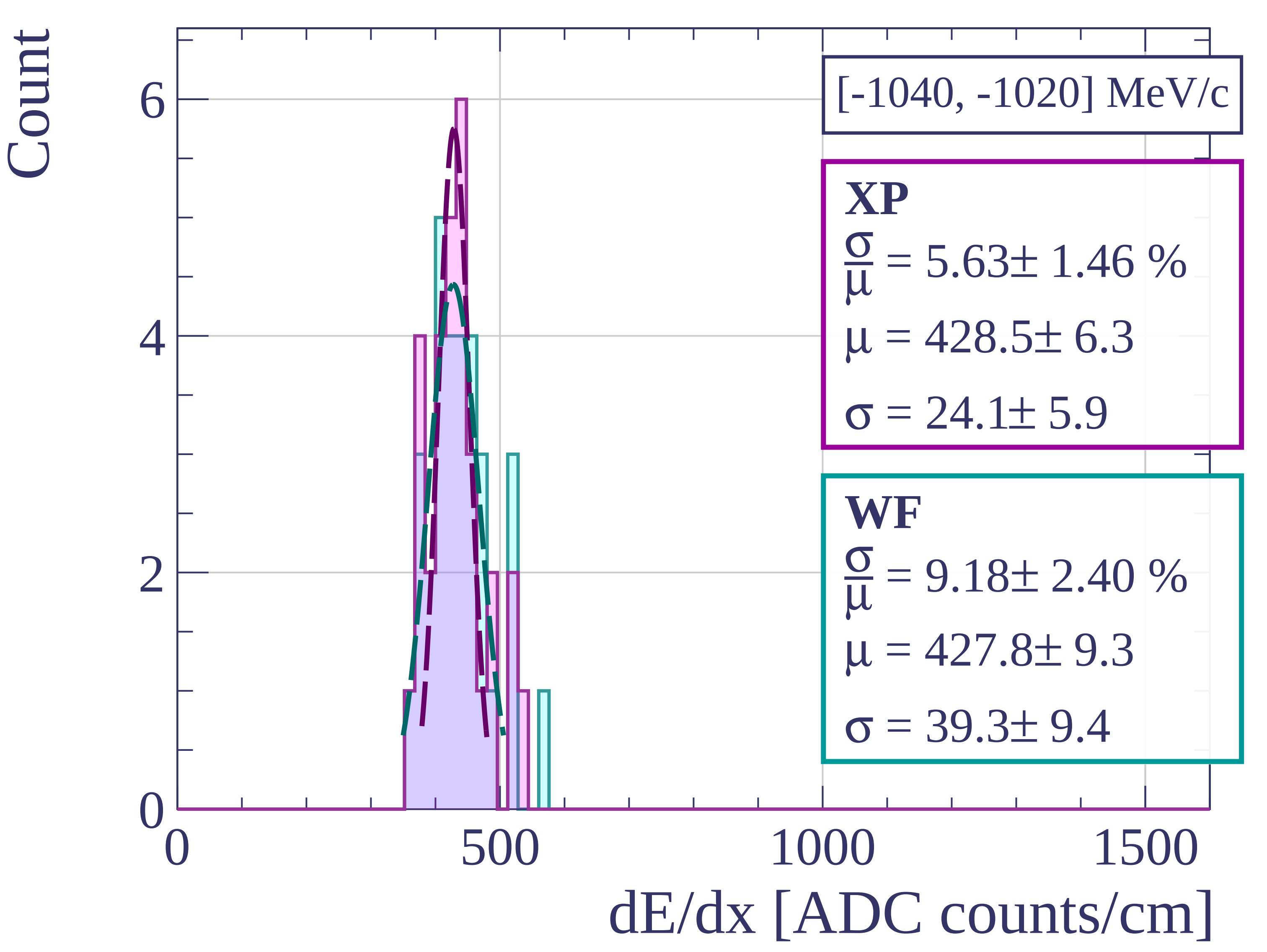


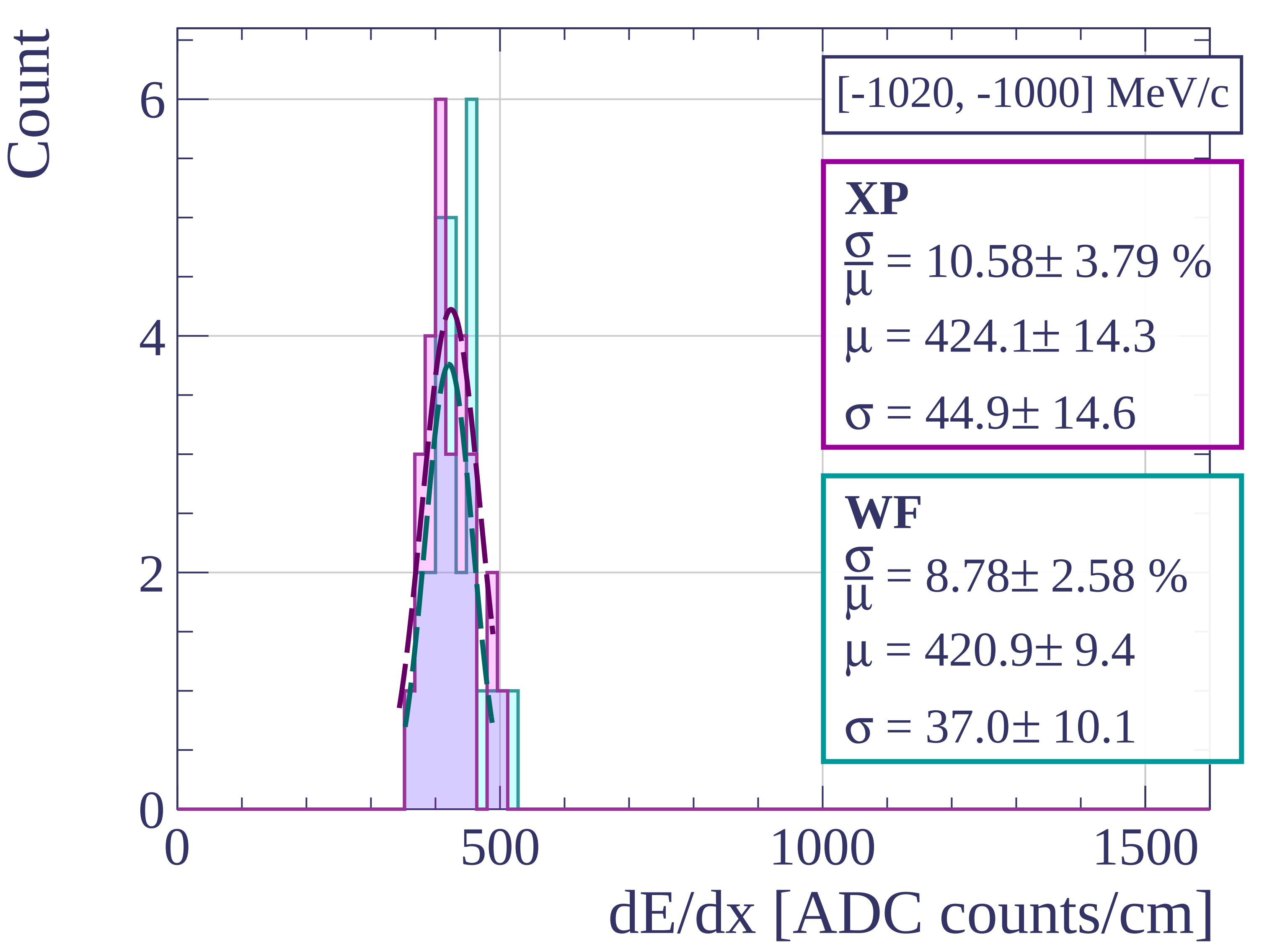
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1100, -1080] MeV/c
XP
σ/μ = 6.85± 2.35 %
μ = 426.9± 8.7
σ = 29.3± 9.4
WF
σ/μ = 24.21± 40.74 %
μ = 391.0± 179.9
σ = 94.7± 115.7


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1080, -1060] MeV/c
XP
σ/μ = 10.20± 3.80 %
μ = 439.9± 12.8
σ = 44.9± 15.4
WF
σ/μ = 19.20± 10.05 %
μ = 427.0± 28.0
σ = 82.0± 37.5

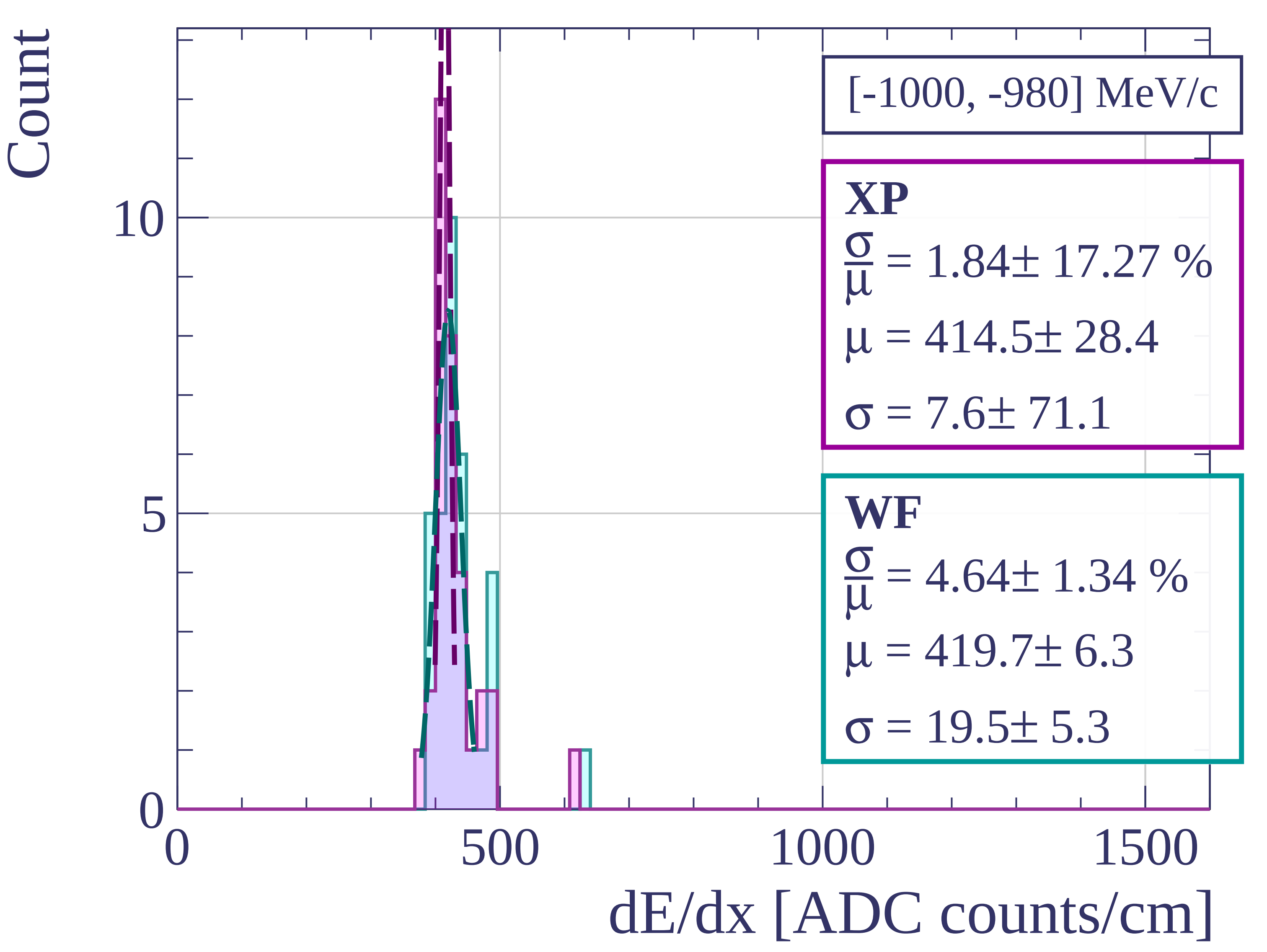


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1060, -1040] MeV/c
XP
σ/μ = 8.38± 3.10 %
μ = 450.4± 11.3
σ = 37.7± 13.0
WF
σ/μ = 6.29± 1.43 %
μ = 446.8± 6.2
σ = 28.1± 6.0

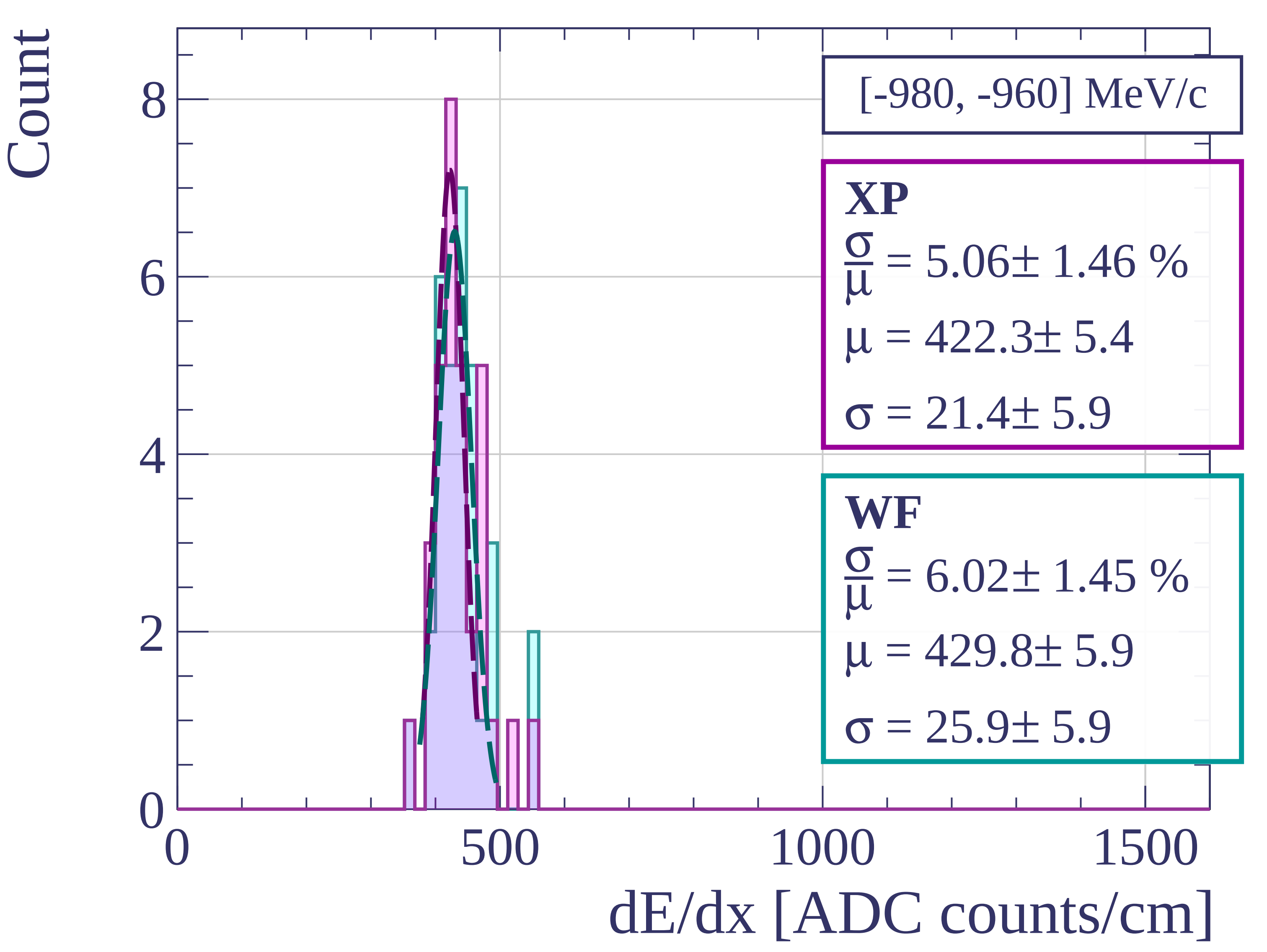

Count
dE/dx [ADC counts/cm]
[-1040, -1020] MeV/c
XP
$\frac{\sigma}{\mu} = 5.63 \pm 1.46$ %
$\mu = 428.5 \pm 6.3$
$\sigma = 24.1 \pm 5.9$
WF
$\frac{\sigma}{\mu} = 9.18 \pm 2.40$ %
$\mu = 427.8 \pm 9.3$
$\sigma = 39.3 \pm 9.4$
0
2
4
6
0
500
1000
1500

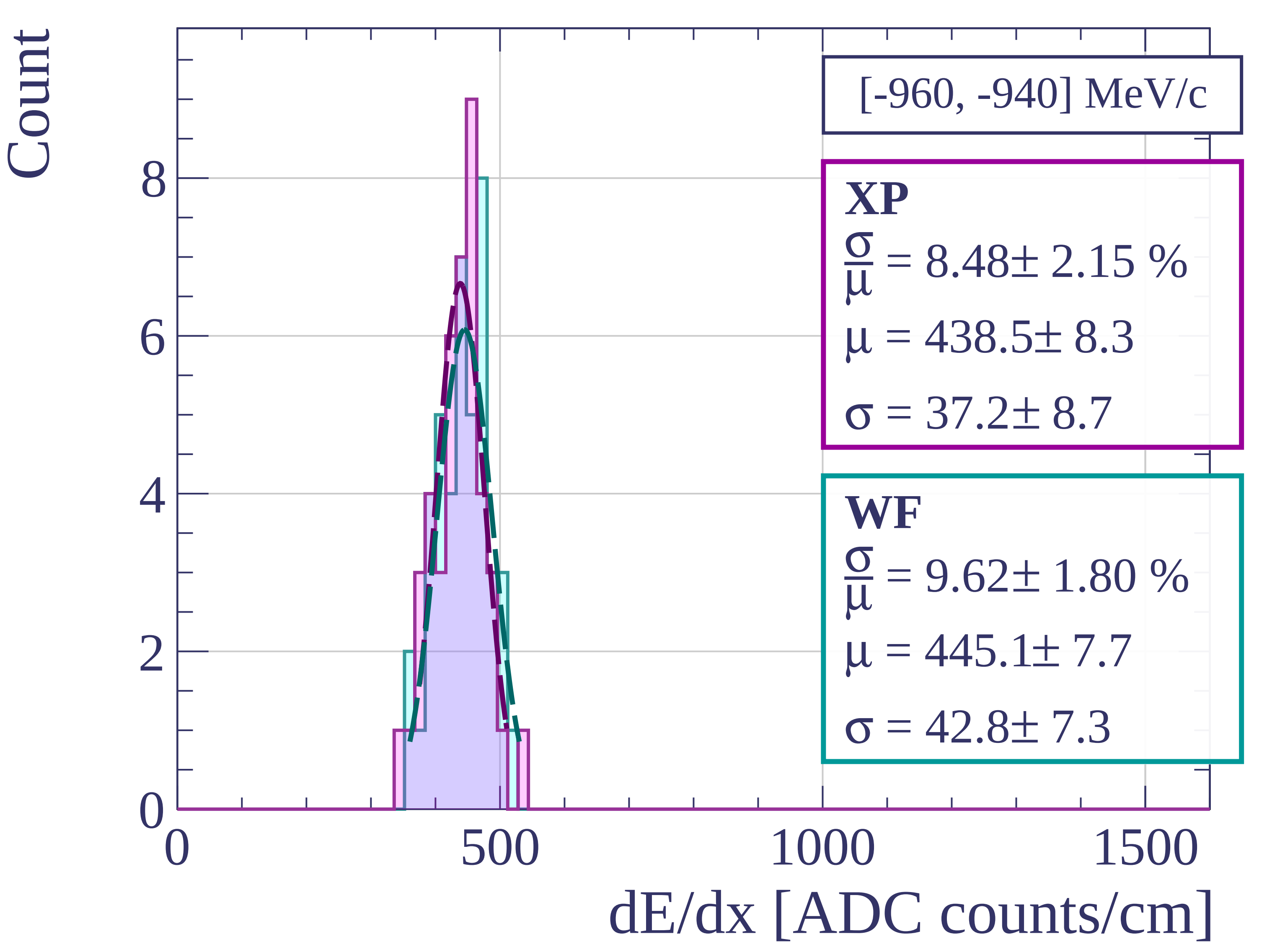


Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1020, -1000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.58± 3.79 %
μ = 424.1± 14.3
σ = 44.9± 14.6
WF
$\frac{\sigma}{\mu}$ = 8.78± 2.58 %
μ = 420.9± 9.4
σ = 37.0± 10.1

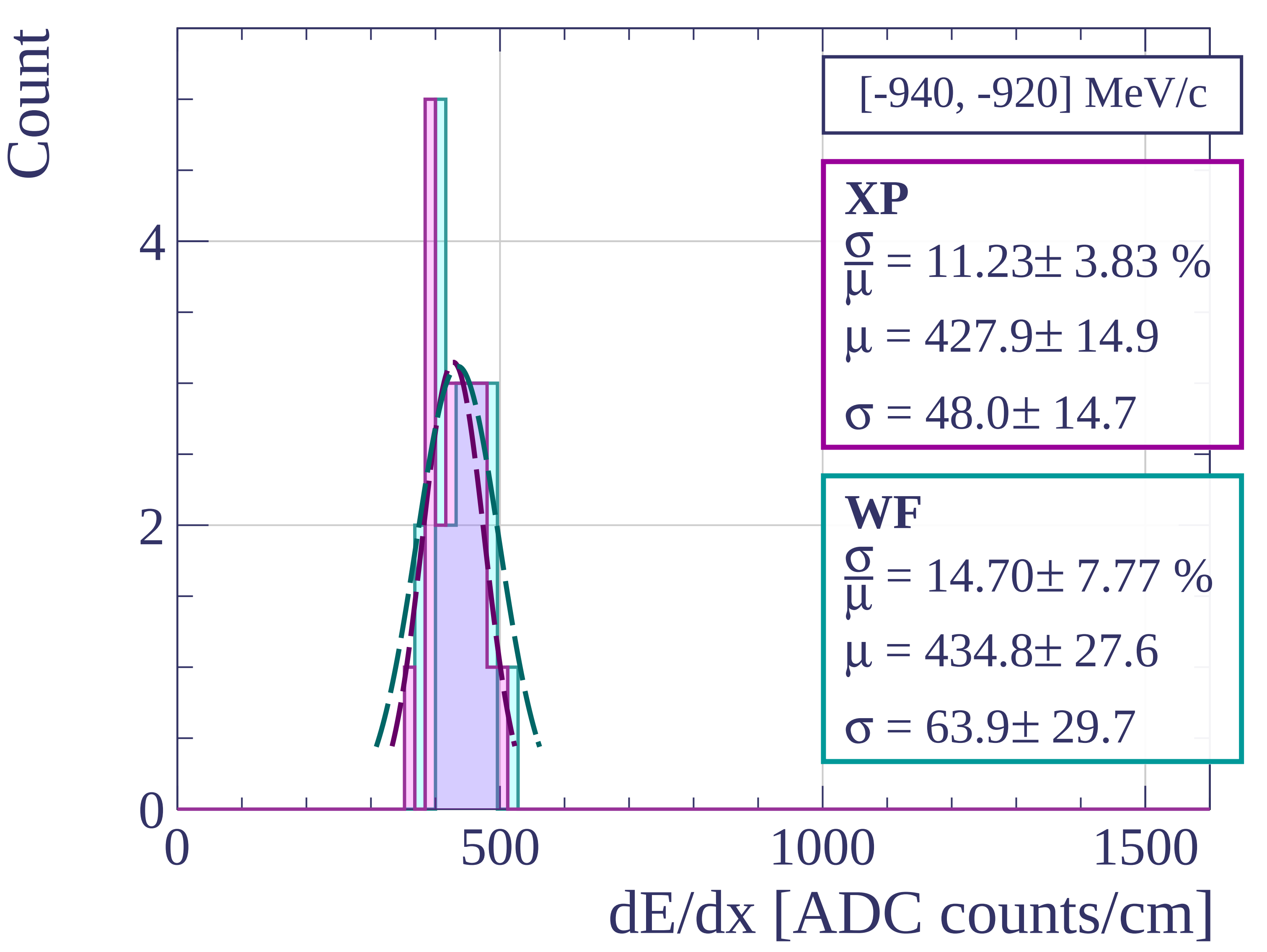


Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
[-1000, -980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 1.84± 17.27 %
μ = 414.5± 28.4
σ = 7.6± 71.1
WF
$\frac{\sigma}{\mu}$ = 4.64± 1.34 %
μ = 419.7± 6.3
σ = 19.5± 5.3

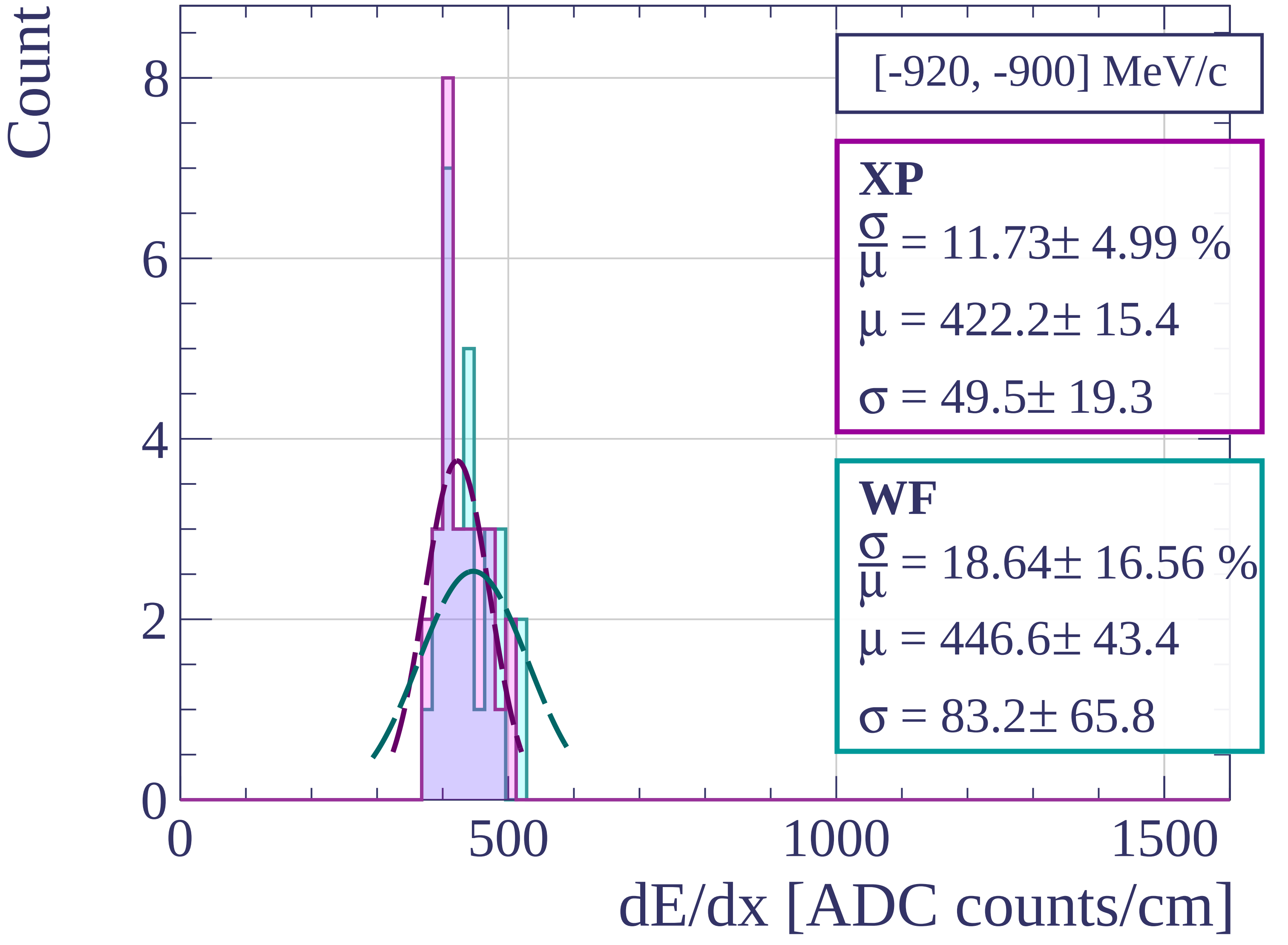

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-980, -960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.06± 1.46 %
μ = 422.3± 5.4
σ = 21.4± 5.9
WF
$\frac{\sigma}{\mu}$ = 6.02± 1.45 %
μ = 429.8± 5.9
σ = 25.9± 5.9

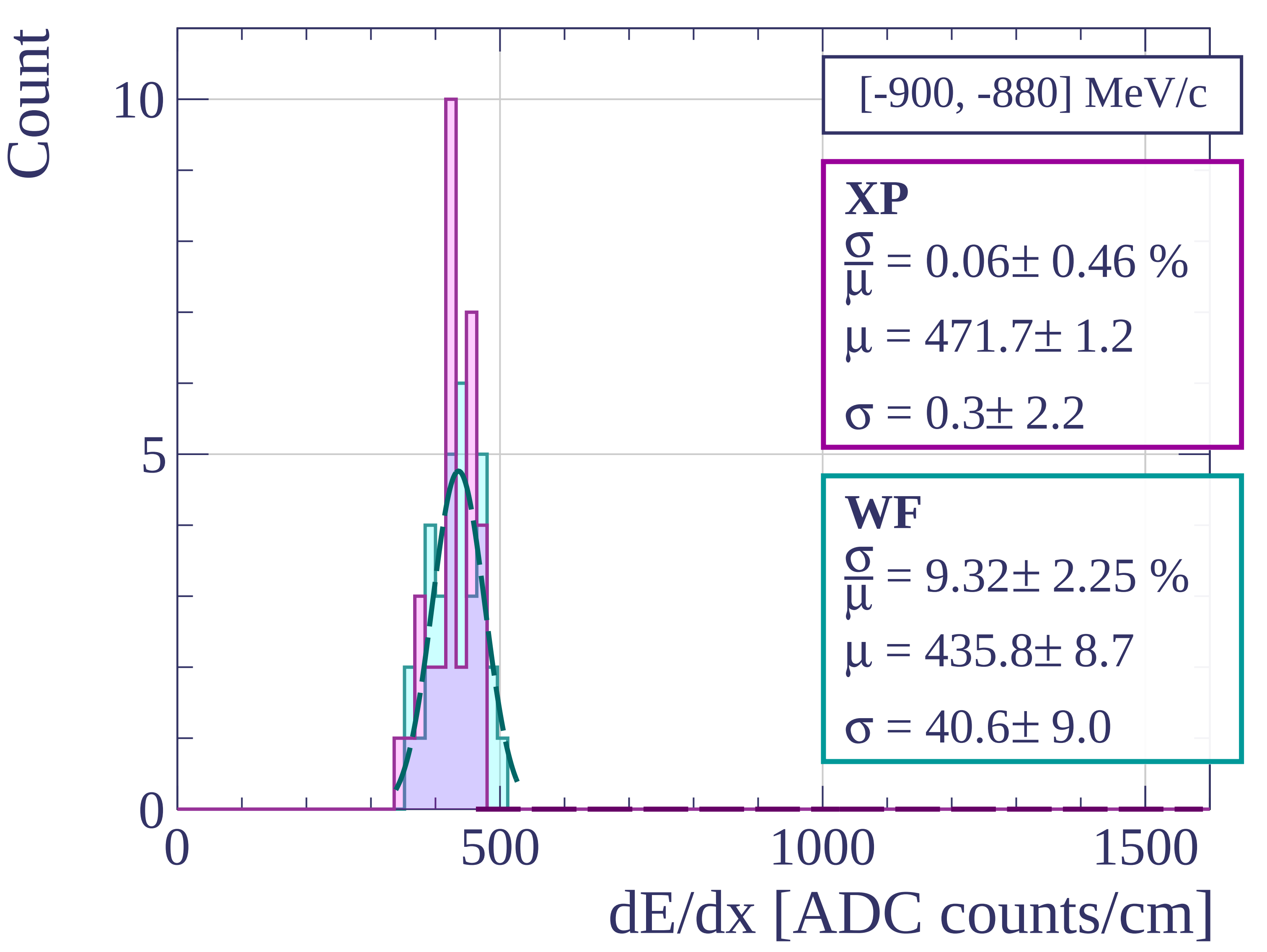


Count
dE/dx [ADC counts/cm]
[-960, -940] MeV/c
XP
σ/μ = 8.48± 2.15 %
μ = 438.5± 8.3
σ = 37.2± 8.7
WF
σ/μ = 9.62± 1.80 %
μ = 445.1± 7.7
σ = 42.8± 7.3

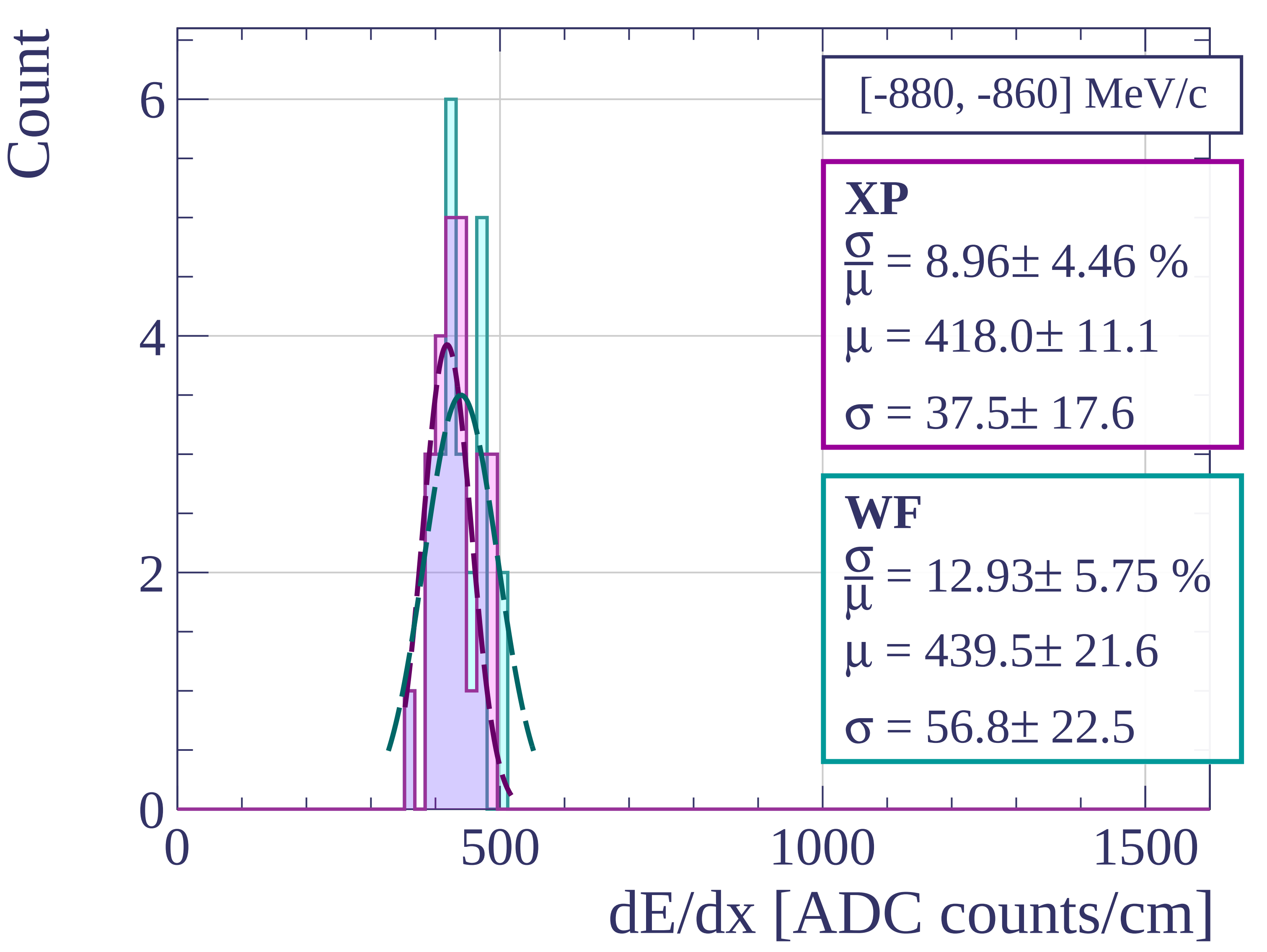


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-940, -920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.23± 3.83 %
μ = 427.9± 14.9
σ = 48.0± 14.7
WF
$\frac{\sigma}{\mu}$ = 14.70± 7.77 %
μ = 434.8± 27.6
σ = 63.9± 29.7

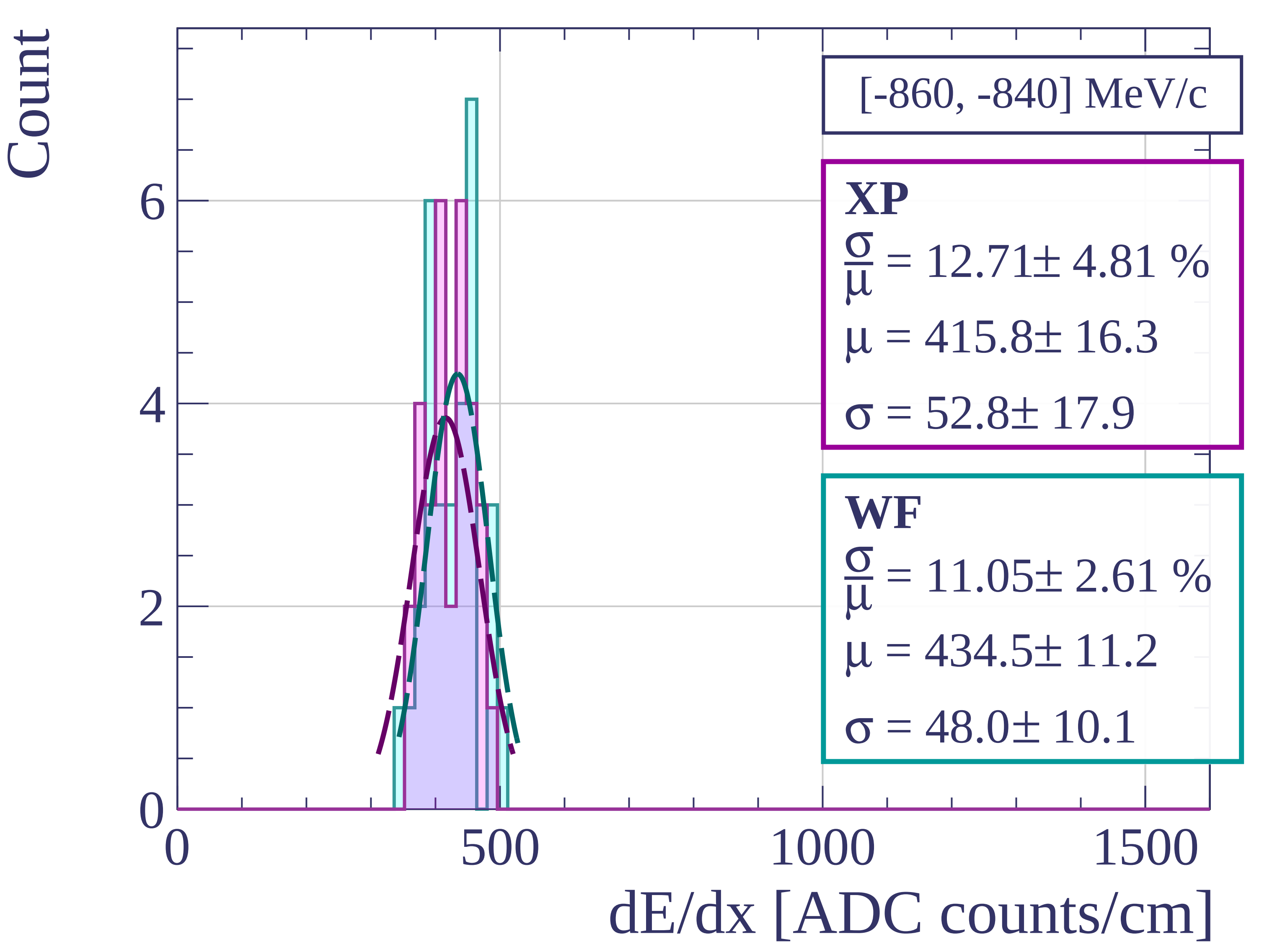

Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-920, -900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.73± 4.99 %
μ = 422.2± 15.4
σ = 49.5± 19.3
WF
$\frac{\sigma}{\mu}$ = 18.64± 16.56 %
μ = 446.6± 43.4
σ = 83.2± 65.8

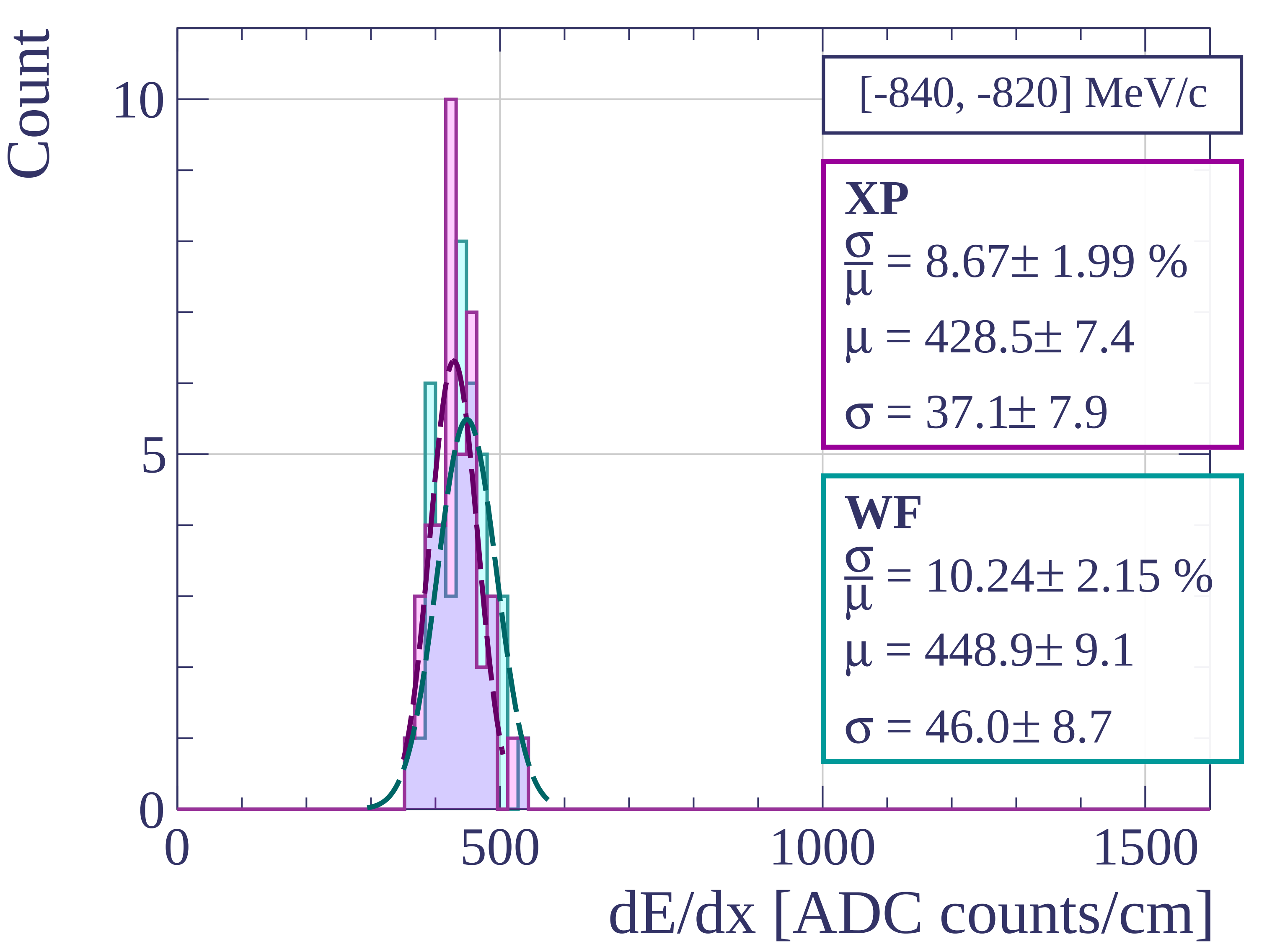


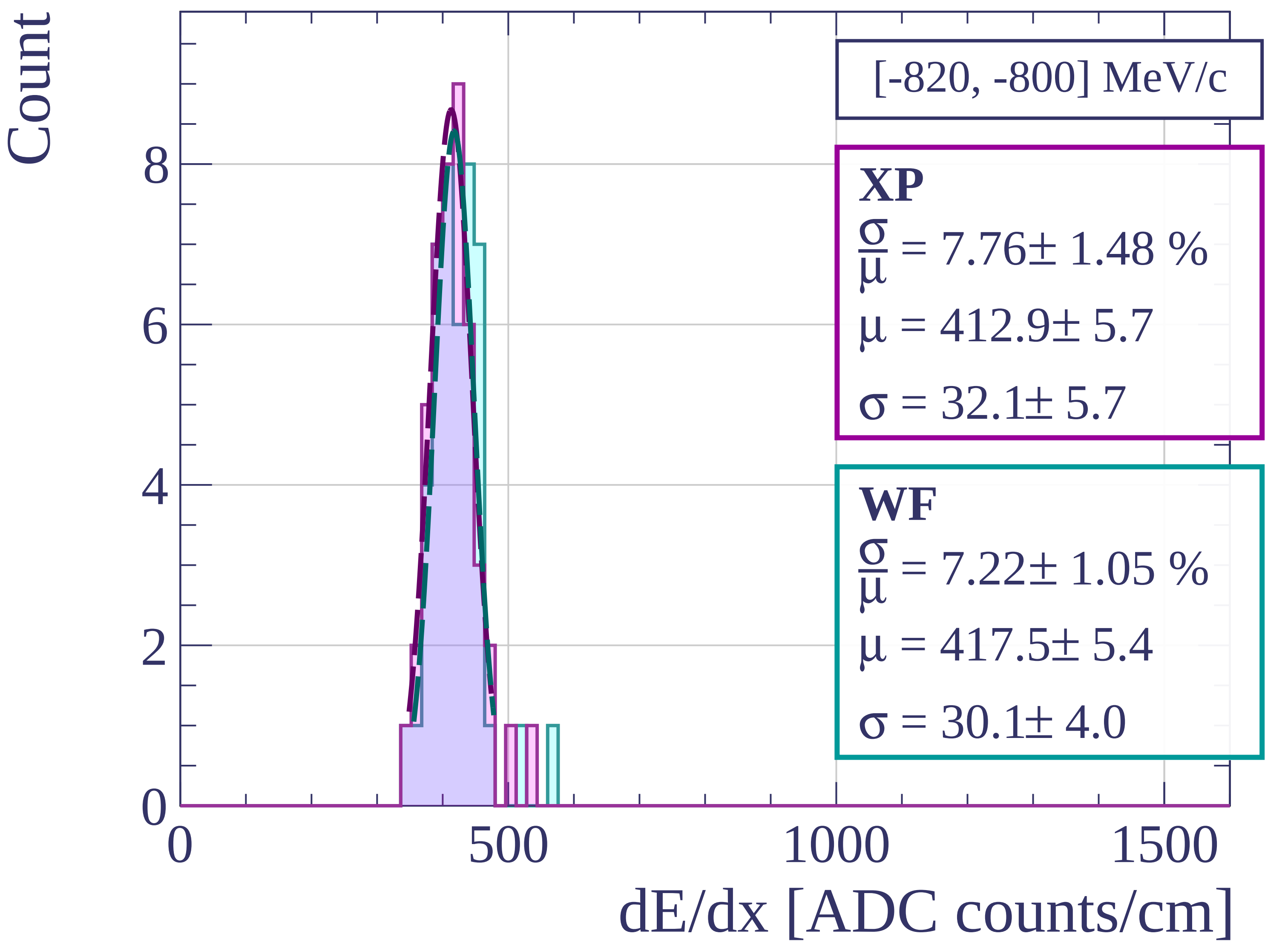
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-900, -880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 0.06± 0.46 %
μ = 471.7± 1.2
σ = 0.3± 2.2
WF
$\frac{\sigma}{\mu}$ = 9.32± 2.25 %
μ = 435.8± 8.7
σ = 40.6± 9.0



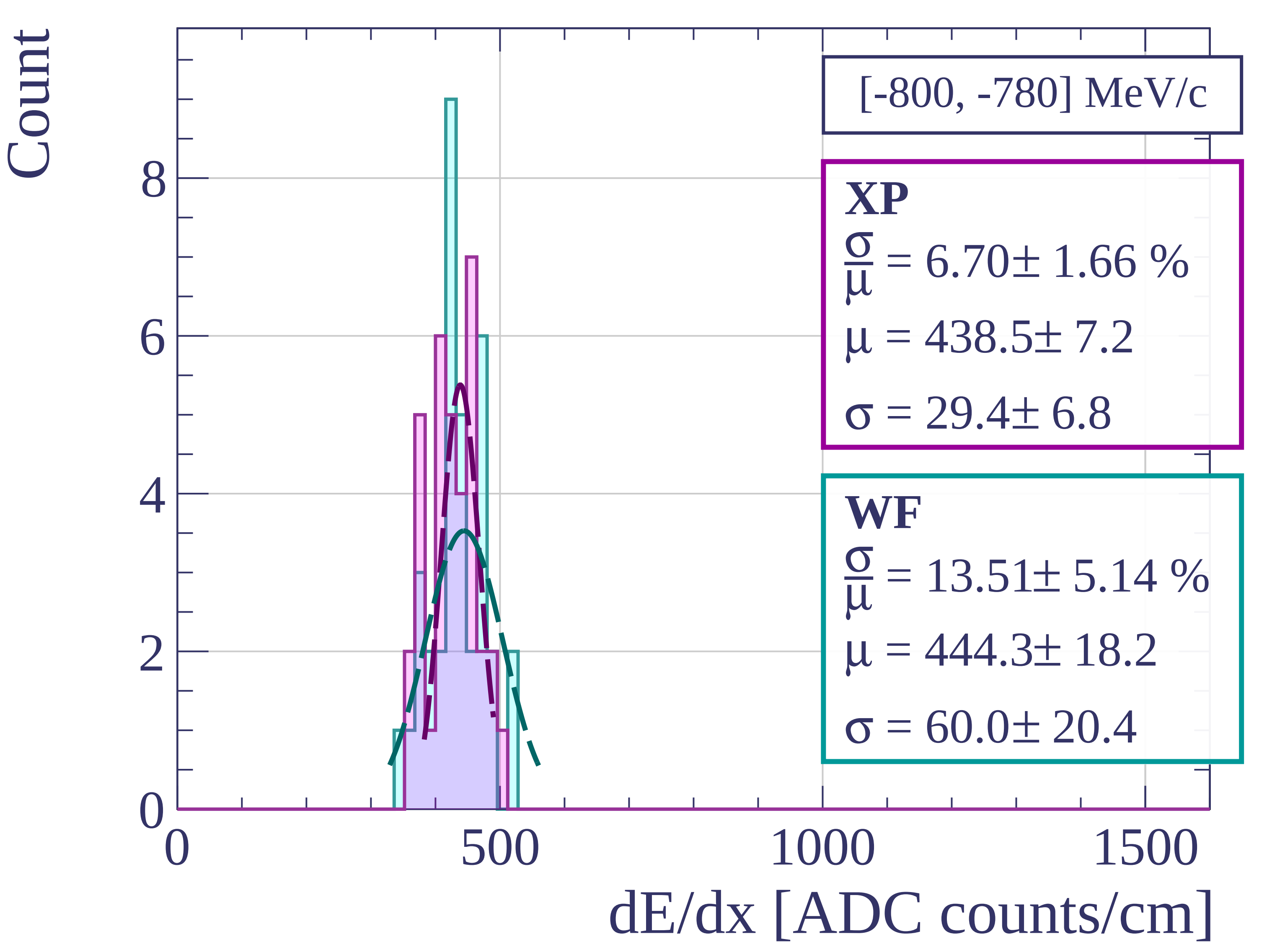
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-880, -860] MeV/c
XP
σ/μ = 8.96± 4.46 %
μ = 418.0± 11.1
σ = 37.5± 17.6
WF
σ/μ = 12.93± 5.75 %
μ = 439.5± 21.6
σ = 56.8± 22.5

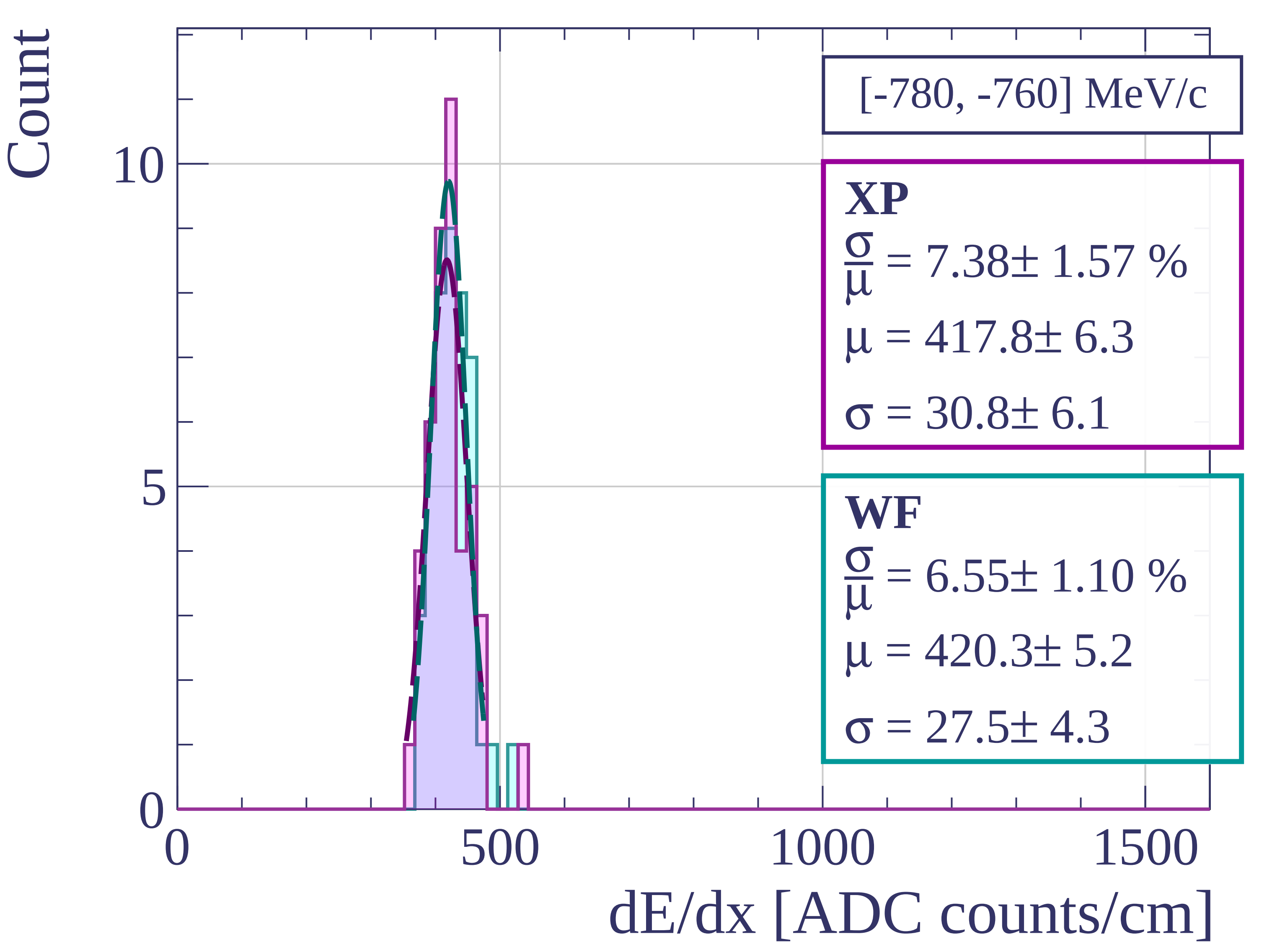

Count
[-860, -840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.71± 4.81 %
μ = 415.8± 16.3
σ = 52.8± 17.9
WF
$\frac{\sigma}{\mu}$ = 11.05± 2.61 %
μ = 434.5± 11.2
σ = 48.0± 10.1
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]

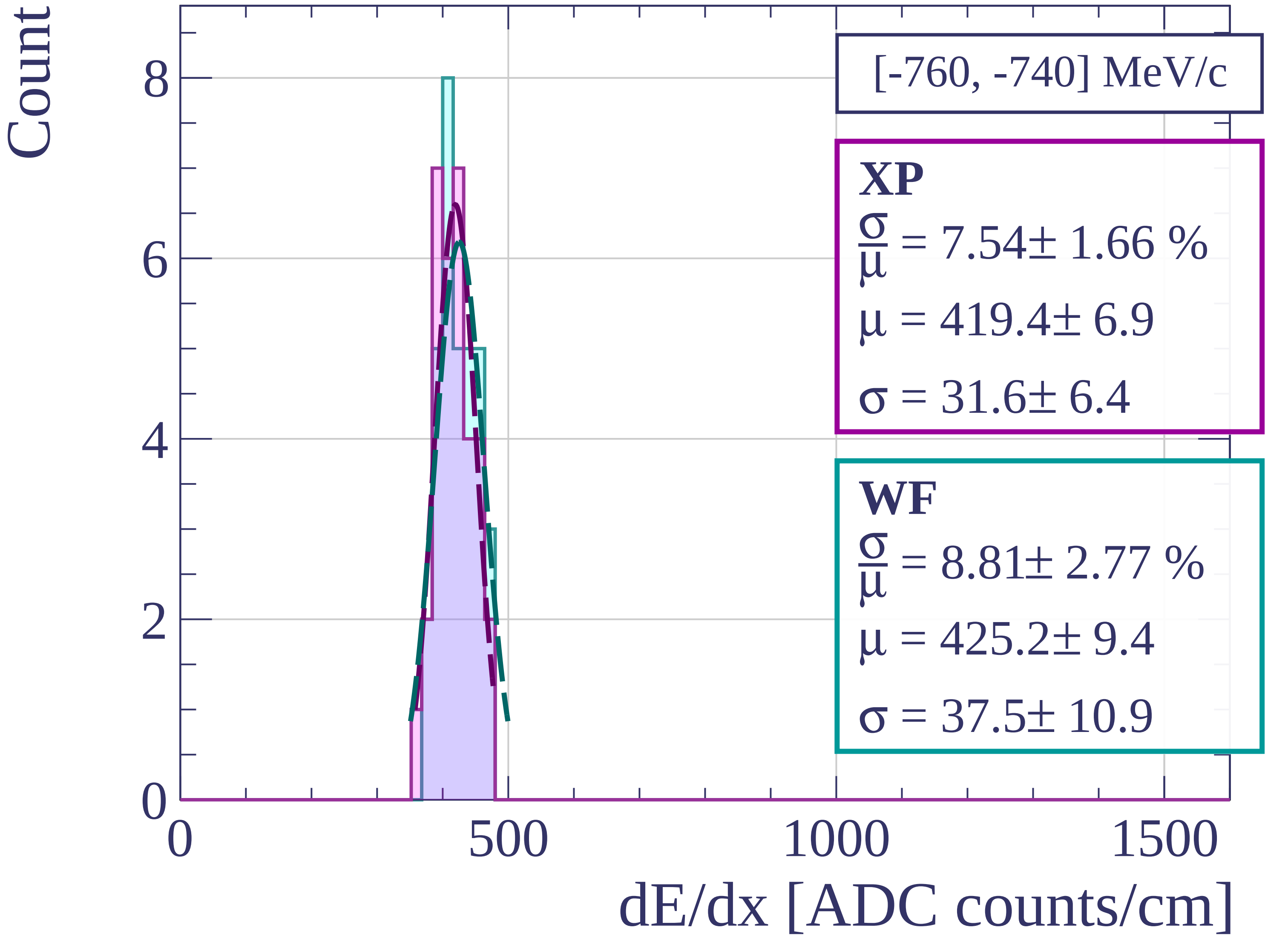

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-840, -820] MeV/c
XP
σ/μ = 8.67± 1.99 %
μ = 428.5± 7.4
σ = 37.1± 7.9
WF
σ/μ = 10.24± 2.15 %
μ = 448.9± 9.1
σ = 46.0± 8.7



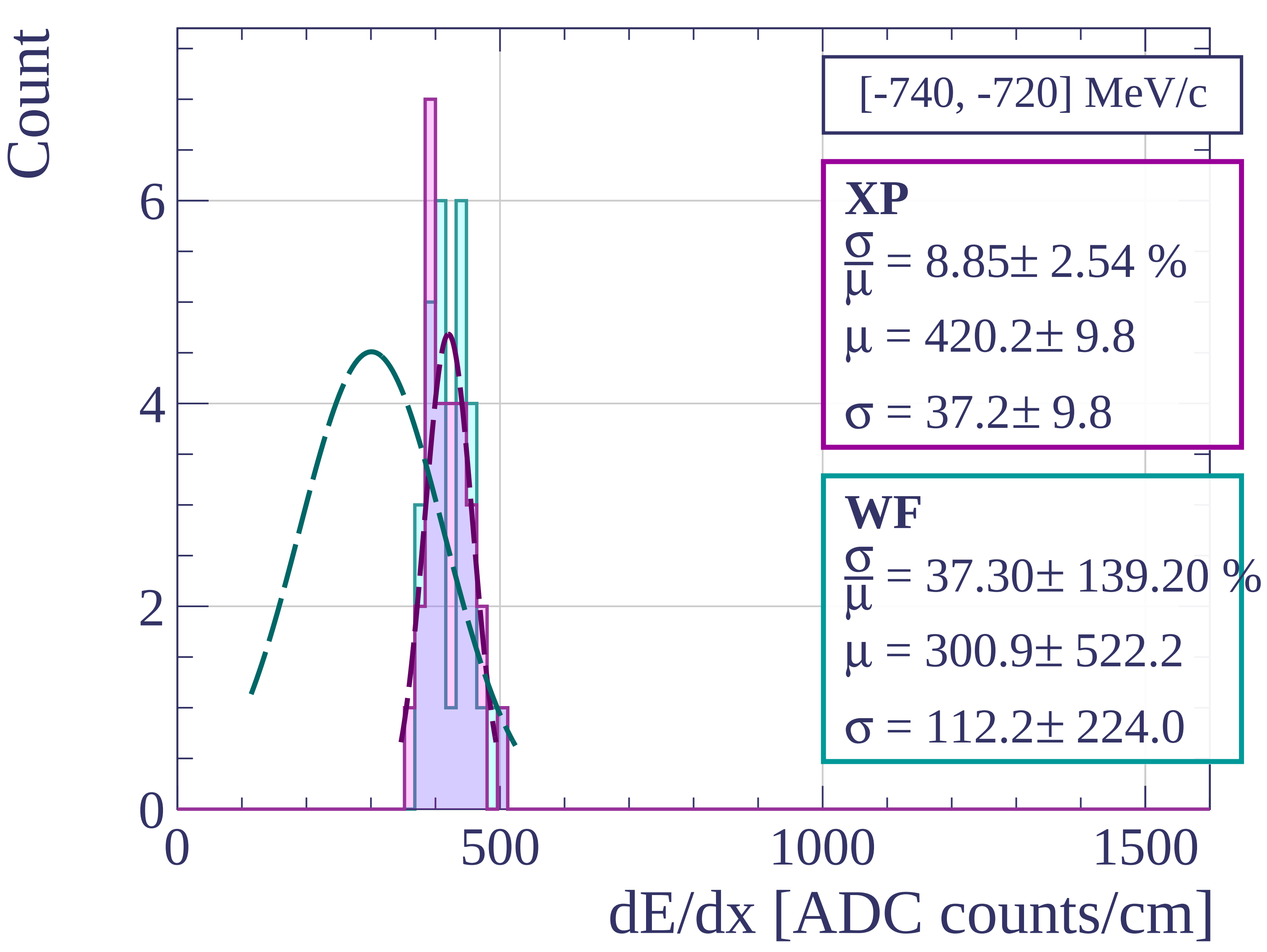
Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[-820, -800] MeV/c
XP
σ/μ = 7.76± 1.48 %
μ = 412.9± 5.7
σ = 32.1± 5.7
WF
σ/μ = 7.22± 1.05 %
μ = 417.5± 5.4
σ = 30.1± 4.0


Count
[-800, -780] MeV/c
XP
σ/μ = 6.70± 1.66 %
μ = 438.5± 7.2
σ = 29.4± 6.8
WF
σ/μ = 13.51± 5.14 %
μ = 444.3± 18.2
σ = 60.0± 20.4
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]

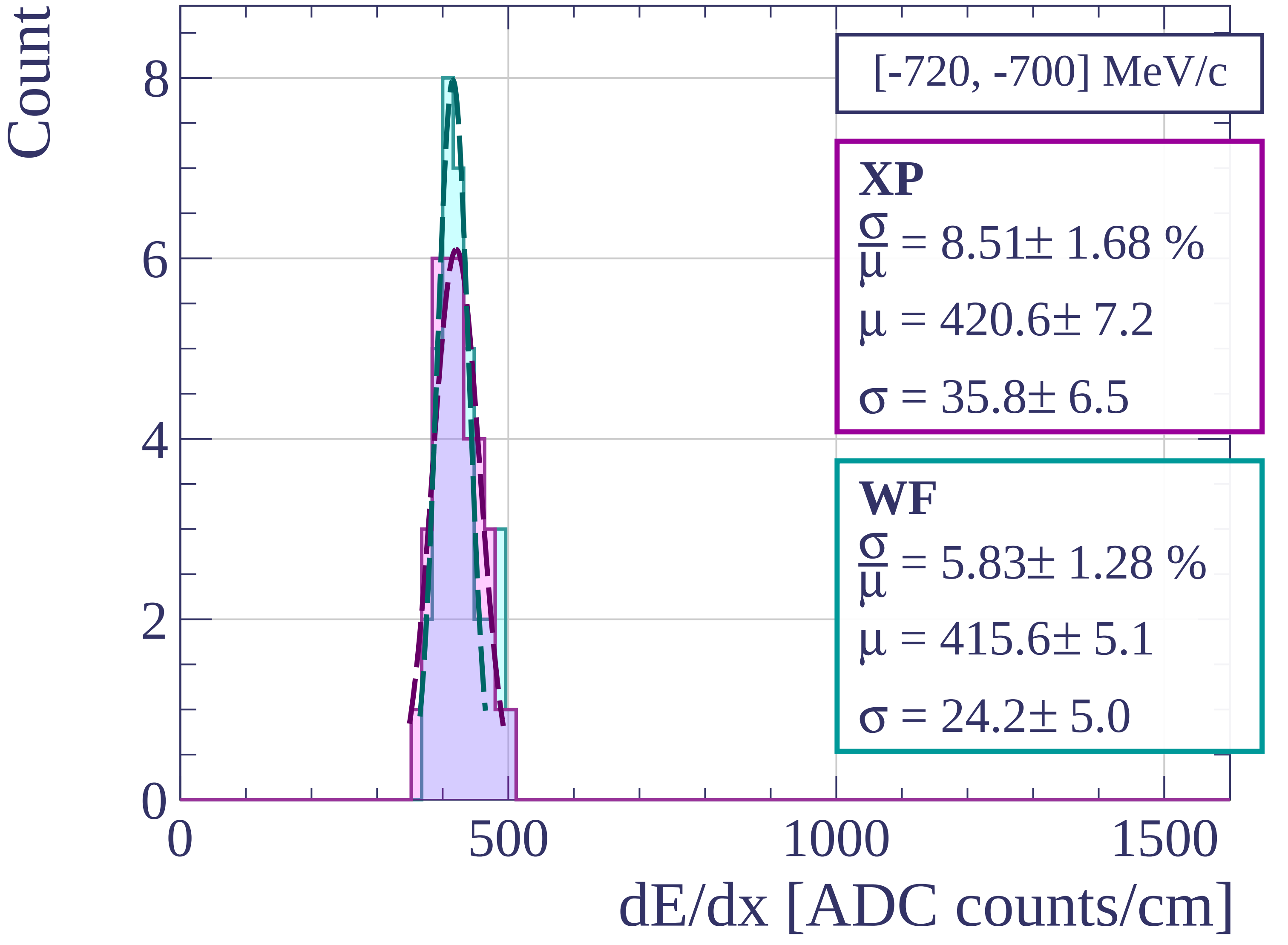

[-780, -760] MeV/c
XP
σ/μ = 7.38± 1.57 %
μ = 417.8± 6.3
σ = 30.8± 6.1
WF
σ/μ = 6.55± 1.10 %
μ = 420.3± 5.2
σ = 27.5± 4.3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]



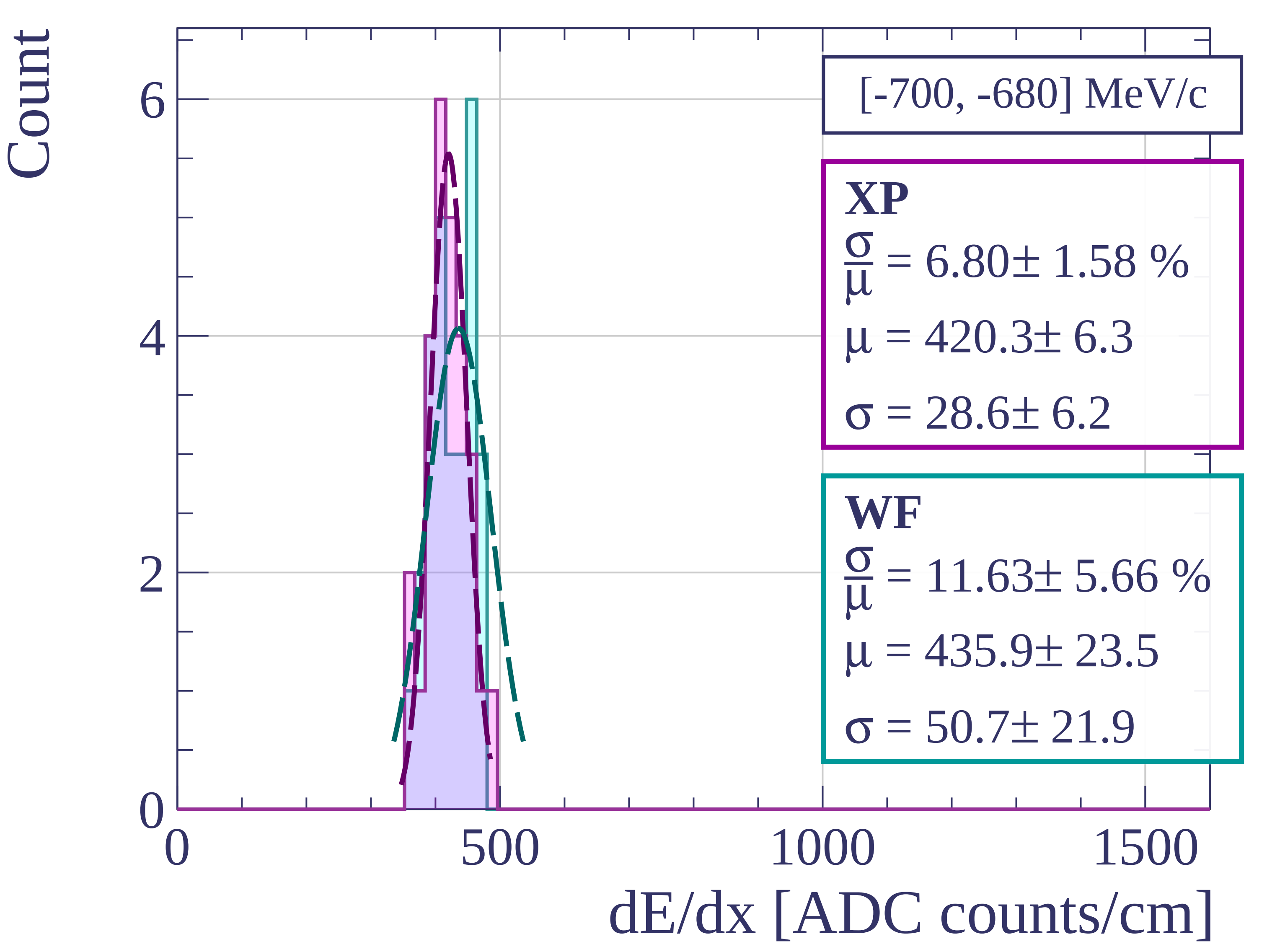
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-760, -740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.54± 1.66 %
μ = 419.4± 6.9
σ = 31.6± 6.4
WF
$\frac{\sigma}{\mu}$ = 8.81± 2.77 %
μ = 425.2± 9.4
σ = 37.5± 10.9



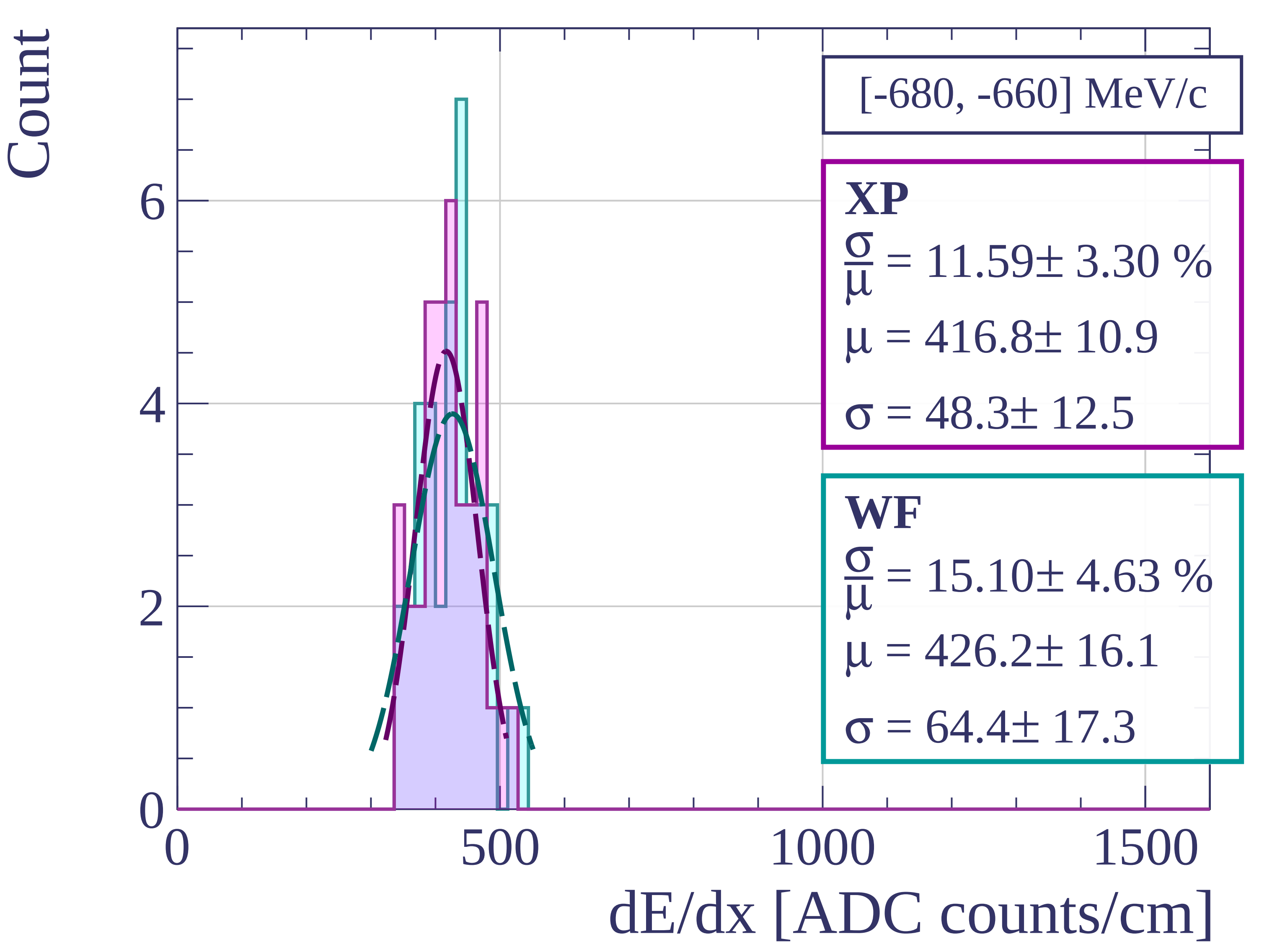
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-740, -720] MeV/c
XP
σ/μ = 8.85± 2.54 %
μ = 420.2± 9.8
σ = 37.2± 9.8
WF
σ/μ = 37.30± 139.20 %
μ = 300.9± 522.2
σ = 112.2± 224.0



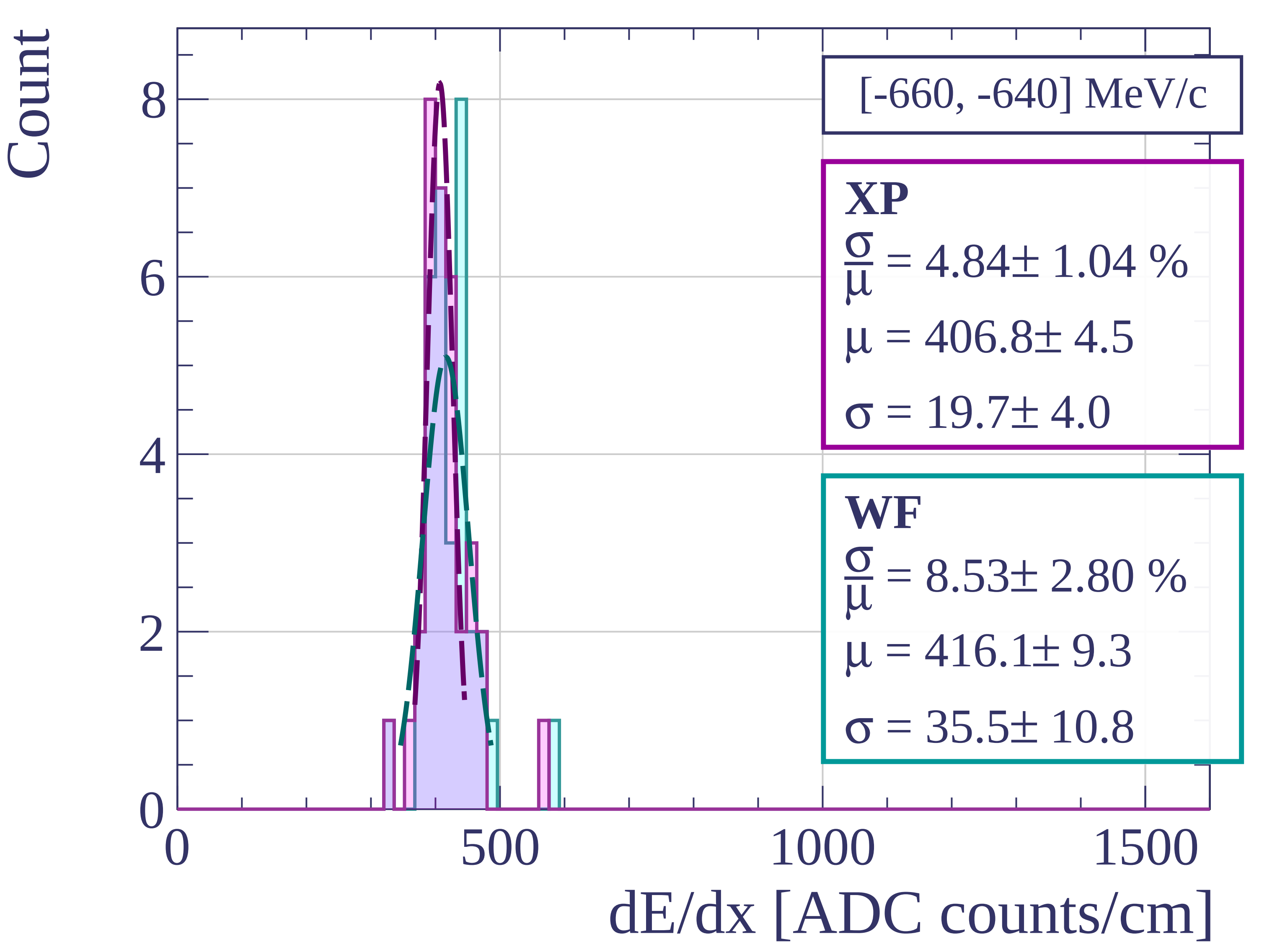
Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[-720, -700] MeV/c
XP
σ/μ = 8.51± 1.68 %
μ = 420.6± 7.2
σ = 35.8± 6.5
WF
σ/μ = 5.83± 1.28 %
μ = 415.6± 5.1
σ = 24.2± 5.0



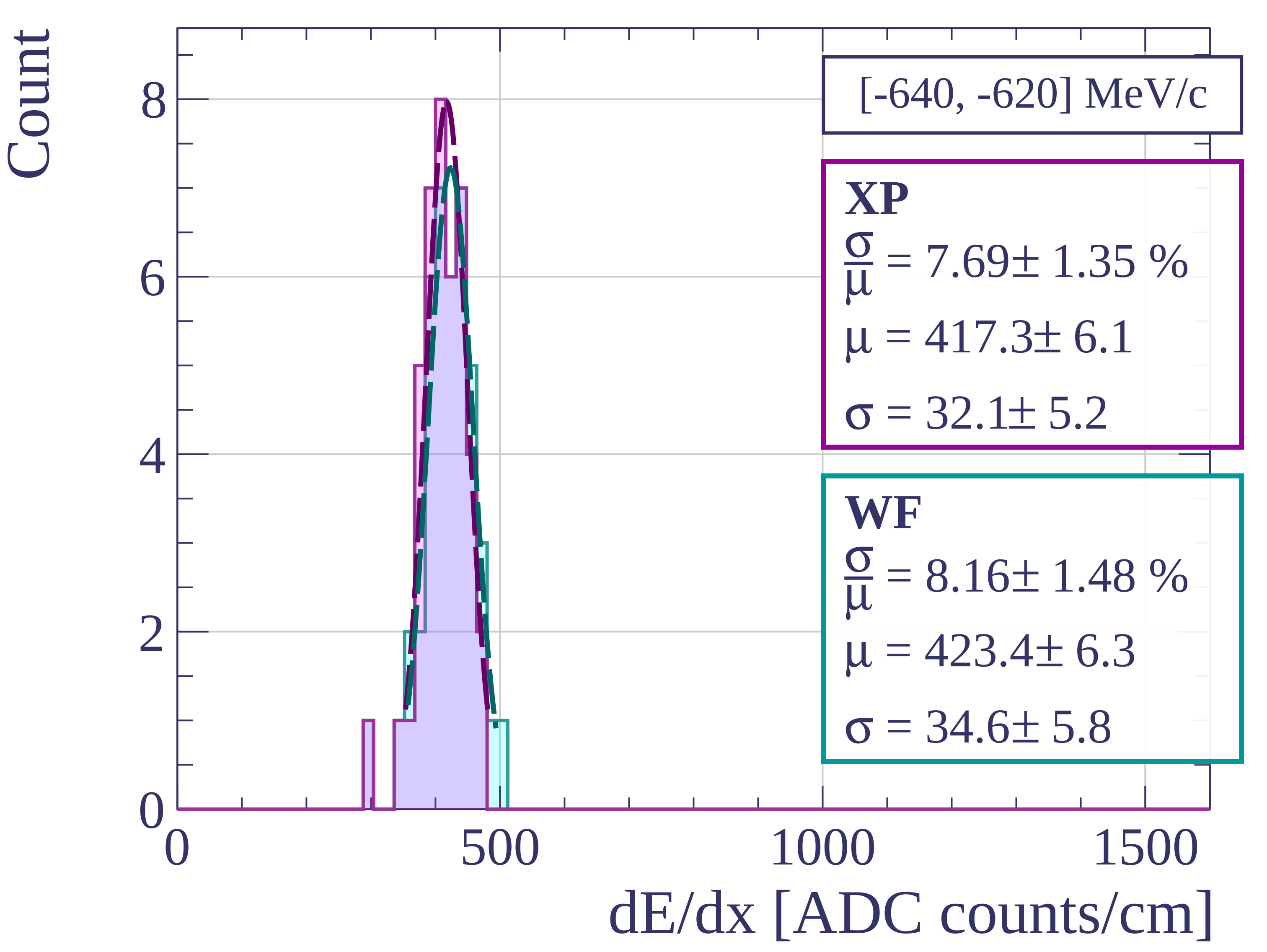
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-700, -680] MeV/c
XP
σ/μ = 6.80± 1.58 %
μ = 420.3± 6.3
σ = 28.6± 6.2
WF
σ/μ = 11.63± 5.66 %
μ = 435.9± 23.5
σ = 50.7± 21.9



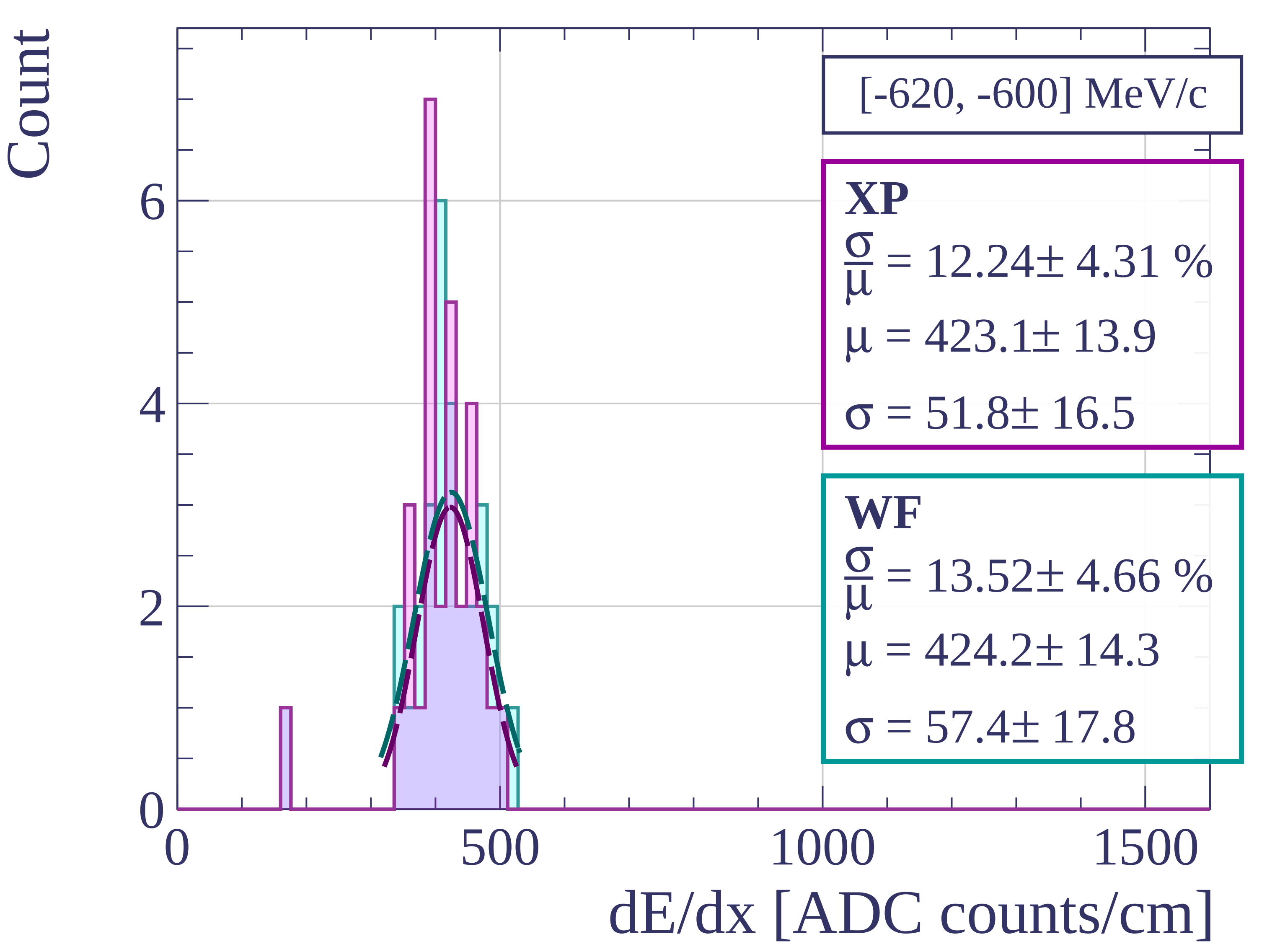
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[-680, -660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.59± 3.30 %
μ = 416.8± 10.9
σ = 48.3± 12.5
WF
$\frac{\sigma}{\mu}$ = 15.10± 4.63 %
μ = 426.2± 16.1
σ = 64.4± 17.3



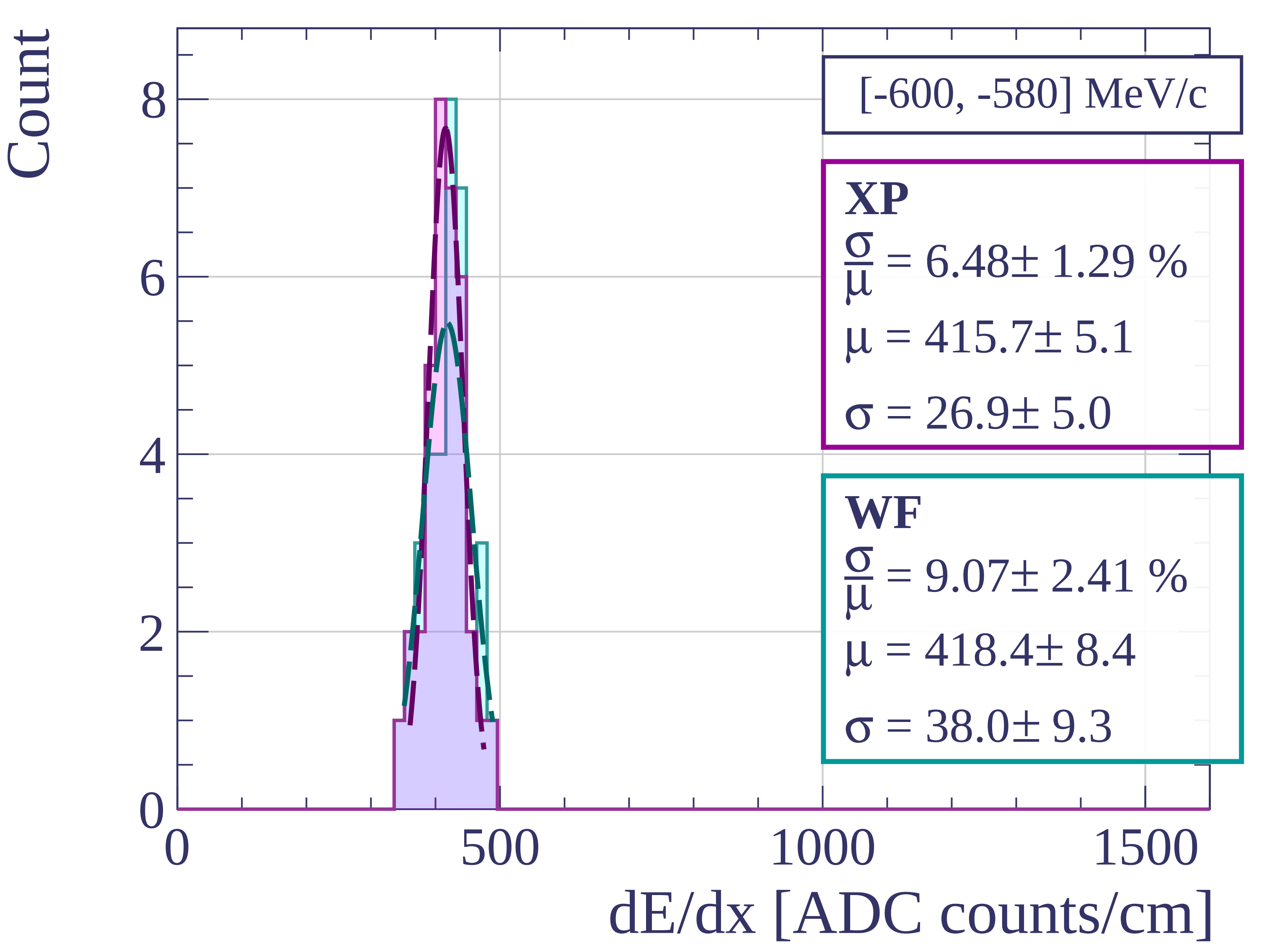
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-660, -640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.84± 1.04 %
μ = 406.8± 4.5
σ = 19.7± 4.0
WF
$\frac{\sigma}{\mu}$ = 8.53± 2.80 %
μ = 416.1± 9.3
σ = 35.5± 10.8



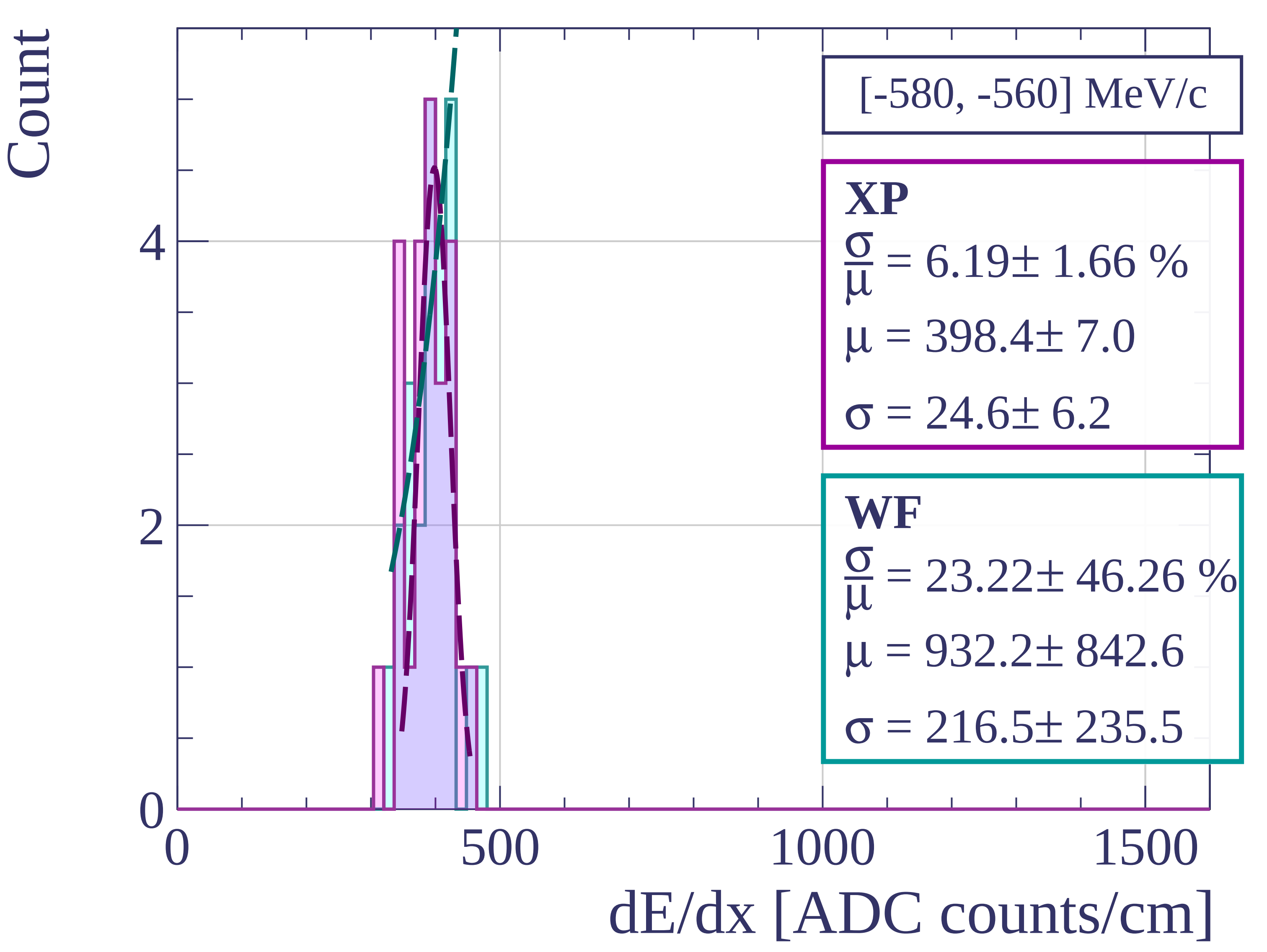
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-640, -620] MeV/c
XP
σ/μ = 7.69± 1.35 %
μ = 417.3± 6.1
σ = 32.1± 5.2
WF
σ/μ = 8.16± 1.48 %
μ = 423.4± 6.3
σ = 34.6± 5.8



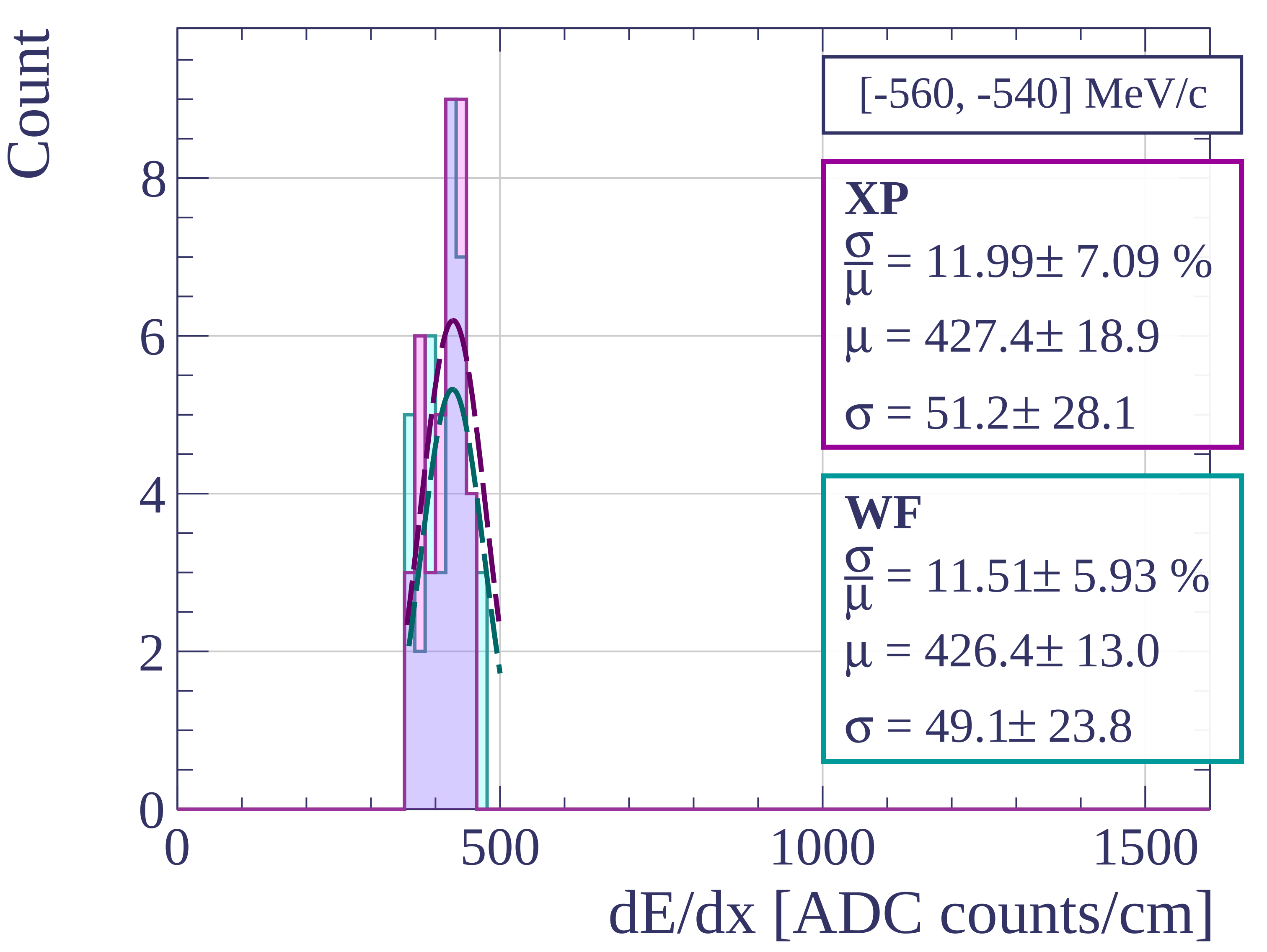
Count
dE/dx [ADC counts/cm]
[-620, -600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.24± 4.31 %
μ = 423.1± 13.9
σ = 51.8± 16.5
WF
$\frac{\sigma}{\mu}$ = 13.52± 4.66 %
μ = 424.2± 14.3
σ = 57.4± 17.8



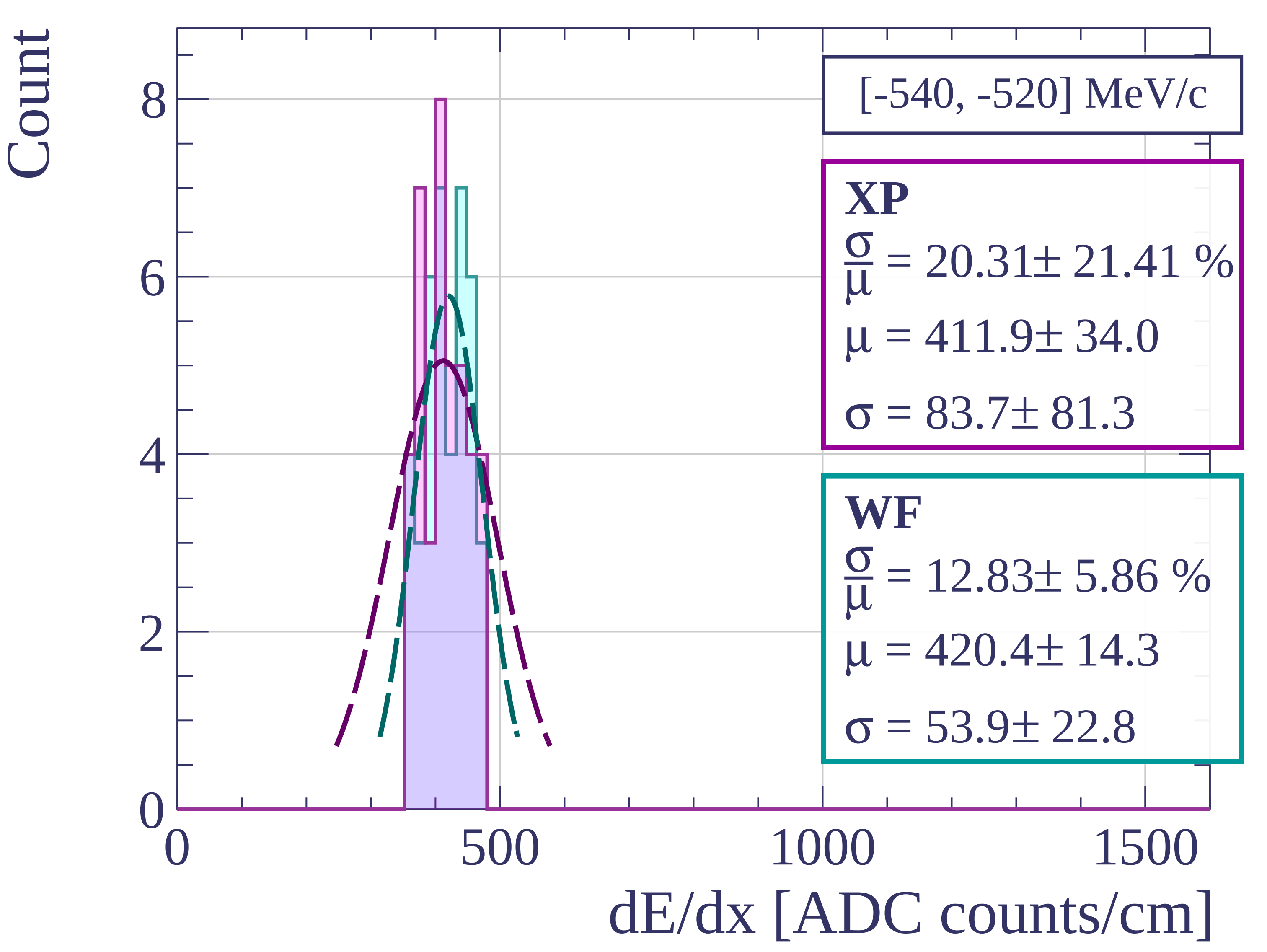
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-600, -580] MeV/c
XP
σ/μ = 6.48± 1.29 %
μ = 415.7± 5.1
σ = 26.9± 5.0
WF
σ/μ = 9.07± 2.41 %
μ = 418.4± 8.4
σ = 38.0± 9.3

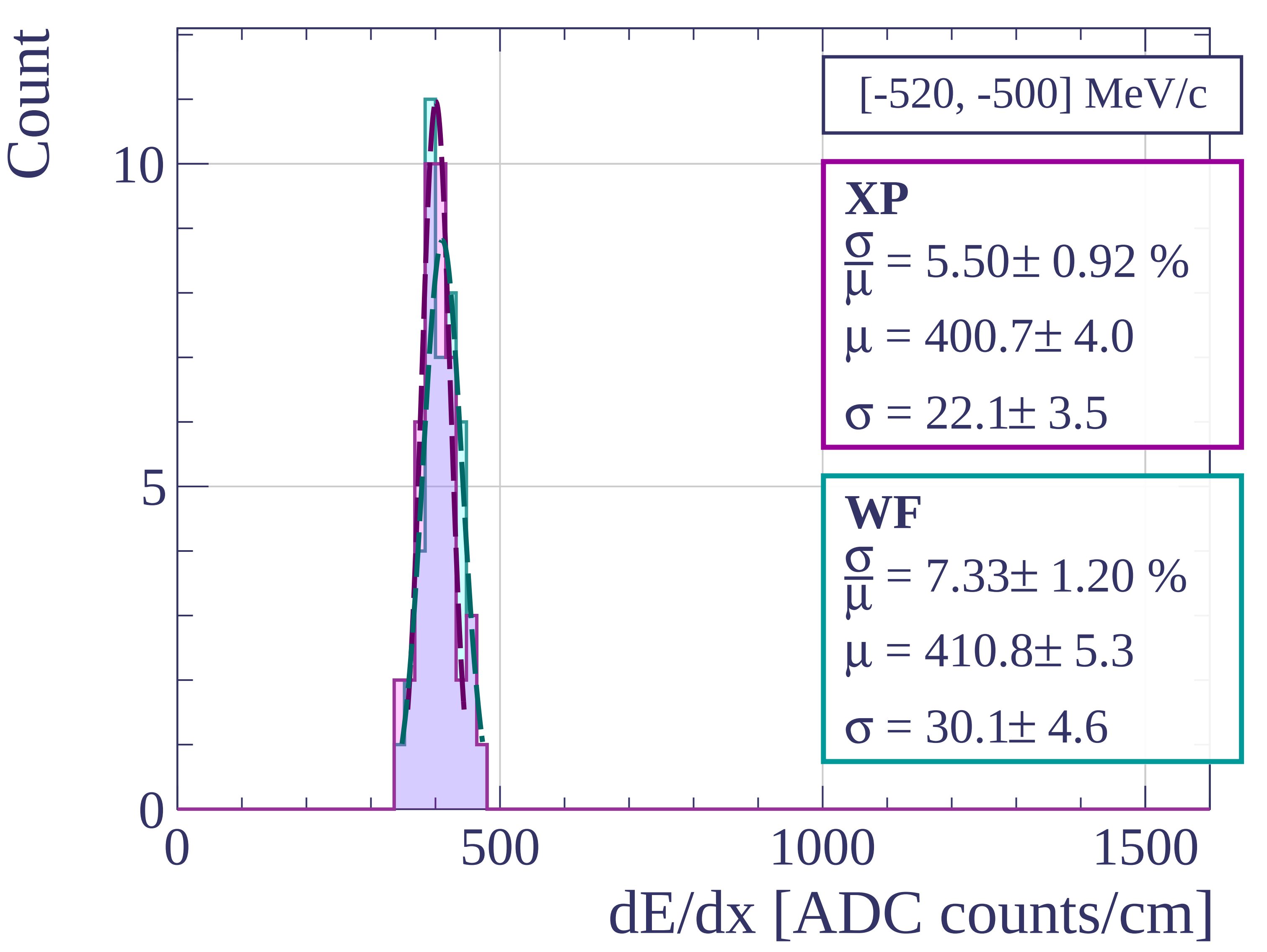


Count
[-580, -560] MeV/c
XP
σ/μ = 6.19± 1.66 %
μ = 398.4± 7.0
σ = 24.6± 6.2
WF
σ/μ = 23.22± 46.26 %
μ = 932.2± 842.6
σ = 216.5± 235.5
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]

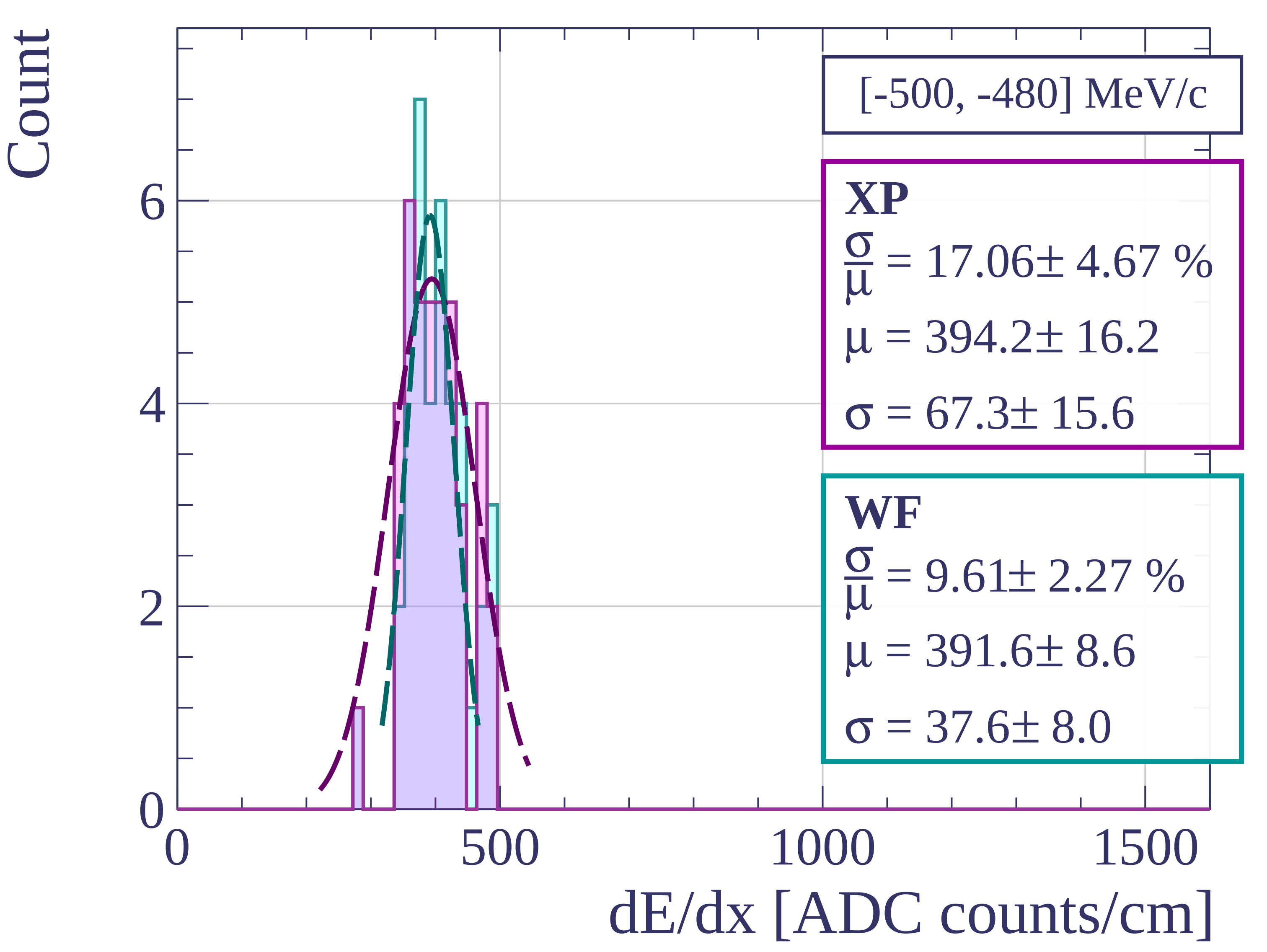


[-560, -540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.99± 7.09 %
μ = 427.4± 18.9
σ = 51.2± 28.1
WF
$\frac{\sigma}{\mu}$ = 11.51± 5.93 %
μ = 426.4± 13.0
σ = 49.1± 23.8
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
500
1000
1500

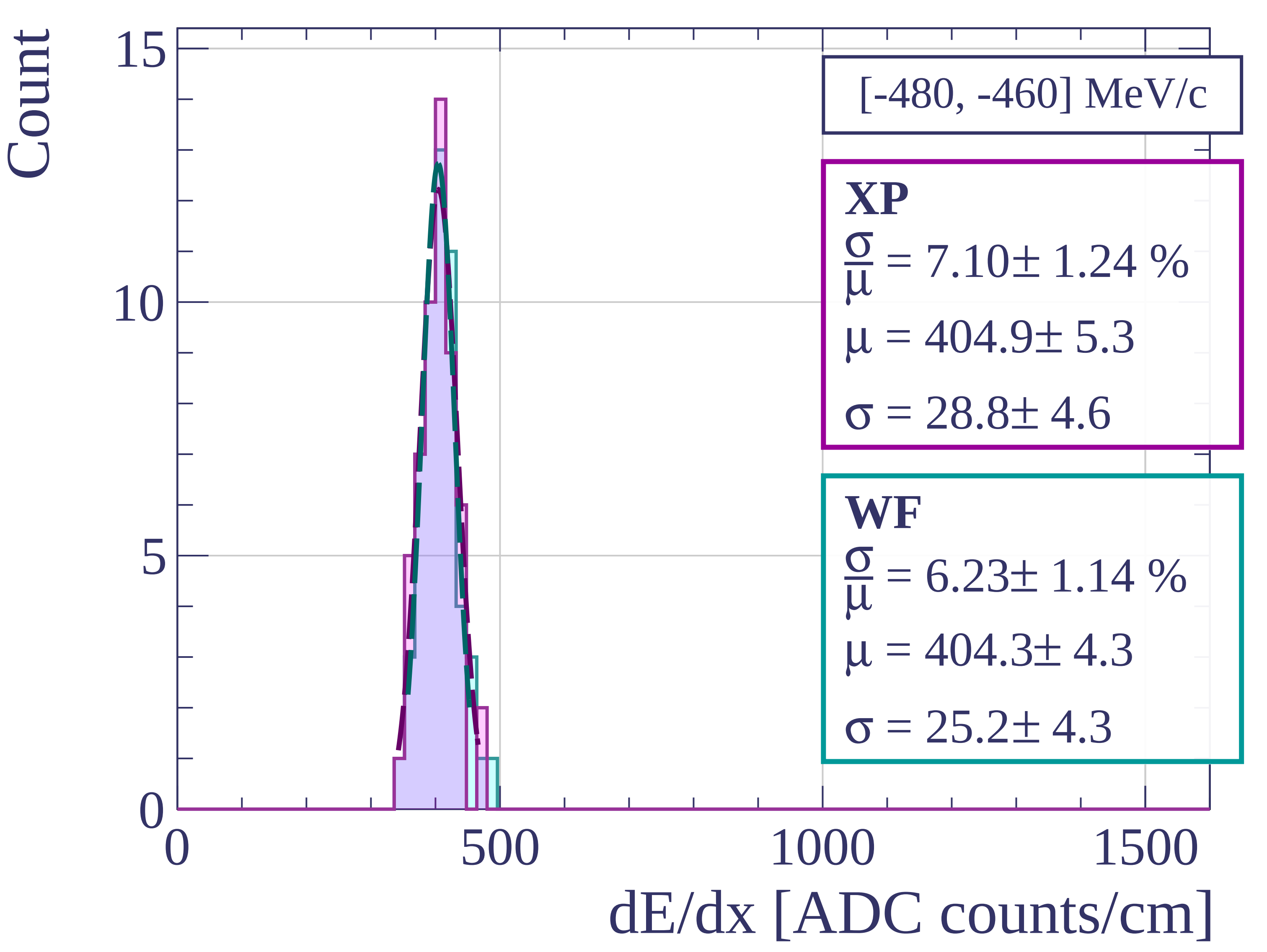

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-540, -520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 20.31± 21.41 %
μ = 411.9± 34.0
σ = 83.7± 81.3
WF
$\frac{\sigma}{\mu}$ = 12.83± 5.86 %
μ = 420.4± 14.3
σ = 53.9± 22.8



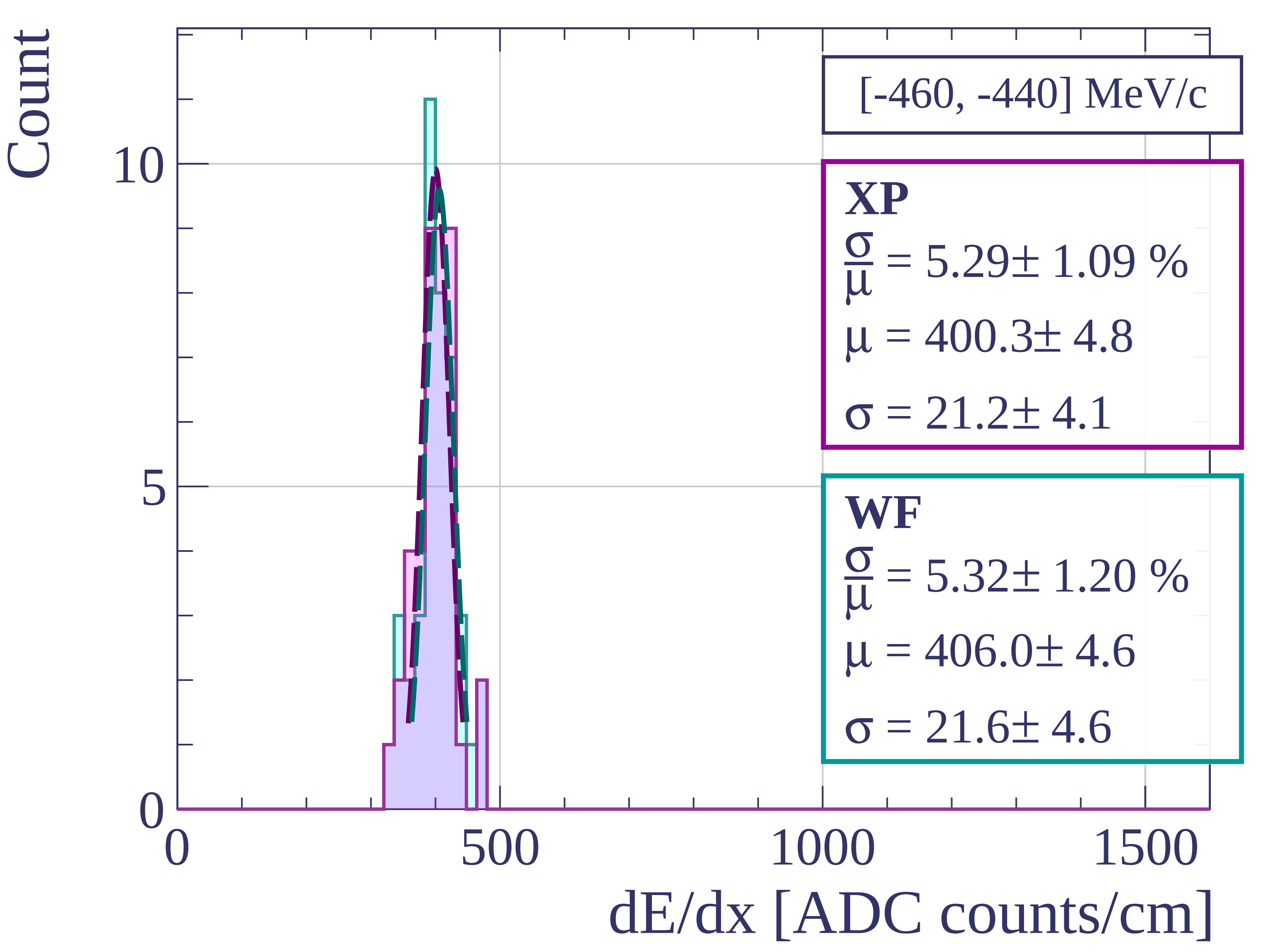
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-520, -500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.50± 0.92 %
μ = 400.7± 4.0
σ = 22.1± 3.5
WF
$\frac{\sigma}{\mu}$ = 7.33± 1.20 %
μ = 410.8± 5.3
σ = 30.1± 4.6



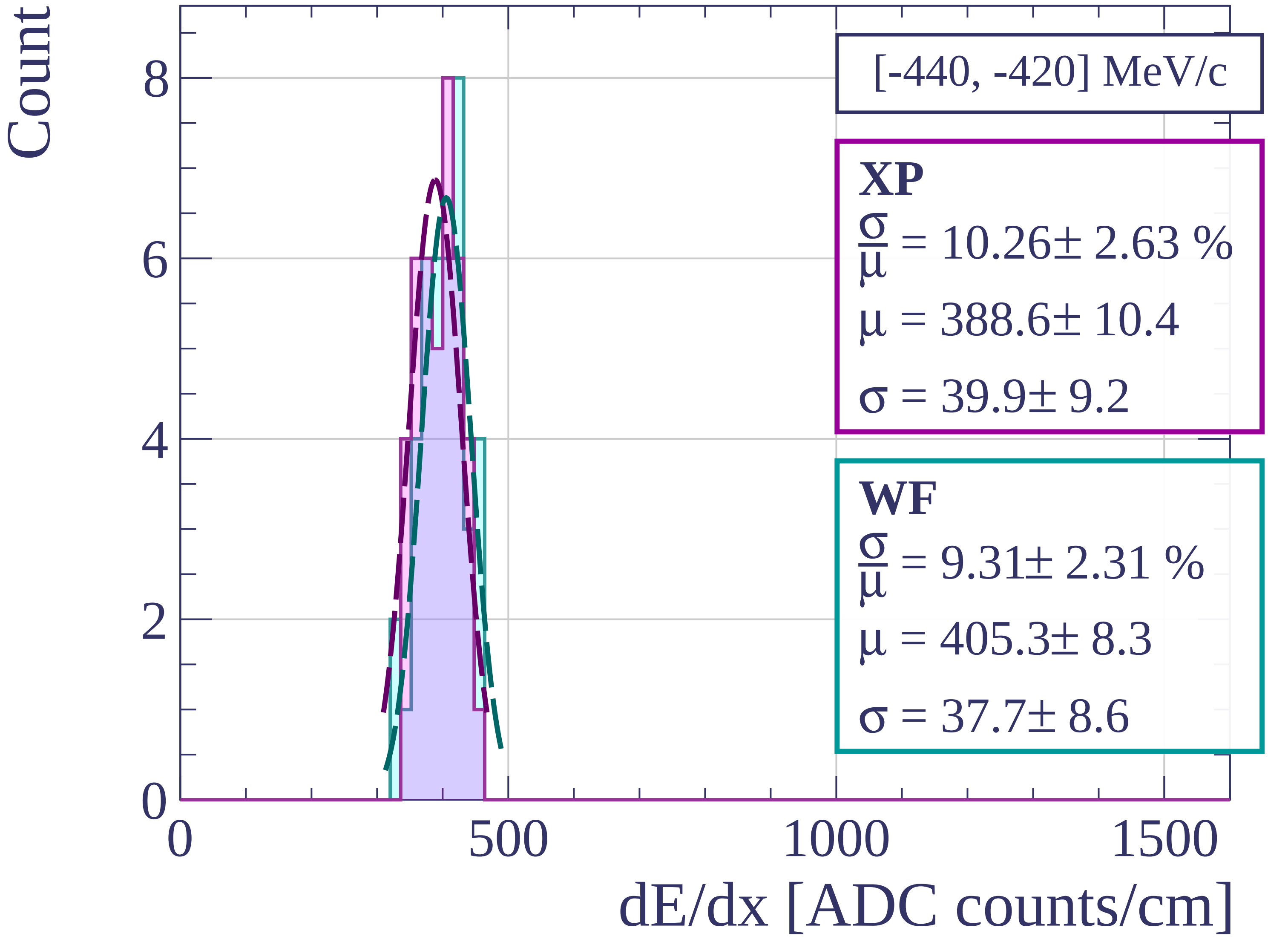
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-500, -480] MeV/c
XP
σ/μ = 17.06± 4.67 %
μ = 394.2± 16.2
σ = 67.3± 15.6
WF
σ/μ = 9.61± 2.27 %
μ = 391.6± 8.6
σ = 37.6± 8.0



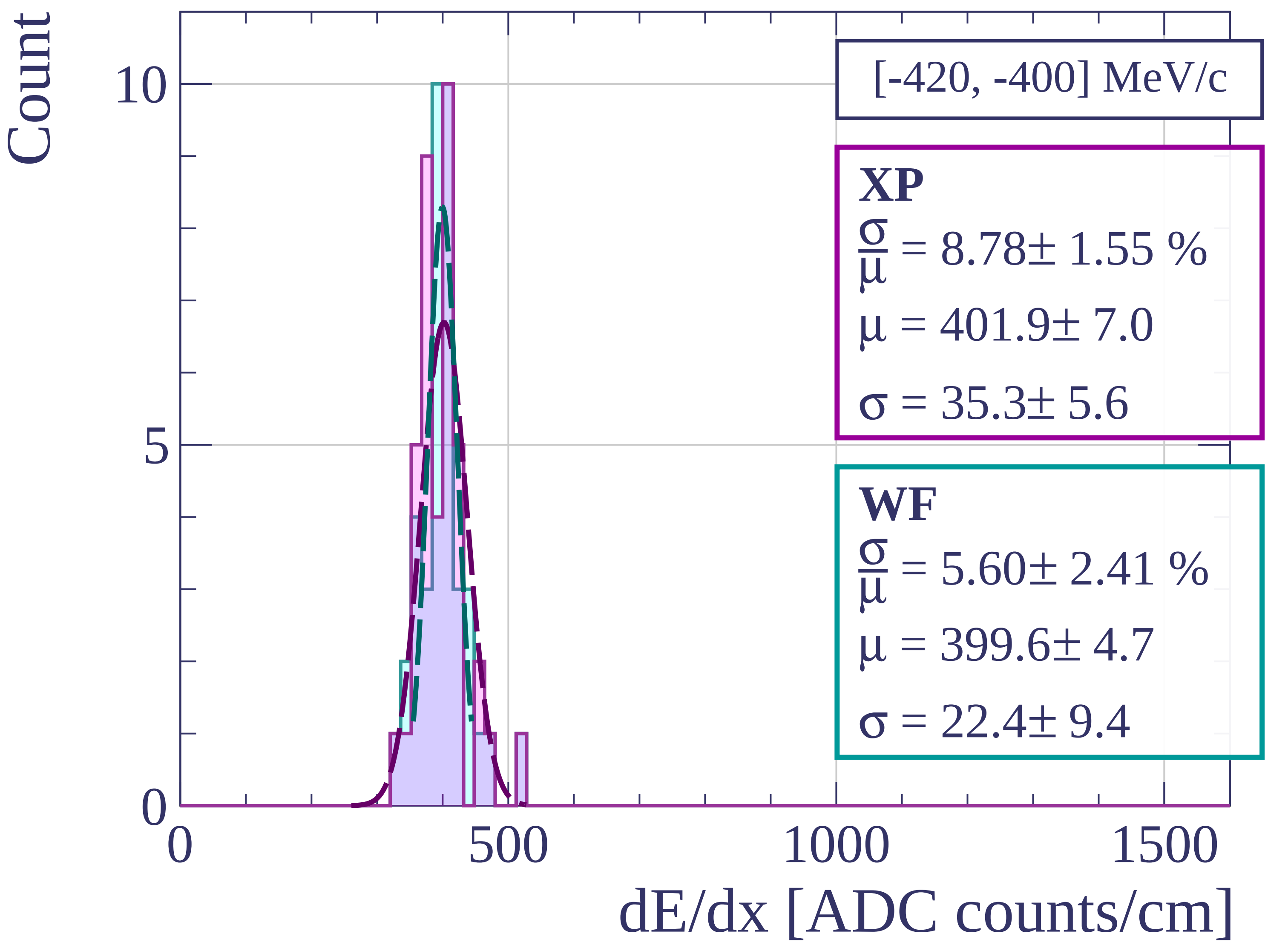
Count
dE/dx [ADC counts/cm]
[-480, -460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.10± 1.24 %
μ = 404.9± 5.3
σ = 28.8± 4.6
WF
$\frac{\sigma}{\mu}$ = 6.23± 1.14 %
μ = 404.3± 4.3
σ = 25.2± 4.3
0
500
1000
1500
5
10
15



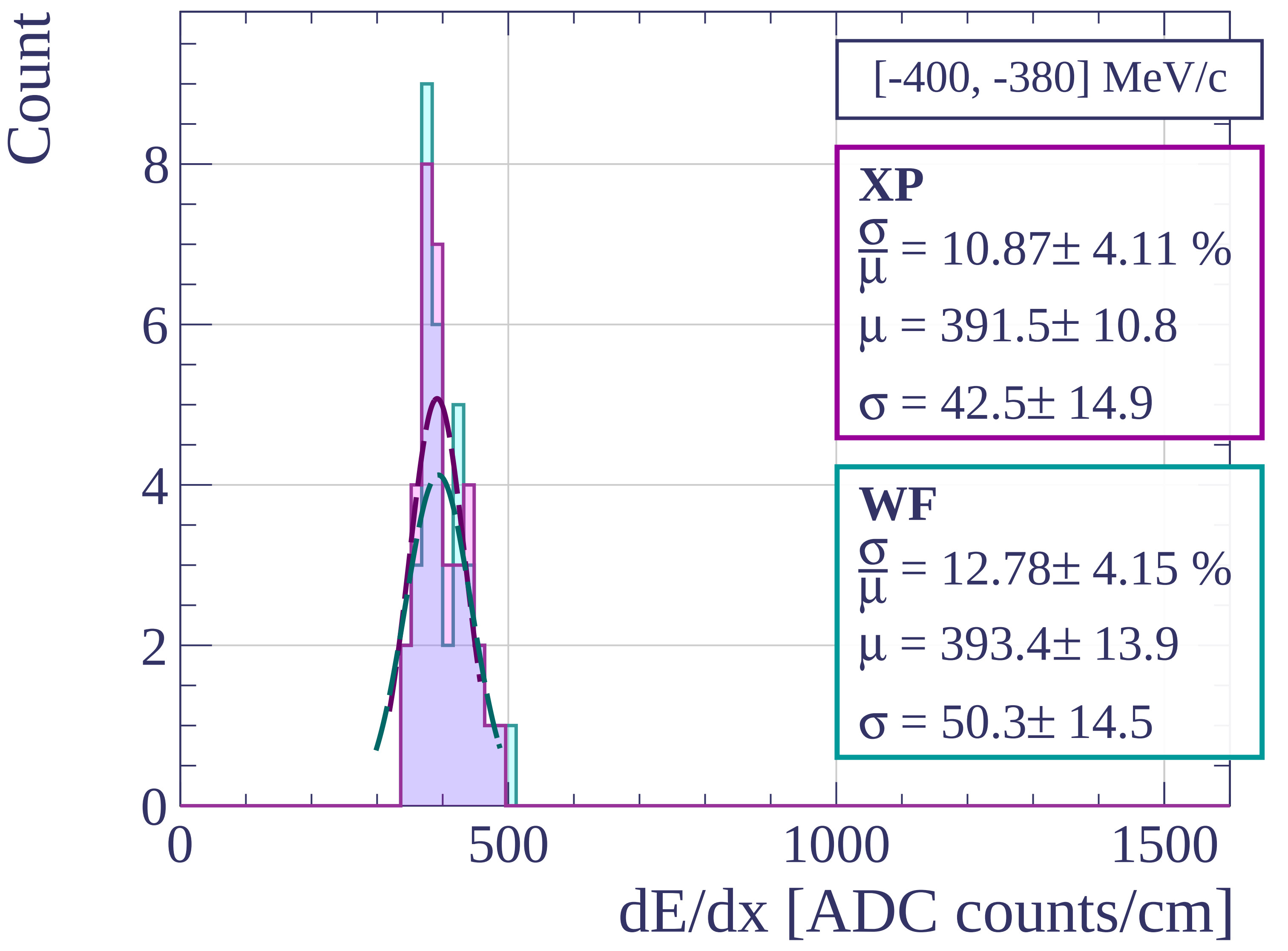
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-460, -440] MeV/c
XP
σ/μ = 5.29± 1.09 %
μ = 400.3± 4.8
σ = 21.2± 4.1
WF
σ/μ = 5.32± 1.20 %
μ = 406.0± 4.6
σ = 21.6± 4.6

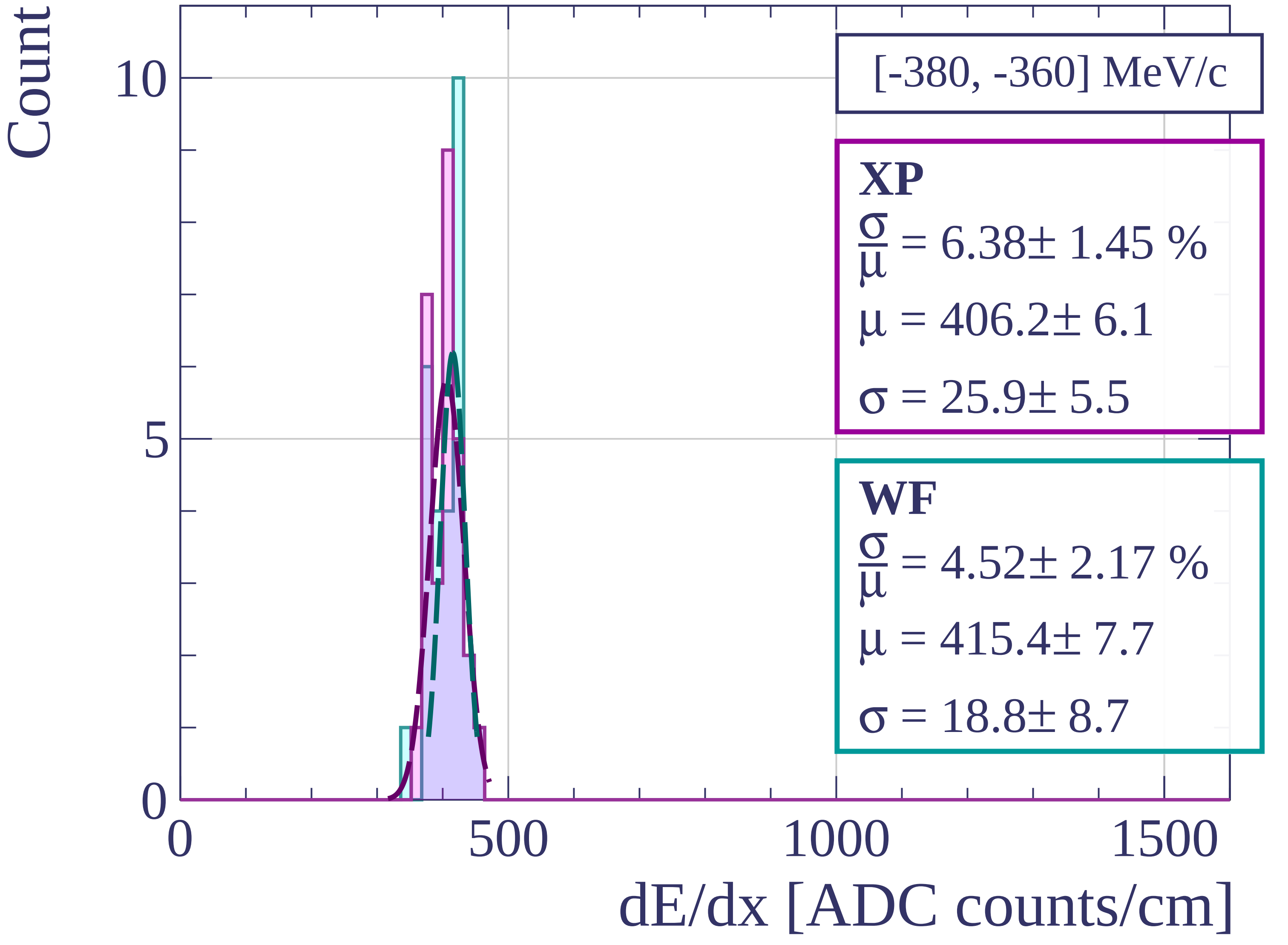


Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-440, -420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.26± 2.63 %
μ = 388.6± 10.4
σ = 39.9± 9.2
WF
$\frac{\sigma}{\mu}$ = 9.31± 2.31 %
μ = 405.3± 8.3
σ = 37.7± 8.6

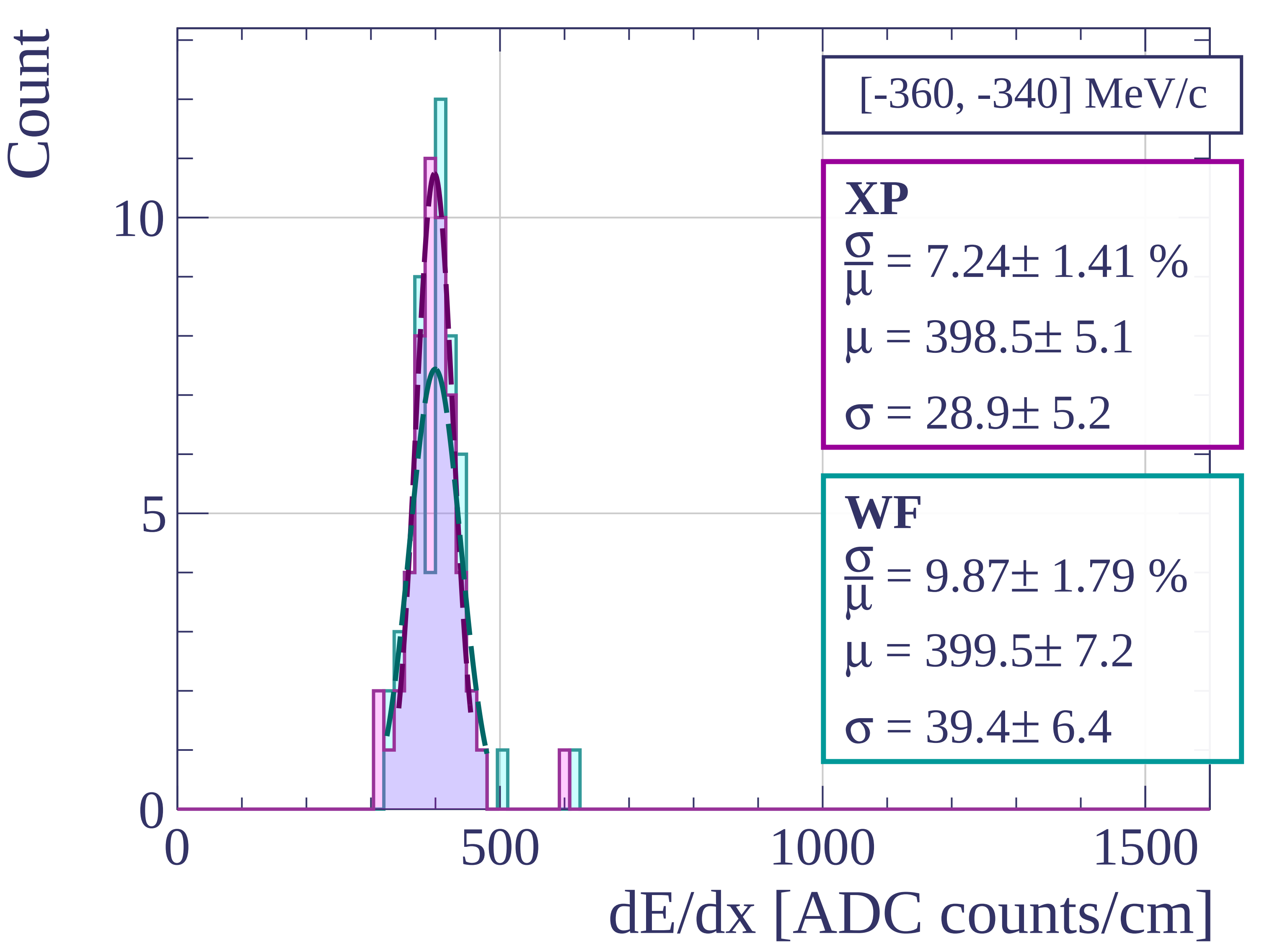


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-420, -400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.78± 1.55 %
μ = 401.9± 7.0
σ = 35.3± 5.6
WF
$\frac{\sigma}{\mu}$ = 5.60± 2.41 %
μ = 399.6± 4.7
σ = 22.4± 9.4

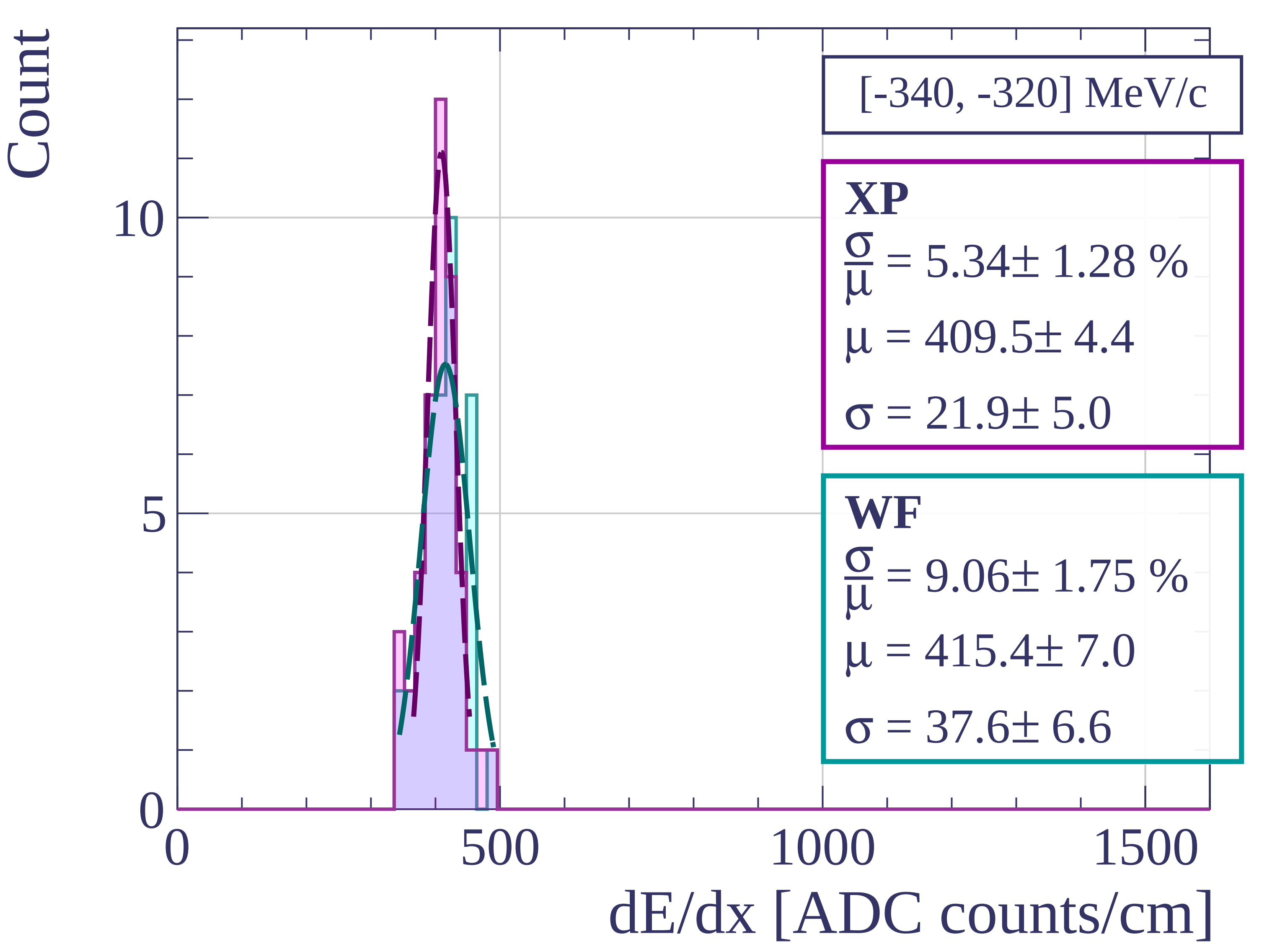

Count
dE/dx [ADC counts/cm]
[-400, -380] MeV/c
XP
σ/μ = 10.87± 4.11 %
μ = 391.5± 10.8
σ = 42.5± 14.9
WF
σ/μ = 12.78± 4.15 %
μ = 393.4± 13.9
σ = 50.3± 14.5
0
2
4
6
8
0
500
1000
1500



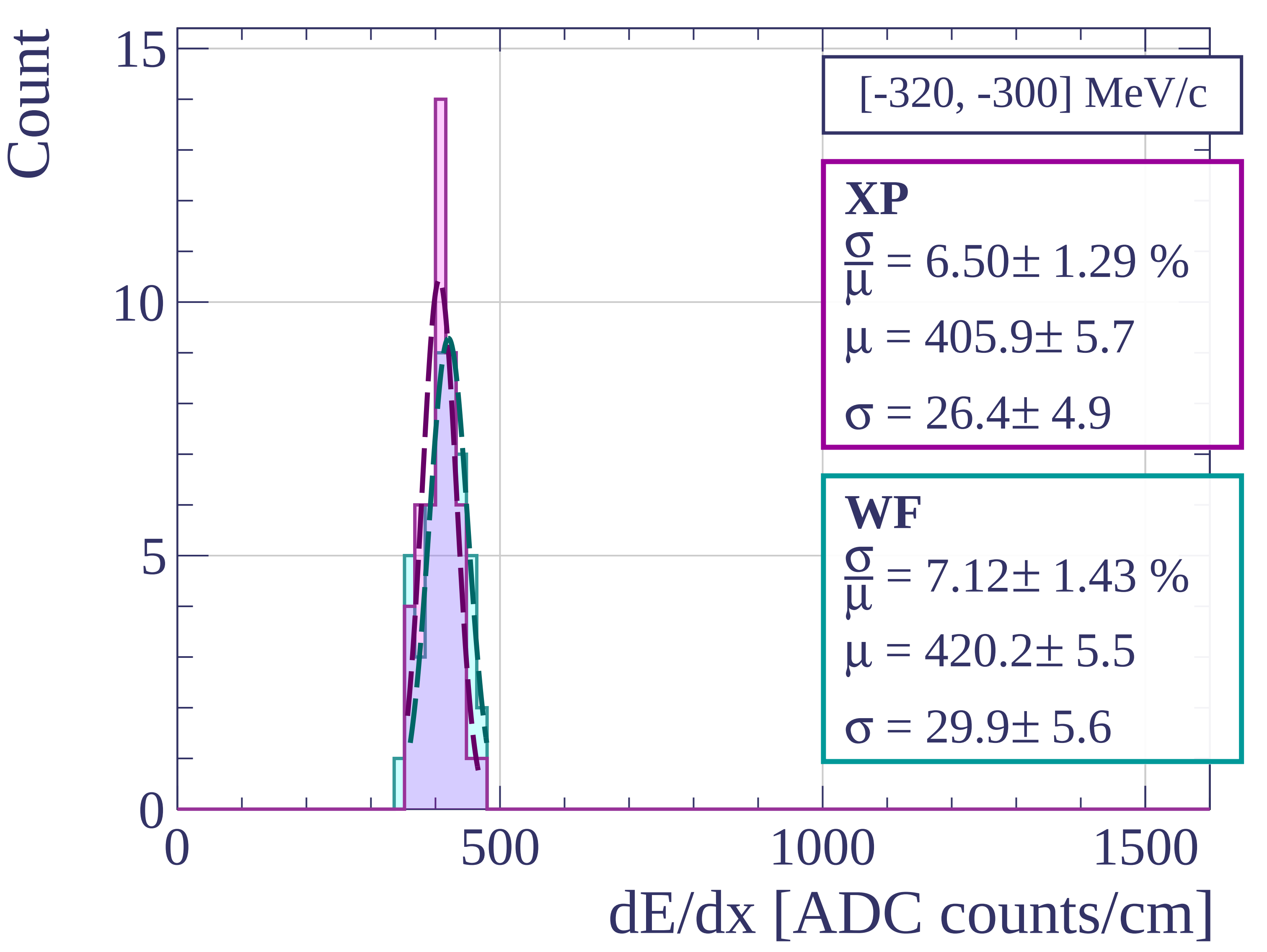
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-380, -360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.38± 1.45 %
μ = 406.2± 6.1
σ = 25.9± 5.5
WF
$\frac{\sigma}{\mu}$ = 4.52± 2.17 %
μ = 415.4± 7.7
σ = 18.8± 8.7



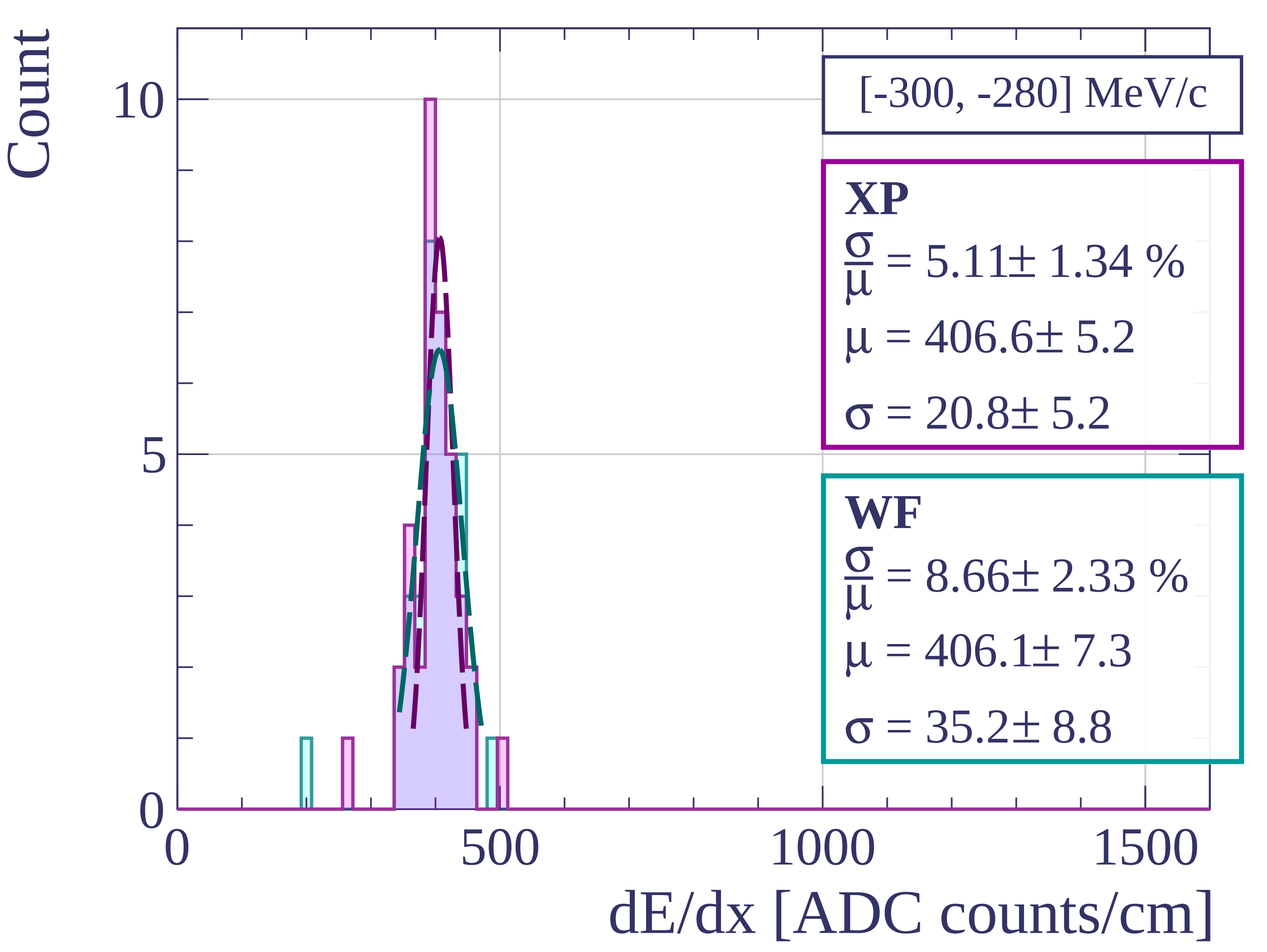
Count
0
5
10
0
500
1000
1500
dE/dx [ADC counts/cm]
[-360, -340] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.24± 1.41 %
μ = 398.5± 5.1
σ = 28.9± 5.2
WF
$\frac{\sigma}{\mu}$ = 9.87± 1.79 %
μ = 399.5± 7.2
σ = 39.4± 6.4



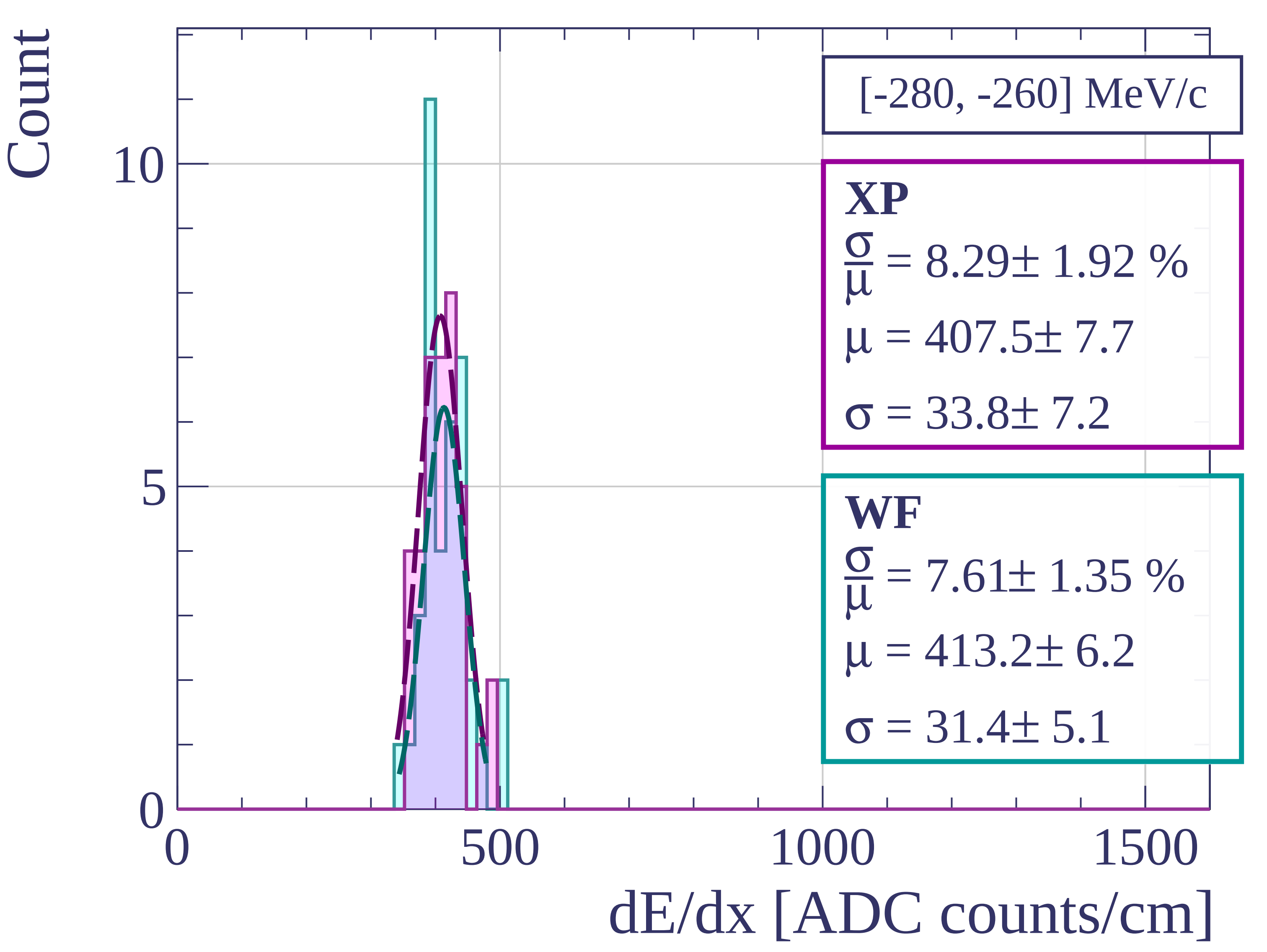
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-340, -320] MeV/c
XP
σ/μ = 5.34± 1.28 %
μ = 409.5± 4.4
σ = 21.9± 5.0
WF
σ/μ = 9.06± 1.75 %
μ = 415.4± 7.0
σ = 37.6± 6.6



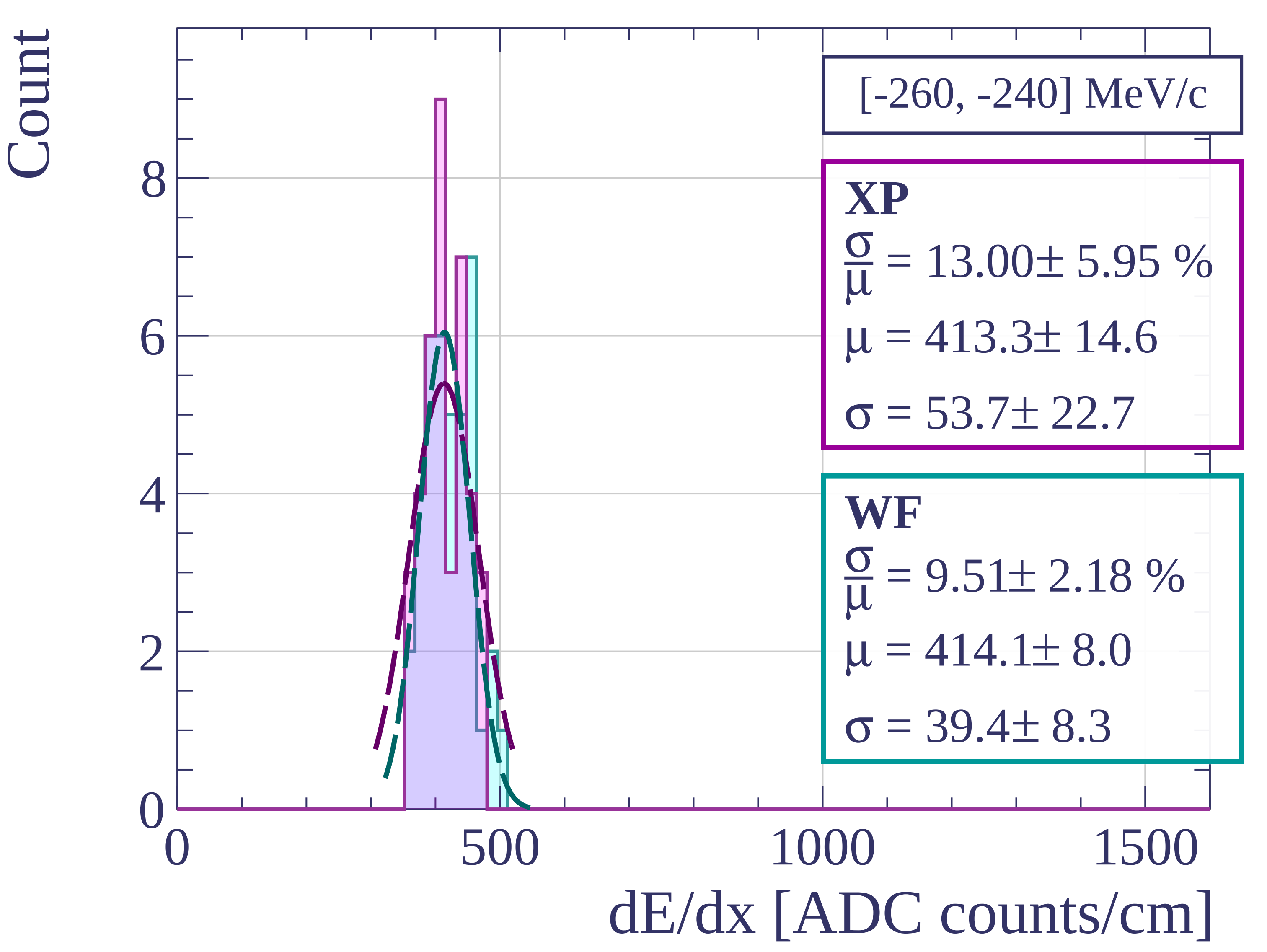
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-320, -300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.50± 1.29 %
μ = 405.9± 5.7
σ = 26.4± 4.9
WF
$\frac{\sigma}{\mu}$ = 7.12± 1.43 %
μ = 420.2± 5.5
σ = 29.9± 5.6



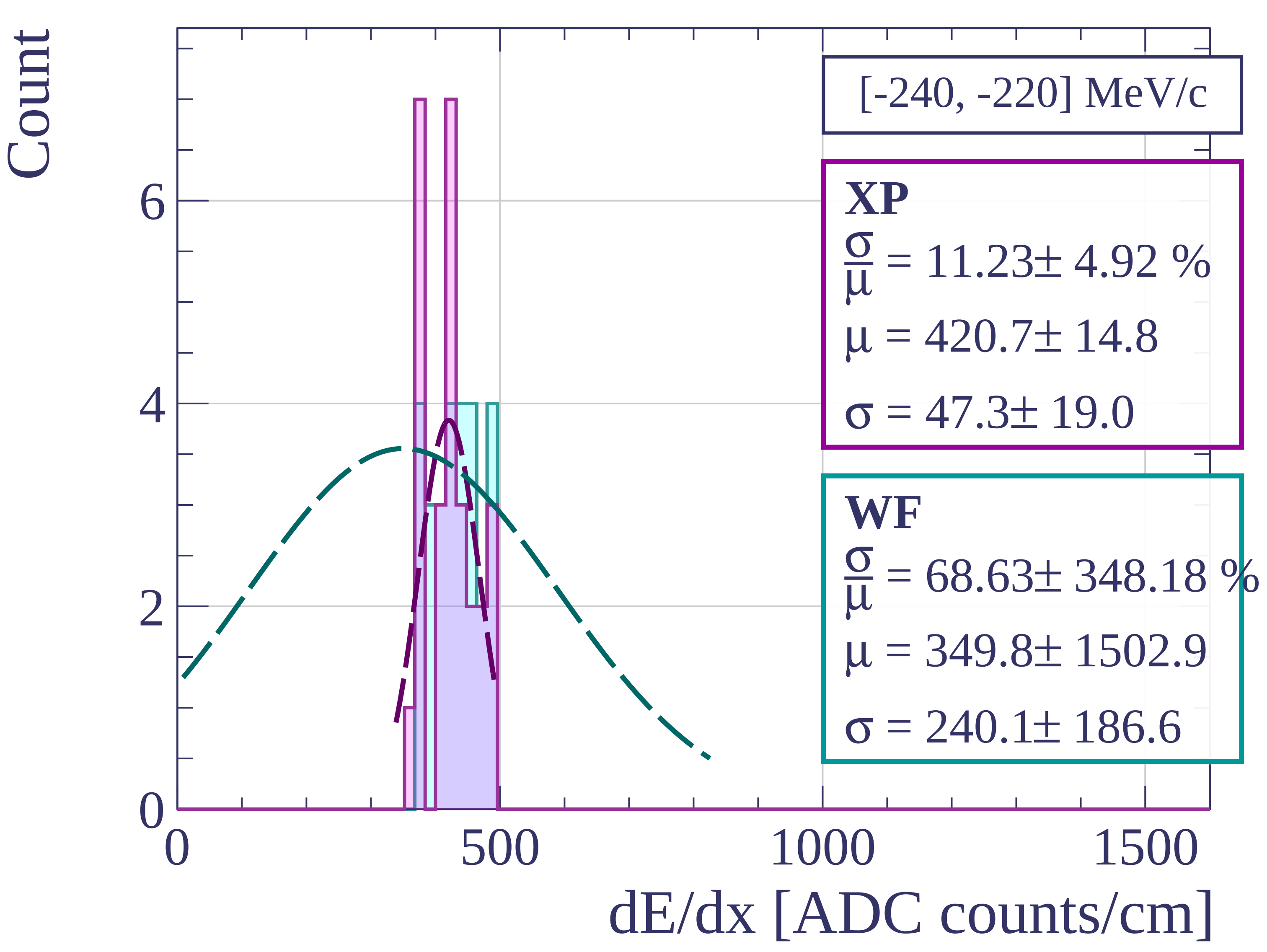
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-300, -280] MeV/c
XP
σ/μ = 5.11± 1.34 %
μ = 406.6± 5.2
σ = 20.8± 5.2
WF
σ/μ = 8.66± 2.33 %
μ = 406.1± 7.3
σ = 35.2± 8.8

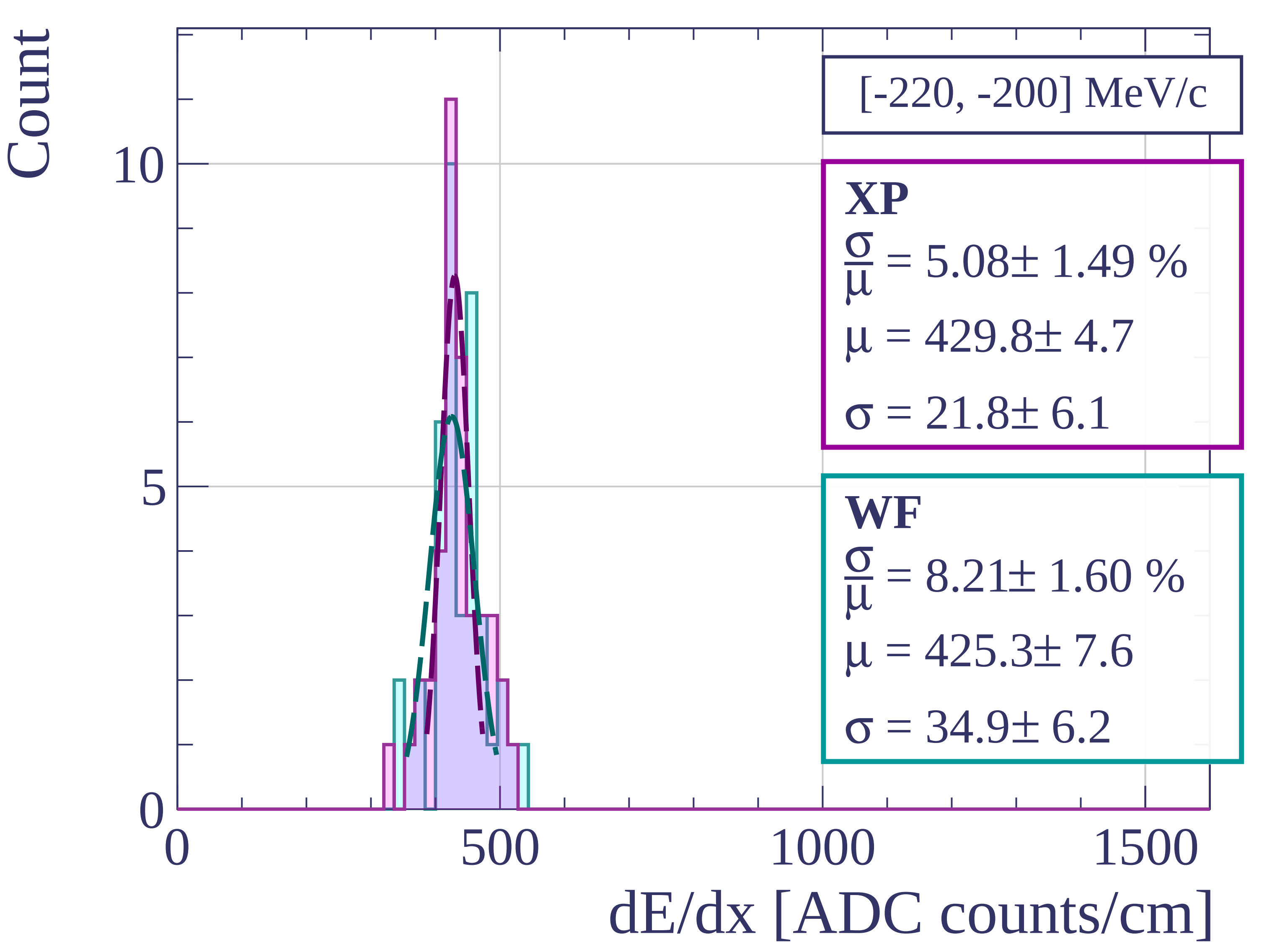


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-280, -260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.29± 1.92 %
μ = 407.5± 7.7
σ = 33.8± 7.2
WF
$\frac{\sigma}{\mu}$ = 7.61± 1.35 %
μ = 413.2± 6.2
σ = 31.4± 5.1

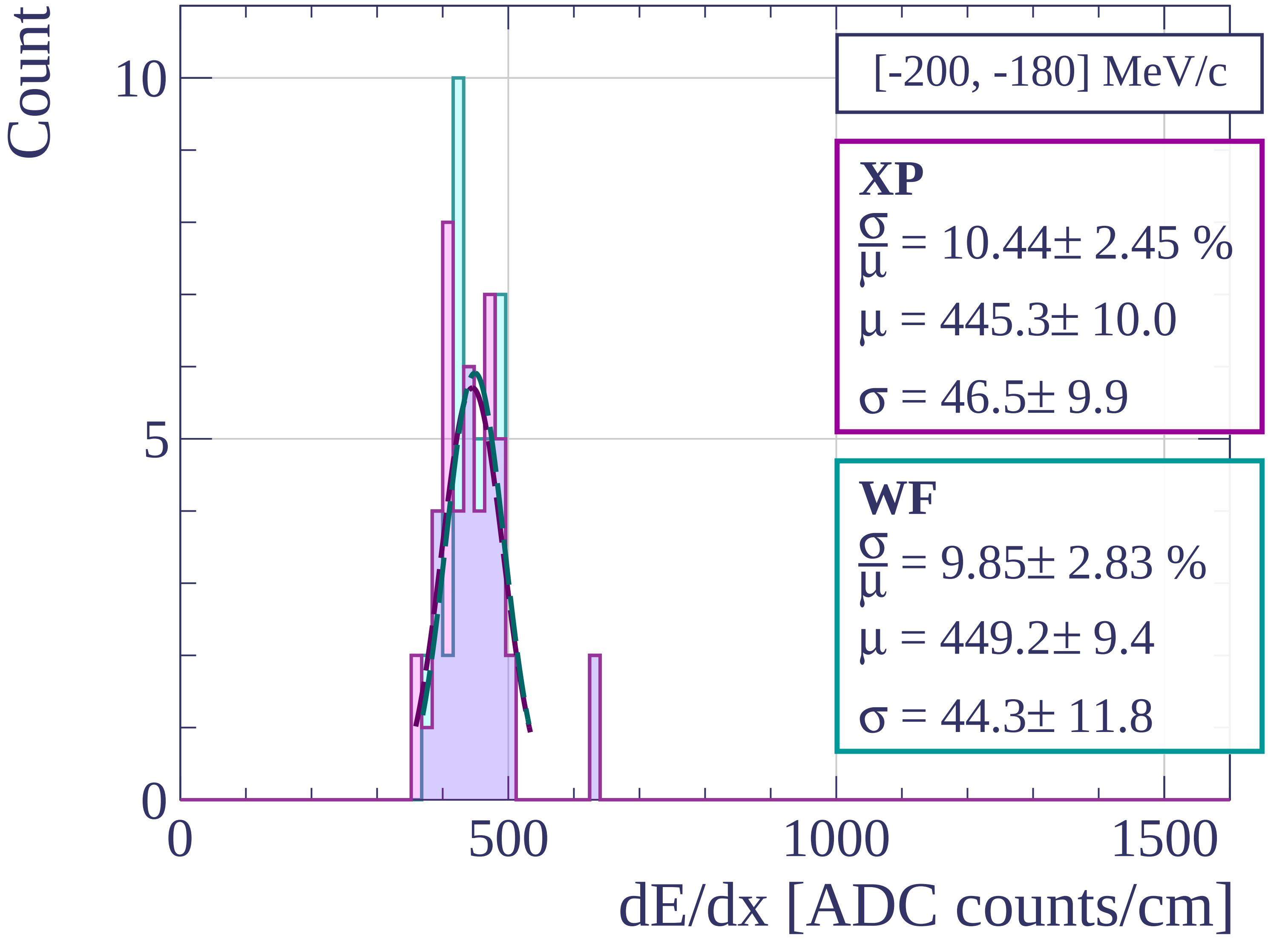


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-260, -240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.00± 5.95 %
μ = 413.3± 14.6
σ = 53.7± 22.7
WF
$\frac{\sigma}{\mu}$ = 9.51± 2.18 %
μ = 414.1± 8.0
σ = 39.4± 8.3

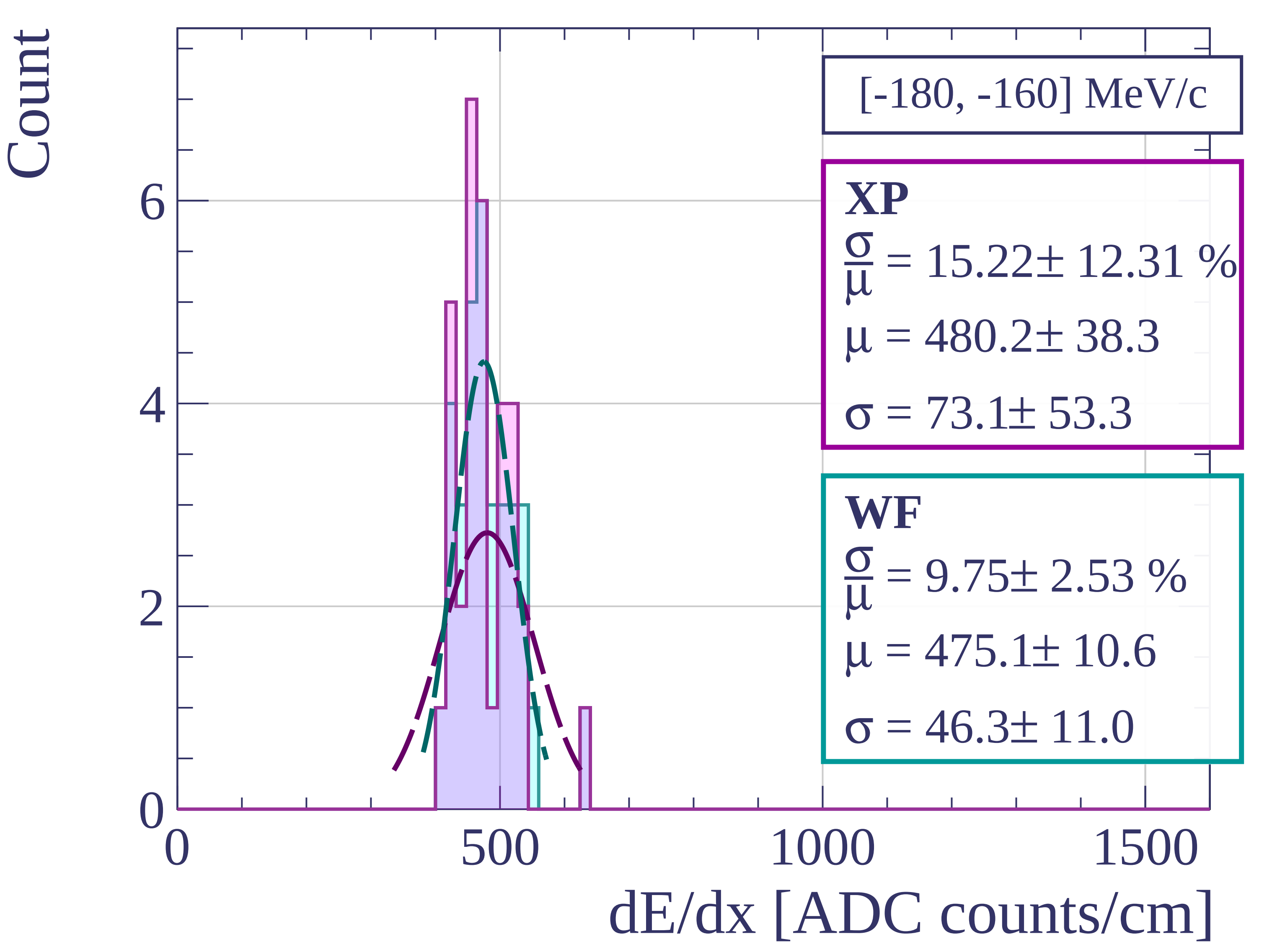

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-240, -220] MeV/c
XP
σ/μ = 11.23± 4.92 %
μ = 420.7± 14.8
σ = 47.3± 19.0
WF
σ/μ = 68.63± 348.18 %
μ = 349.8± 1502.9
σ = 240.1± 186.6

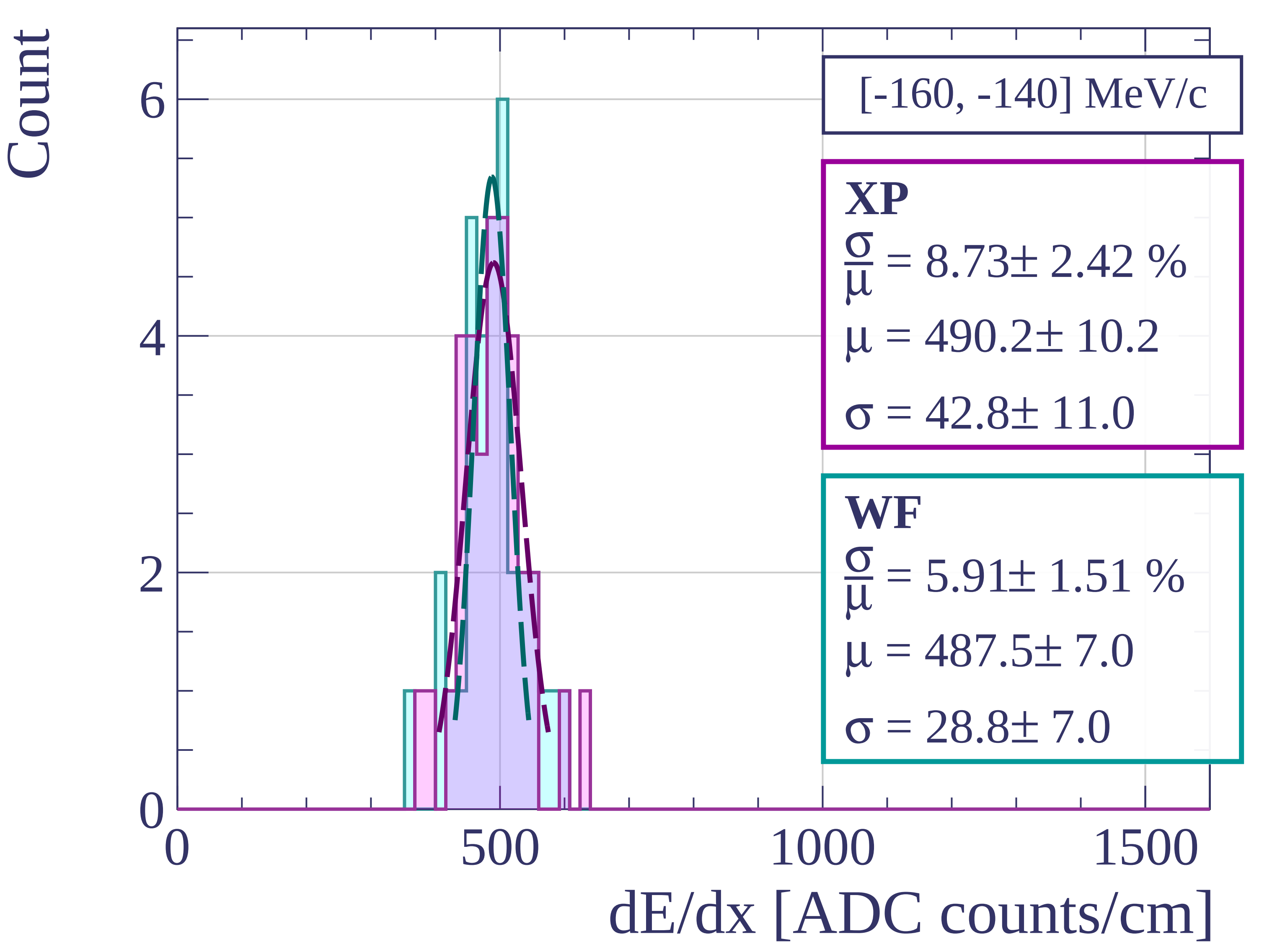


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-220, -200] MeV/c
XP
σ/μ = 5.08± 1.49 %
μ = 429.8± 4.7
σ = 21.8± 6.1
WF
σ/μ = 8.21± 1.60 %
μ = 425.3± 7.6
σ = 34.9± 6.2

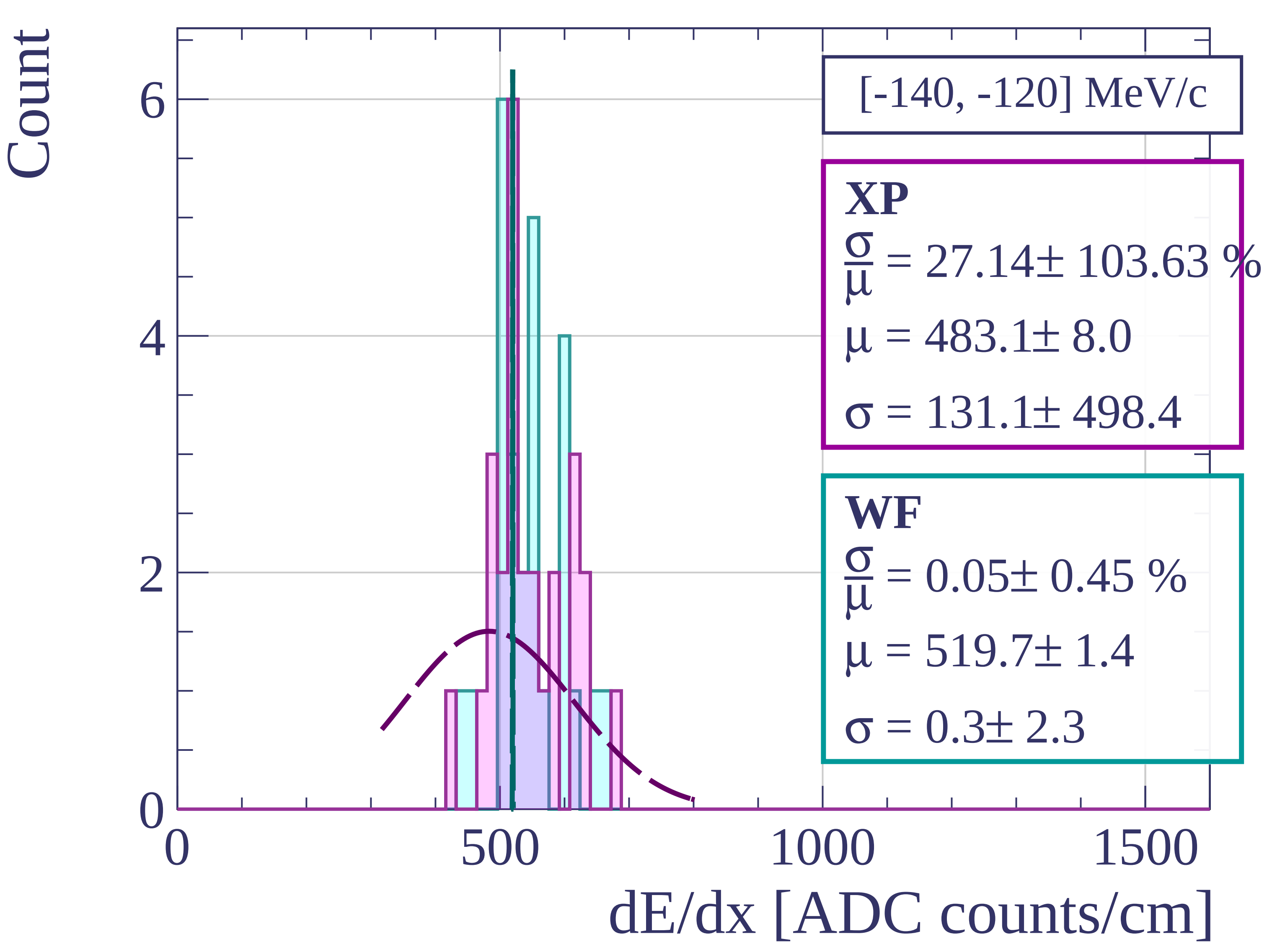


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-200, -180] MeV/c
XP
σ/μ = 10.44± 2.45 %
μ = 445.3± 10.0
σ = 46.5± 9.9
WF
σ/μ = 9.85± 2.83 %
μ = 449.2± 9.4
σ = 44.3± 11.8

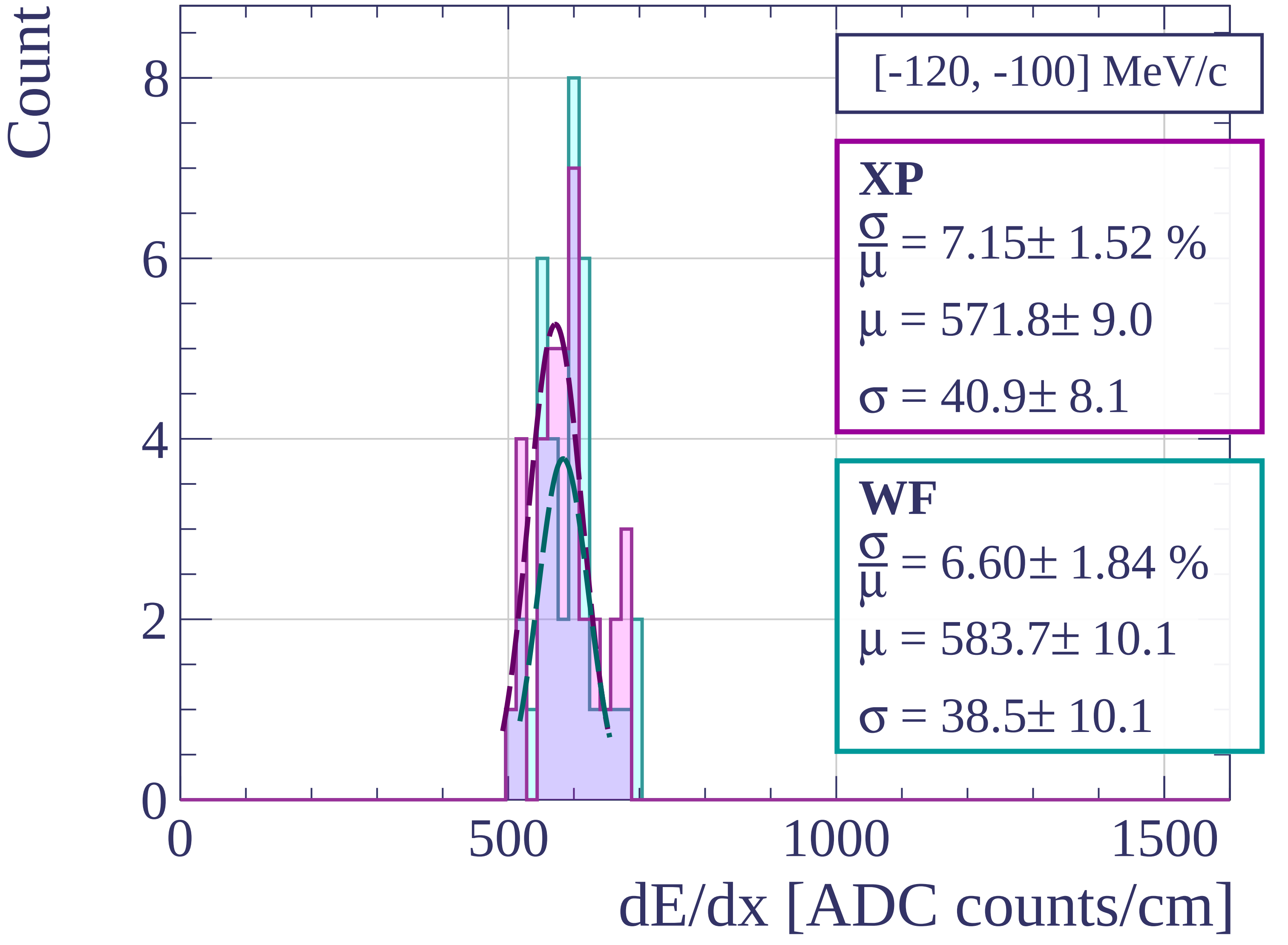

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-180, -160] MeV/c
XP
σ/μ = 15.22± 12.31 %
μ = 480.2± 38.3
σ = 73.1± 53.3
WF
σ/μ = 9.75± 2.53 %
μ = 475.1± 10.6
σ = 46.3± 11.0



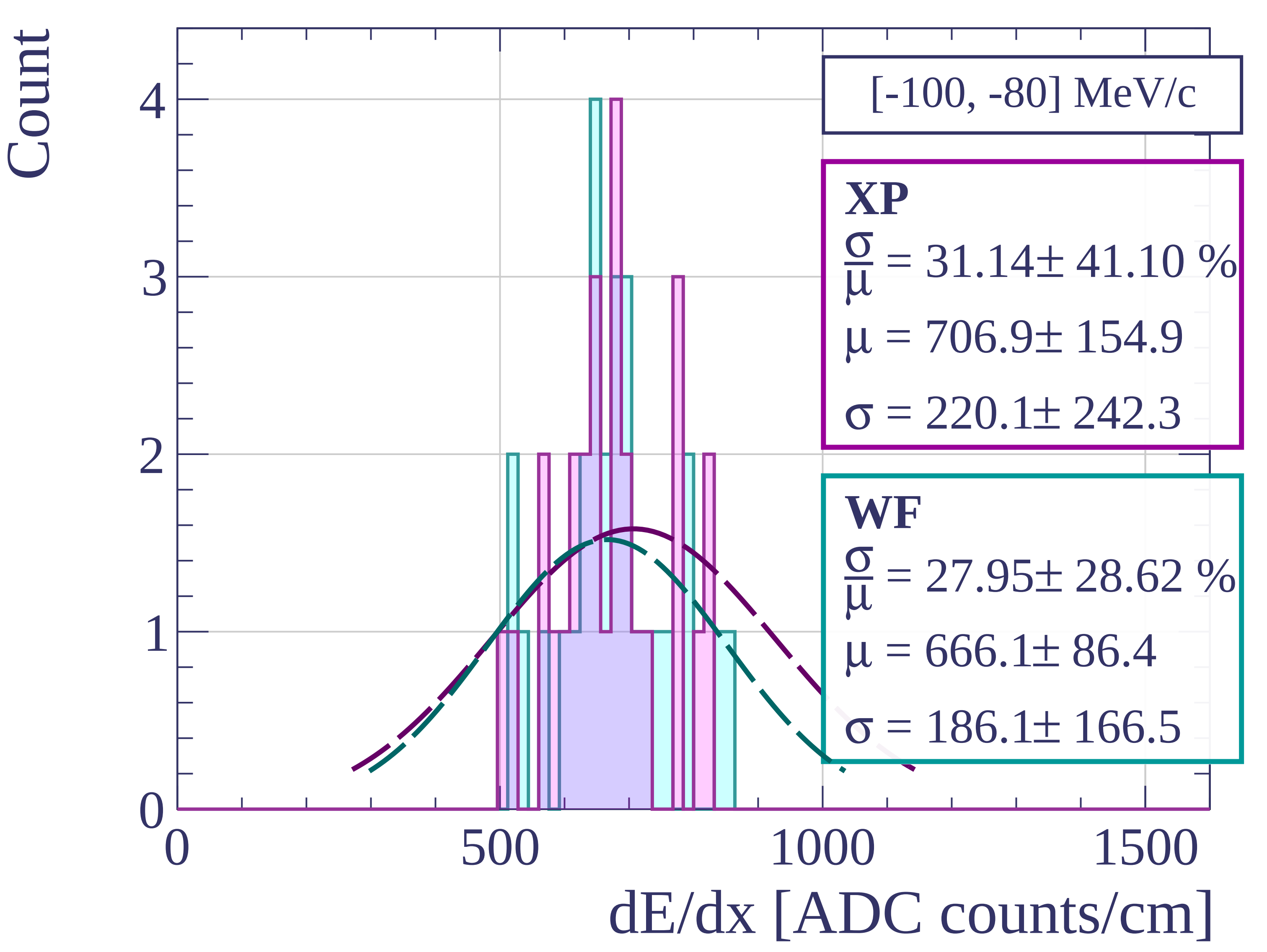
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[-160, -140] MeV/c
XP
σ/μ = 8.73± 2.42 %
μ = 490.2± 10.2
σ = 42.8± 11.0
WF
σ/μ = 5.91± 1.51 %
μ = 487.5± 7.0
σ = 28.8± 7.0

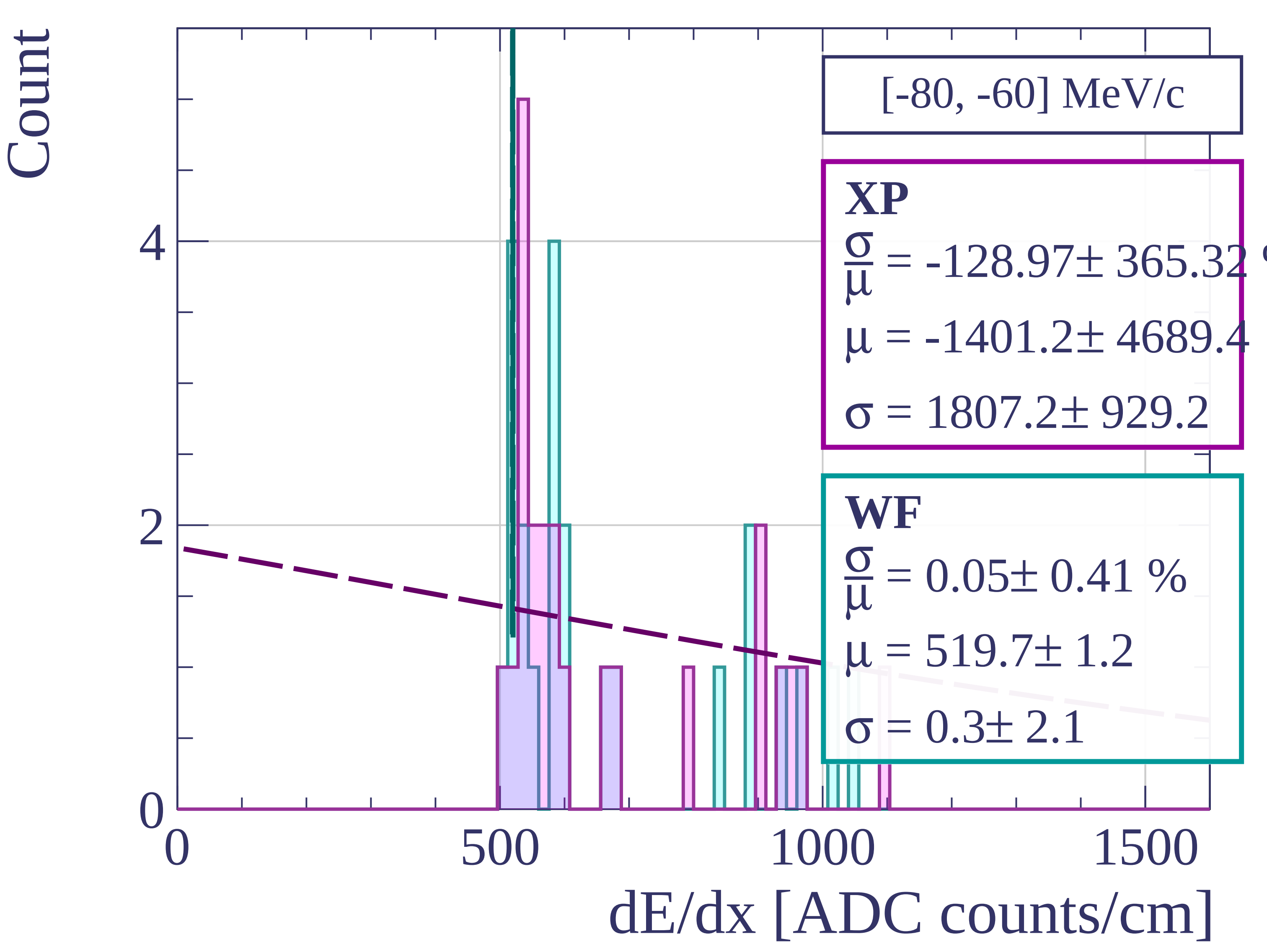


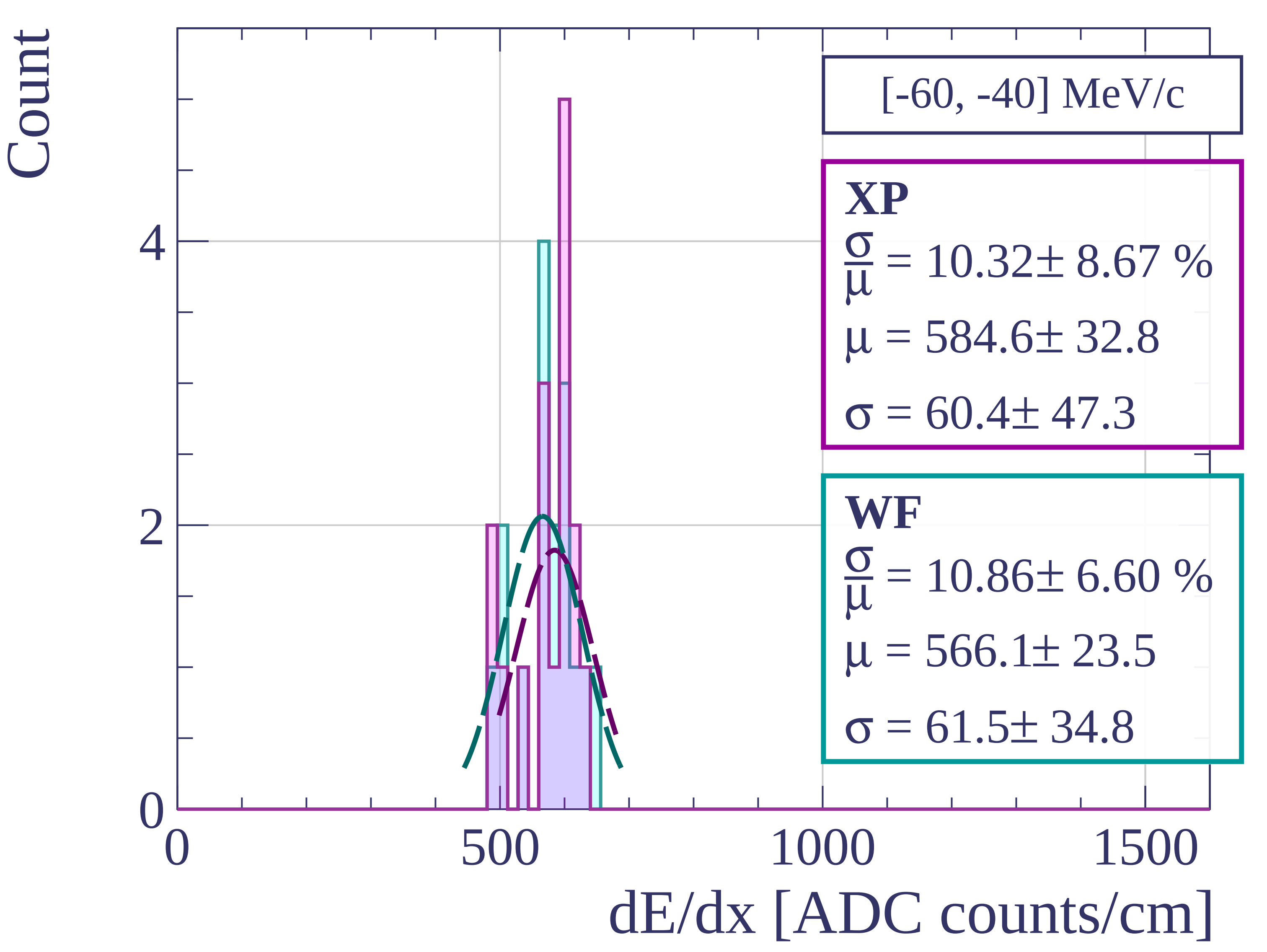
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-140, -120] MeV/c
XP
σ/μ = 27.14± 103.63 %
μ = 483.1± 8.0
σ = 131.1± 498.4
WF
σ/μ = 0.05± 0.45 %
μ = 519.7± 1.4
σ = 0.3± 2.3

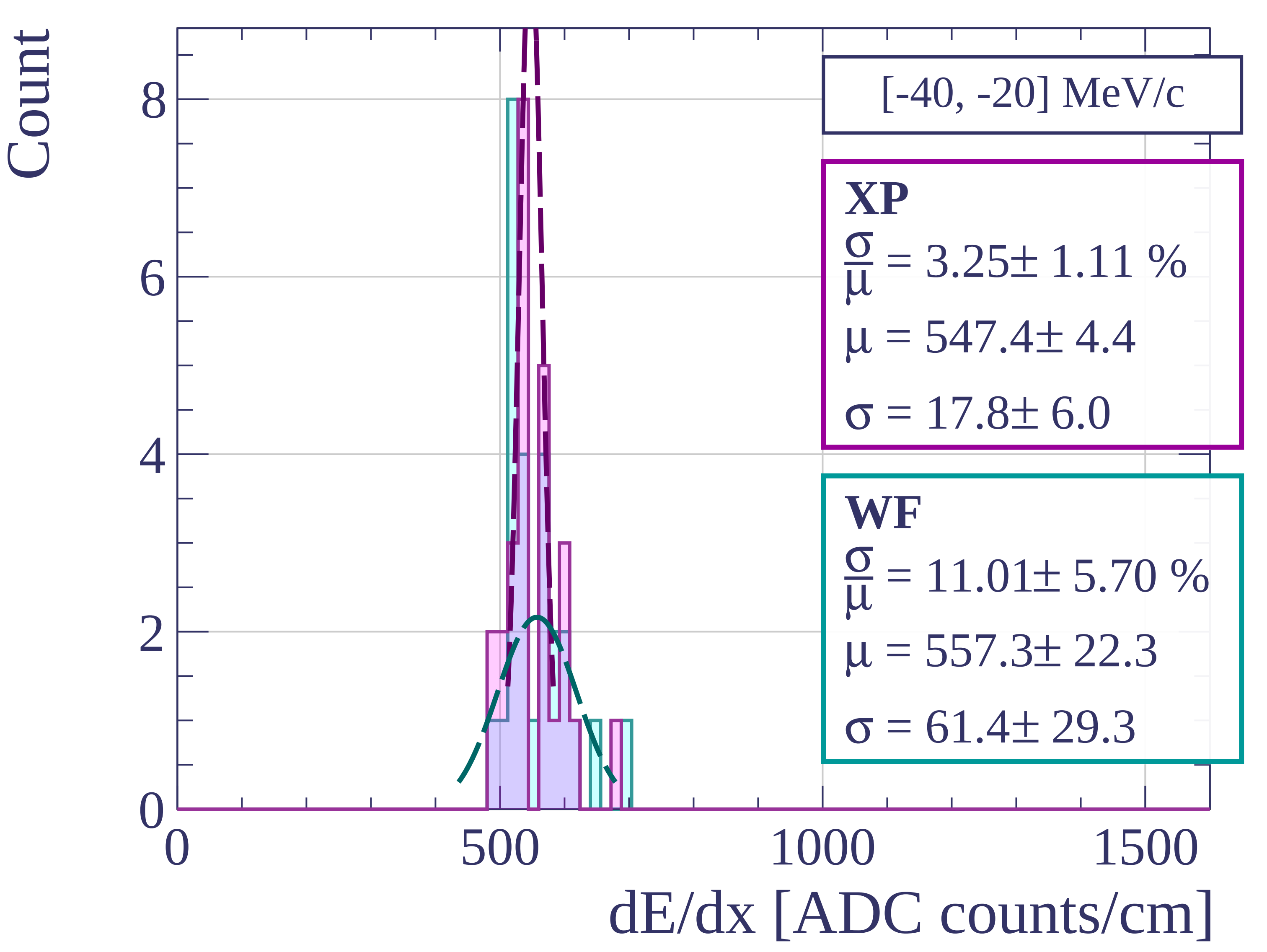


Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-120, -100] MeV/c
XP
σ/μ = 7.15± 1.52 %
μ = 571.8± 9.0
σ = 40.9± 8.1
WF
σ/μ = 6.60± 1.84 %
μ = 583.7± 10.1
σ = 38.5± 10.1

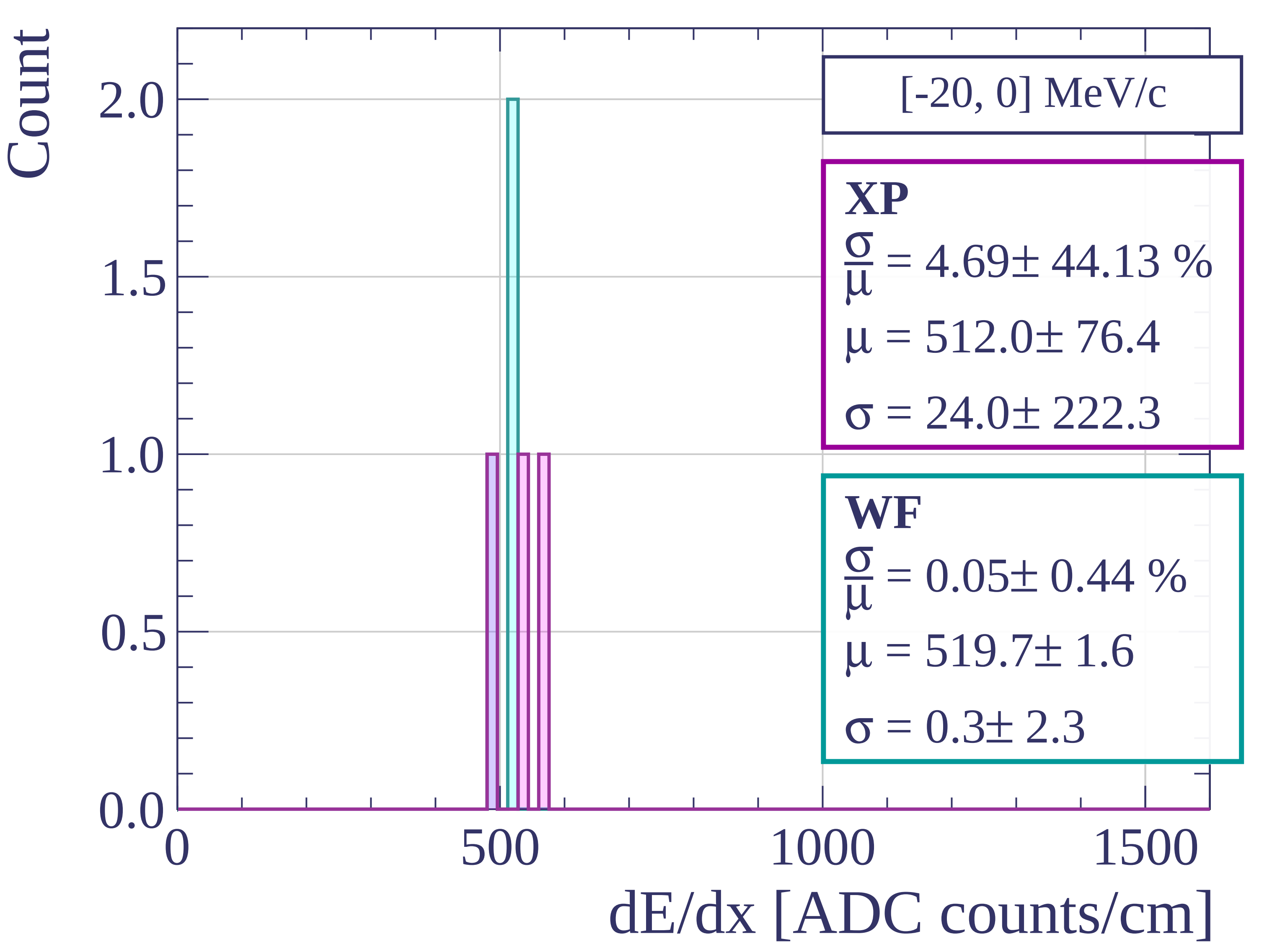

Count
0
1
2
3
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-100, -80] MeV/c
XP
σ/μ = 31.14± 41.10 %
μ = 706.9± 154.9
σ = 220.1± 242.3
WF
σ/μ = 27.95± 28.62 %
μ = 666.1± 86.4
σ = 186.1± 166.5


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-80, -60] MeV/c
XP
σ/μ = -128.97± 365.32
μ = -1401.2± 4689.4
σ = 1807.2± 929.2
WF
σ/μ = 0.05± 0.41 %
μ = 519.7± 1.2
σ = 0.3± 2.1

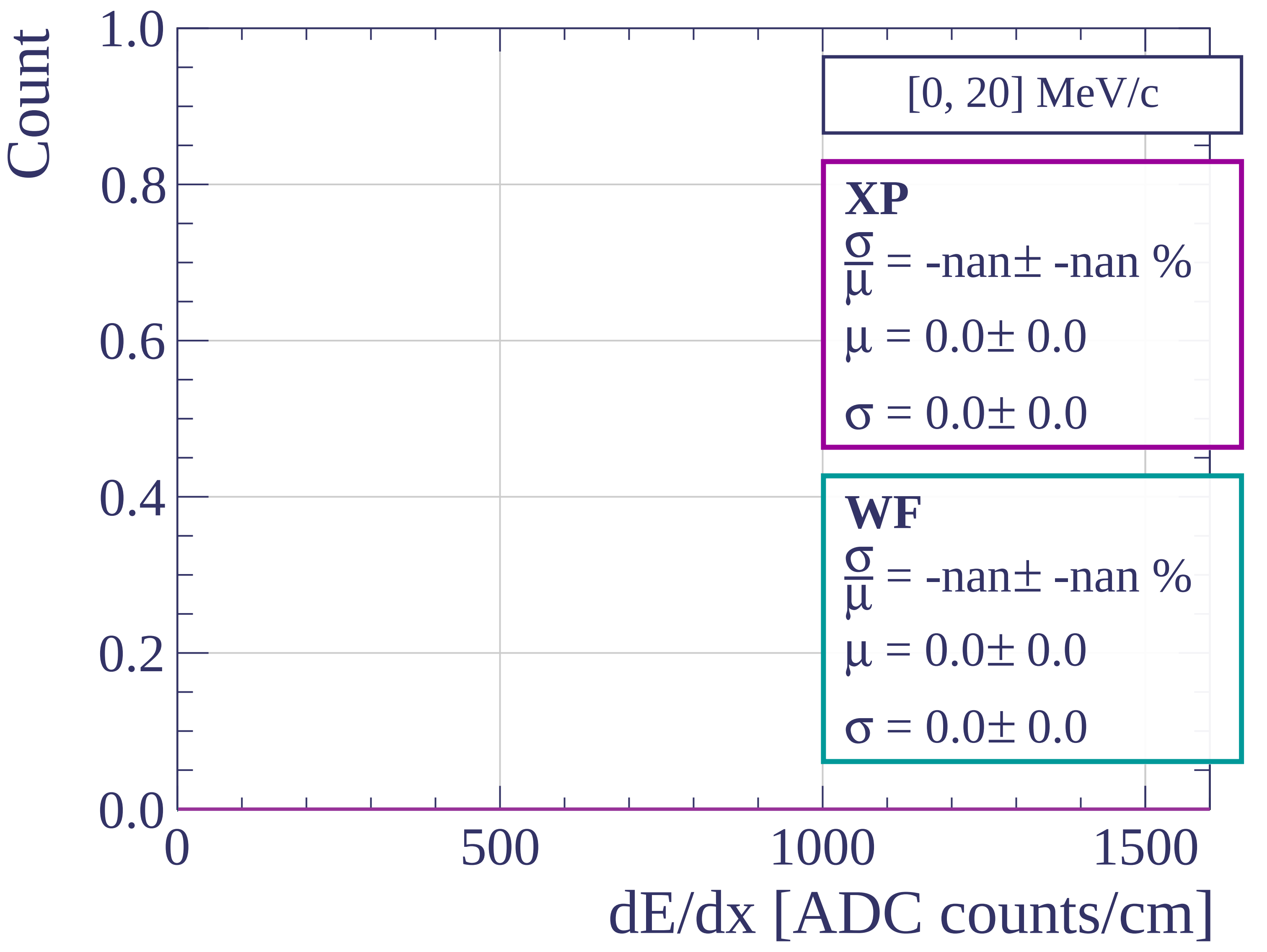

[-60, -40] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.32± 8.67 %
μ = 584.6± 32.8
σ = 60.4± 47.3
WF
$\frac{\sigma}{\mu}$ = 10.86± 6.60 %
μ = 566.1± 23.5
σ = 61.5± 34.8
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500



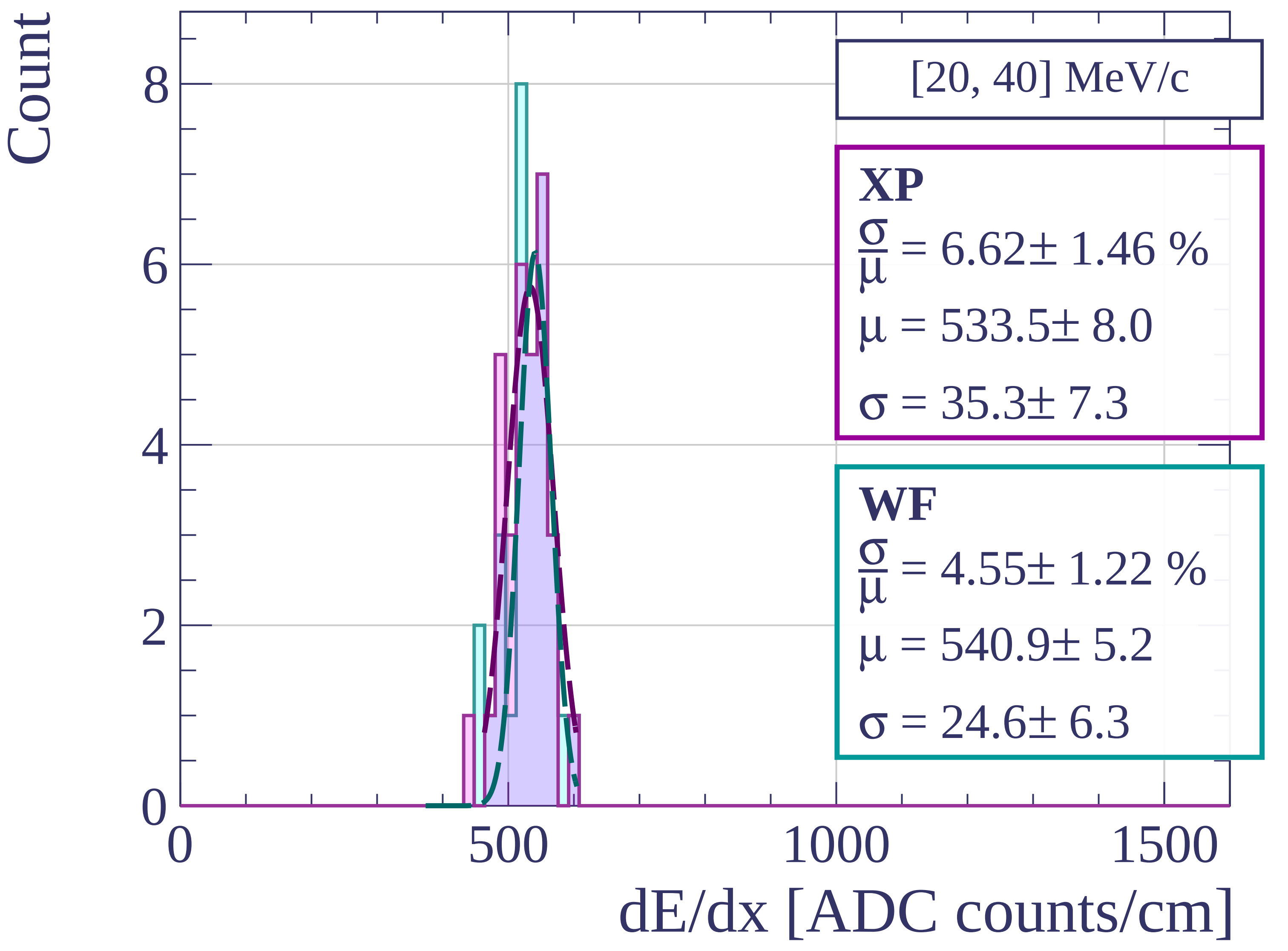
Count
dE/dx [ADC counts/cm]
[-40, -20] MeV/c
XP
σ/μ = 3.25± 1.11 %
μ = 547.4± 4.4
σ = 17.8± 6.0
WF
σ/μ = 11.01± 5.70 %
μ = 557.3± 22.3
σ = 61.4± 29.3
0
2
4
6
8
0
500
1000
1500



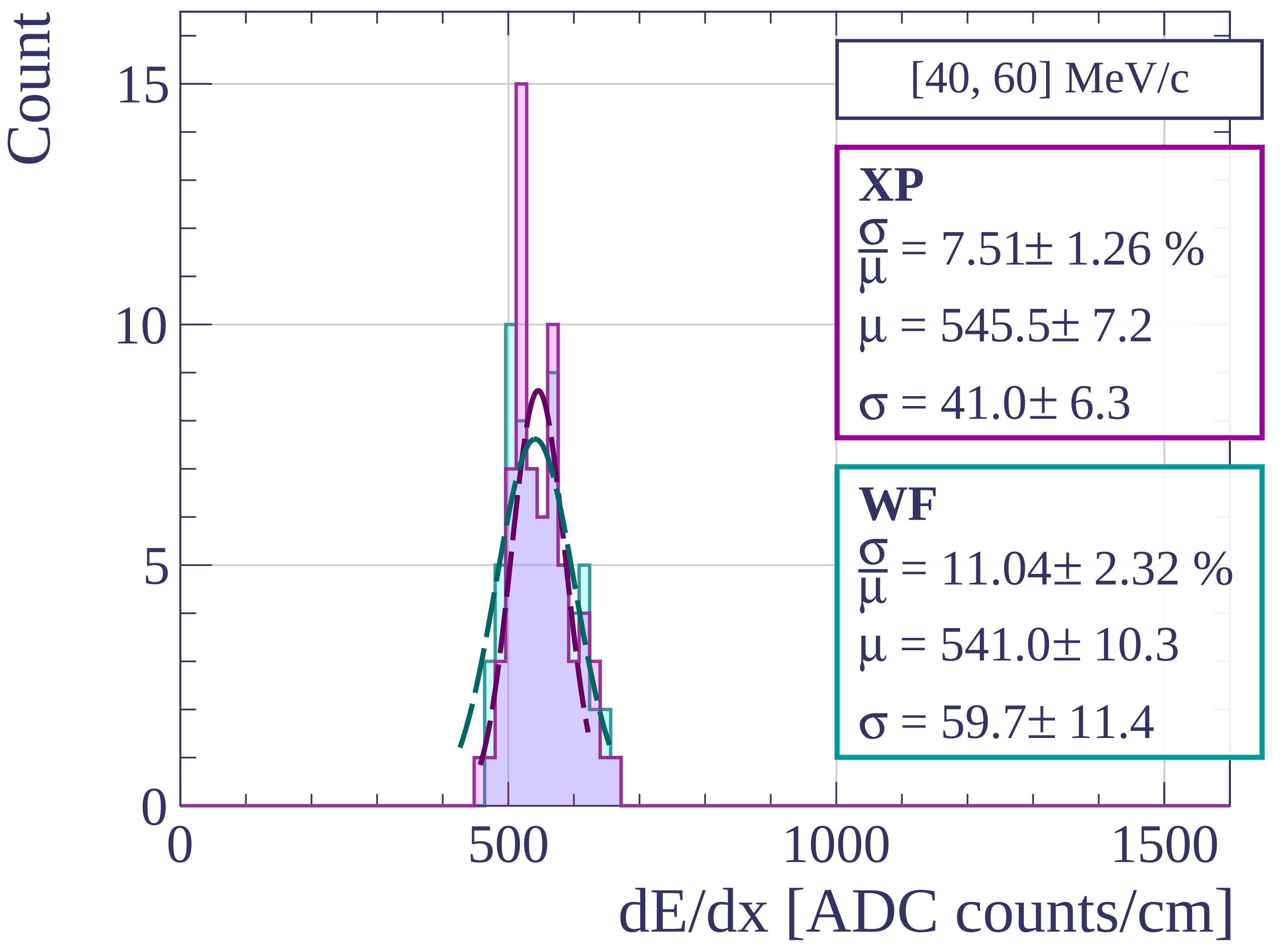
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-20, 0] MeV/c
XP
σ/μ = 4.69± 44.13 %
μ = 512.0± 76.4
σ = 24.0± 222.3
WF
σ/μ = 0.05± 0.44 %
μ = 519.7± 1.6
σ = 0.3± 2.3

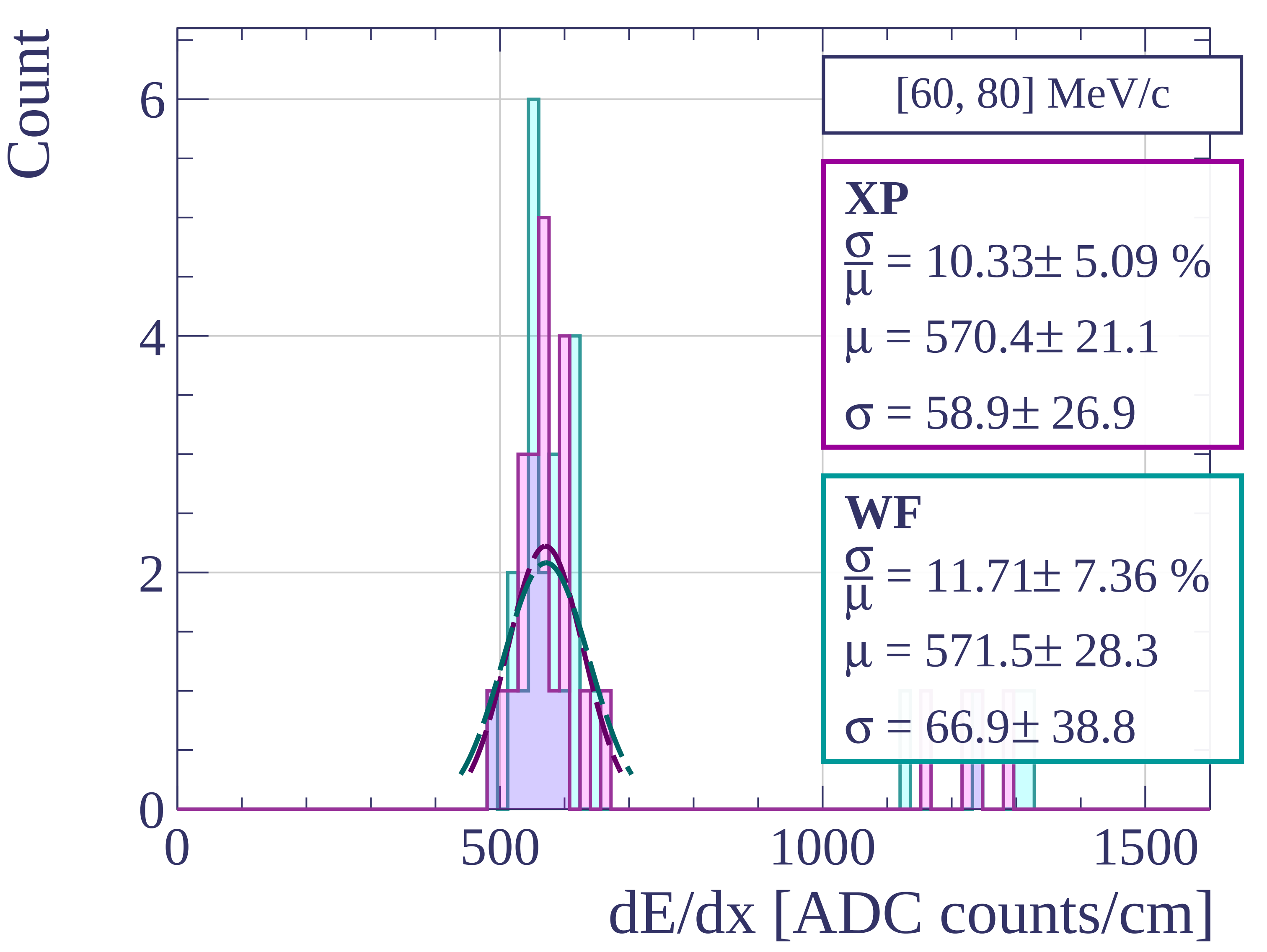


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 20] MeV/c
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

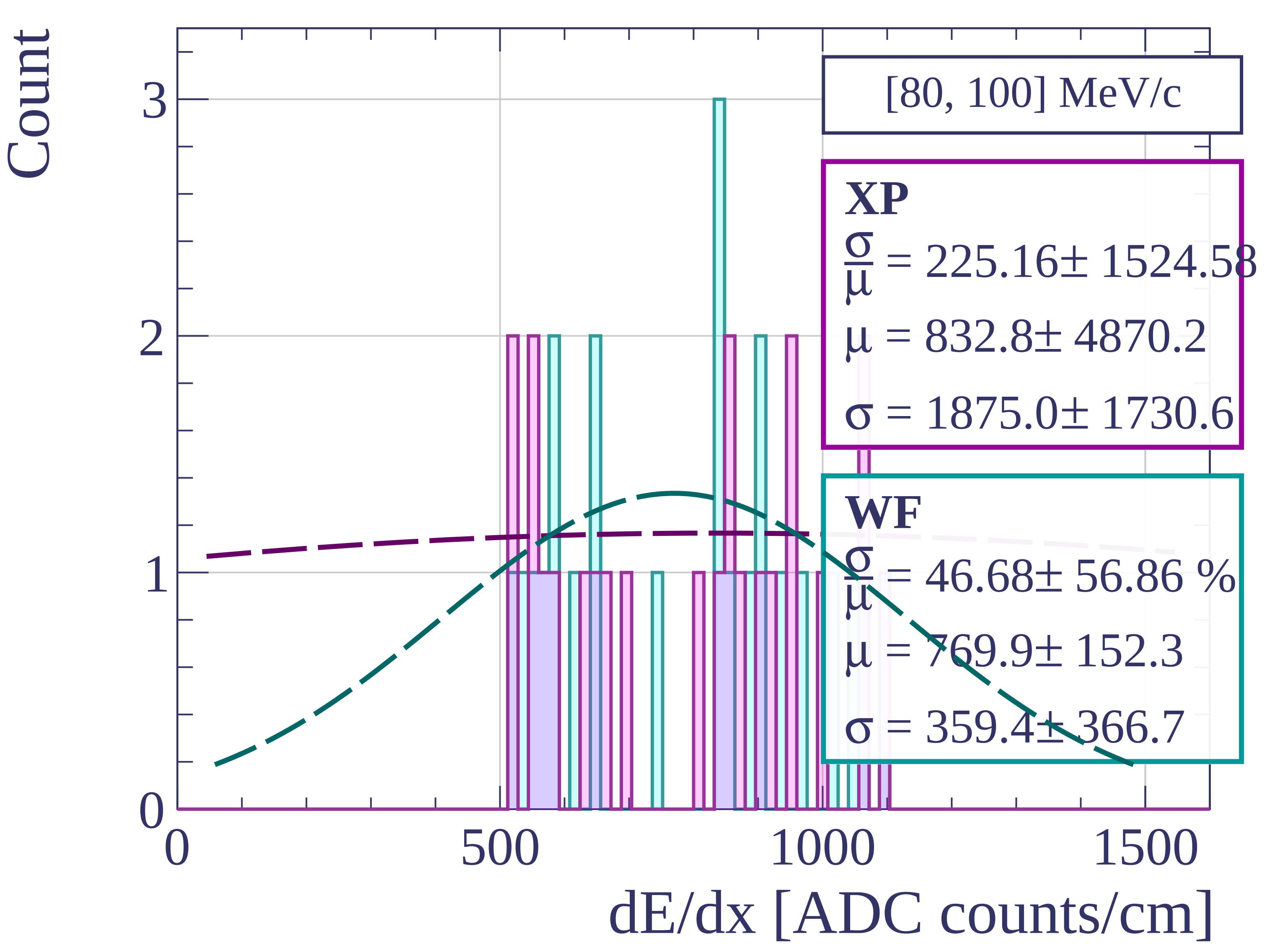


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[20, 40] MeV/c
XP
σ/μ = 6.62± 1.46 %
μ = 533.5± 8.0
σ = 35.3± 7.3
WF
σ/μ = 4.55± 1.22 %
μ = 540.9± 5.2
σ = 24.6± 6.3

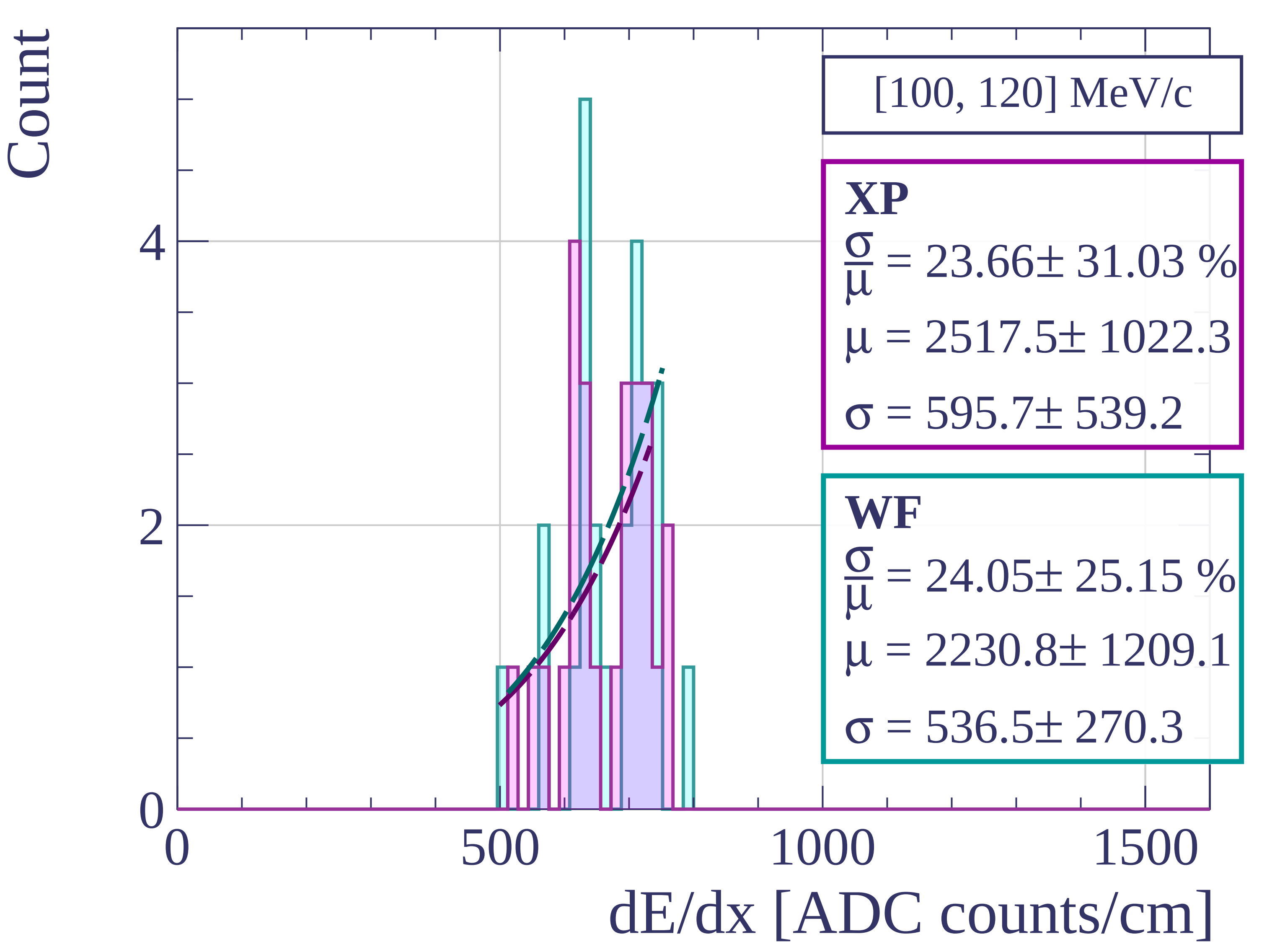

Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[40, 60] MeV/c
XP
σ/μ = 7.51± 1.26 %
μ = 545.5± 7.2
σ = 41.0± 6.3
WF
σ/μ = 11.04± 2.32 %
μ = 541.0± 10.3
σ = 59.7± 11.4



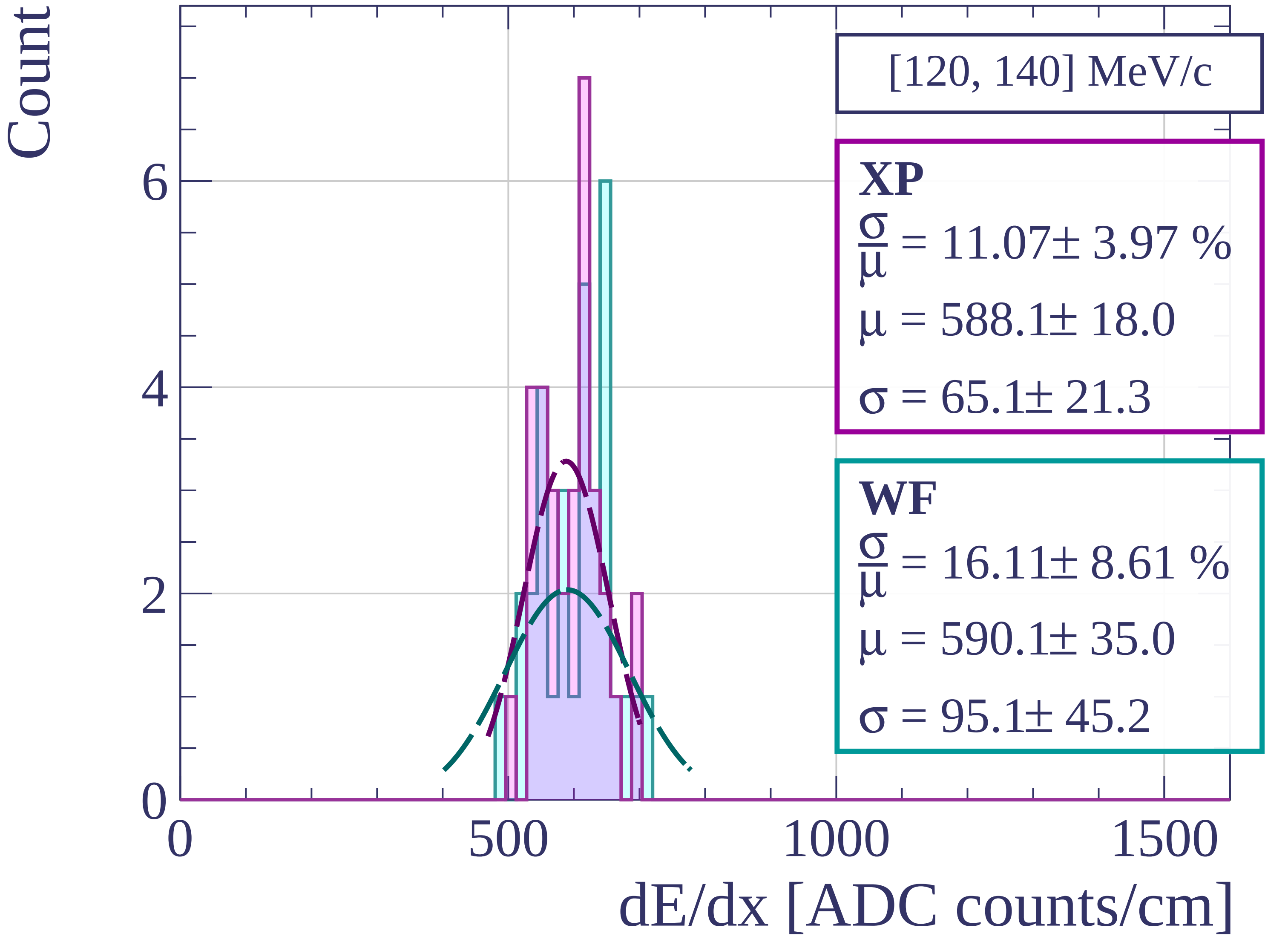
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[60, 80] MeV/c
XP
σ/μ = 10.33± 5.09 %
μ = 570.4± 21.1
σ = 58.9± 26.9
WF
σ/μ = 11.71± 7.36 %
μ = 571.5± 28.3
σ = 66.9± 38.8



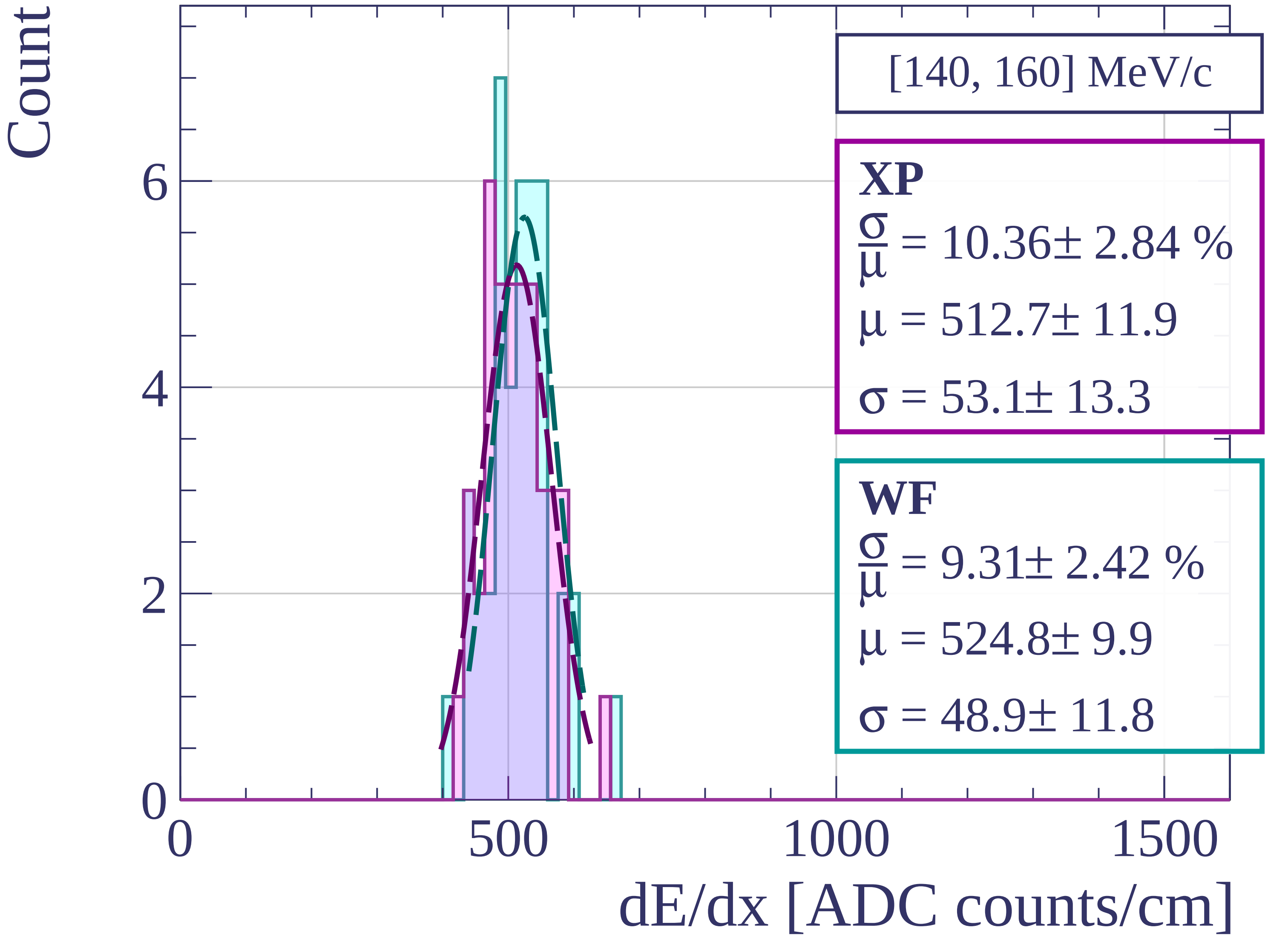
Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 225.16± 1524.58
μ = 832.8± 4870.2
σ = 1875.0± 1730.6
WF
$\frac{\sigma}{\mu}$ = 46.68± 56.86 %
μ = 769.9± 152.3
σ = 359.4± 366.7



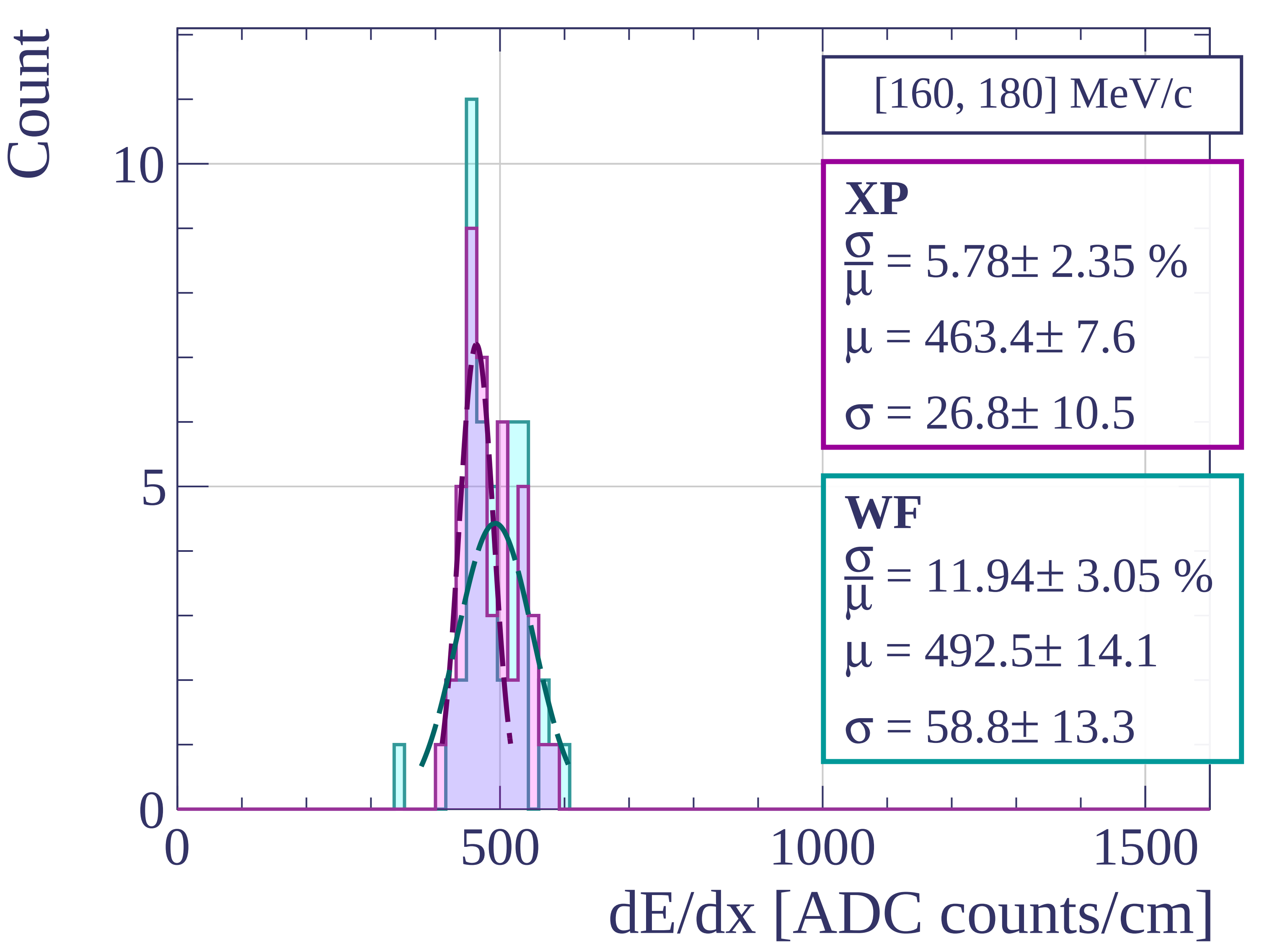
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[100, 120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 23.66± 31.03 %
μ = 2517.5± 1022.3
σ = 595.7± 539.2
WF
$\frac{\sigma}{\mu}$ = 24.05± 25.15 %
μ = 2230.8± 1209.1
σ = 536.5± 270.3



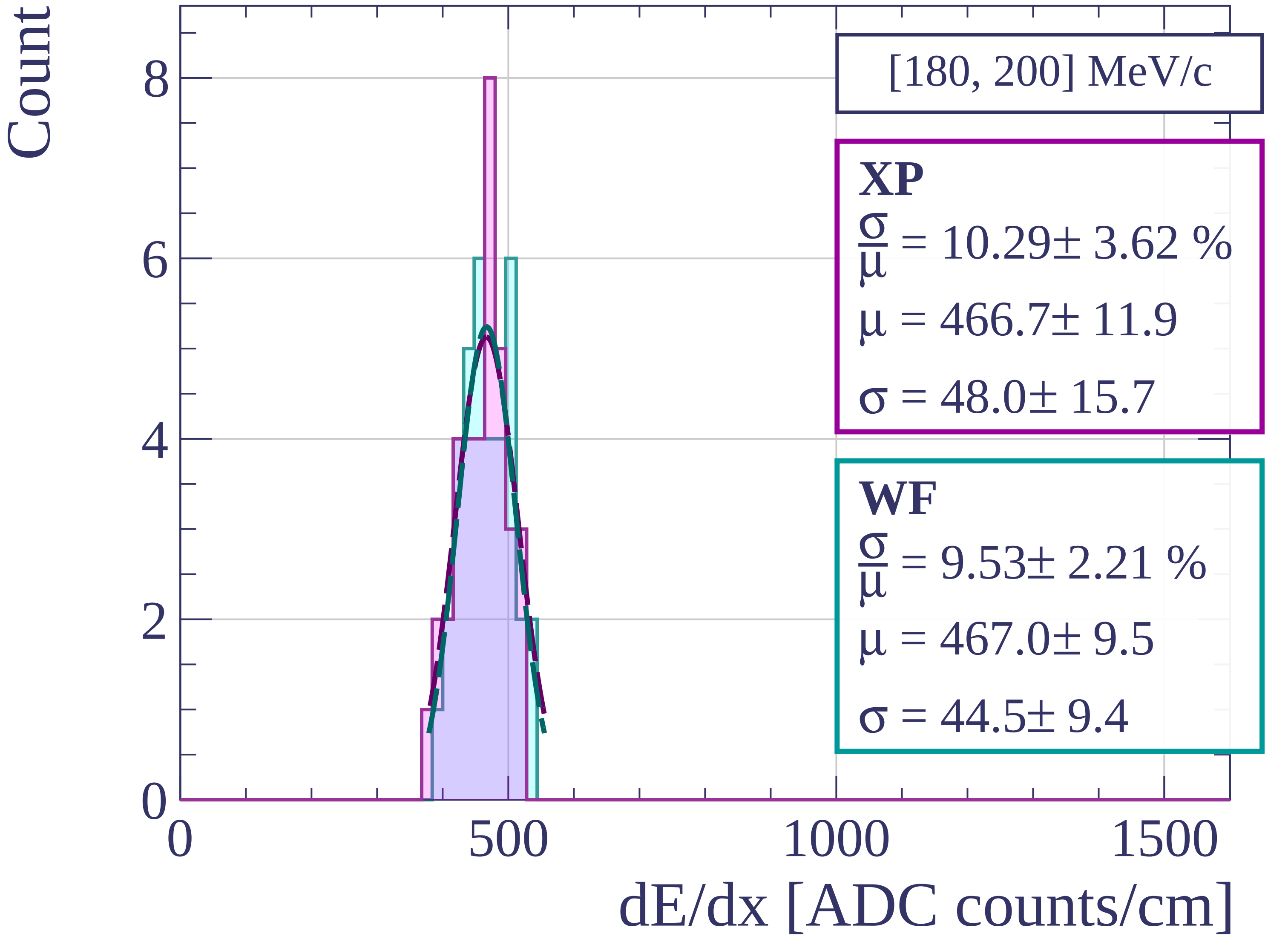
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[120, 140] MeV/c
XP
σ/μ = 11.07± 3.97 %
μ = 588.1± 18.0
σ = 65.1± 21.3
WF
σ/μ = 16.11± 8.61 %
μ = 590.1± 35.0
σ = 95.1± 45.2

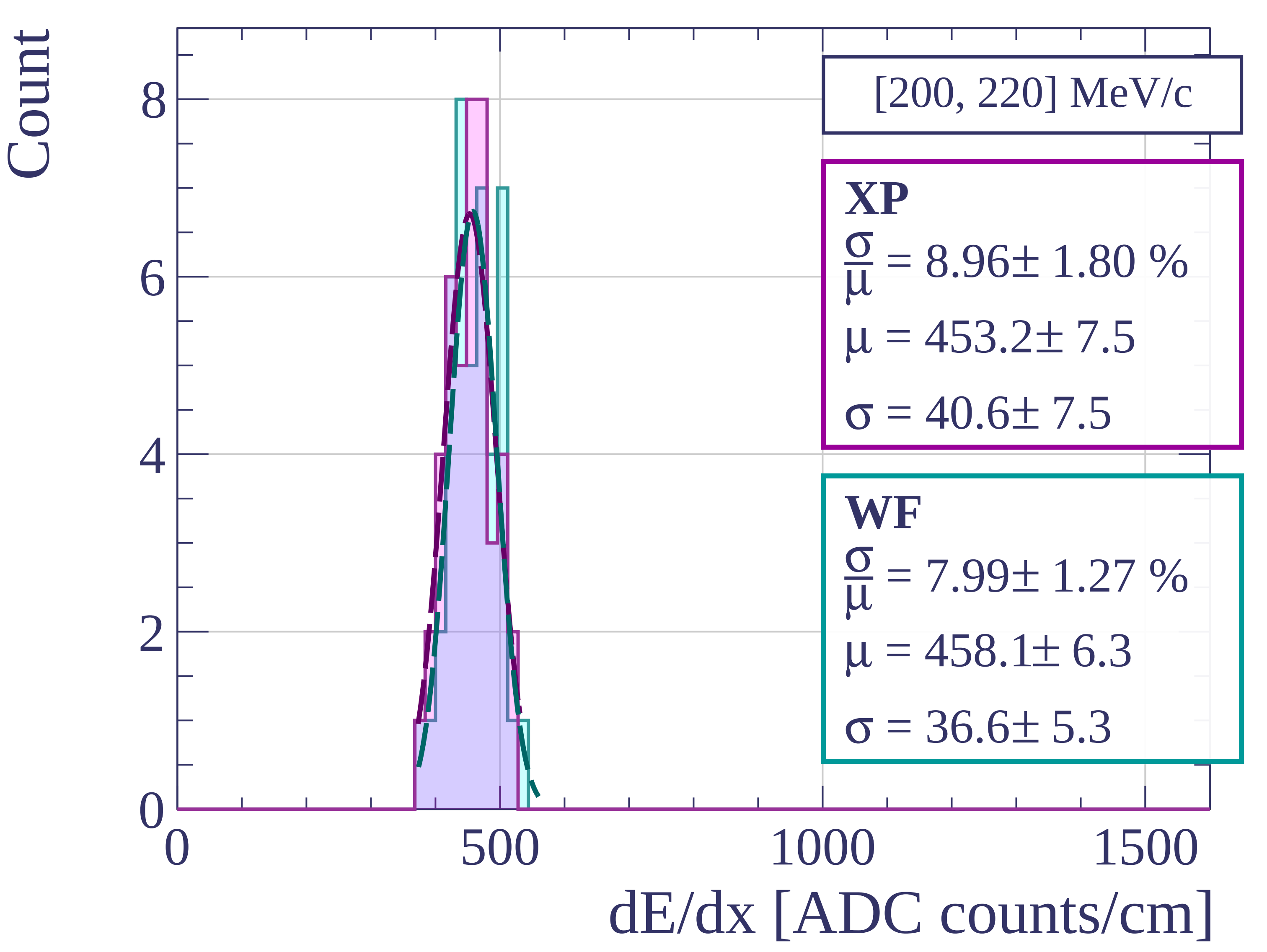


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[140, 160] MeV/c
XP
σ/μ = 10.36± 2.84 %
μ = 512.7± 11.9
σ = 53.1± 13.3
WF
σ/μ = 9.31± 2.42 %
μ = 524.8± 9.9
σ = 48.9± 11.8

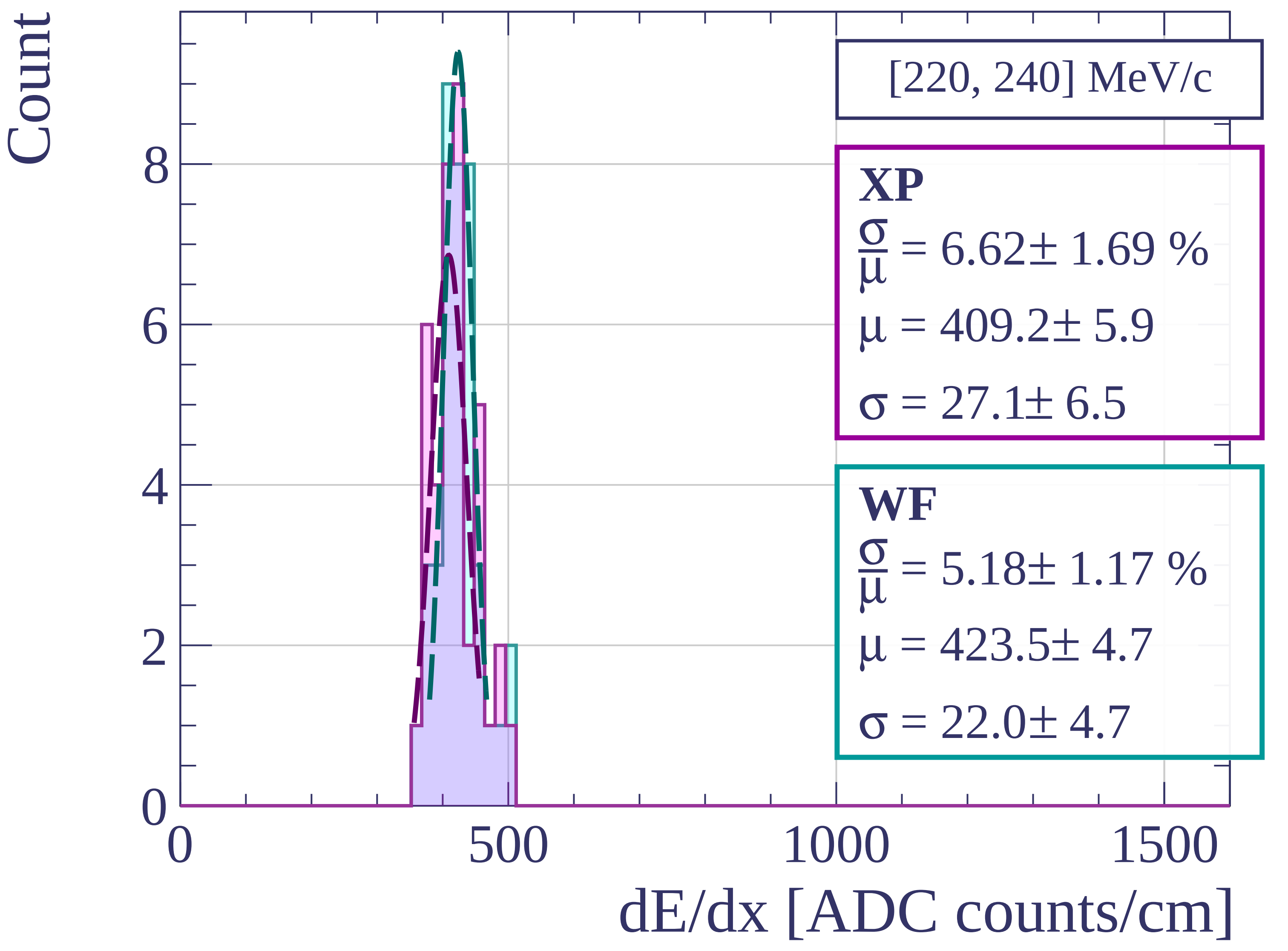


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 180] MeV/c
XP
σ/μ = 5.78± 2.35 %
μ = 463.4± 7.6
σ = 26.8± 10.5
WF
σ/μ = 11.94± 3.05 %
μ = 492.5± 14.1
σ = 58.8± 13.3

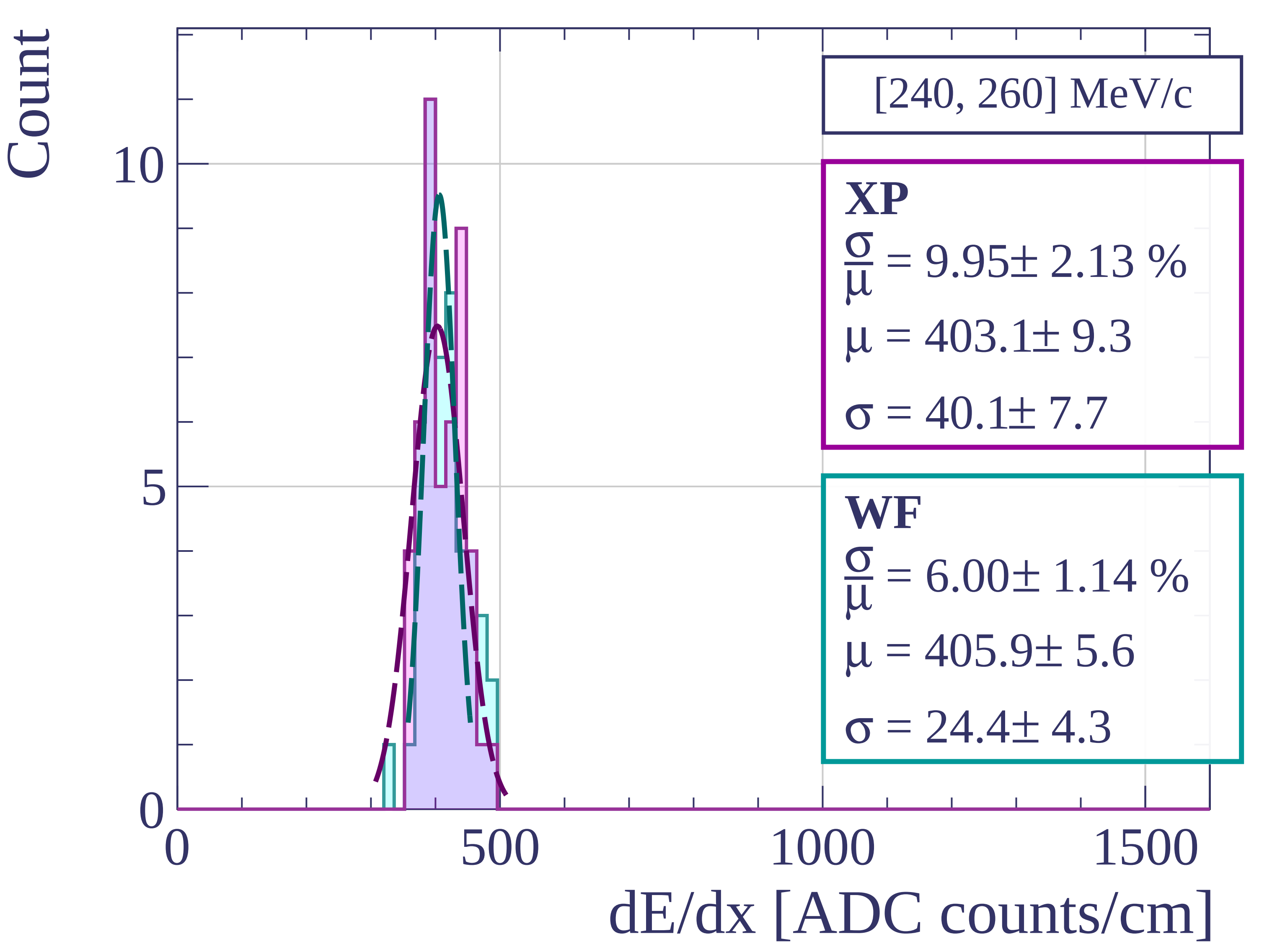

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[180, 200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.29± 3.62 %
μ = 466.7± 11.9
σ = 48.0± 15.7
WF
$\frac{\sigma}{\mu}$ = 9.53± 2.21 %
μ = 467.0± 9.5
σ = 44.5± 9.4

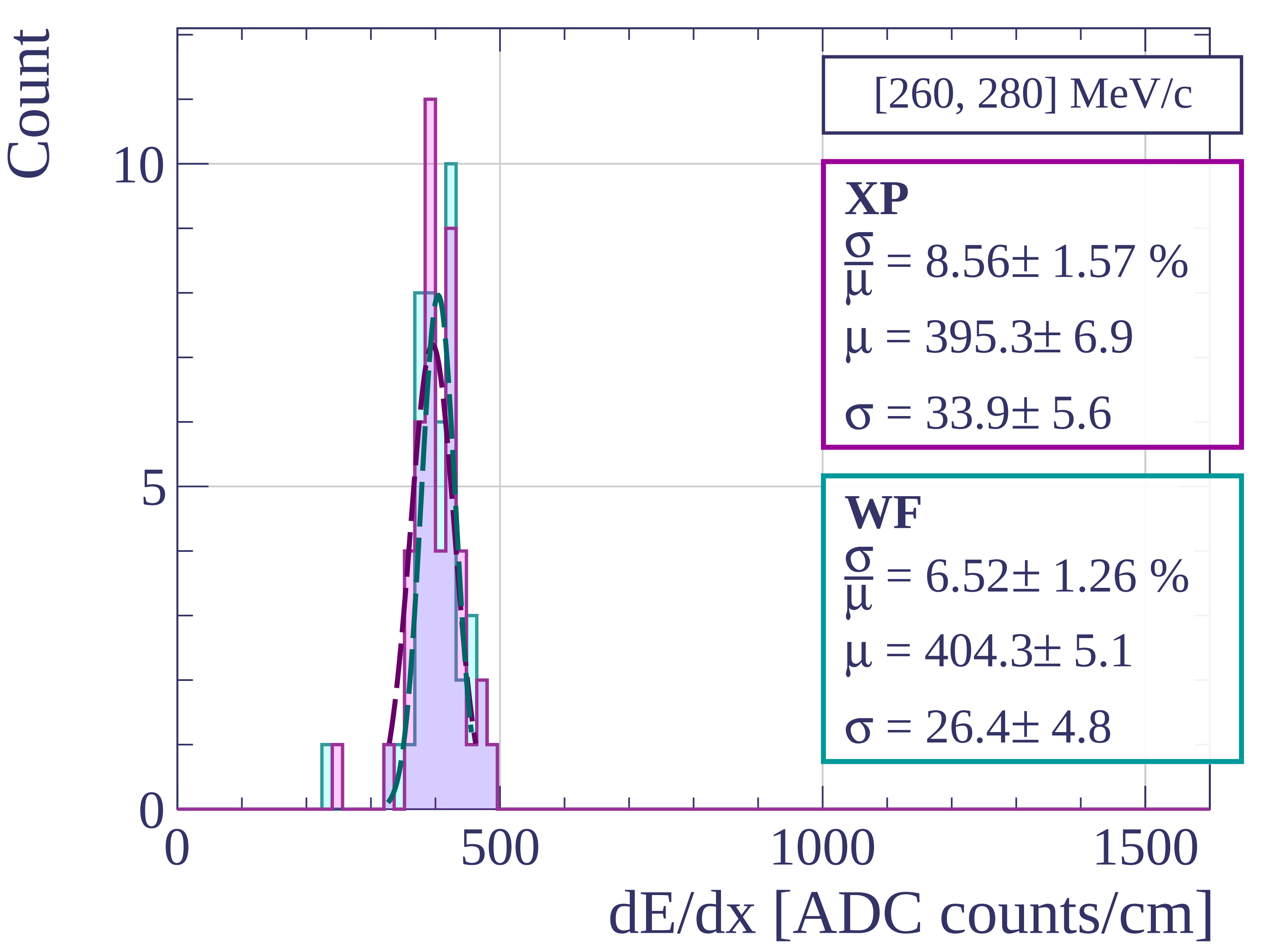


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[200, 220] MeV/c
XP
σ/μ = 8.96± 1.80 %
μ = 453.2± 7.5
σ = 40.6± 7.5
WF
σ/μ = 7.99± 1.27 %
μ = 458.1± 6.3
σ = 36.6± 5.3

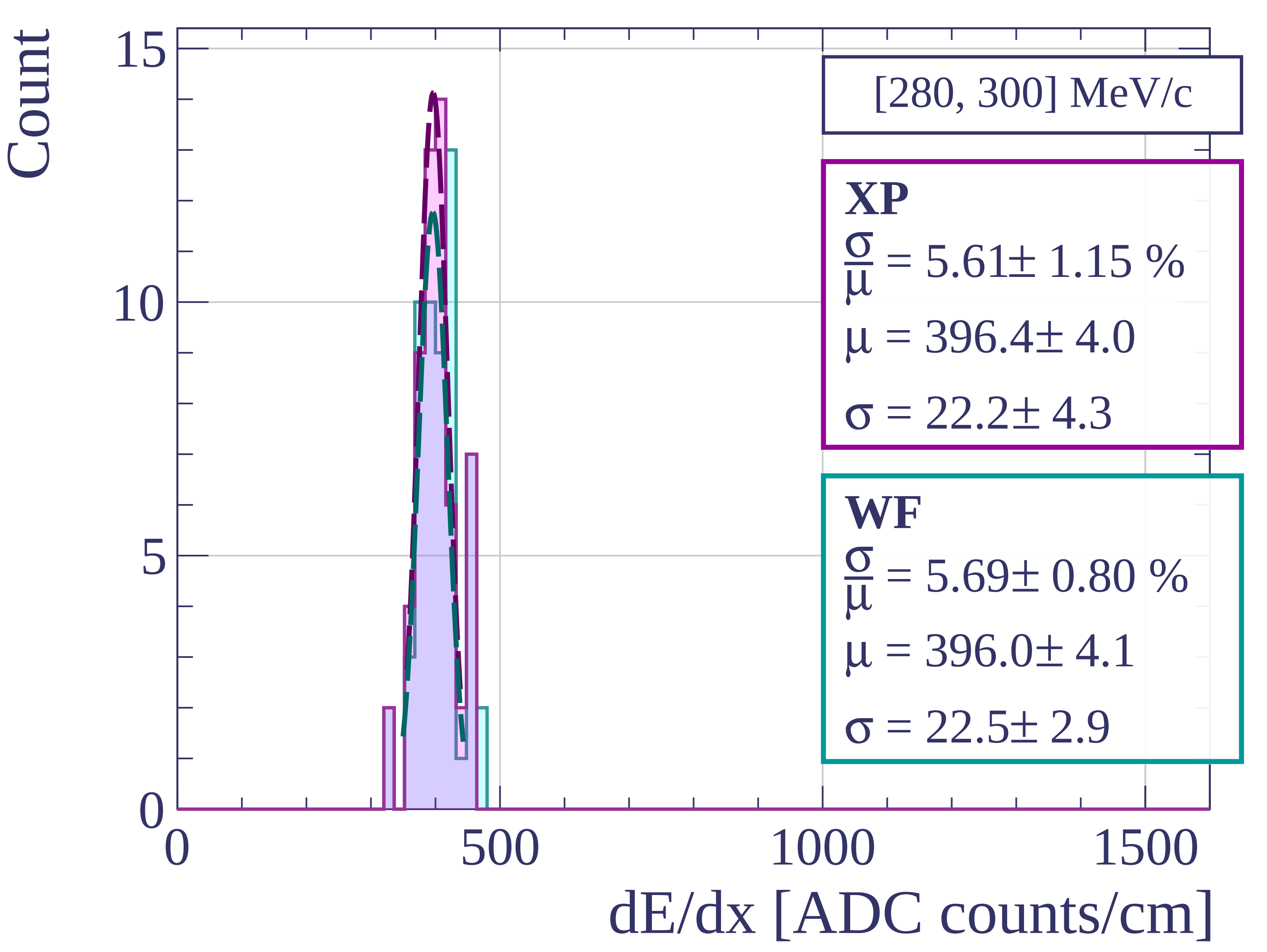


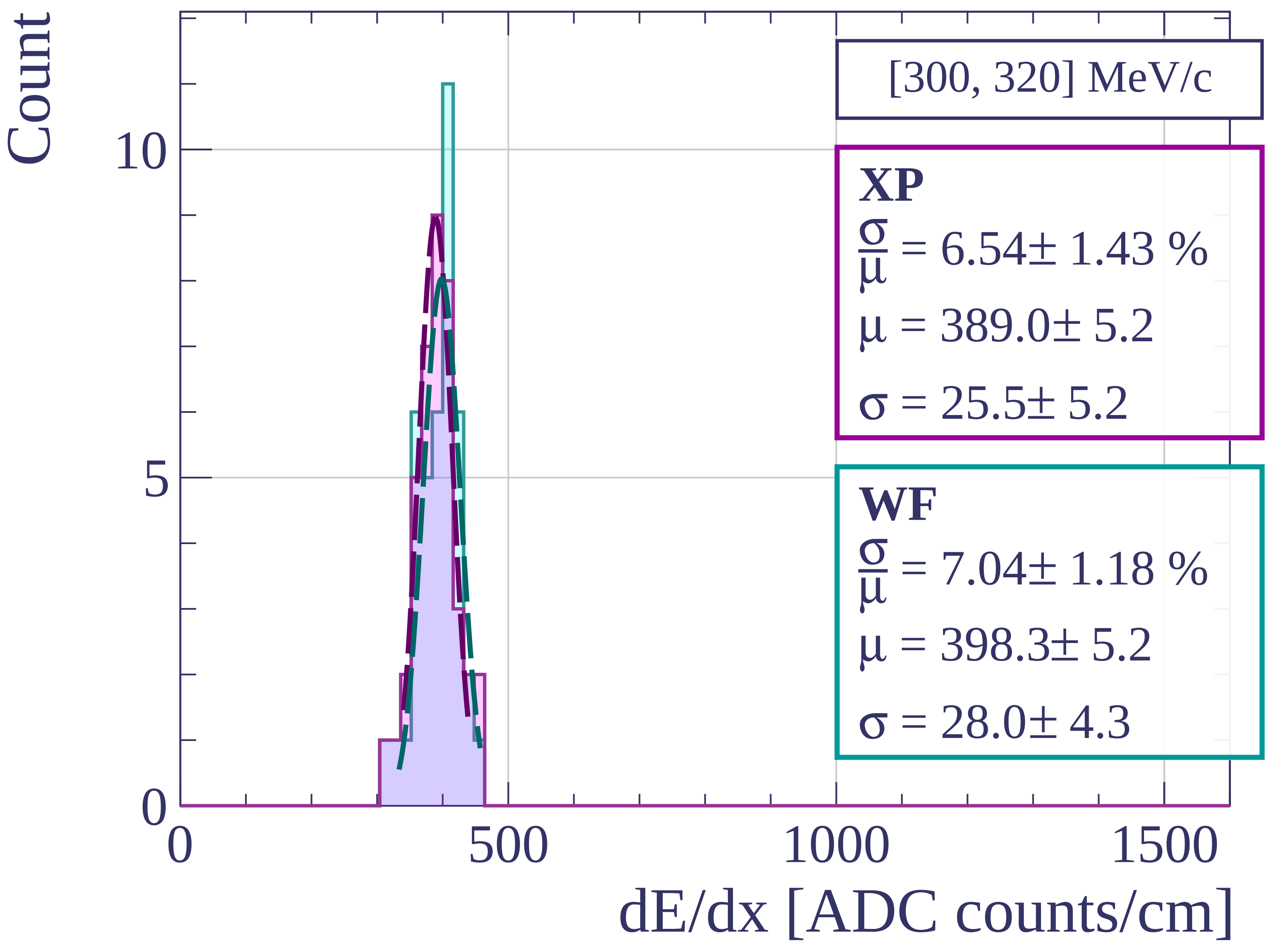
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[220, 240] MeV/c
XP
σ/μ = 6.62± 1.69 %
μ = 409.2± 5.9
σ = 27.1± 6.5
WF
σ/μ = 5.18± 1.17 %
μ = 423.5± 4.7
σ = 22.0± 4.7


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.95± 2.13 %
μ = 403.1± 9.3
σ = 40.1± 7.7
WF
$\frac{\sigma}{\mu}$ = 6.00± 1.14 %
μ = 405.9± 5.6
σ = 24.4± 4.3

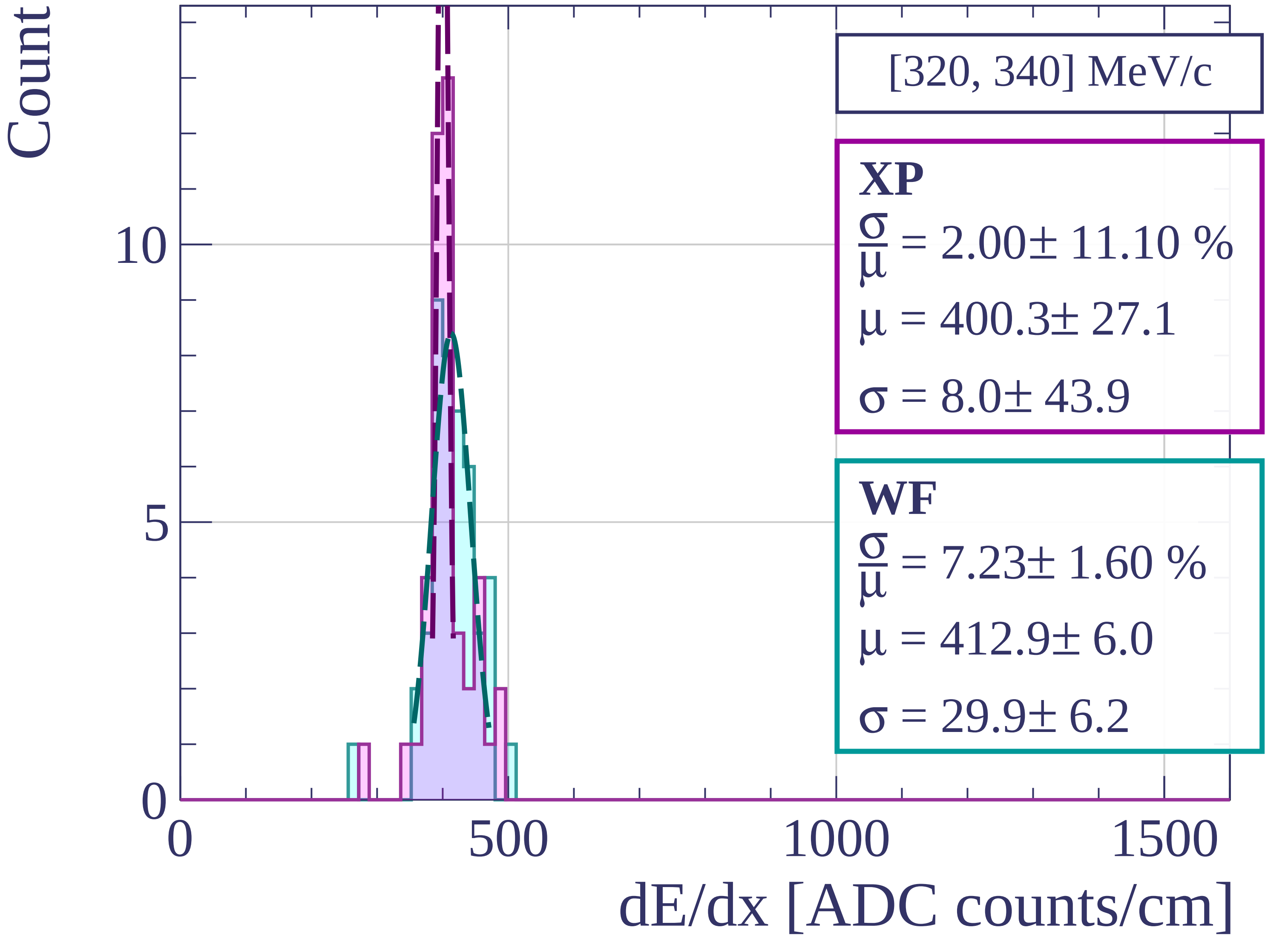


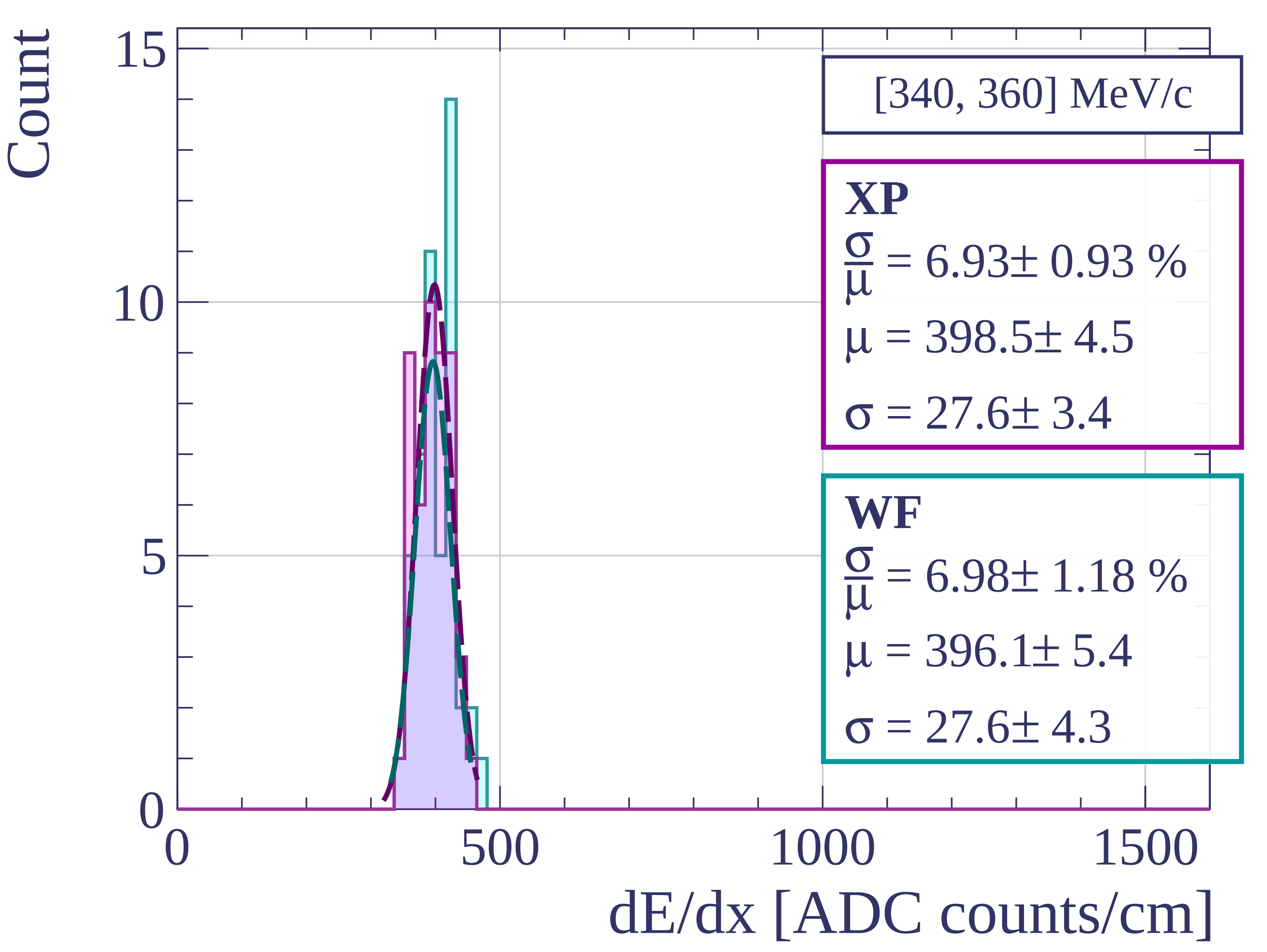
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[260, 280] MeV/c
XP
σ/μ = 8.56± 1.57 %
μ = 395.3± 6.9
σ = 33.9± 5.6
WF
σ/μ = 6.52± 1.26 %
μ = 404.3± 5.1
σ = 26.4± 4.8

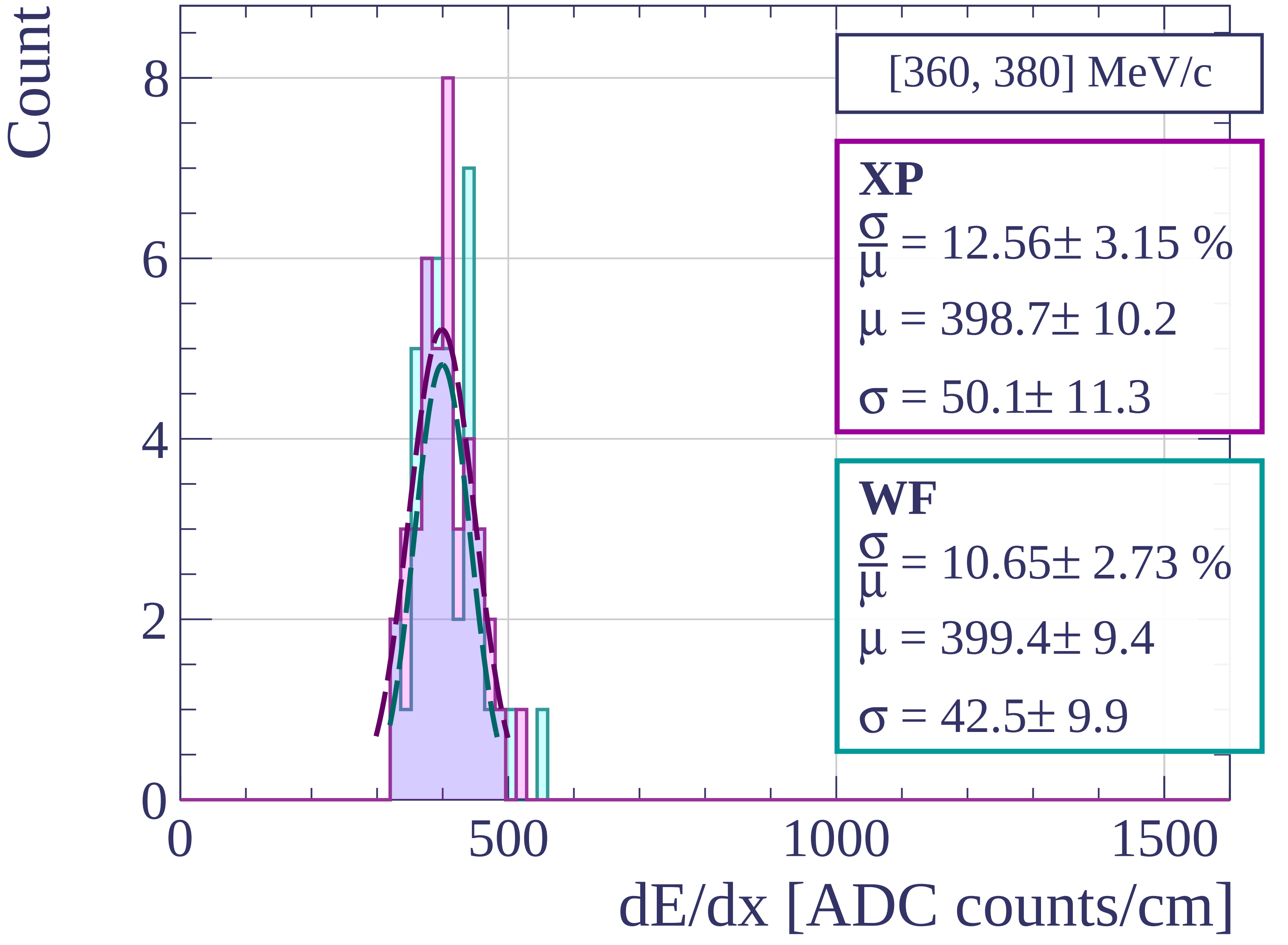

Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[280, 300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.61± 1.15 %
μ = 396.4± 4.0
σ = 22.2± 4.3
WF
$\frac{\sigma}{\mu}$ = 5.69± 0.80 %
μ = 396.0± 4.1
σ = 22.5± 2.9



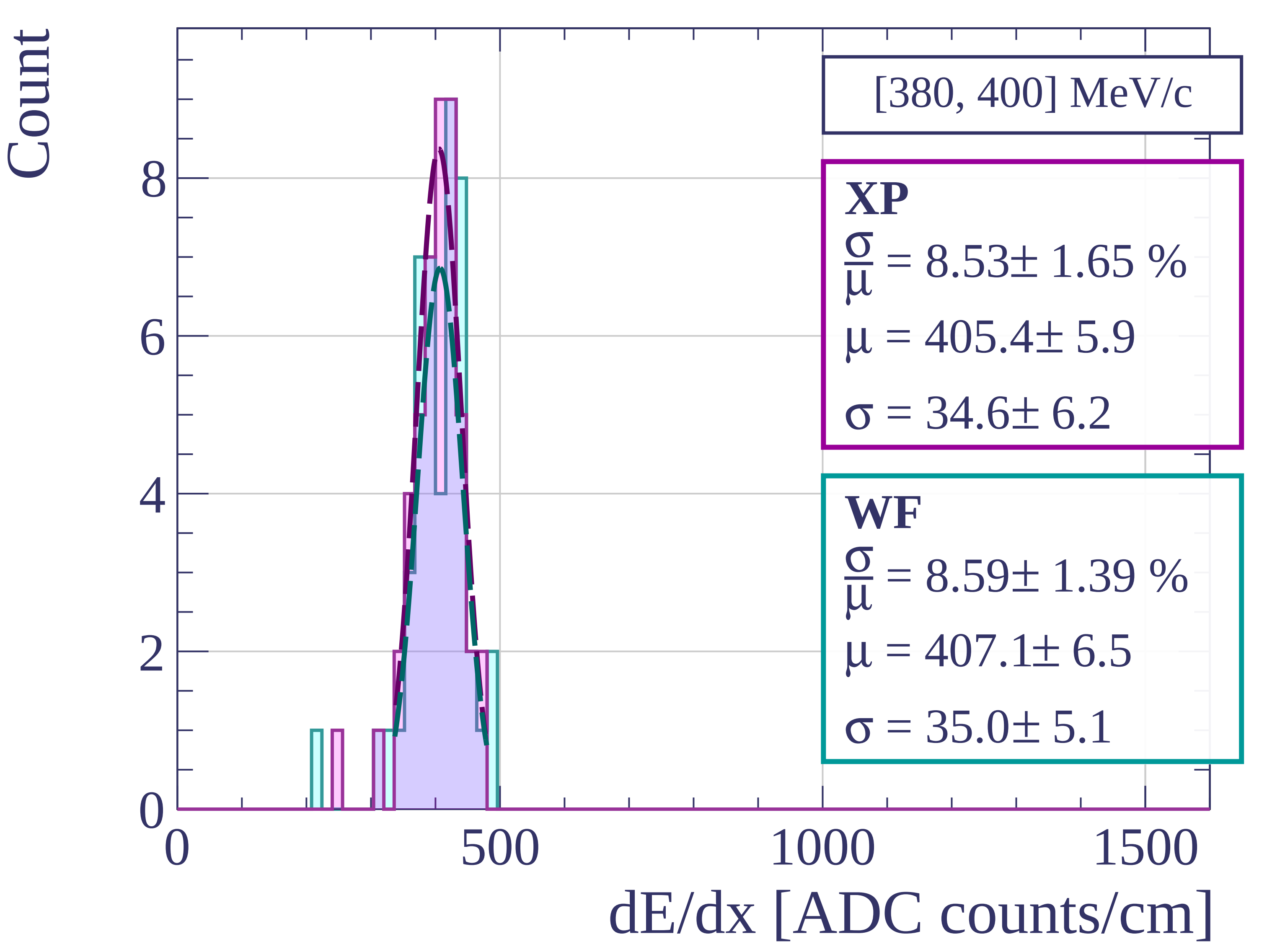
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[300, 320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.54± 1.43 %
μ = 389.0± 5.2
σ = 25.5± 5.2
WF
$\frac{\sigma}{\mu}$ = 7.04± 1.18 %
μ = 398.3± 5.2
σ = 28.0± 4.3


Count
[320, 340] MeV/c
XP
σ/μ = 2.00± 11.10 %
μ = 400.3± 27.1
σ = 8.0± 43.9
WF
σ/μ = 7.23± 1.60 %
μ = 412.9± 6.0
σ = 29.9± 6.2
0
5
10
0
500
1000
1500
dE/dx [ADC counts/cm]

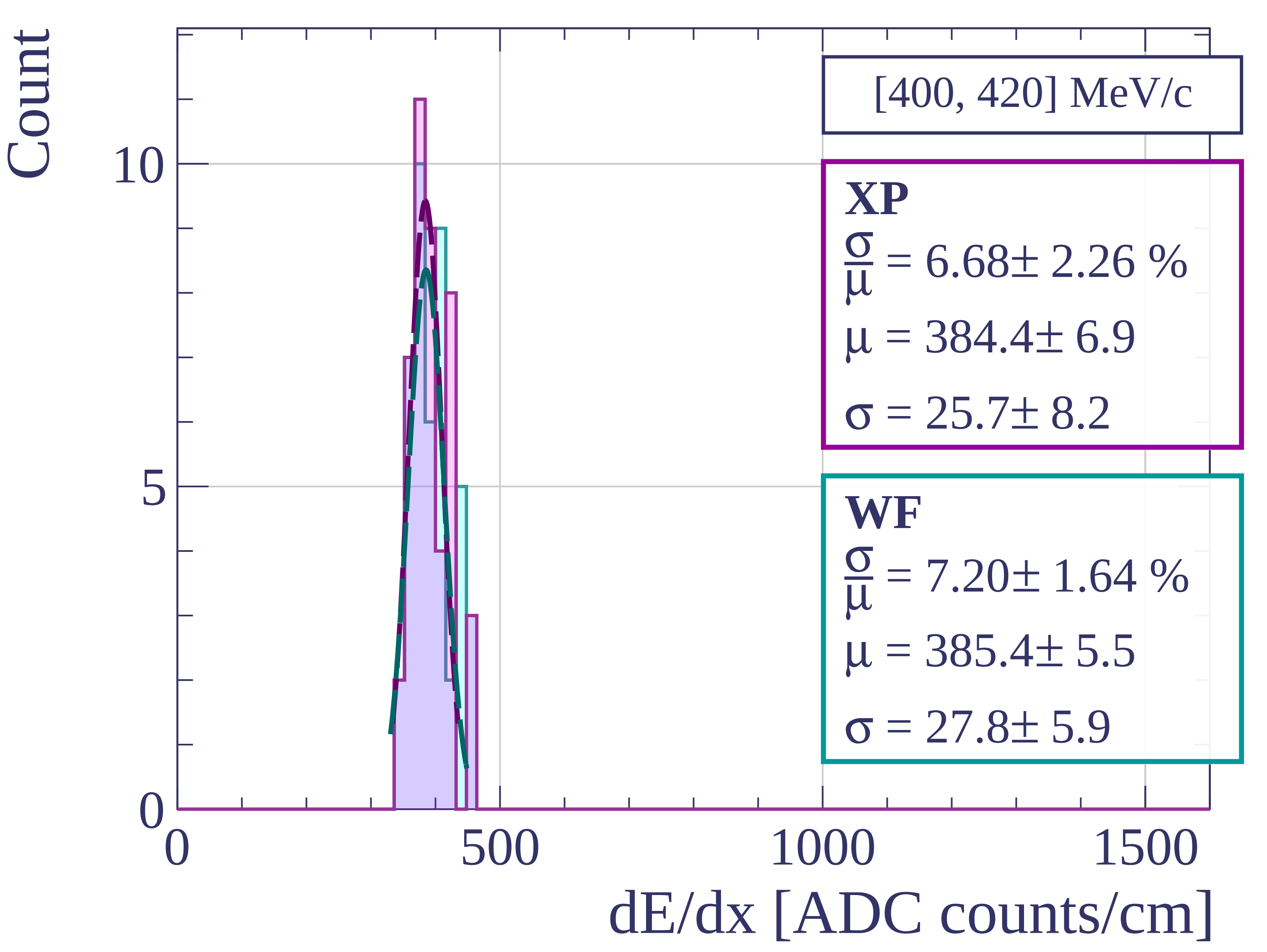

Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[340, 360] MeV/c
XP
σ/μ = 6.93± 0.93 %
μ = 398.5± 4.5
σ = 27.6± 3.4
WF
σ/μ = 6.98± 1.18 %
μ = 396.1± 5.4
σ = 27.6± 4.3

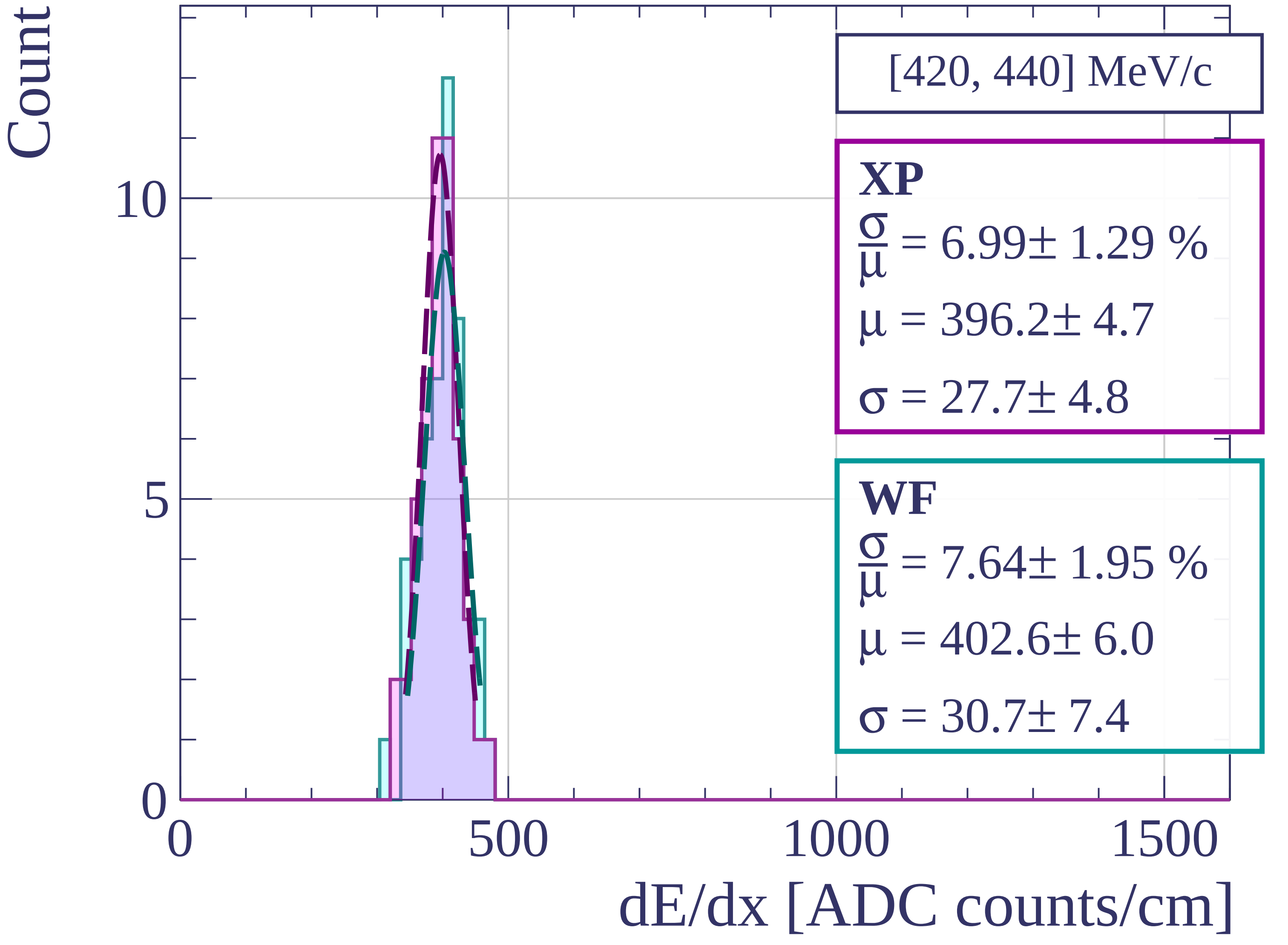


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[360, 380] MeV/c
XP
σ/μ = 12.56± 3.15 %
μ = 398.7± 10.2
σ = 50.1± 11.3
WF
σ/μ = 10.65± 2.73 %
μ = 399.4± 9.4
σ = 42.5± 9.9

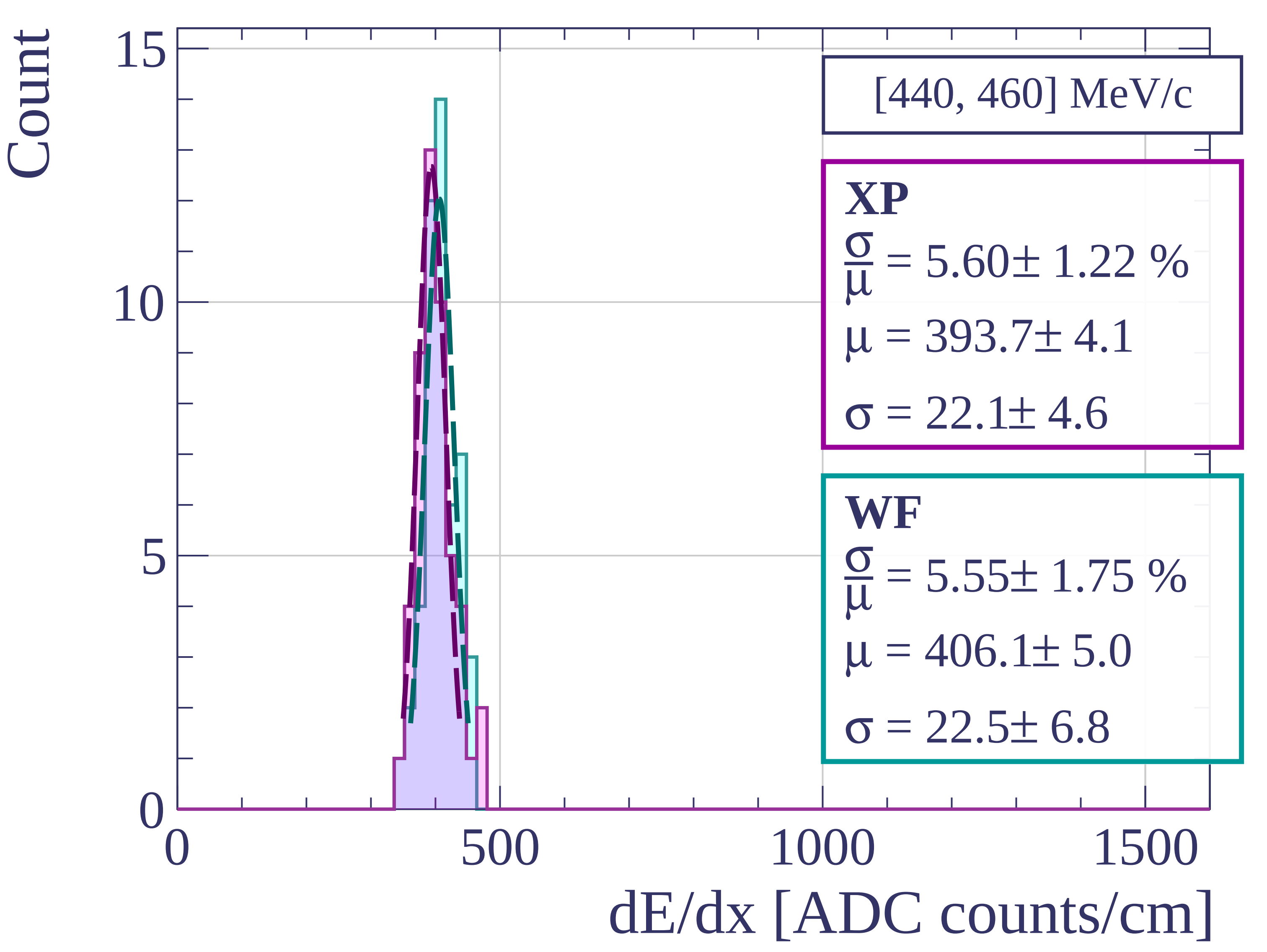


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[380, 400] MeV/c
XP
σ/μ = 8.53± 1.65 %
μ = 405.4± 5.9
σ = 34.6± 6.2
WF
σ/μ = 8.59± 1.39 %
μ = 407.1± 6.5
σ = 35.0± 5.1

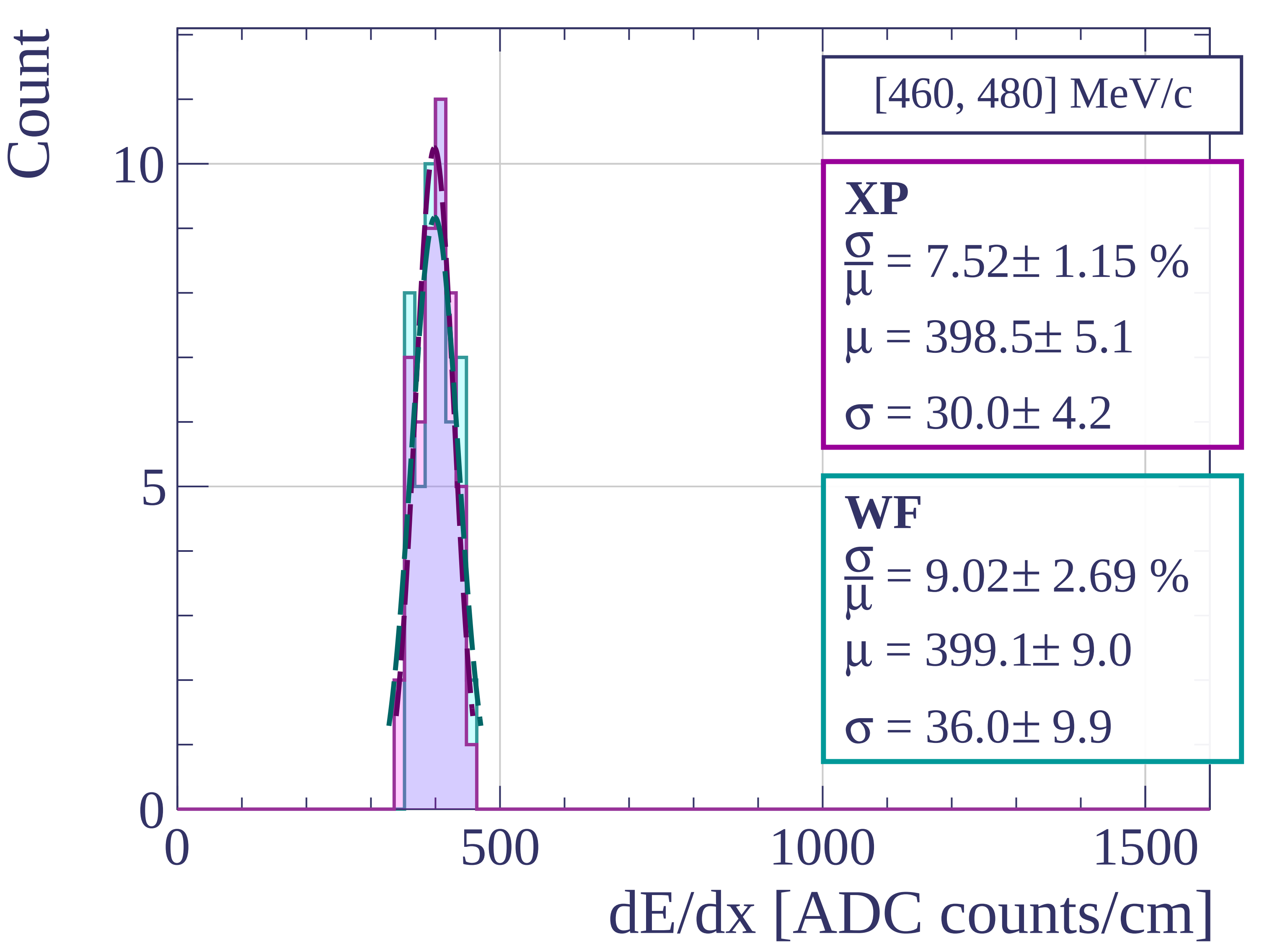

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.68± 2.26 %
μ = 384.4± 6.9
σ = 25.7± 8.2
WF
$\frac{\sigma}{\mu}$ = 7.20± 1.64 %
μ = 385.4± 5.5
σ = 27.8± 5.9



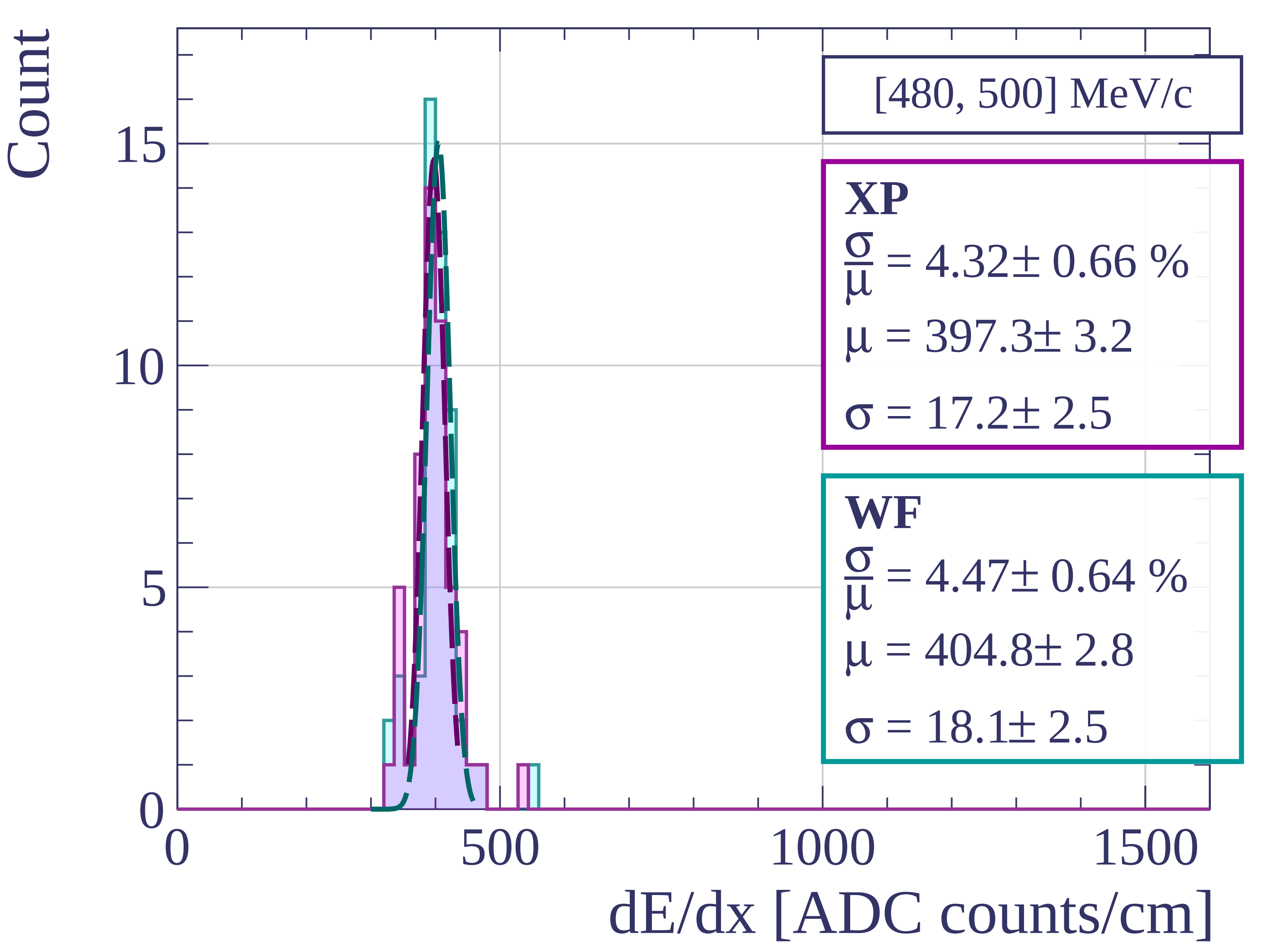
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[420, 440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.99± 1.29 %
μ = 396.2± 4.7
σ = 27.7± 4.8
WF
$\frac{\sigma}{\mu}$ = 7.64± 1.95 %
μ = 402.6± 6.0
σ = 30.7± 7.4



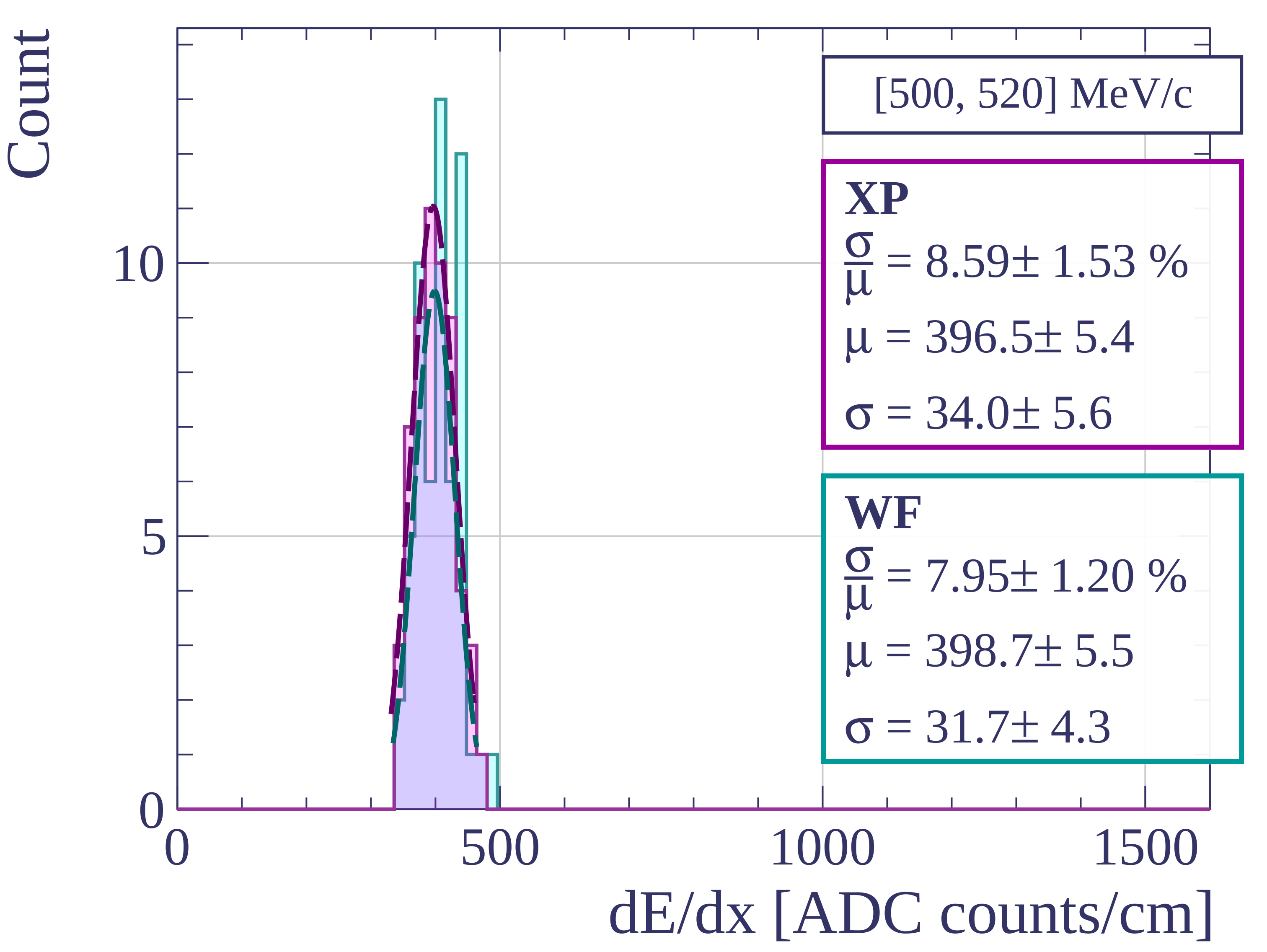
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[440, 460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.60± 1.22 %
μ = 393.7± 4.1
σ = 22.1± 4.6
WF
$\frac{\sigma}{\mu}$ = 5.55± 1.75 %
μ = 406.1± 5.0
σ = 22.5± 6.8



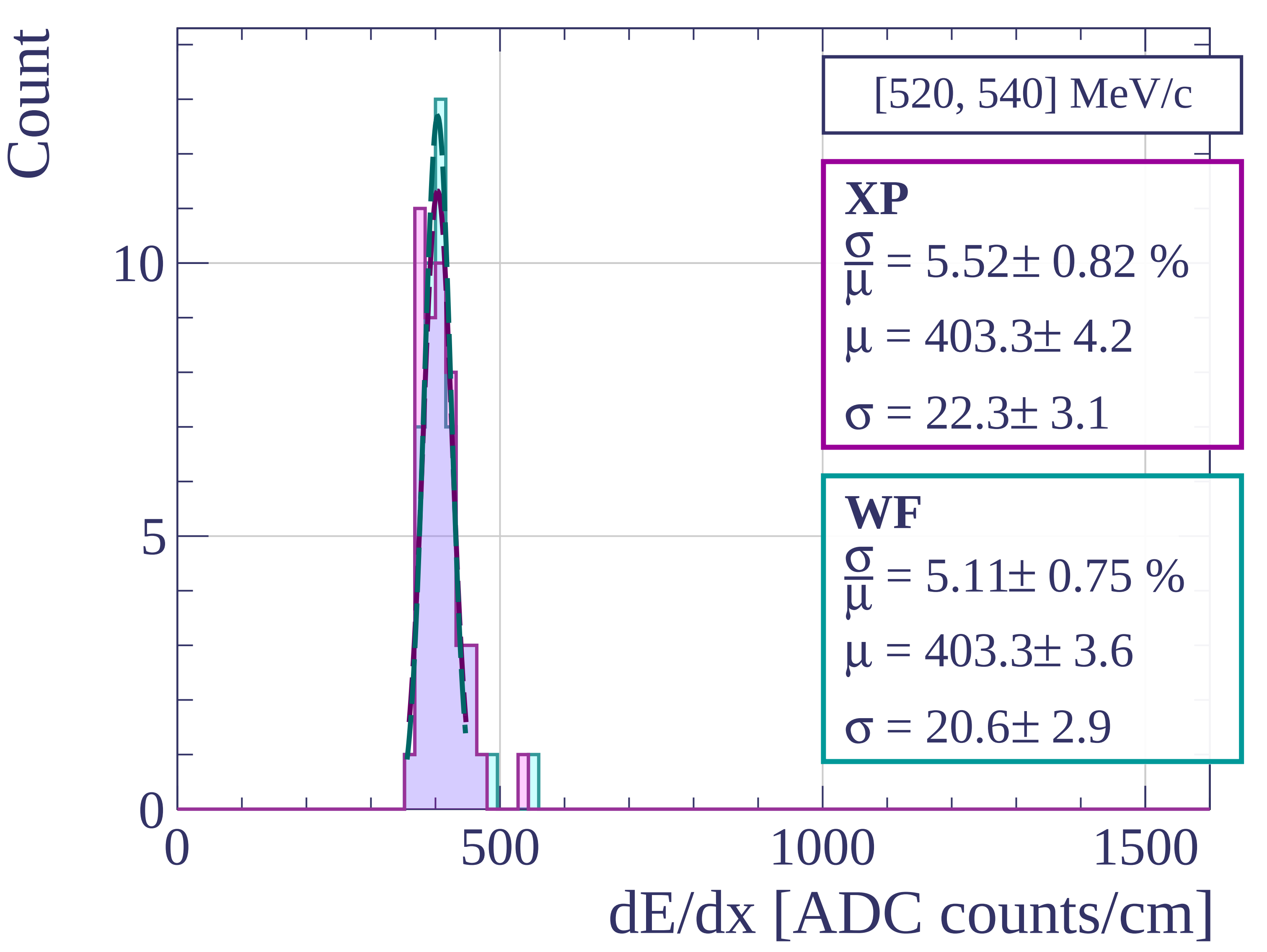
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[460, 480] MeV/c
XP
σ/μ = 7.52± 1.15 %
μ = 398.5± 5.1
σ = 30.0± 4.2
WF
σ/μ = 9.02± 2.69 %
μ = 399.1± 9.0
σ = 36.0± 9.9



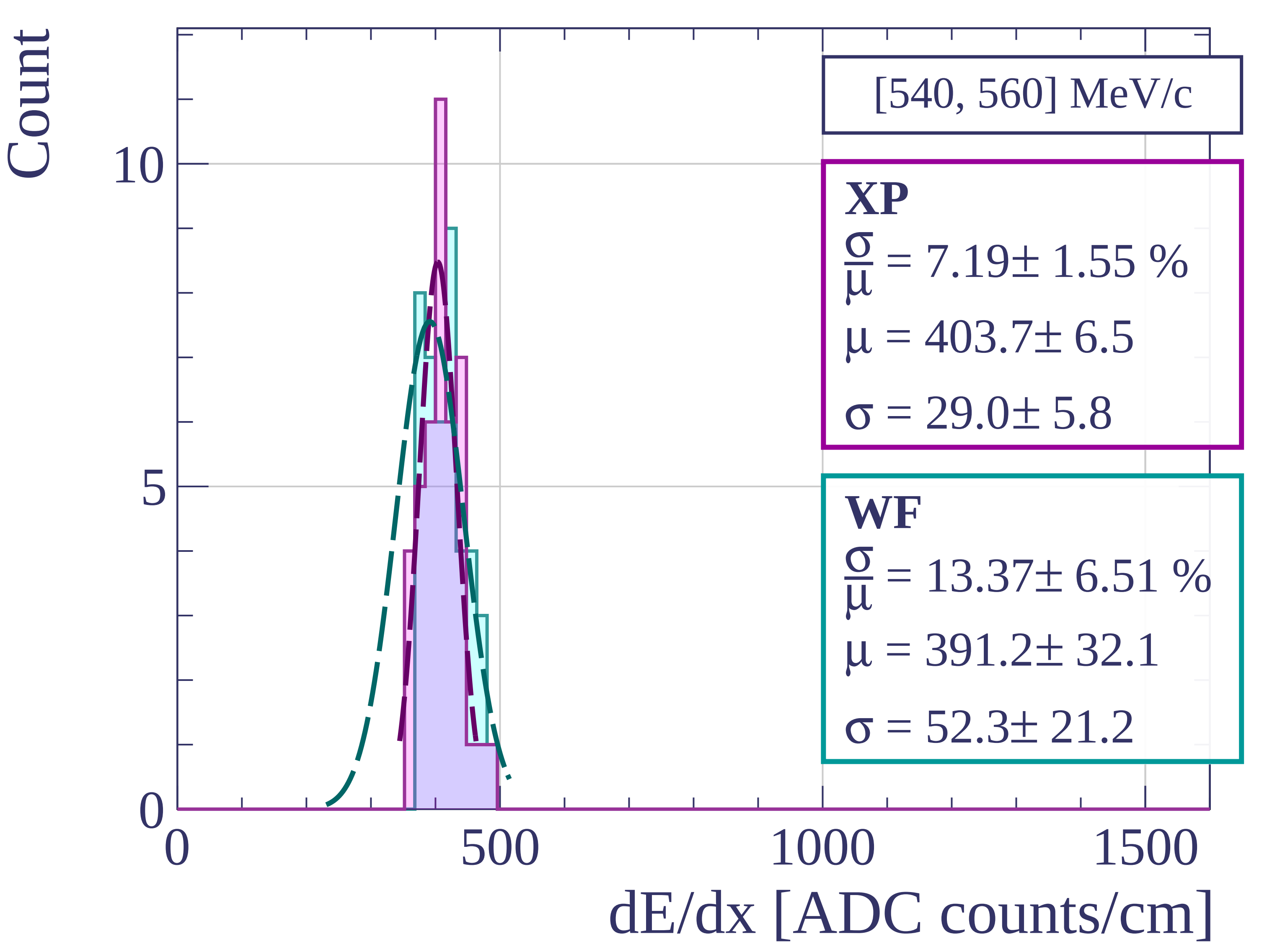
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.32± 0.66 %
μ = 397.3± 3.2
σ = 17.2± 2.5
WF
$\frac{\sigma}{\mu}$ = 4.47± 0.64 %
μ = 404.8± 2.8
σ = 18.1± 2.5



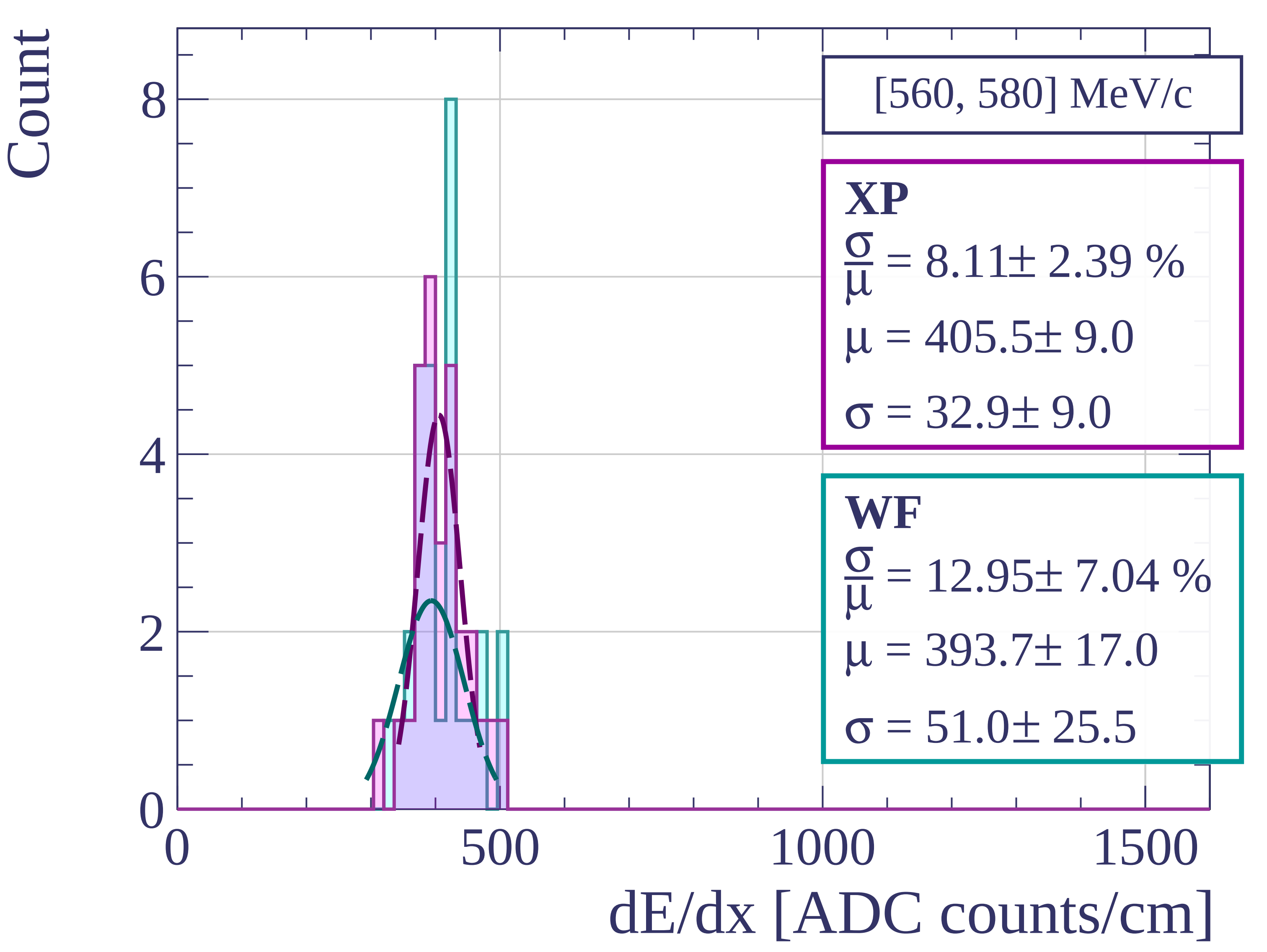
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
[500, 520] MeV/c
XP
σ/μ = 8.59± 1.53 %
μ = 396.5± 5.4
σ = 34.0± 5.6
WF
σ/μ = 7.95± 1.20 %
μ = 398.7± 5.5
σ = 31.7± 4.3

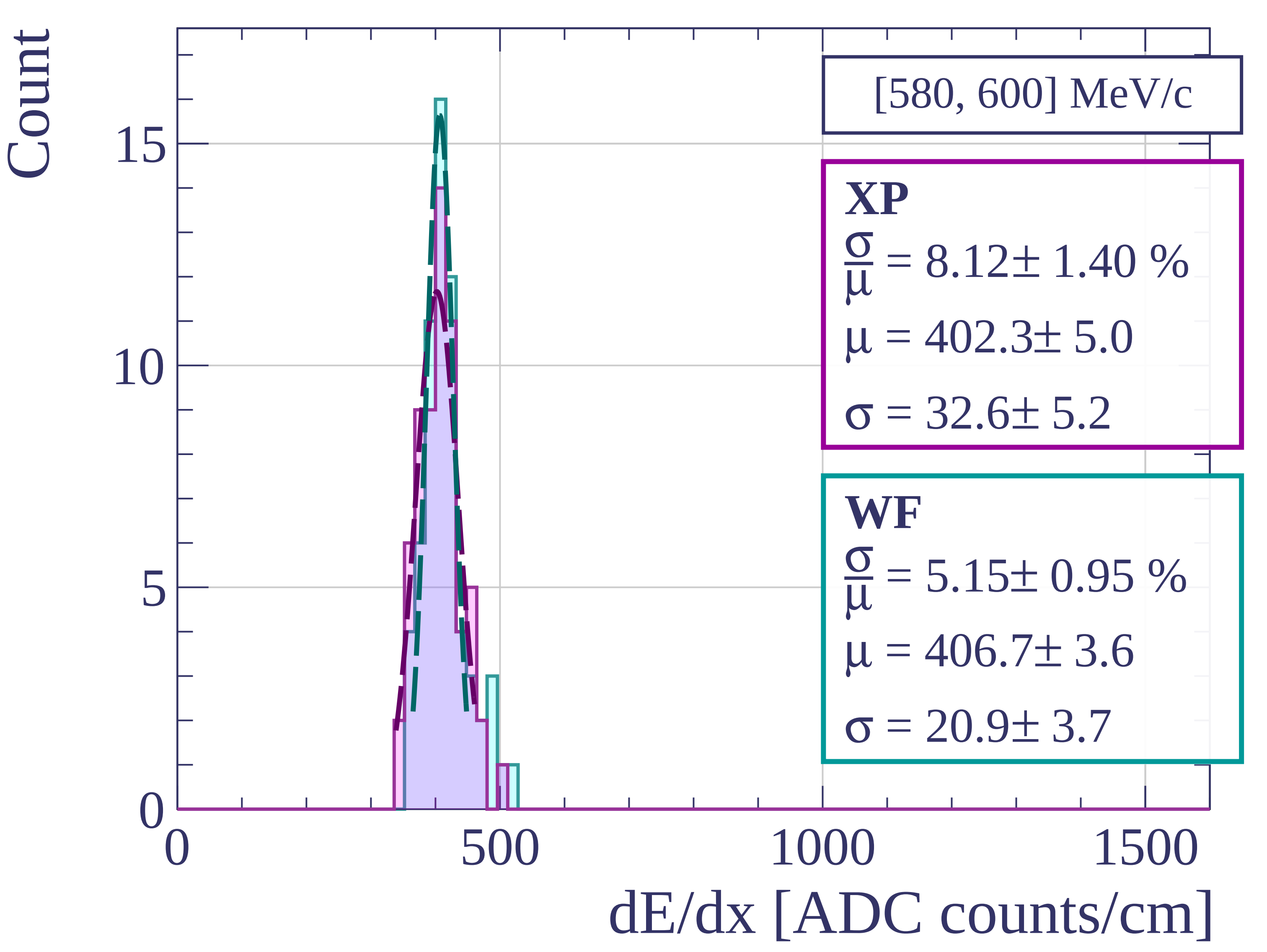


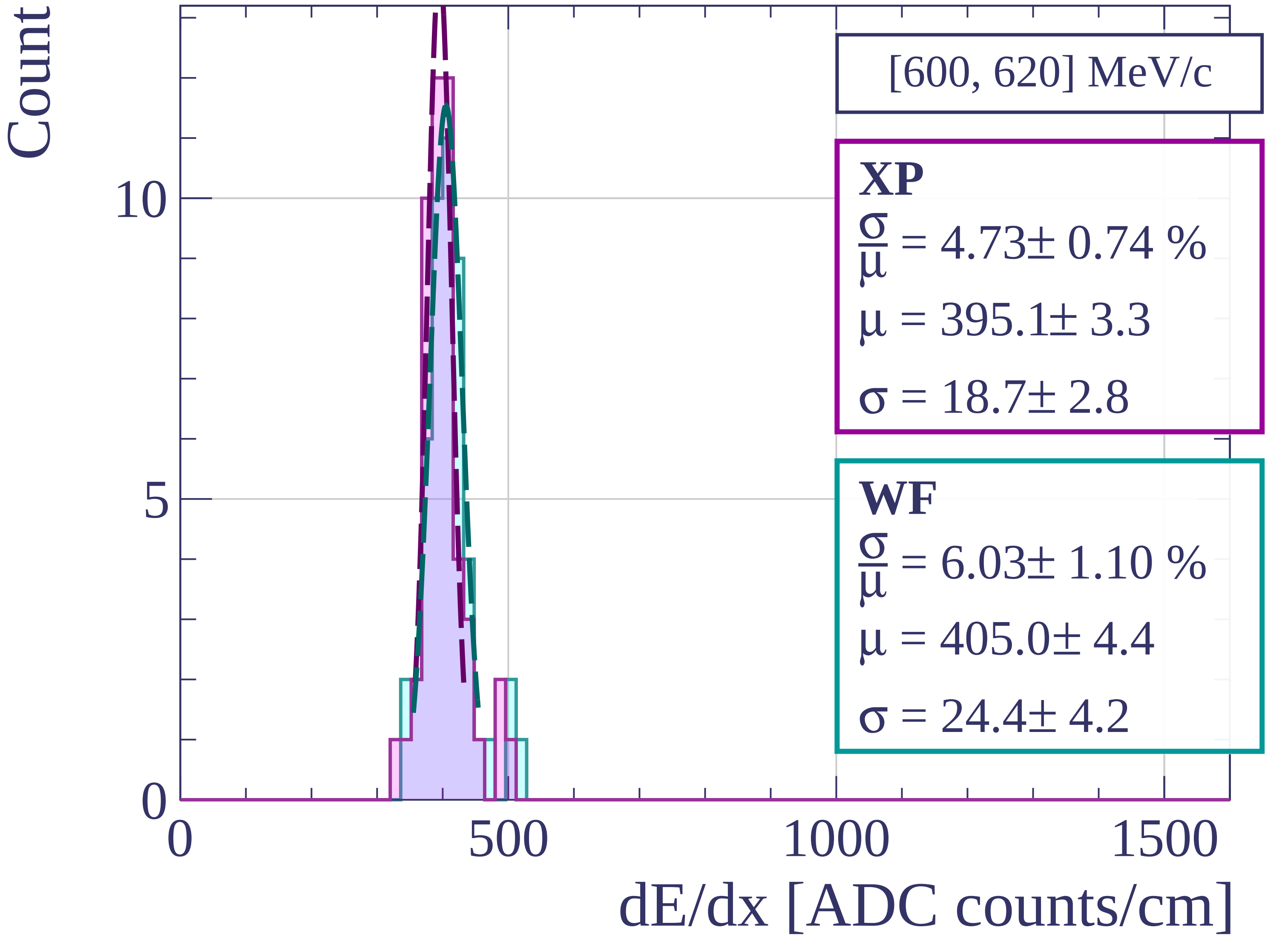
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[520, 540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.52± 0.82 %
μ = 403.3± 4.2
σ = 22.3± 3.1
WF
$\frac{\sigma}{\mu}$ = 5.11± 0.75 %
μ = 403.3± 3.6
σ = 20.6± 2.9



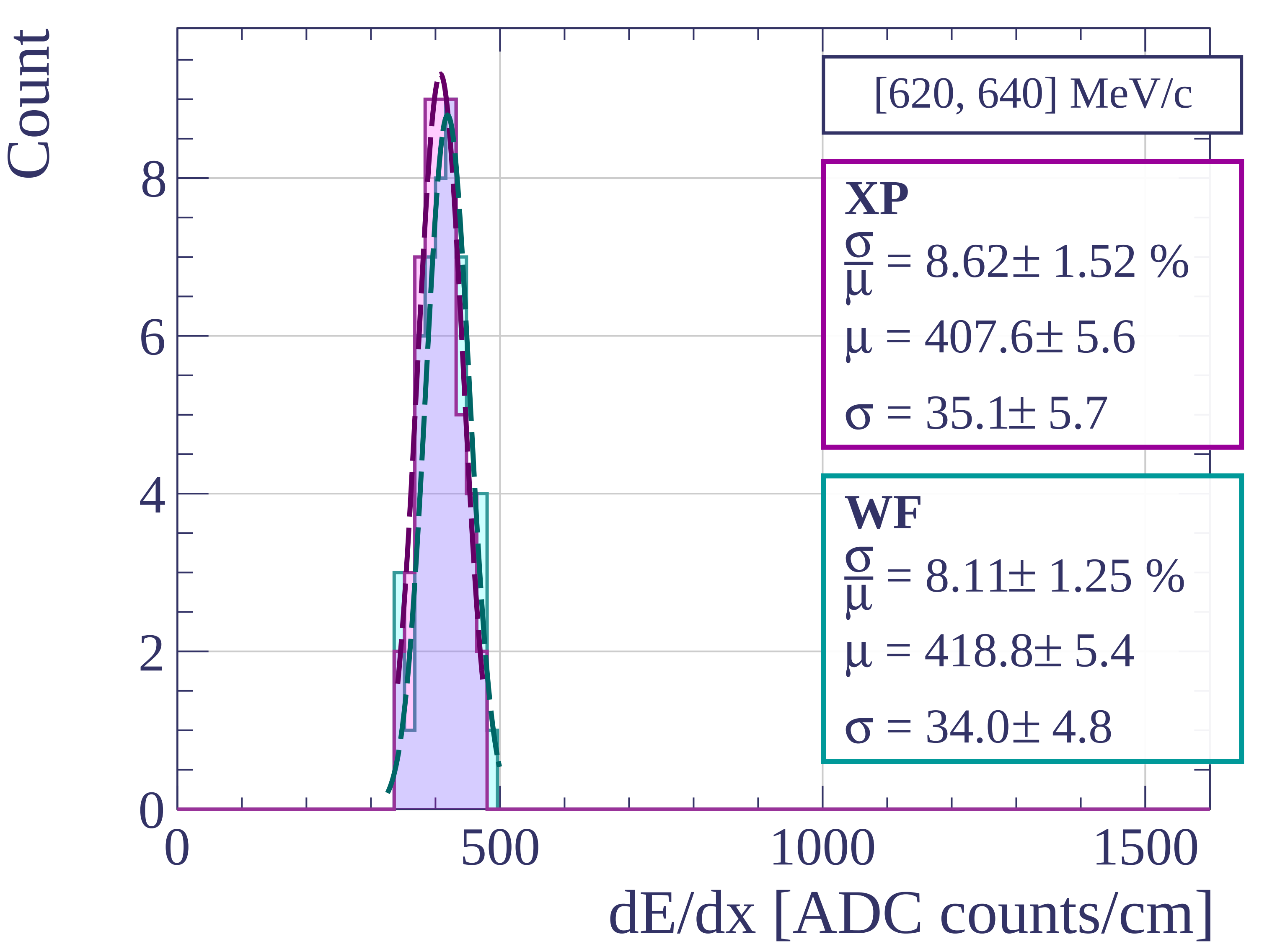
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[540, 560] MeV/c
XP
σ/μ = 7.19± 1.55 %
μ = 403.7± 6.5
σ = 29.0± 5.8
WF
σ/μ = 13.37± 6.51 %
μ = 391.2± 32.1
σ = 52.3± 21.2


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 580] MeV/c
XP
σ/μ = 8.11± 2.39 %
μ = 405.5± 9.0
σ = 32.9± 9.0
WF
σ/μ = 12.95± 7.04 %
μ = 393.7± 17.0
σ = 51.0± 25.5

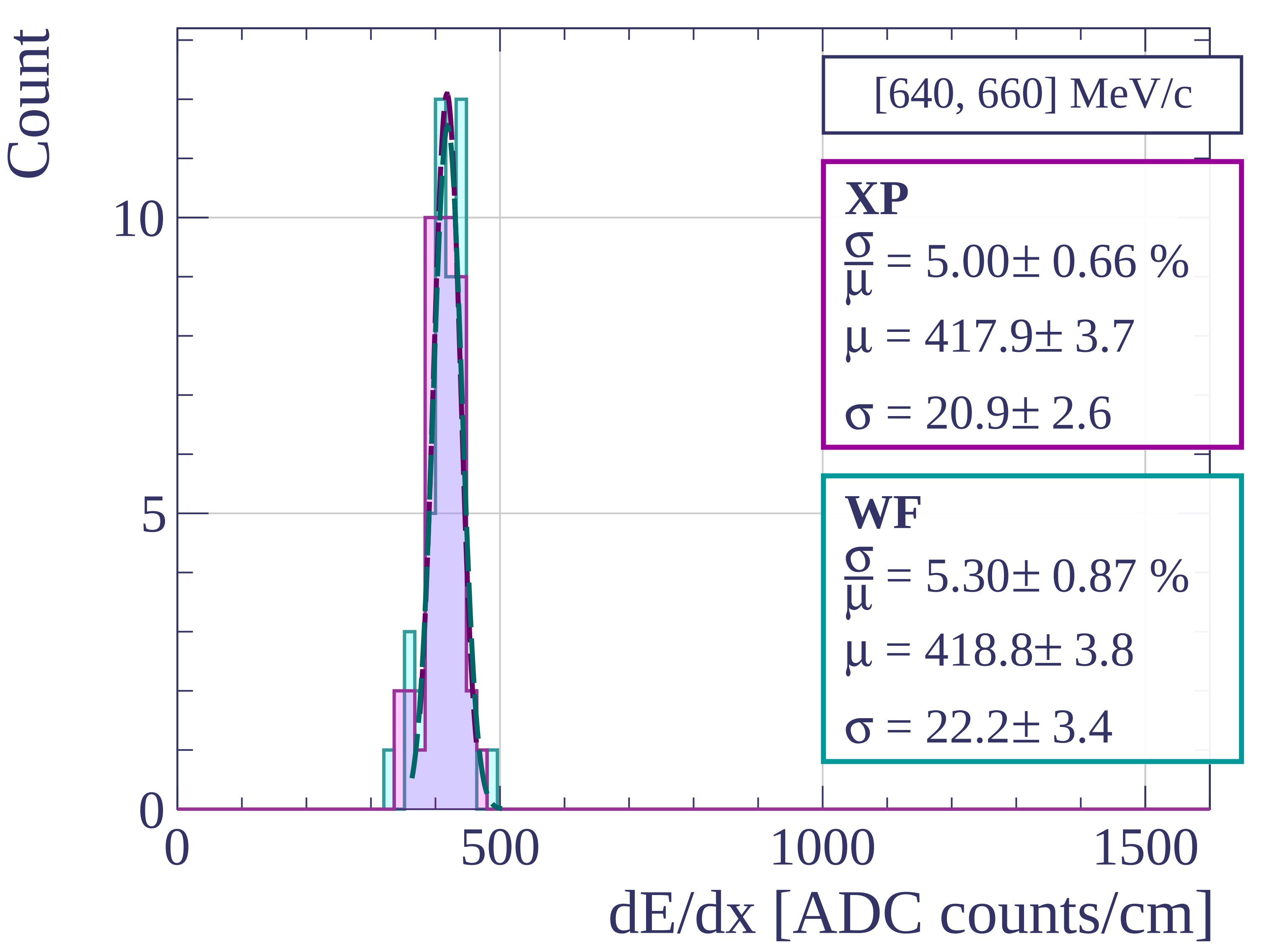

Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
[580, 600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.12± 1.40 %
μ = 402.3± 5.0
σ = 32.6± 5.2
WF
$\frac{\sigma}{\mu}$ = 5.15± 0.95 %
μ = 406.7± 3.6
σ = 20.9± 3.7



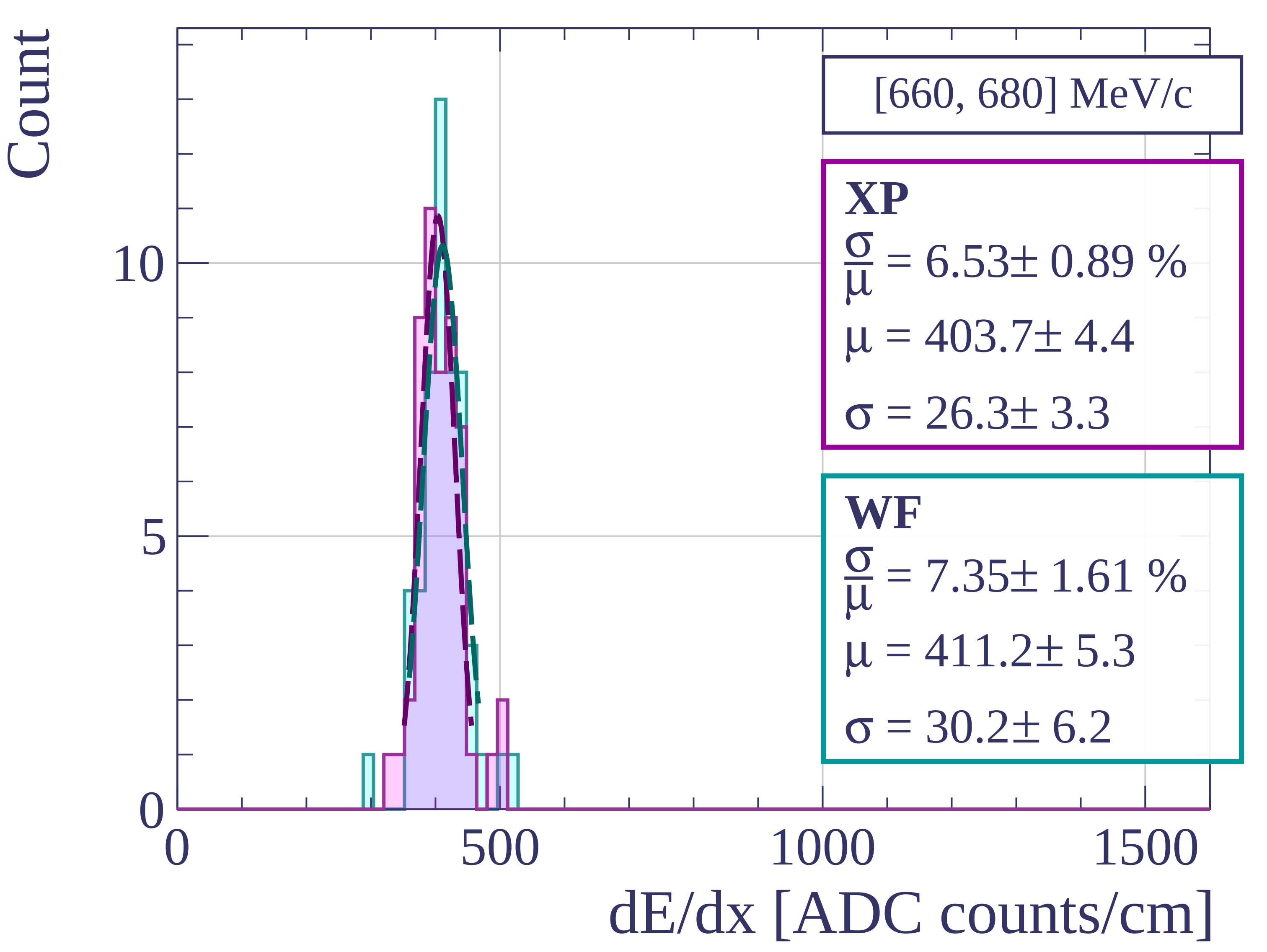
Count
0
5
10
0
500
1000
1500
dE/dx [ADC counts/cm]
[600, 620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.73± 0.74 %
μ = 395.1± 3.3
σ = 18.7± 2.8
WF
$\frac{\sigma}{\mu}$ = 6.03± 1.10 %
μ = 405.0± 4.4
σ = 24.4± 4.2

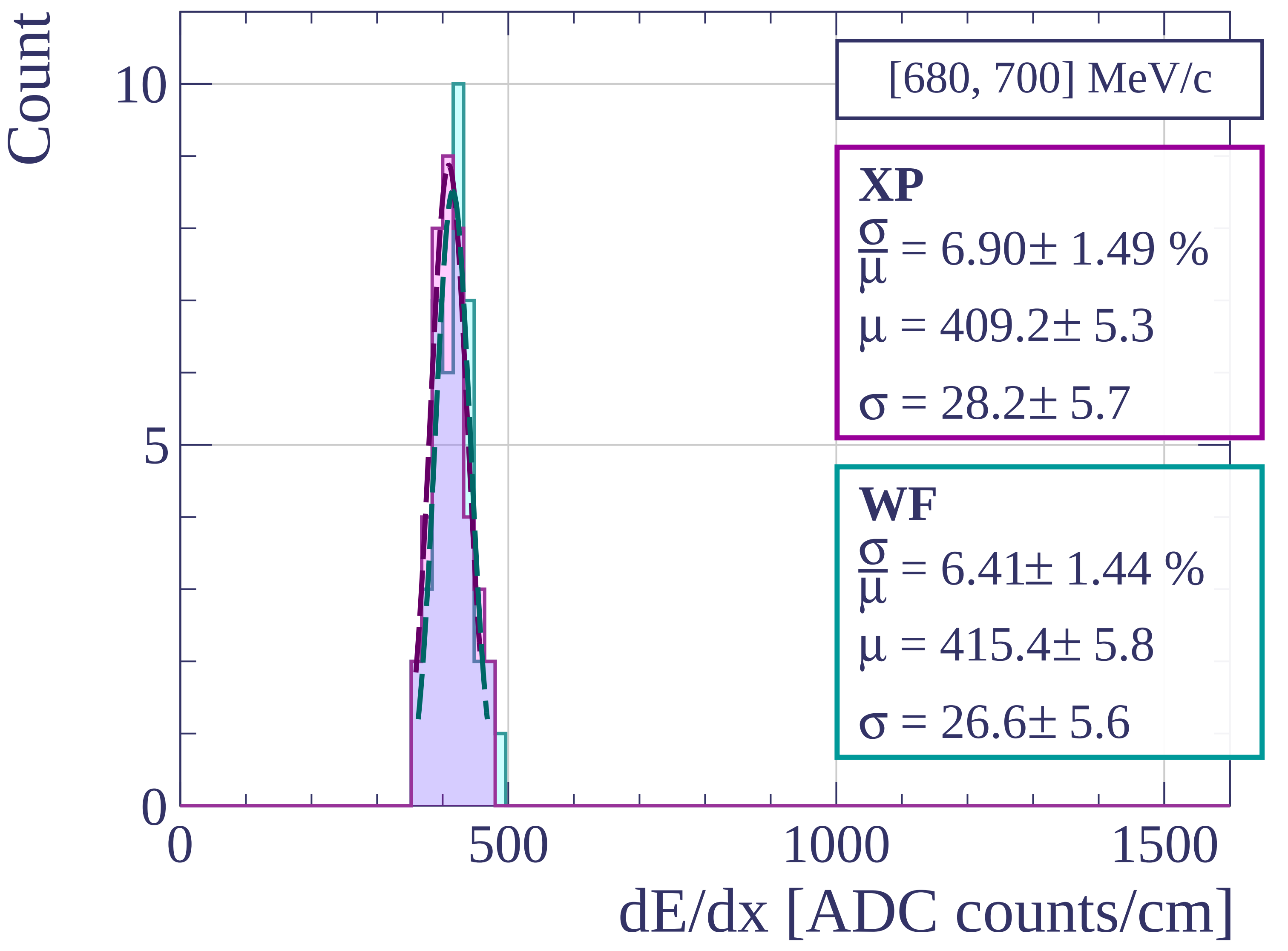


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[620, 640] MeV/c
XP
σ/μ = 8.62± 1.52 %
μ = 407.6± 5.6
σ = 35.1± 5.7
WF
σ/μ = 8.11± 1.25 %
μ = 418.8± 5.4
σ = 34.0± 4.8

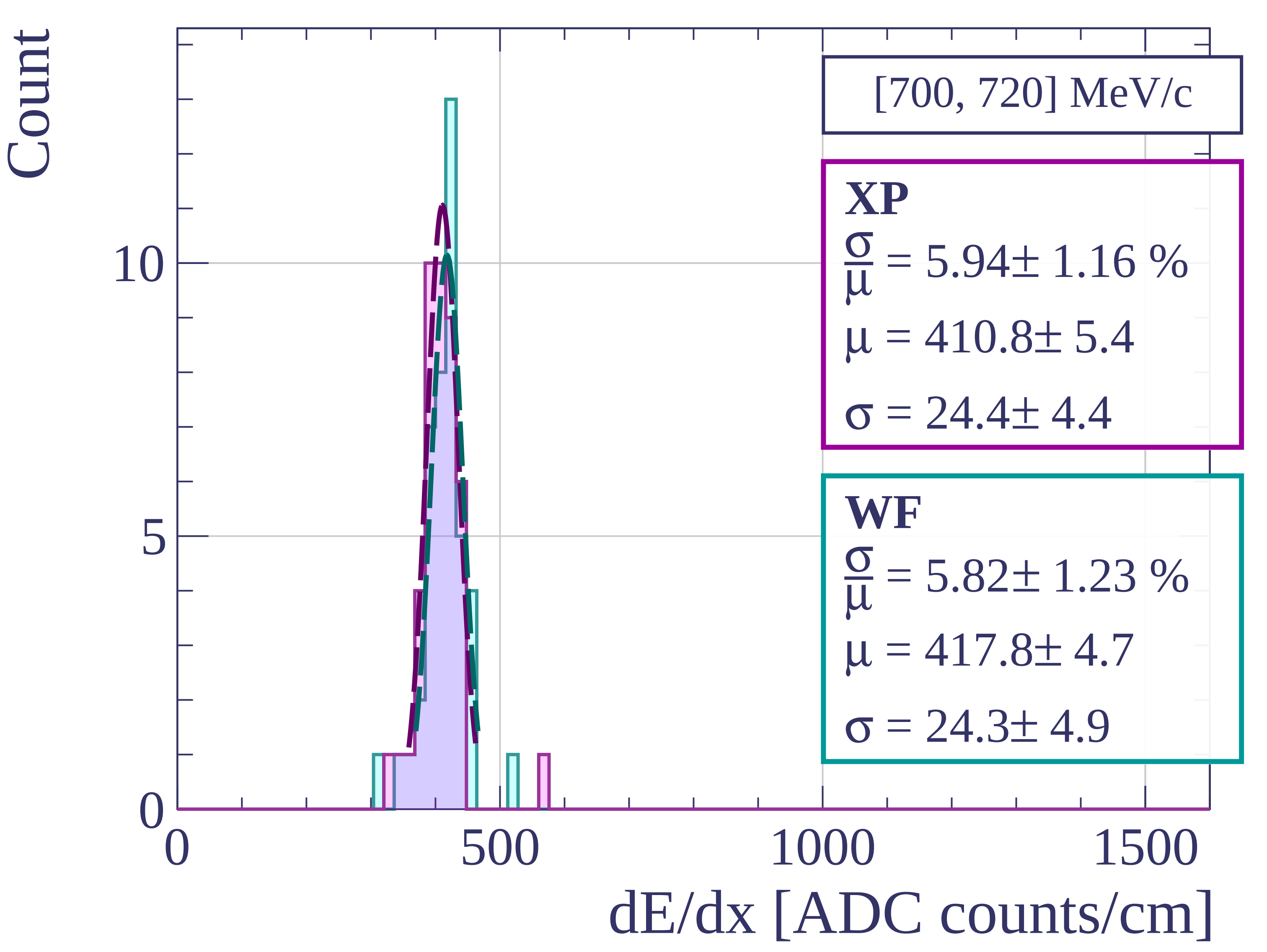


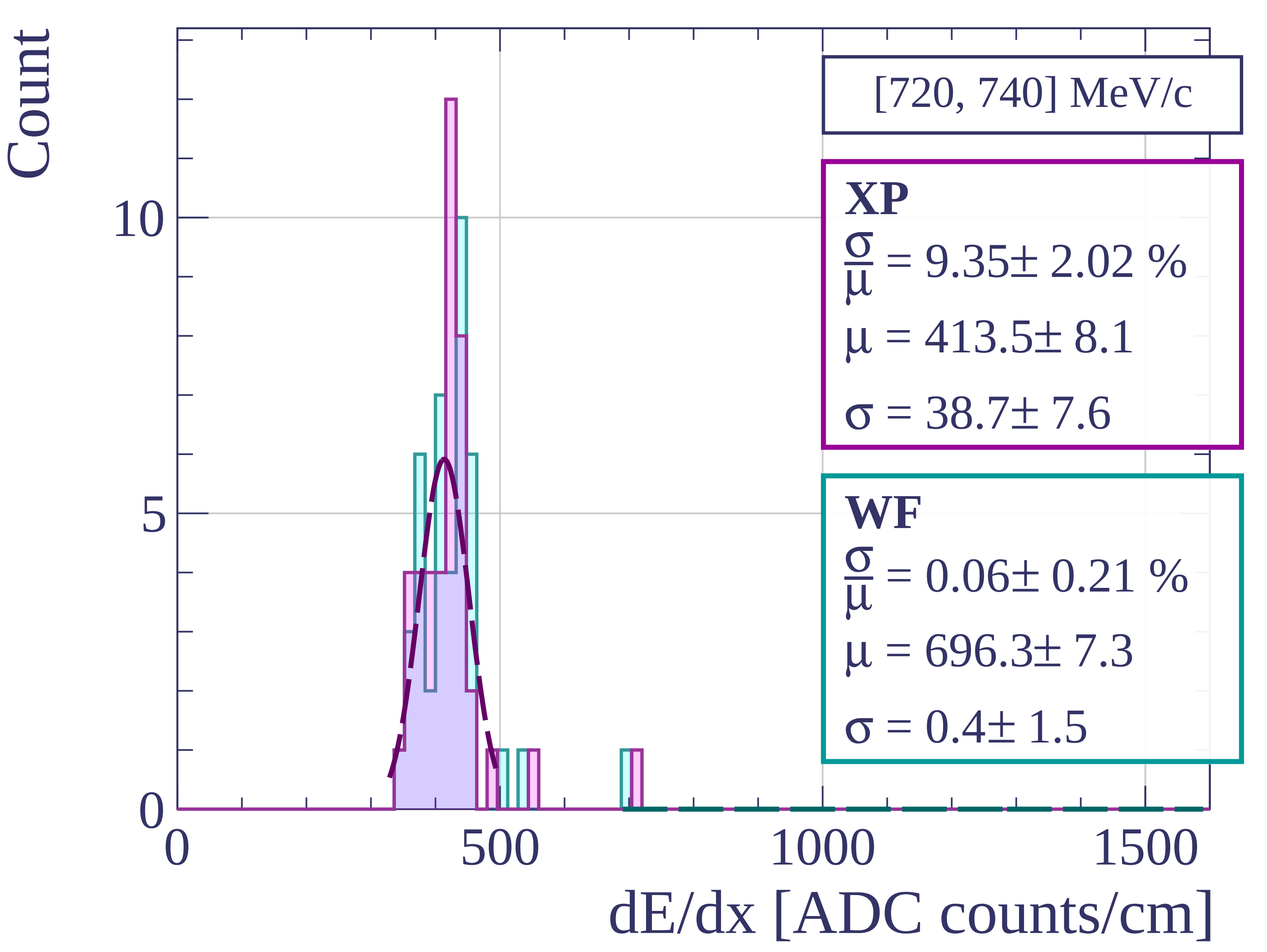
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[640, 660] MeV/c
XP
σ/μ = 5.00± 0.66 %
μ = 417.9± 3.7
σ = 20.9± 2.6
WF
σ/μ = 5.30± 0.87 %
μ = 418.8± 3.8
σ = 22.2± 3.4


Count
dE/dx [ADC counts/cm]
[660, 680] MeV/c
XP
σ/μ = 6.53± 0.89 %
μ = 403.7± 4.4
σ = 26.3± 3.3
WF
σ/μ = 7.35± 1.61 %
μ = 411.2± 5.3
σ = 30.2± 6.2
0
5
10
0
500
1000
1500

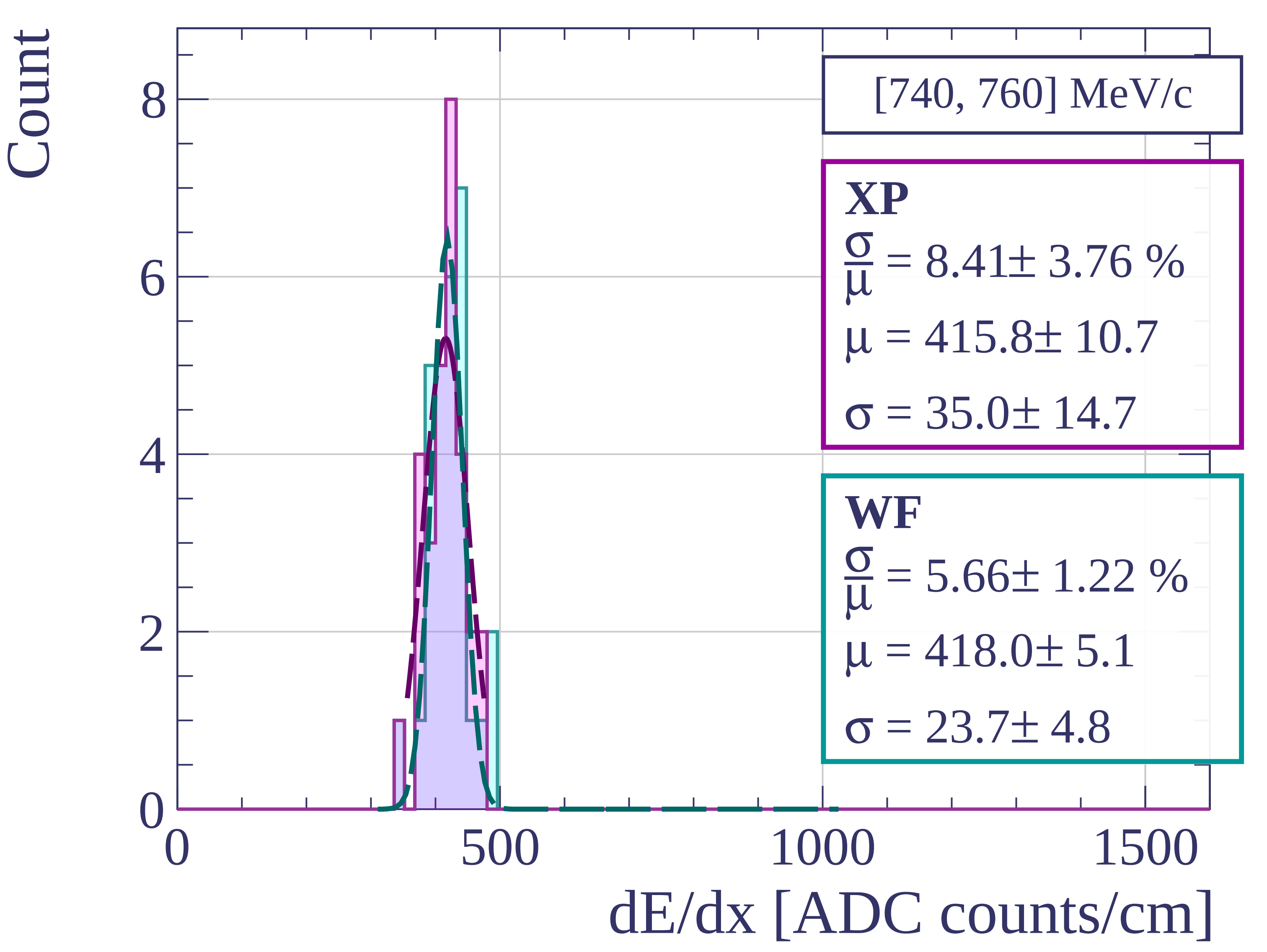


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[680, 700] MeV/c
XP
σ/μ = 6.90± 1.49 %
μ = 409.2± 5.3
σ = 28.2± 5.7
WF
σ/μ = 6.41± 1.44 %
μ = 415.4± 5.8
σ = 26.6± 5.6

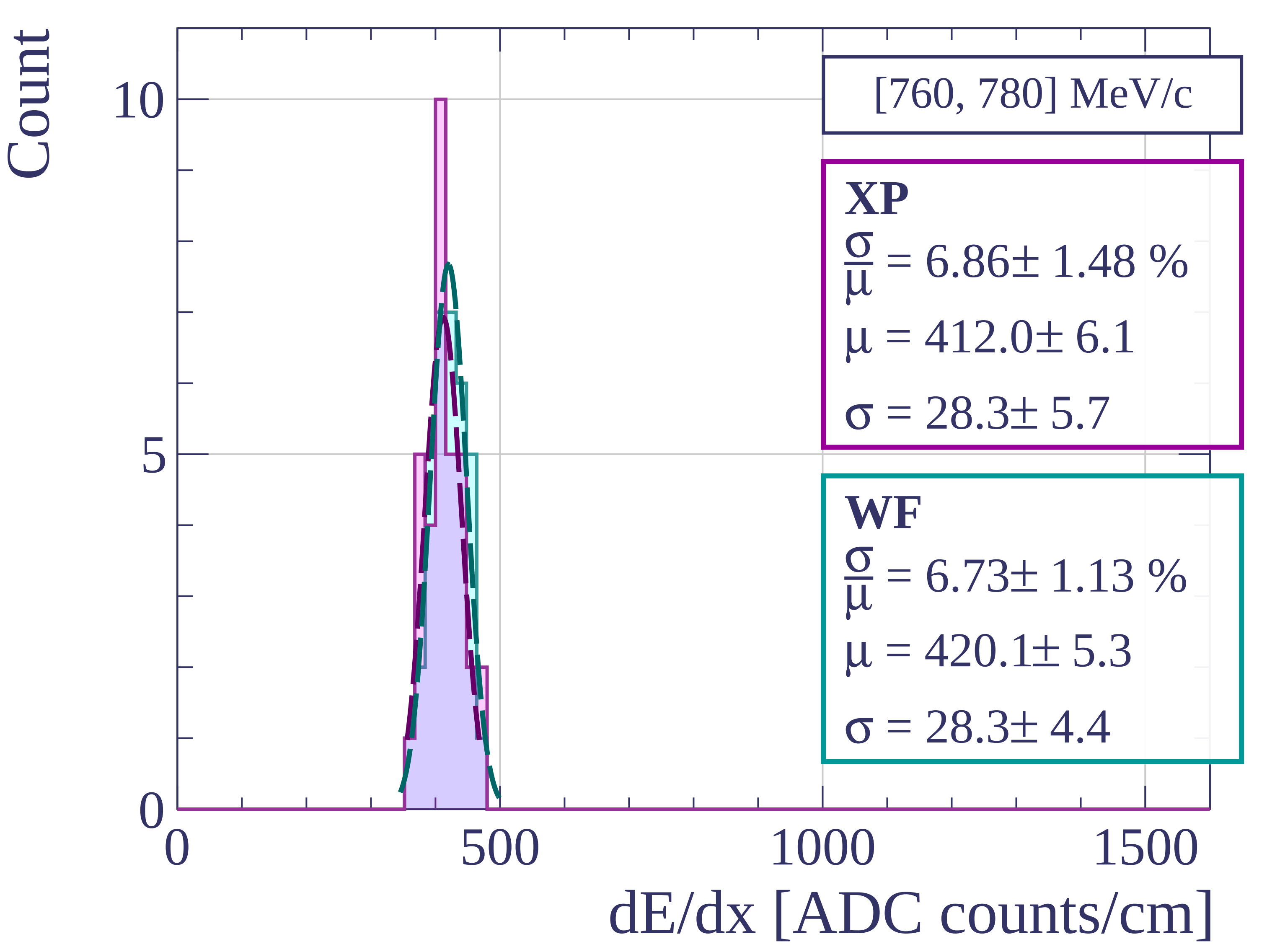

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[700, 720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.94± 1.16 %
μ = 410.8± 5.4
σ = 24.4± 4.4
WF
$\frac{\sigma}{\mu}$ = 5.82± 1.23 %
μ = 417.8± 4.7
σ = 24.3± 4.9



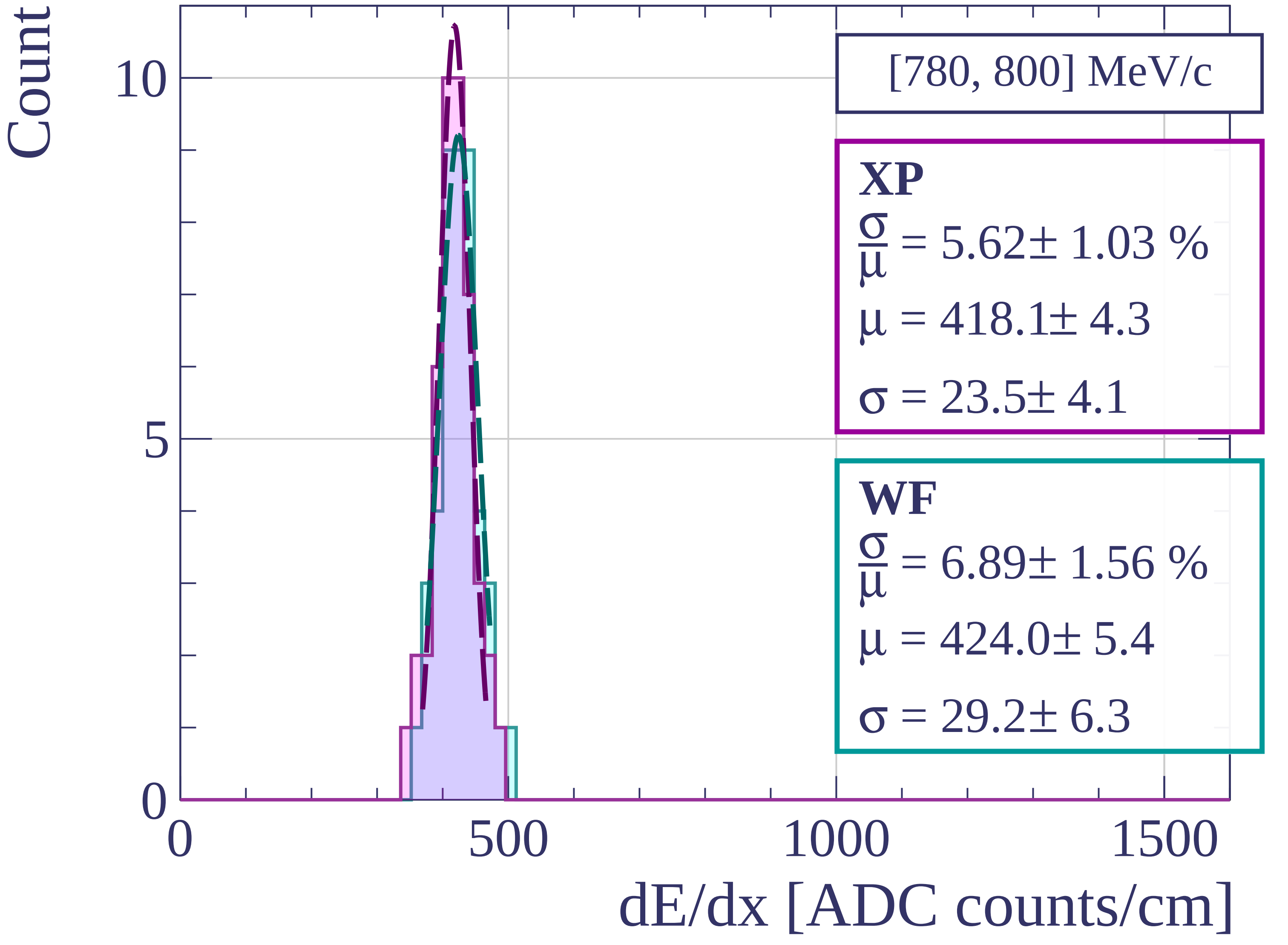
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.35± 2.02 %
μ = 413.5± 8.1
σ = 38.7± 7.6
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.21 %
μ = 696.3± 7.3
σ = 0.4± 1.5



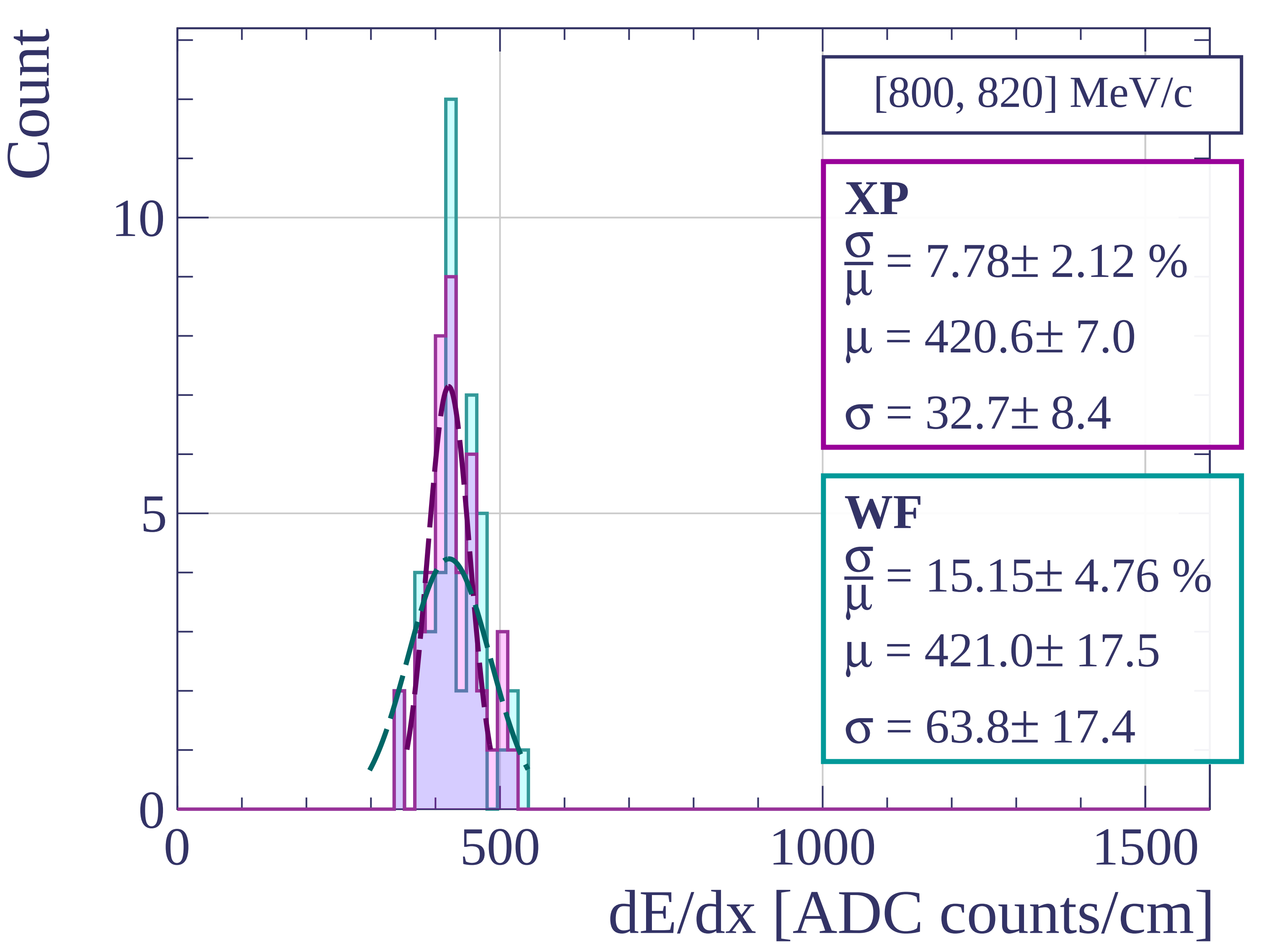
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[740, 760] MeV/c
XP
σ/μ = 8.41± 3.76 %
μ = 415.8± 10.7
σ = 35.0± 14.7
WF
σ/μ = 5.66± 1.22 %
μ = 418.0± 5.1
σ = 23.7± 4.8



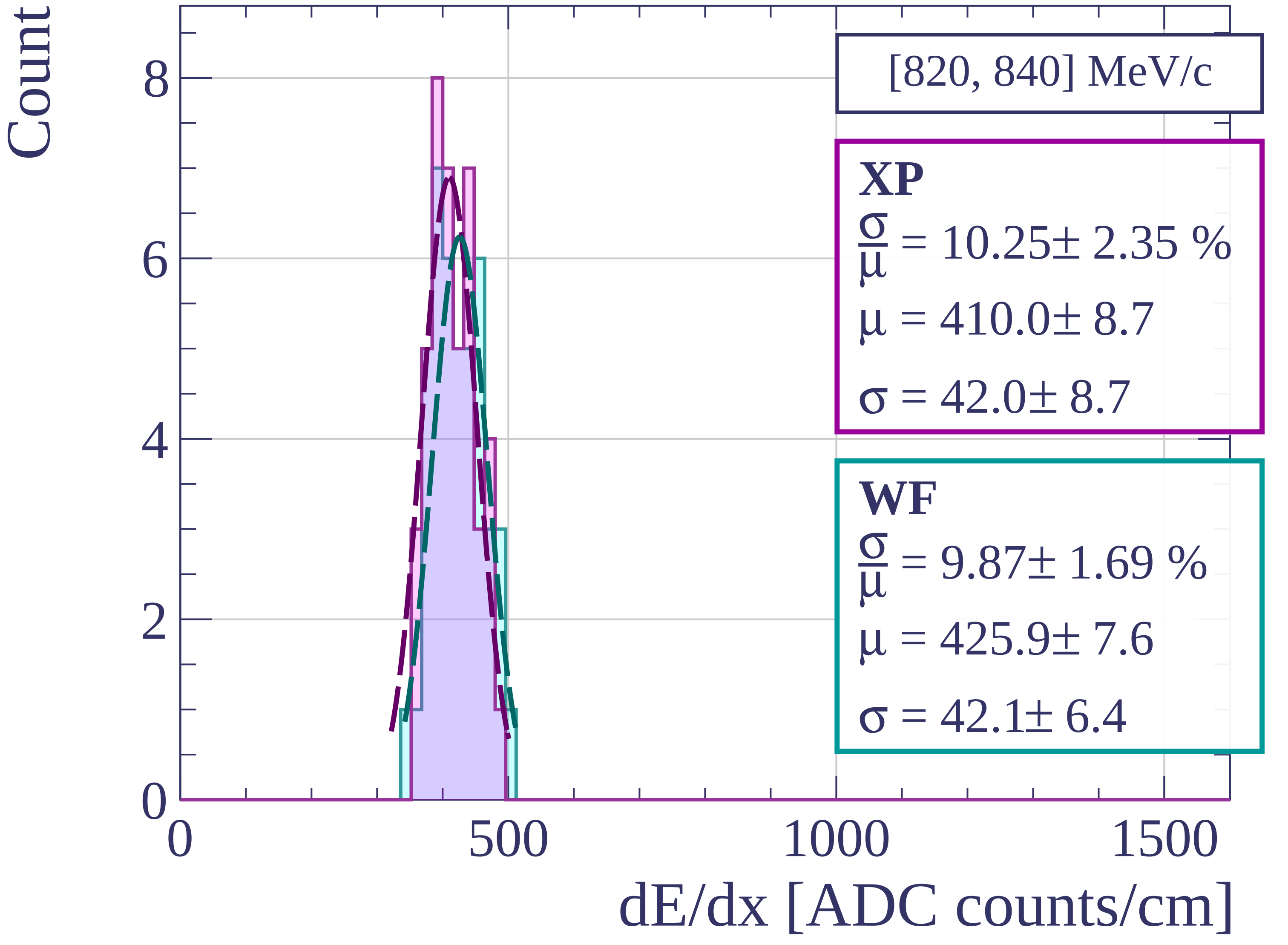
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[760, 780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.86± 1.48 %
μ = 412.0± 6.1
σ = 28.3± 5.7
WF
$\frac{\sigma}{\mu}$ = 6.73± 1.13 %
μ = 420.1± 5.3
σ = 28.3± 4.4



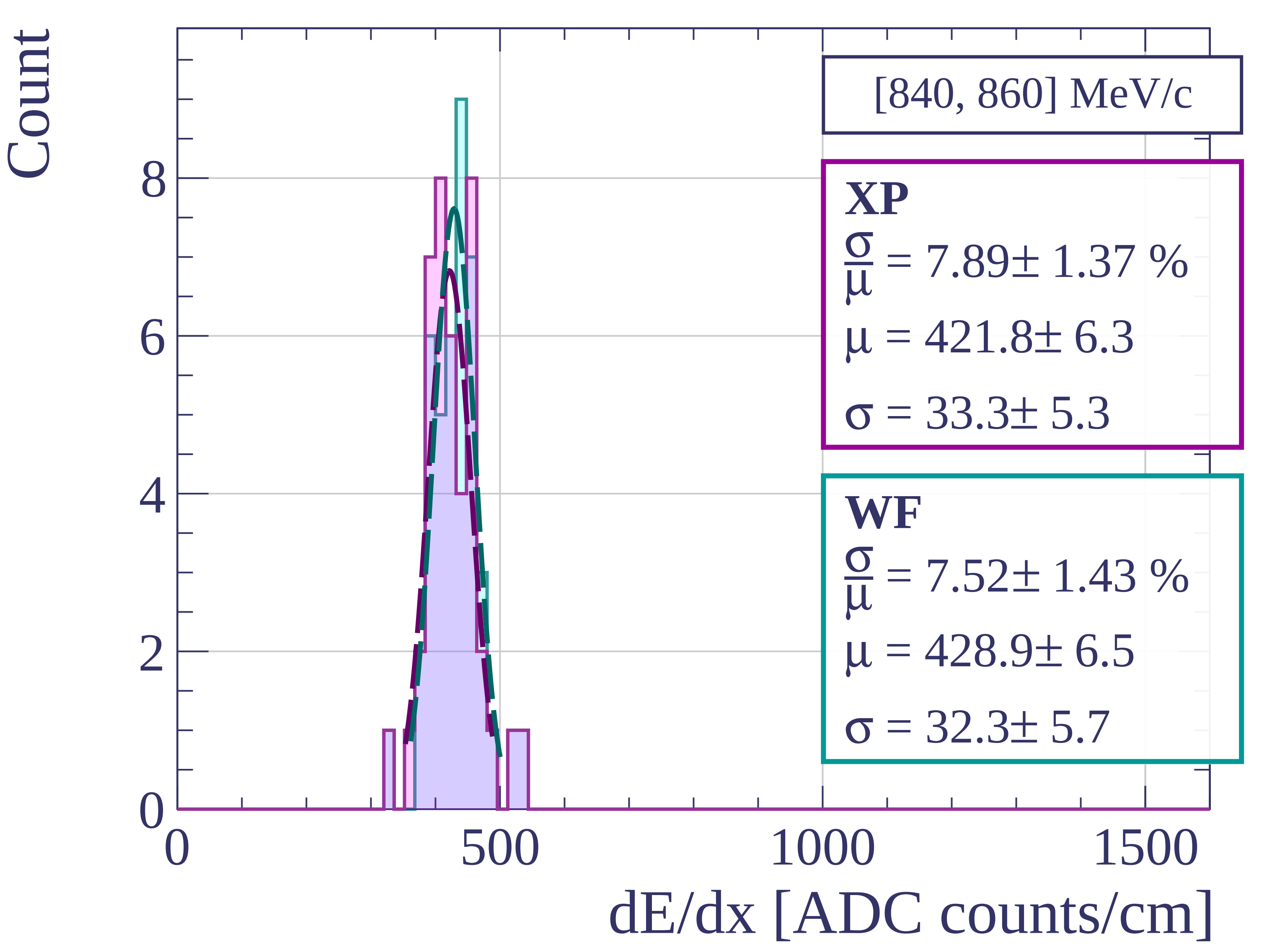
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[780, 800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.62± 1.03 %
μ = 418.1± 4.3
σ = 23.5± 4.1
WF
$\frac{\sigma}{\mu}$ = 6.89± 1.56 %
μ = 424.0± 5.4
σ = 29.2± 6.3



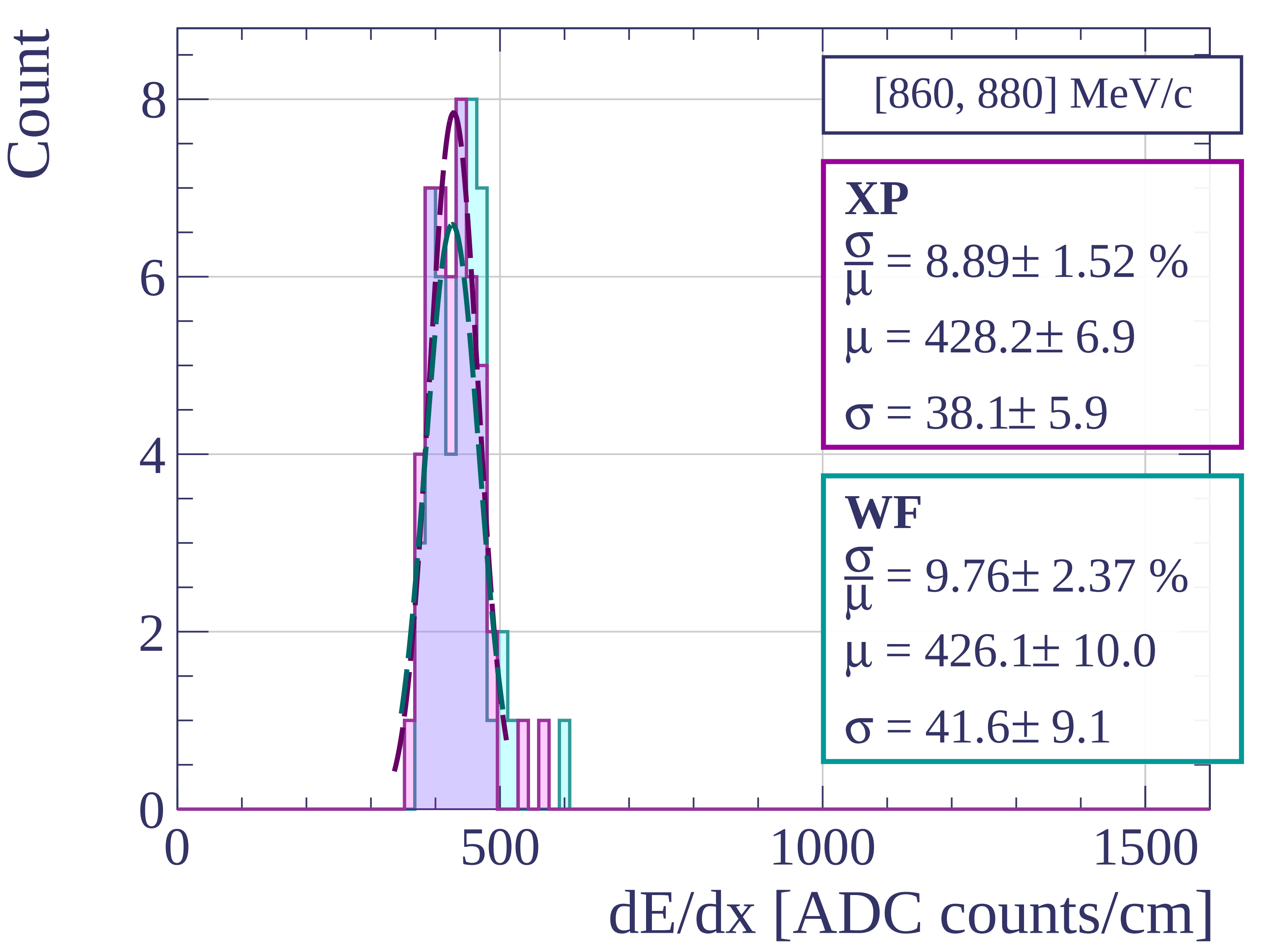
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.78± 2.12 %
μ = 420.6± 7.0
σ = 32.7± 8.4
WF
$\frac{\sigma}{\mu}$ = 15.15± 4.76 %
μ = 421.0± 17.5
σ = 63.8± 17.4

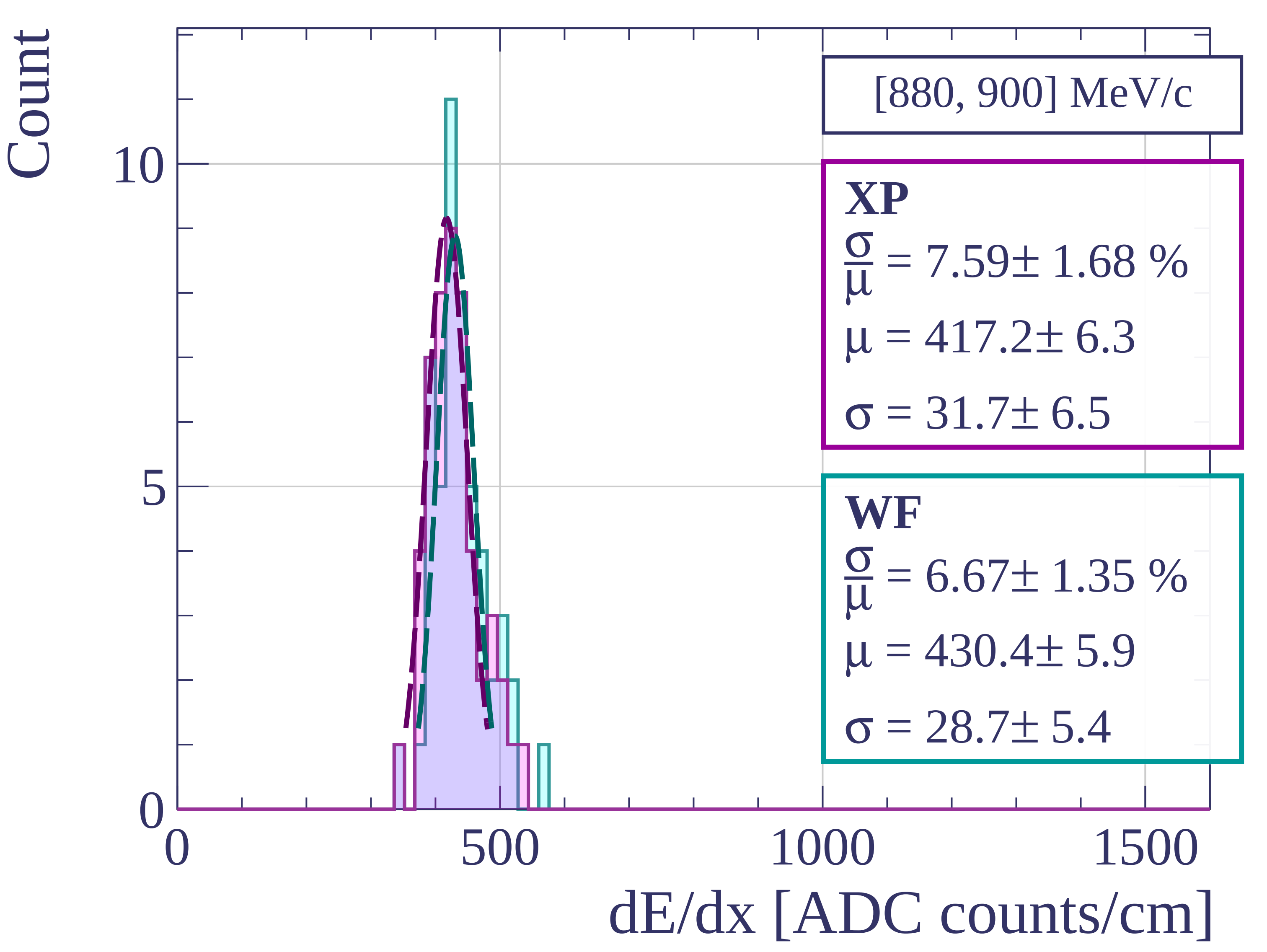


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[820, 840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.25± 2.35 %
μ = 410.0± 8.7
σ = 42.0± 8.7
WF
$\frac{\sigma}{\mu}$ = 9.87± 1.69 %
μ = 425.9± 7.6
σ = 42.1± 6.4

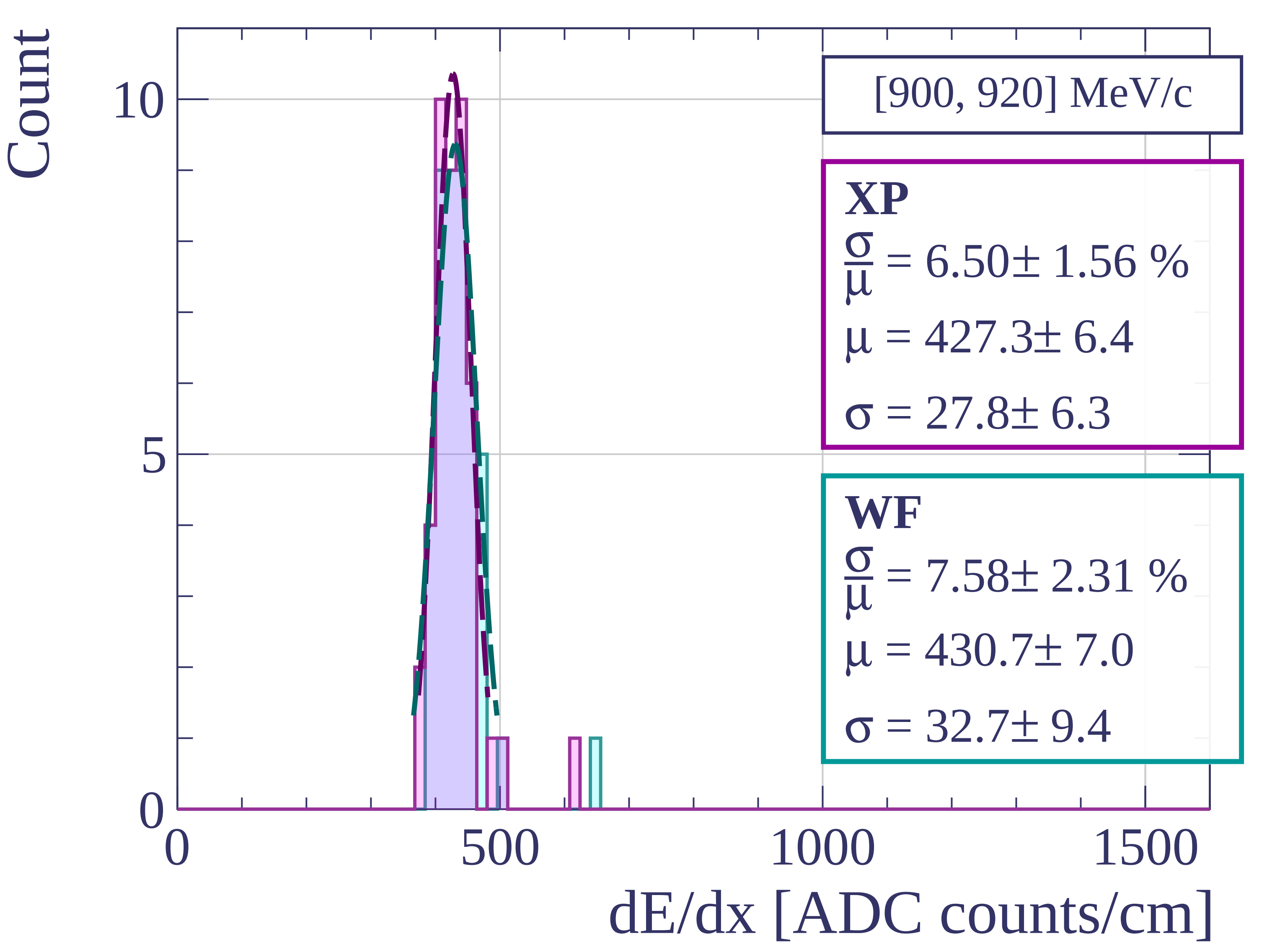


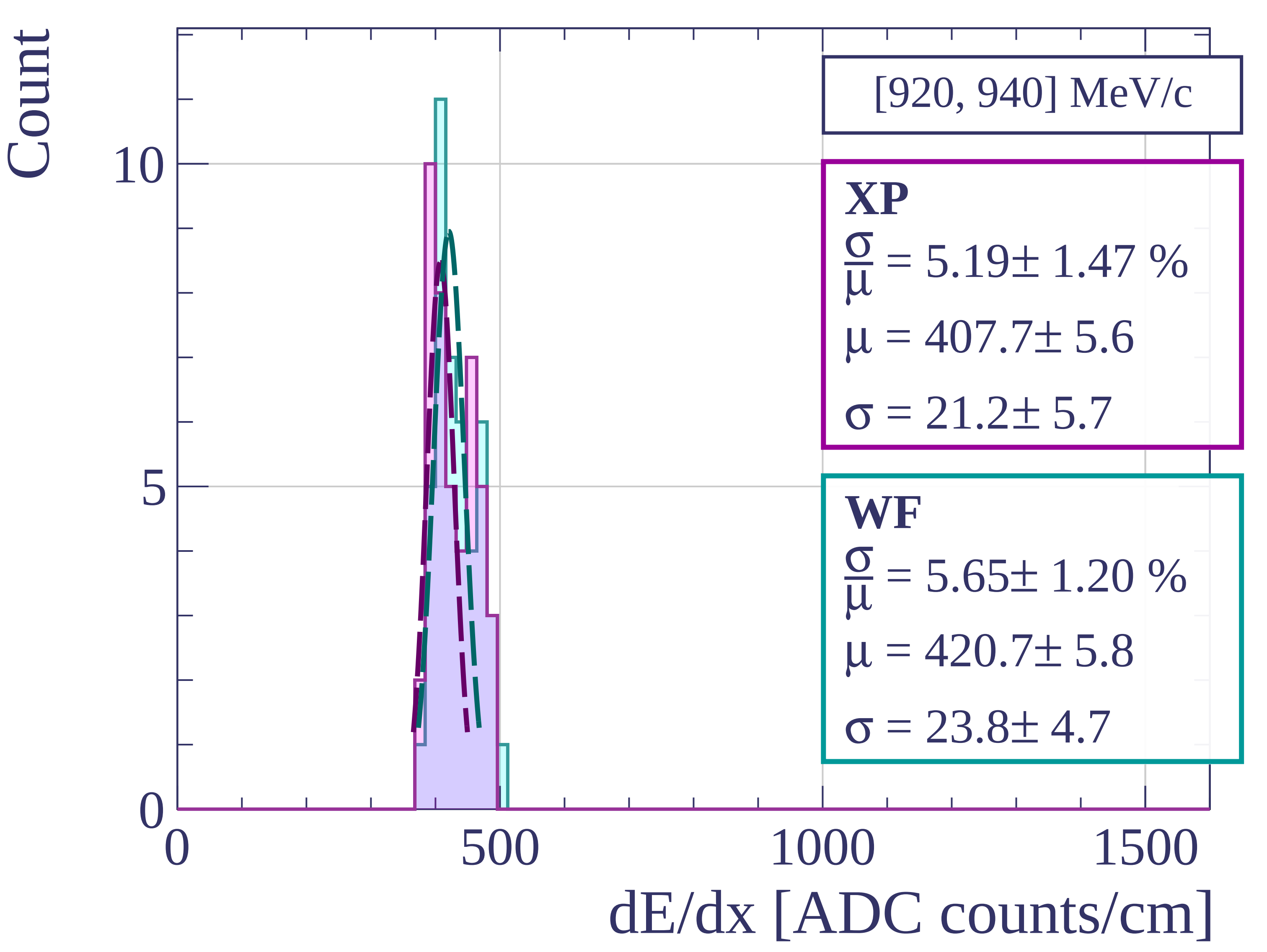
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[840, 860] MeV/c
XP
σ/μ = 7.89± 1.37 %
μ = 421.8± 6.3
σ = 33.3± 5.3
WF
σ/μ = 7.52± 1.43 %
μ = 428.9± 6.5
σ = 32.3± 5.7


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[860, 880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.89± 1.52 %
μ = 428.2± 6.9
σ = 38.1± 5.9
WF
$\frac{\sigma}{\mu}$ = 9.76± 2.37 %
μ = 426.1± 10.0
σ = 41.6± 9.1

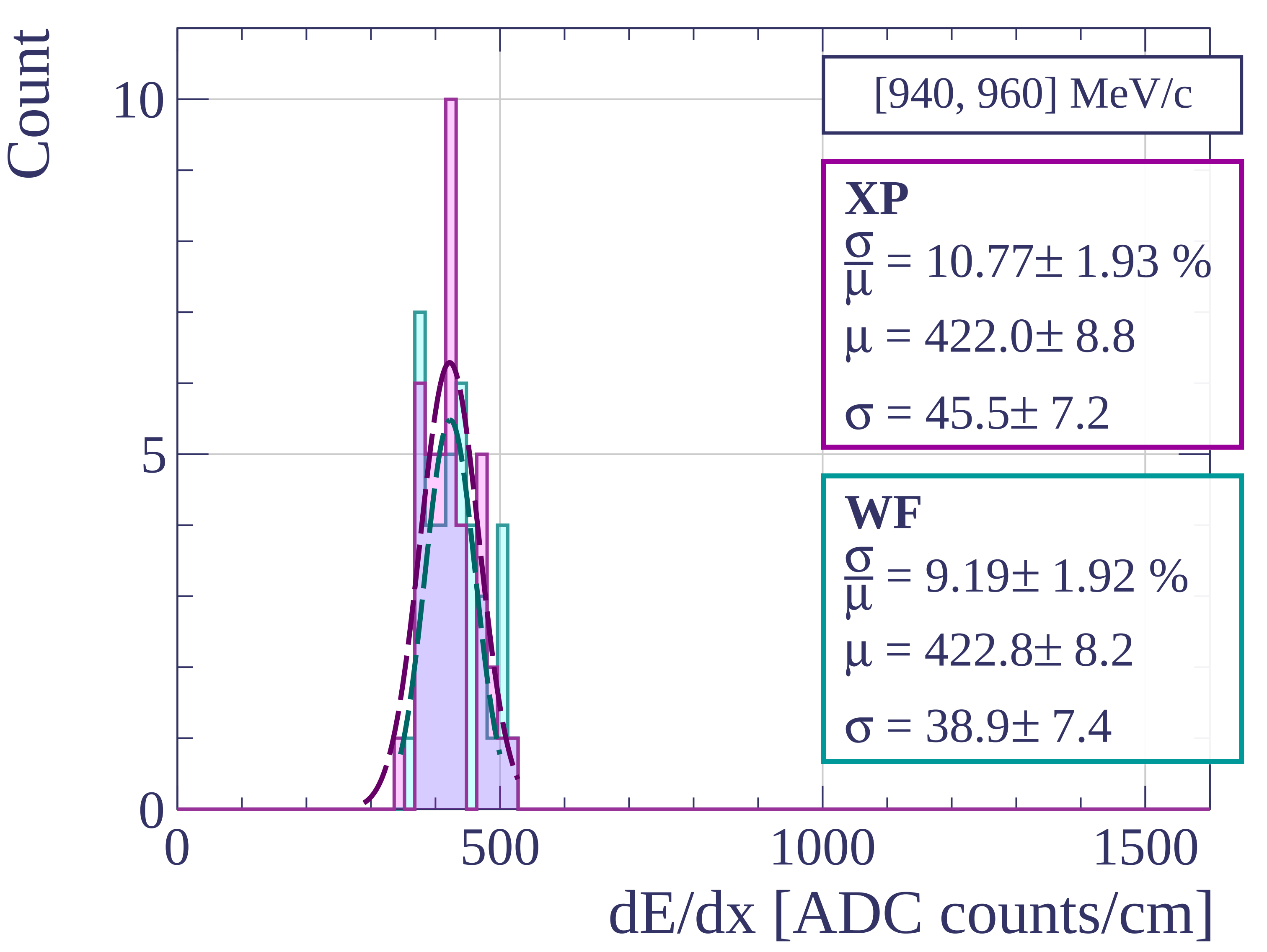


[880, 900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.59± 1.68 %
μ = 417.2± 6.3
σ = 31.7± 6.5
WF
$\frac{\sigma}{\mu}$ = 6.67± 1.35 %
μ = 430.4± 5.9
σ = 28.7± 5.4
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500

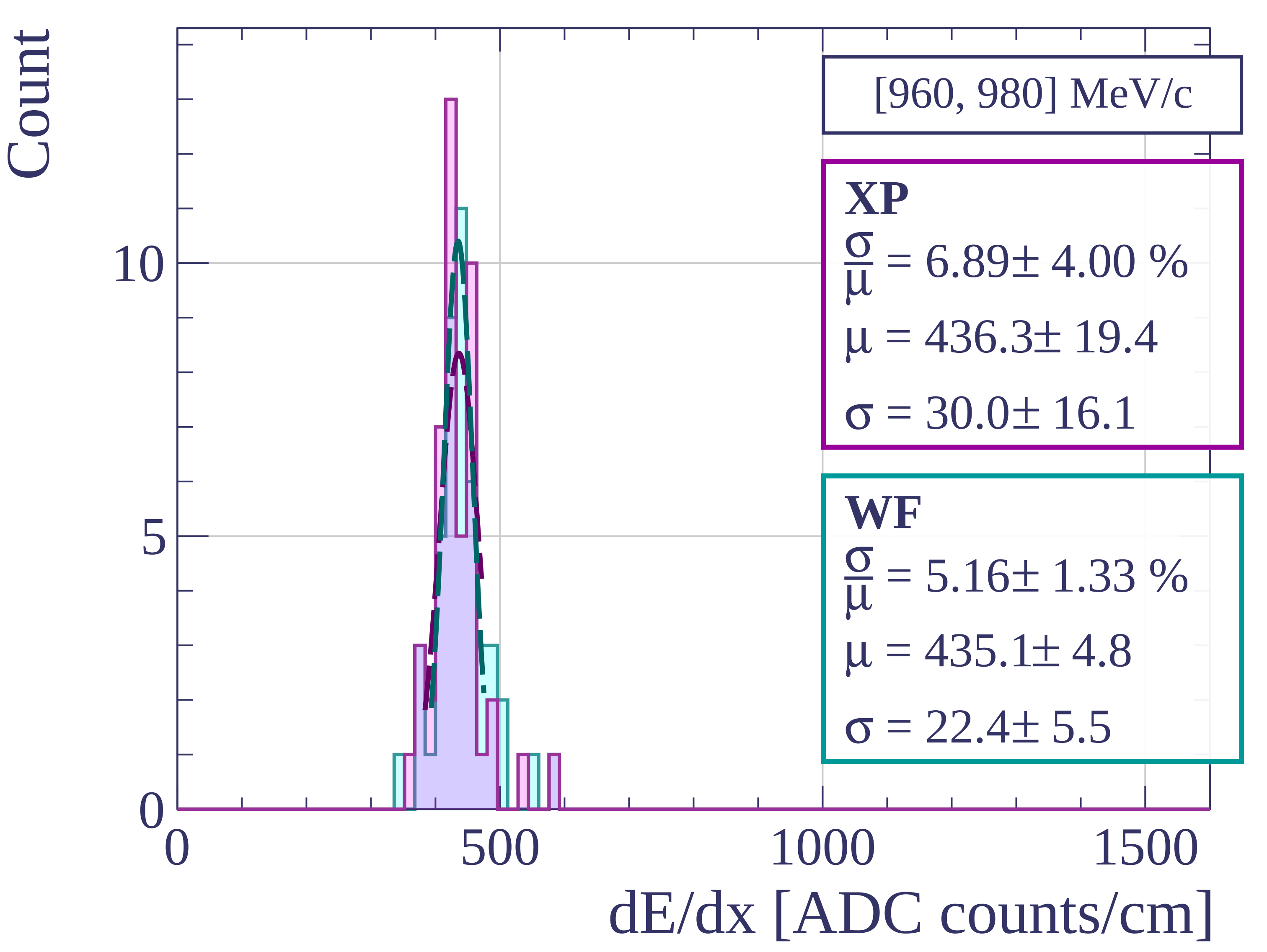

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[900, 920] MeV/c
XP
σ/μ = 6.50± 1.56 %
μ = 427.3± 6.4
σ = 27.8± 6.3
WF
σ/μ = 7.58± 2.31 %
μ = 430.7± 7.0
σ = 32.7± 9.4

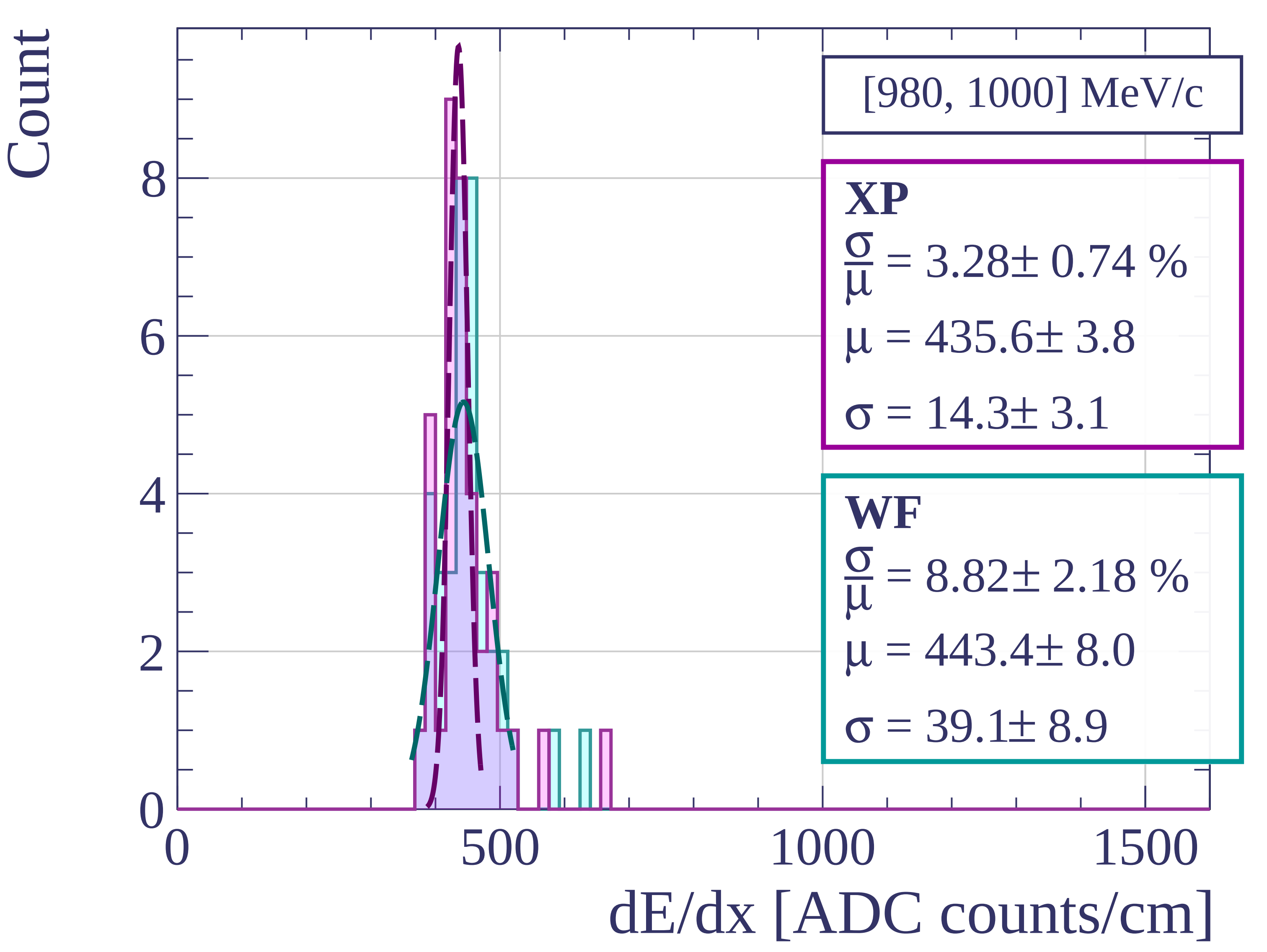


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[920, 940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.19± 1.47 %
μ = 407.7± 5.6
σ = 21.2± 5.7
WF
$\frac{\sigma}{\mu}$ = 5.65± 1.20 %
μ = 420.7± 5.8
σ = 23.8± 4.7

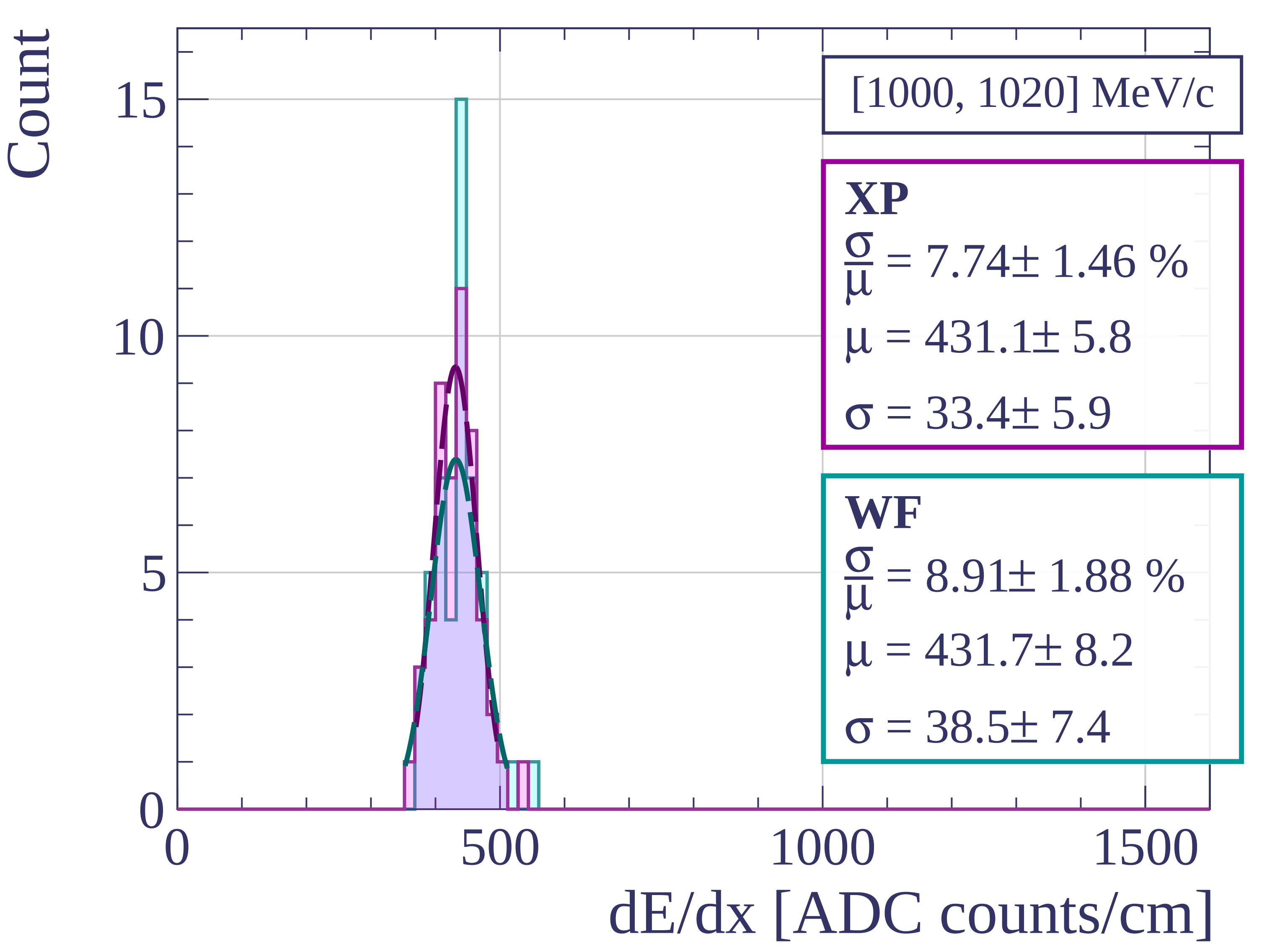


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[940, 960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.77± 1.93 %
μ = 422.0± 8.8
σ = 45.5± 7.2
WF
$\frac{\sigma}{\mu}$ = 9.19± 1.92 %
μ = 422.8± 8.2
σ = 38.9± 7.4

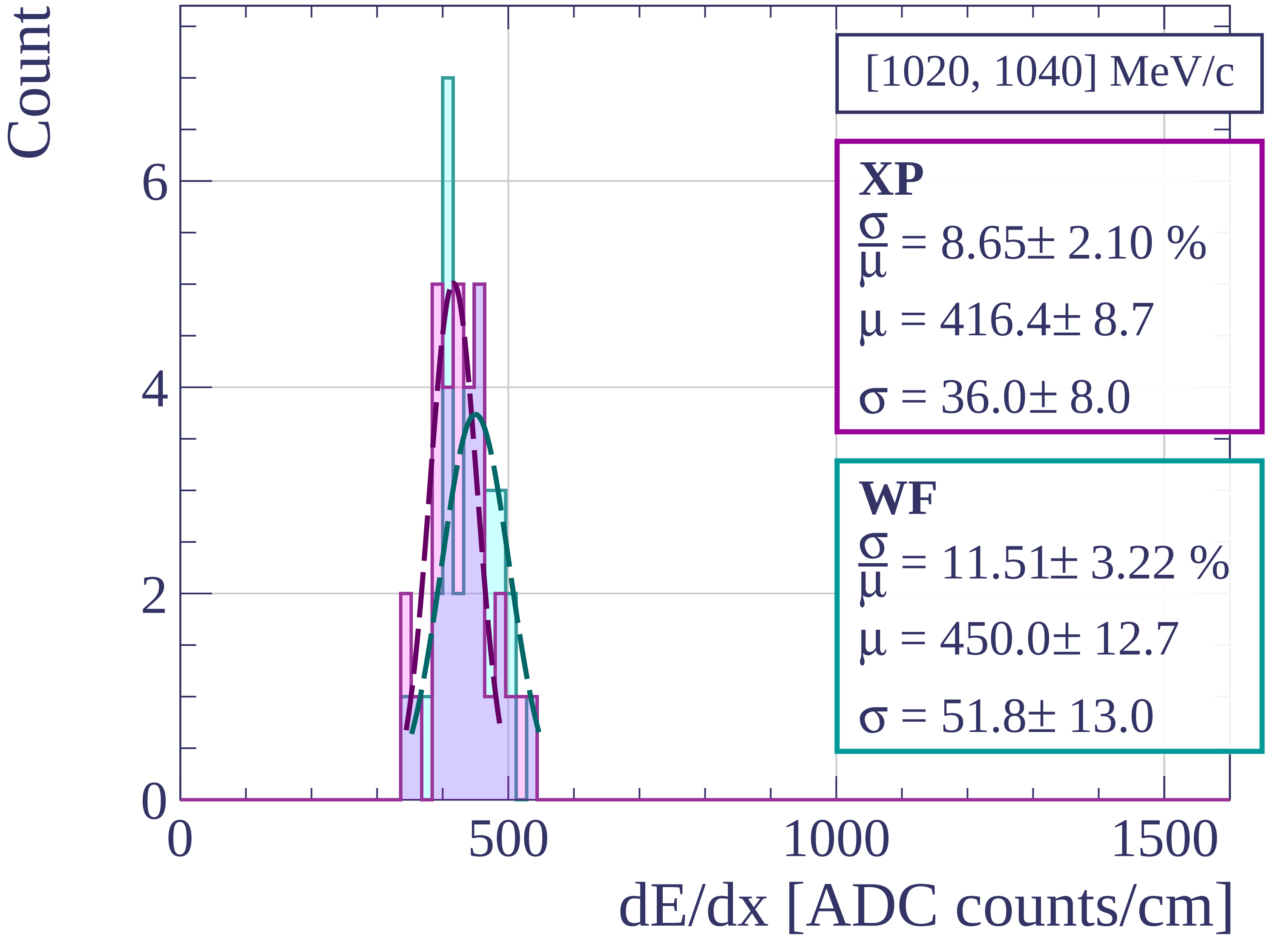

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.89± 4.00 %
μ = 436.3± 19.4
σ = 30.0± 16.1
WF
$\frac{\sigma}{\mu}$ = 5.16± 1.33 %
μ = 435.1± 4.8
σ = 22.4± 5.5

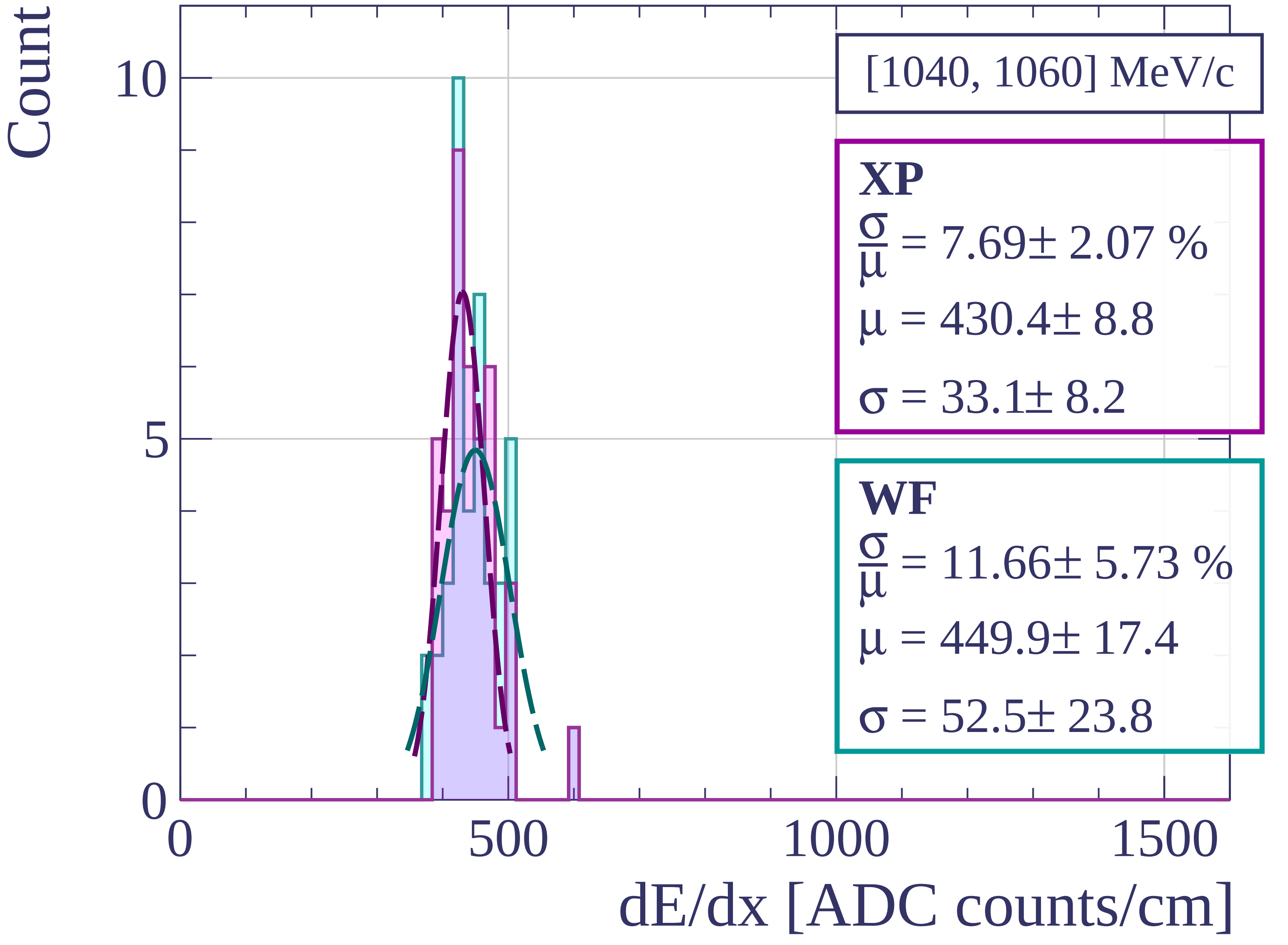


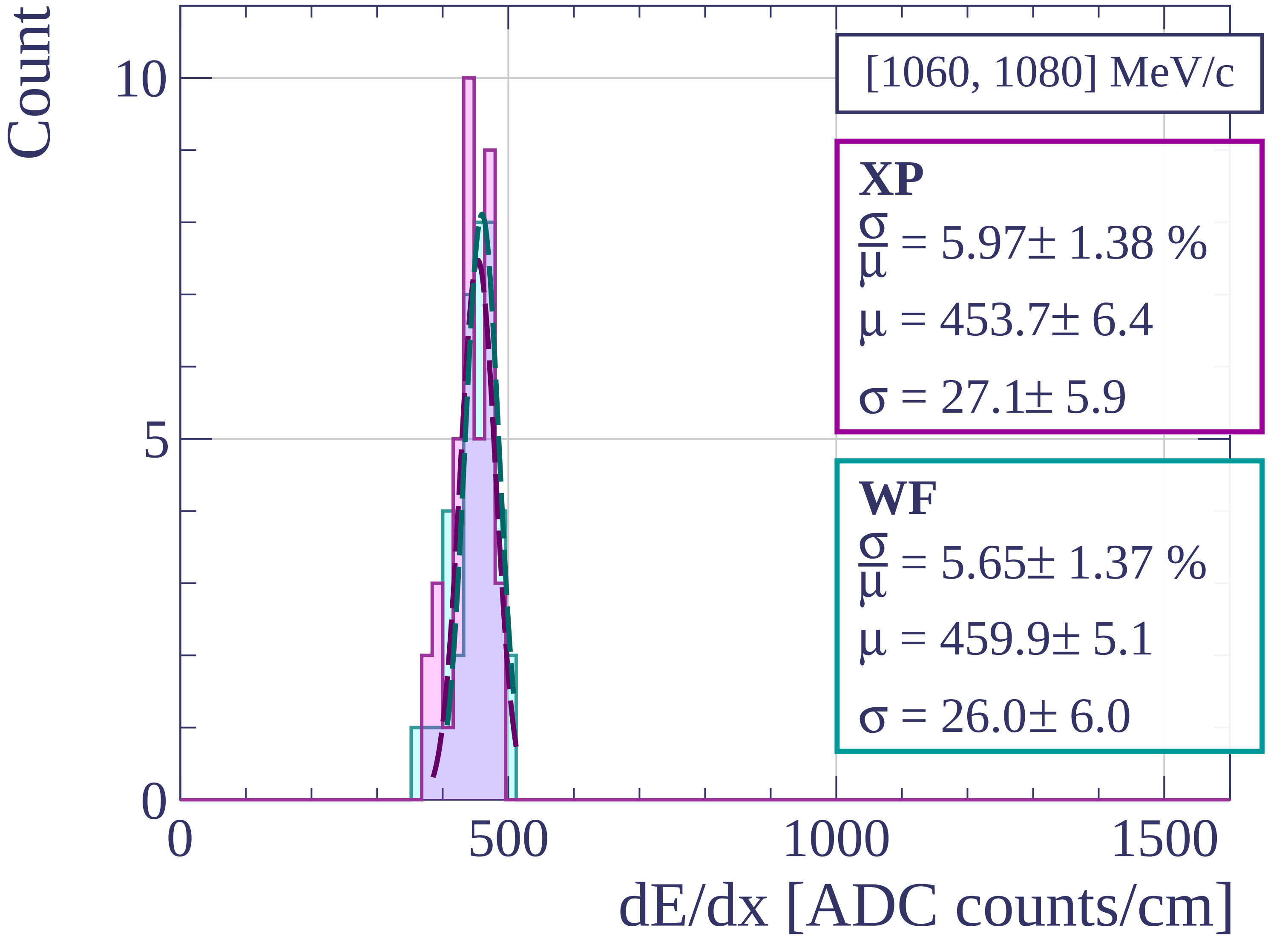
Count
dE/dx [ADC counts/cm]
[980, 1000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 3.28± 0.74 %
μ = 435.6± 3.8
σ = 14.3± 3.1
WF
$\frac{\sigma}{\mu}$ = 8.82± 2.18 %
μ = 443.4± 8.0
σ = 39.1± 8.9
0
500
1000
1500
0
2
4
6
8



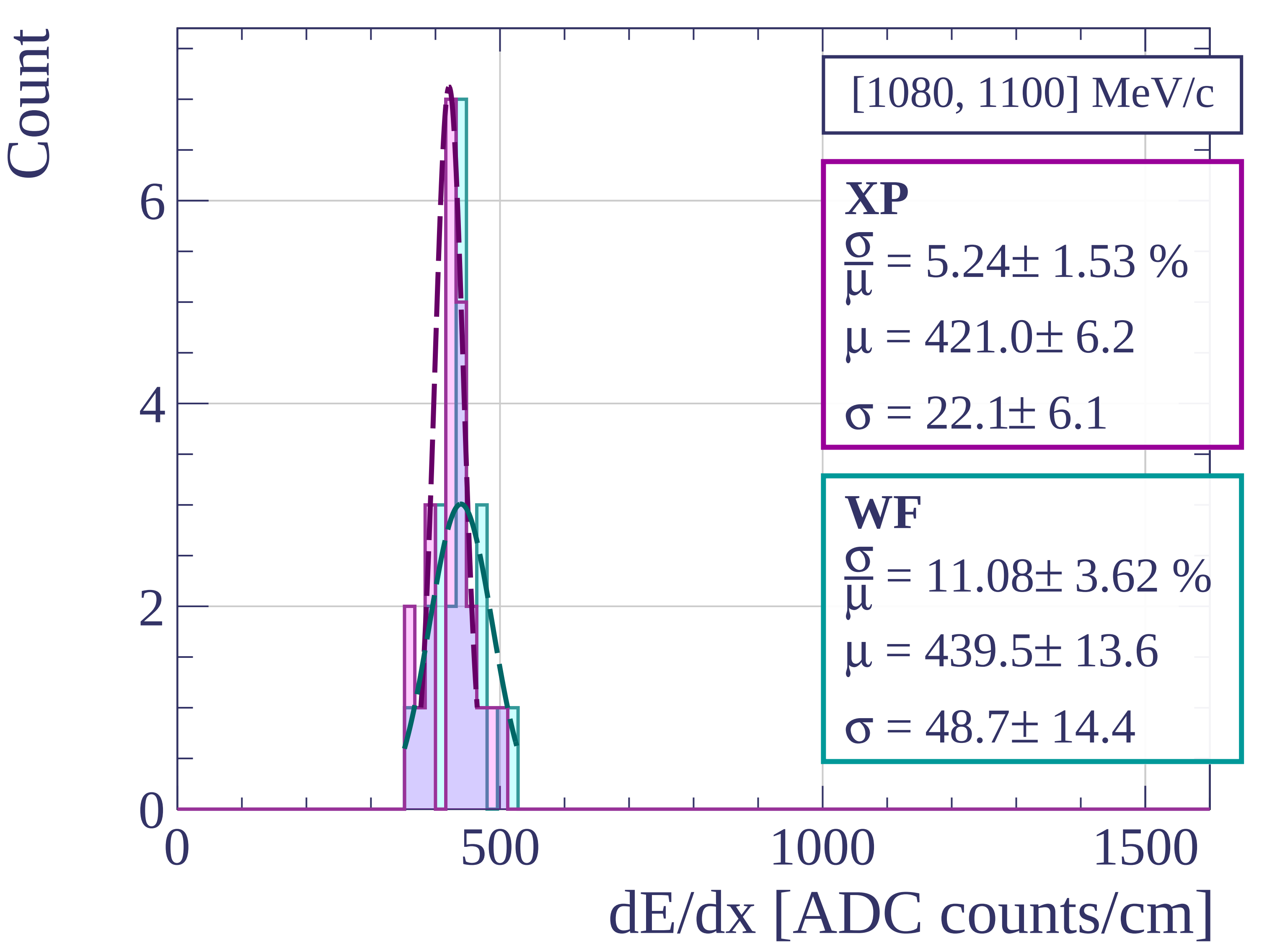
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1000, 1020] MeV/c
XP
σ/μ = 7.74± 1.46 %
μ = 431.1± 5.8
σ = 33.4± 5.9
WF
σ/μ = 8.91± 1.88 %
μ = 431.7± 8.2
σ = 38.5± 7.4


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1020, 1040] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.65± 2.10 %
μ = 416.4± 8.7
σ = 36.0± 8.0
WF
$\frac{\sigma}{\mu}$ = 11.51± 3.22 %
μ = 450.0± 12.7
σ = 51.8± 13.0

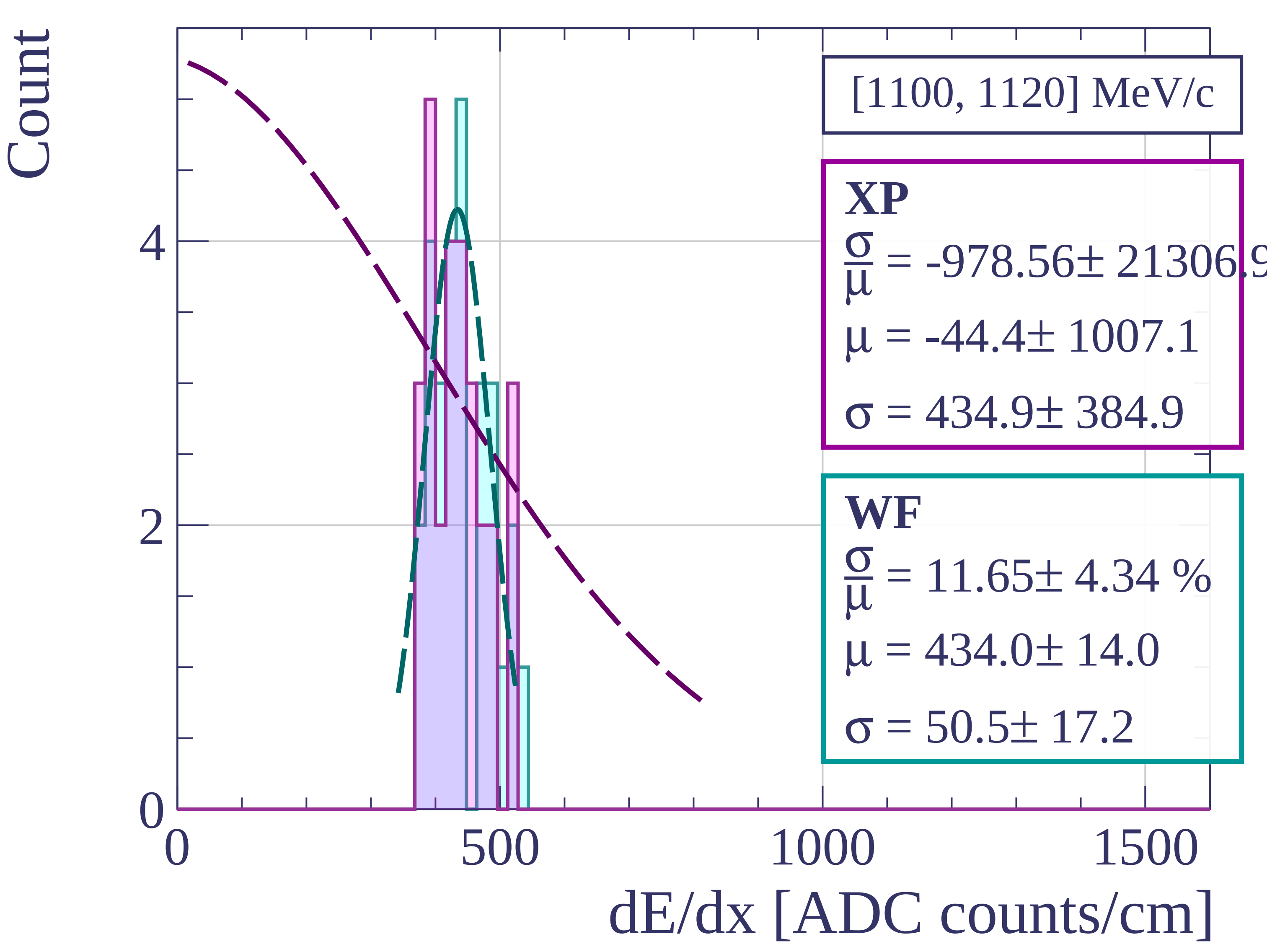

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1040, 1060] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.69± 2.07 %
μ = 430.4± 8.8
σ = 33.1± 8.2
WF
$\frac{\sigma}{\mu}$ = 11.66± 5.73 %
μ = 449.9± 17.4
σ = 52.5± 23.8



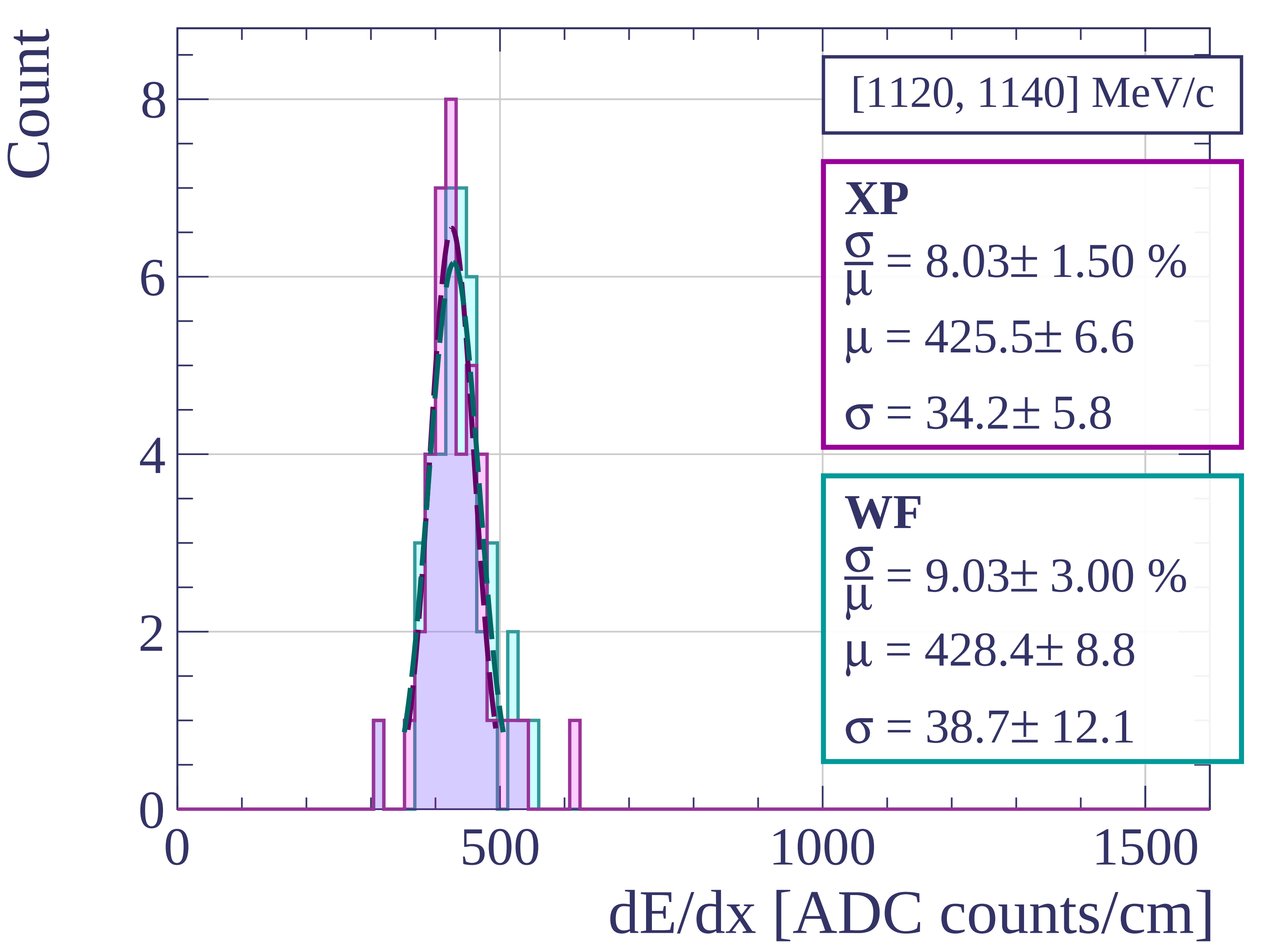
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1060, 1080] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.97± 1.38 %
μ = 453.7± 6.4
σ = 27.1± 5.9
WF
$\frac{\sigma}{\mu}$ = 5.65± 1.37 %
μ = 459.9± 5.1
σ = 26.0± 6.0

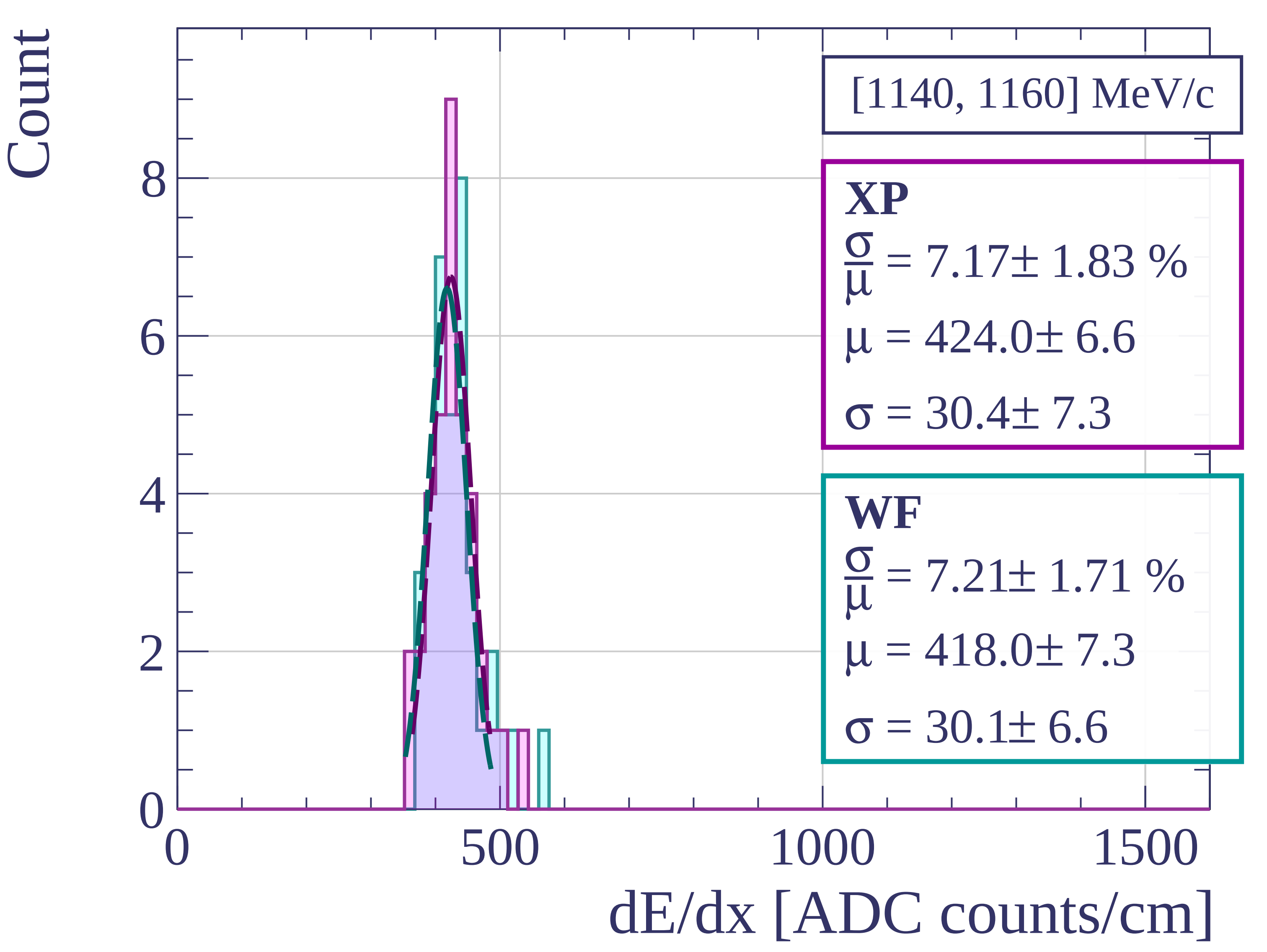


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1080, 1100] MeV/c
XP
σ/μ = 5.24± 1.53 %
μ = 421.0± 6.2
σ = 22.1± 6.1
WF
σ/μ = 11.08± 3.62 %
μ = 439.5± 13.6
σ = 48.7± 14.4

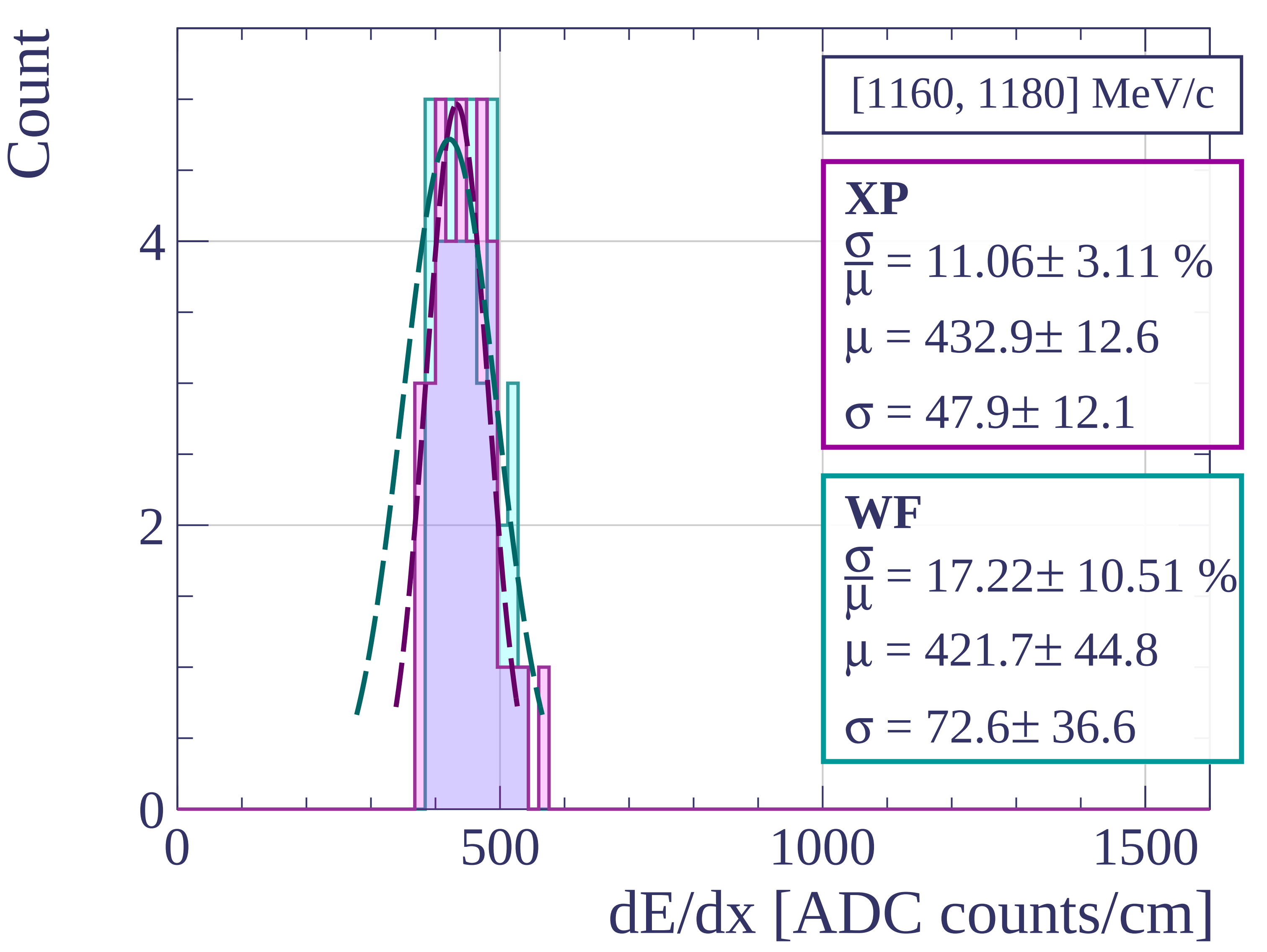


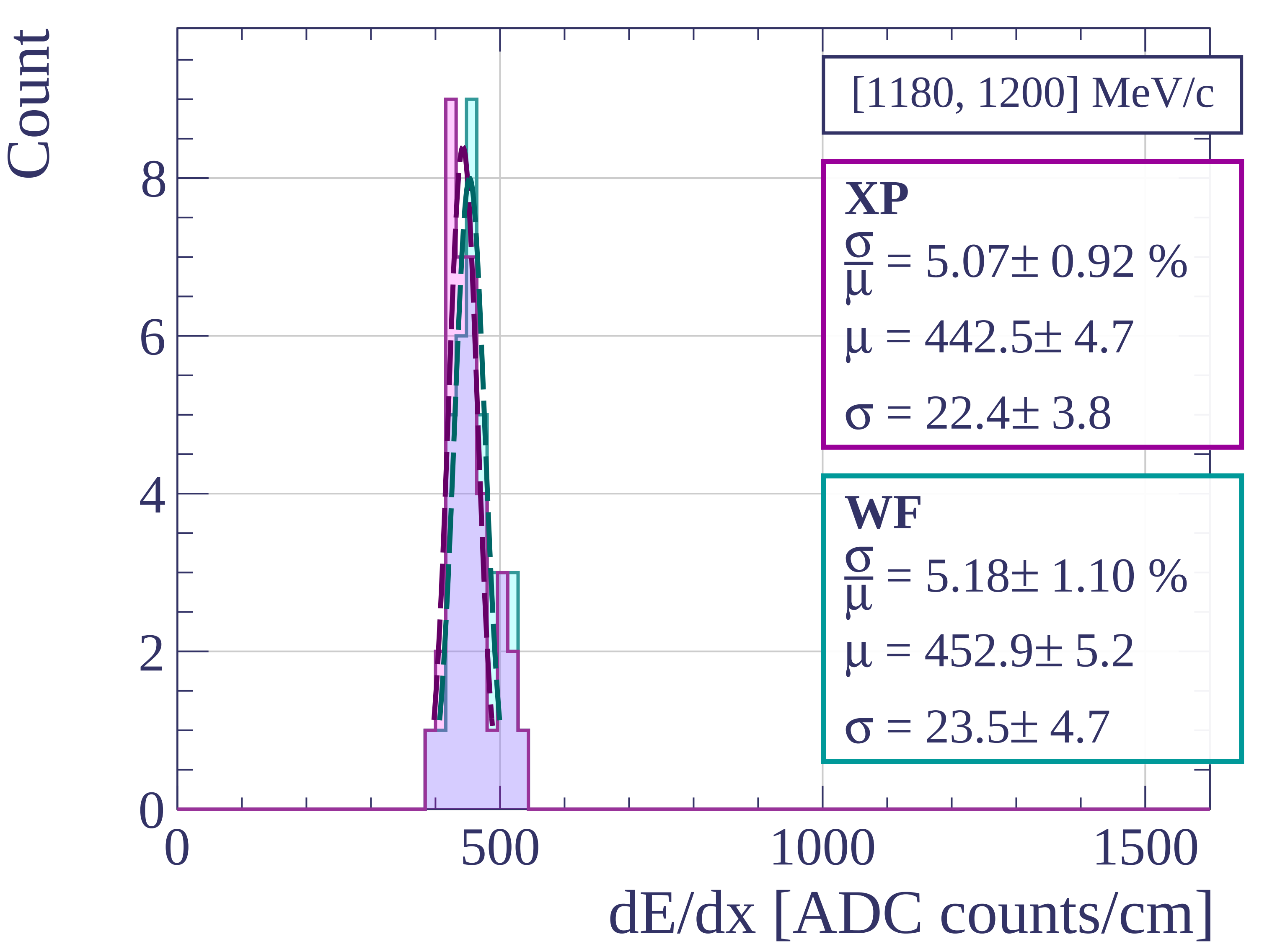
Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[1100, 1120] MeV/c
XP
σ/μ = -978.56± 21306.9
μ = -44.4± 1007.1
σ = 434.9± 384.9
WF
σ/μ = 11.65± 4.34 %
μ = 434.0± 14.0
σ = 50.5± 17.2


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1140] MeV/c
XP
σ/μ = 8.03± 1.50 %
μ = 425.5± 6.6
σ = 34.2± 5.8
WF
σ/μ = 9.03± 3.00 %
μ = 428.4± 8.8
σ = 38.7± 12.1

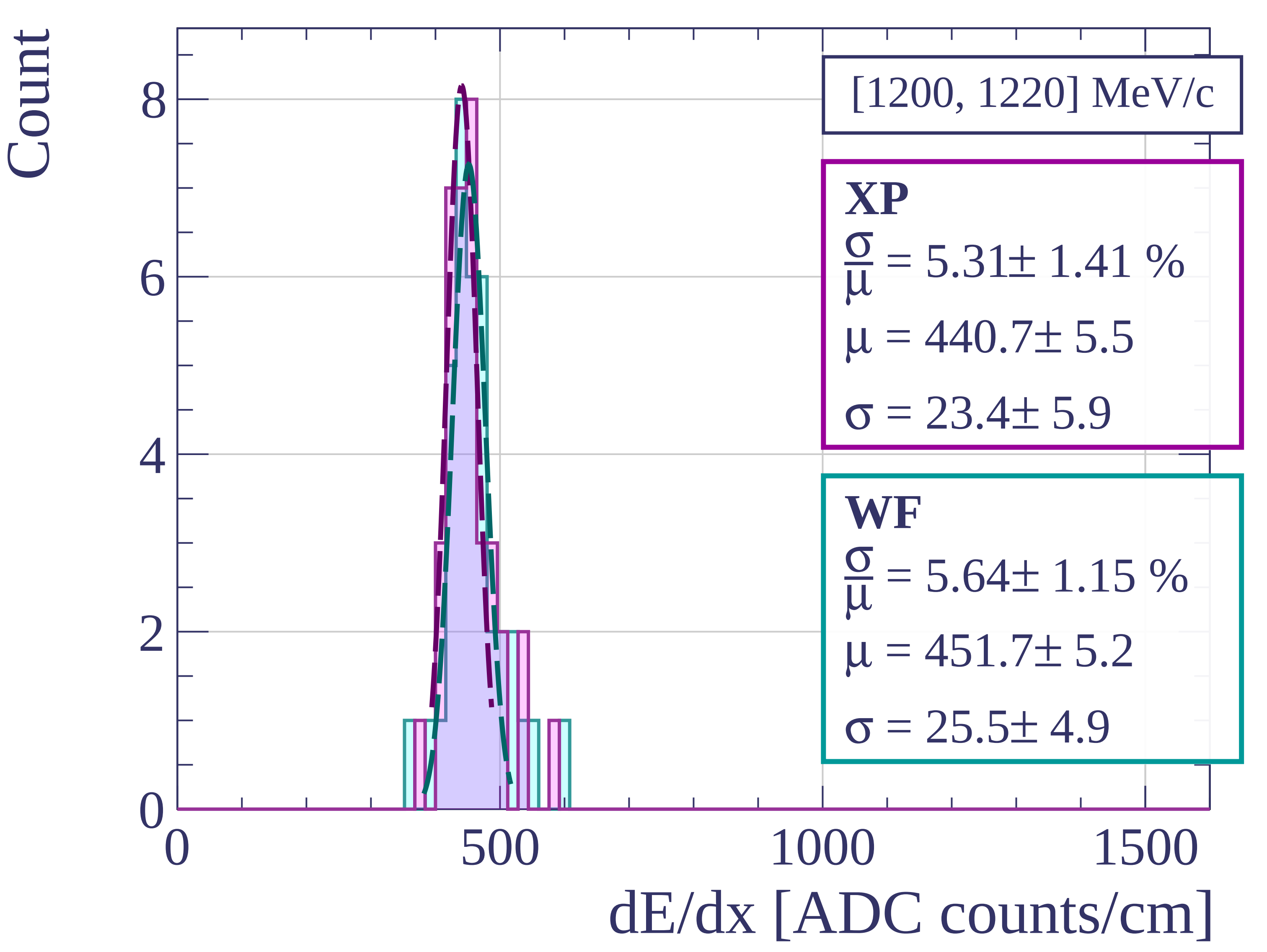


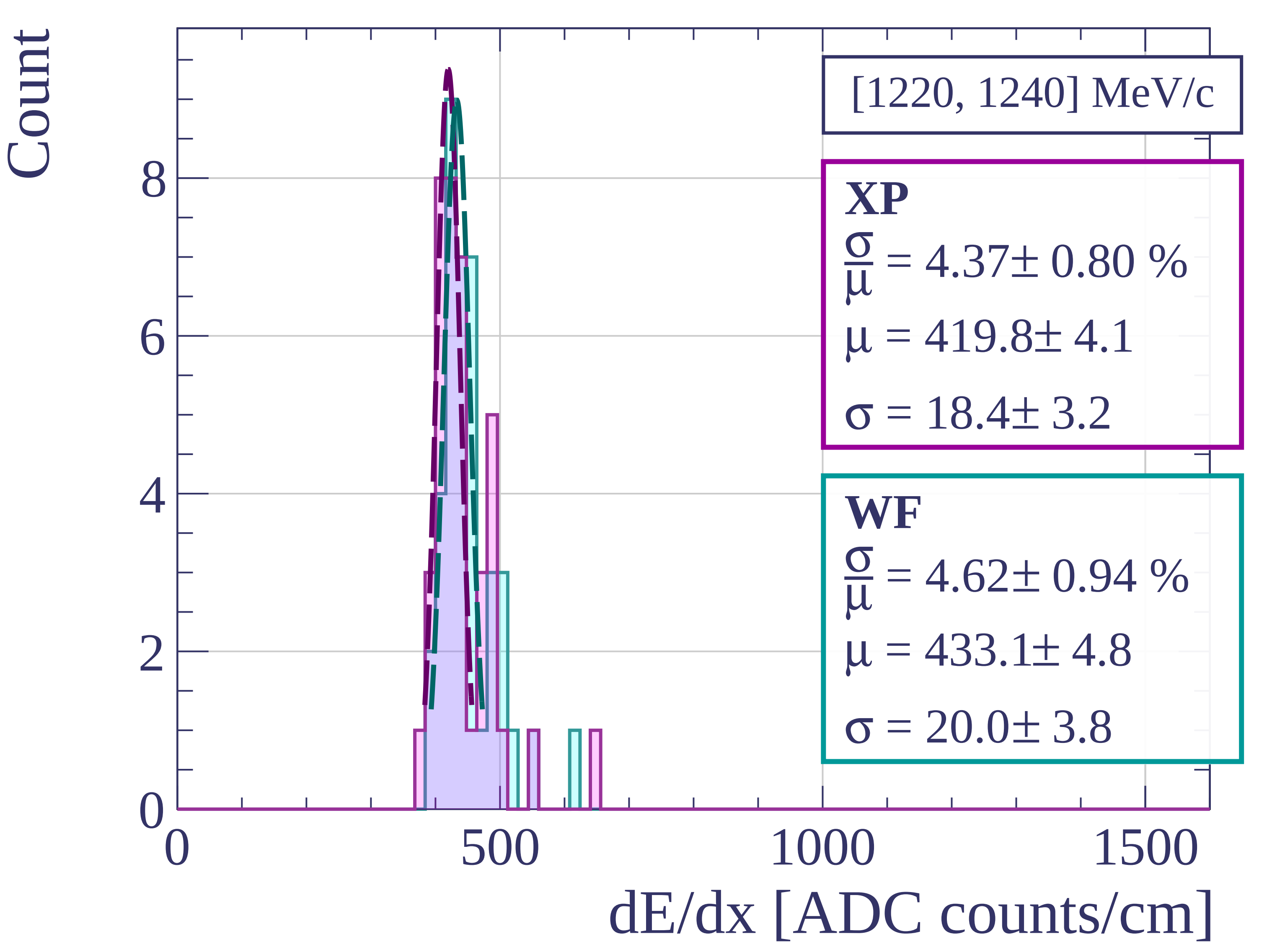
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[1140, 1160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.17± 1.83 %
μ = 424.0± 6.6
σ = 30.4± 7.3
WF
$\frac{\sigma}{\mu}$ = 7.21± 1.71 %
μ = 418.0± 7.3
σ = 30.1± 6.6


Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[1160, 1180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.06± 3.11 %
μ = 432.9± 12.6
σ = 47.9± 12.1
WF
$\frac{\sigma}{\mu}$ = 17.22± 10.51 %
μ = 421.7± 44.8
σ = 72.6± 36.6

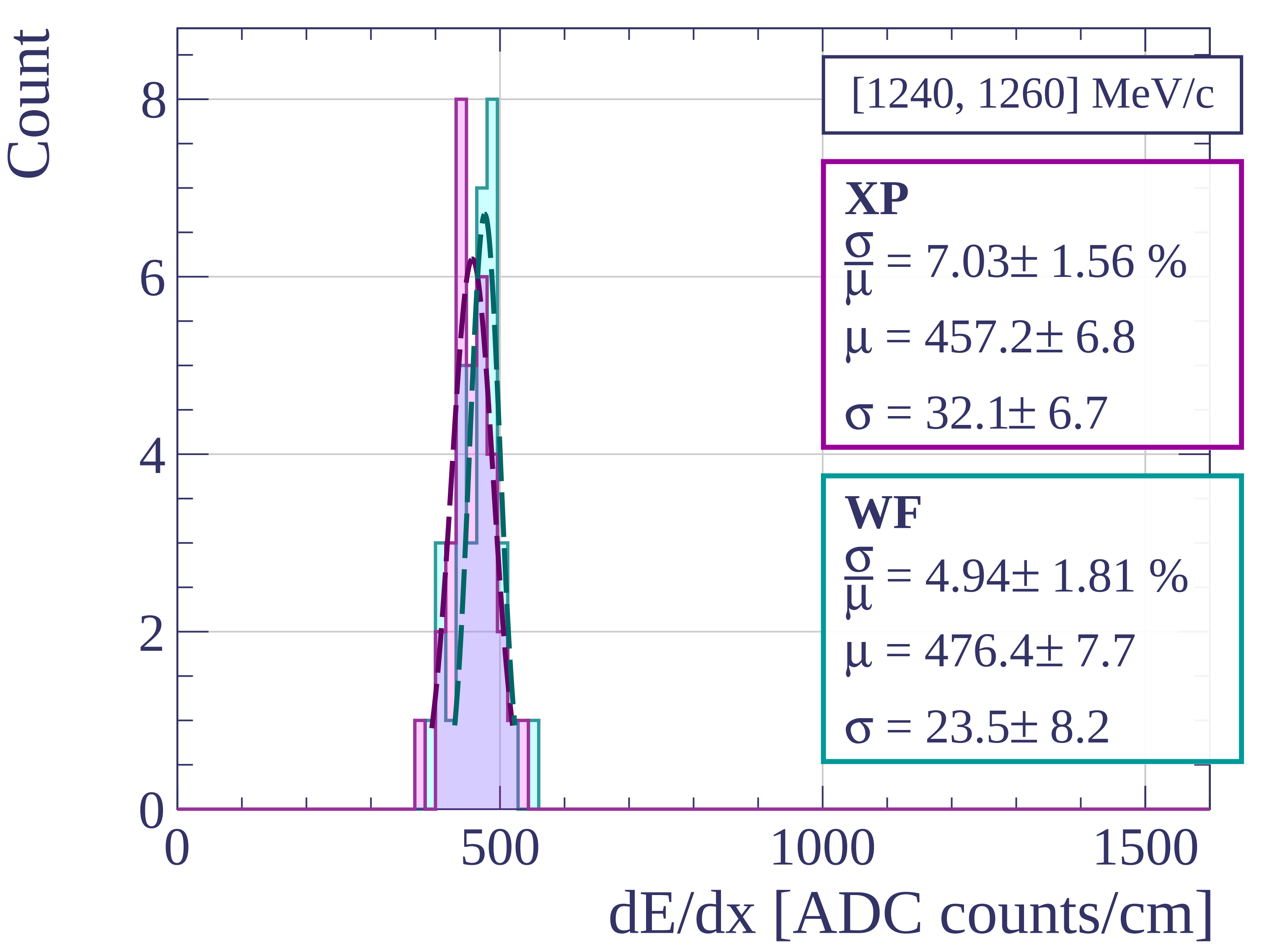


Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[1180, 1200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.07± 0.92 %
μ = 442.5± 4.7
σ = 22.4± 3.8
WF
$\frac{\sigma}{\mu}$ = 5.18± 1.10 %
μ = 452.9± 5.2
σ = 23.5± 4.7

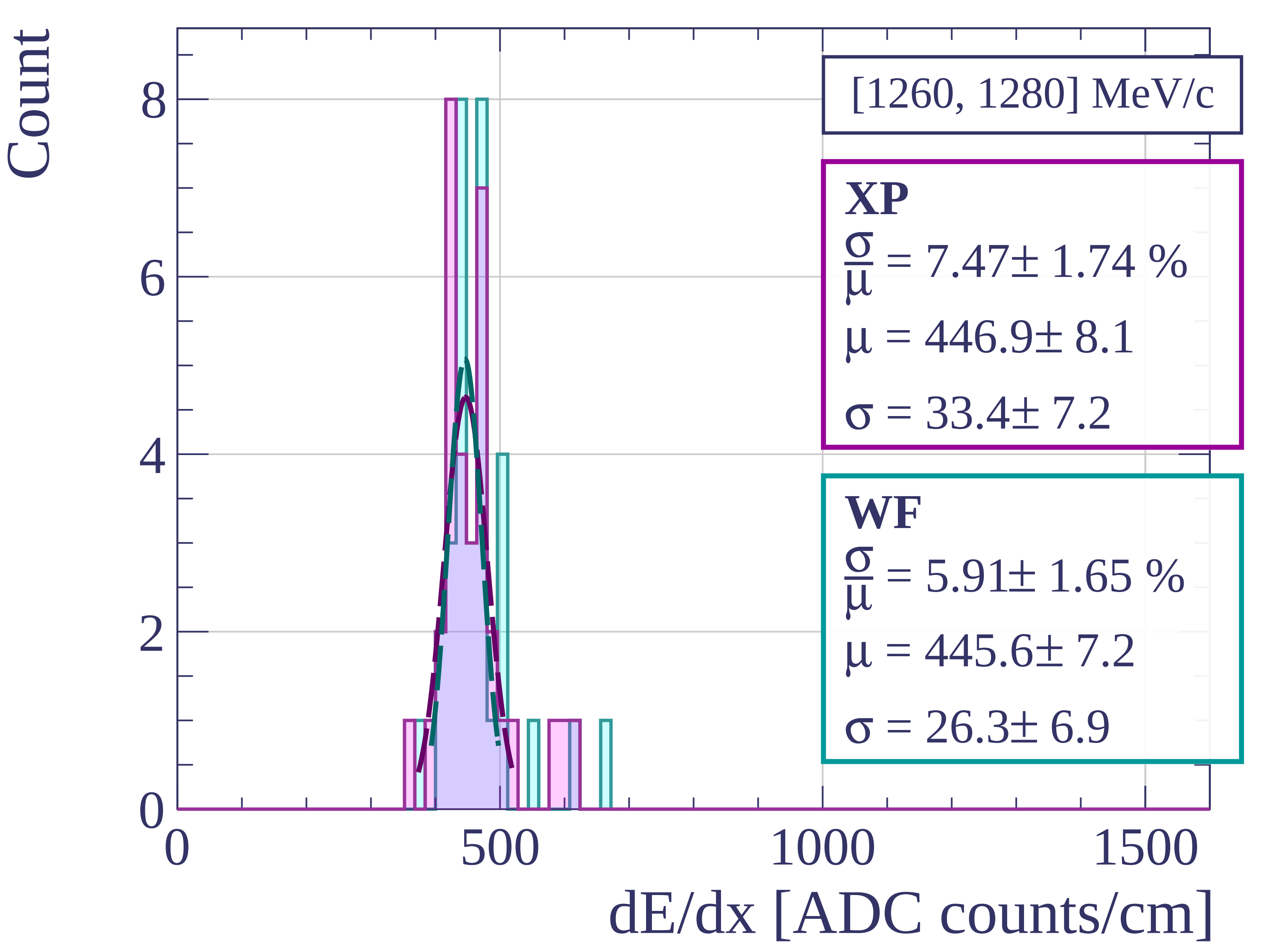

Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[1200, 1220] MeV/c
XP
σ/μ = 5.31± 1.41 %
μ = 440.7± 5.5
σ = 23.4± 5.9
WF
σ/μ = 5.64± 1.15 %
μ = 451.7± 5.2
σ = 25.5± 4.9



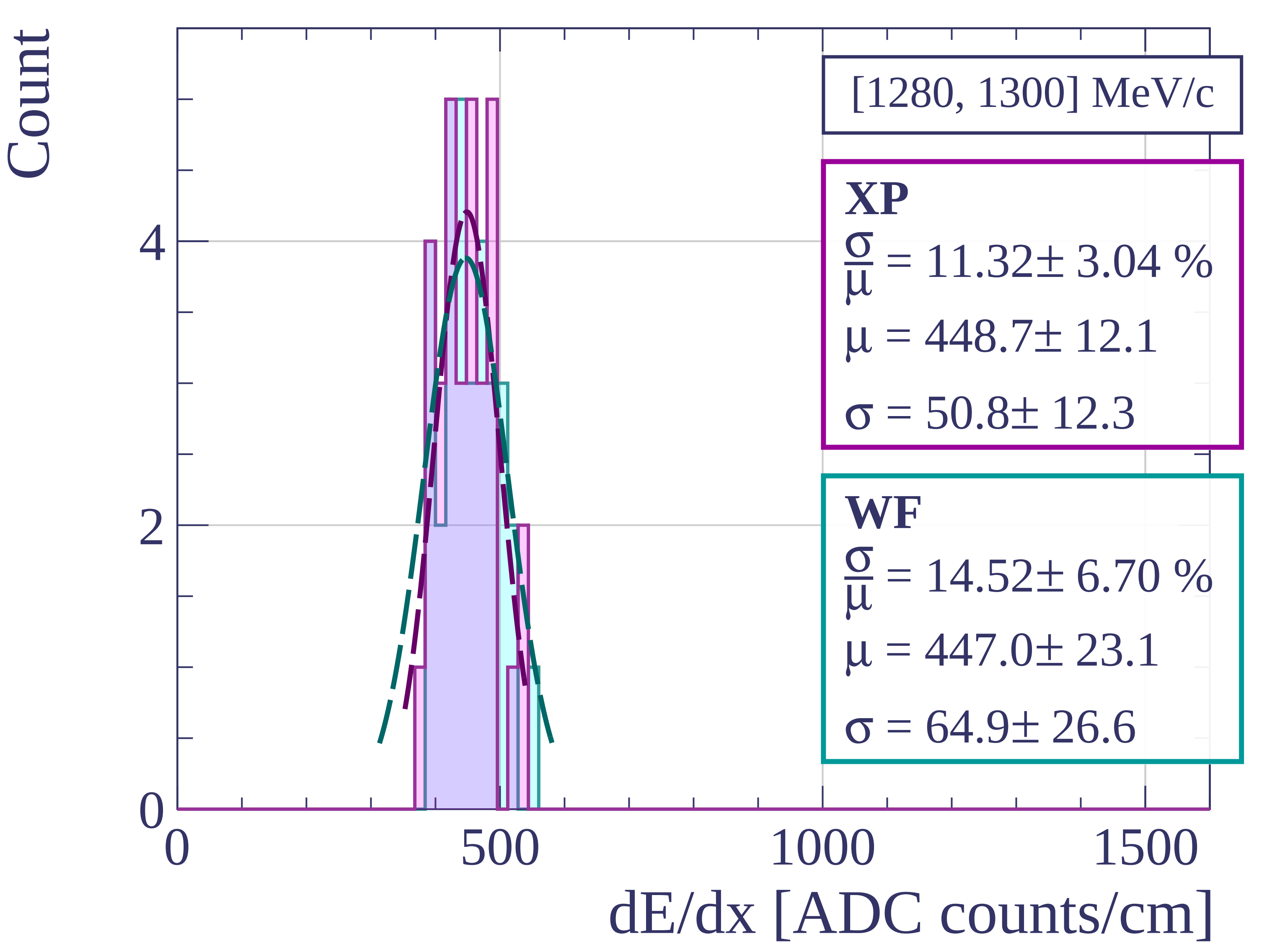
Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[1220, 1240] MeV/c
XP
σ/μ = 4.37± 0.80 %
μ = 419.8± 4.1
σ = 18.4± 3.2
WF
σ/μ = 4.62± 0.94 %
μ = 433.1± 4.8
σ = 20.0± 3.8



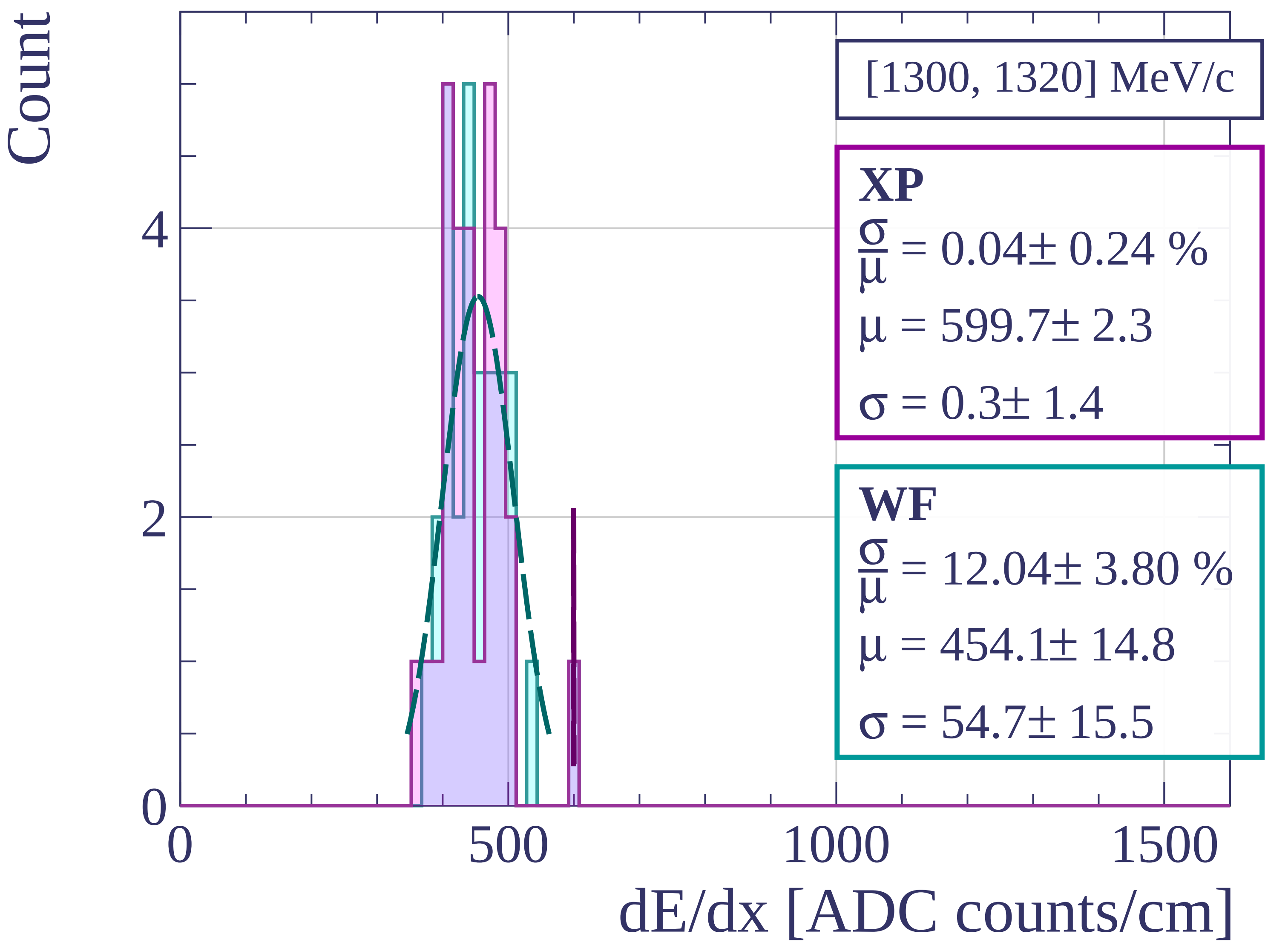
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[1240, 1260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.03± 1.56 %
μ = 457.2± 6.8
σ = 32.1± 6.7
WF
$\frac{\sigma}{\mu}$ = 4.94± 1.81 %
μ = 476.4± 7.7
σ = 23.5± 8.2



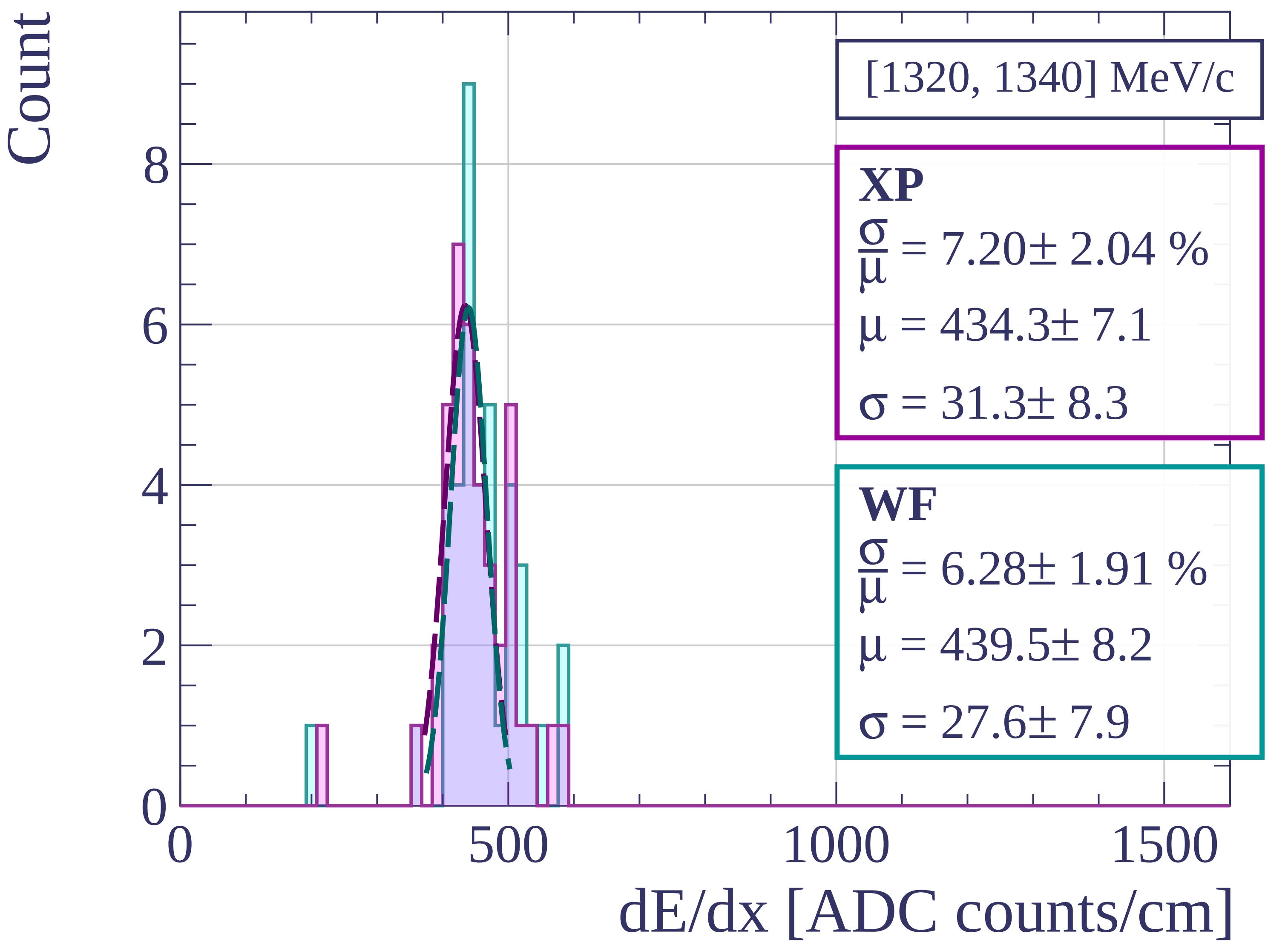
Count
dE/dx [ADC counts/cm]
[1260, 1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.47± 1.74 %
μ = 446.9± 8.1
σ = 33.4± 7.2
WF
$\frac{\sigma}{\mu}$ = 5.91± 1.65 %
μ = 445.6± 7.2
σ = 26.3± 6.9
0
2
4
6
8
0
500
1000
1500

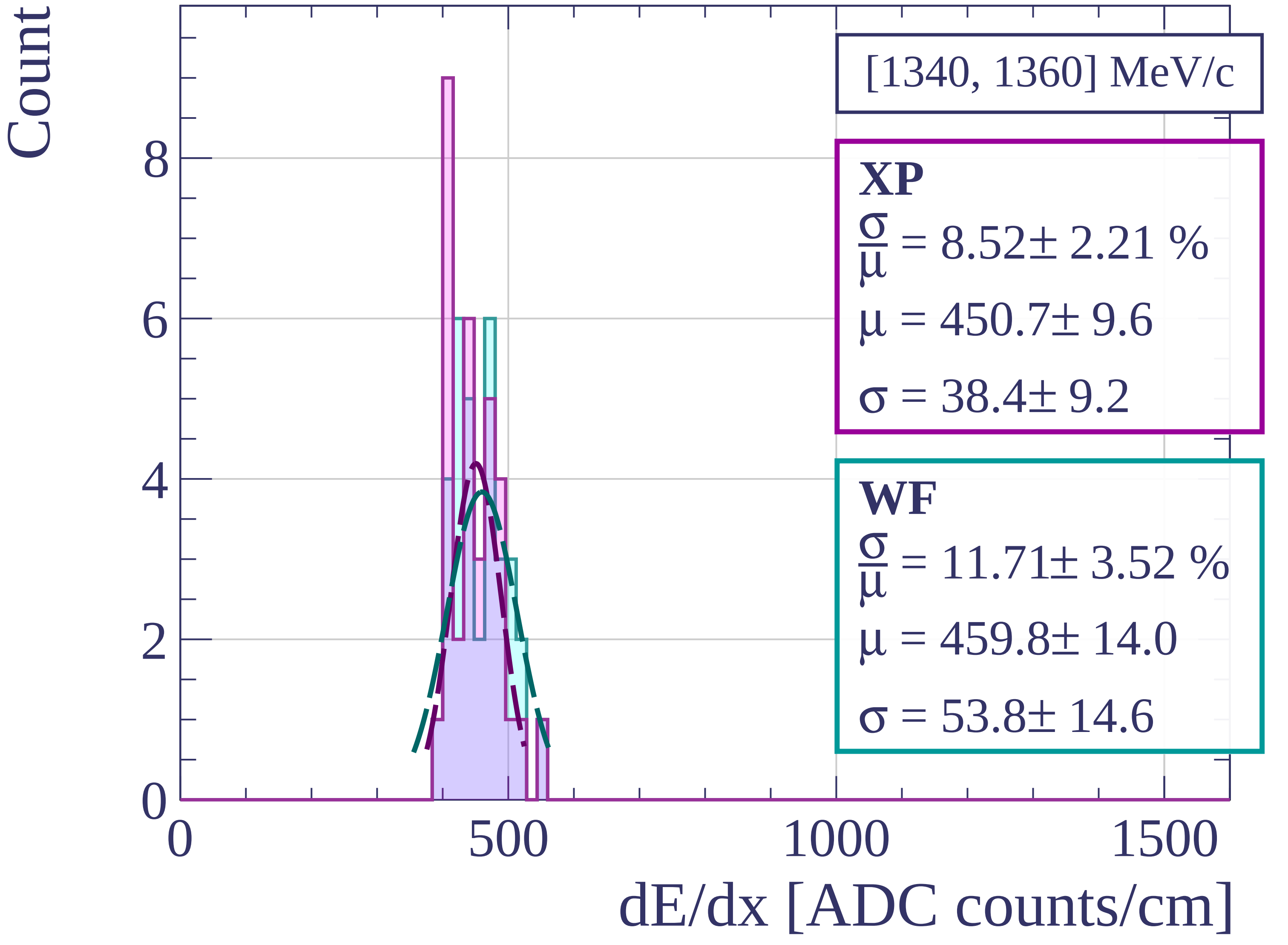


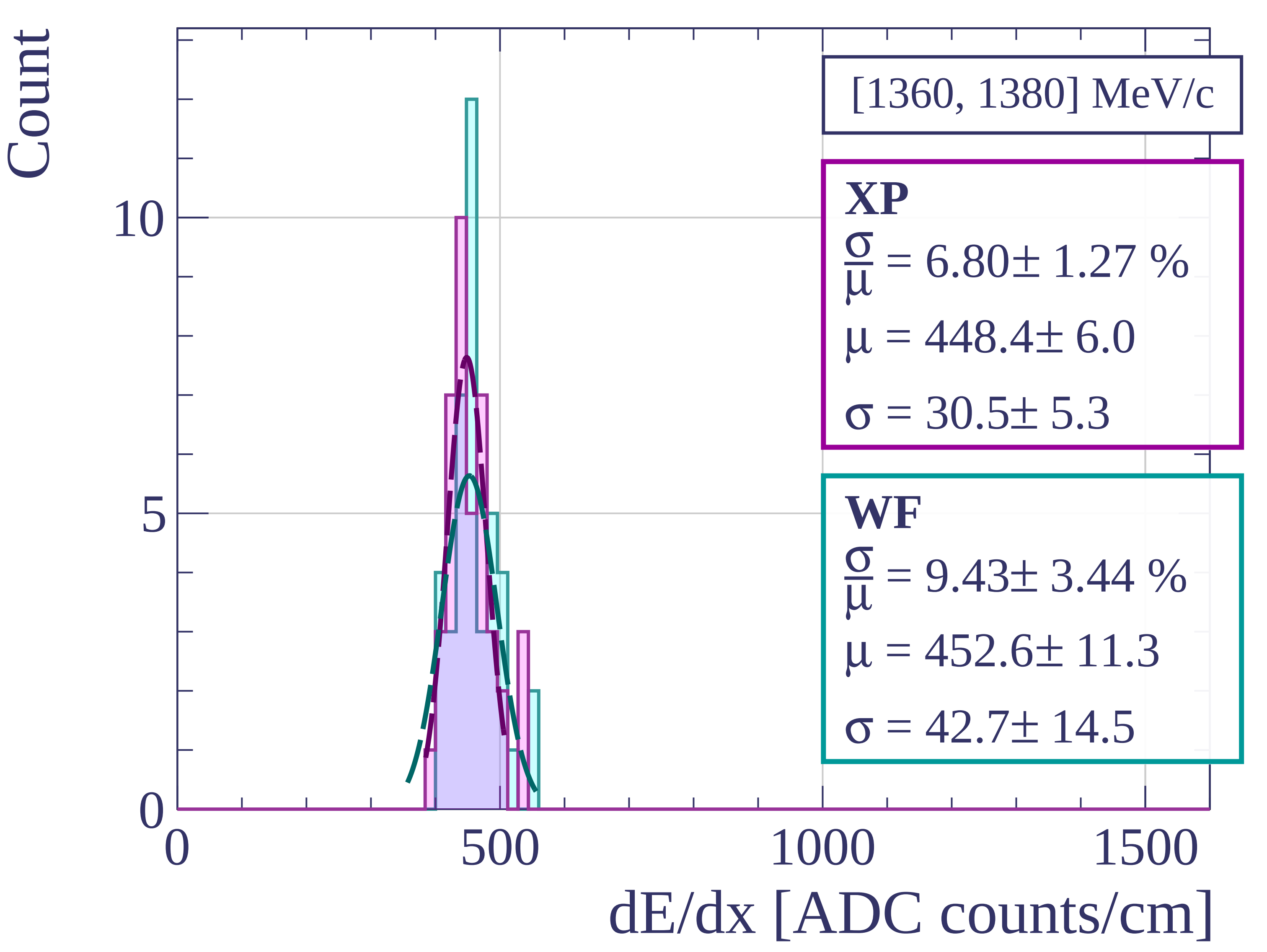
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.32± 3.04 %
μ = 448.7± 12.1
σ = 50.8± 12.3
WF
$\frac{\sigma}{\mu}$ = 14.52± 6.70 %
μ = 447.0± 23.1
σ = 64.9± 26.6

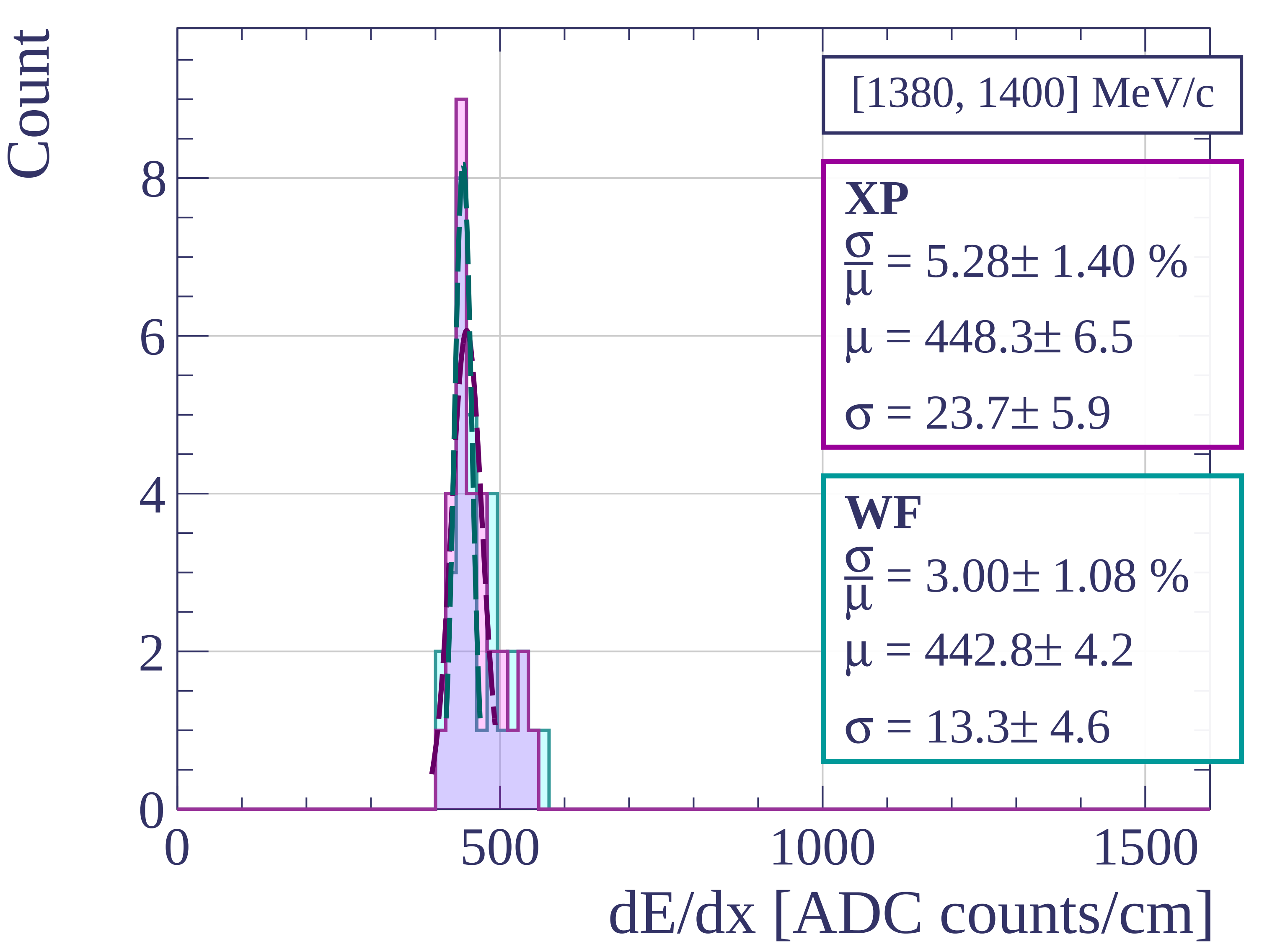


Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[1300, 1320] MeV/c
XP
σ/μ = 0.04± 0.24 %
μ = 599.7± 2.3
σ = 0.3± 1.4
WF
σ/μ = 12.04± 3.80 %
μ = 454.1± 14.8
σ = 54.7± 15.5

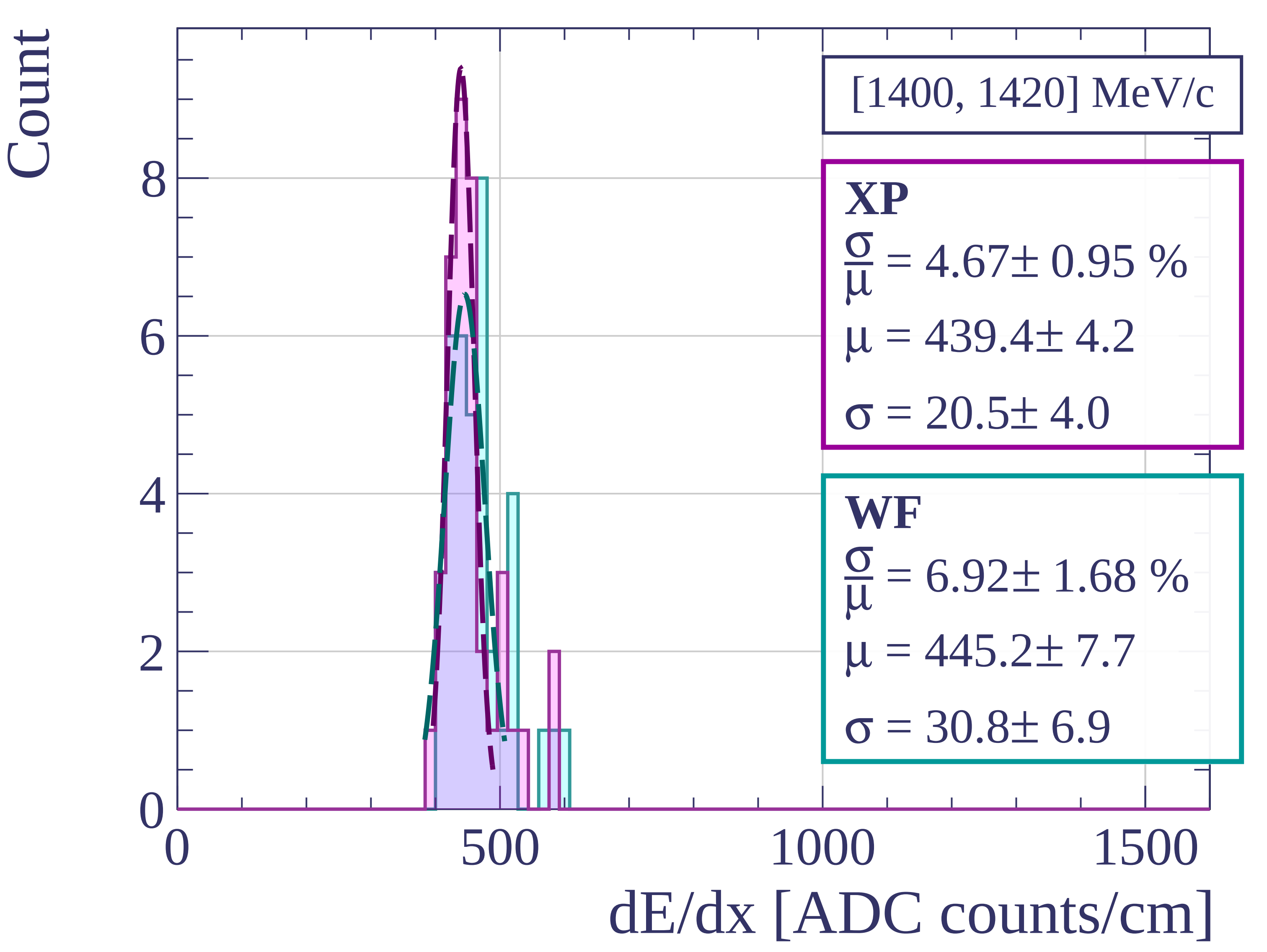

Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[1320, 1340] MeV/c
XP
σ/μ = 7.20± 2.04 %
μ = 434.3± 7.1
σ = 31.3± 8.3
WF
σ/μ = 6.28± 1.91 %
μ = 439.5± 8.2
σ = 27.6± 7.9

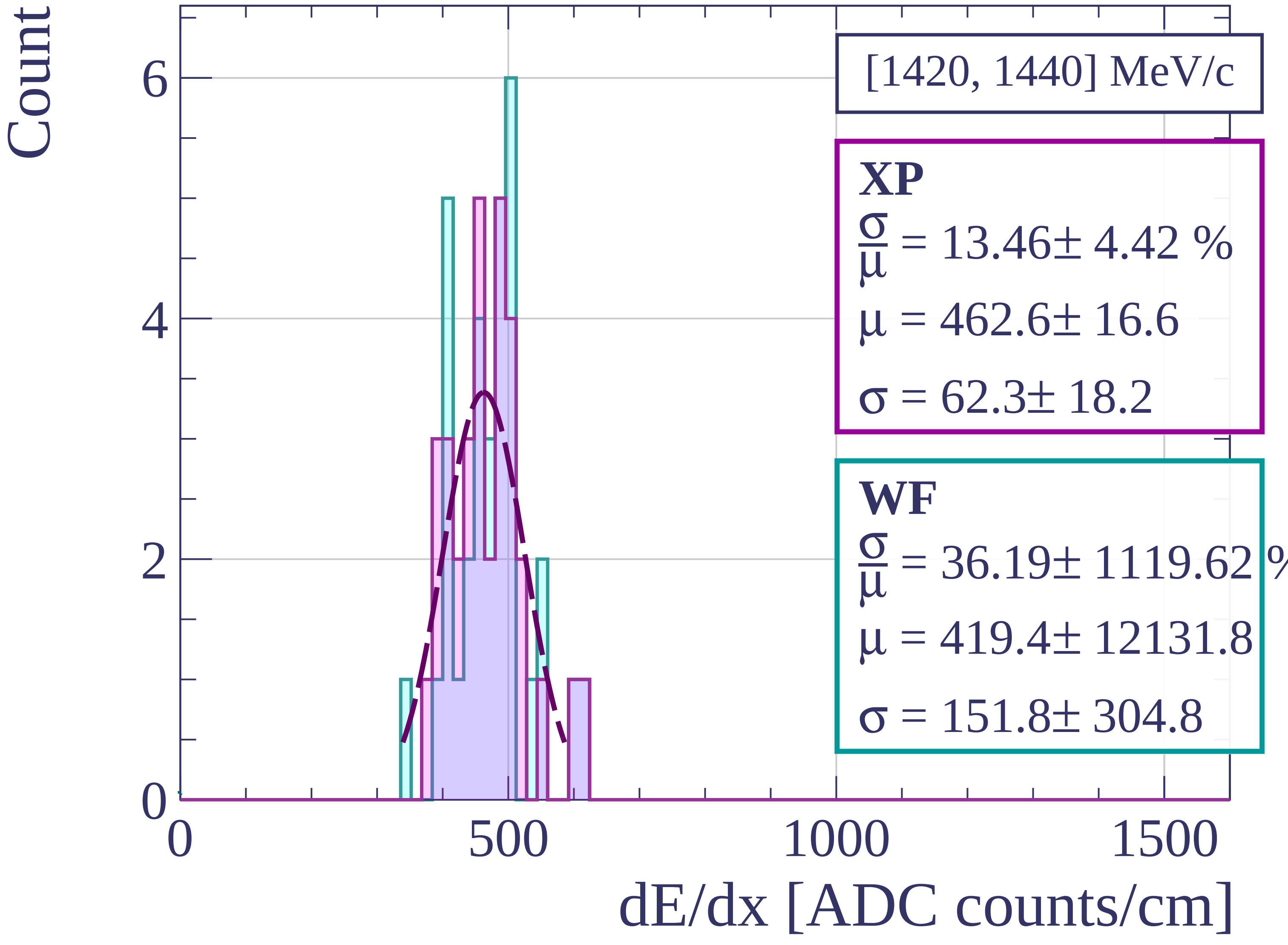

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1340, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.52± 2.21 %
μ = 450.7± 9.6
σ = 38.4± 9.2
WF
$\frac{\sigma}{\mu}$ = 11.71± 3.52 %
μ = 459.8± 14.0
σ = 53.8± 14.6


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1380] MeV/c
XP
σ/μ = 6.80± 1.27 %
μ = 448.4± 6.0
σ = 30.5± 5.3
WF
σ/μ = 9.43± 3.44 %
μ = 452.6± 11.3
σ = 42.7± 14.5

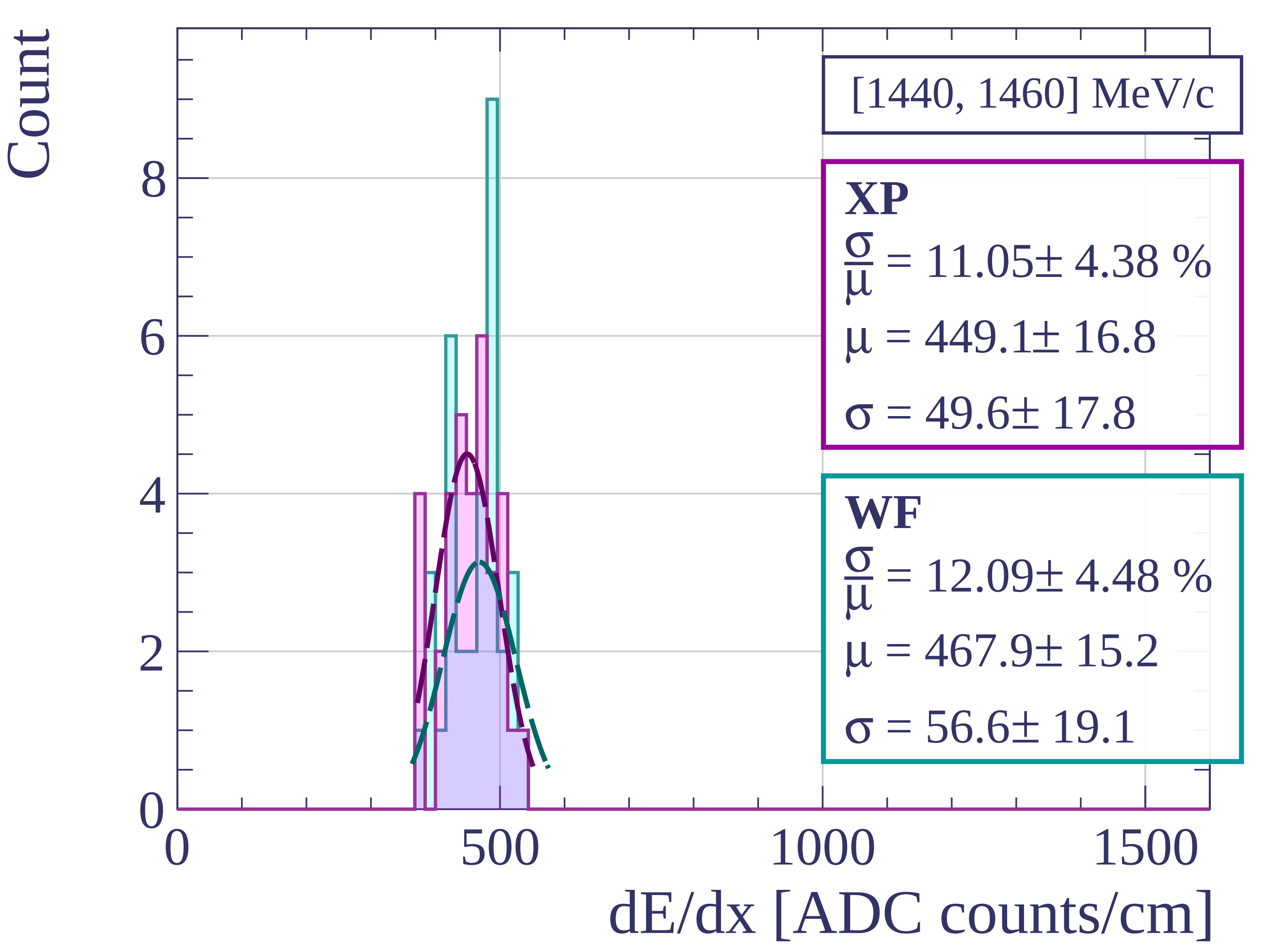


Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[1380, 1400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.28± 1.40 %
μ = 448.3± 6.5
σ = 23.7± 5.9
WF
$\frac{\sigma}{\mu}$ = 3.00± 1.08 %
μ = 442.8± 4.2
σ = 13.3± 4.6

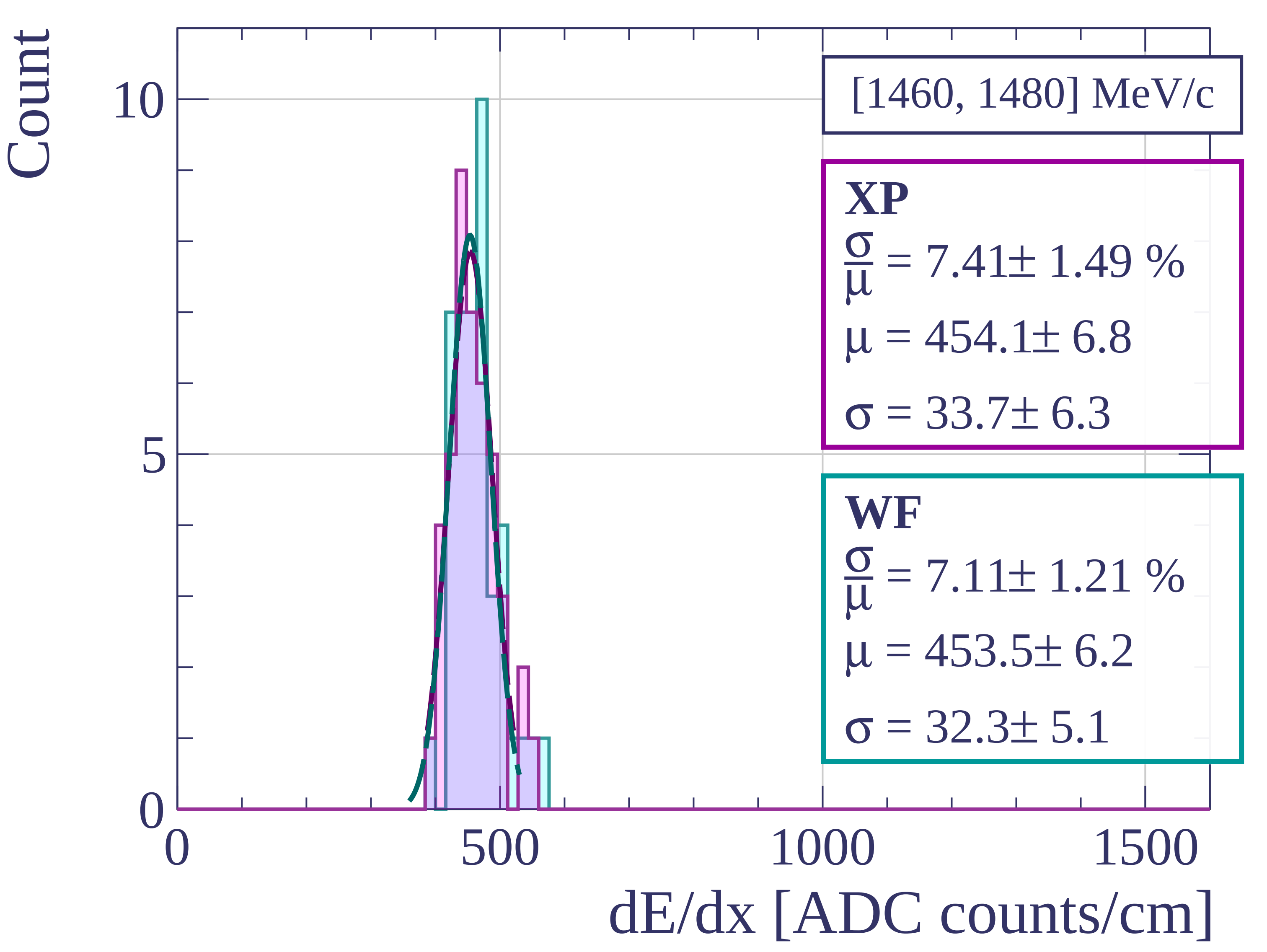

Count
[1400, 1420] MeV/c
XP
σ/μ = 4.67± 0.95 %
μ = 439.4± 4.2
σ = 20.5± 4.0
WF
σ/μ = 6.92± 1.68 %
μ = 445.2± 7.7
σ = 30.8± 6.9
dE/dx [ADC counts/cm]



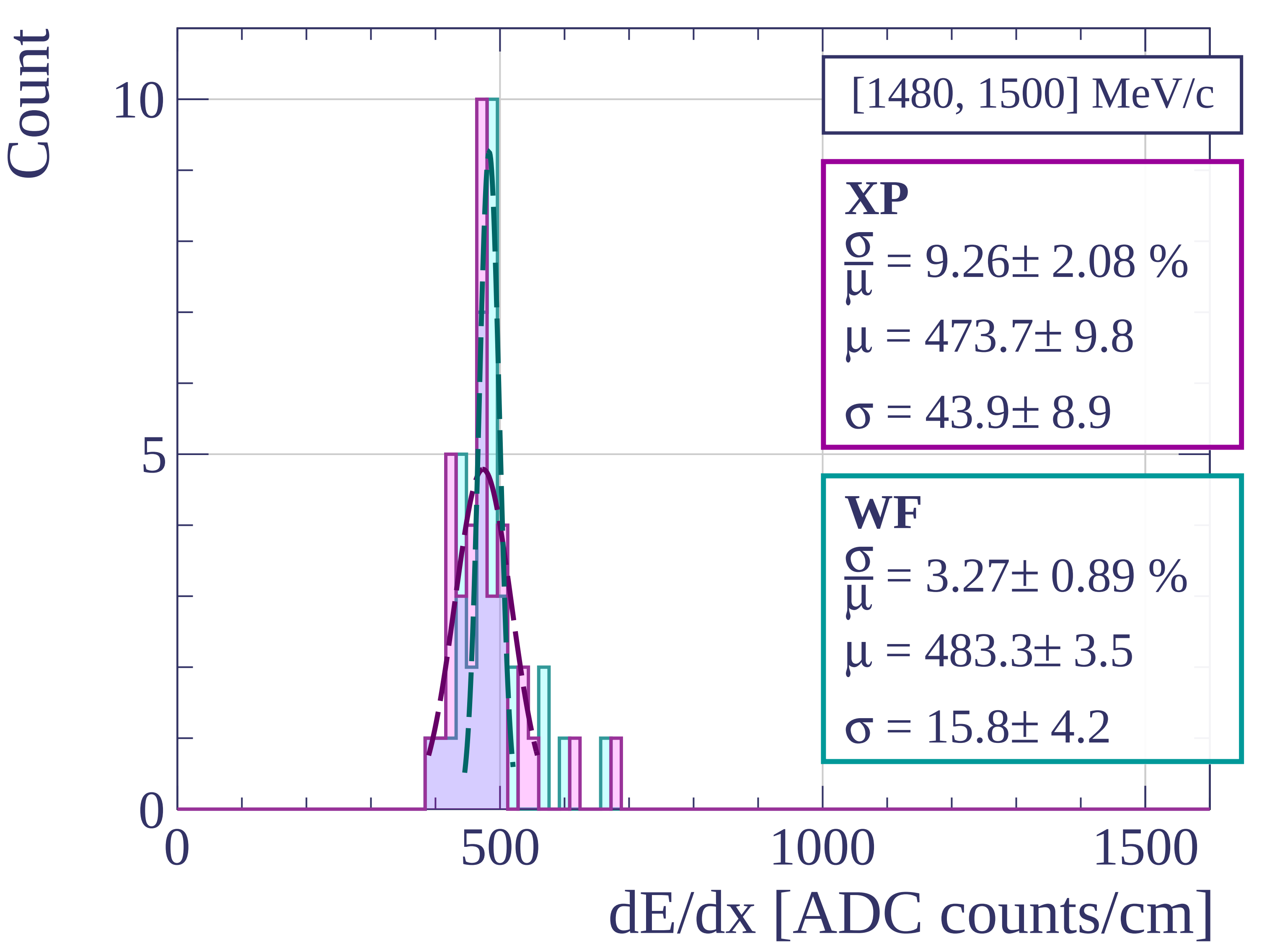
Count
dE/dx [ADC counts/cm]
[1420, 1440] MeV/c
XP
σ/μ = 13.46± 4.42 %
μ = 462.6± 16.6
σ = 62.3± 18.2
WF
σ/μ = 36.19± 1119.62 %
μ = 419.4± 12131.8
σ = 151.8± 304.8
0
2
4
6
0
500
1000
1500

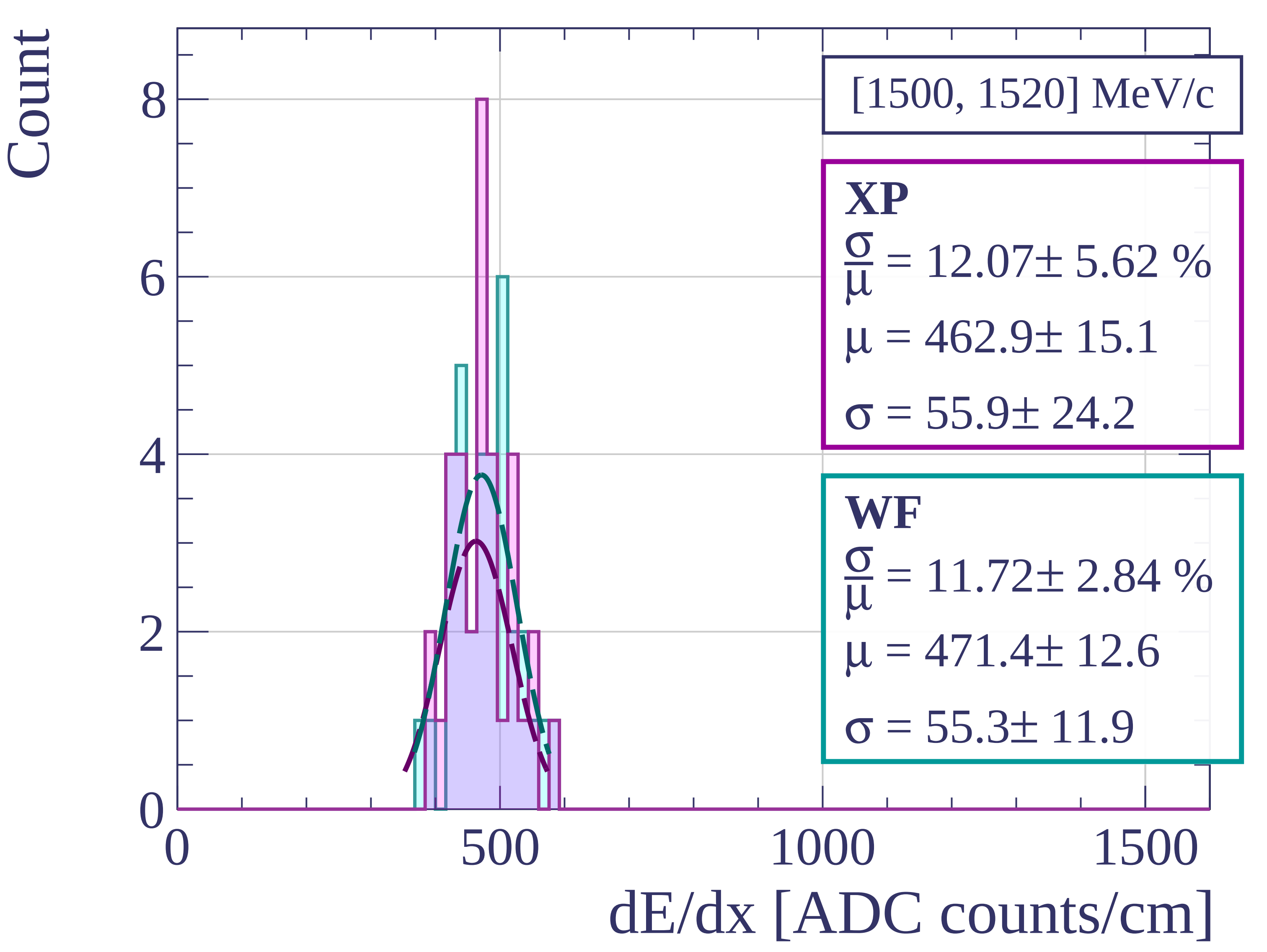


Count
dE/dx [ADC counts/cm]
[1440, 1460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.05± 4.38 %
μ = 449.1± 16.8
σ = 49.6± 17.8
WF
$\frac{\sigma}{\mu}$ = 12.09± 4.48 %
μ = 467.9± 15.2
σ = 56.6± 19.1
0
2
4
6
8
0
500
1000
1500

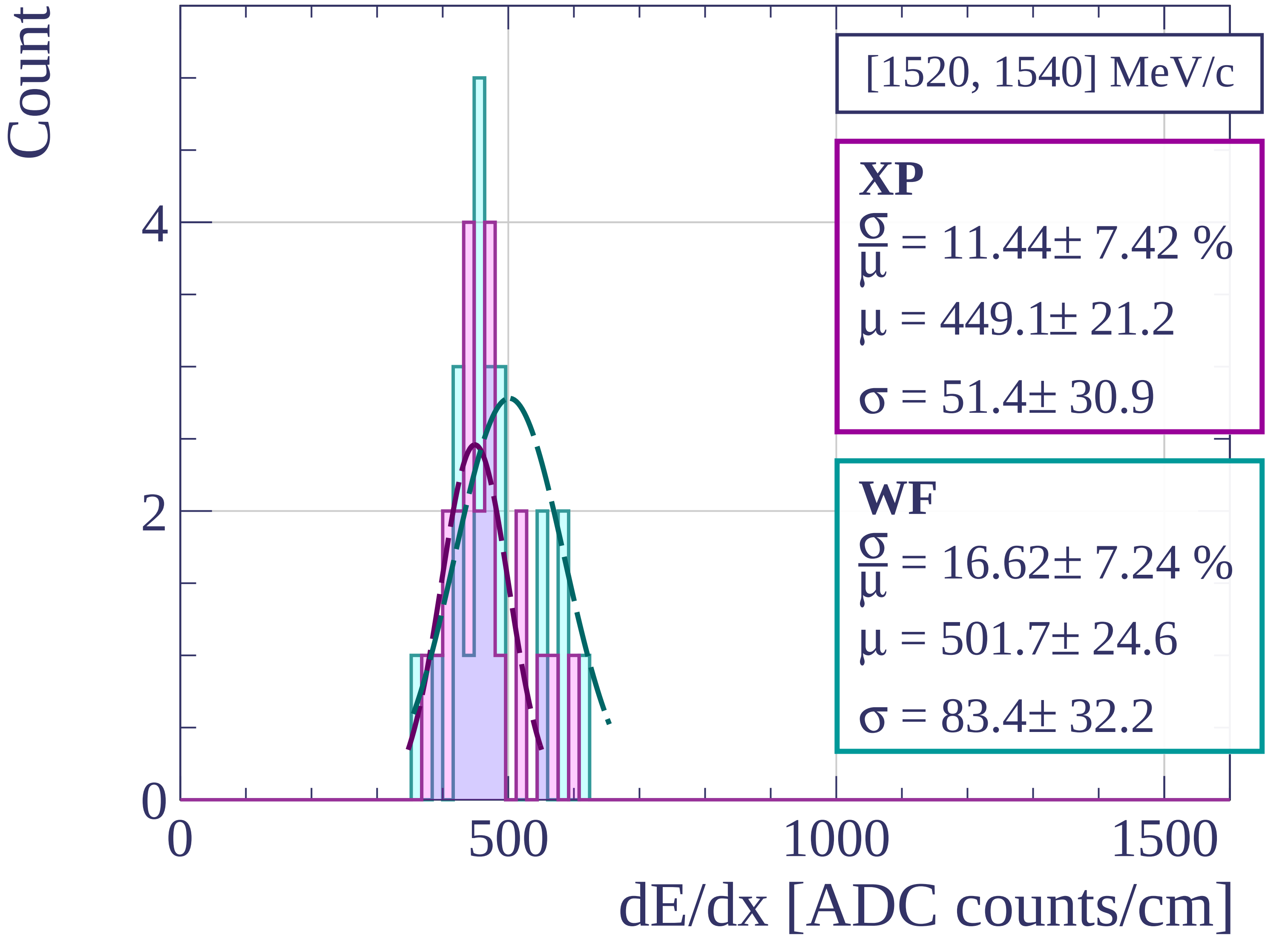


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1460, 1480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.41± 1.49 %
μ = 454.1± 6.8
σ = 33.7± 6.3
WF
$\frac{\sigma}{\mu}$ = 7.11± 1.21 %
μ = 453.5± 6.2
σ = 32.3± 5.1

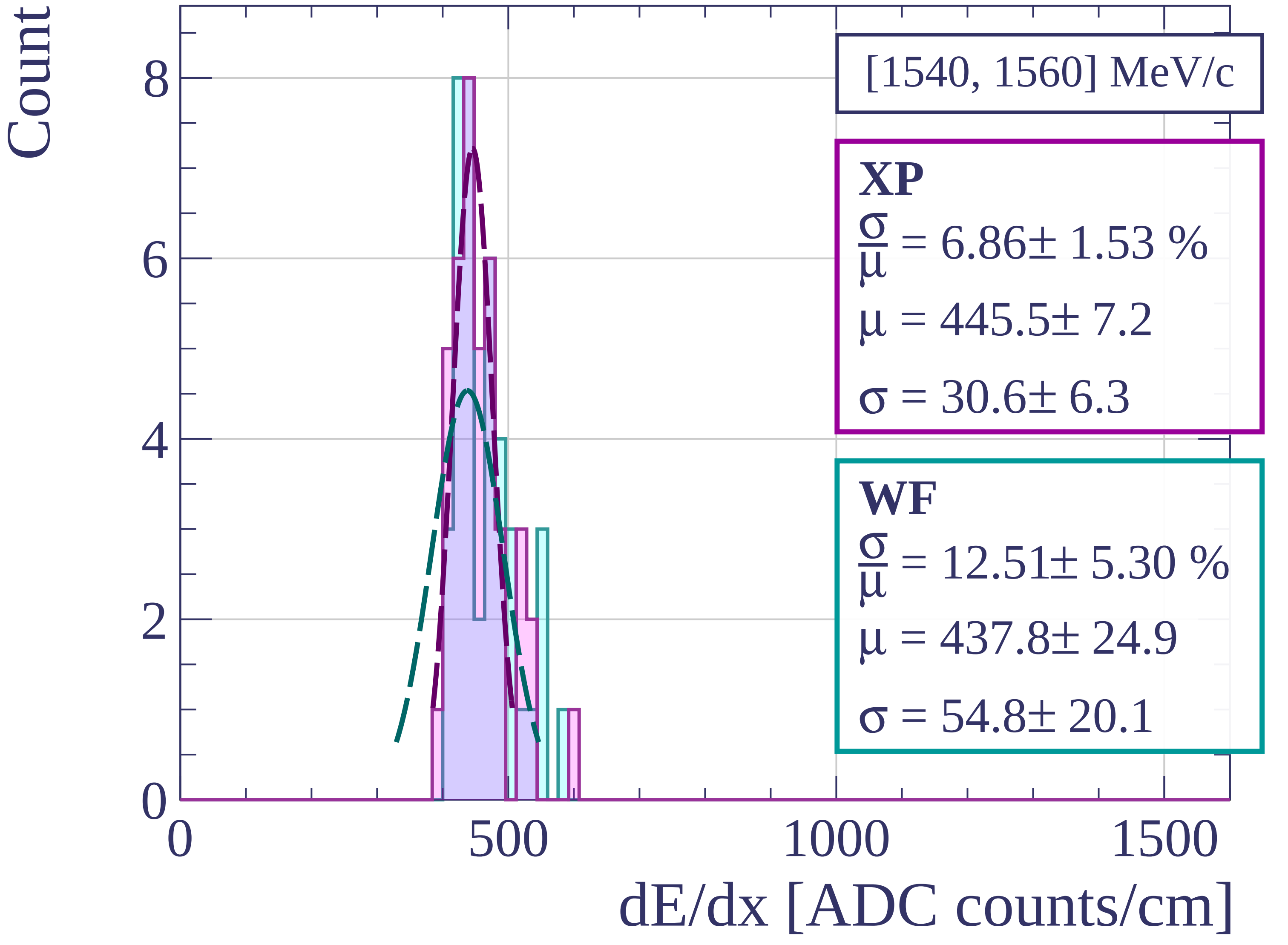

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1480, 1500] MeV/c
XP
σ/μ = 9.26± 2.08 %
μ = 473.7± 9.8
σ = 43.9± 8.9
WF
σ/μ = 3.27± 0.89 %
μ = 483.3± 3.5
σ = 15.8± 4.2



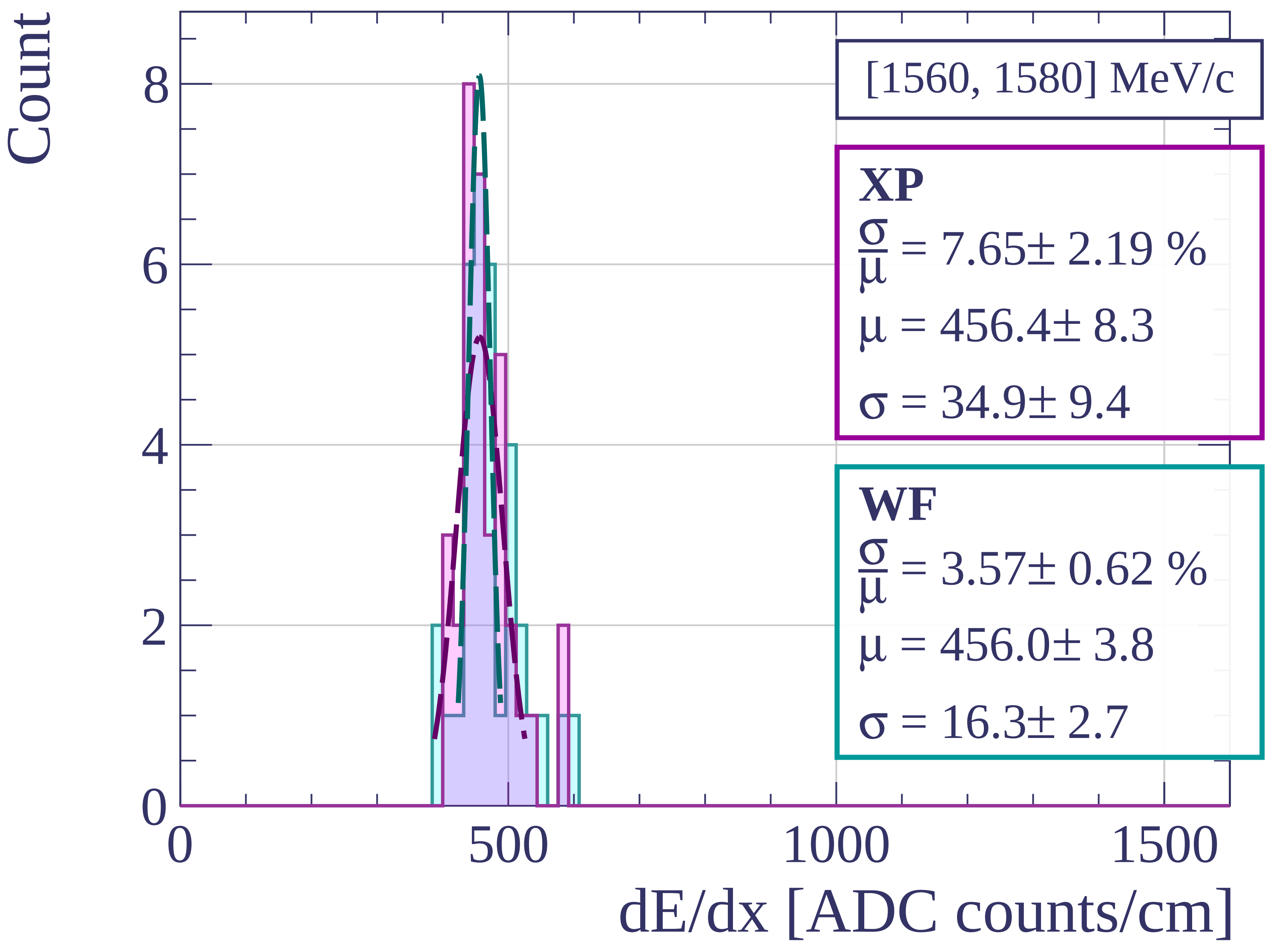
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1500, 1520] MeV/c
XP
σ/μ = 12.07± 5.62 %
μ = 462.9± 15.1
σ = 55.9± 24.2
WF
σ/μ = 11.72± 2.84 %
μ = 471.4± 12.6
σ = 55.3± 11.9

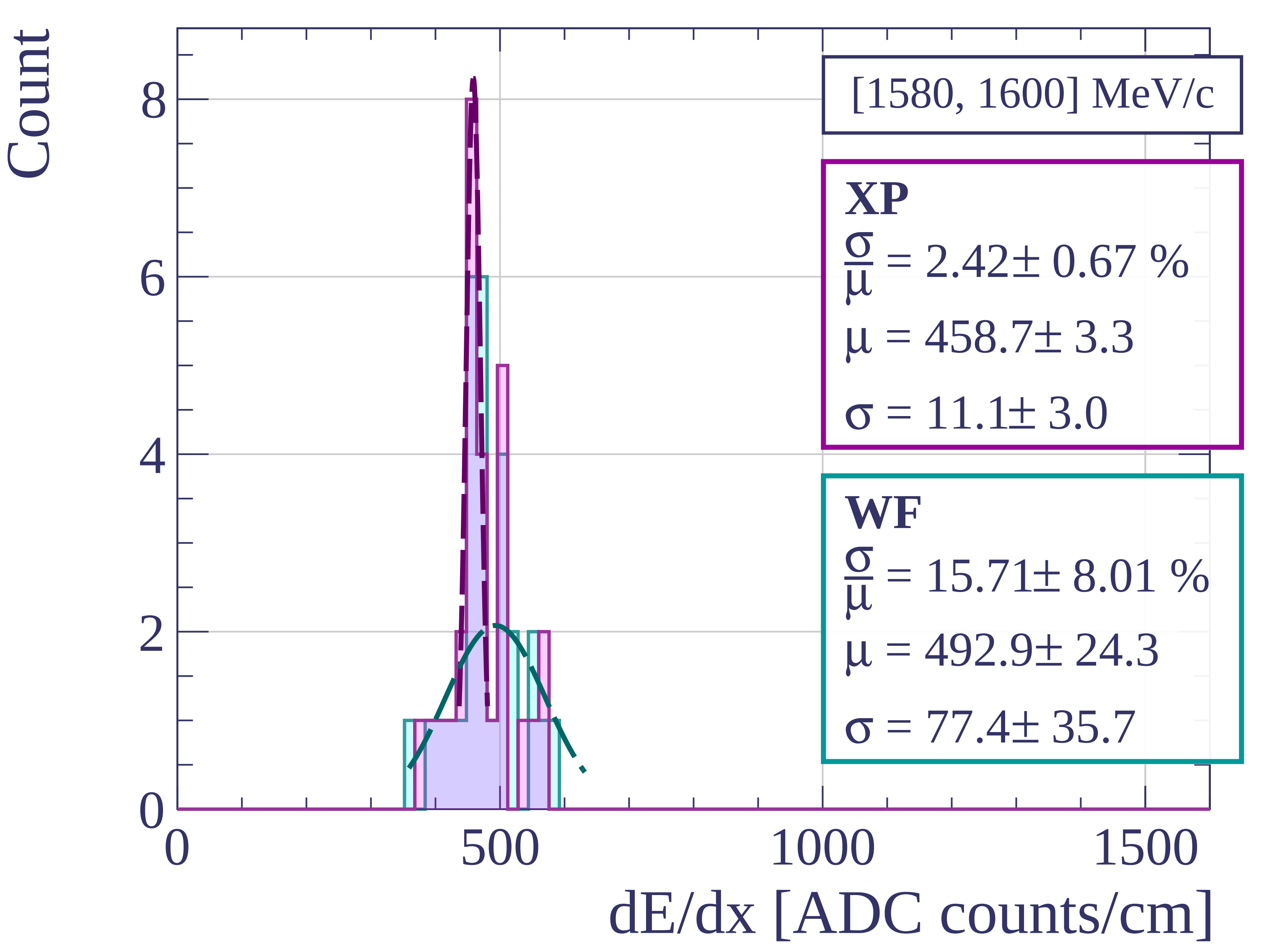


Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[1520, 1540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.44± 7.42 %
μ = 449.1± 21.2
σ = 51.4± 30.9
WF
$\frac{\sigma}{\mu}$ = 16.62± 7.24 %
μ = 501.7± 24.6
σ = 83.4± 32.2

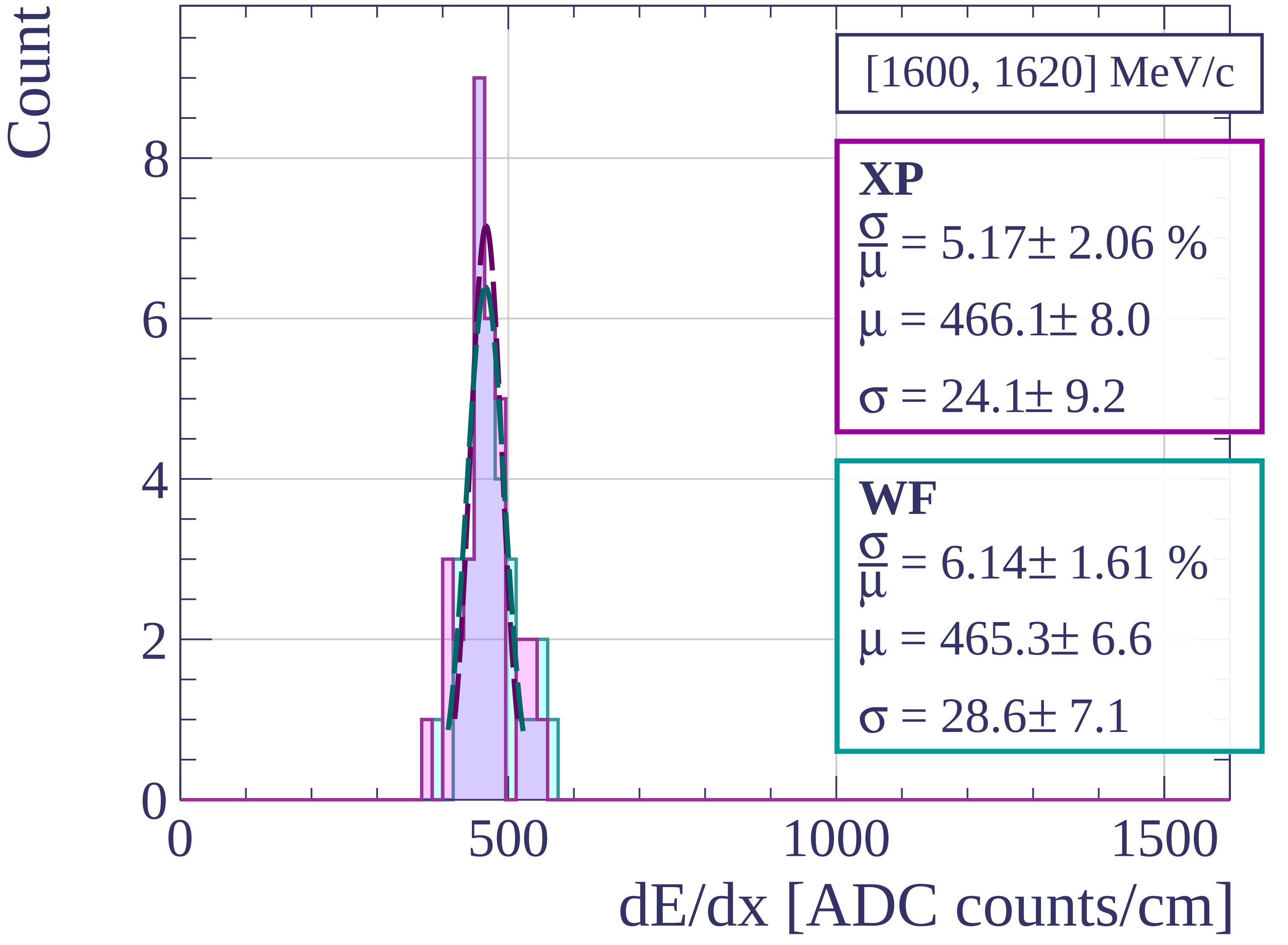


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1540, 1560] MeV/c
XP
σ/μ = 6.86± 1.53 %
μ = 445.5± 7.2
σ = 30.6± 6.3
WF
σ/μ = 12.51± 5.30 %
μ = 437.8± 24.9
σ = 54.8± 20.1

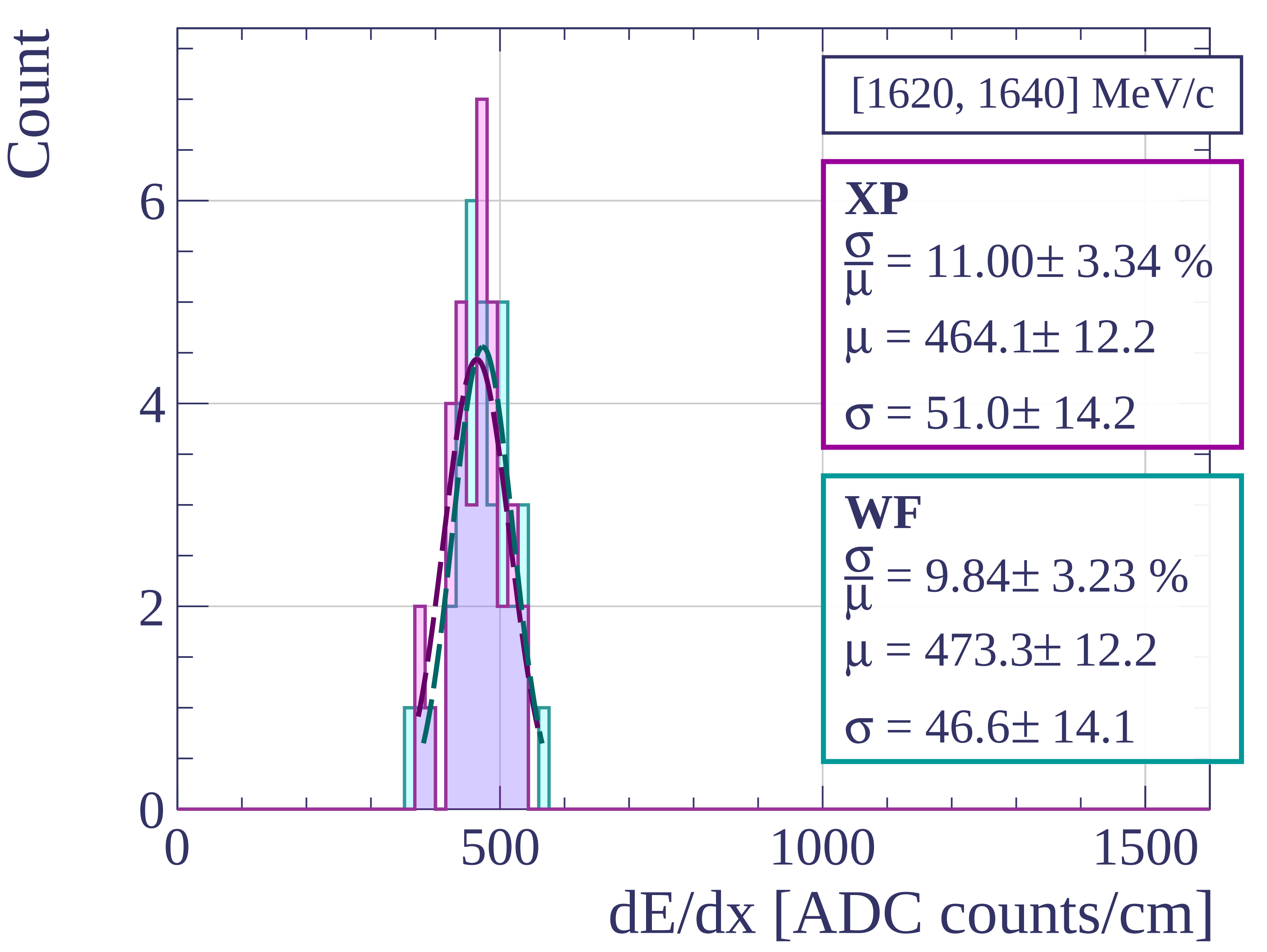

Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[1560, 1580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.65± 2.19 %
μ = 456.4± 8.3
σ = 34.9± 9.4
WF
$\frac{\sigma}{\mu}$ = 3.57± 0.62 %
μ = 456.0± 3.8
σ = 16.3± 2.7



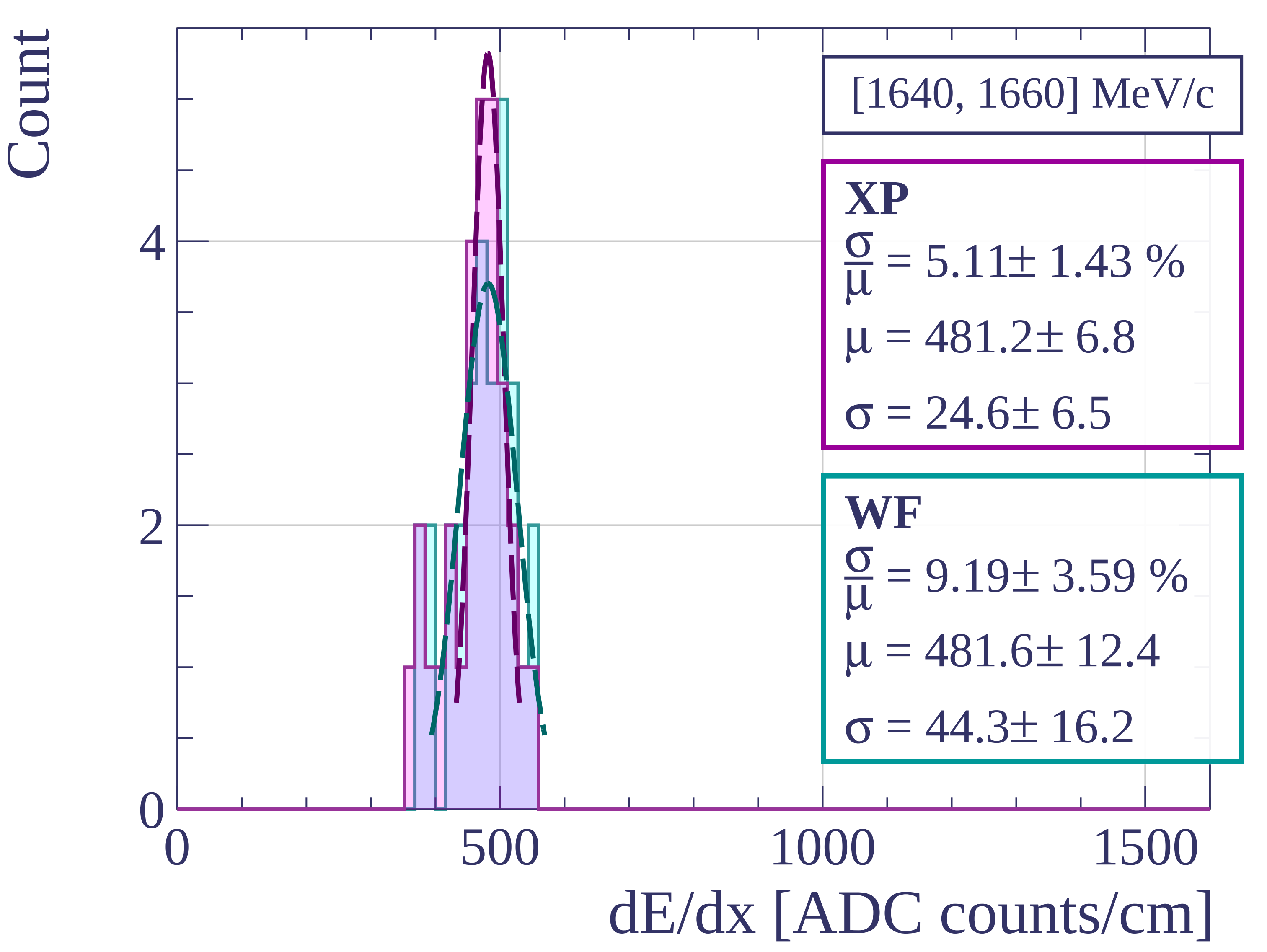
Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[1580, 1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 2.42± 0.67 %
μ = 458.7± 3.3
σ = 11.1± 3.0
WF
$\frac{\sigma}{\mu}$ = 15.71± 8.01 %
μ = 492.9± 24.3
σ = 77.4± 35.7



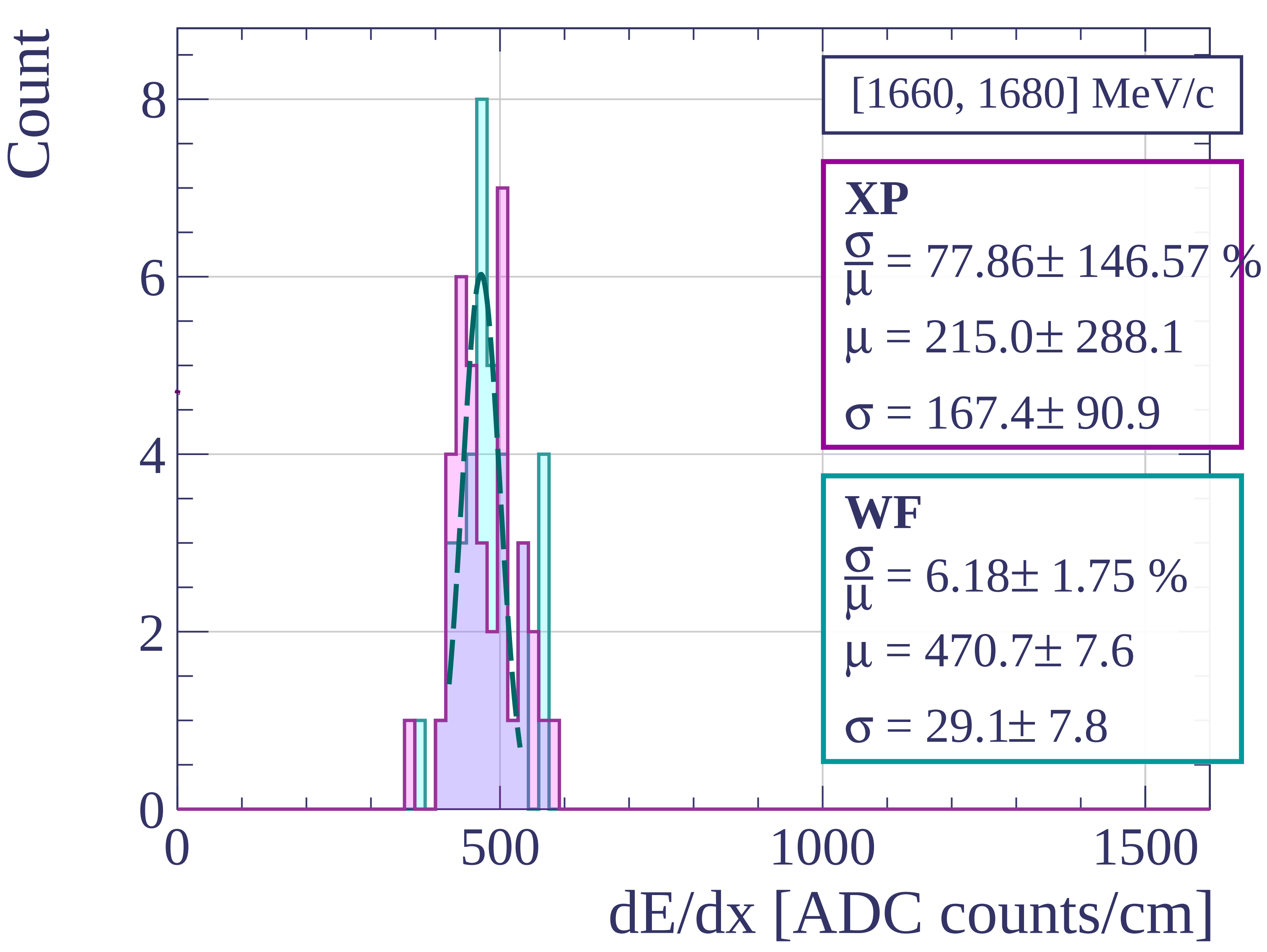
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1620] MeV/c
XP
σ/μ = 5.17± 2.06 %
μ = 466.1± 8.0
σ = 24.1± 9.2
WF
σ/μ = 6.14± 1.61 %
μ = 465.3± 6.6
σ = 28.6± 7.1



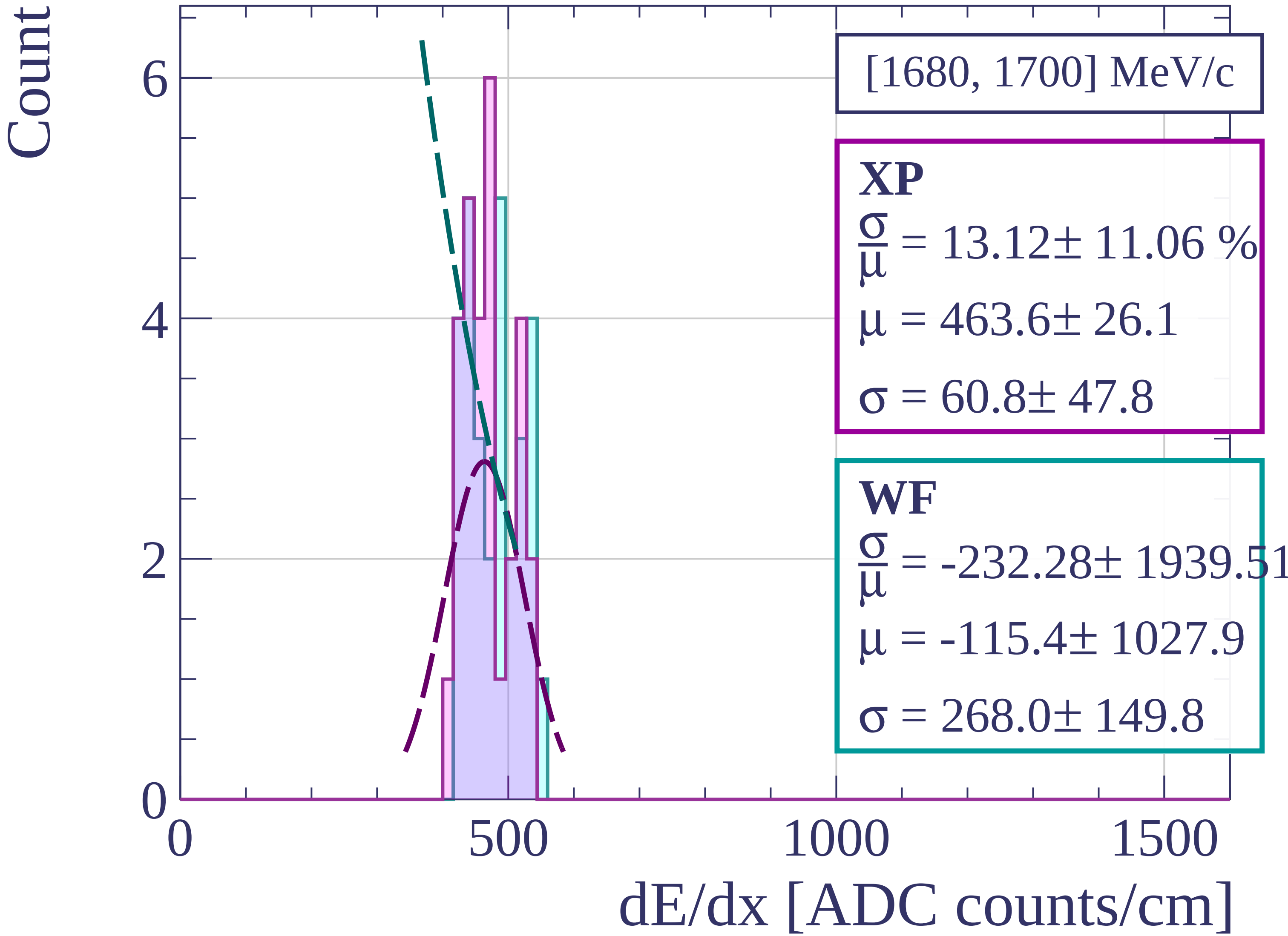
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
[1620, 1640] MeV/c
XP
σ/μ = 11.00± 3.34 %
μ = 464.1± 12.2
σ = 51.0± 14.2
WF
σ/μ = 9.84± 3.23 %
μ = 473.3± 12.2
σ = 46.6± 14.1



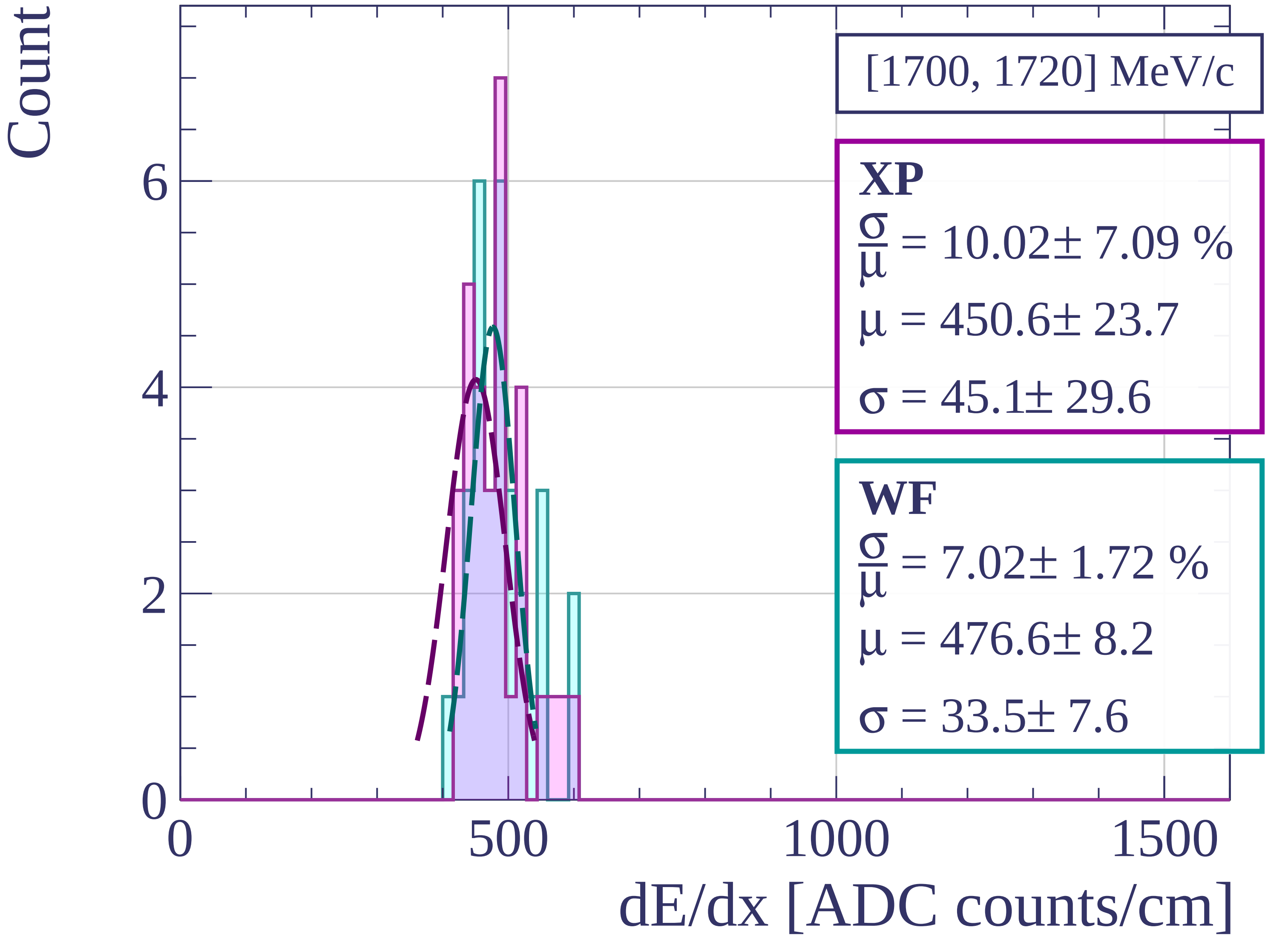
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[1640, 1660] MeV/c
XP
σ/μ = 5.11± 1.43 %
μ = 481.2± 6.8
σ = 24.6± 6.5
WF
σ/μ = 9.19± 3.59 %
μ = 481.6± 12.4
σ = 44.3± 16.2

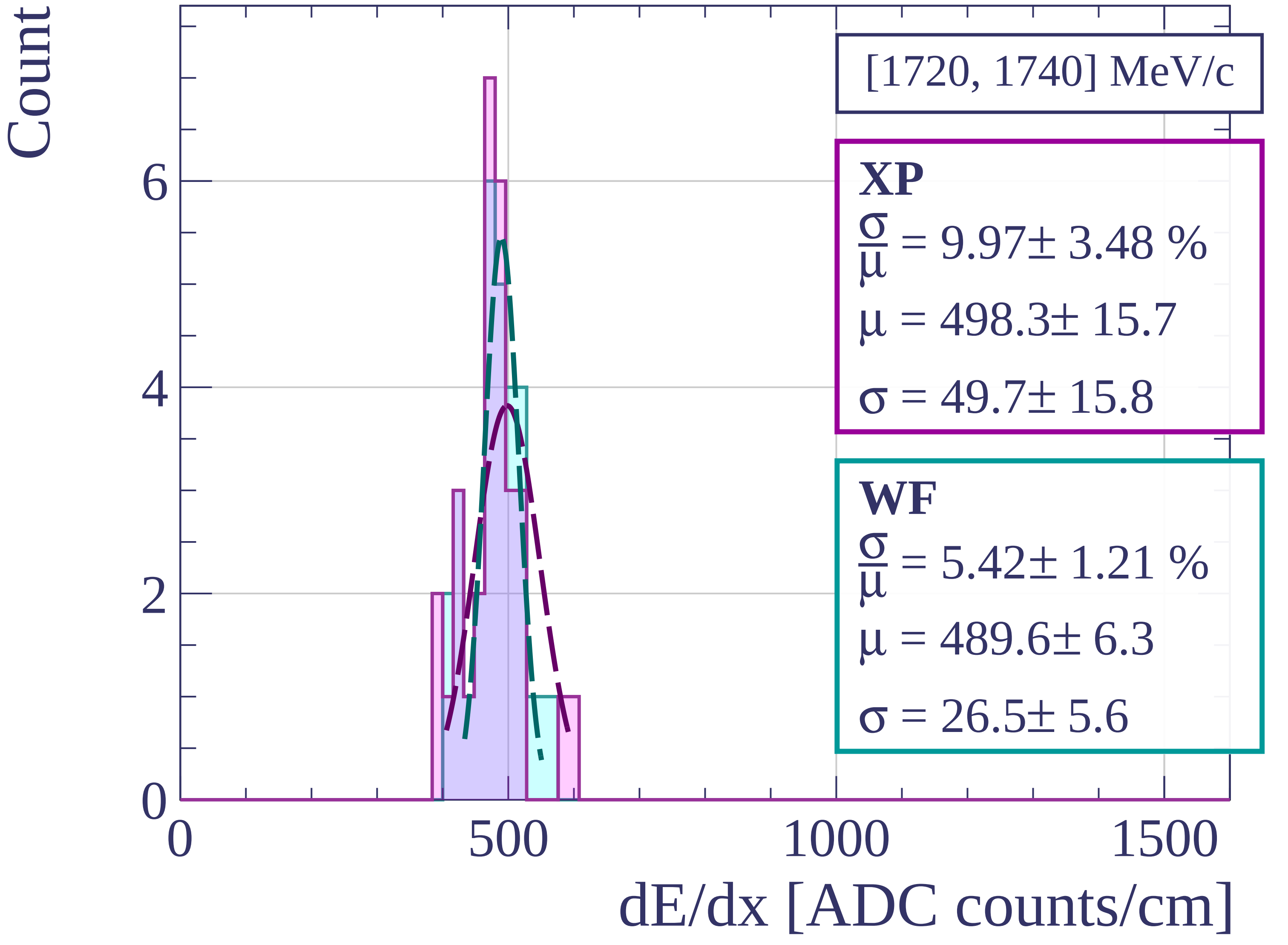


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1660, 1680] MeV/c
XP
σ/μ = 77.86± 146.57 %
μ = 215.0± 288.1
σ = 167.4± 90.9
WF
σ/μ = 6.18± 1.75 %
μ = 470.7± 7.6
σ = 29.1± 7.8

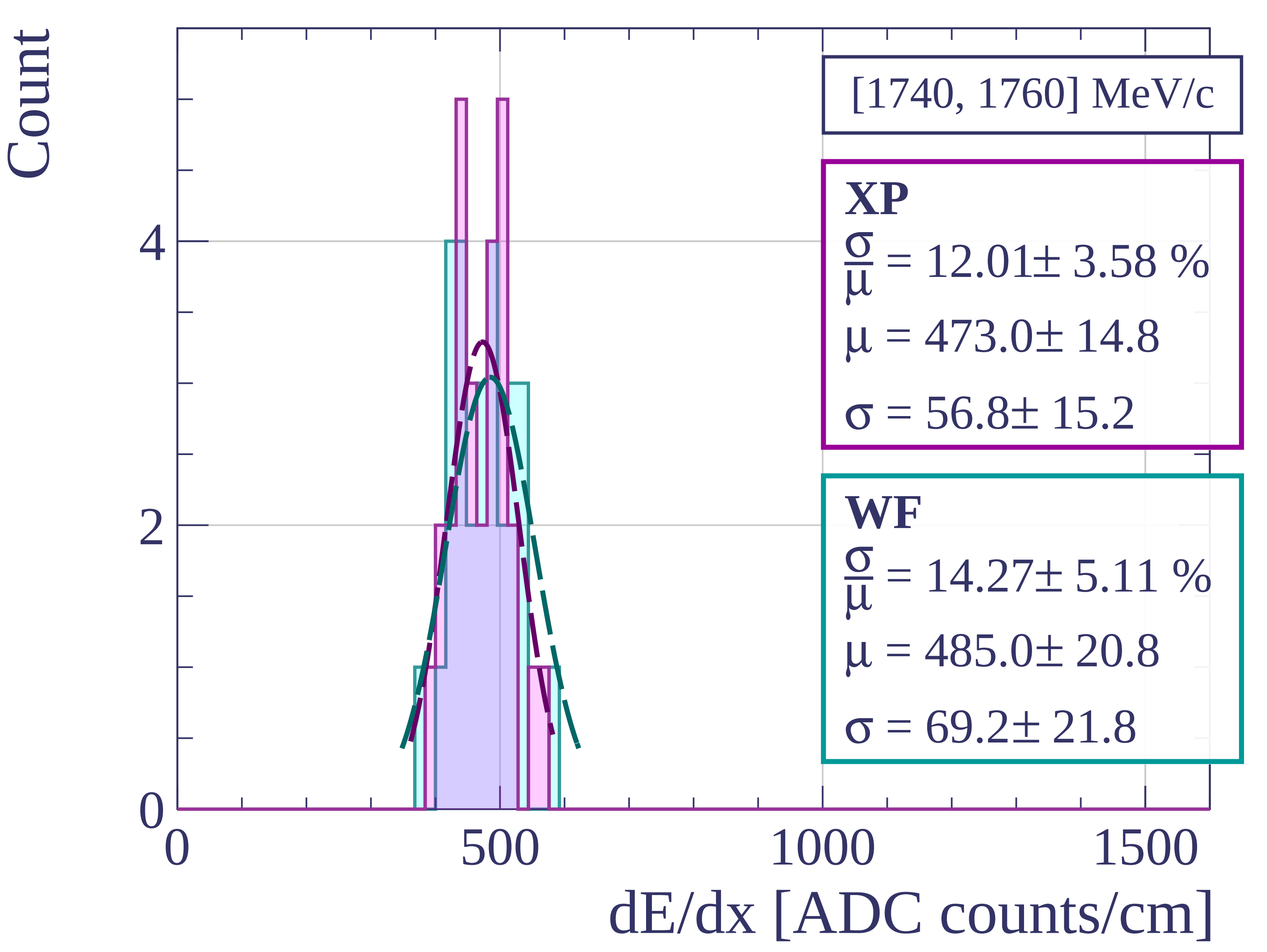


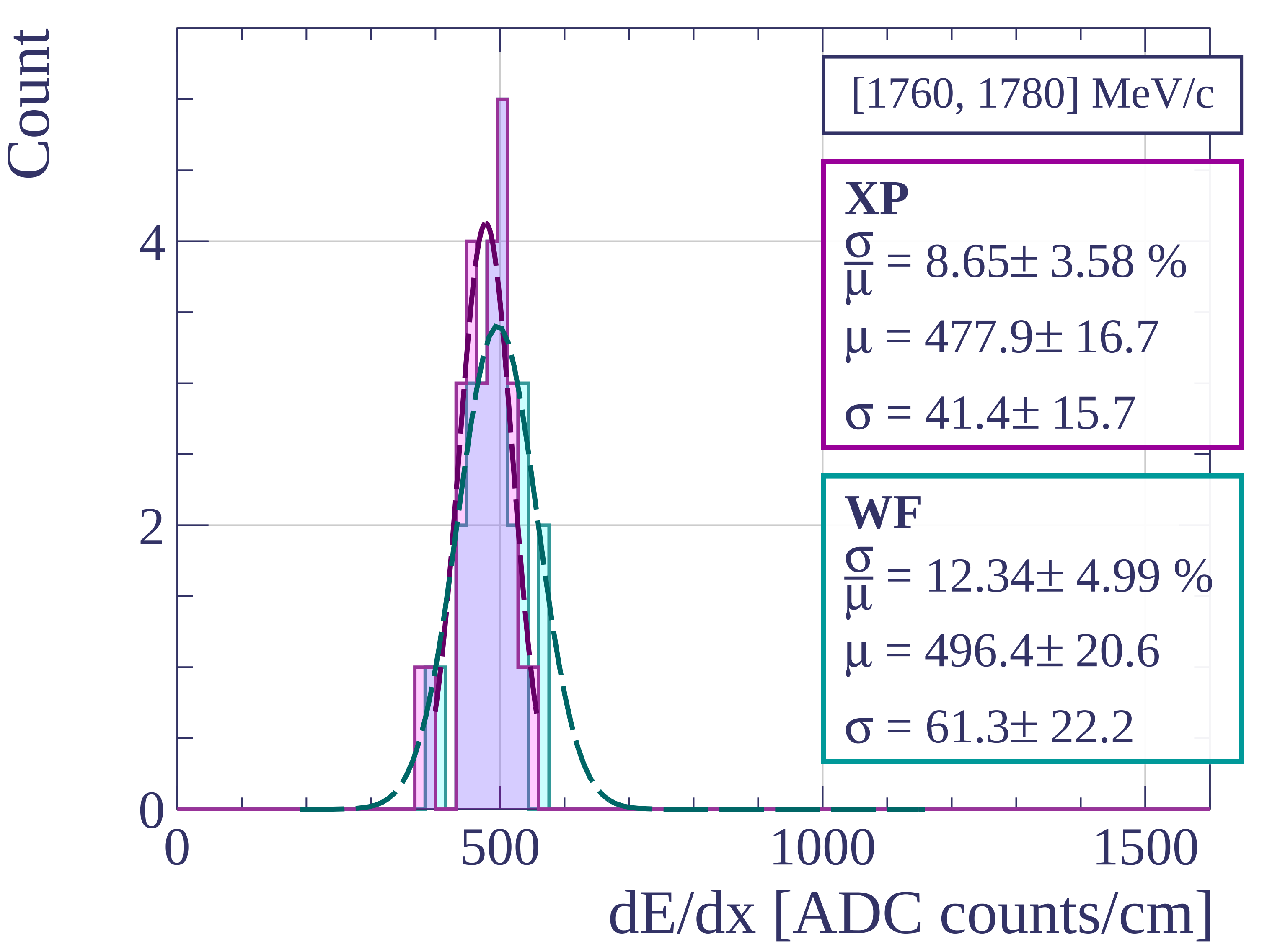
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[1680, 1700] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.12± 11.06 %
μ = 463.6± 26.1
σ = 60.8± 47.8
WF
$\frac{\sigma}{\mu}$ = -232.28± 1939.51
μ = -115.4± 1027.9
σ = 268.0± 149.8


Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1700, 1720] MeV/c
XP
σ/μ = 10.02± 7.09 %
μ = 450.6± 23.7
σ = 45.1± 29.6
WF
σ/μ = 7.02± 1.72 %
μ = 476.6± 8.2
σ = 33.5± 7.6

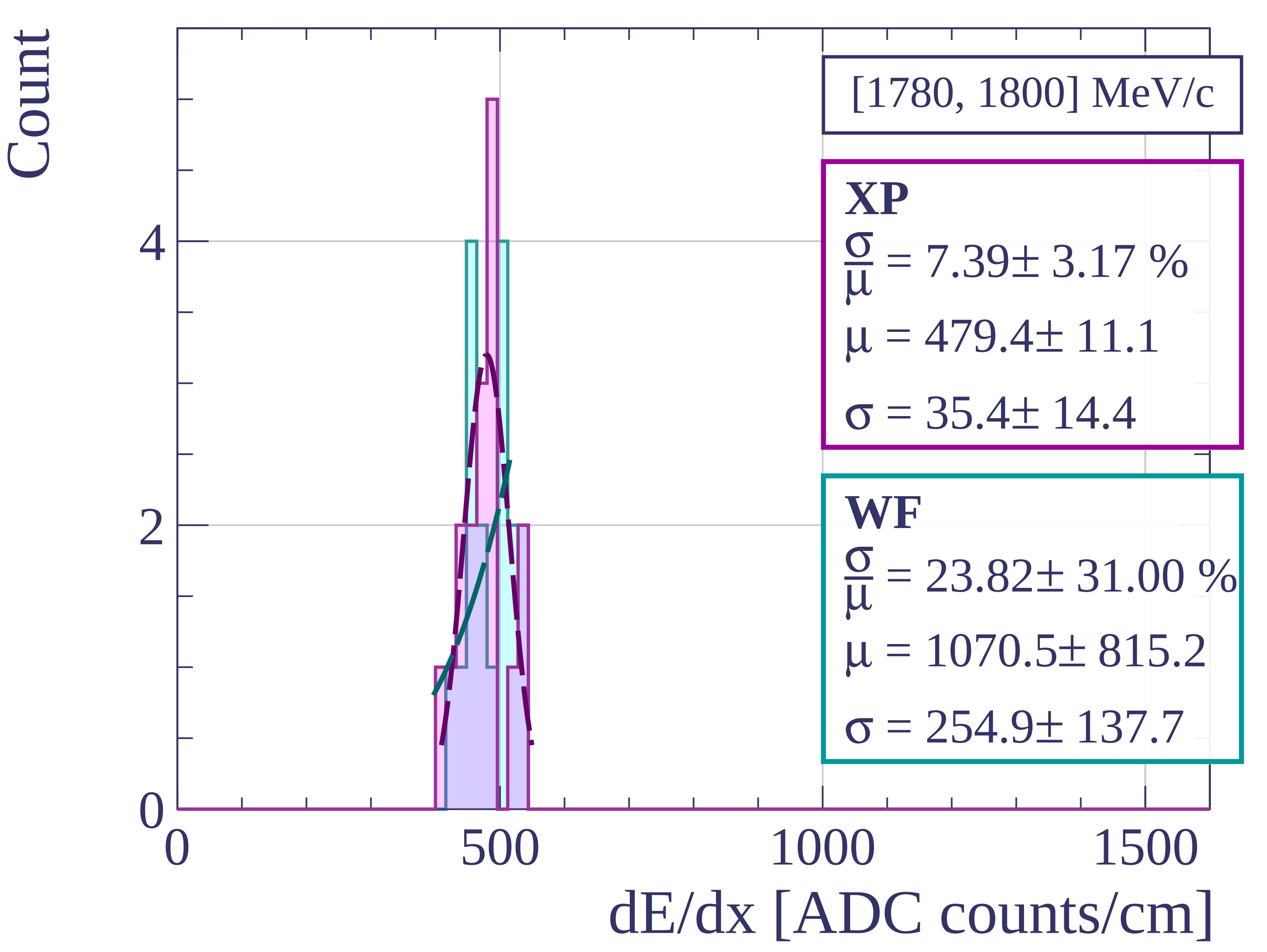


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1720, 1740] MeV/c
XP
σ/μ = 9.97± 3.48 %
μ = 498.3± 15.7
σ = 49.7± 15.8
WF
σ/μ = 5.42± 1.21 %
μ = 489.6± 6.3
σ = 26.5± 5.6

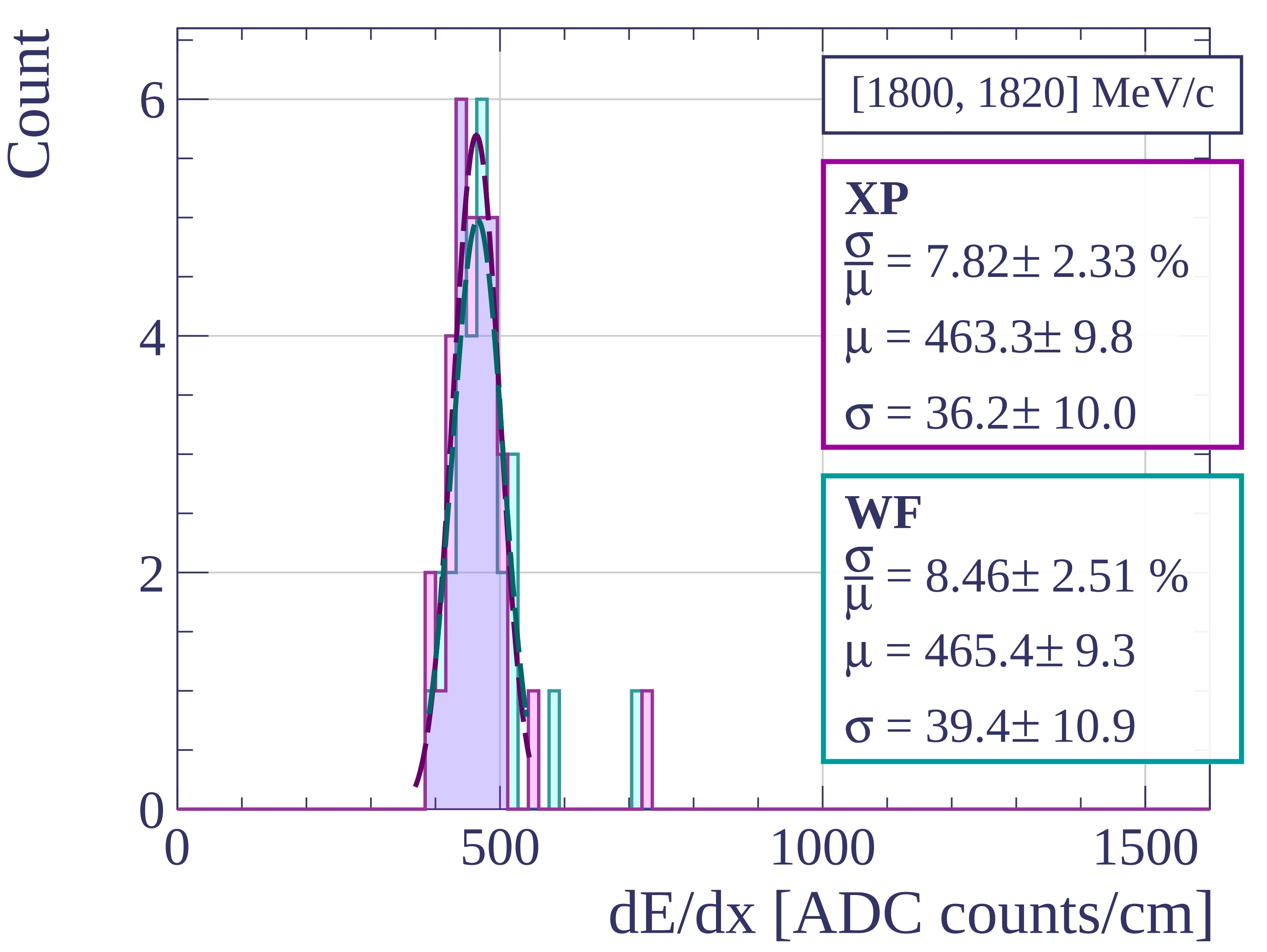

Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[1740, 1760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.01± 3.58 %
μ = 473.0± 14.8
σ = 56.8± 15.2
WF
$\frac{\sigma}{\mu}$ = 14.27± 5.11 %
μ = 485.0± 20.8
σ = 69.2± 21.8



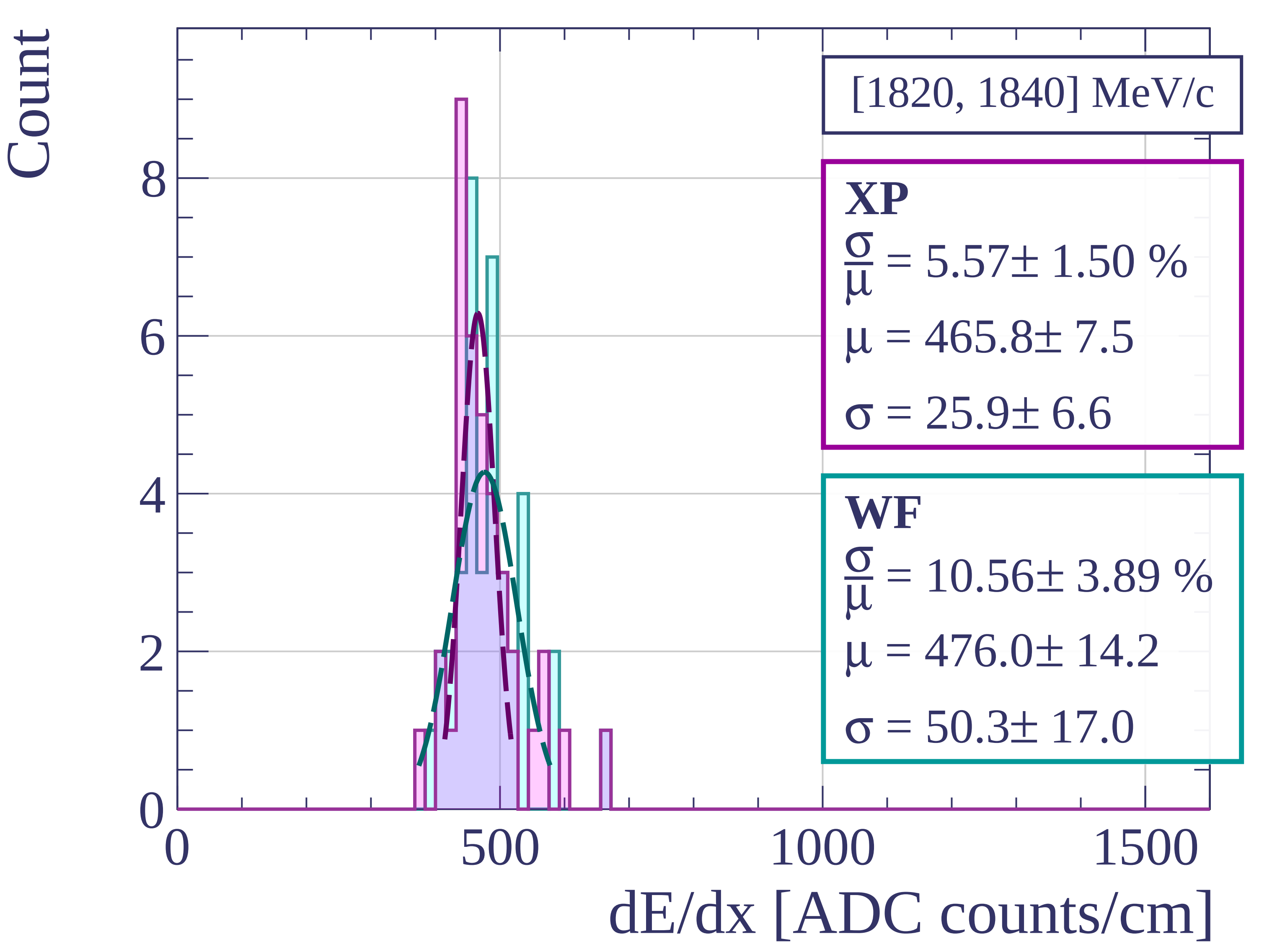
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1760, 1780] MeV/c
XP
σ/μ = 8.65± 3.58 %
μ = 477.9± 16.7
σ = 41.4± 15.7
WF
σ/μ = 12.34± 4.99 %
μ = 496.4± 20.6
σ = 61.3± 22.2



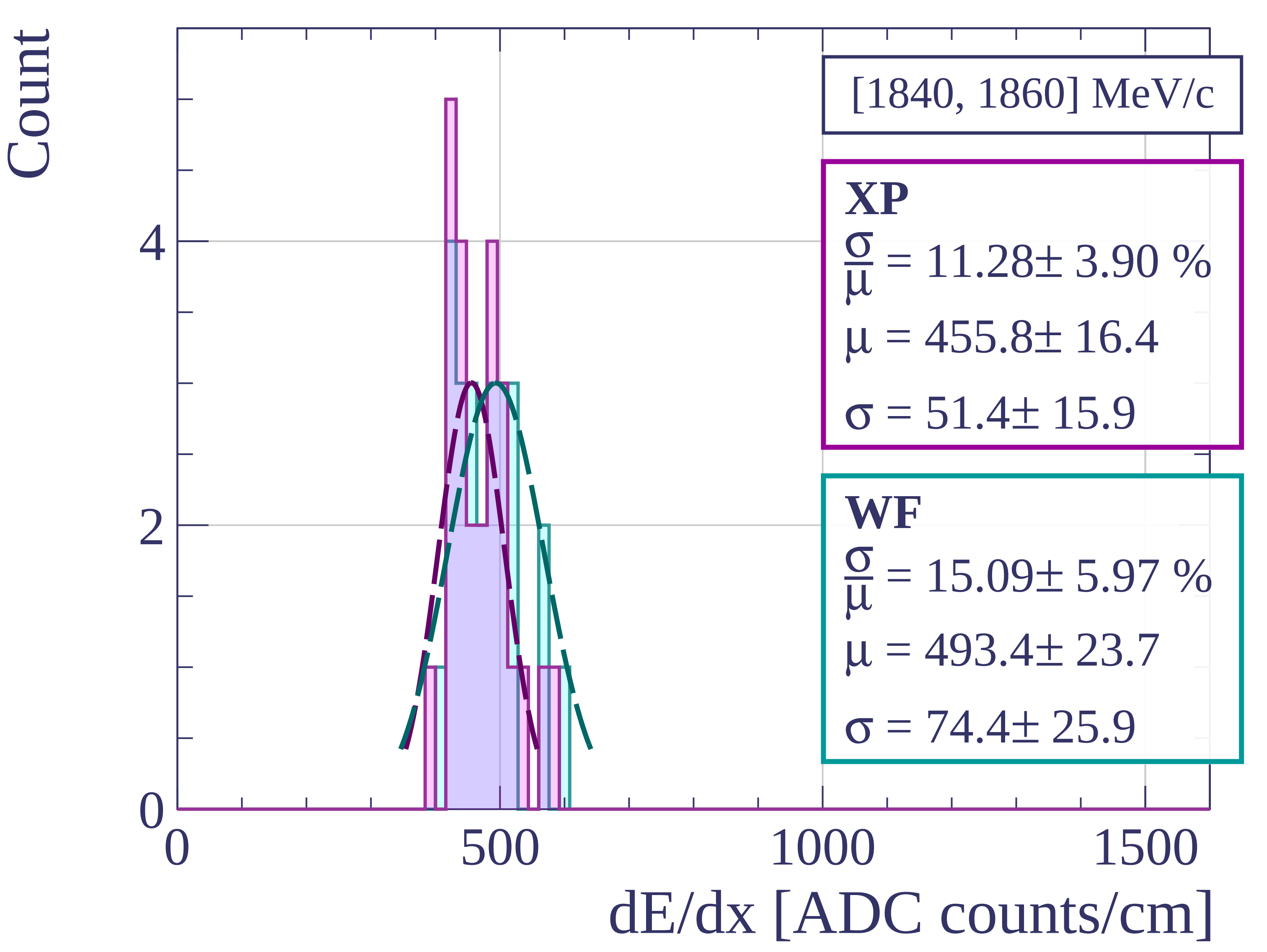
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1780, 1800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.39± 3.17 %
μ = 479.4± 11.1
σ = 35.4± 14.4
WF
$\frac{\sigma}{\mu}$ = 23.82± 31.00 %
μ = 1070.5± 815.2
σ = 254.9± 137.7



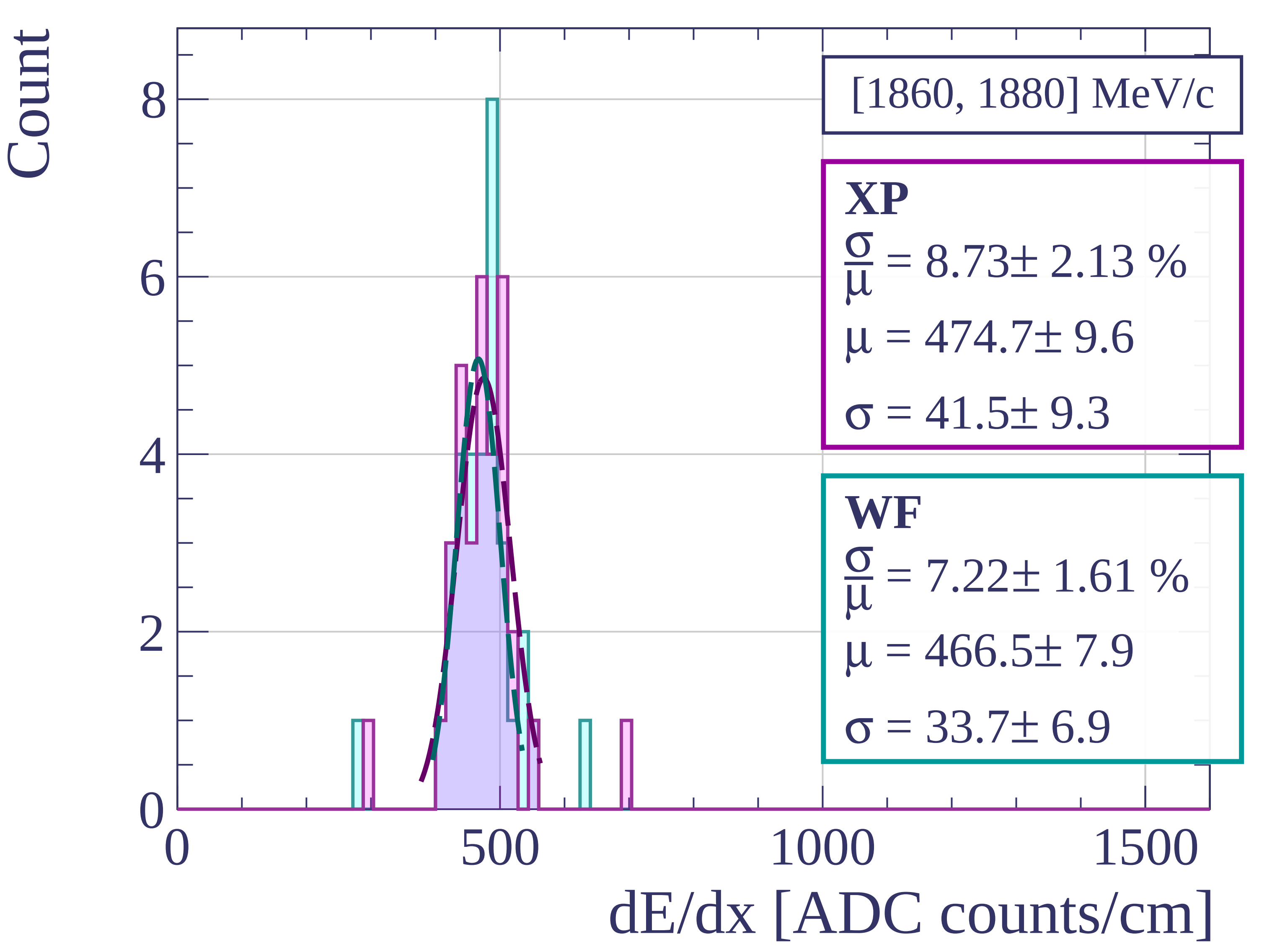
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1800, 1820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.82± 2.33 %
μ = 463.3± 9.8
σ = 36.2± 10.0
WF
$\frac{\sigma}{\mu}$ = 8.46± 2.51 %
μ = 465.4± 9.3
σ = 39.4± 10.9



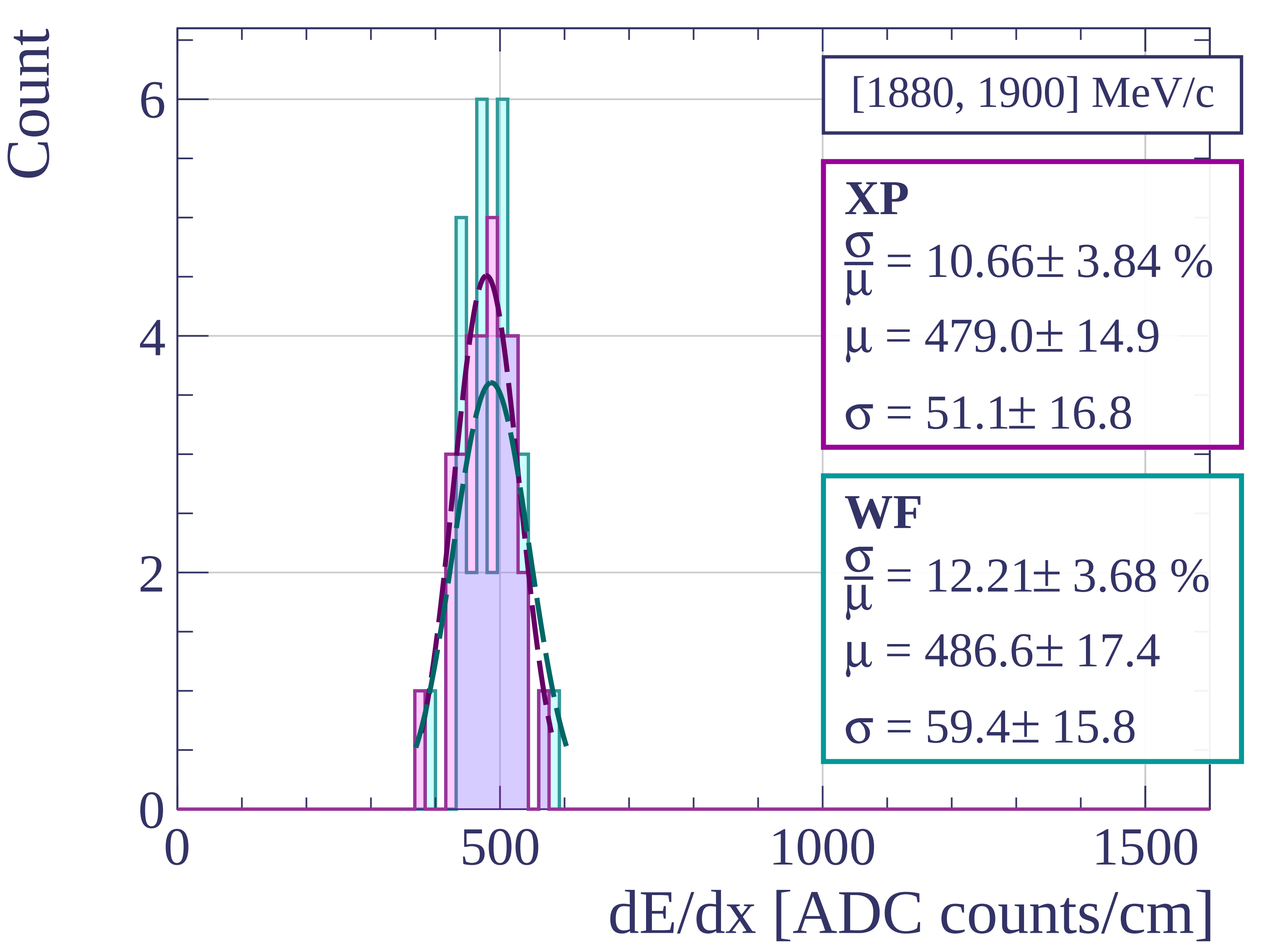
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1820, 1840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.57± 1.50 %
μ = 465.8± 7.5
σ = 25.9± 6.6
WF
$\frac{\sigma}{\mu}$ = 10.56± 3.89 %
μ = 476.0± 14.2
σ = 50.3± 17.0



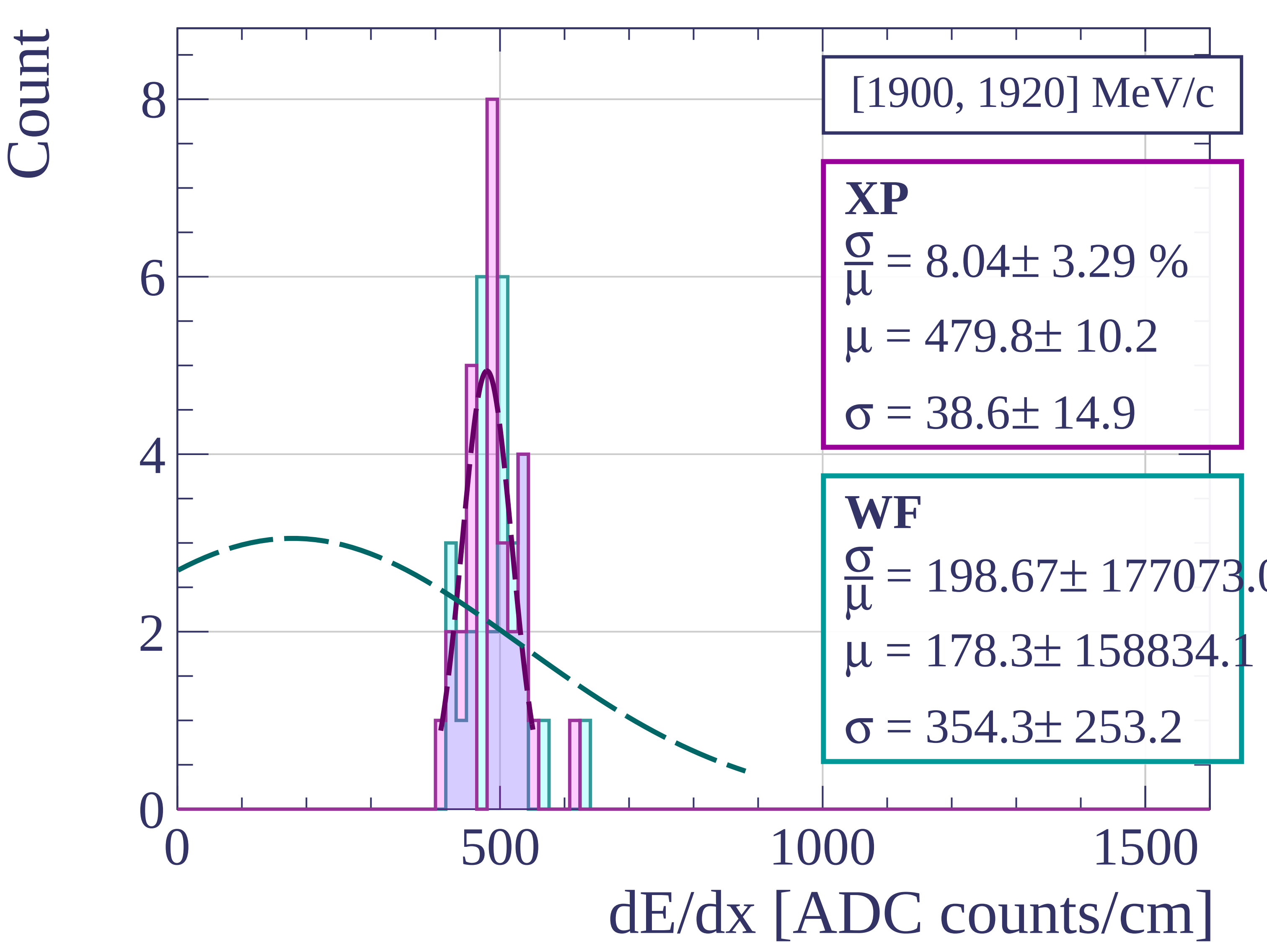
Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[1840, 1860] MeV/c
XP
σ/μ = 11.28± 3.90 %
μ = 455.8± 16.4
σ = 51.4± 15.9
WF
σ/μ = 15.09± 5.97 %
μ = 493.4± 23.7
σ = 74.4± 25.9

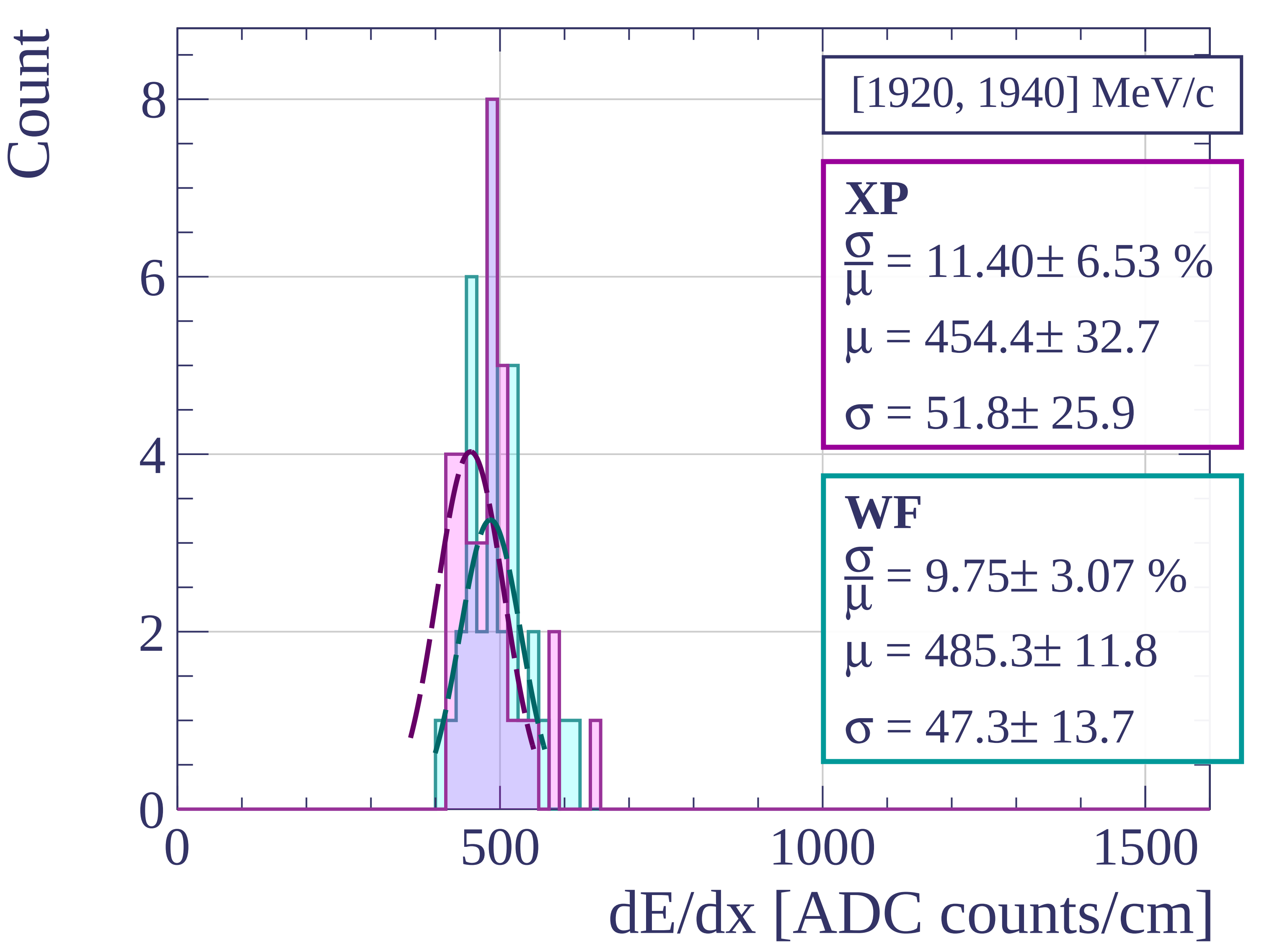


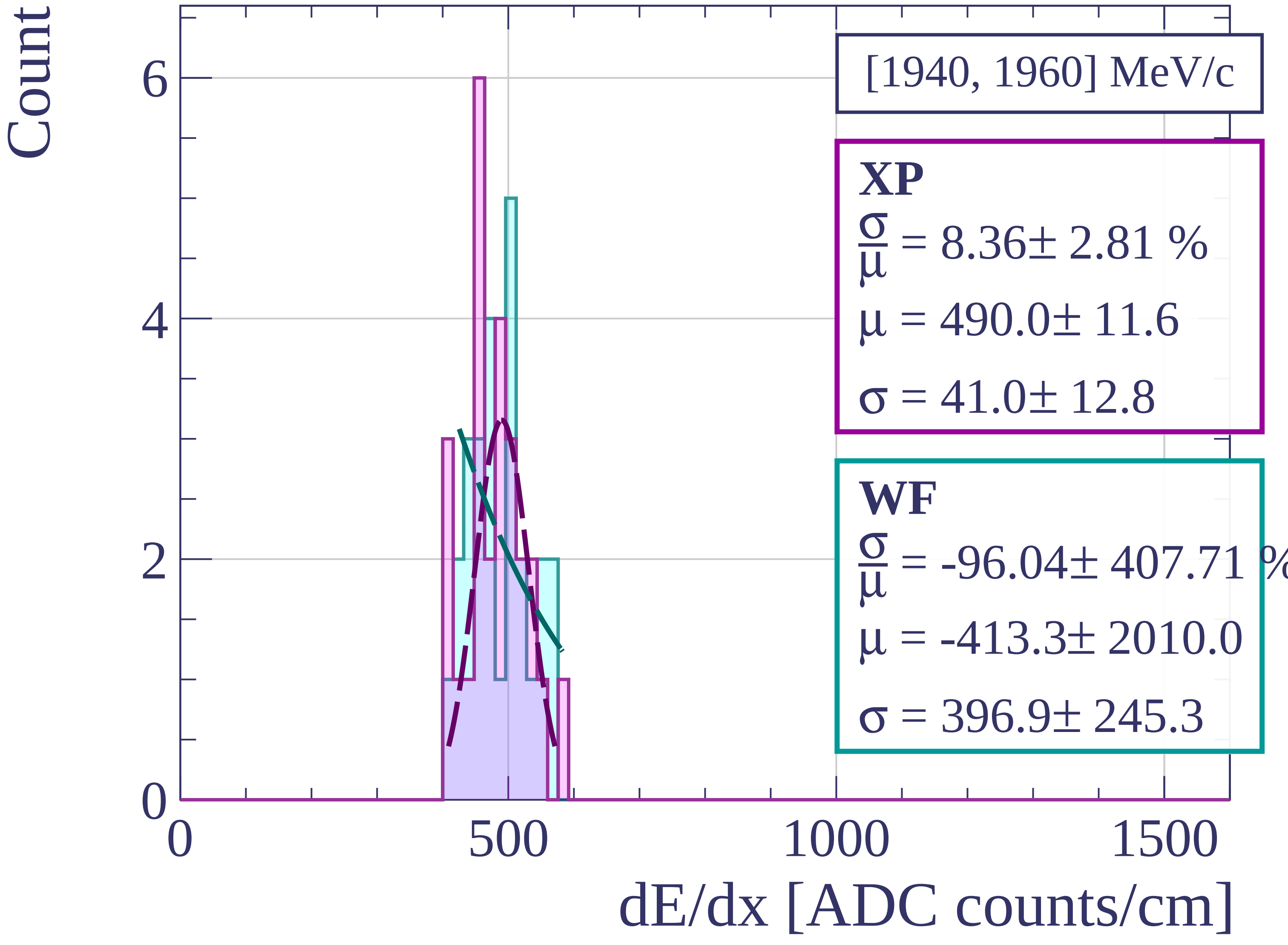
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1860, 1880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.73± 2.13 %
μ = 474.7± 9.6
σ = 41.5± 9.3
WF
$\frac{\sigma}{\mu}$ = 7.22± 1.61 %
μ = 466.5± 7.9
σ = 33.7± 6.9



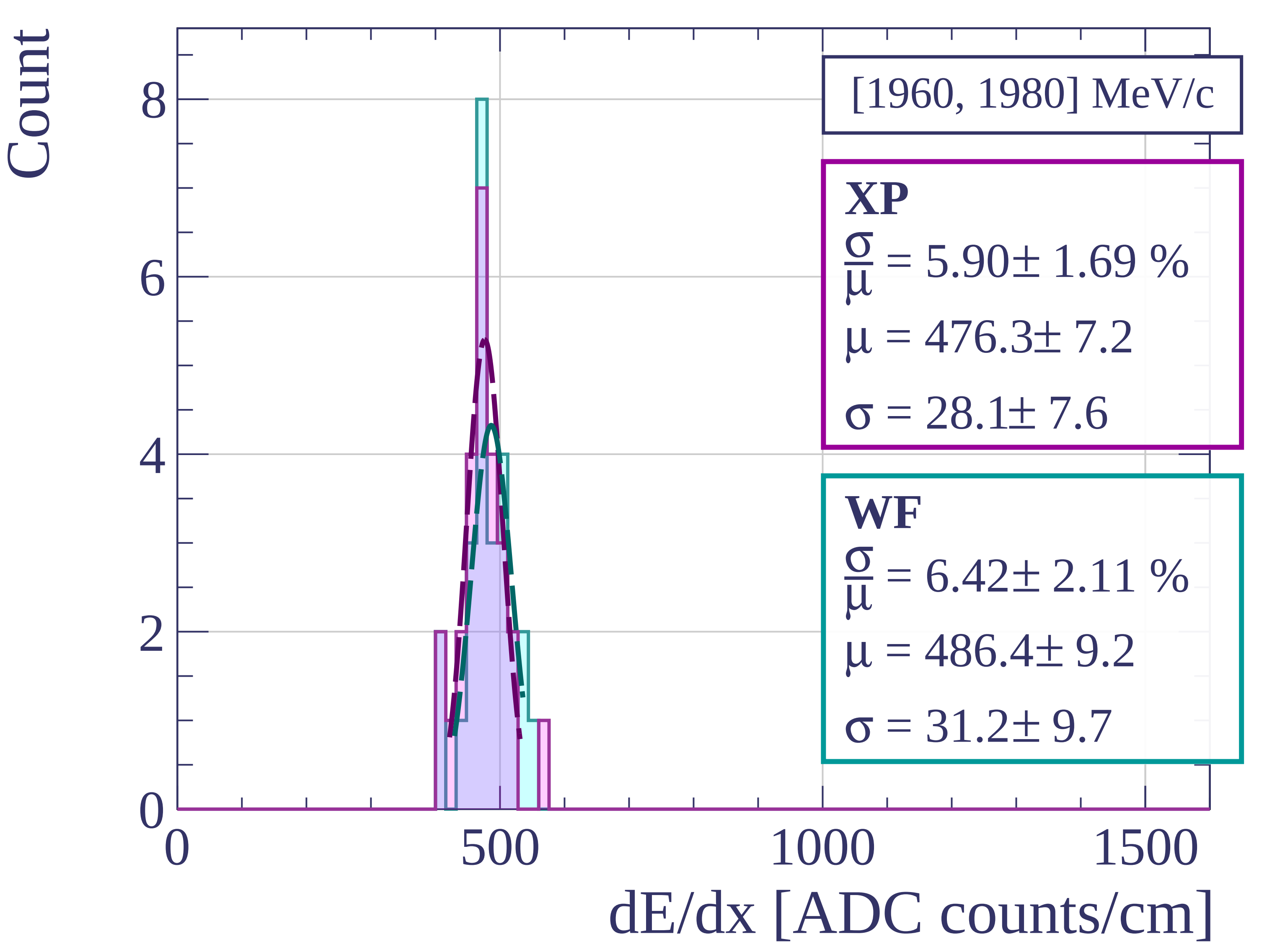
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
[1880, 1900] MeV/c
XP
σ/μ = 10.66± 3.84 %
μ = 479.0± 14.9
σ = 51.1± 16.8
WF
σ/μ = 12.21± 3.68 %
μ = 486.6± 17.4
σ = 59.4± 15.8

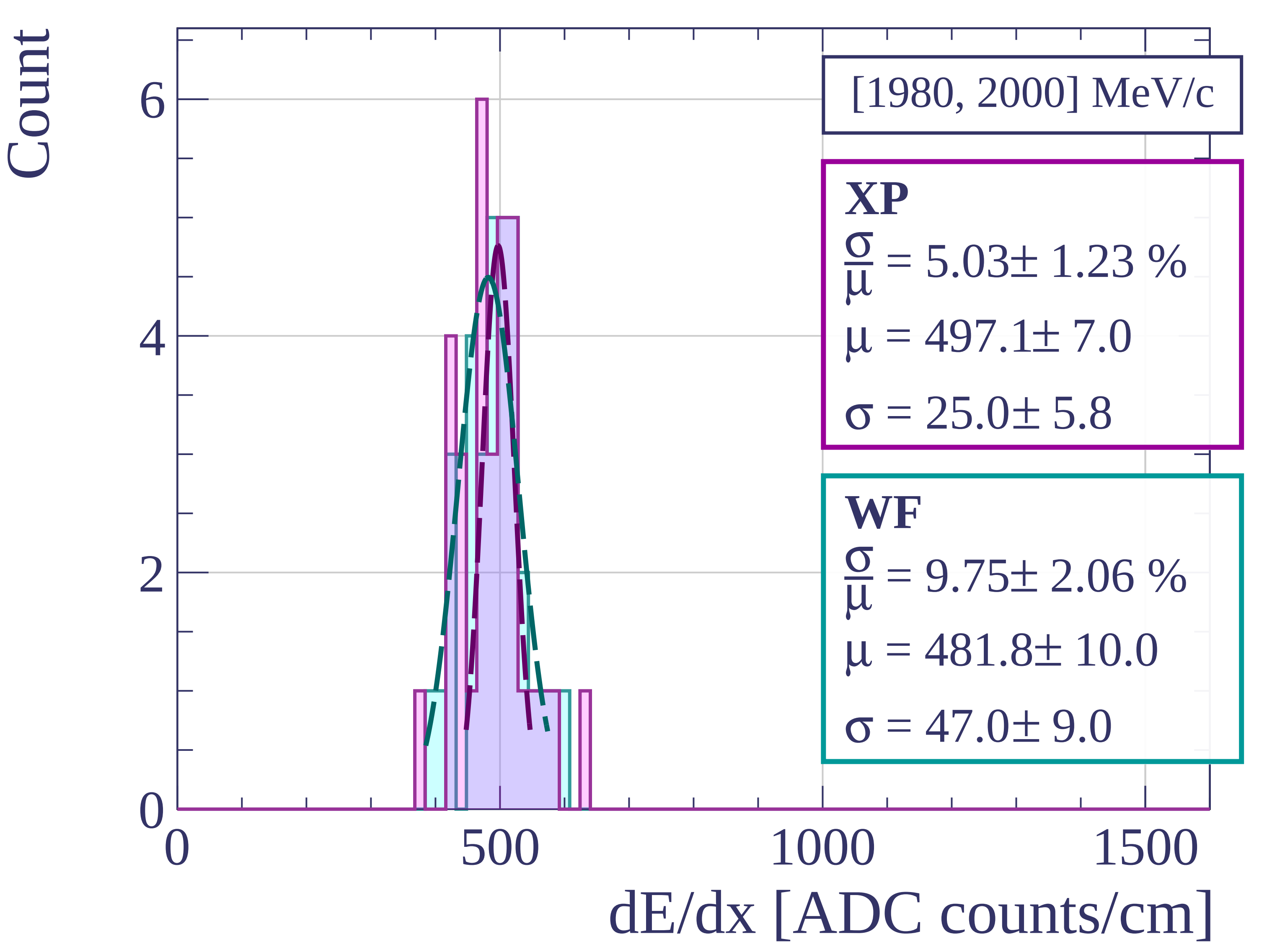

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1900, 1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.04± 3.29 %
μ = 479.8± 10.2
σ = 38.6± 14.9
WF
$\frac{\sigma}{\mu}$ = 198.67± 177073.0
μ = 178.3± 158834.1
σ = 354.3± 253.2


Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[1920, 1940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.40± 6.53 %
μ = 454.4± 32.7
σ = 51.8± 25.9
WF
$\frac{\sigma}{\mu}$ = 9.75± 3.07 %
μ = 485.3± 11.8
σ = 47.3± 13.7

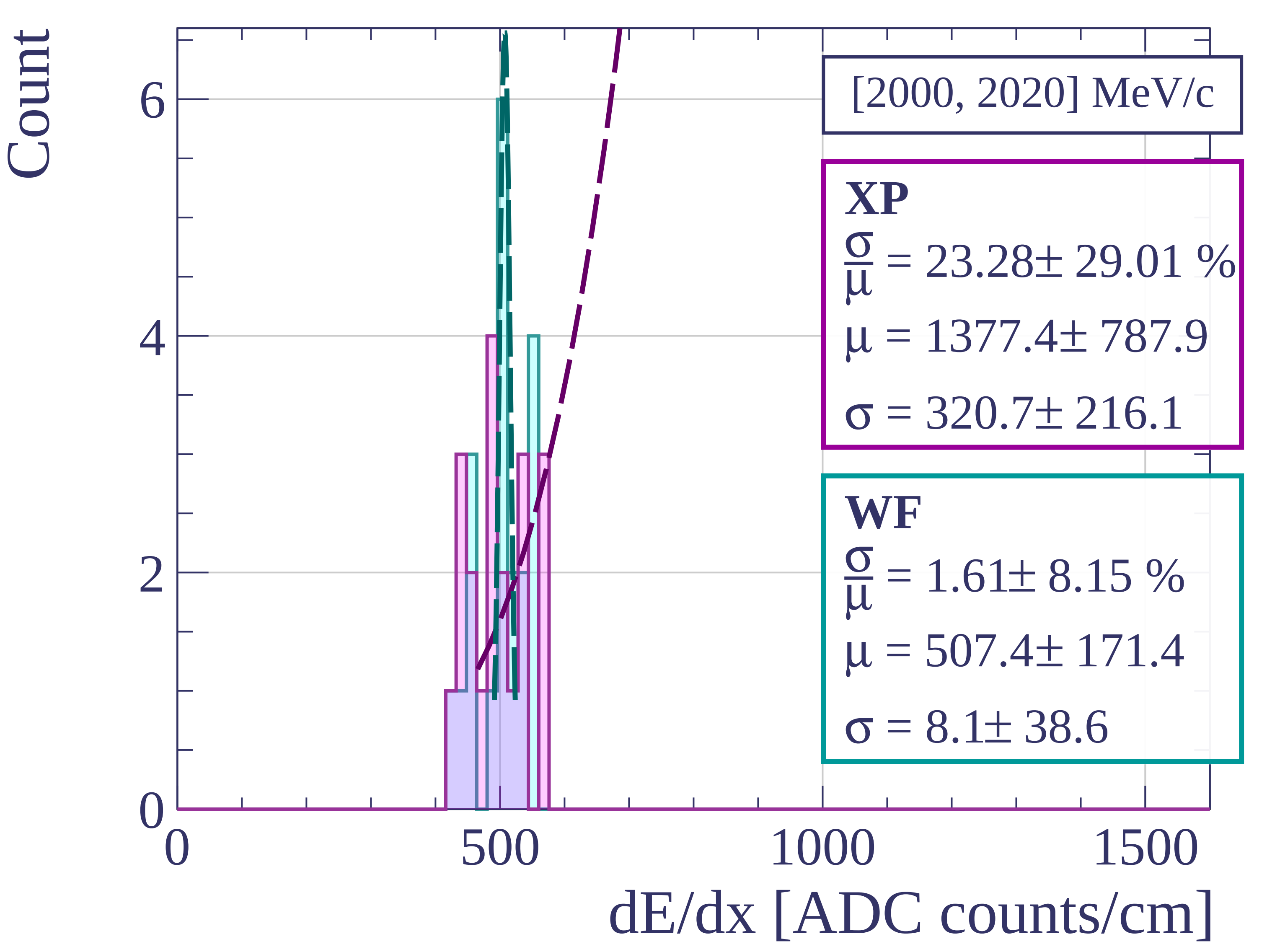


Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1940, 1960] MeV/c
XP
σ/μ = 8.36± 2.81 %
μ = 490.0± 11.6
σ = 41.0± 12.8
WF
σ/μ = -96.04± 407.71 %
μ = -413.3± 2010.0
σ = 396.9± 245.3

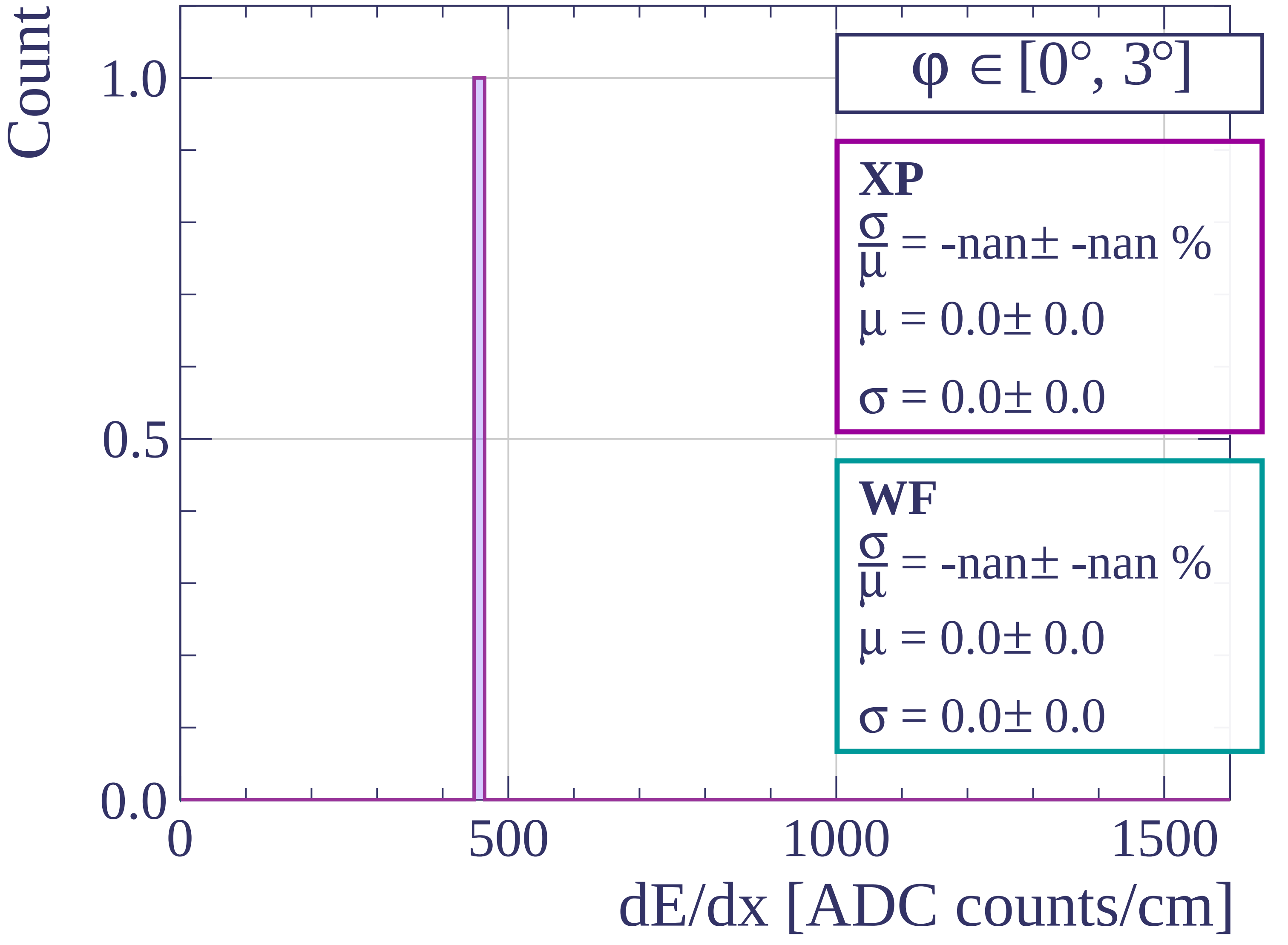

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1960, 1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.90± 1.69 %
μ = 476.3± 7.2
σ = 28.1± 7.6
WF
$\frac{\sigma}{\mu}$ = 6.42± 2.11 %
μ = 486.4± 9.2
σ = 31.2± 9.7



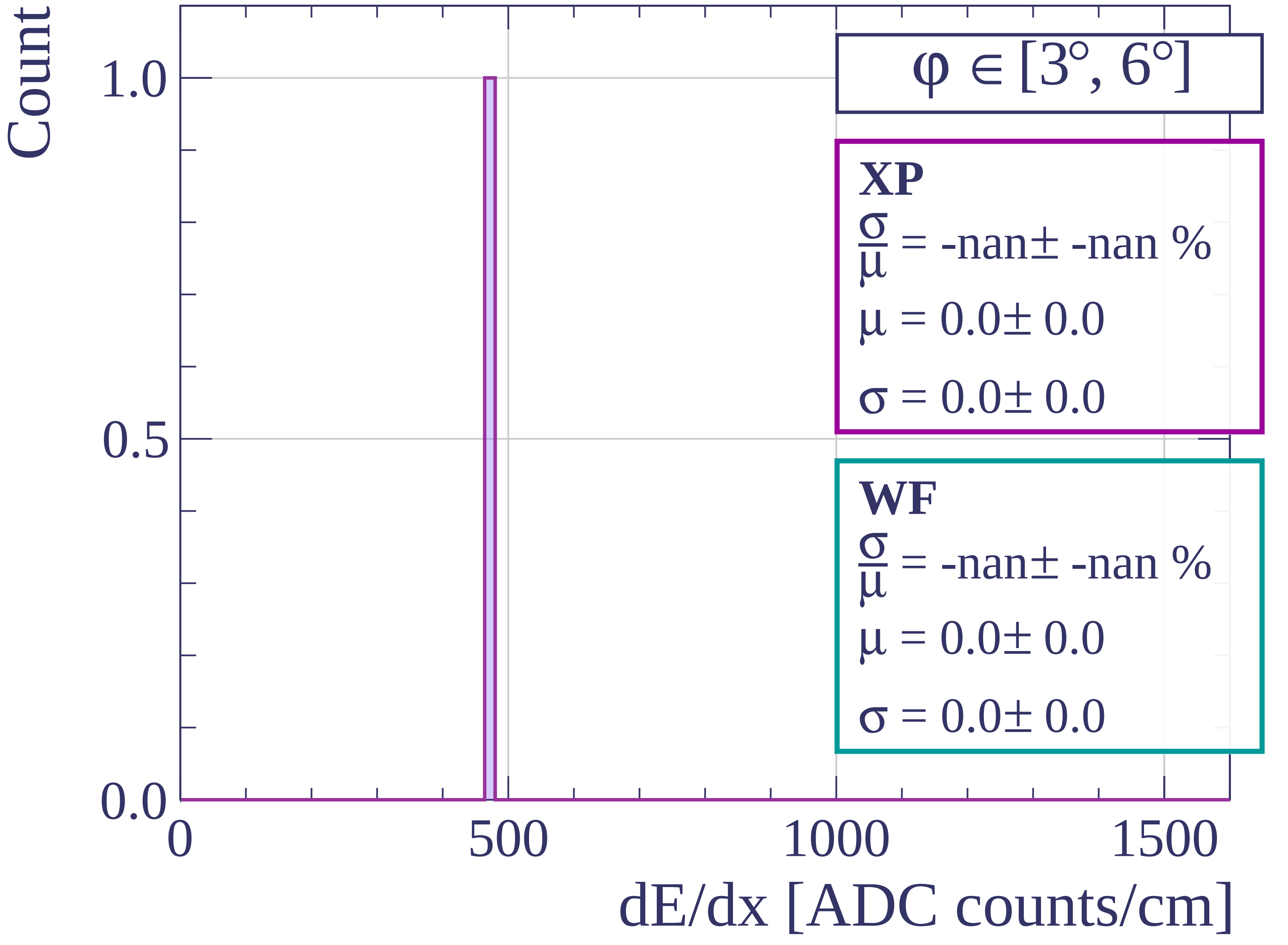
Count
dE/dx [ADC counts/cm]
[1980, 2000] MeV/c
XP
σ/μ = 5.03± 1.23 %
μ = 497.1± 7.0
σ = 25.0± 5.8
WF
σ/μ = 9.75± 2.06 %
μ = 481.8± 10.0
σ = 47.0± 9.0



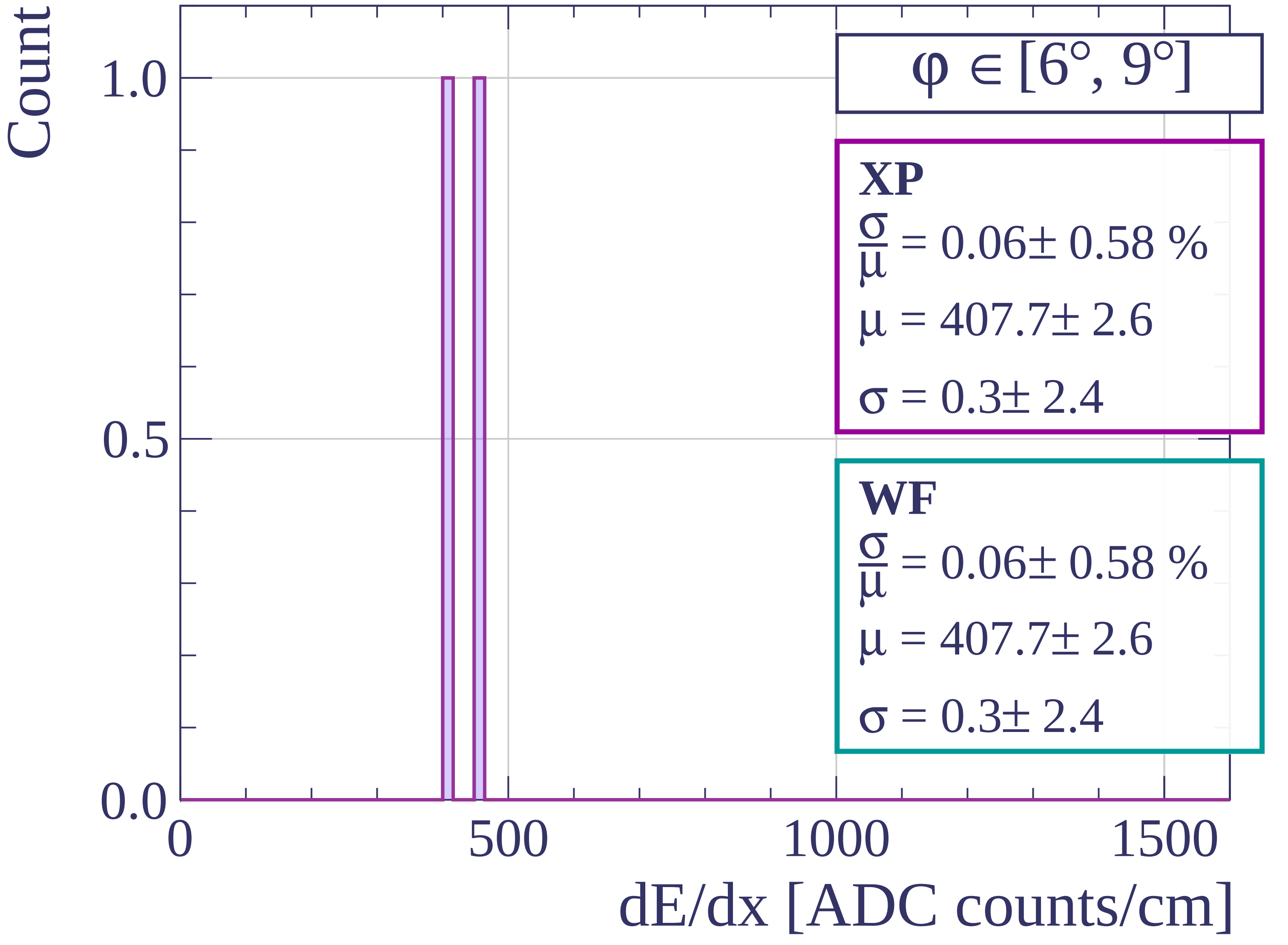
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[2000, 2020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 23.28± 29.01 %
μ = 1377.4± 787.9
σ = 320.7± 216.1
WF
$\frac{\sigma}{\mu}$ = 1.61± 8.15 %
μ = 507.4± 171.4
σ = 8.1± 38.6



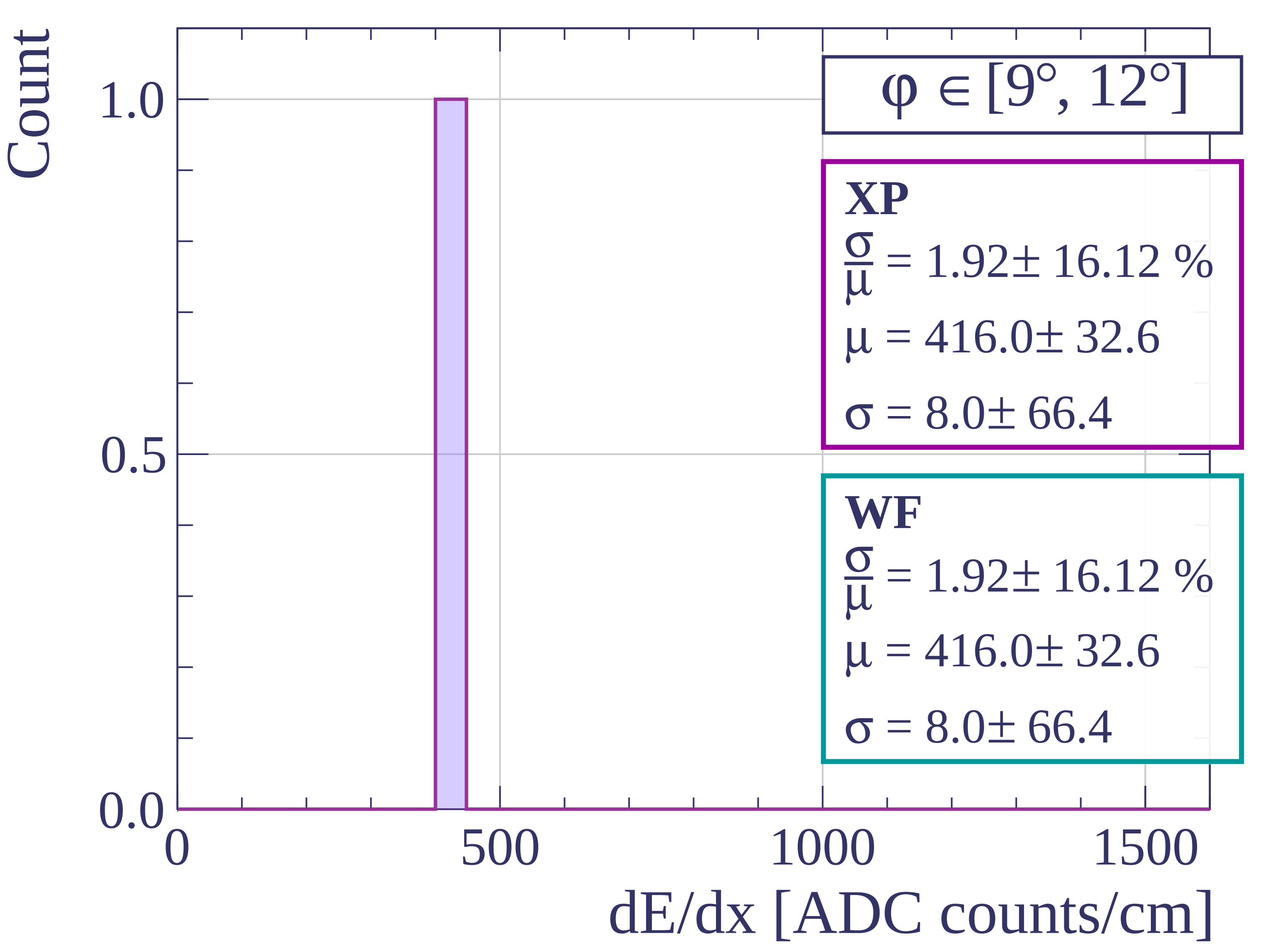
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [0°, 3°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

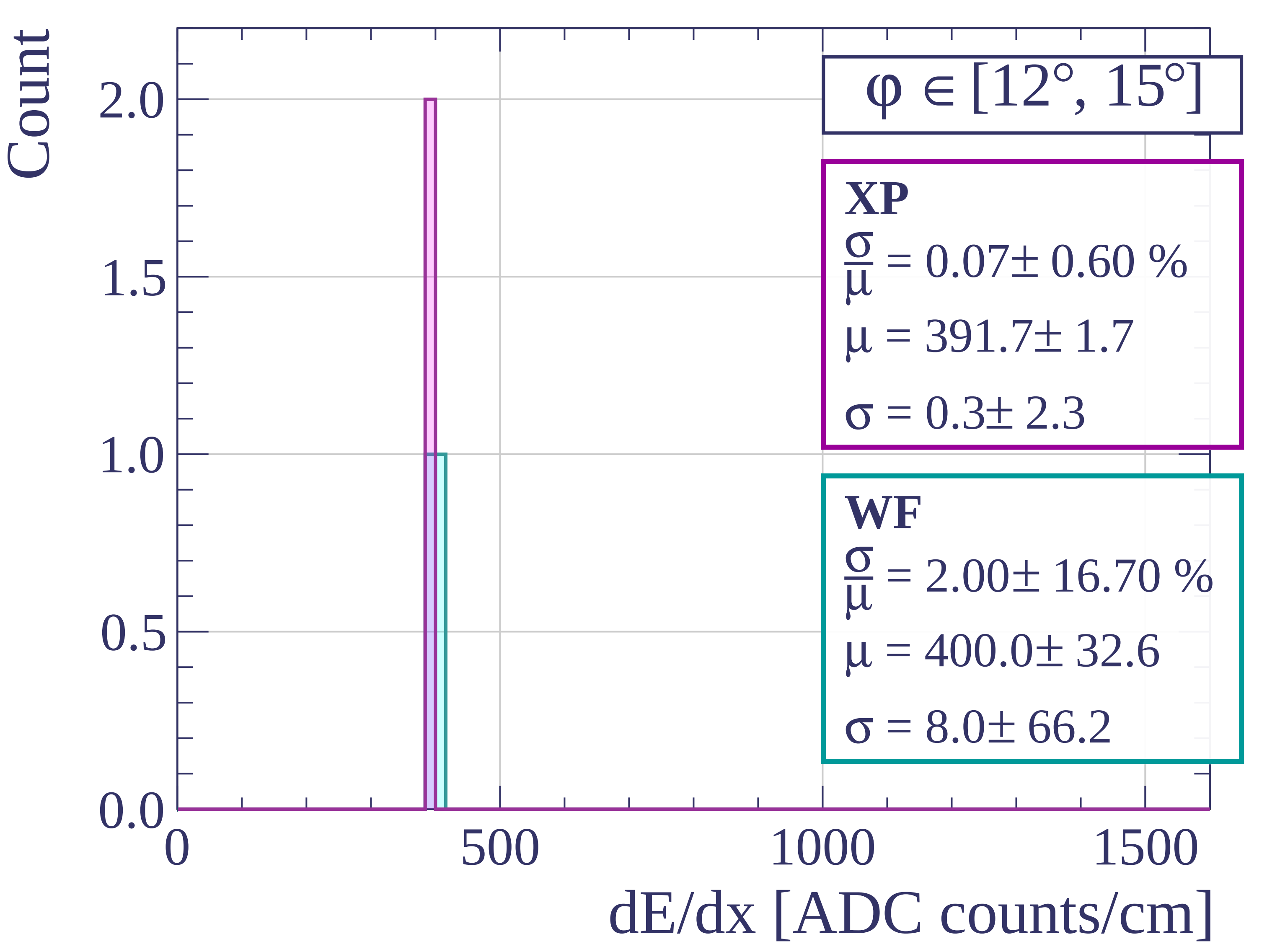


Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [3°, 6°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

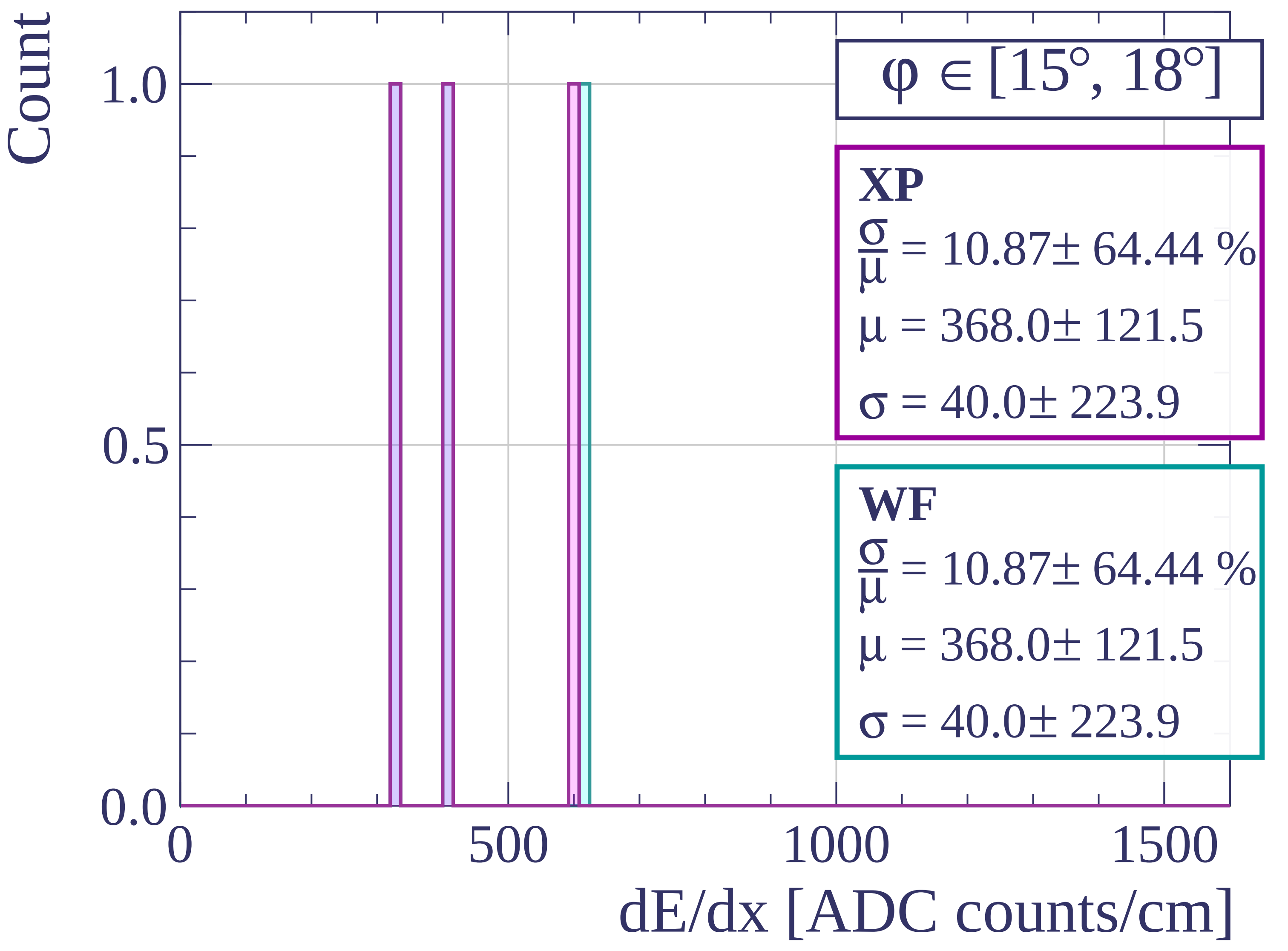


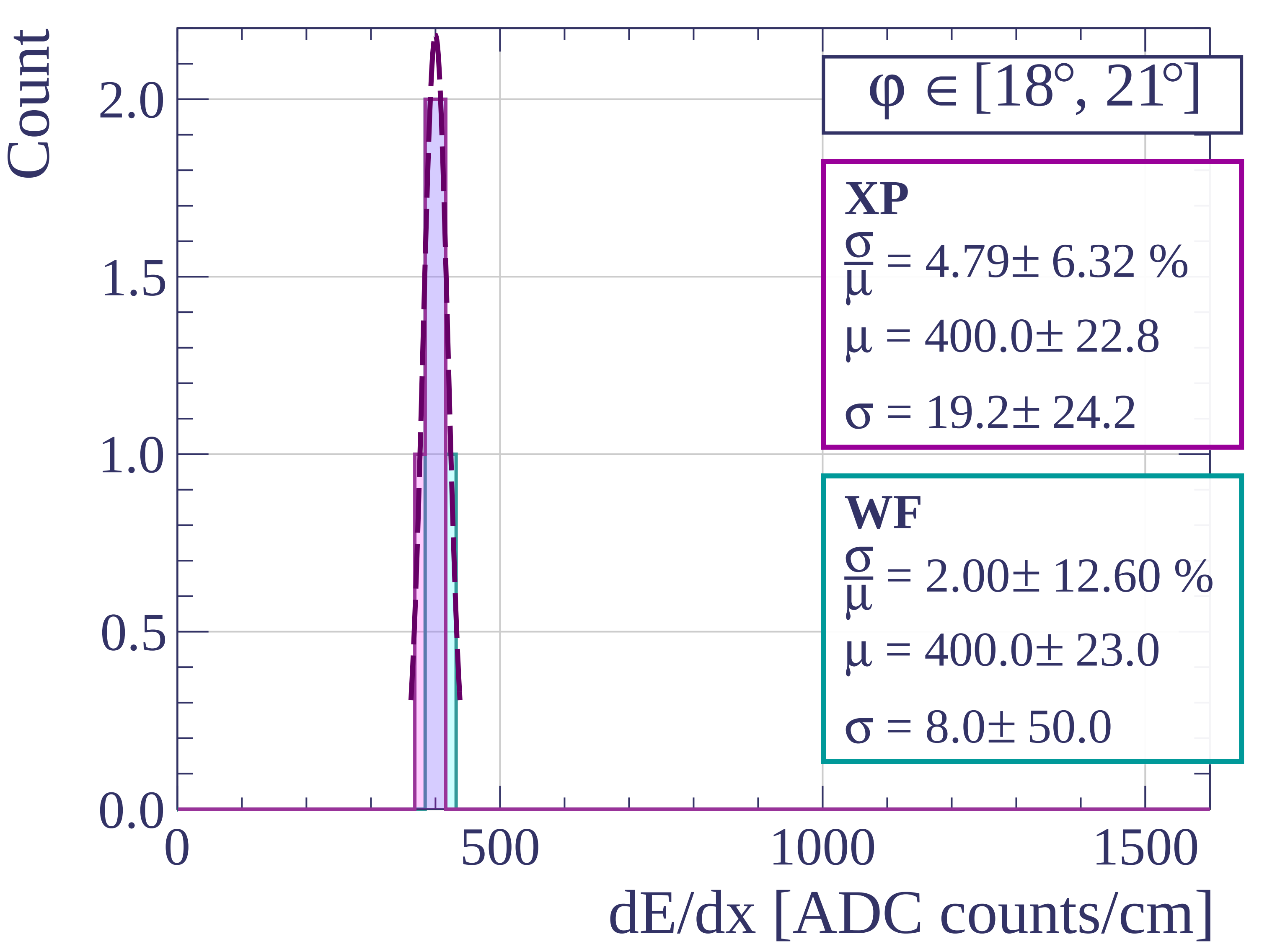
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [6°, 9°]
XP
σ/μ = 0.06± 0.58 %
μ = 407.7± 2.6
σ = 0.3± 2.4
WF
σ/μ = 0.06± 0.58 %
μ = 407.7± 2.6
σ = 0.3± 2.4

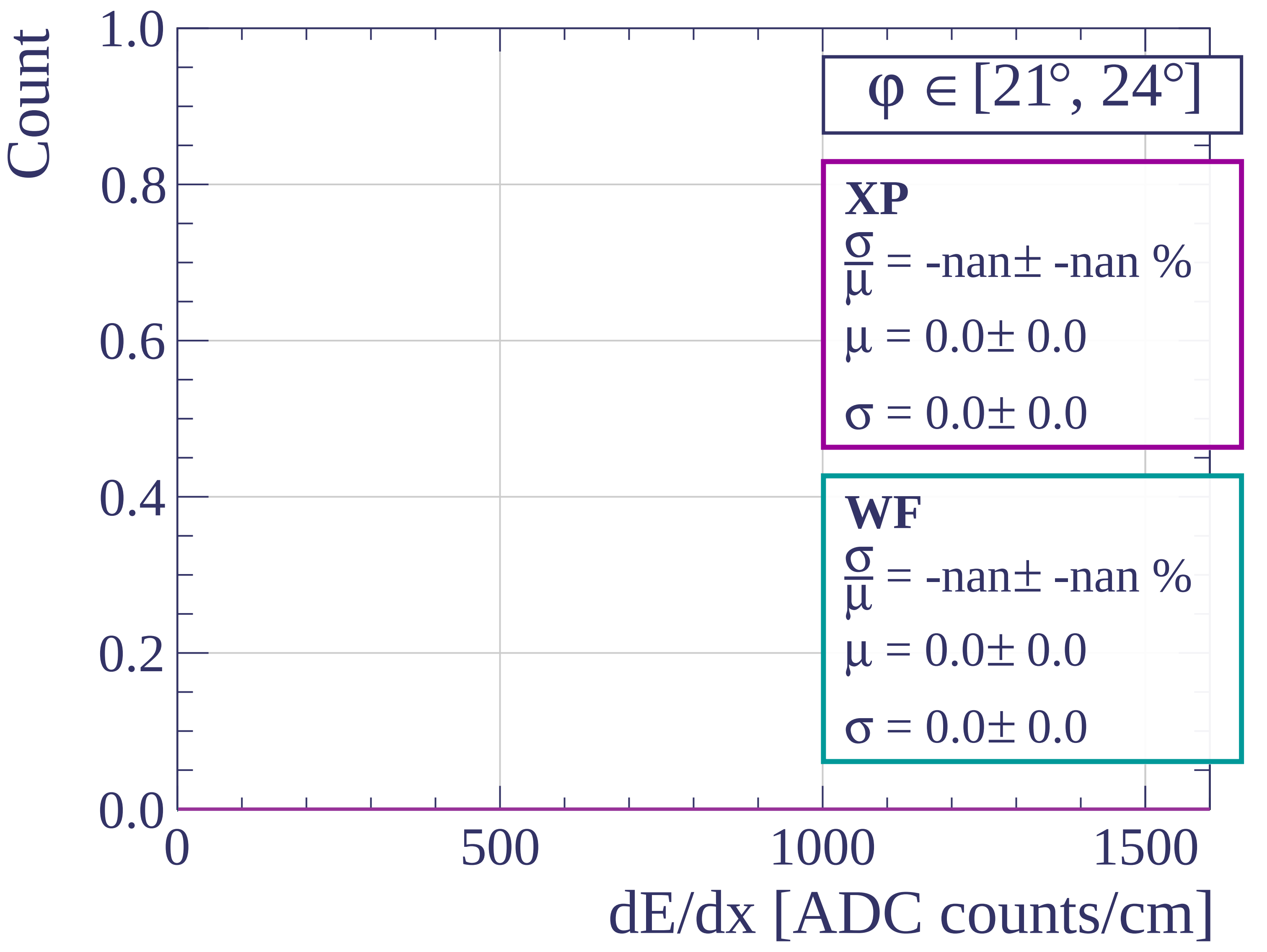

Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [9°, 12°]
XP
σ/μ = 1.92± 16.12 %
μ = 416.0± 32.6
σ = 8.0± 66.4
WF
σ/μ = 1.92± 16.12 %
μ = 416.0± 32.6
σ = 8.0± 66.4



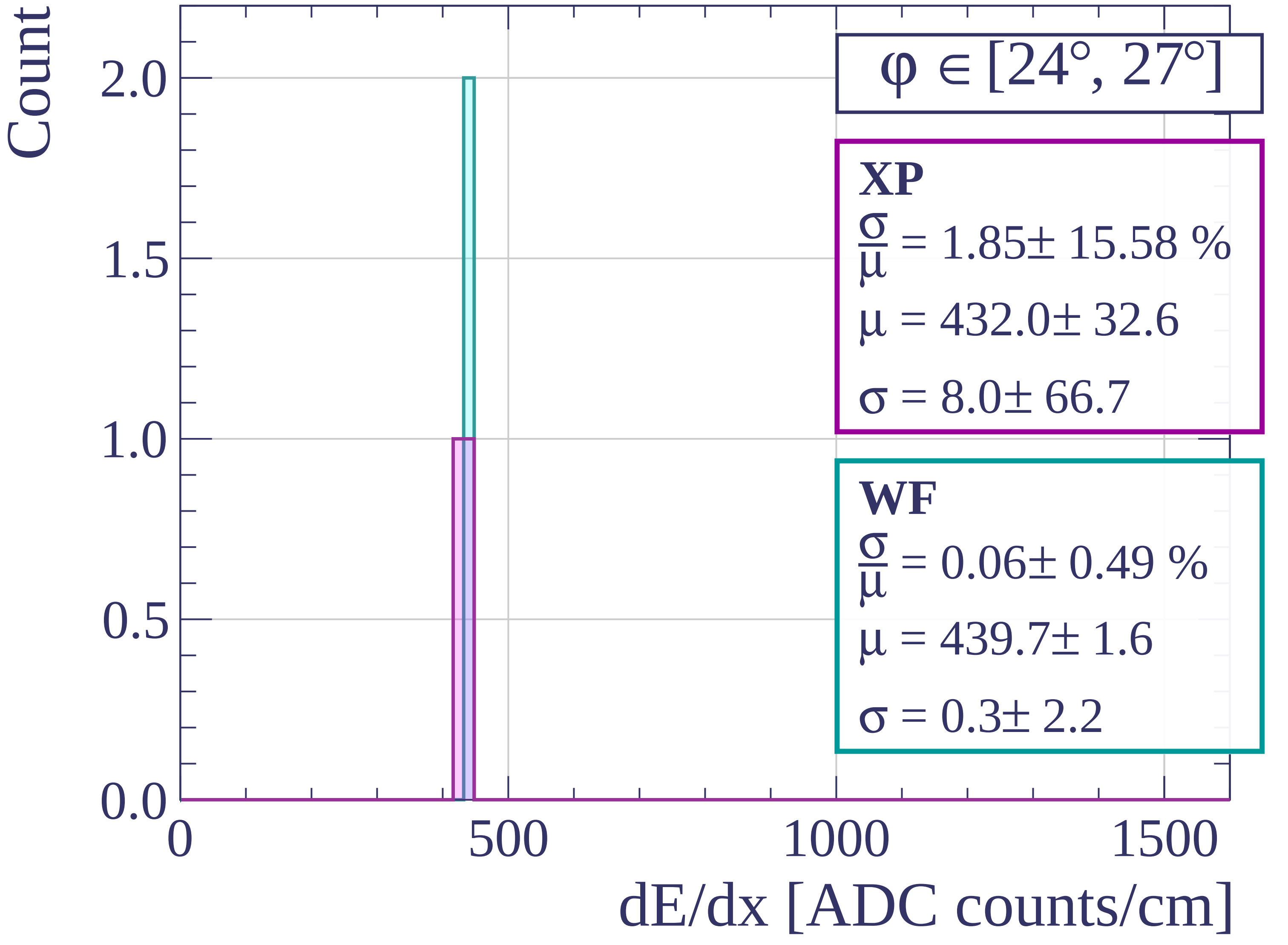
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [12°, 15°]
XP
σ/μ = 0.07± 0.60 %
μ = 391.7± 1.7
σ = 0.3± 2.3
WF
σ/μ = 2.00± 16.70 %
μ = 400.0± 32.6
σ = 8.0± 66.2

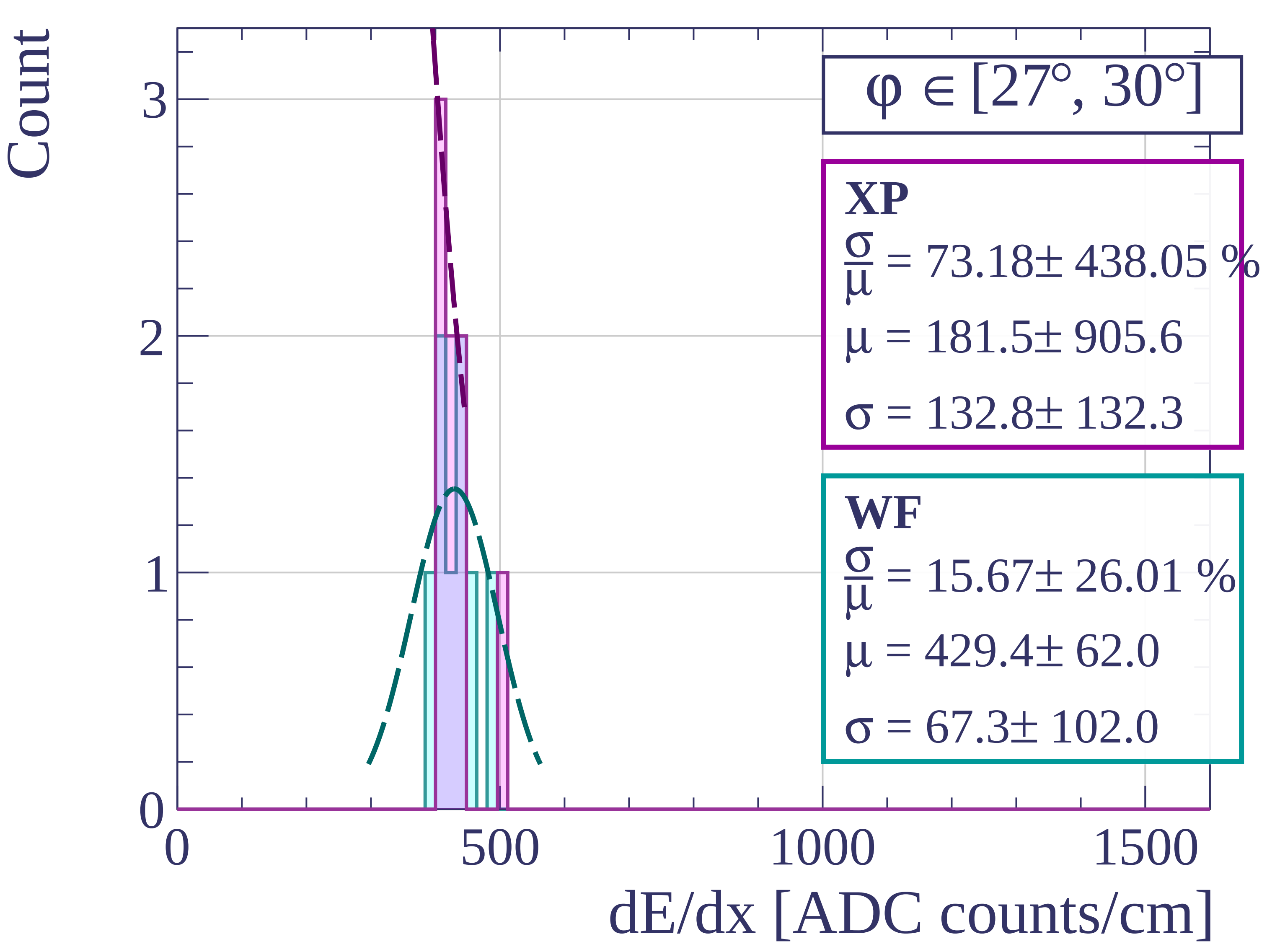

Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [15°, 18°]
XP
σ/μ = 10.87± 64.44 %
μ = 368.0± 121.5
σ = 40.0± 223.9
WF
σ/μ = 10.87± 64.44 %
μ = 368.0± 121.5
σ = 40.0± 223.9


Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [18°, 21°]
XP
σ/μ = 4.79± 6.32 %
μ = 400.0± 22.8
σ = 19.2± 24.2
WF
σ/μ = 2.00± 12.60 %
μ = 400.0± 23.0
σ = 8.0± 50.0

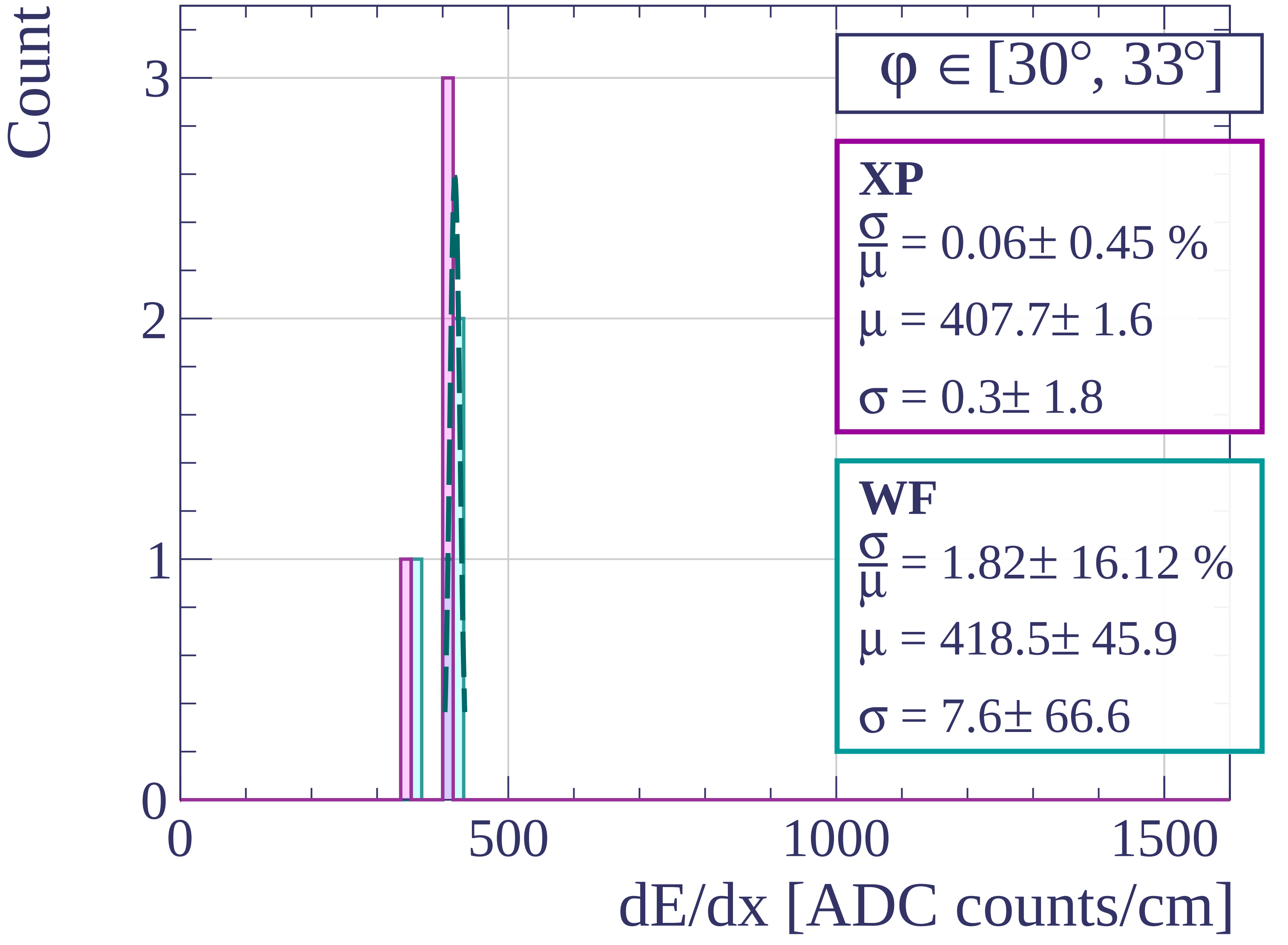


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [21°, 24°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

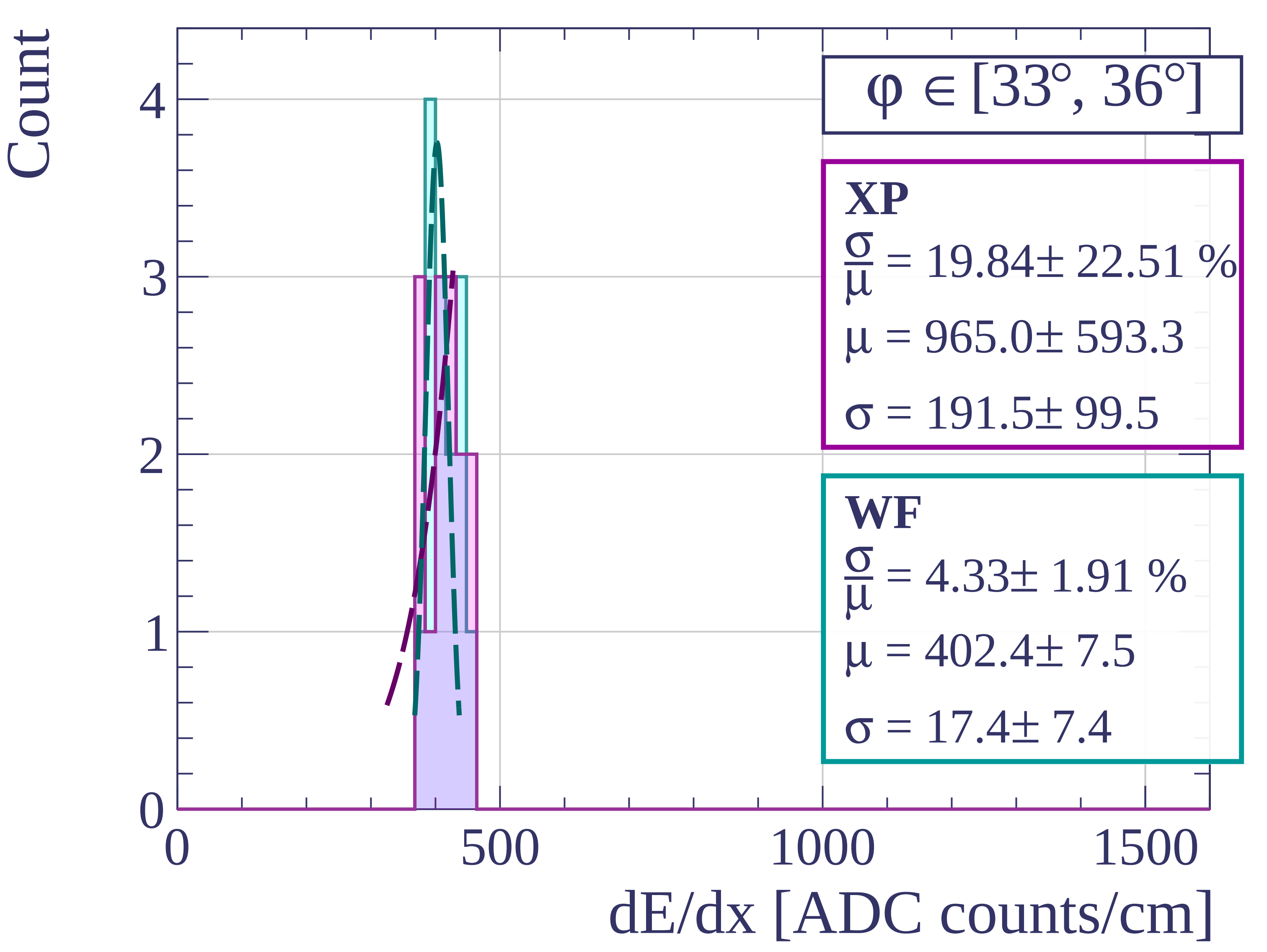

Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [24°, 27°]
XP
σ/μ = 1.85± 15.58 %
μ = 432.0± 32.6
σ = 8.0± 66.7
WF
σ/μ = 0.06± 0.49 %
μ = 439.7± 1.6
σ = 0.3± 2.2



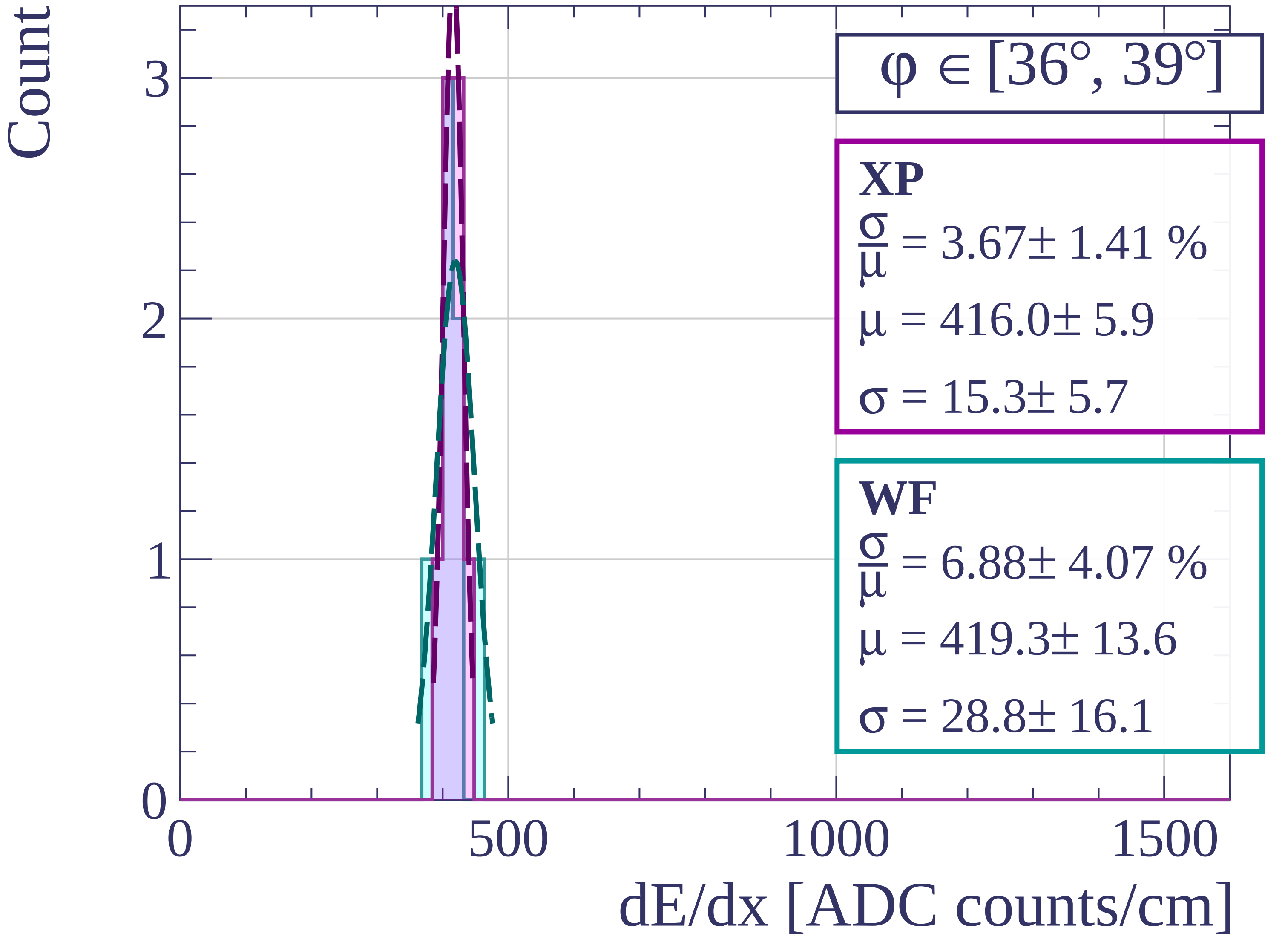
Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [27°, 30°]
XP
σ/μ = 73.18± 438.05 %
μ = 181.5± 905.6
σ = 132.8± 132.3
WF
σ/μ = 15.67± 26.01 %
μ = 429.4± 62.0
σ = 67.3± 102.0



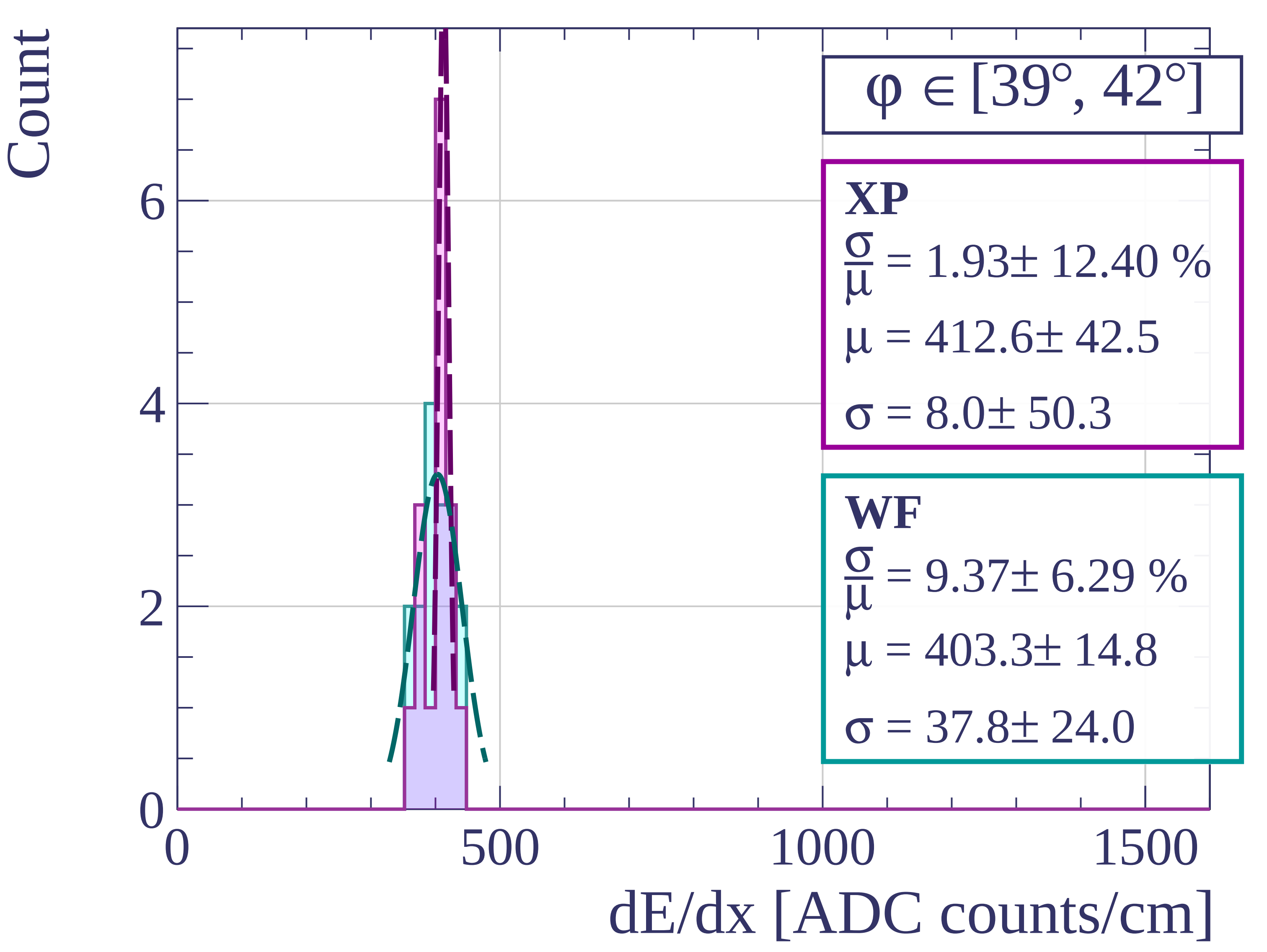
Count
0
1
2
3
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [30°, 33°]
XP
σ/μ = 0.06± 0.45 %
μ = 407.7± 1.6
σ = 0.3± 1.8
WF
σ/μ = 1.82± 16.12 %
μ = 418.5± 45.9
σ = 7.6± 66.6



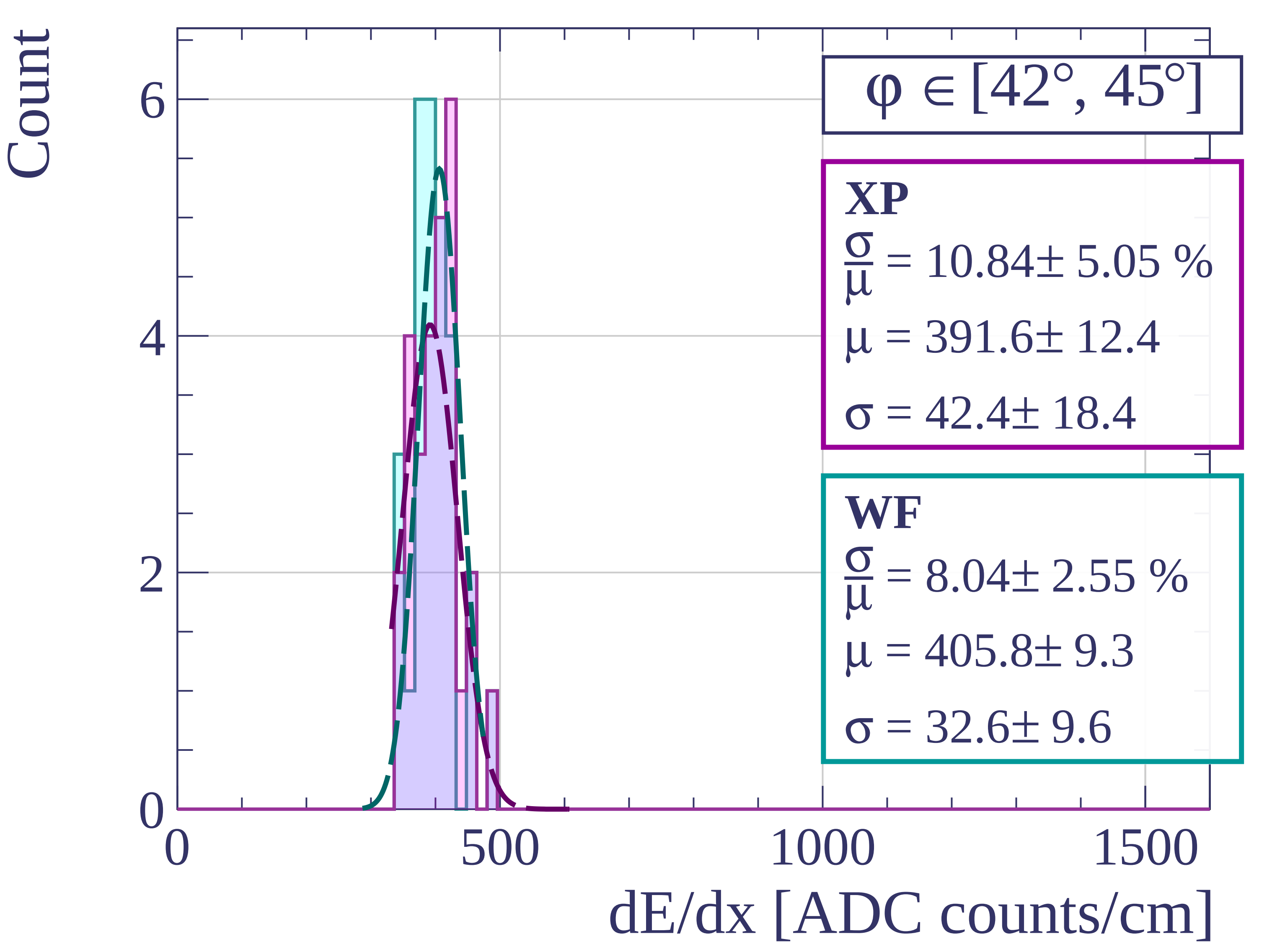
Count
0
1
2
3
4
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [33°, 36°]
XP
σ/μ = 19.84± 22.51 %
μ = 965.0± 593.3
σ = 191.5± 99.5
WF
σ/μ = 4.33± 1.91 %
μ = 402.4± 7.5
σ = 17.4± 7.4



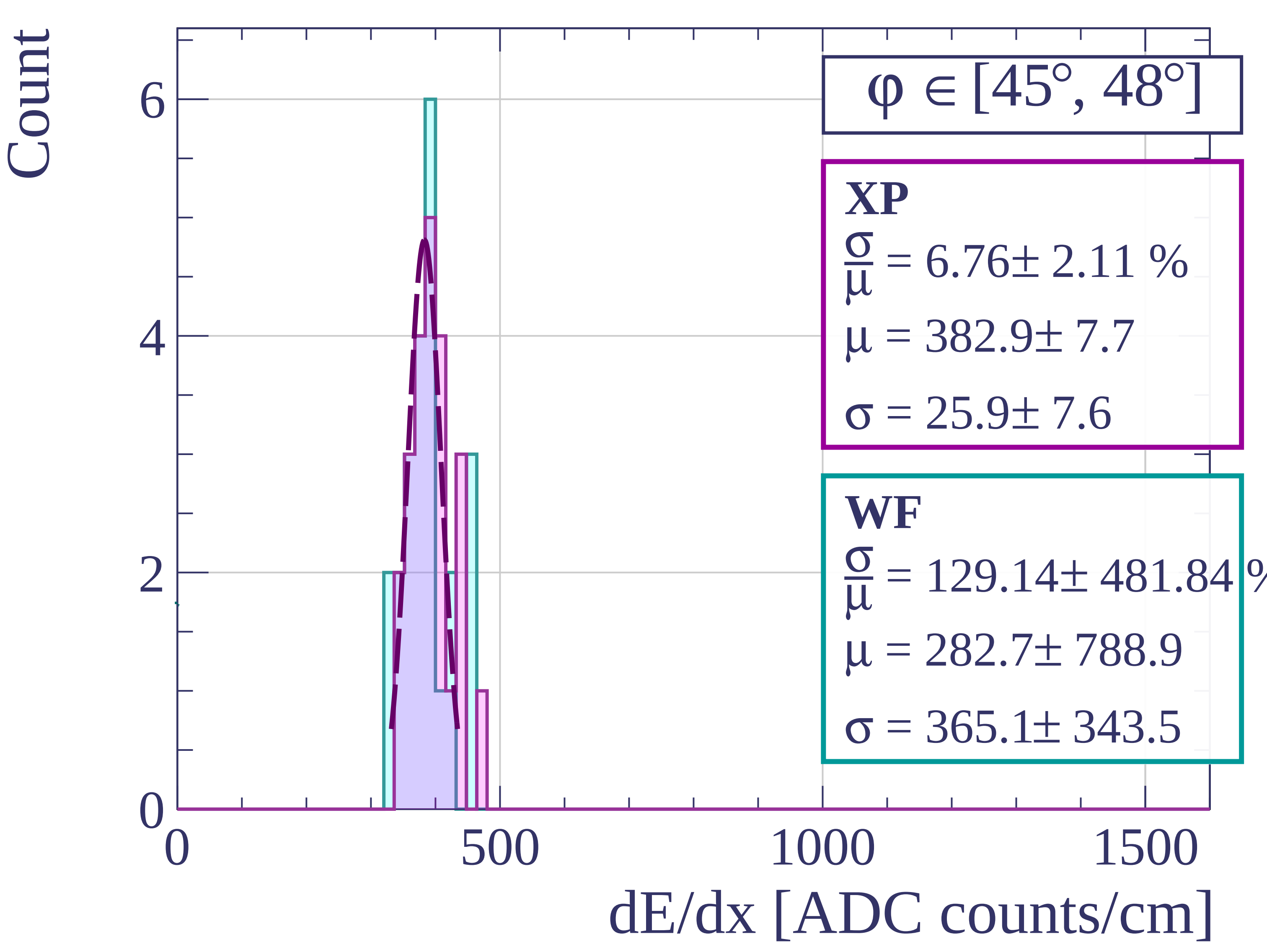
Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [36°, 39°]
XP
σ/μ = 3.67± 1.41 %
μ = 416.0± 5.9
σ = 15.3± 5.7
WF
σ/μ = 6.88± 4.07 %
μ = 419.3± 13.6
σ = 28.8± 16.1

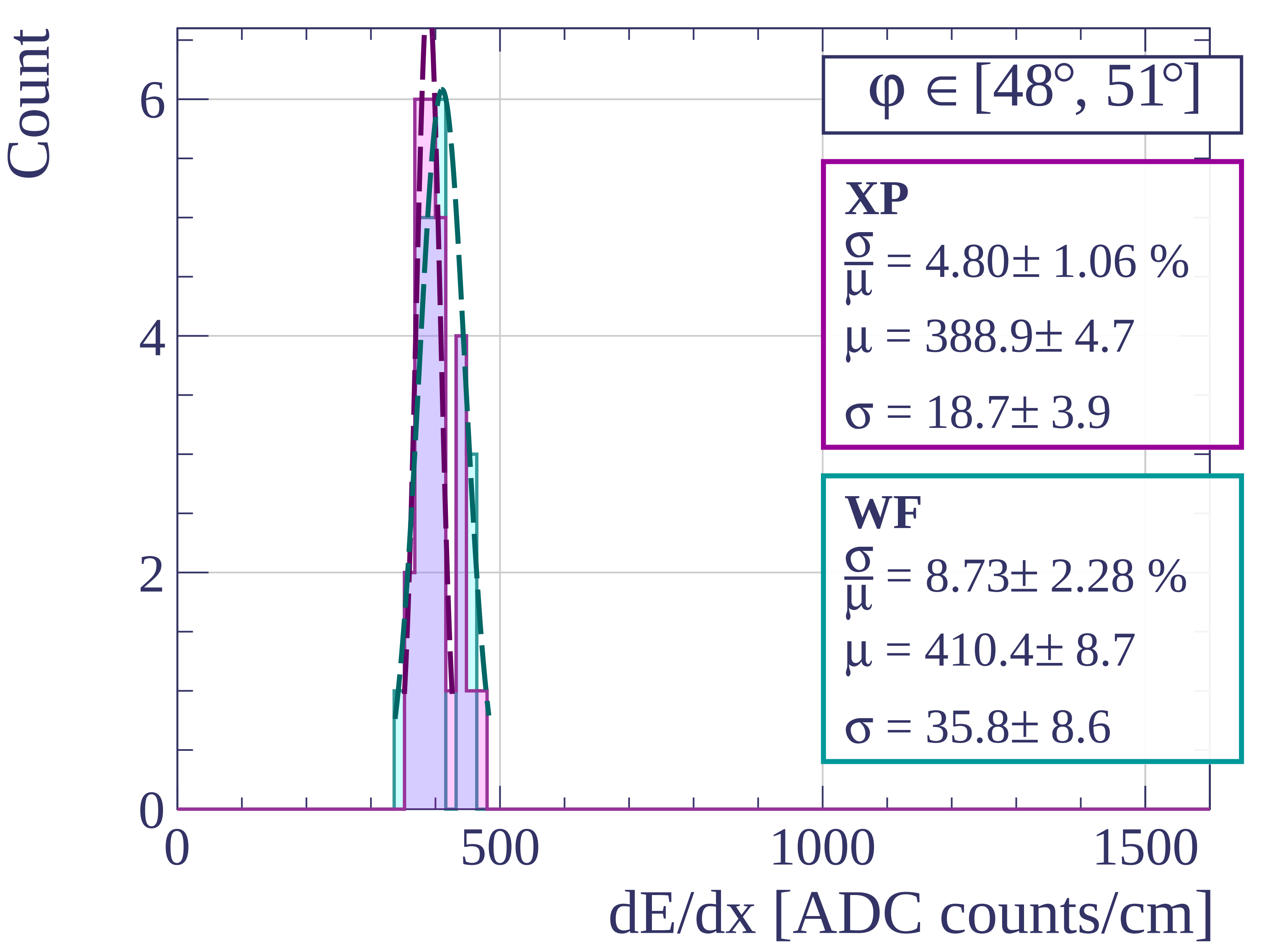


Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [39°, 42°]
XP
$\frac{\sigma}{\mu}$ = 1.93± 12.40 %
μ = 412.6± 42.5
σ = 8.0± 50.3
WF
$\frac{\sigma}{\mu}$ = 9.37± 6.29 %
μ = 403.3± 14.8
σ = 37.8± 24.0

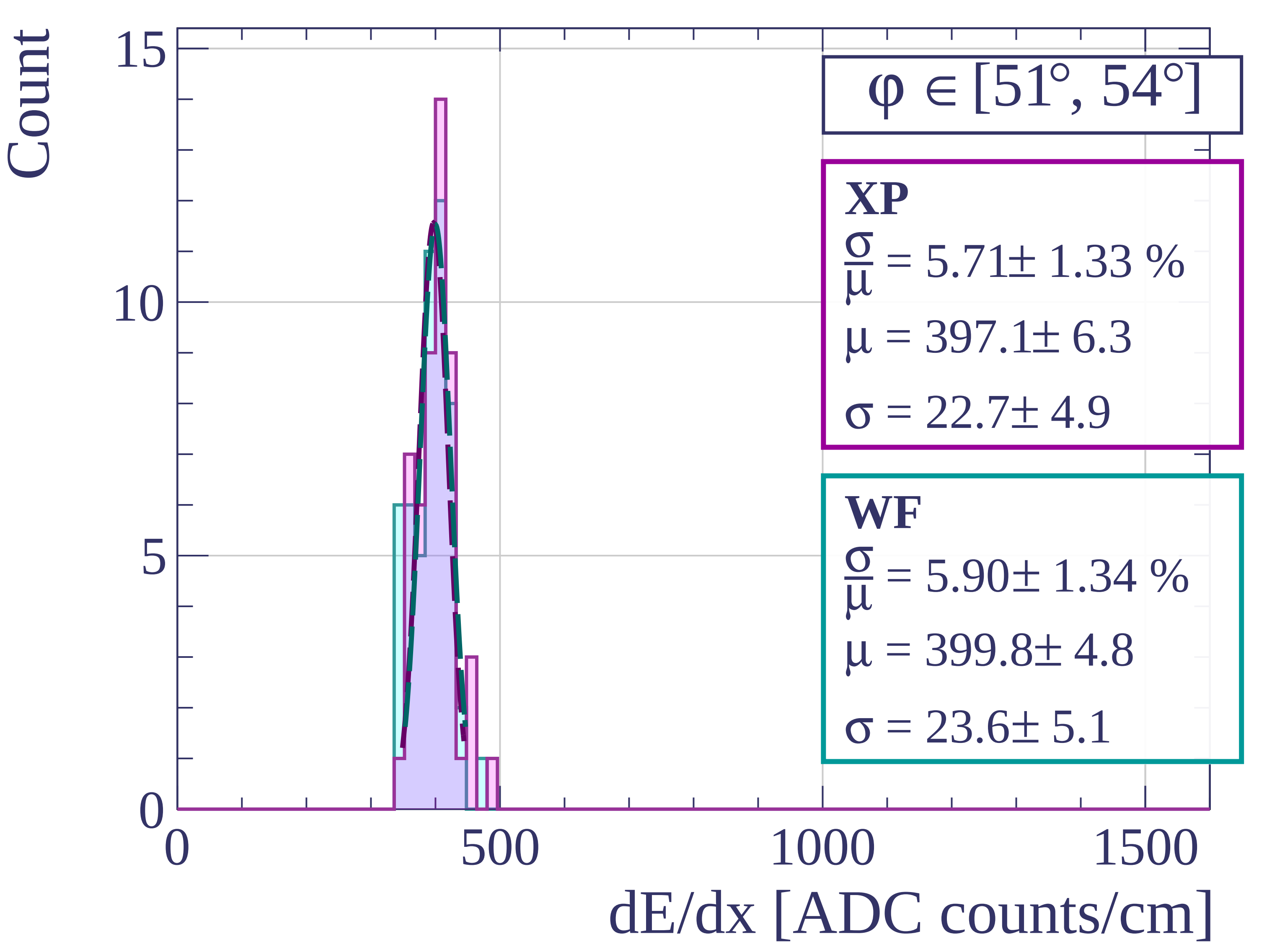


Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [42°, 45°]
XP
σ/μ = 10.84± 5.05 %
μ = 391.6± 12.4
σ = 42.4± 18.4
WF
σ/μ = 8.04± 2.55 %
μ = 405.8± 9.3
σ = 32.6± 9.6

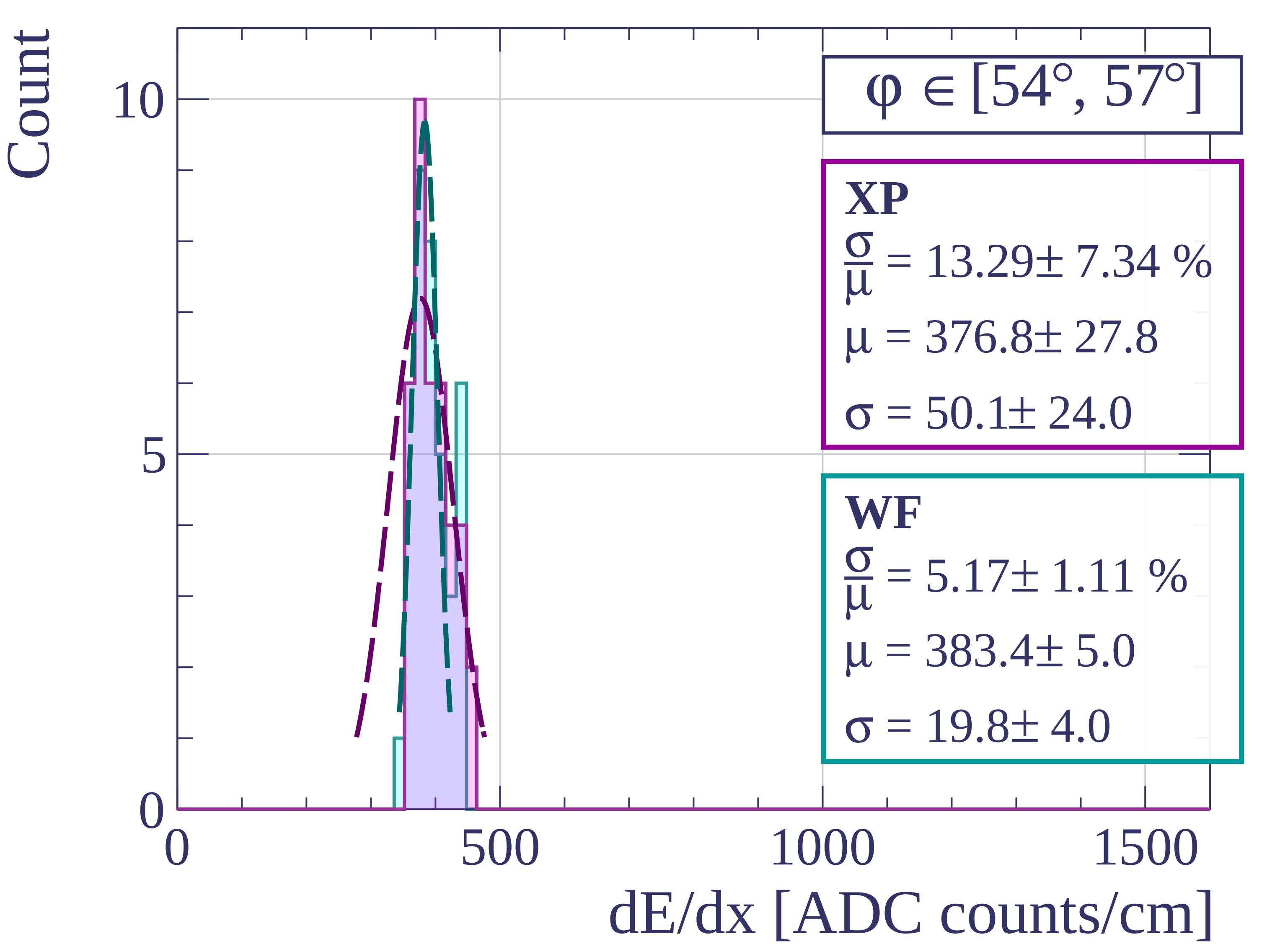

φ ∈ [45°, 48°]
XP
σ/μ = 6.76± 2.11 %
μ = 382.9± 7.7
σ = 25.9± 7.6
WF
σ/μ = 129.14± 481.84 %
μ = 282.7± 788.9
σ = 365.1± 343.5
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]



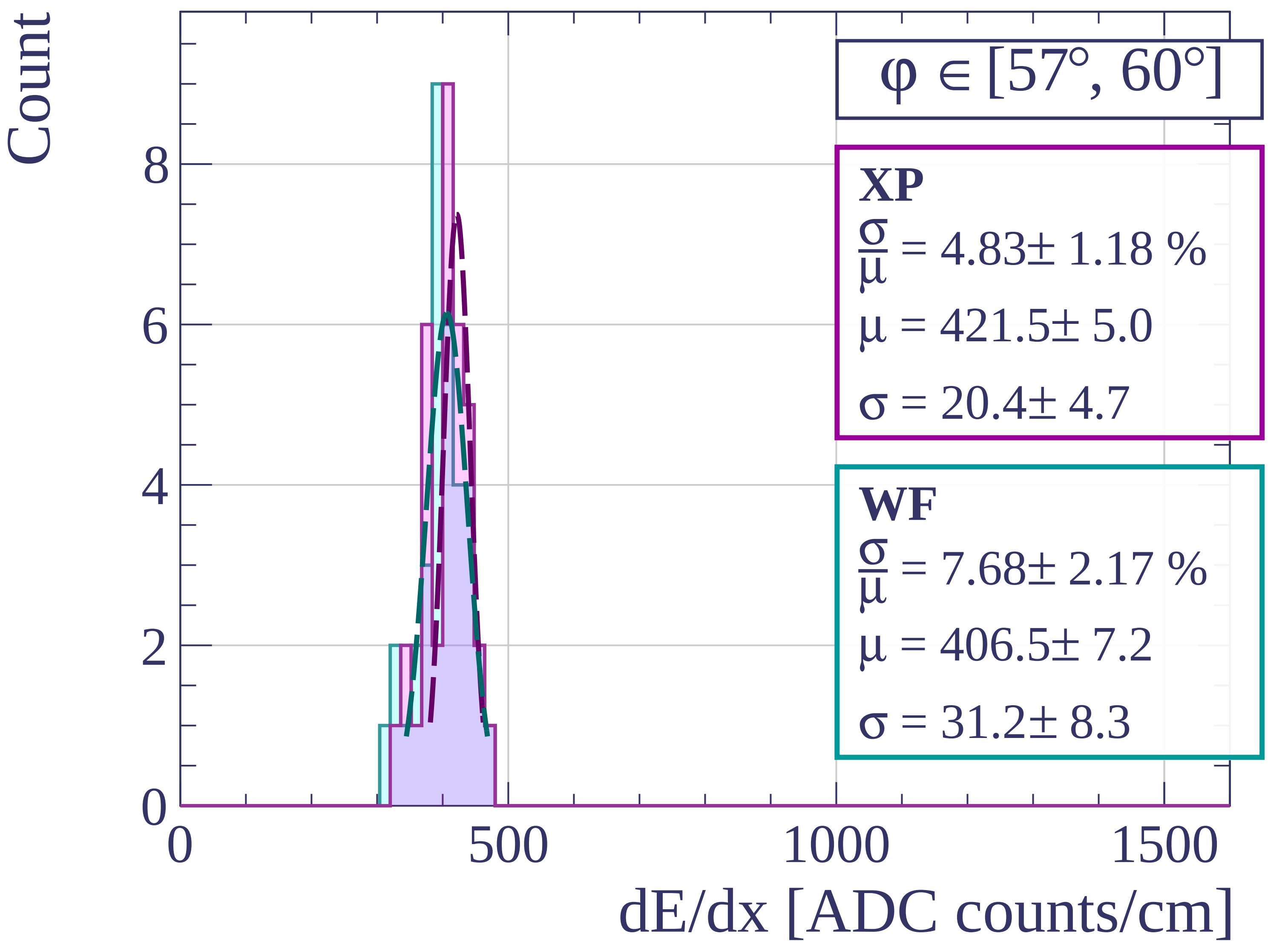
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [48°, 51°]
XP
σ/μ = 4.80± 1.06 %
μ = 388.9± 4.7
σ = 18.7± 3.9
WF
σ/μ = 8.73± 2.28 %
μ = 410.4± 8.7
σ = 35.8± 8.6

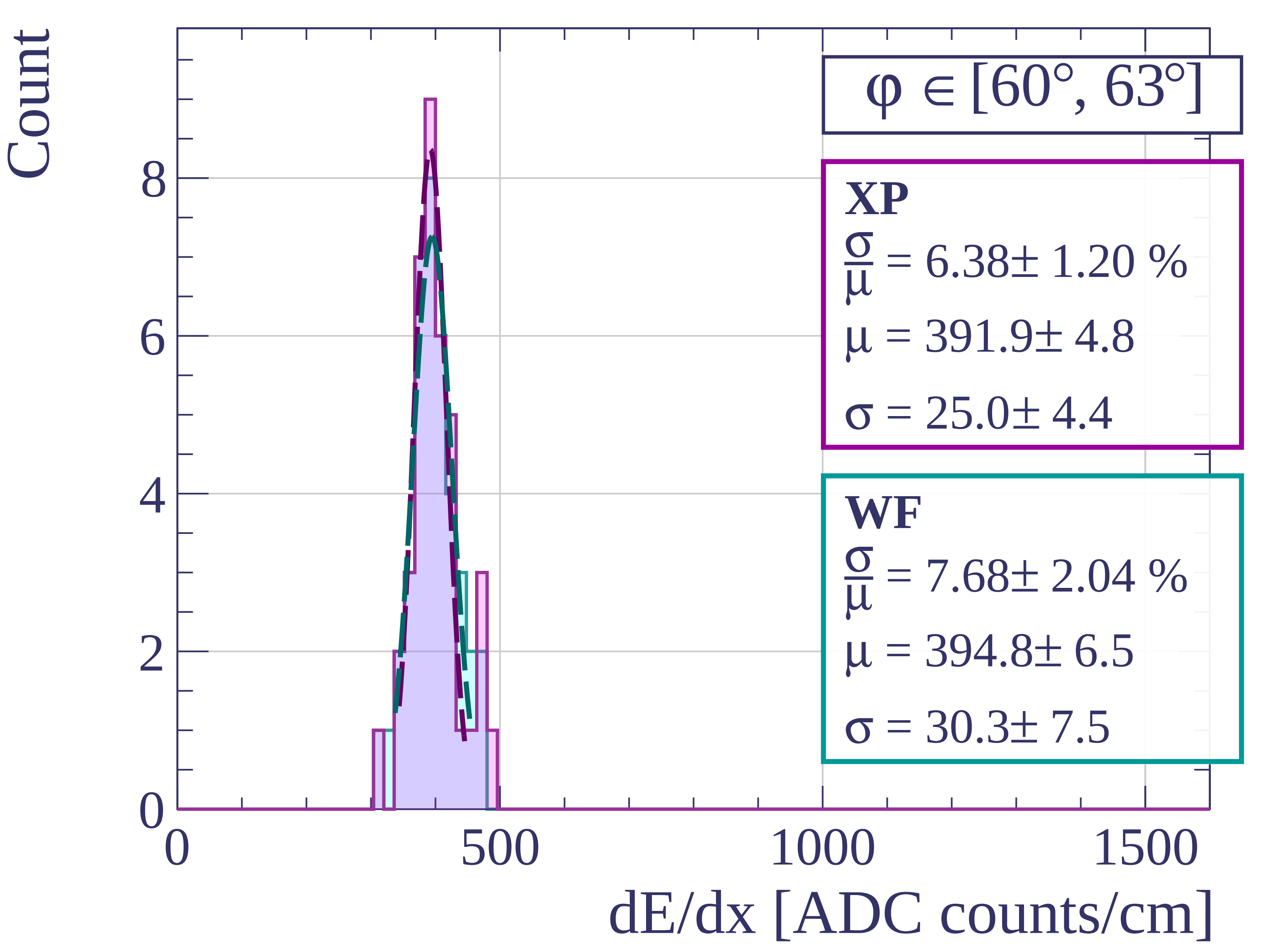


Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [51°, 54°]
XP
σ/μ = 5.71± 1.33 %
μ = 397.1± 6.3
σ = 22.7± 4.9
WF
σ/μ = 5.90± 1.34 %
μ = 399.8± 4.8
σ = 23.6± 5.1

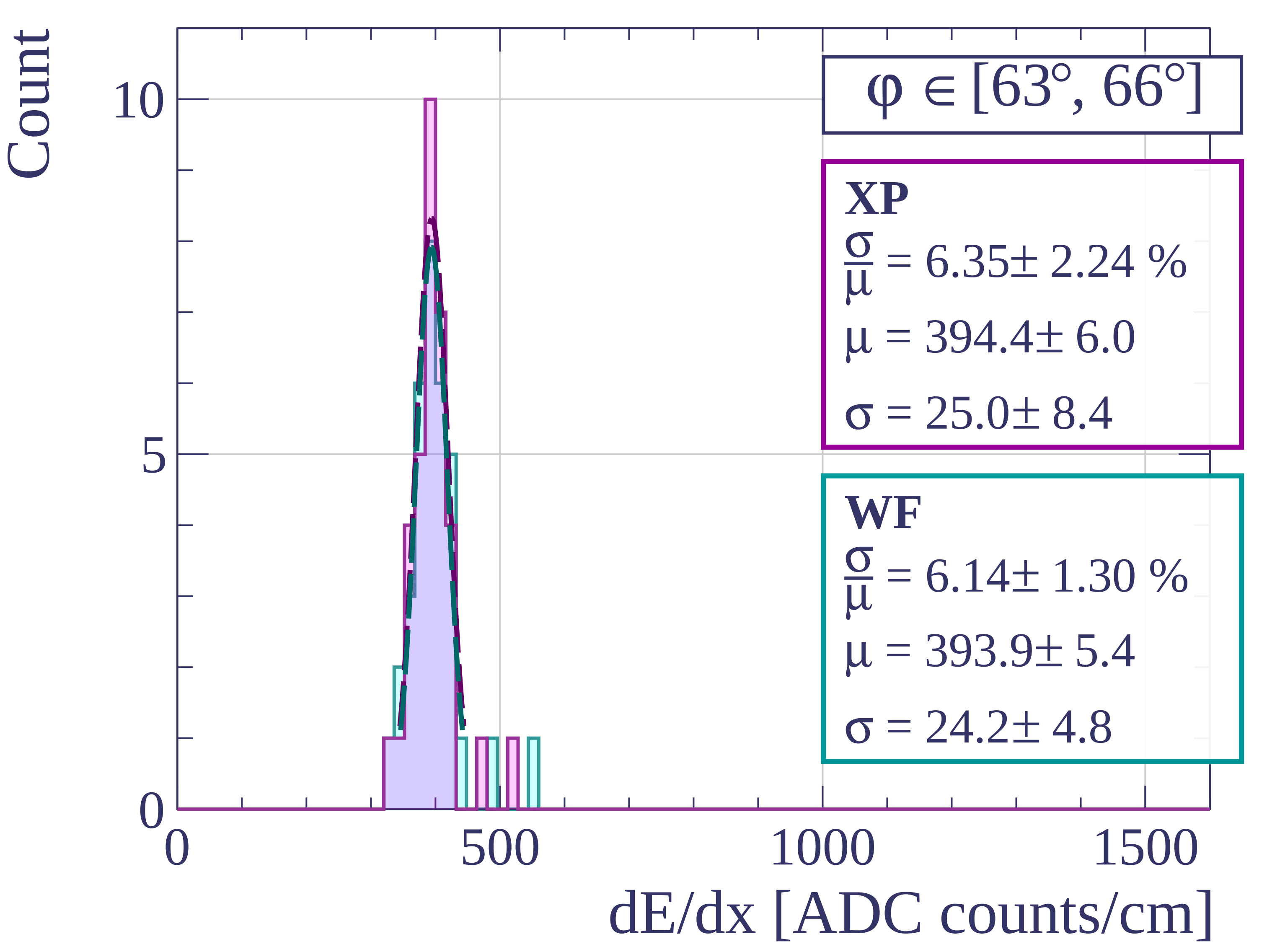


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [54°, 57°]
XP
σ/μ = 13.29± 7.34 %
μ = 376.8± 27.8
σ = 50.1± 24.0
WF
σ/μ = 5.17± 1.11 %
μ = 383.4± 5.0
σ = 19.8± 4.0

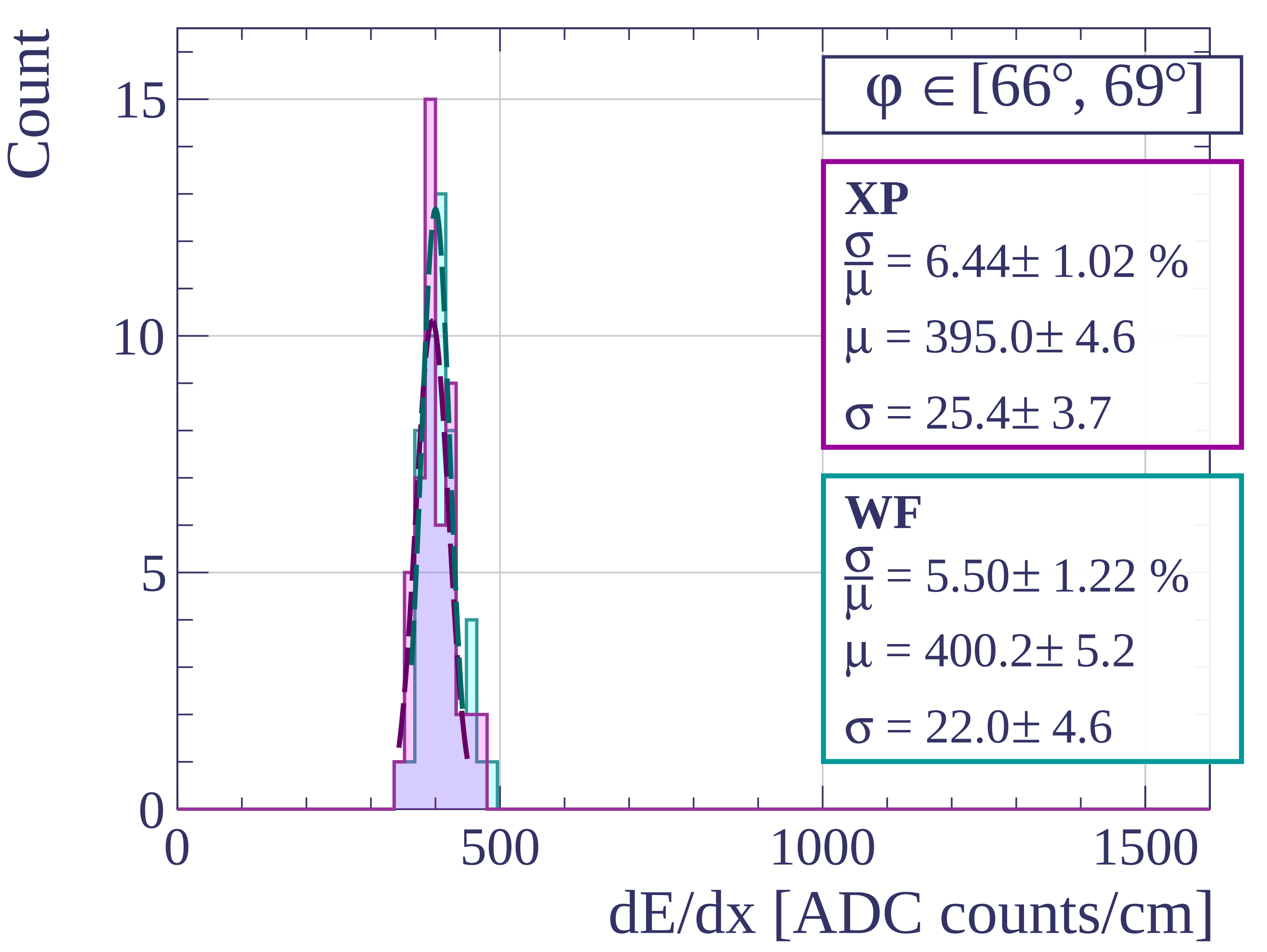

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [57°, 60°]
XP
σ/μ = 4.83± 1.18 %
μ = 421.5± 5.0
σ = 20.4± 4.7
WF
σ/μ = 7.68± 2.17 %
μ = 406.5± 7.2
σ = 31.2± 8.3



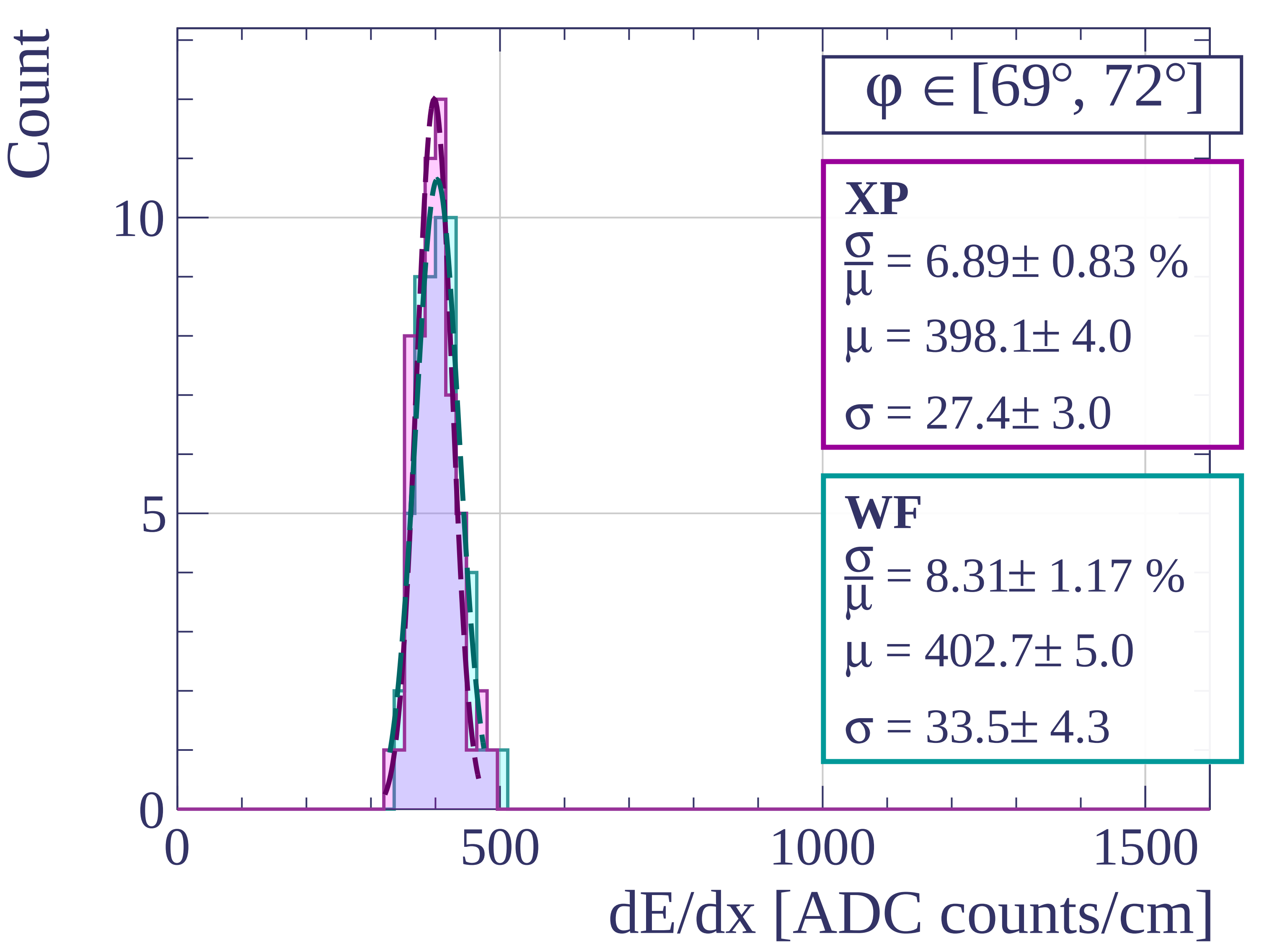
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [60°, 63°]
XP
$\frac{\sigma}{\mu}$ = 6.38± 1.20 %
μ = 391.9± 4.8
σ = 25.0± 4.4
WF
$\frac{\sigma}{\mu}$ = 7.68± 2.04 %
μ = 394.8± 6.5
σ = 30.3± 7.5



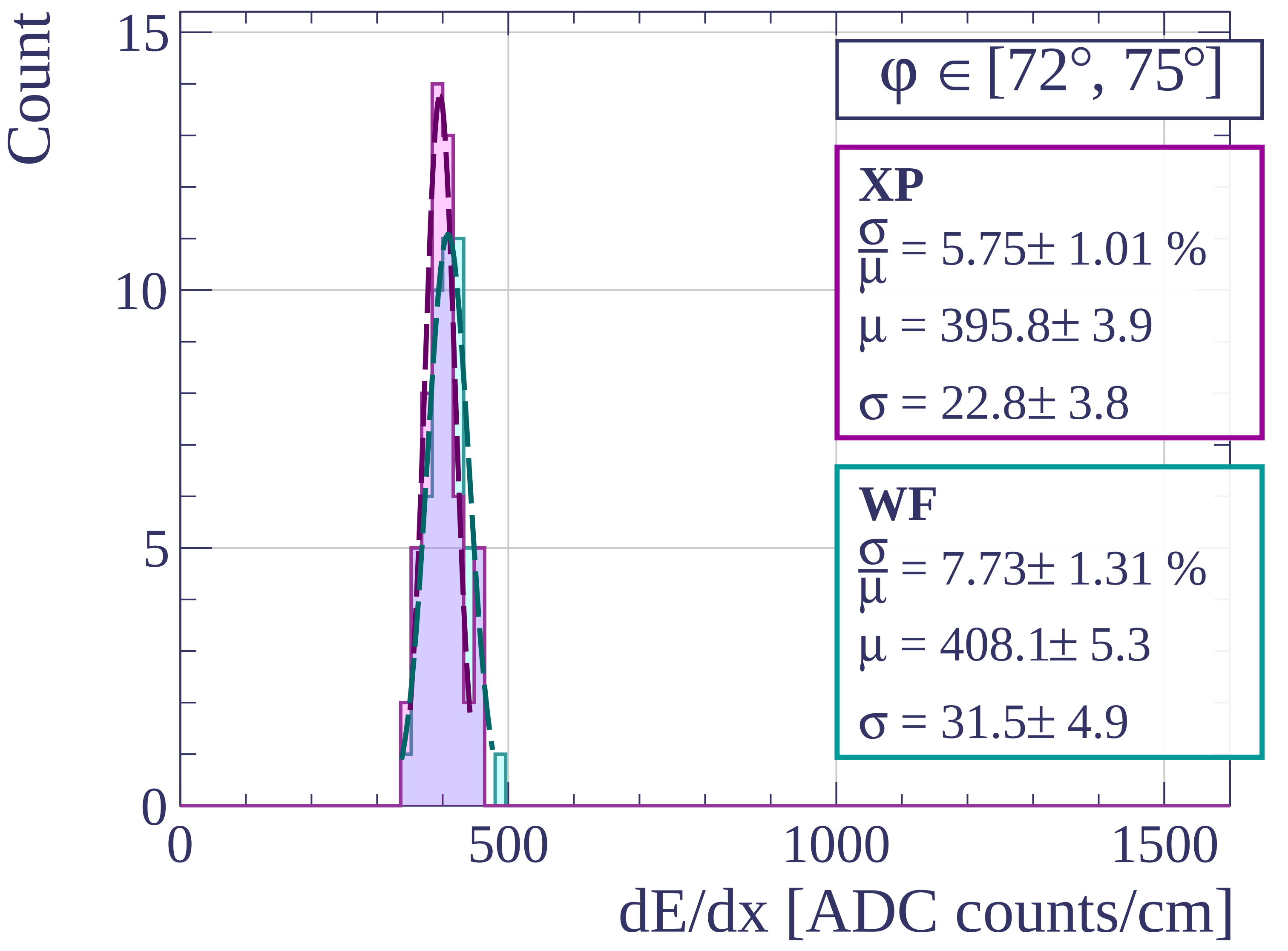
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [63°, 66°]
XP
σ/μ = 6.35± 2.24 %
μ = 394.4± 6.0
σ = 25.0± 8.4
WF
σ/μ = 6.14± 1.30 %
μ = 393.9± 5.4
σ = 24.2± 4.8

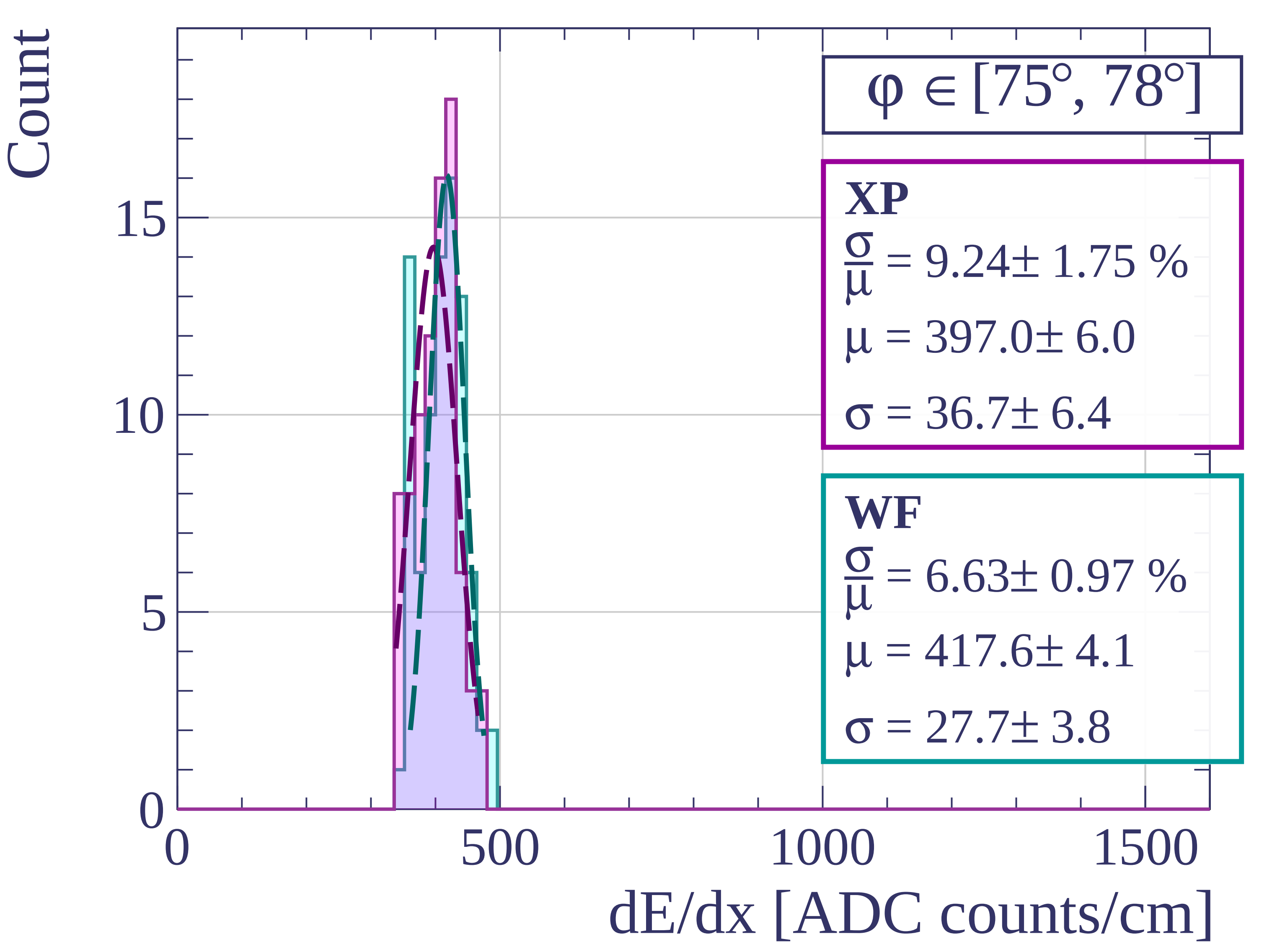


Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [66°, 69°]
XP
σ/μ = 6.44± 1.02 %
μ = 395.0± 4.6
σ = 25.4± 3.7
WF
σ/μ = 5.50± 1.22 %
μ = 400.2± 5.2
σ = 22.0± 4.6

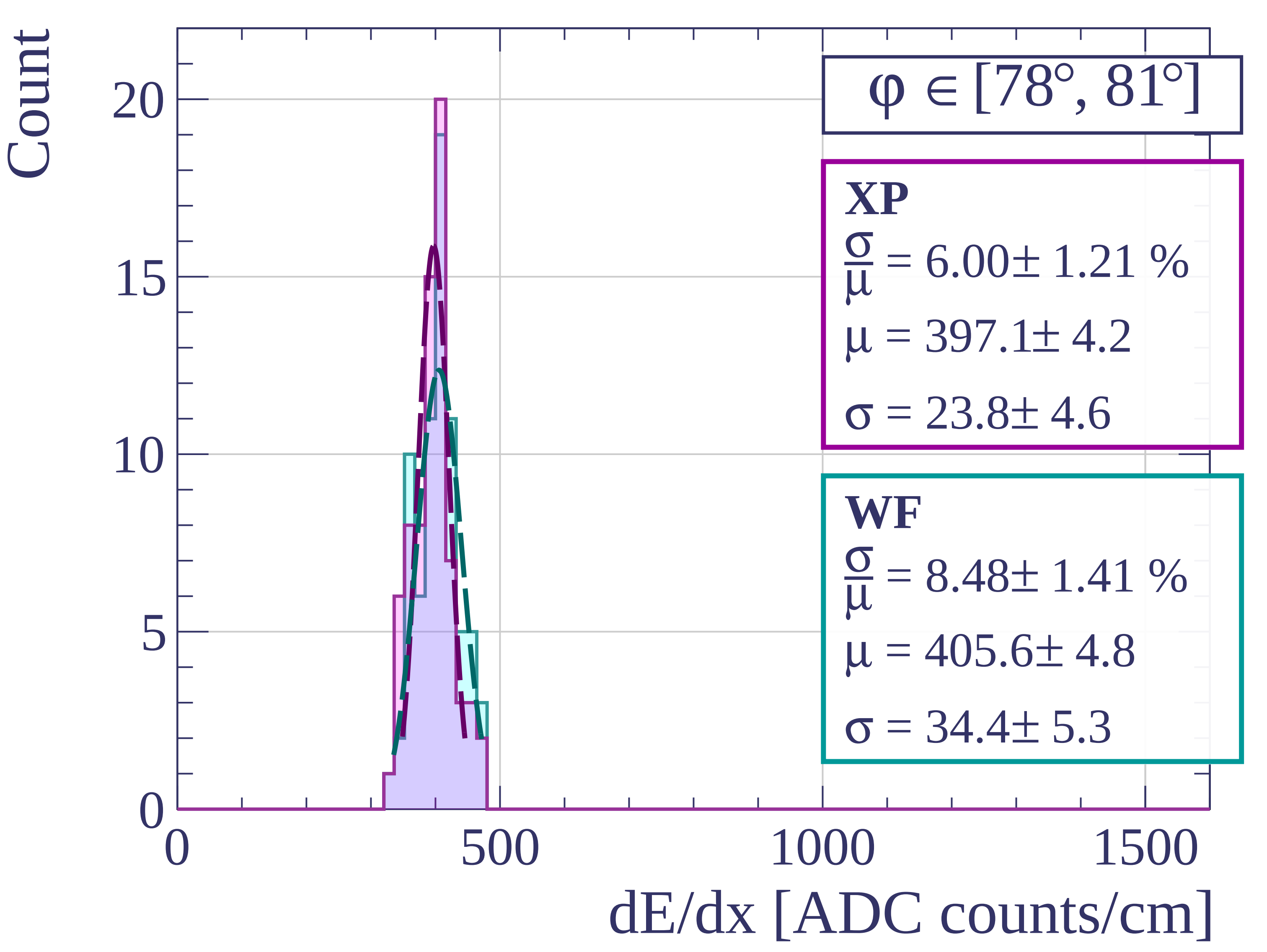


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [69°, 72°]
XP
σ/μ = 6.89± 0.83 %
μ = 398.1± 4.0
σ = 27.4± 3.0
WF
σ/μ = 8.31± 1.17 %
μ = 402.7± 5.0
σ = 33.5± 4.3

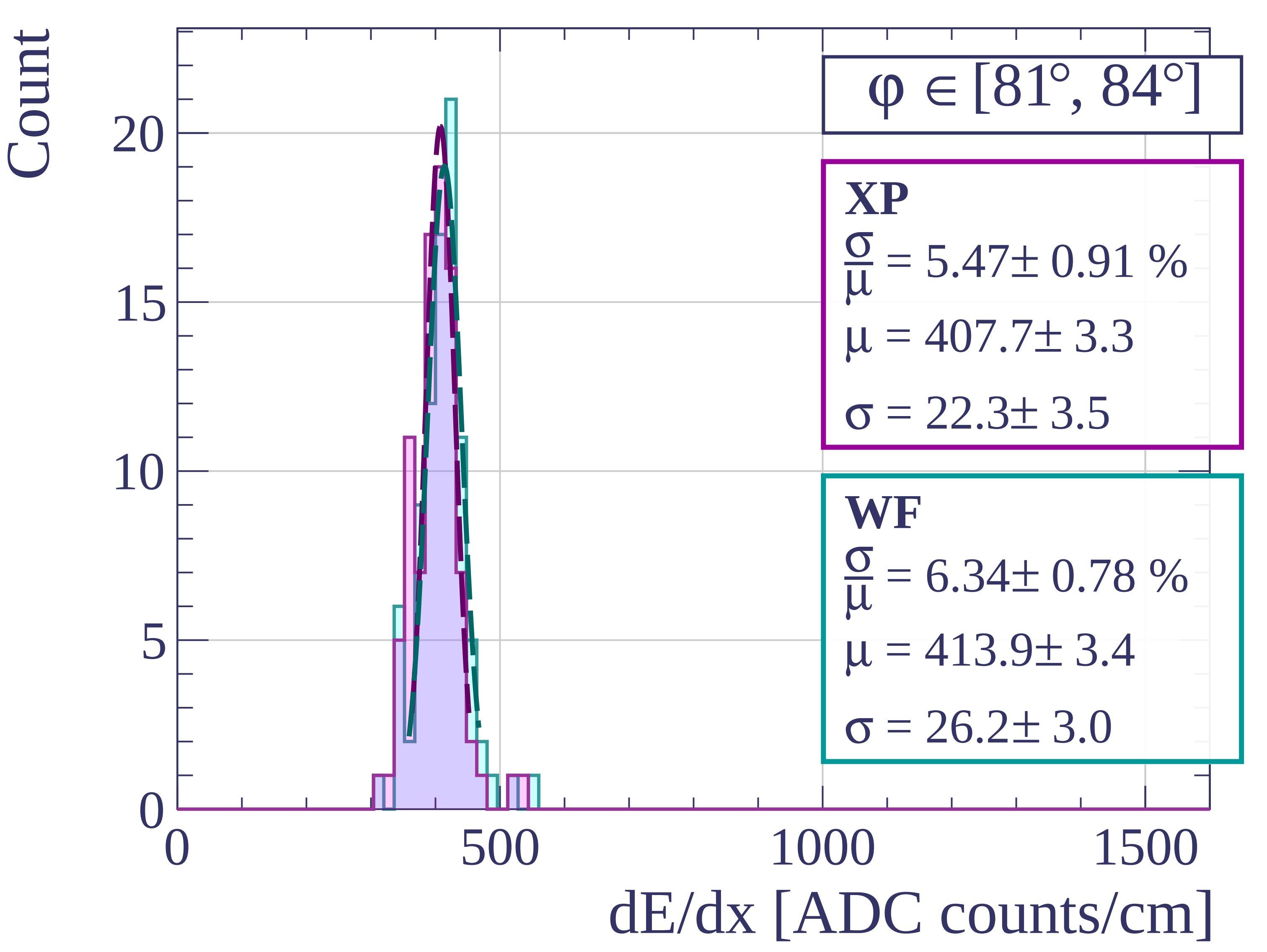

Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
φ ∈ [72°, 75°]
XP
σ/μ = 5.75± 1.01 %
μ = 395.8± 3.9
σ = 22.8± 3.8
WF
σ/μ = 7.73± 1.31 %
μ = 408.1± 5.3
σ = 31.5± 4.9

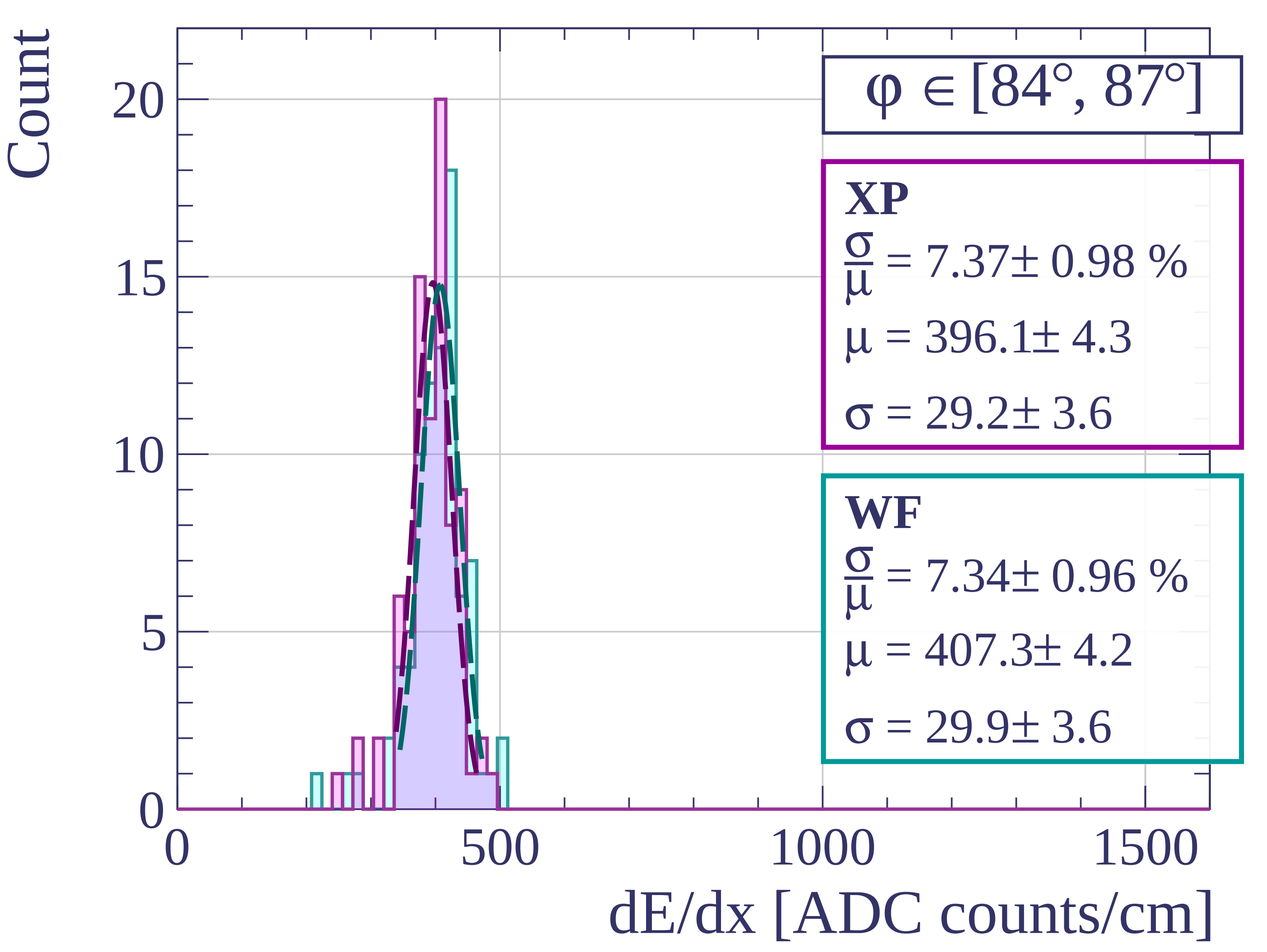


Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
φ ∈ [75°, 78°]
XP
σ/μ = 9.24± 1.75 %
μ = 397.0± 6.0
σ = 36.7± 6.4
WF
σ/μ = 6.63± 0.97 %
μ = 417.6± 4.1
σ = 27.7± 3.8

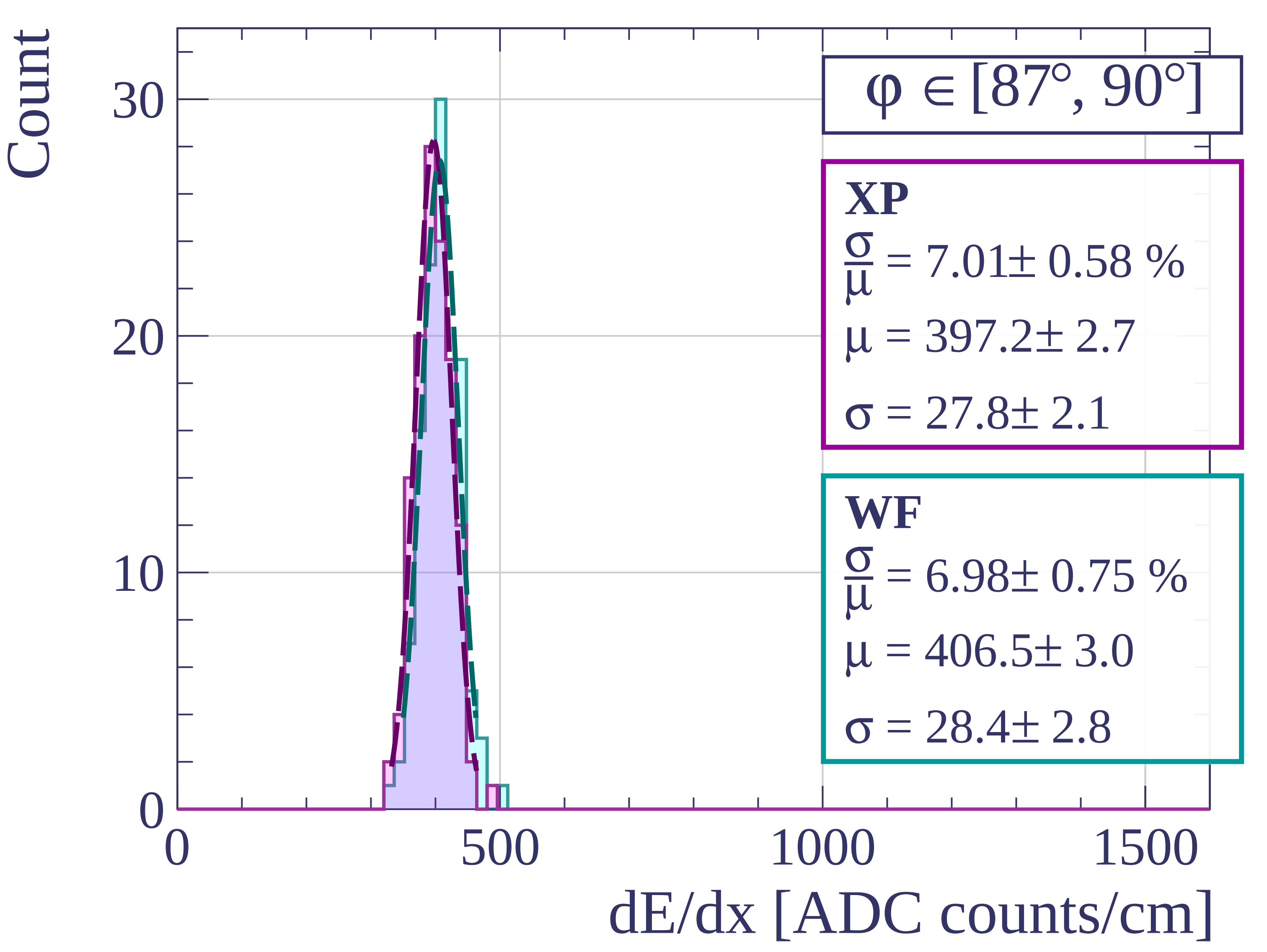


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
5
10
15
20
φ ∈ [78°, 81°]
XP
σ/μ = 6.00± 1.21 %
μ = 397.1± 4.2
σ = 23.8± 4.6
WF
σ/μ = 8.48± 1.41 %
μ = 405.6± 4.8
σ = 34.4± 5.3

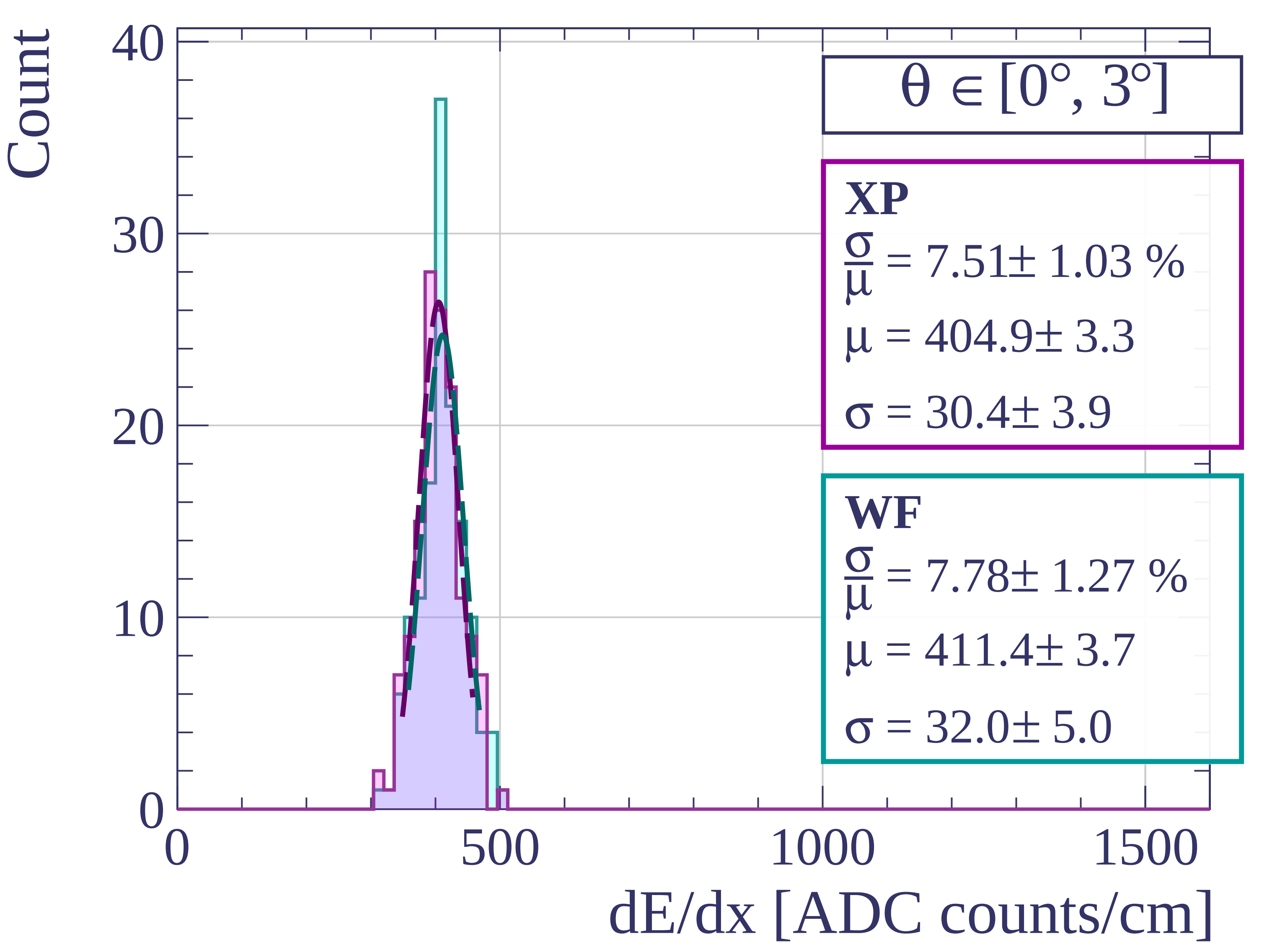

Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [81°, 84°]
XP
σ/μ = 5.47± 0.91 %
μ = 407.7± 3.3
σ = 22.3± 3.5
WF
σ/μ = 6.34± 0.78 %
μ = 413.9± 3.4
σ = 26.2± 3.0



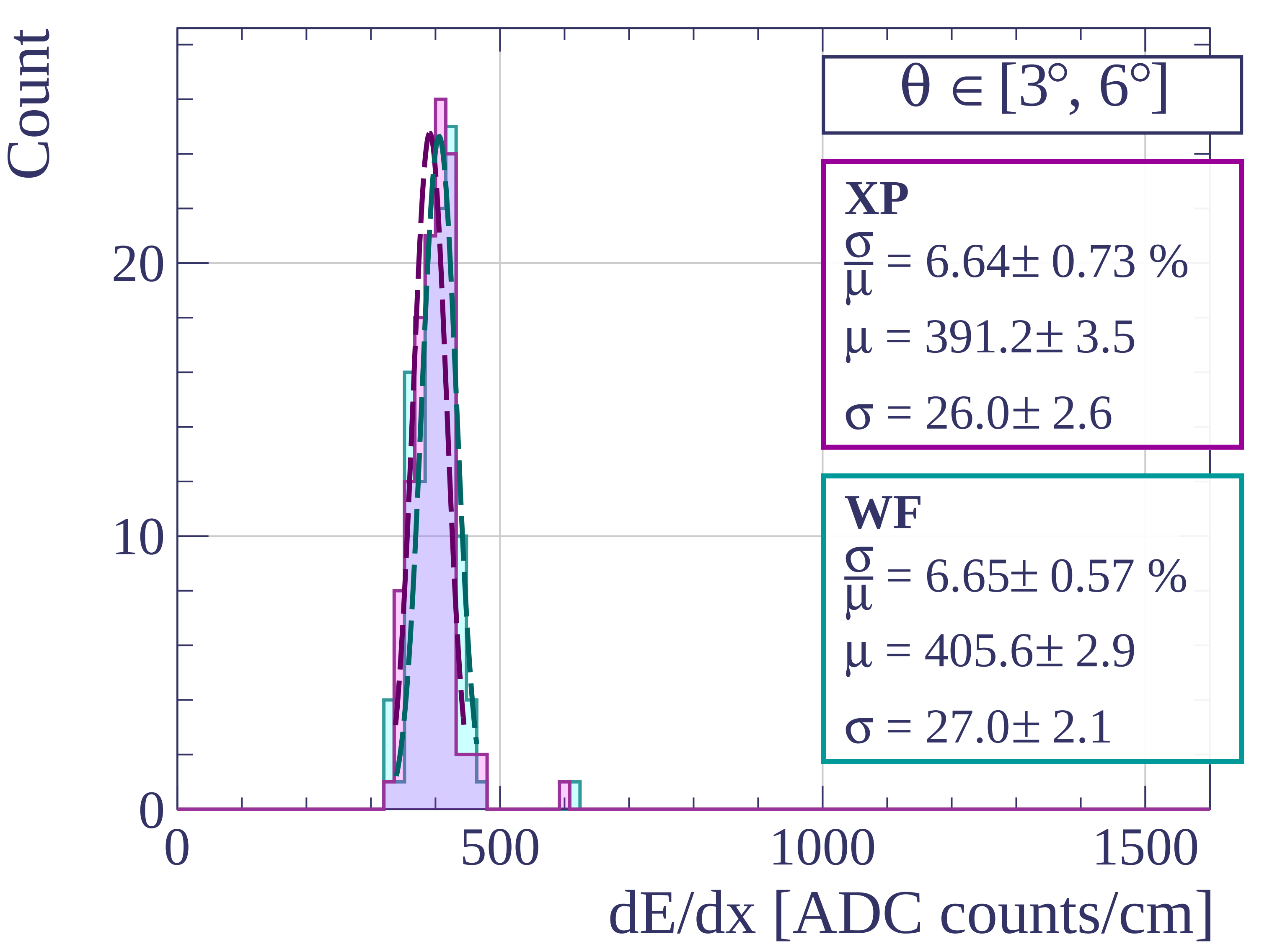
Count
0
5
10
15
20
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [84°, 87°]
XP
σ/μ = 7.37± 0.98 %
μ = 396.1± 4.3
σ = 29.2± 3.6
WF
σ/μ = 7.34± 0.96 %
μ = 407.3± 4.2
σ = 29.9± 3.6



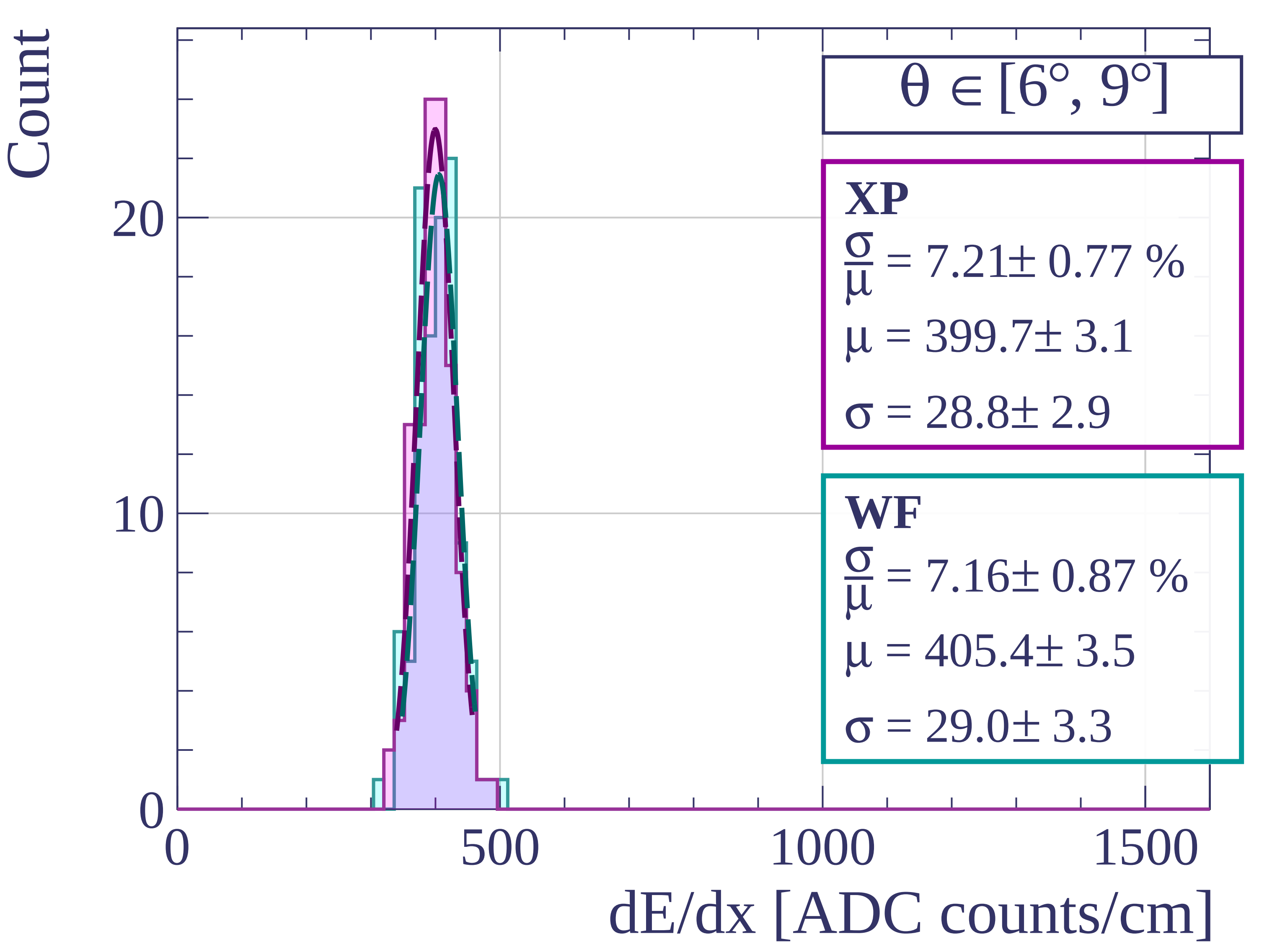
Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [87°, 90°]
XP
σ/μ = 7.01± 0.58 %
μ = 397.2± 2.7
σ = 27.8± 2.1
WF
σ/μ = 6.98± 0.75 %
μ = 406.5± 3.0
σ = 28.4± 2.8



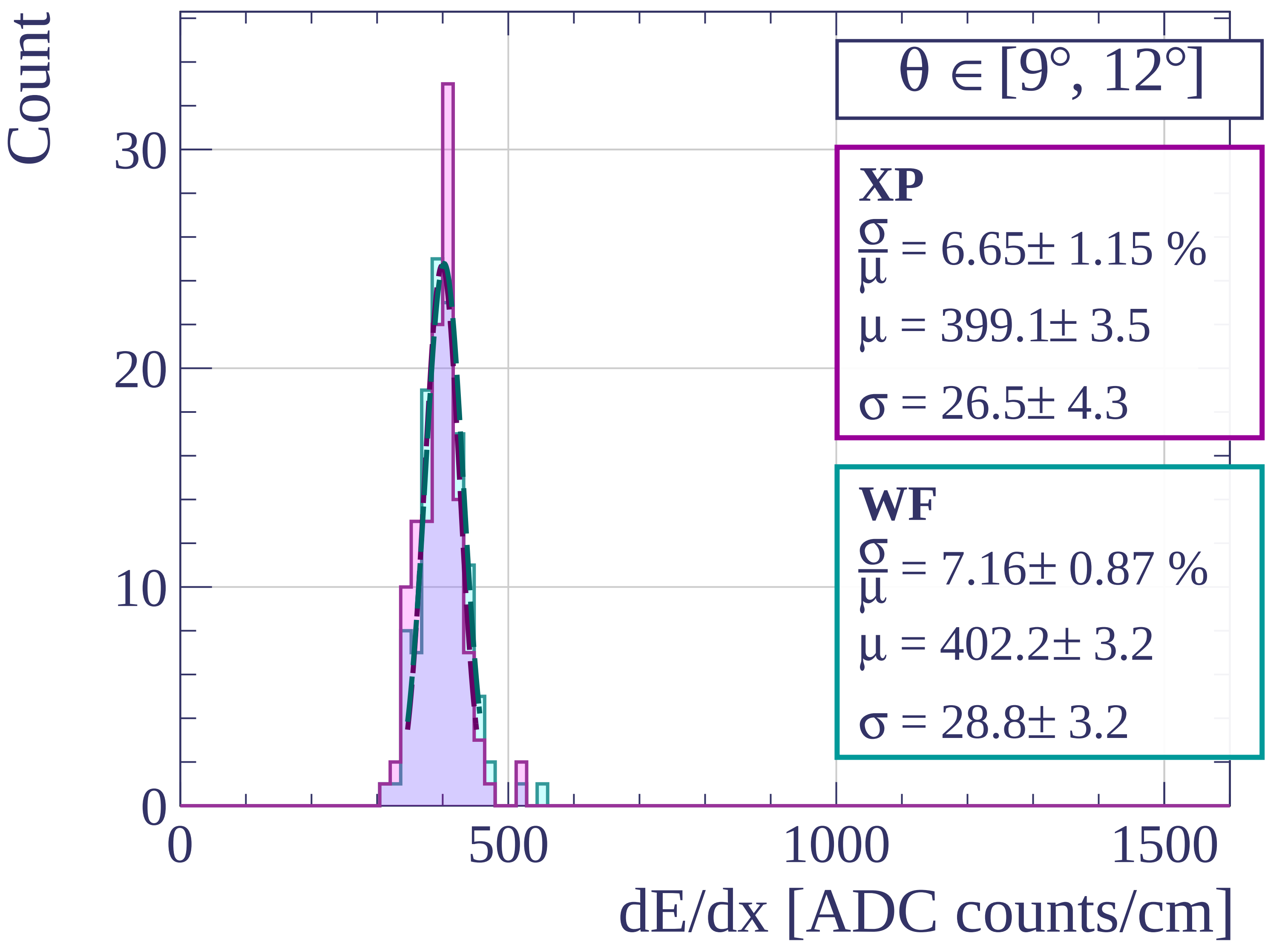
Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [0°, 3°]
XP
σ/μ = 7.51± 1.03 %
μ = 404.9± 3.3
σ = 30.4± 3.9
WF
σ/μ = 7.78± 1.27 %
μ = 411.4± 3.7
σ = 32.0± 5.0



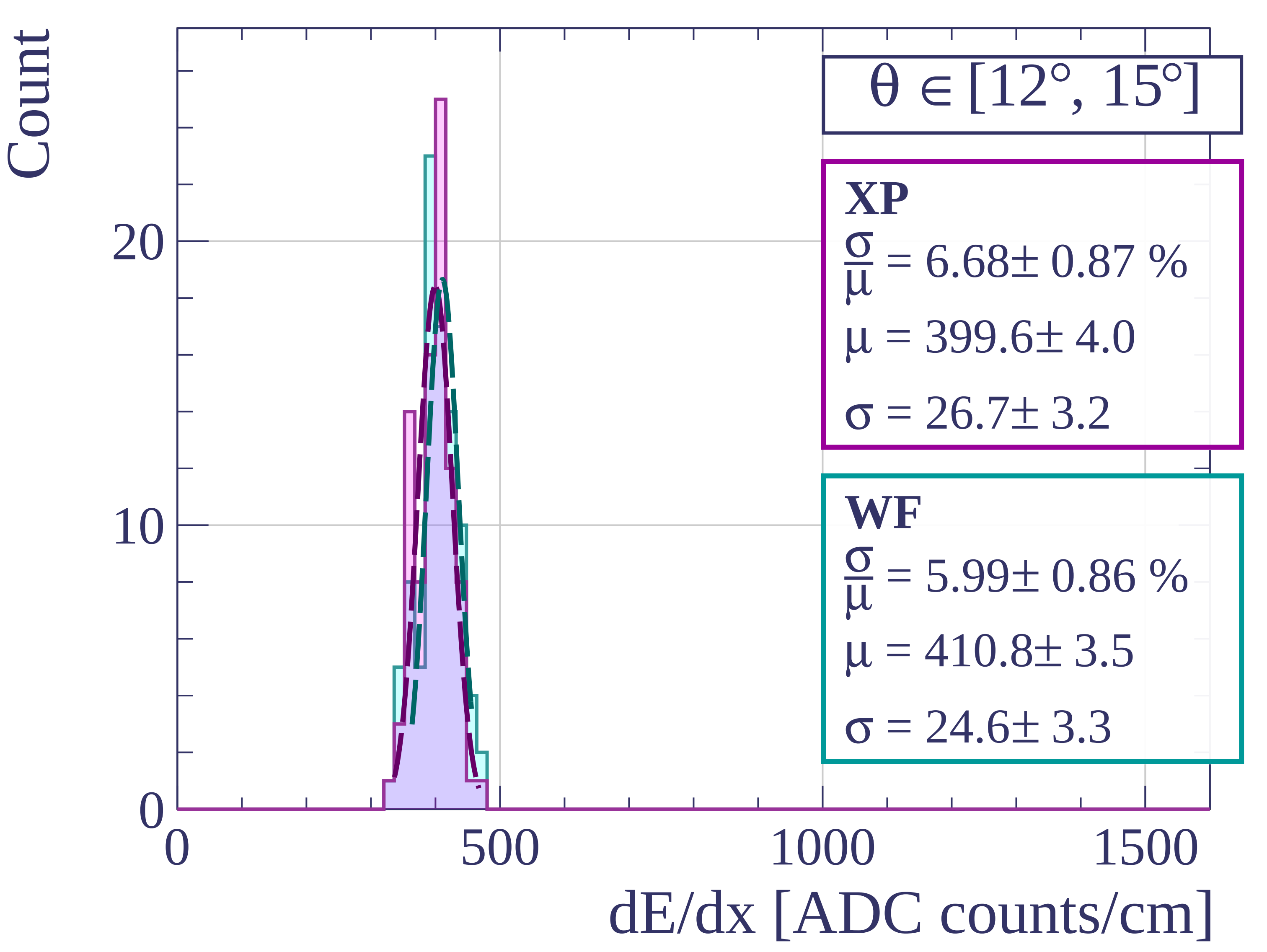
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
10
20
θ ∈ [3°, 6°]
XP
σ/μ = 6.64± 0.73 %
μ = 391.2± 3.5
σ = 26.0± 2.6
WF
σ/μ = 6.65± 0.57 %
μ = 405.6± 2.9
σ = 27.0± 2.1



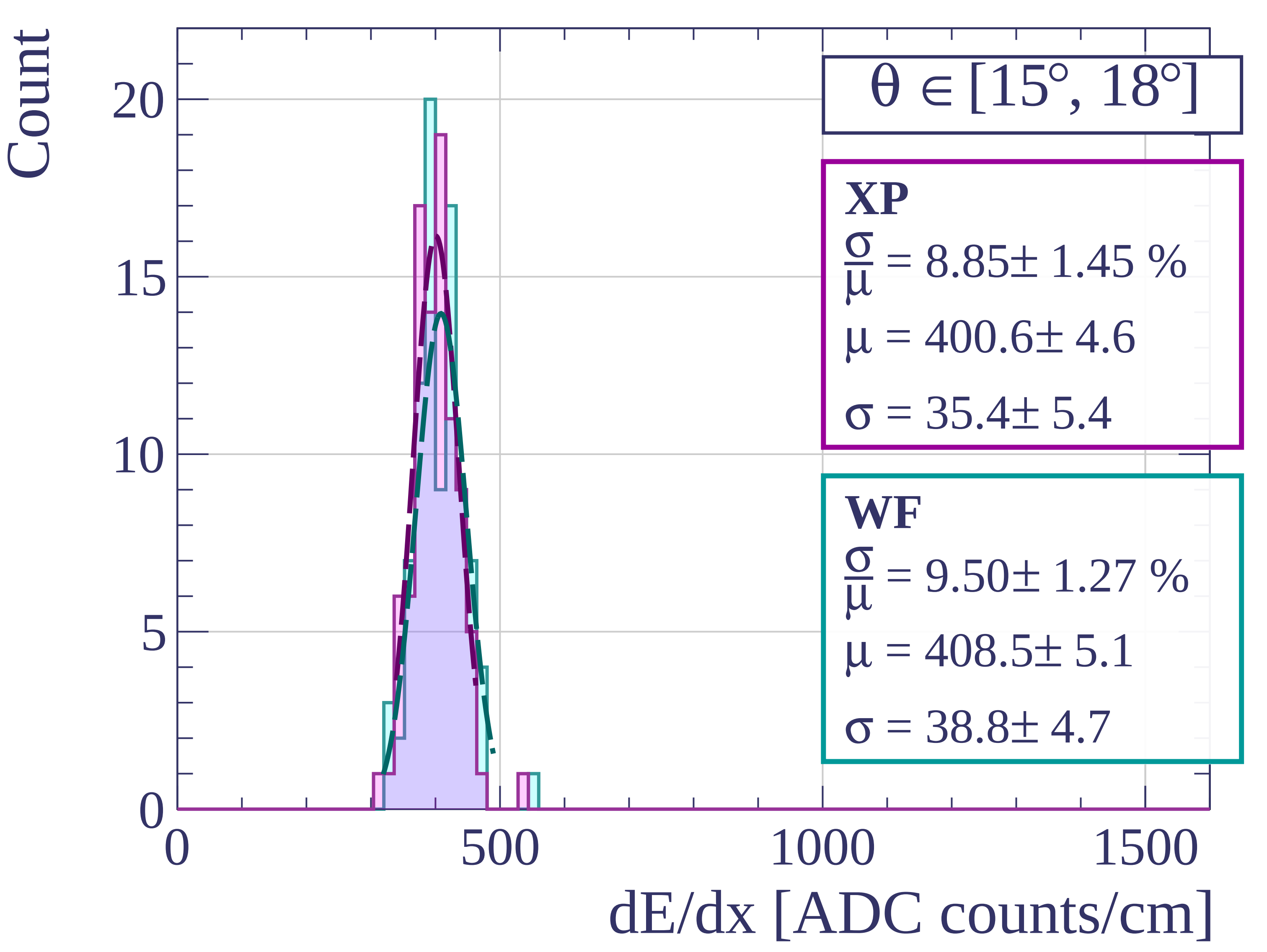
Count
dE/dx [ADC counts/cm]
0
10
20
0
500
1000
1500
θ ∈ [6°, 9°]
XP
σ/μ = 7.21± 0.77 %
μ = 399.7± 3.1
σ = 28.8± 2.9
WF
σ/μ = 7.16± 0.87 %
μ = 405.4± 3.5
σ = 29.0± 3.3

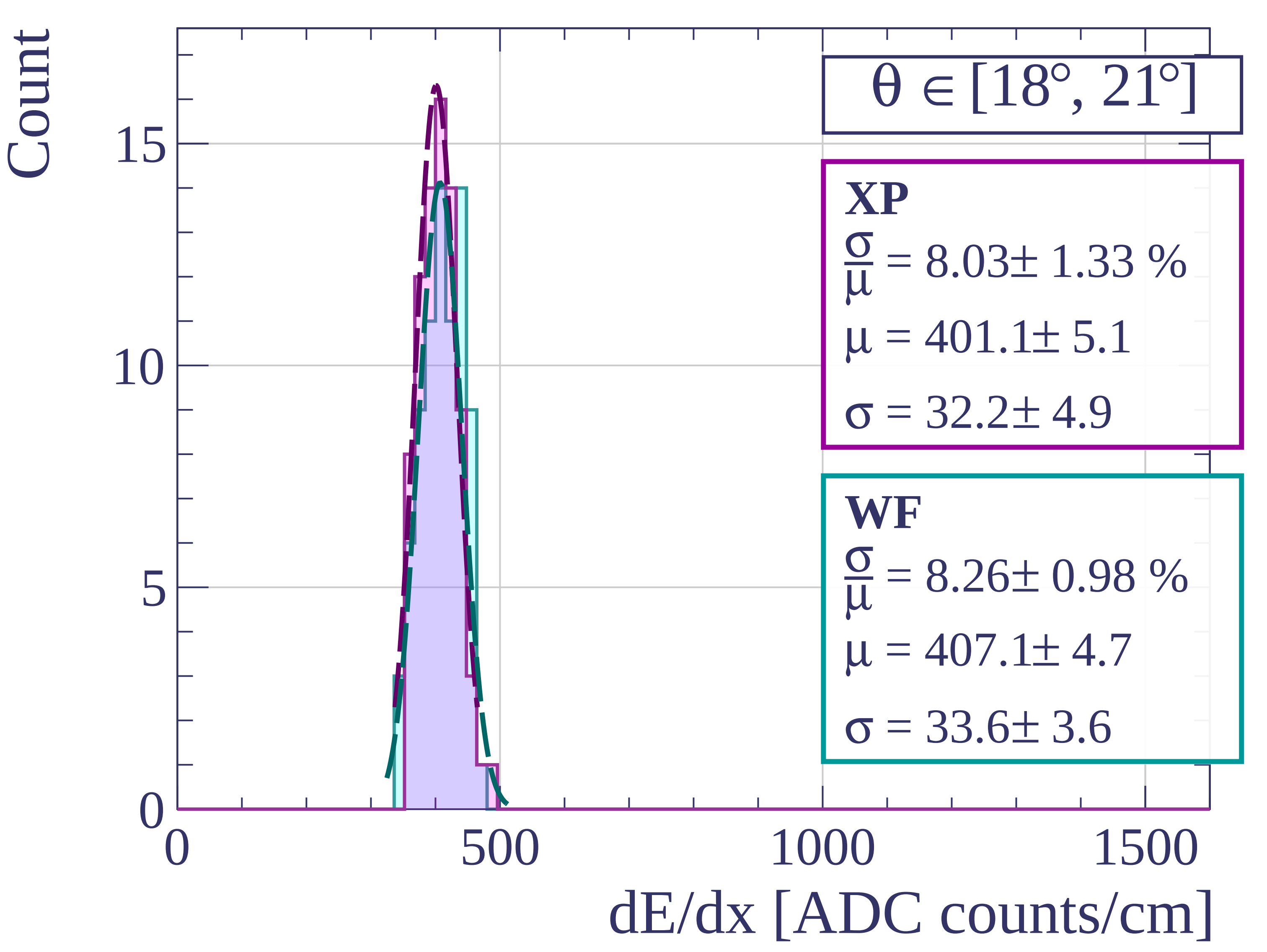


Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [9°, 12°]
XP
σ/μ = 6.65± 1.15 %
μ = 399.1± 3.5
σ = 26.5± 4.3
WF
σ/μ = 7.16± 0.87 %
μ = 402.2± 3.2
σ = 28.8± 3.2

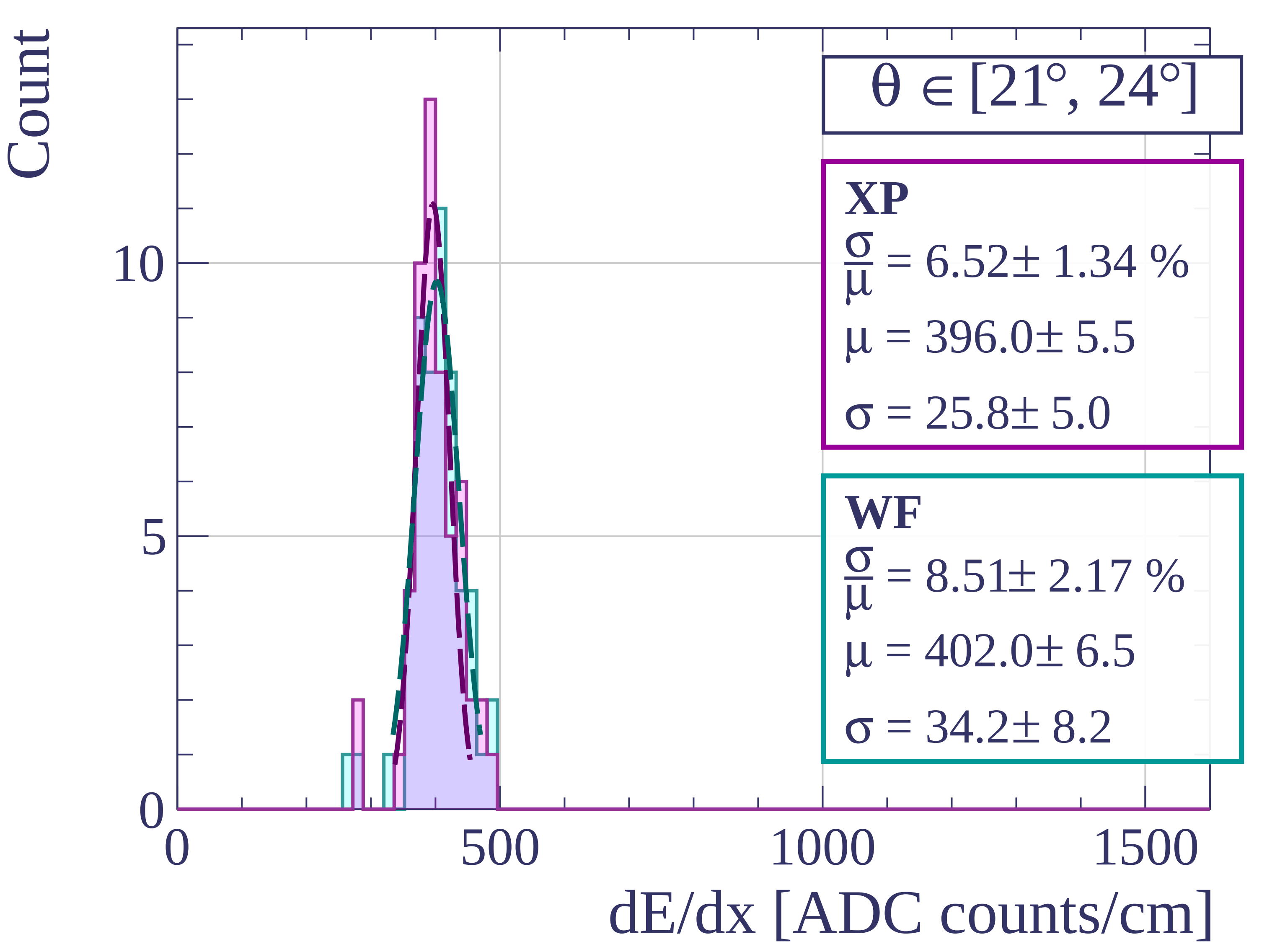


Count
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [12°, 15°]
XP
σ/μ = 6.68± 0.87 %
μ = 399.6± 4.0
σ = 26.7± 3.2
WF
σ/μ = 5.99± 0.86 %
μ = 410.8± 3.5
σ = 24.6± 3.3

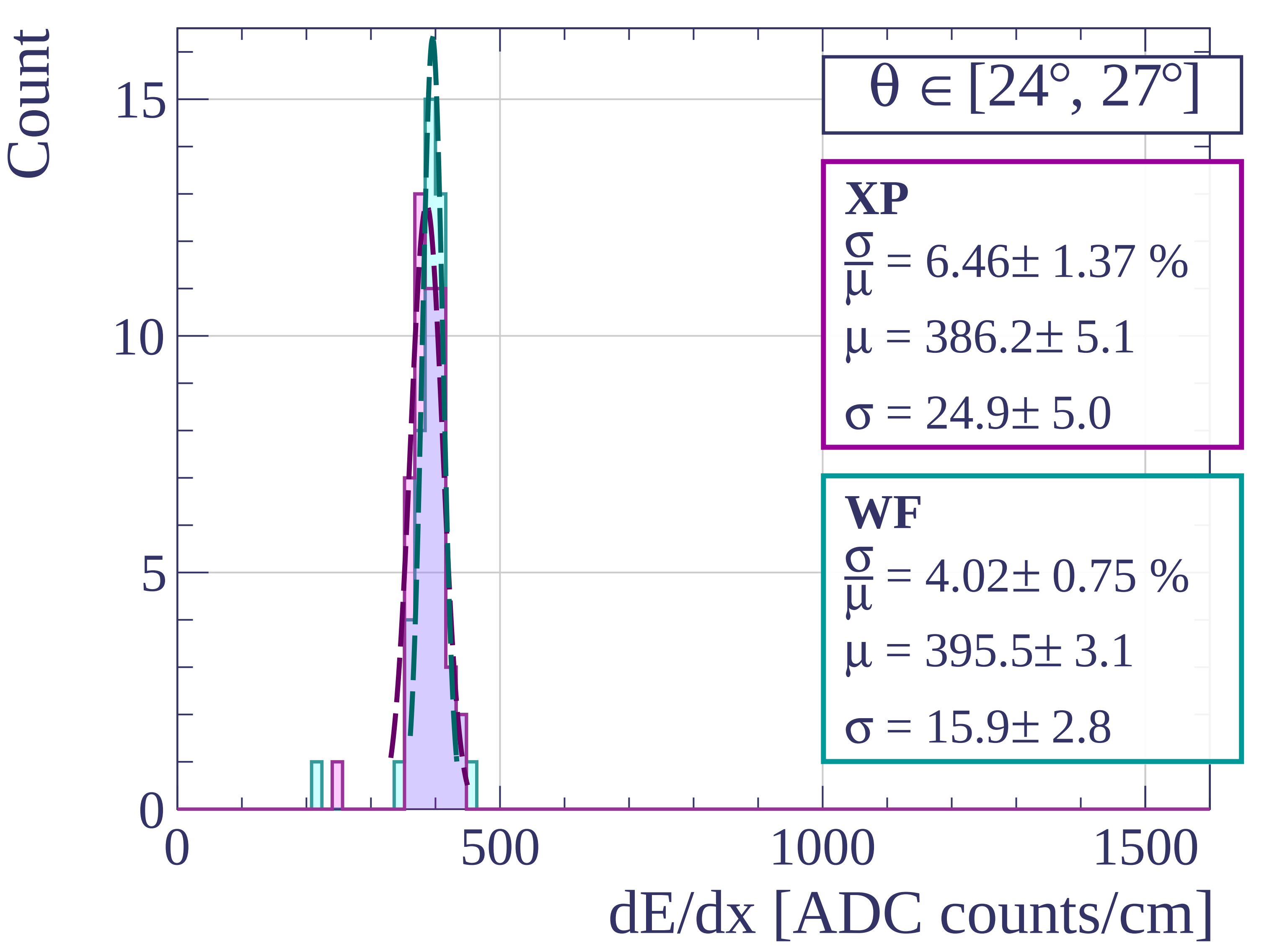

Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [15°, 18°]
XP
σ/μ = 8.85± 1.45 %
μ = 400.6± 4.6
σ = 35.4± 5.4
WF
σ/μ = 9.50± 1.27 %
μ = 408.5± 5.1
σ = 38.8± 4.7



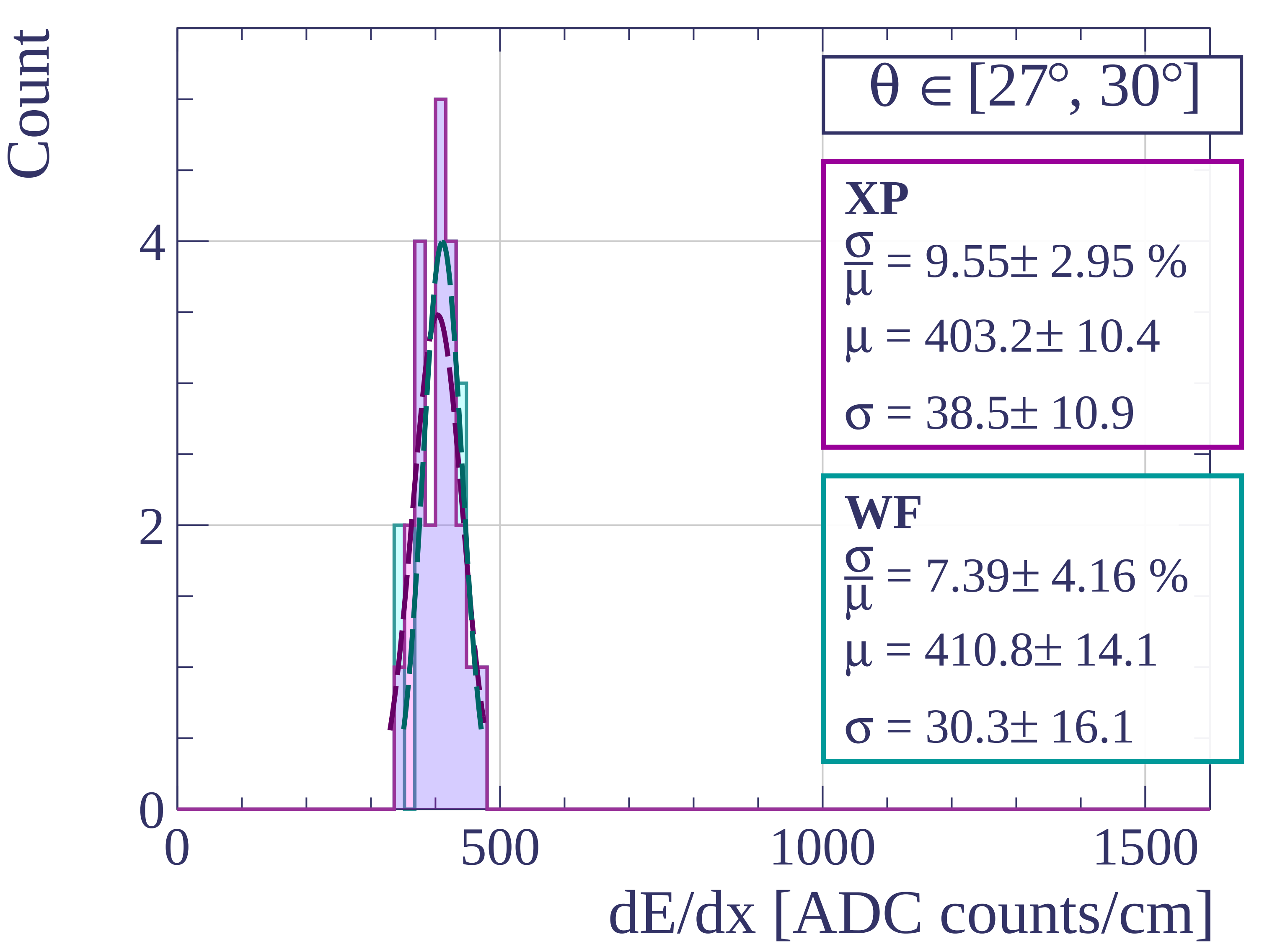
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [18°, 21°]
XP
σ/μ = 8.03± 1.33 %
μ = 401.1± 5.1
σ = 32.2± 4.9
WF
σ/μ = 8.26± 0.98 %
μ = 407.1± 4.7
σ = 33.6± 3.6

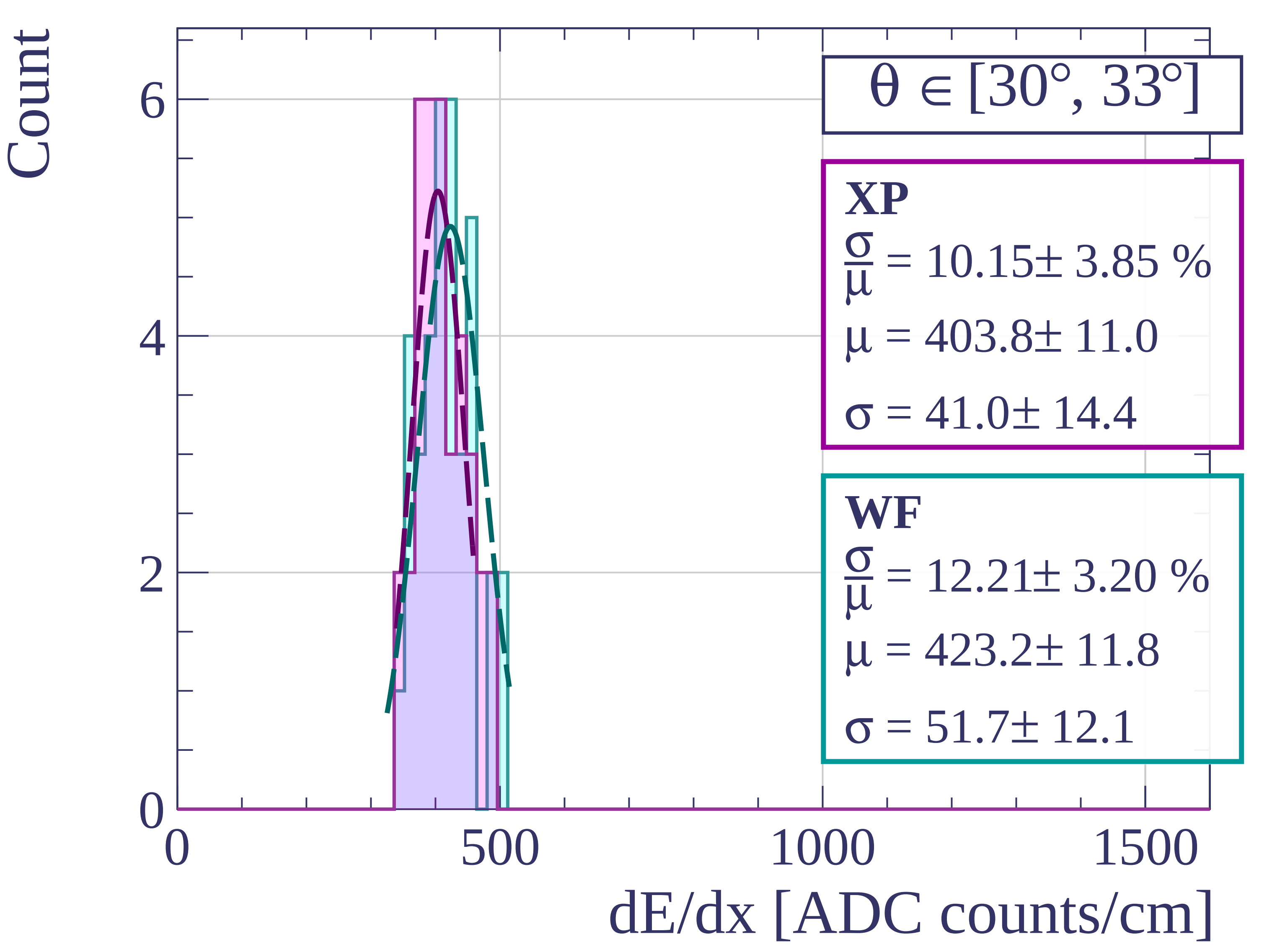


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [21°, 24°]
XP
σ/μ = 6.52± 1.34 %
μ = 396.0± 5.5
σ = 25.8± 5.0
WF
σ/μ = 8.51± 2.17 %
μ = 402.0± 6.5
σ = 34.2± 8.2

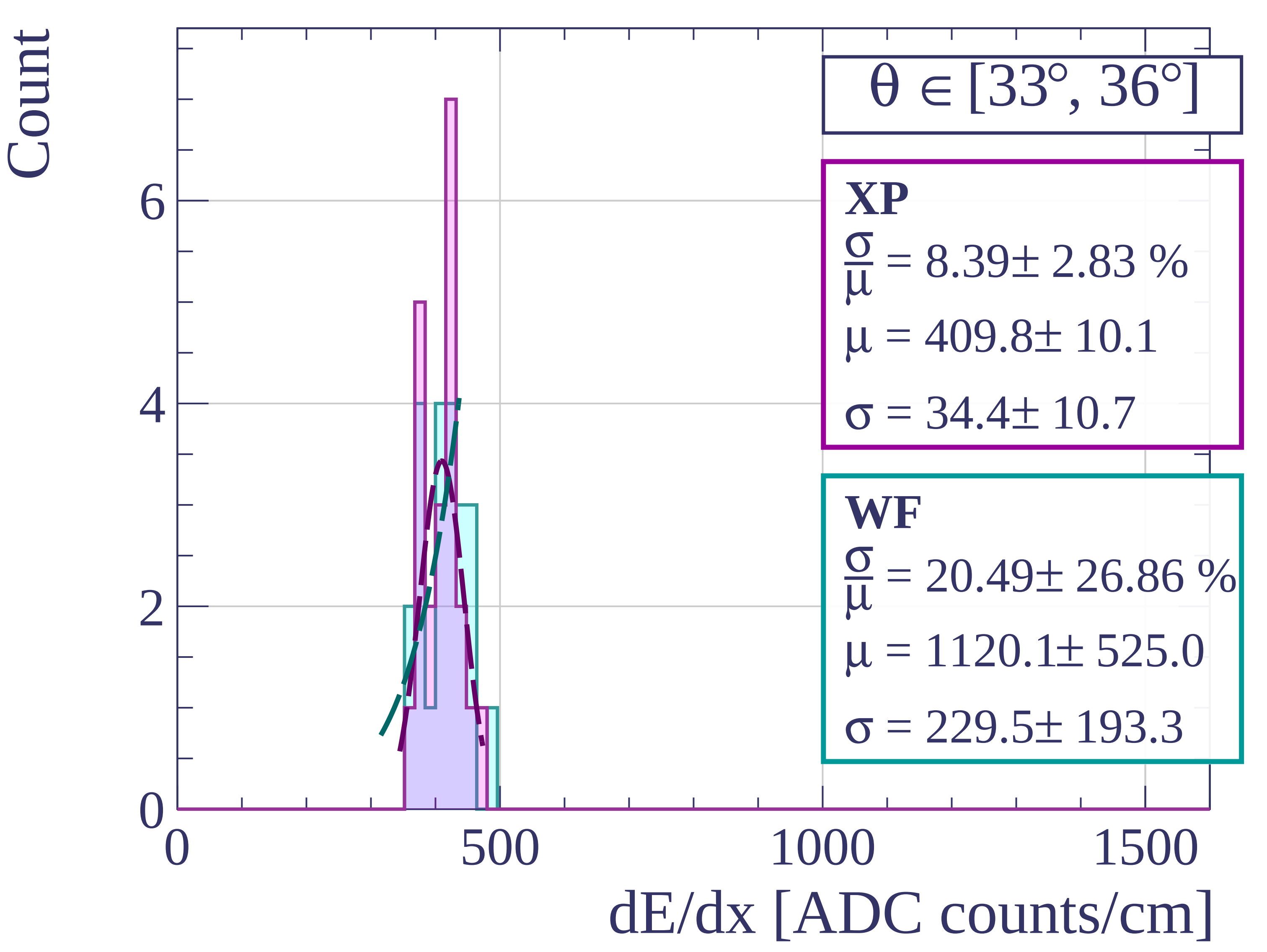


Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [24°, 27°]
XP
σ/μ = 6.46± 1.37 %
μ = 386.2± 5.1
σ = 24.9± 5.0
WF
σ/μ = 4.02± 0.75 %
μ = 395.5± 3.1
σ = 15.9± 2.8

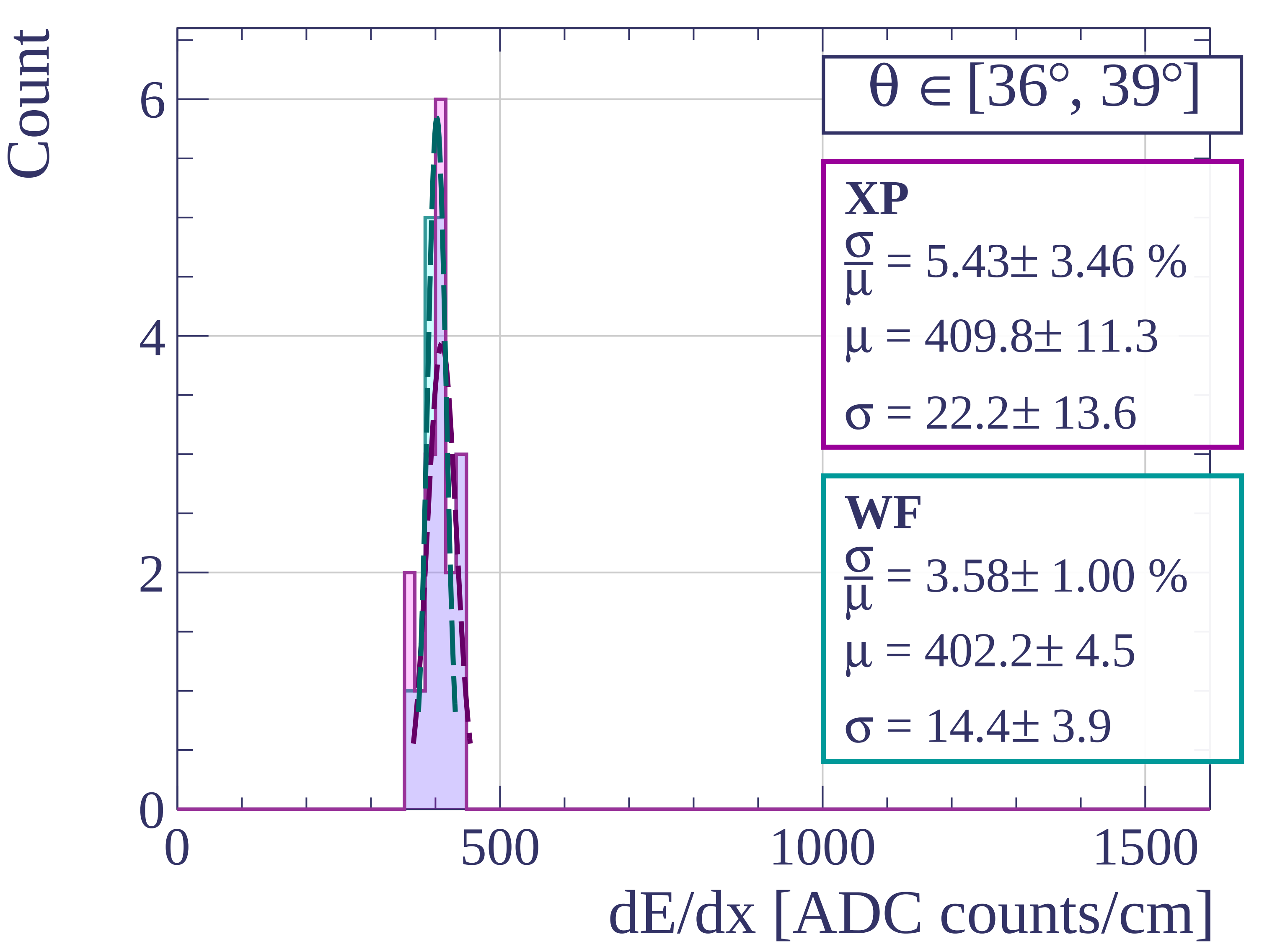

θ ∈ [27°, 30°]
XP
σ/μ = 9.55± 2.95 %
μ = 403.2± 10.4
σ = 38.5± 10.9
WF
σ/μ = 7.39± 4.16 %
μ = 410.8± 14.1
σ = 30.3± 16.1
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500



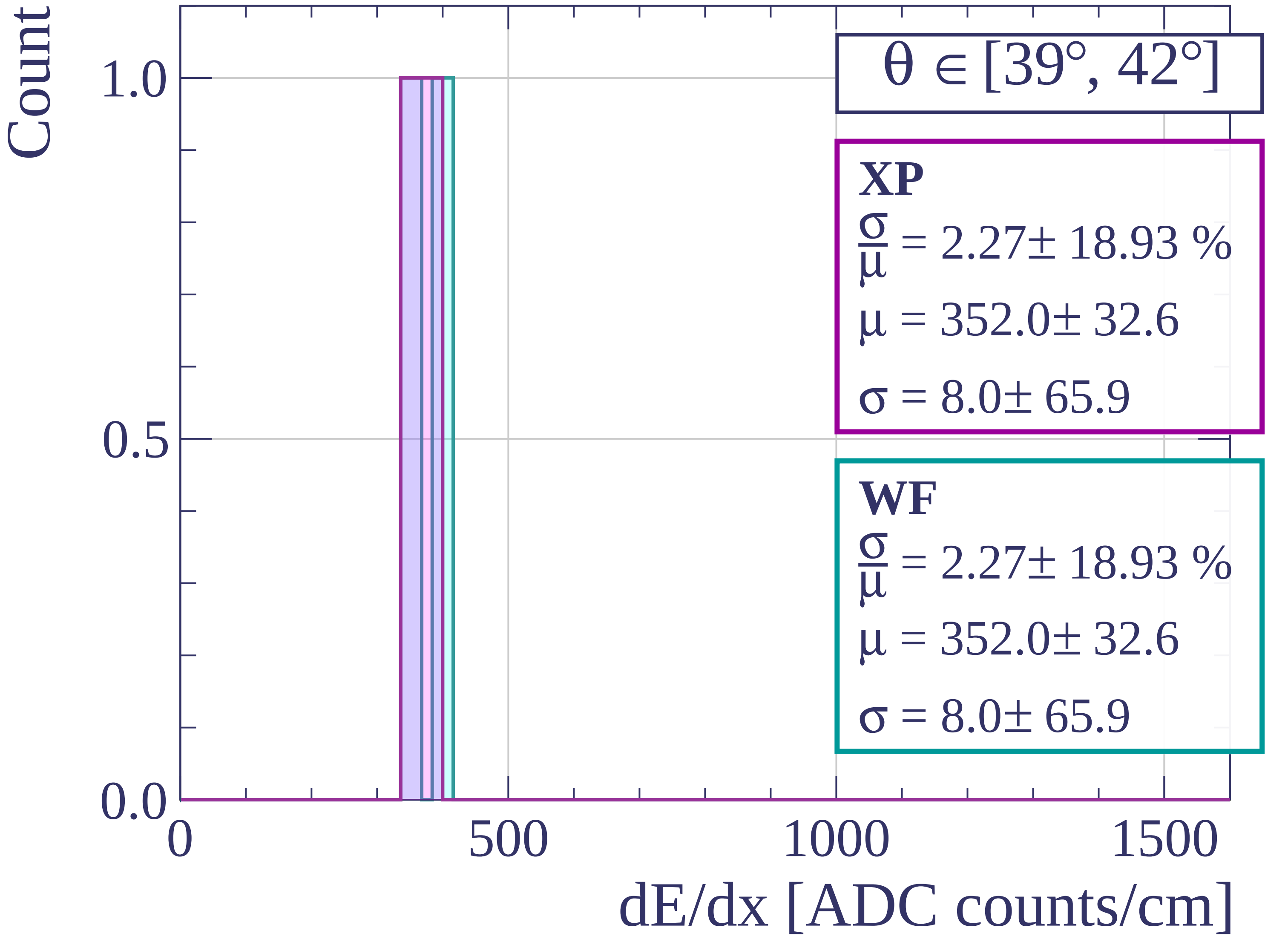
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
θ ∈ [30°, 33°]
XP
σ/μ = 10.15± 3.85 %
μ = 403.8± 11.0
σ = 41.0± 14.4
WF
σ/μ = 12.21± 3.20 %
μ = 423.2± 11.8
σ = 51.7± 12.1



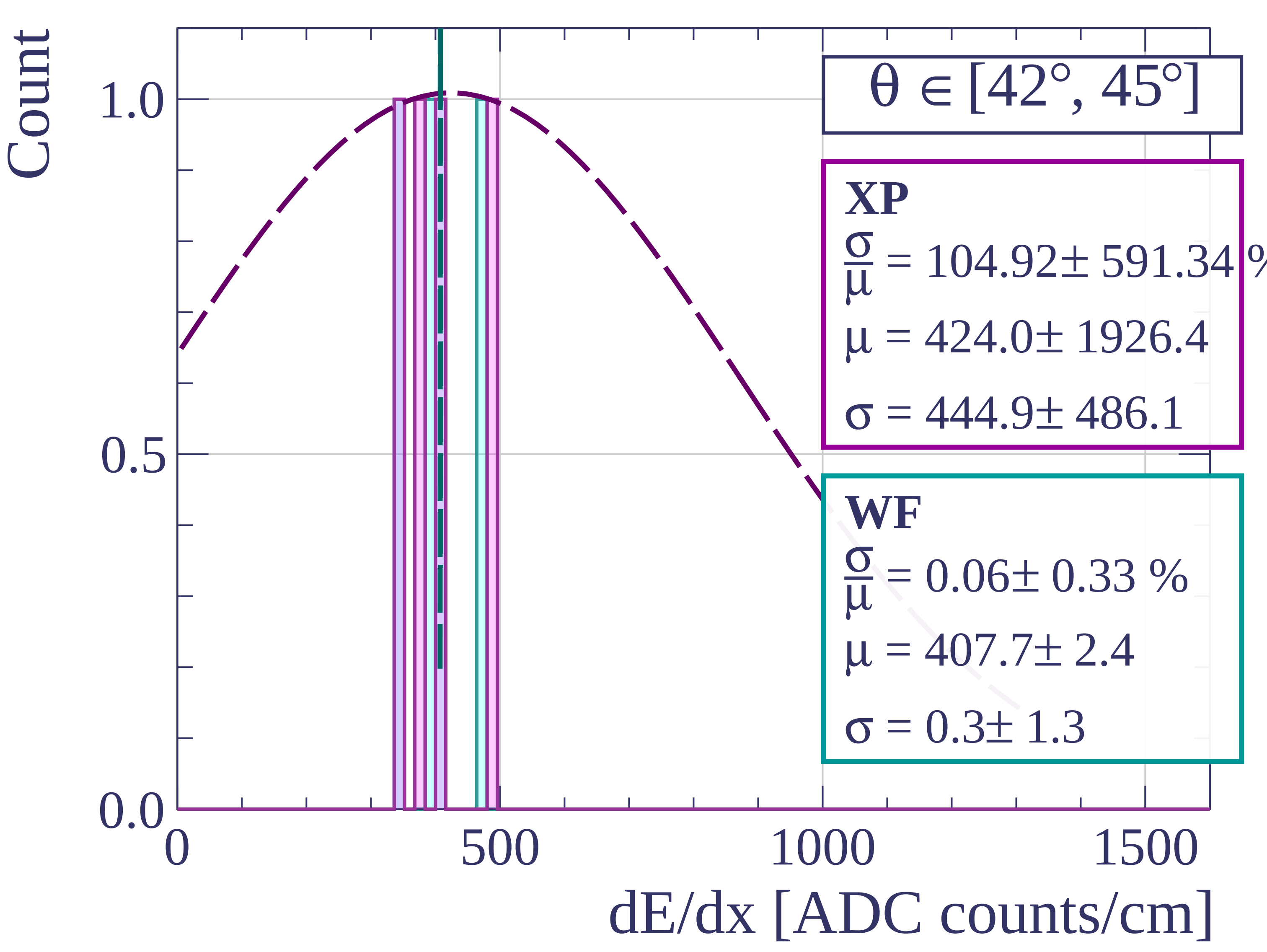
Count
dE/dx [ADC counts/cm]
θ ∈ [33°, 36°]
XP
σ/μ = 8.39± 2.83 %
μ = 409.8± 10.1
σ = 34.4± 10.7
WF
σ/μ = 20.49± 26.86 %
μ = 1120.1± 525.0
σ = 229.5± 193.3
0
2
4
6
0
500
1000
1500

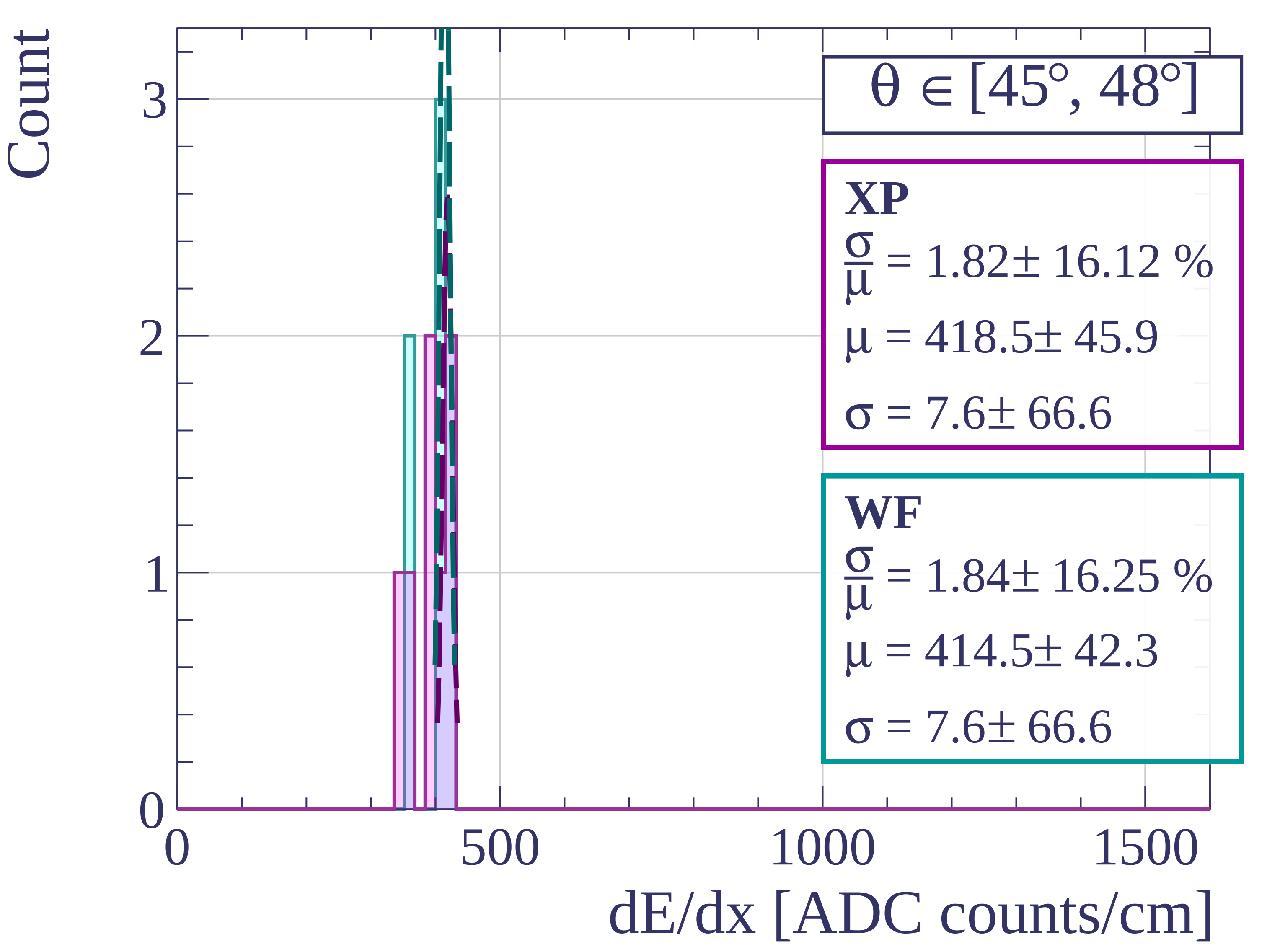


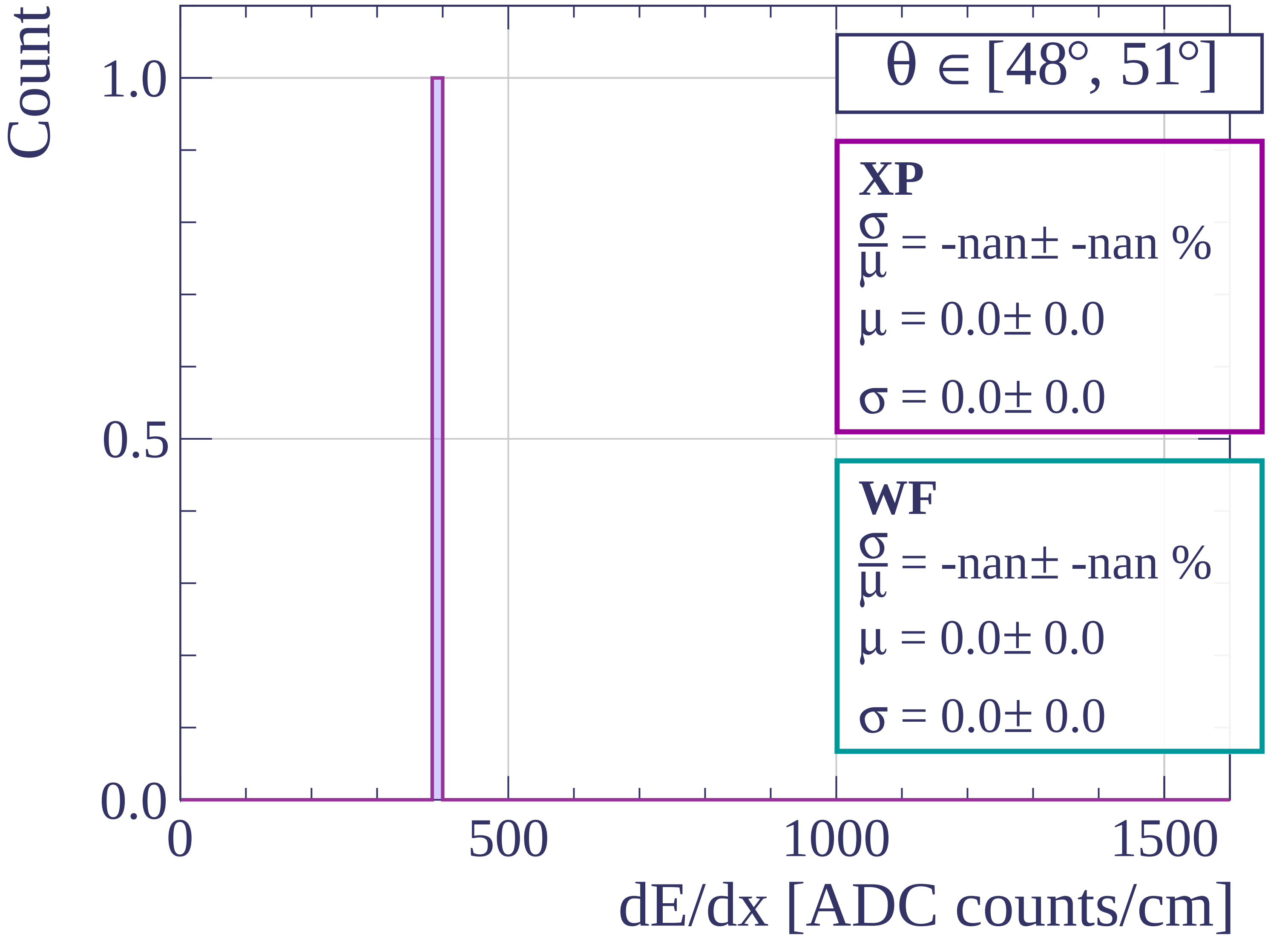
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
θ ∈ [36°, 39°]
XP
σ/μ = 5.43± 3.46 %
μ = 409.8± 11.3
σ = 22.2± 13.6
WF
σ/μ = 3.58± 1.00 %
μ = 402.2± 4.5
σ = 14.4± 3.9

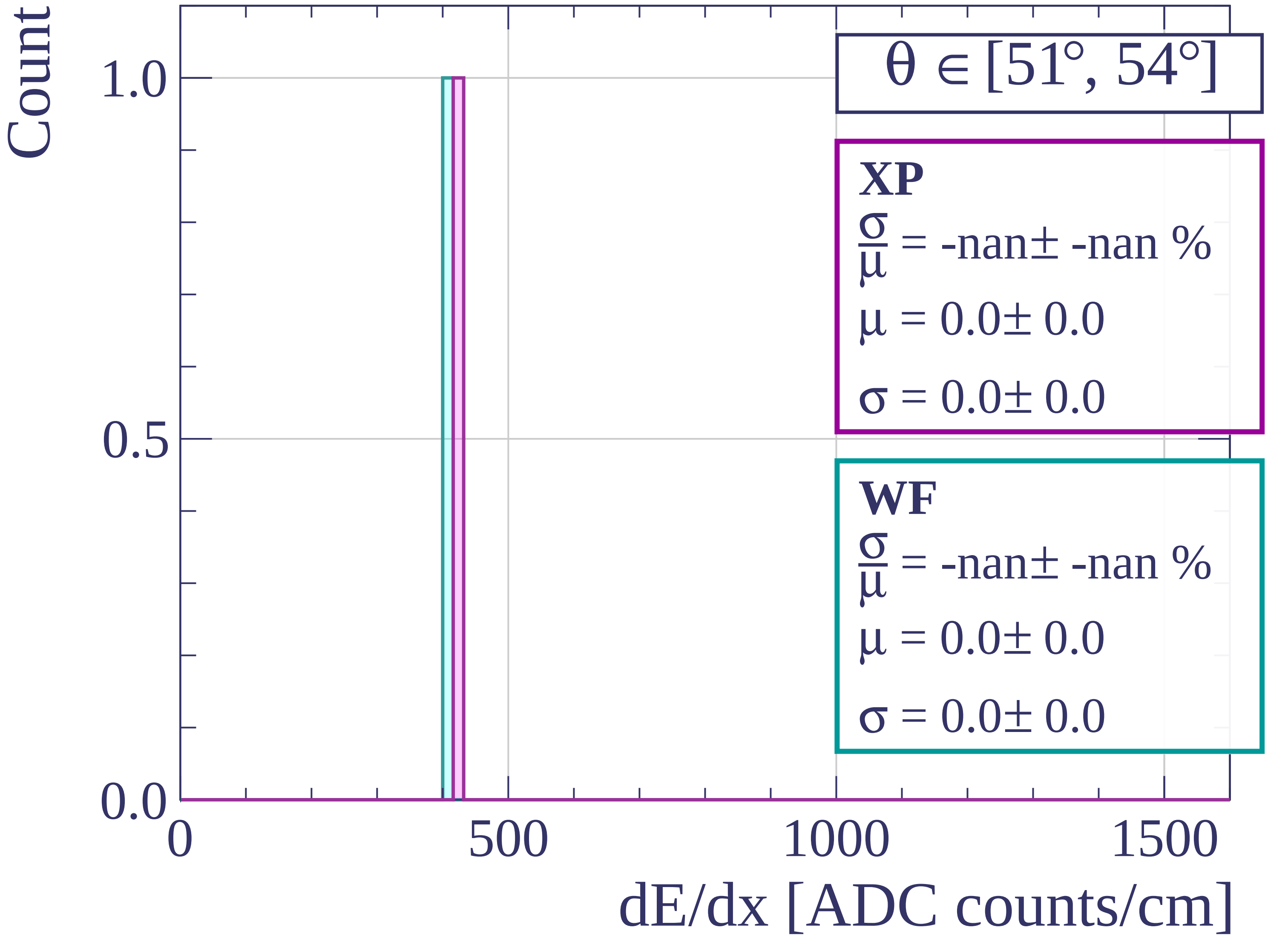


Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [39°, 42°]
XP
σ/μ = 2.27± 18.93 %
μ = 352.0± 32.6
σ = 8.0± 65.9
WF
σ/μ = 2.27± 18.93 %
μ = 352.0± 32.6
σ = 8.0± 65.9

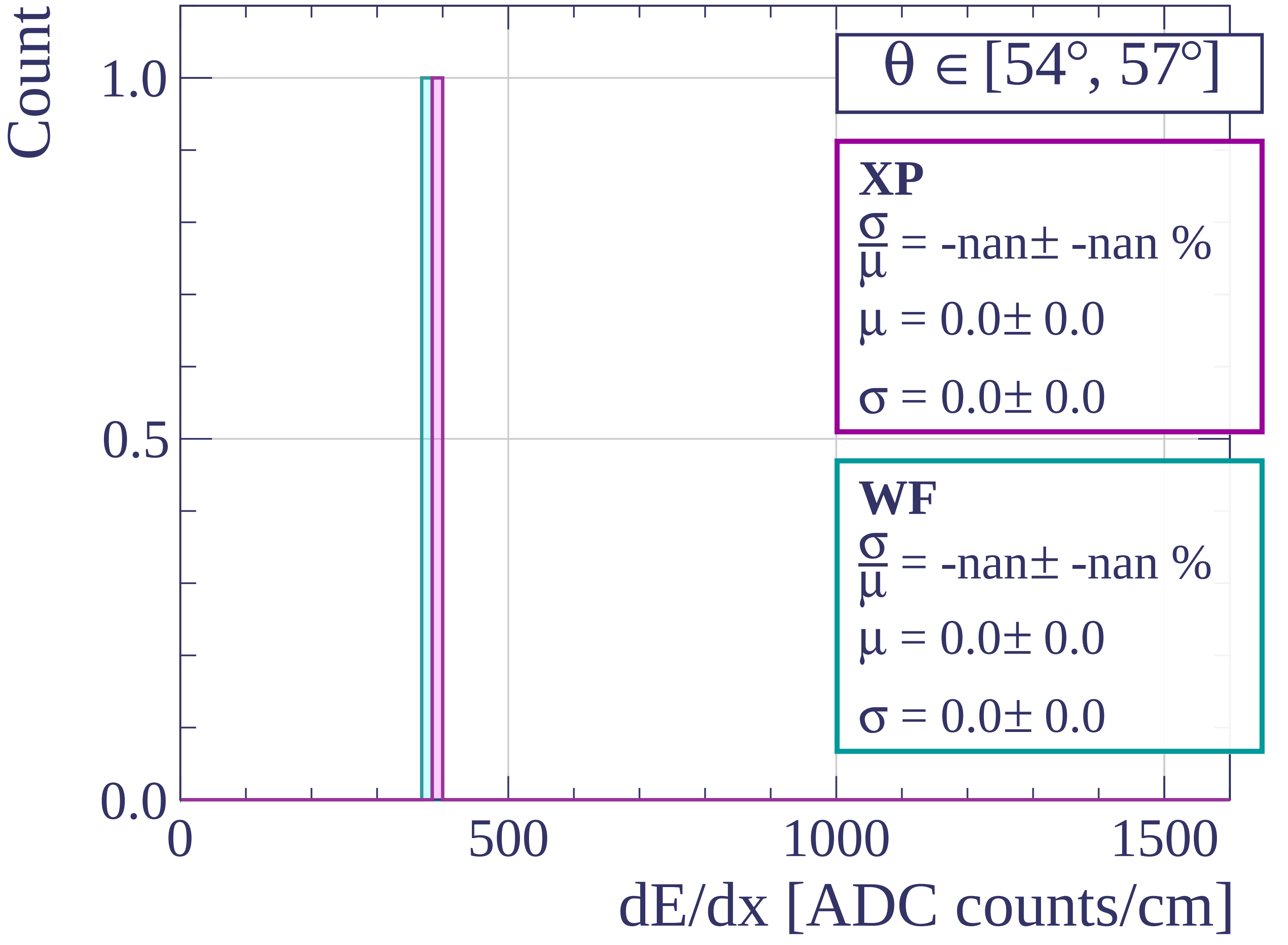

Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [42°, 45°]
XP
σ/μ = 104.92± 591.34 %
μ = 424.0± 1926.4
σ = 444.9± 486.1
WF
σ/μ = 0.06± 0.33 %
μ = 407.7± 2.4
σ = 0.3± 1.3

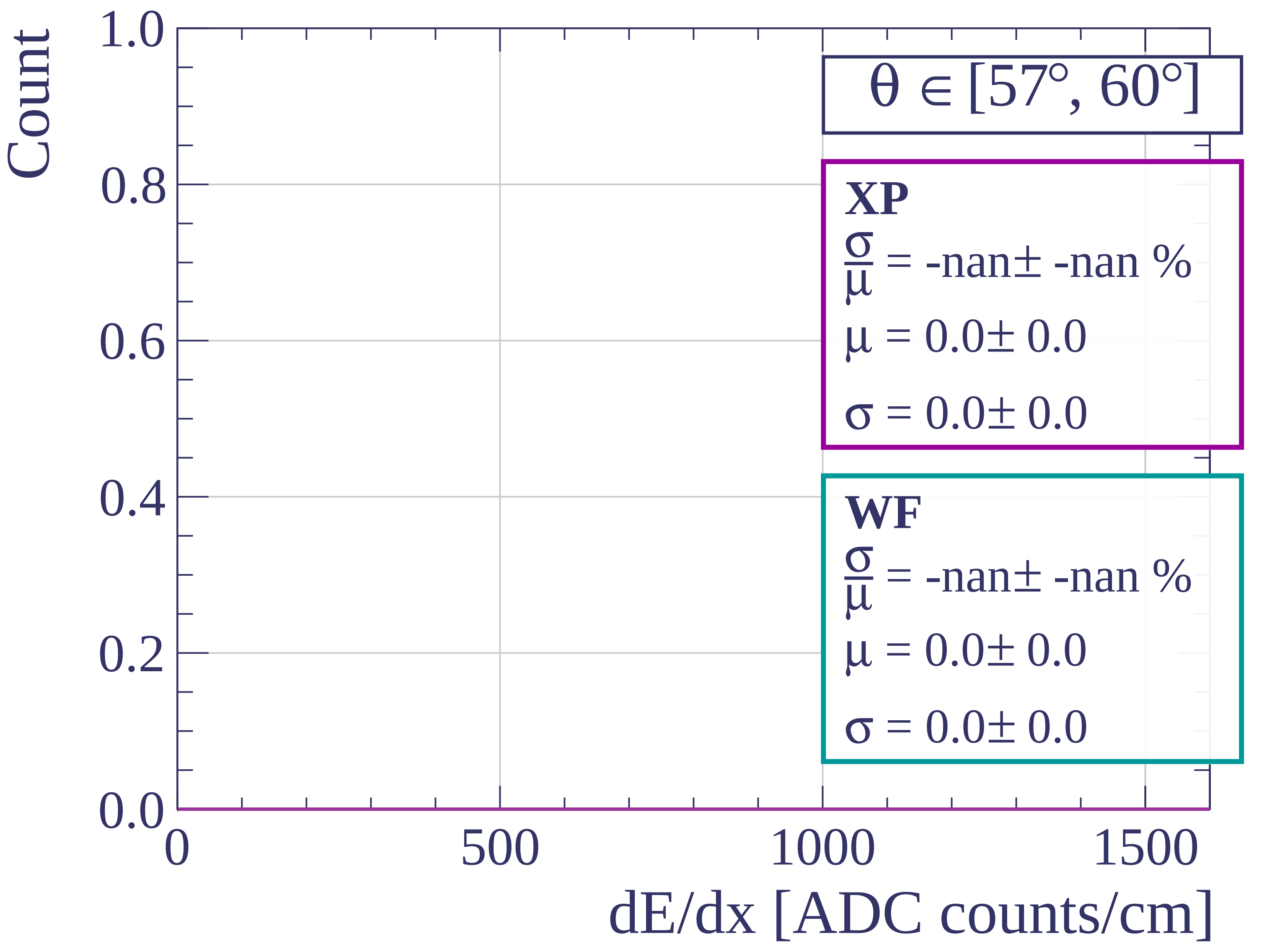

Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [45°, 48°]
XP
σ/μ = 1.82± 16.12 %
μ = 418.5± 45.9
σ = 7.6± 66.6
WF
σ/μ = 1.84± 16.25 %
μ = 414.5± 42.3
σ = 7.6± 66.6

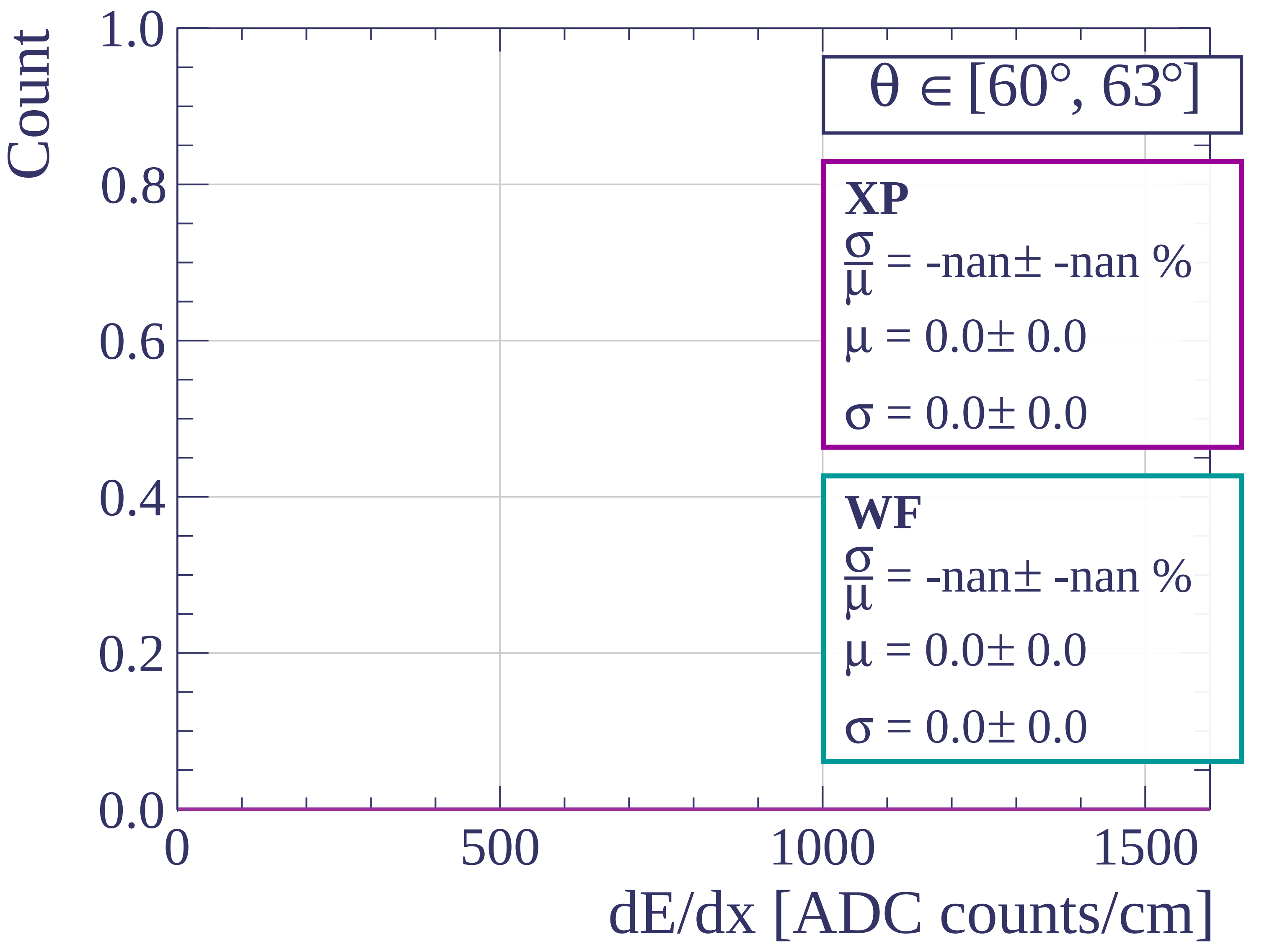

Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [48°, 51°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



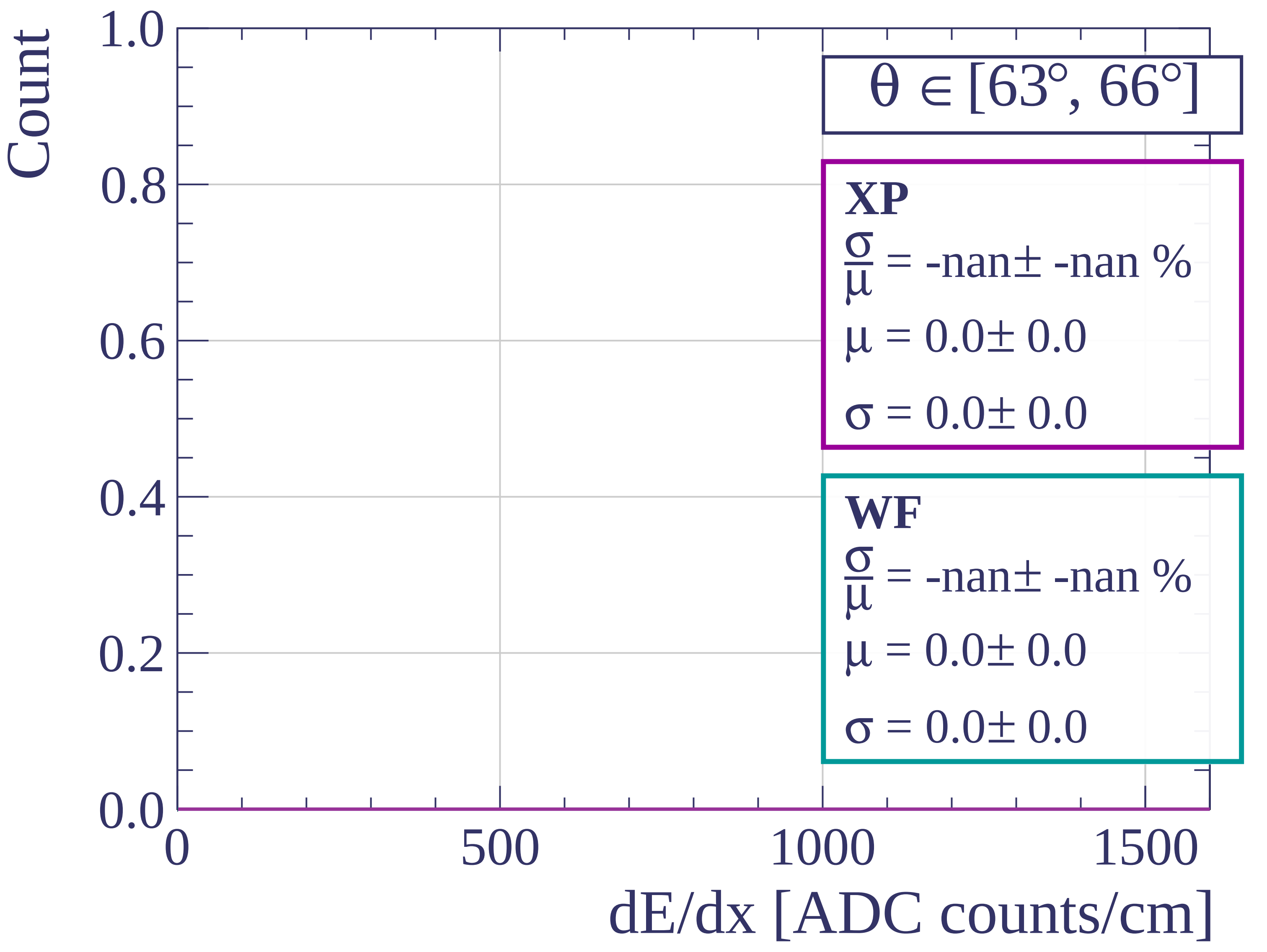
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [51°, 54°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [54°, 57°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

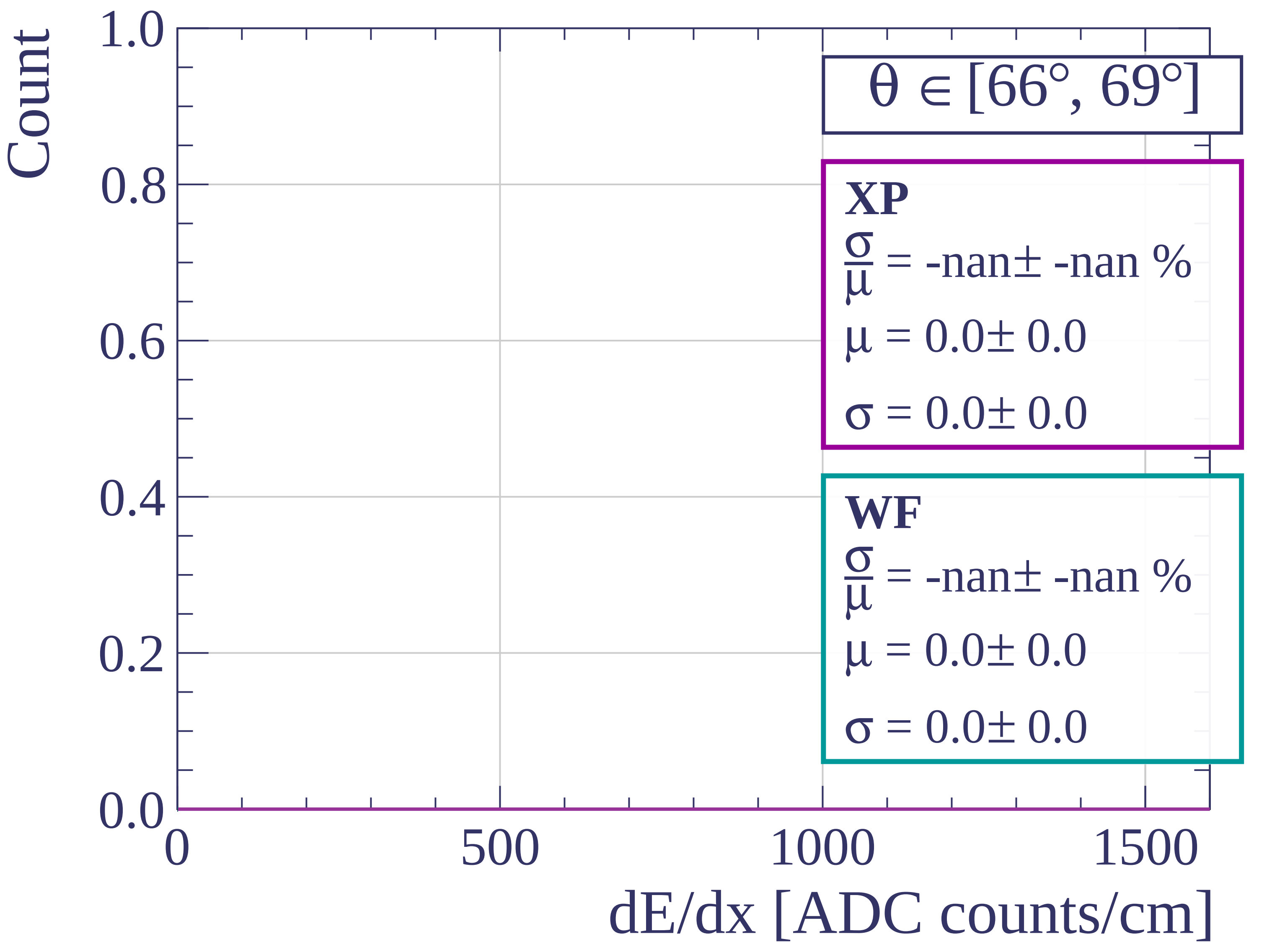

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [57°, 60°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



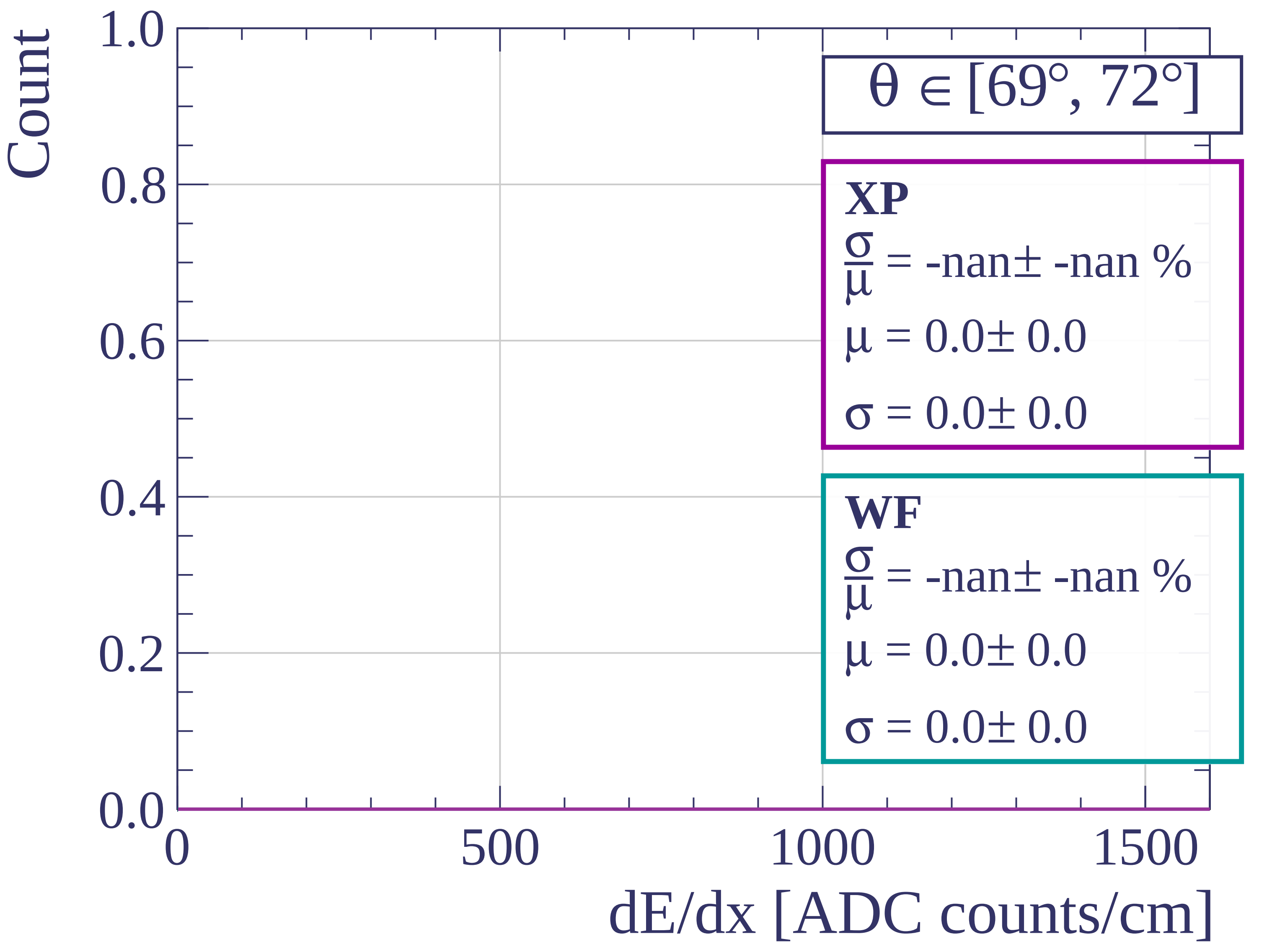
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [60°, 63°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



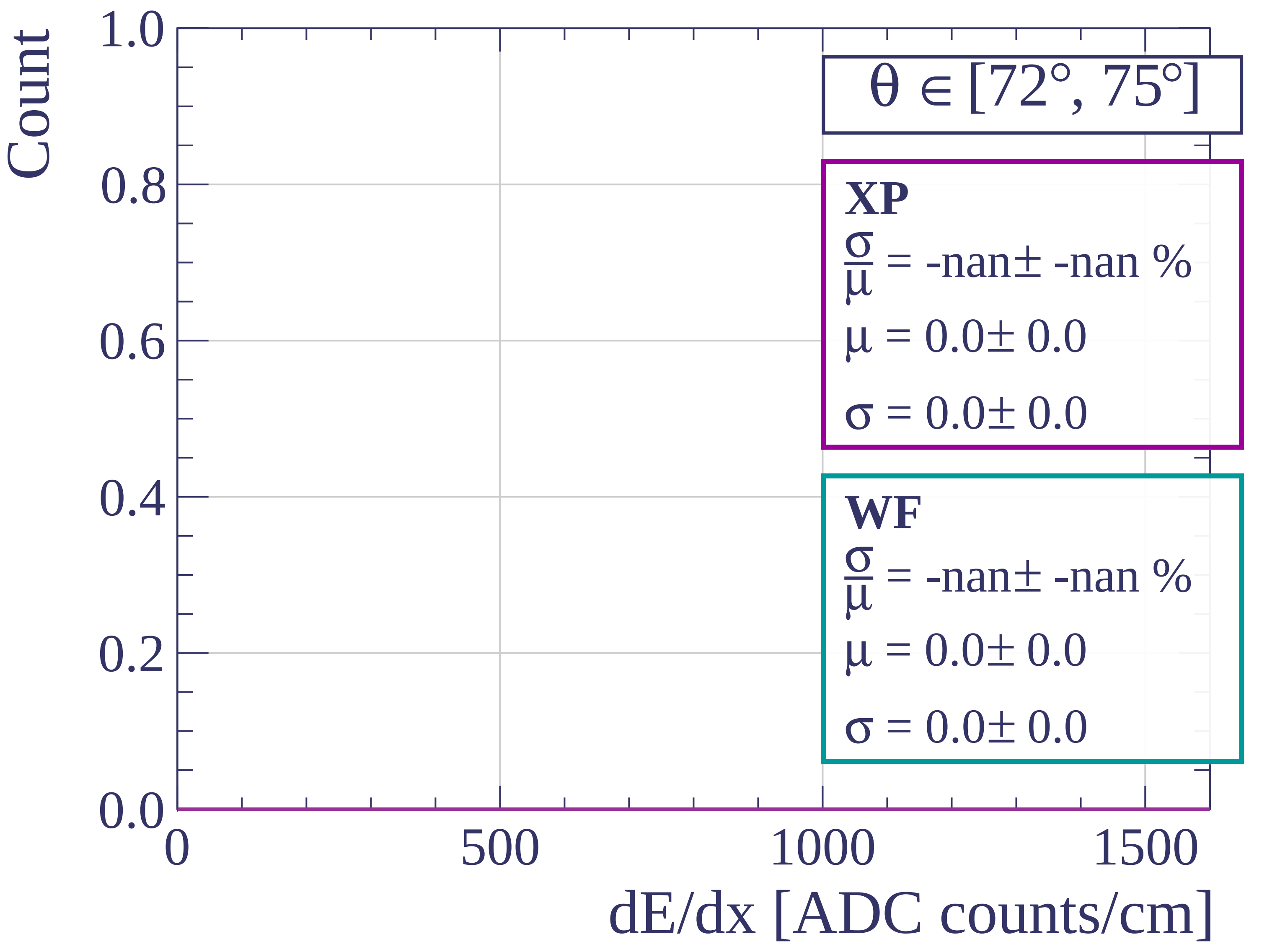
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [63°, 66°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



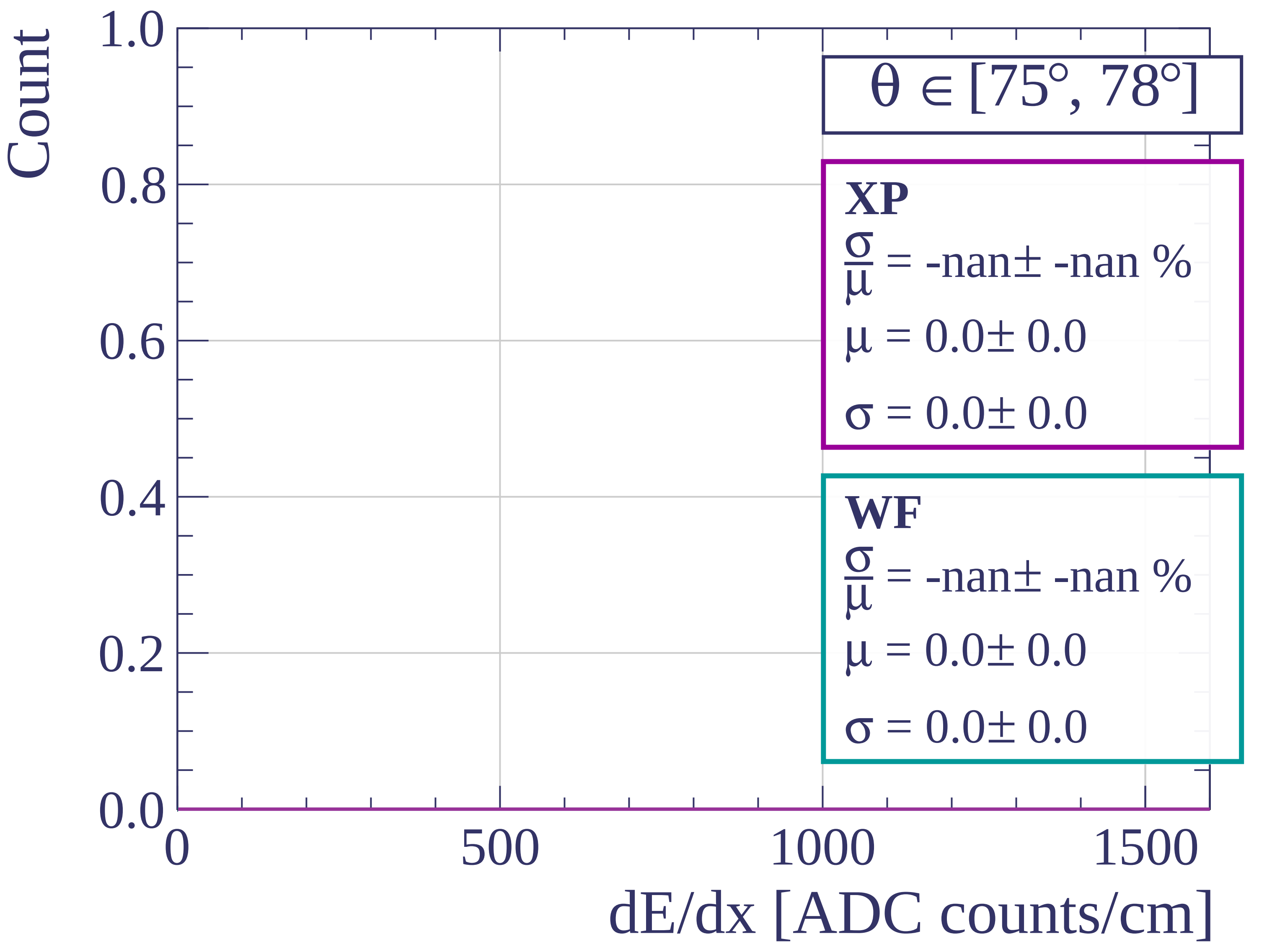
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [66°, 69°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

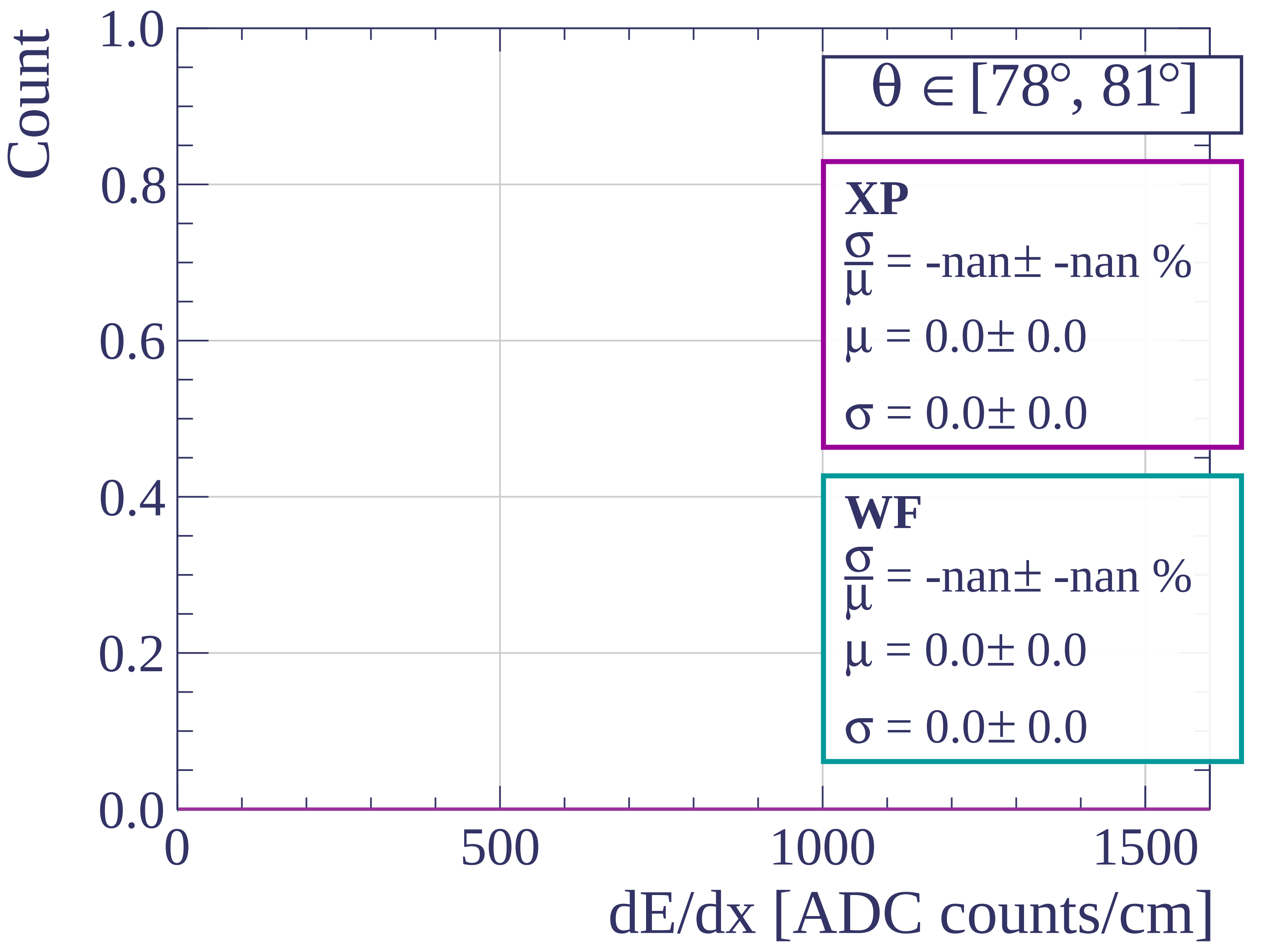


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [69°, 72°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

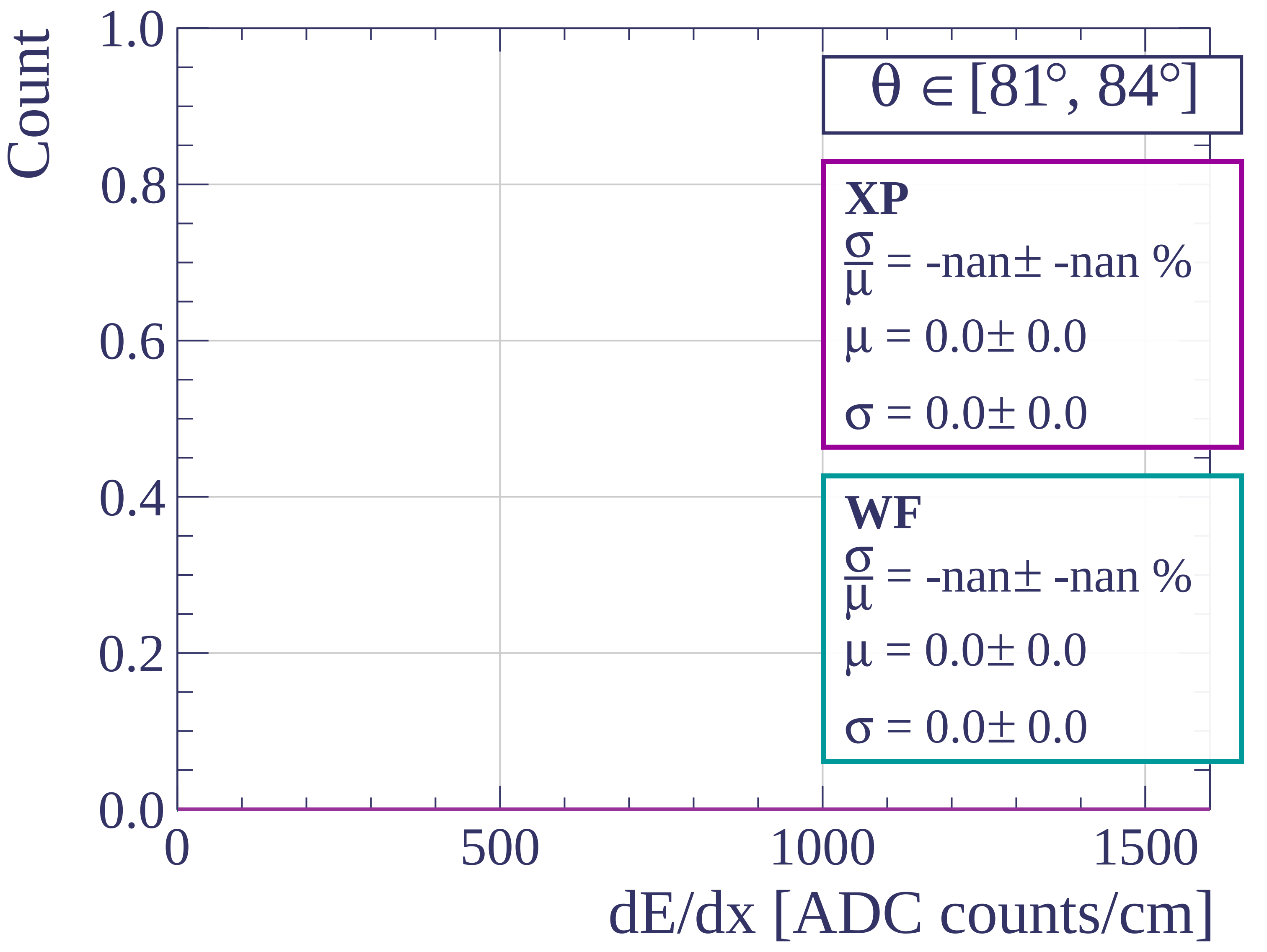


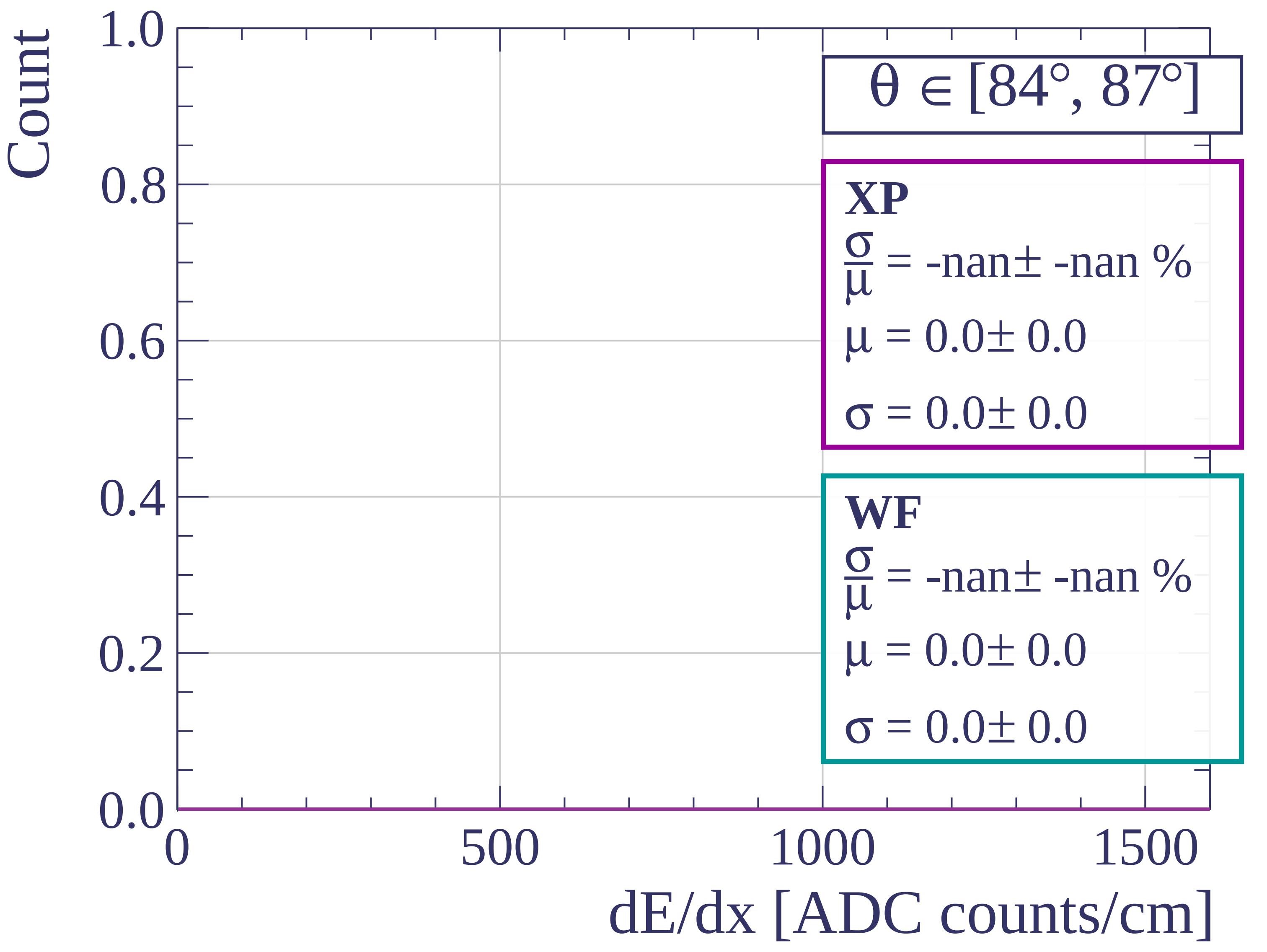
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [72°, 75°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

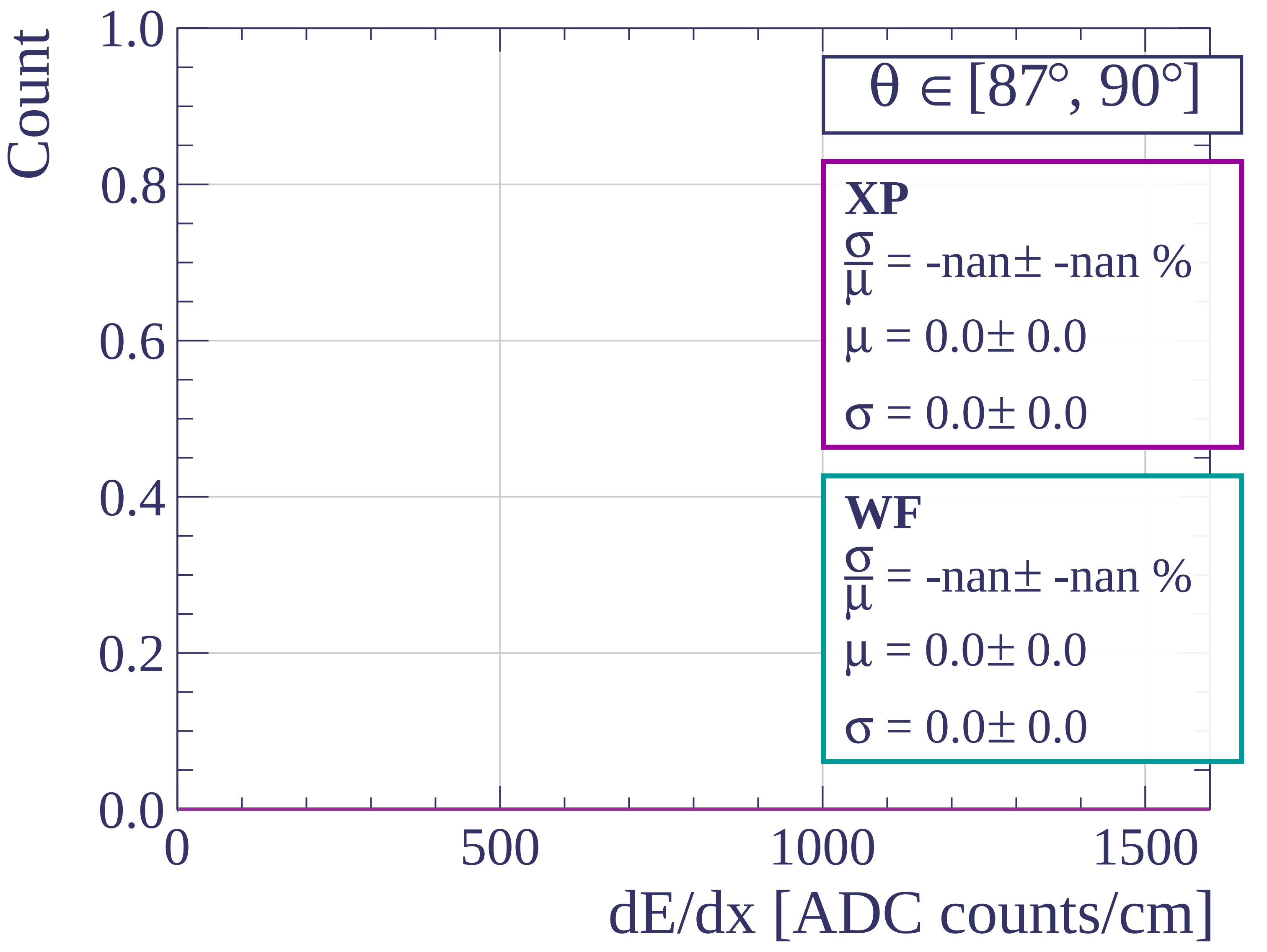

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [75°, 78°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

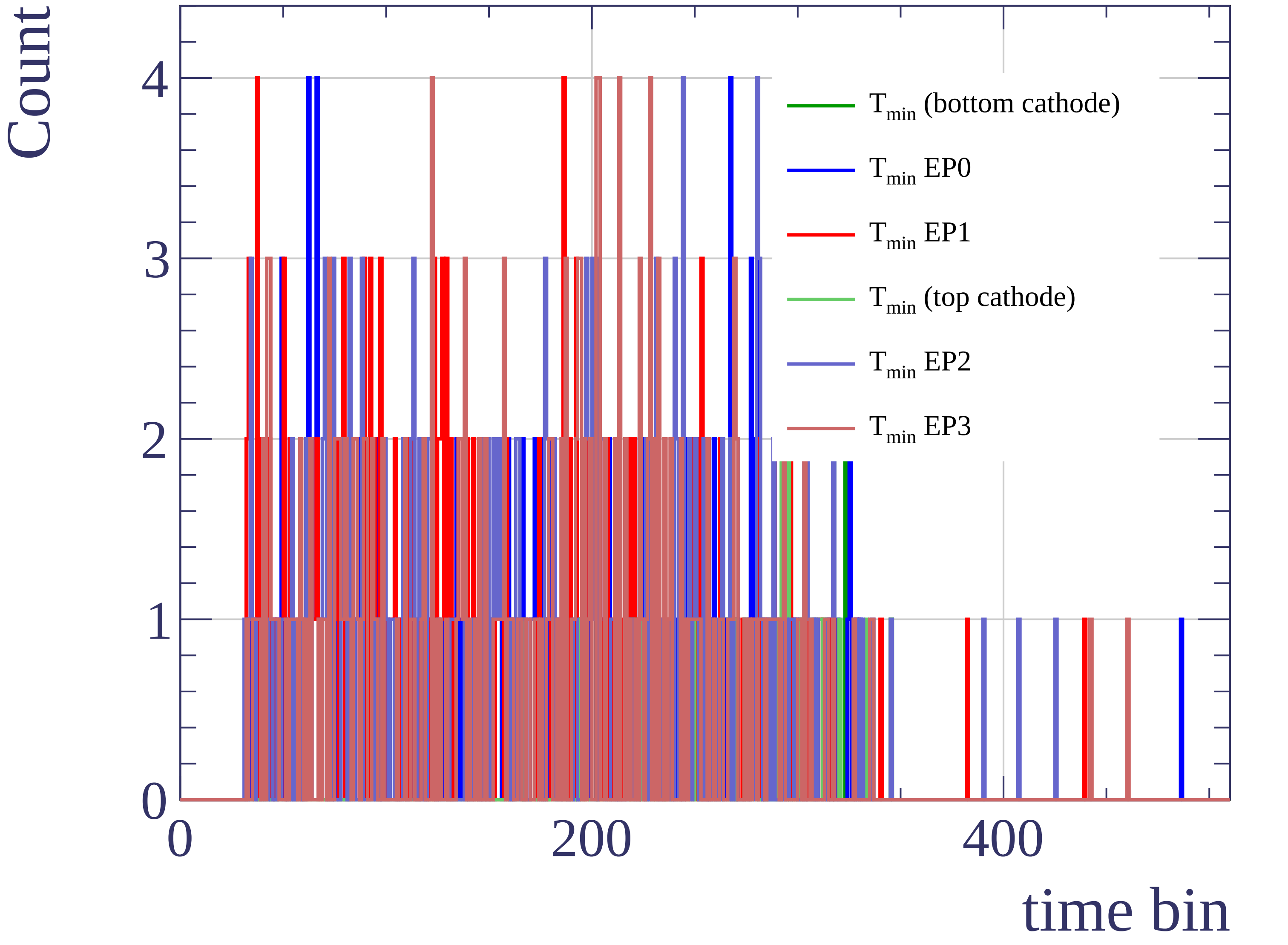


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [78°, 81°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

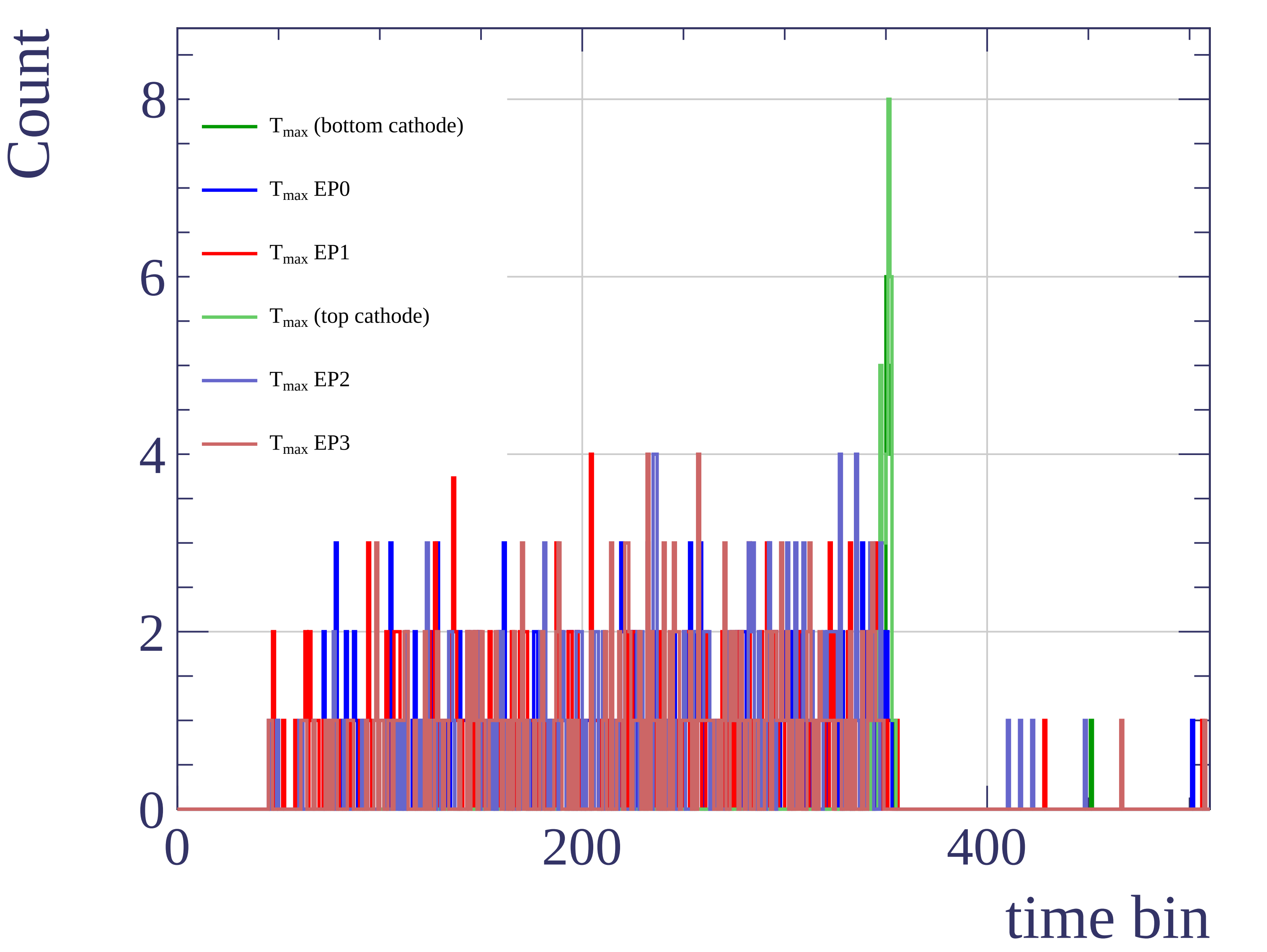

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [81°, 84°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [84°, 87°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [87°, 90°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
200
400
0
1
2
3
4
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



Count
time bin
0
2
4
6
8
0
200
400
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3